CUSAT CAT

2023 · 2024 · 2025

Chapter-Wise Question Bank

Physics · Chemistry · Mathematics

3147	14	3 Years	75
Total Questions	Exam Shifts	2023–2025	Chapters

PHYSICS	1049 questions · Q1–75 per shift · 29 chapters
CHEMISTRY	839 questions · Q76–135 per shift · 29 chapters
MATHEMATICS	1259 questions · Q136–225 per shift · 17 chapters

Every question tagged with year & shift · 2×2 option grid · Answer key after each chapter

PHYSICS

NCERT Chapter-Wise Compilation

CUSAT CAT 2023 · 2024 · 2025 — All 14 Shifts

Units & Measurements

51 questions

Q1. *CUSAT CAT 2023 - Shift 1*

In an instrument used for measuring angle, 29 divisions of the main scale exactly coincides with 30 divisions of the Vernier scale. If the smallest division on the main scale is 0.5°, then the least count of the instrument will be

(A) one degree	**(B)** half minute
(C) half degree	**(D)** one minute

Q2. *CUSAT CAT 2023 - Shift 1*

The damping force of an oscillator is directly proportional to the velocity. The unit of constant of proportionality is –1

(A) kgs	**(B)** kgs –1
(C) kgms –2	**(D)** kgms

Q3. *CUSAT CAT 2023 - Shift 1*

The SI unit of luminous intensity is

(A) Lux	**(B)** Lumen
(C) Candela	**(D)** Candela power

Q4. *CUSAT CAT 2023 - Shift 1*

The dimension of torque is –2

(A) MLT 2 –2	**(B)** ML T 2 2 –2
(C) M L T 2 –2	**(D)** M LT

Q5. *CUSAT CAT 2023 - Shift 1*

The unit of angular acceleration in the SI system is –1

(A) N kg –2	**(B)** m s –2
(C) rad s –1	**(D)** m kg K CHEMISTRY (UG) – SHIFT I (FINAL) 5

Q6. *CUSAT CAT 2023 - Shift 2*

If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale, the least count of the screw gauge is

(A) 0.001 cm	**(B)** 0.01 cm
(C) 0.001 mm	**(D)** 1 mm

Q7. *CUSAT CAT 2023 - Shift 2*

The unit of specific resistance is

(A) Ω 2	**(B)** Ω
(C) Ω metre	**(D)** Ω/m

Q8. *CUSAT CAT 2023 - Shift 3*

Which one of the following is TRUE for a conductor?

(A) The resistivity of a conducting wire is proportional to its length	**(B)** The resistivity of a conducting wire is inversely proportional to its area of cross section
(C) The resistivity of a conducting wire is dependent both on its length and area of cross section	**(D)** The resistivity is a property of the material which does not depend on dimensions of the conductor but depends on nature and temperature of the conductor

Q9. *CUSAT CAT 2023 - Shift 3*

The velocity of a particle in time t is given by v(t)=a+bt+ct . What will be the dimension of ‘c’?

(A) T –2	**(B)** L 2
(C) LT –4	**(D)** LT –5

Q10. *CUSAT CAT 2023 - Shift 3*

Specific resistance is numerically equal to the resistance offered by

(A) 1 cm length of a conductor	**(B)** A conductor of unit cross-section 2
(C) 1 cm length of conductor of 1 cm of cross-section 3	**(D)** 1 cm of a conductor CHEMISTRY (UG)– SHIFT III (FINAL)

Q11. *CUSAT CAT 2023 - Shift 4*

The magnitude of a vector E is 5 unit, F has a magnitude of 6 unit and the cross product of E and F has a magnitude of 15 units. The angle between E and F is

(A) 0°	**(B)** 30°
(C) 60°	**(D)** 90°

Q12. *CUSAT CAT 2023 - Shift 4*

Which of the following is smallest unit?

(A) Millimetre	**(B)** Angstrom
(C) Fermi	**(D)** Micrometre

Q13. *CUSAT CAT 2023 - Shift 5*

If L, C and R represent inductance, capacitance and resistance respectively, which of the following combinations has the dimension of frequency? C

(A) L L	**(B)** R 1
(C) RC 1	**(D)** LC

Q14. *CUSAT CAT 2023 - Shift 5*

The SI unit of luminous intensity (I) is

(A) Candela	**(B)** Weber
(C) Henry	**(D)** Gauss's

Q15. *CUSAT CAT 2023 - Shift 5*

The dimensions of Planck's constant is same as that of

(A) angular momentum	**(B)** linear momentum
(C) work	**(D)** coefficient of viscosity A +

Q16. *CUSAT CAT 2023 - Shift 5*

Young's modulus of a material has the same unit as

(A) Pressure	**(B)** Strain
(C) Compressibility	**(D)** Force

Q17. *CUSAT CAT 2024 - Shift 1*

Accuracy of measurement in an experiment expresses

(A) correctness of the measurement	**(B)** deviation of the measured value from the predicted value
(C) feasibility of the experiment	**(D)** reproducibility of the experiment

Q18. *CUSAT CAT 2024 - Shift 1*

The Vernier scale is divided into 30 divisions, which coincides with 29 main scale 1 divisions. If each division is mm, the least count by the instrument is 2 1

(A) mm 60	**(B)** 1 mm 1
(C) mm 30	**(D)** 0.5 mm

Q19. *CUSAT CAT 2024 - Shift 1*

At constant volume the temperature of a gas in a container is increased. Then

(A) collisions on walls will be less	**(B)** number of collisions per unit time will increase
(C) collisions will not change	**(D)** flow of heat occurs in a particular direction

Q20. *CUSAT CAT 2024 - Shift 1*

The diameter of the solid ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 1.5%, the relative percentage error in the density is

(A) 1.5%	**(B)** 2.6%
(C) 3.1%	**(D)** 4.3%

Q21. *CUSAT CAT 2024 - Shift 1*

Gray is a unit of

(A) Radiation dose	**(B)** Photon energy
(C) Phonon energy	**(D)** Magnetic field

Q22. *CUSAT CAT 2024 - Shift 2*

In measurements of physical quantities, the standard deviation σ and the probable error ‘r’ are related by

(A) r = ± 0.6745 σ	**(B)** r = σ
(C) σ = ± 0.6745 r	**(D)** σ= r

Q23. *CUSAT CAT 2024 - Shift 2*

Which of the following is a dimensionless quantity?

(A) Meter (m)	**(B)** Kilogram (kg)
(C) Radian (rad)	**(D)** Second (s)

Q24. *CUSAT CAT 2024 - Shift 2*

The average number of atoms per unit cell in a body centered cubic structure is

(A) 6	**(B)** 4
(C) 2	**(D)** 8

Q25. *CUSAT CAT 2024 - Shift 2*

The energy difference between the ground state and the first excited state of an electron in a box of length 1.0Å is –17

(A) 1.81 × 10 eV –17	**(B)** 1.81 × 10 Cal –17
(C) 1.81 × 10 J –17	**(D)** 1.81 × 10 KCal 72 A measure of the repeatability of measurement of a quantity is

Q26. *CUSAT CAT 2024 - Shift 3*

Identify the unit vector in the following

(A) i^+ ^j i^	**(B)** 2 ^j ^
(C) k– 2 i^+ ^j	**(D)** 2

Q27. *CUSAT CAT 2024 - Shift 3*

The digits of a measured number that are known to be correct are called

(A) accuracy digits	**(B)** precision digits
(C) significant digits	**(D)** correct digits

Q28. *CUSAT CAT 2024 - Shift 4*

The dimensional formula for modulus of rigidity is 2 –2

(A) [ML T] –1 –2	**(B)** [ML T] –1 –3
(C) [ML T] –2 –2	**(D)** [ML T]

Q29. *CUSAT CAT 2024 - Shift 4*

The dimensions ML T may correspond to

(A) Work done by a force	**(B)** Linear momentum
(C) Pressure	**(D)** Frequency

Q30. *CUSAT CAT 2024 - Shift 4*

The SI unit of specific heat capacity is –1 –1

(A) J kg K –1 –2	**(B)** J kg K –2 –1
(C) J kg K –1 –1	**(D)** J kgK

Q31. *CUSAT CAT 2024 - Shift 5*

What is the unit of self-inductance?

(A) Henry	**(B)** Weber
(C) Volt	**(D)** Fermi

Q32. *CUSAT CAT 2024 - Shift 5*

The pair of physical quantities that has the different dimensions is

(A) Reynolds number and coefficient of friction	**(B)** Curie and frequency of a light wave
(C) Latent heat and gravitational potential	**(D)** Planck’s constant and torque 2 3

Q33. *CUSAT CAT 2024 - Shift 5*

Which one of the following is the unit of solid angle?

(A) Degree	**(B)** Radian
(C) Steradian	**(D)** Second

Q34. *CUSAT CAT 2024 - Shift 5*

The energy radiated by unit area in unit time of a planet is 5.67 × 10 Watt. Its –8 2 4 surface temperature is then (Given Stefan's constant σ = 5.67 × 10 W/m K)

(A) 1273°C	**(B)** 1000°C
(C) 727°C	**(D)** 727 K

Q35. *CUSAT CAT 2025 - Shift 1*

According to uncertainty principle for an electron, time measurement will become uncertain if …….. is measured with high certainty.

(A) Energy	**(B)** Position
(C) Momentum	**(D)** Velocity

Q36. *CUSAT CAT 2025 - Shift 1*

Given S = a + bt + ct2, if S is measured in metres and t in seconds. The unit of c is

(A) m2s	**(B)** m
(C) ms– 1	**(D)** ms–2

Q37. *CUSAT CAT 2025 - Shift 1*

A screw gauge gives the following readings when used to measure the diameter of a wire. Main scale reading : 0 mm, Circular scale reading : 52 divisions Given that, 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is

(A) 0.52 cm	**(B)** 0.026 cm
(C) 0.26 cm	**(D)** 0.052 cm

Q38. *CUSAT CAT 2025 - Shift 1*

A rod is measured to have a length of 5.6 cm using a ruler with the smallest division of 0.1 cm. Calculate the percentage of error in the measurement.

(A) 1.79%	**(B)** 2.80%
(C) 0.79%	**(D)** 3.10%

Q39. *CUSAT CAT 2025 - Shift 1*

Convert a velocity of 72 kmh–1 into ms–1 with the help of dimensional analysis.

(A) 20 ms–1	**(B)** 40 ms–1
(C) 100 ms–1	**(D)** 200 ms–1

Q40. *CUSAT CAT 2025 - Shift 2*

The dimensional formula of latent heat and gravitational constant G are respectively

(A) ML2T–2 and M2L1T–1	**(B)** ML–2T–2 and M2L3T–1
(C) ML2T–1 and M2L0T–1	**(D)** M0L2T–2 and M–1L3T–2

Q41. *CUSAT CAT 2025 - Shift 2*

The period of oscillation of a simple pendulum of length l is about 0.5sec. The time of 100 oscillations has been measured with a stop watch of 1 sec resolution. The length l is 10cm measured with the accuracy of 1mm. Then the percentage error in the g measurement is

(A) 1%	**(B)** 2%
(C) 3%	**(D)** 5%

Q42. *CUSAT CAT 2025 - Shift 2*

According to Stefan's law the total radiant energy emitted per second per unit surface area of a black body is proportional to

(A) T–1	**(B)** T
(C) T–4	**(D)** T4

Q43. *CUSAT CAT 2025 - Shift 2*

An electromagnetic wave is represented by the electric field E = E nsin ωt+(6y–8z) . Taking unit vectors in x, y and z directions to be l, j, k , 0 the direction of propagation s is 4j–3k

(A) s= 5 3i–4j	**(B)** s = 5 –3j+3k
(C) s = 5 –4k +3j	**(D)** s = 5

Q44. *CUSAT CAT 2025 - Shift 2*

If the errors in the measurement of l and g are 2% and 4% respectively, the percentage of error in the time period of pendulum is

(A) 3	**(B)** 4
(C) 6	**(D)** 5

Q45. *CUSAT CAT 2025 - Shift 3*

A particle of unit mass undergoes one dimensional motion such that its velocity varies –2n according to v(x)=bx where b and n are constants and x is the position of the particle. The acceleration of the particle as a function of x is given by

(A) –2nb 2 x –4n–1	**(B)** –2nb 2 x –2n–1
(C) –2nb 2 x –2n+1	**(D)** –2nbx –4n+1

Q46. *CUSAT CAT 2025 - Shift 3*

An ideal fluid is flowing in a non-uniform cross-sectional tube XY (as shown in the figure) from end X to end Y. If K and K are the kinetic energy per unit volume of the 1 2 fluid at X and Y respectively, then the correct option is K

(A) K = 2 1 2	**(B)** K = K 1 2
(C) K < K 1 2	**(D)** K > K 1 2

Q47. *CUSAT CAT 2025 - Shift 3*

The current density is a A

(A) Scalar with SI units m A	**(B)** Vector with SI units 2 m A
(C) Scalar with SI units 2 m A	**(D)** Vector with SI units m

Q48. *CUSAT CAT 2025 - Shift 3*

Which of the following have same dimensions?

(A) Planck's constant and momentum	**(B)** Torque and force
(C) Potential energy and force	**(D)** Torque and potential energy

Q49. *CUSAT CAT 2025 - Shift 4*

The dimension and unit of coefficient of viscosity is

(A) [ML–1T –1] and poise	**(B)** [ML–1T –2] and poise
(C) [ML–2T –2] and newton	**(D)** [ML–1T –2] and newton

Q50. *CUSAT CAT 2025 - Shift 4*

In a cuboid of dimension 2L × 2L × L, a charge q is placed at the centre of the surface 'S' having area of 4 L2. The flux through the opposite surface to 'S' is given by

(A) 12■	**(B)** 3■
(C) 2■	**(D)** 6■

Q51. *CUSAT CAT 2025 - Shift 4*

Which of the following physical quantities have the same dimensions?

(A) Electronic displacement	**(B)** —
(C) —	**(D)** and surface charge density

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	C	4	B	5	C	6	A	7	C	8	D	9	D	10	C
11	B	12	C	13	C	14	A	15	A	16	A	17	A	18	A	19	B	20	B
21	A	22	A	23	C	24	C	25	C	26	D	27	C	28	B	29	C	30	A
31	A	32	D	33	C	34	C	35	A	36	D	37	D	38	A	39	A	40	D
41	D	42	D	43	C	44	A	45	A	46	D	47	B	48	D	49	A	50	D
51	A																		

Motion in a Straight Line

3 questions

Q1. *CUSAT CAT 2024 - Shift 2*

In uniformly accelerated motion, how many variables are required to fully describe the system?

(A) 1	**(B)** 3
(C) 4	**(D)** 2

Q2. *CUSAT CAT 2024 - Shift 3*

A body starts from rest and moves with an acceleration of 20 cm/s . Distance travelled by the body in 10 s is

(A) 200 cm	**(B)** 2000 cm
(C) 1000 cm	**(D)** 20 cm

Q3. *CUSAT CAT 2024 - Shift 3*

A body moving in circular motion with constant speed has

(A) constant velocity	**(B)** constant acceleration
(C) constant kinetic energy	**(D)** constant displacement

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	C														

Motion in a Plane

15 questions

Q1. *CUSAT CAT 2023 - Shift 2*

In the formation of rainbows, the violet light emerges at an angle of

(A) 10°	**(B)** 20°
(C) 30°	**(D)** 40°

Q2. *CUSAT CAT 2023 - Shift 3*

Two spherical rain drops reach the surface of the earth with terminal velocities having ratio 16 : 9. The ratio of their surface area is

(A) 4 : 3	**(B)** 16 : 9
(C) 9 : 16	**(D)** 3 : 4

Q3. *CUSAT CAT 2023 - Shift 4*

The focus of all particles in a medium vibrating in the same phase is called

(A) wavelet	**(B)** wavefront
(C) wave train	**(D)** wave function

Q4. *CUSAT CAT 2023 - Shift 4*

An electron and a proton enter a magnetic field perpendicularly. Both have same kinetic energy. Which of the following is TRUE?

(A) Trajectory of electron is less curved	**(B)** Trajectory of proton is less curved
(C) Both trajectories are equally curved	**(D)** Both move on straight path

Q5. *CUSAT CAT 2023 - Shift 5*

A projectile thrown at an angle of 45° with horizontal attains a maximum height of h . 1 Another projectile thrown with the same velocity, at an angle of 30° attains a maximum height h . The relation between h and h is 2 1 2 h

(A) h = 2 1 2	**(B)** h =h 1 2 h
(C) h = 2 1 2	**(D)** h =2h 1 2

Q6. *CUSAT CAT 2024 - Shift 1*

If R and H represent horizontal range and maximum height of the projectile, then the angle of projection with the horizontal is H –1

(A) tan R 2H –1	**(B)** tan R 4H –1
(C) tan R 4R –1	**(D)** tan H 2

Q7. *CUSAT CAT 2024 - Shift 1*

In projectile motion, the maximum height attained by the projectile depends on

(A) Initial speed	**(B)** Angle of projection
(C) Both initial speed and angle of projection	**(D)** Mass of the projectile

Q8. *CUSAT CAT 2024 - Shift 2*

A projectile is launched with a speed of 20 m/s at an angle θ from the ground. At what point in its trajectory does the projectile have the maximum kinetic energy?

(A) At the apex of its trajectory	**(B)** At the initial point
(C) Upon reaching the ground	**(D)** Halfway between the initial and final points

Q9. *CUSAT CAT 2024 - Shift 4*

Two rain drops reach the earth with the terminal velocities in the ratio of 4 : 9. The ratio of their masses is

(A) 4 2 : 9 2	**(B)** 2 : 3
(C) 4 : 9	**(D)** 8 : 27

Q10. *CUSAT CAT 2025 - Shift 1*

Two cars A and B move in the same direction with velocities V = 20 i m/s and A V = 15 i m/s. Then the relative velocity of A with respect to B is B

(A) 35 i m/s	**(B)** 5 i m/s
(C) 35 j m/s	**(D)** 5 j m/s

Q11. *CUSAT CAT 2025 - Shift 1*

A particle moves in xy-plane with coordinates given by x = a sin ωt and y = a cos ωt. The particle follows

(A) an elliptical path	**(B)** a circular path
(C) a parabolic path	**(D)** a straight-line path inclined equally to x and y-axis

Q12. *CUSAT CAT 2025 - Shift 1*

A train of 150 m length is going towards North direction at a speed of 10 m/s. A parrot flies at the speed of 5 m/s towards South direction parallel to the railways track. The time taken by the parrot to cross the train is

(A) 12 s	**(B)** 8 s
(C) 15 s	**(D)** 10 s

Q13. *CUSAT CAT 2025 - Shift 2*

A particle is projected at 60° to the horizontal with a kinetic energy K. The kinetic energy at highest point is

(A) Zero K	**(B)** 4 K
(C) 2	**(D)** K

Q14.

CUSAT CAT 2025 - Shift 2

Pick the WRONG statement in the context of rainbow formation.

(A) The order of colours is reversed in the secondary rainbow	**(B)** An observer can see a rainbow when his front is towards the sun
(C) Rainbow is a combined effect of dispersion, refraction and reflection of sunlight	**(D)** When the light rays undergo total internal reflections in a water drop, a secondary rainbow is formed

Q15.

CUSAT CAT 2025 - Shift 3

In which of the following devices, the eddy current effect is NOT used?

(A) Magnetic braking in train	**(B)** Electromagnet
(C) Electric heater	**(D)** Induction furnace

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	B	3	B	4	B	5	D	6	C	7	C	8	B	9	D	10	B
11	B	12	D	13	B	14	B	15	C										

Laws of Motion

66 questions

Q1. *CUSAT CAT 2023 - Shift 1*

If a spring extends by x on loading, the energy stored in the spring is (T is the tension in the spring and K is the spring constant) 2 T

(A) 2K 2K	**(B)** 2 T T
(C) 2K T	**(D)** K

Q2. *CUSAT CAT 2023 - Shift 1*

Match the following. Technology Scientific Principle (a) Production of ultra high (i) Bernoulli's principle magnetic fields (b) Cyclotron (ii) Superconductivity (c) Aeroplane (iii) Motion of charged particles in electromagnetic fields (d) Rocket propulsion (iv) Faraday's law of induction (e) Electric generator (v) Newton's law of motion

(A) (a)-(iv); (b)-(v); (c)-(i); (d)-(ii); (e)-(iii)	**(B)** (a)-(ii); (b)-(v); (c)- (iv); (d)-(iii); (e)-(i)
(C) (a)-(ii); (b)-(iii); (c)-(i); (d)-(v); (e)- (iv)	**(D)** (a)-(v); (b)-(iv); (c)-(iii); (d)-(ii); (e)-(i)

Q3. *CUSAT CAT 2023 - Shift 1*

The motion of a rocket is based on the principle of conservation of

(A) linear momentum	**(B)** angular momentum
(C) kinetic energy	**(D)** mass

Q4. *CUSAT CAT 2023 - Shift 2*

If the surface tension of a liquid is 5 Nm , then the surface energy of this liquid film 2 on a ring of area 0.15 m is

(A) 0.75 J	**(B)** 1.5 J
(C) 2.25 J	**(D)** 3.0 J

Q5. *CUSAT CAT 2023 - Shift 2*

IIIfff ttthhheee uuunnnccceeerrrttaaaiiinnnttyyy iiinnn ttthhheee mmmeeeaaasssuuurrreeemmmeeennnttt ooofff pppoooss siiittiiiooonnn ooofff aaannn eeellleeeccctttrrrooonnn iiisss 111000 m, what is the uunncceerrttaaiinnttyy iinn tthhee mmeeaassuurreemmeenntt ooff iittss momentum? –24 –1

(A) 3.33 × 10 kkggms –25 –1	**(B)** 1.65 × 10 kkggmmss –24 –1
(C) 6.6 × 10 kgmmss –20 –1	**(D)** 6.6 × 10 kgmmss

Q6. *CUSAT CAT 2023 - Shift 2*

Lenz's law is a consequence of the law of conservation of

(A) energy	**(B)** charge
(C) electric flux	**(D)** linear momentum

Q7. *CUSAT CAT 2023 - Shift 2*

A light and a heavy body have equal kinetic energy. Which one has a greater momentum?

(A) The heavy body	**(B)** The light body
(C) Both	**(D)** Cannot be said

Q8. *CUSAT CAT 2023 - Shift 2*

A block slides down an incline of angle 30° with an acceleration . Find the kinetic 4 friction co-efficient 3

(A) 2 1	**(B)** 2 1
(C) 2 3 1	**(D)** 2 22

Q9. *CUSAT CAT 2023 - Shift 2*

If the linear momentum is increased by 50%, then kinetic energy will be increased by

(A) 50%	**(B)** 20%
(C) 125%	**(D)** 100%

Q10. *CUSAT CAT 2023 - Shift 2*

Which one of the following is NOT a conservative force?

(A) Magnetic force	**(B)** Force of friction
(C) Gravitational force	**(D)** Electrostatic force

Q11. *CUSAT CAT 2023 - Shift 2*

In a nuclear reaction, which of the following is/are conserved?

(A) Mass only	**(B)** Energy only
(C) Momentum only	**(D)** Mass, energy and momentum

Q12. *CUSAT CAT 2023 - Shift 3*

A particle of mass m is tied to a light string and rotated with a speed v along a circular path of radius r. If T represents the tension in the string and mg represents the gravitational force acting on the particle, then actual/real forces acting on the particle are

(A) mg and T only 2	**(B)** mg, T and an additional force of mv /r directed inwards 2
(C) mg, T and an additional force of mv /r directed outwards 2	**(D)** only a force mv /r directed inwards

Q13. *CUSAT CAT 2023 - Shift 3*

Bernoulli theorem represents the law of conservation of

(A) mass	**(B)** momentum
(C) energy	**(D)** angular momentum 11 2

Q14. *CUSAT CAT 2023 - Shift 3*

When a charged particle moves perpendicular to a uniform magnetic field, its

(A) energy changes but momentum remains unchanged	**(B)** momentum changes but energy remains unchanged
(C) energy and momentum both remain unchanged	**(D)** both energy and momentum are changed

Q15. *CUSAT CAT 2023 - Shift 3*

A ballet dancer stretches her hands out for a slowing down; this is based on the principle of

(A) conservation of force	**(B)** conservation of energy
(C) conservation of linear momentum	**(D)** conservation of angular momentum

Q16. *CUSAT CAT 2023 - Shift 3*

Kinetic energy of a body of mass m and momentum p is given by 2

(A) p m 2 m	**(B)** 2p
(C) mp 2 p	**(D)** 2m

Q17. *CUSAT CAT 2023 - Shift 3*

A stone of mass 1.3 kg slides on ice with velocity of 3.12 m/s. The stone stops due to friction in 10 seconds. Assuming the force of friction to be a constant, it is

(A) –0.41 N	**(B)** 0.41 N
(C) –0.82 N	**(D)** 0.82 N

Q18. *CUSAT CAT 2023 - Shift 3*

Kepler's second is a consequence of

(A) conservation of energy	**(B)** conservation of linear momentum
(C) conservation of angular momentum	**(D)** conservation of mass

Q19. *CUSAT CAT 2023 - Shift 3*

The moment of inertia of a uniform semicircular wire of mass M and radius r about a line passing through its ends is

(A) Mr 2 1 2	**(B)** Mr 2 1 2
(C) Mr 4 2 2	**(D)** Mr 5

Q20. *CUSAT CAT 2023 - Shift 4*

Air is pushed into a soap bubble of radius r to double its radius. If the surface tension of the soap solution is S, the work done in the process is

(A) 8πr 2 S	**(B)** 24πr 2 S
(C) 16πr 2 S	**(D)** 4πr 2 S

Q21. *CUSAT CAT 2023 - Shift 4*

The moment of inertia of a circular loop of radius R at a distance of around a 2 rotating axis parallel to horizontal diameter of loop is

(A) MR 2 1 2	**(B)** MR 2
(C) 2MR 2 3 2	**(D)** MR 4

Q22. *CUSAT CAT 2023 - Shift 4*

Choose the INCORRECT statement.

(A) Newton proposed corpuscular theory of light	**(B)** Huygens replaced Newton's theory by wave theory of light
(C) Einstein put forward quantum theory of light	**(D)** de Broglie gave the concept of dual nature of matter

Q23. *CUSAT CAT 2023 - Shift 4*

The internal energy of a monoatomic ideal gas is 1.5 nRT. One mole of helium is 2 kept in a cylinder of cross-section 8.5 cm . The cylinder is closed by a light frictionless piston. The gas is heated slowly in a process during which a total of 42 J heat is given to the gas. If the temperature rises through 2°C, find the distance moved by the piston. (Atmospheric pressure = 100 kPa)

(A) 0.2 m	**(B)** 0.4 m
(C) 0.6 m	**(D)** 0.8 m

Q24. *CUSAT CAT 2023 - Shift 5*

A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90°. The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is 2 ML

(A) 24 2 ML	**(B)** 12 2 ML
(C) 6 2 2ML	**(D)** 24 2

Q25. *CUSAT CAT 2023 - Shift 5*

The motion of planets in the solar system is an example of the conservation of

(A) energy	**(B)** linear momentum
(C) angular momentum	**(D)** mass CHEMISTRY (UG)– SHIFT V (FINAL)

Q26. *CUSAT CAT 2024 - Shift 1*

In a tug of war contest, two men pull on a horizontal rope from opposite sides. The winner will be the man who

(A) exerts greater force on the rope with larger angle with the vertical	**(B)** exerts greater force on the ground
(C) exerts force on the rope which is greater than the tension in the rope	**(D)** makes a smaller angle with the vertical

Q27. *CUSAT CAT 2024 - Shift 1*

The kinetic energy of a body of moment of inertia I and angular momentum L is 2 L

(A) I L	**(B)** 2I 2 L
(C) 2I 2 IL	**(D)** 2

Q28. *CUSAT CAT 2024 - Shift 1*

The Paint gun works on the principle of

(A) Bernoulli's principle	**(B)** Boyle's law
(C) Archimedes' principle	**(D)** Newton's law of motion

Q29. *CUSAT CAT 2024 - Shift 1*

A 2 kg object moving with a velocity of 4 m/s collides with a stationary object of mass 3 kg. After the collision, both objects move together. What is their common velocity?

(A) 0.4 m/s	**(B)** 0.8 m/s
(C) 1.2 m/s	**(D)** 1.6 m/s

Q30. *CUSAT CAT 2024 - Shift 2*

Two bodies having masses in the ratio of 3 : 1 possess same kinetic energy. The ratio of linear momentum of B to A is

(A) 1:3	**(B)** 3:1
(C) 1: 3	**(D)** 3:1

Q31. *CUSAT CAT 2024 - Shift 2*

The angular momentum of the electron results in

(A) Zeeman effect	**(B)** Magnetic moment
(C) Stark effect	**(D)** Emission of light and emission spectrum

Q32. *CUSAT CAT 2024 - Shift 2*

A step down transformer with a primary winding of 1,74,000 turns and secondary windings of 1000 turns operates from a 40,000 V high tension line and supplies a load of 60 A. Assuming 100% efficiency, the power output of the transformer is

(A) 2400 KW	**(B)** 13.8 KW
(C) 24,000 KW	**(D)** 40,000 KW

Q33. *CUSAT CAT 2024 - Shift 2*

Bernoulli theorem is based on conservation of

(A) momentum	**(B)** mass
(C) energy	**(D)** angular momentum

Q34. *CUSAT CAT 2024 - Shift 2*

A particle of mass m is observed from an inertial frame of reference and is found to move in a circle of radius r with a uniform speed v. The centrifugal force on it is 2 mv

(A) towards the centre r 2 mv	**(B)** away from centre r 2 mv
(C) along the tangent through the particle r	**(D)** zero

Q35. *CUSAT CAT 2024 - Shift 2*

The motion of a rocket is based on the principle of conservation of

(A) linear momentum	**(B)** angular momentum
(C) kinetic energy	**(D)** mass

Q36. *CUSAT CAT 2024 - Shift 2*

According to kinetic theory, the pressure exerted by an ideal gas on the walls of its container is due to

(A) Collisions between gas molecules	**(B)** Gravitational attraction between gas molecules
(C) Magnetic forces between gas molecules	**(D)** Electrostatic repulsion between gas molecules

Q37. *CUSAT CAT 2024 - Shift 2*

Identify the incorrect statement from the following.

(A) The direction of momentum and the velocity of an object is the same	**(B)** Momentum is a scalar quantity
(C) The object has a large momentum if either mass or velocity is large	**(D)** Momentum describes the motion of an object

Q38. *CUSAT CAT 2024 - Shift 2*

Just before striking the ground, a 2.0 kg mass has 400 J of KE. If friction can be ignored, from what height was it dropped?

(A) 200 m	**(B)** 20 m
(C) 40.8 m	**(D)** 40 m

Q39. *CUSAT CAT 2024 - Shift 3*

A particle moving in a circular path has an angular momentum L. If the frequency of rotation is halved, then the angular momentum becomes

(A) L L	**(B)** 2 L
(C) 3 L	**(D)** 4

Q40. *CUSAT CAT 2024 - Shift 3*

Which phenomenon best supports the theory that matter has a wave nature?

(A) Electron momentum	**(B)** Photon diffraction
(C) Electron diffraction	**(D)** Photon momentum

Q41. *CUSAT CAT 2024 - Shift 3*

Which one of the following statements describes Newton's First Law of Motion?

(A) An object will accelerate if acted upon by an unbalanced force.	**(B)** An object remains at rest or in uniform motion unless acted upon by a net external force
(C) Force is equal to the rate of change of momentum	**(D)** Every action has an equal and opposite reaction

Q42. *CUSAT CAT 2024 - Shift 3*

The change in momentum of an object is given by which of the following equation?

(A) $\Delta p = m/(v - v)$ f i	**(B)** $\Delta p = (v - v)/m$ f i
(C) $\Delta p = (m - m)$ v f i	**(D)** $\Delta p = m (v - v)$ f i

Q43. *CUSAT CAT 2024 - Shift 4*

The linear momentum of a system of particles is equal to

(A) total mass × acceleration	**(B)** total mass×velocity
(C) total mass × velocity of center of mass of the particles	**(D)** total mass ×acceleration of center of mass of the particles

Q44. *CUSAT CAT 2024 - Shift 5*

A marble block of mass 2 kg lying on ice when given a velocity of 6 m/s is stopped by friction in 10 s. Then the coefficient of friction is

(A) 0.01	**(B)** 0.02
(C) 0.03	**(D)** 0.06

Q45. *CUSAT CAT 2024 - Shift 5*

If an electron and a photon propagate in the form of waves having same wavelength, it implies that they have same

(A) speed	**(B)** momentum
(C) energy	**(D)** All of the above

Q46. *CUSAT CAT 2024 - Shift 5*

A book of mass m, placed on an inclined table, making an angle of 30° with the g horizontal, slides down with an acceleration of . The frictional force acting on the 3 book is

(A) 1 mg 2	**(B)** mg 3 1
(C) mg 3 1	**(D)** mg 6

Q47. *CUSAT CAT 2024 - Shift 5*

A ball of mass 50 g is dropped from a height of 10 m. It reaches the ground with a speed of 1.5 m/s. The work done by the air-friction is

(A) 5.056 J	**(B)** 4.954 J
(C) –4.954 J	**(D)** –5.056 J

Q48.

CUSAT CAT 2025 - Shift 1

Three liquids of densities ρ,ρ and ρ (with ρ >ρ >ρ), having the same value 1 2 3 1 2 3 of surface tension T, rise to the same height in three identical capillaries. The angles of contact θ,θ andθ obey 1 2 3 π

(A) >θ >θ >θ ≥0 1 2 3 2 π	**(B)** 0≤θ <θ <θ < 1 2 3 2 π
(C) <θ <θ <θ <π 1 2 3 2 π	**(D)** π>θ >θ >θ > 1 2 3 2

Q49.

CUSAT CAT 2025 - Shift 1

Which among the following forces is a contact force?

(A) Gravitational force	**(B)** Magnetic force
(C) Frictional force	**(D)** Electrostatic force

Q50.

CUSAT CAT 2025 - Shift 1

A block of mass 5 kg is placed at rest and a table of rough surface. Now, if a force of 30 N is applied in the direction parallel to surface of the table, the block slides through a distance of 50 m in an interval of time 10 s. Coefficient of kinetic friction is (given, g = 10 ms–2)

(A) 0.60	**(B)** 0.75
(C) 0.50	**(D)** 0.25

Q51.

CUSAT CAT 2025 - Shift 1

A system consists of two identical spheres each of mass 1.5 kg and radius 50 cm at the ends of a light rod. The distance between the centres of the two spheres is 5 m. What will be the moment of inertia of the system about an axis perpendicular to the rod passing through its midpoint ?

(A) 1.905 × 10–5 kgm2	**(B)** 18.75 × 10–5 kgm2
(C) 19.05 kgm2	**(D)** 1.875 × 10–5 kgm2

Q52.

CUSAT CAT 2025 - Shift 2

What is the angle between the frictional force and the instantaneous velocity of a vehicle plying on a rough road?

(A) 0°	**(B)** 45°
(C) 90°	**(D)** 180°

Q53.

CUSAT CAT 2025 - Shift 2

When the kinetic energy of a body increases by 4 times,then its momentum increases by

(A) 2 times	**(B)** 16 times
(C) 2times	**(D)** 4 times

Q54. *CUSAT CAT 2025 - Shift 2*

Which one of the following statements is INCORRECT?

(A) Frictional force opposes the relative motion	**(B)** Limiting value of static friction is directly proportional to normal reaction
(C) Rolling friction is smaller than sliding friction	**(D)** Coefficient of sliding friction has dimensions of length

Q55. *CUSAT CAT 2025 - Shift 2*

A piece of iron is heated in a flame. If first becomes dull red, then becomes reddish yellow and finally turns to white hot. The correct explanation for the above observation is possible by using

(A) Stefan's law	**(B)** Wien's displacement law
(C) Kirchhoff's law	**(D)** Newton's law of cooling

Q56. *CUSAT CAT 2025 - Shift 2*

What percentage of kinetic energy is transferred to a stationary particle when a moving particle strikes a stationary particle of 5 times its mass? (Assume the collision to be head-on elastic collision)

(A) 50.0%	**(B)** 66.6%
(C) 55.5%	**(D)** 33.3%

Q57. *CUSAT CAT 2025 - Shift 3*

A rigid ball of mass m strikes a rigid wall and gets reflected without loss of speed as shown in figure. The value of impulse imparted by the wall to the ball will be v 60o 60o v mv

(A) 3	**(B)** mv
(C) 2mv mv	**(D)** 2

Q58. *CUSAT CAT 2025 - Shift 3*

Kepler's second law is based on which one of the following physical quantity?

(A) Energy	**(B)** Mass
(C) Linear momentum	**(D)** Angular momentum 2 2

Q59. *CUSAT CAT 2025 - Shift 3*

The moment of inertia of a sphere about a diameter of a sphere is MR , M is the 5 mass and R is the radius, then the moment of inertia of a sphere about a tangent to the sphere is 2 2

(A) MR 7 2 2	**(B)** MR 3 3 2
(C) MR 5 7 2	**(D)** MR 5

Q60. *CUSAT CAT 2025 - Shift 3*

The extension in a string, obeying Hook's law is x. The speed of sound in the stretched string is v. If the extension in the string is increased to 1.5x, the speed of sound will be

(A) 1.22 v	**(B)** 0.61 v
(C) 1.50 v	**(D)** 0.75 v

Q61. *CUSAT CAT 2025 - Shift 3*

If a body moves in such a way that its linear momentum is constant, then

(A) its kinetic energy is zero	**(B)** the net force acting on it must be zero
(C) its acceleration must be non-zero and constant	**(D)** its center of mass must remain at rest

Q62. *CUSAT CAT 2025 - Shift 3*

Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? (g = 10 m/s2)

(A) 10.2 kW	**(B)** 8.1 kW
(C) 12.3 kW	**(D)** 7.0 kW

Q63. *CUSAT CAT 2025 - Shift 3*

If the tension in a stretched string is increased, what happens to the velocity of a wave traveling through the string?

(A) Increases	**(B)** Decreases
(C) Remains the same	**(D)** Becomes zero

Q64. *CUSAT CAT 2025 - Shift 3*

A string of length 1m and mass 5 kg is fixed at both ends. The tension in the string is 8.0 N. The string is set into vibration using an external vibrator of frequency 100 Hz. The separation between successive nodes and the string is close to

(A) 16.6 cm	**(B)** 20.0 cm
(C) 10.0 cm	**(D)** 33.3 cm

Q65. *CUSAT CAT 2025 - Shift 4*

Moment of inertia of square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by 2 Ml

(A) 6 2 2	**(B)** Ml 3
(C) Ml 2 2 Ml	**(D)** 12

Q66.

CUSAT CAT 2025 - Shift 4

A particle of mass 20 g is released with an initial velocity 5 m/s along the curve from the point A as shown in the figure. The point A is at height h from the point B. The particle slides along the frictionless surface. When the particle reaches point B, its angular momentum about O will be

(A) 8 kg-m2/s	**(B)** 6 kg-m2/s
(C) 3 kg-m2/s	**(D)** 2 kg-m2/s 1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	C	3	A	4	B	5	B	6	A	7	A	8	C	9	C	10	B
11	D	12	A	13	C	14	B	15	D	16	D	17	A	18	C	19	B	20	B
21	D	22	C	23	A	24	B	25	C	26	B	27	C	28	A	29	D	30	C
31	B	32	B	33	C	34	D	35	A	36	A	37	B	38	B	39	B	40	C
41	B	42	D	43	D	44	D	45	B	46	D	47	C	48	B	49	C	50	C
51	C	52	D	53	A	54	D	55	B	56	C	57	B	58	D	59	D	60	A
61	B	62	B	63	A	64	B	65	B	66	B								

Work, Energy & Power

58 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A small signal AC voltage V(t) = V sin ωt applied across an ideal capacitor. Then o which one of the following is TRUE?

(A) Current I(t)leads the voltage V(t) by 90°	**(B)** Current I(t) lags the voltage V(t) and average power dissipated in the capacitor is zero
(C) Current I(t) is in phase with voltage V(t)	**(D)** Current I(t)leads voltage V(t) by 180°

Q2. *CUSAT CAT 2023 - Shift 1*

A point charge q is moved on an equipotential surface with potential V. The work done is

(A) Zero	**(B)** qV qV
(C) 2	**(D)** 2qV

Q3. *CUSAT CAT 2023 - Shift 1*

A standard 100 watt incandescent light bulb emits approximately

(A) 17 lumens	**(B)** 170 lumens
(C) 1700 lumens	**(D)** 1.7 lumens

Q4. *CUSAT CAT 2023 - Shift 1*

The spring constant of a spring balance is 5 × 10 Nm . It is initially stretched by 5 cm from the unstretched position. Then the work done to stretch it further by another 5 cm is

(A) 18.75 J	**(B)** 25.0 J
(C) 6.25 J	**(D)** 12.5 J

Q5. *CUSAT CAT 2023 - Shift 1*

Two sound waves move in the same direction in the same medium. The pressure amplitudes of the waves are equal but the wavelength of the first wave is double the second wave. Let the average power transmitted across a cross section by the first wave be P and that by the second wave be P . Then 1 2

(A) P = P 1 2	**(B)** P = 4P 1 2
(C) P = 2P 2 1 P	**(D)** P = 2 1 2

Q6. *CUSAT CAT 2023 - Shift 2*

A wire fixed at the upper end and stretched by length l by applying a force F. The work done in stretching the wire is

(A) Fl	**(B)** 2Fl Fl
(C) 2 F	**(D)** 2l –1

Q7. *CUSAT CAT 2023 - Shift 2*

To increase the angular magnification of a simple microscope, one should increase the

(A) focal length of the lens	**(B)** power of the lens
(C) aperture of the lens	**(D)** object size

Q8. *CUSAT CAT 2023 - Shift 3*

Consider a telescope whose objective has a focal length of 100 cm and the eyepiece has a focal length of 1 cm. The magnifying power of this telescope is

(A) 100	**(B)** 200
(C) 10	**(D)** 5

Q9. *CUSAT CAT 2023 - Shift 4*

A particle is moving in a circle of radius r under the action of a forceF =αr which is directed towards centre of the circle. Total mechanical energy (which is sum of kinetic energy and potential energy), of the particle is (Take potential energy is = 0 for r = 0) 1 3

(A) αr 2 5 3	**(B)** αr 6 4 3
(C) αr 3	**(D)** αr 3

Q10. *CUSAT CAT 2023 - Shift 4*

An elevator weighing 500 kg is to be lifted up at a constant velocity of 0.20 m/s. The minimum power of the motor that required is

(A) 500 W	**(B)** 1000 W
(C) 1500 W	**(D)** 10000 W 3

Q11. *CUSAT CAT 2023 - Shift 4*

Work-Energy theorem is valid in the case of

(A) external forces only	**(B)** internal forces only
(C) conservation forces only	**(D)** all types of forces

Q12. *CUSAT CAT 2023 - Shift 5*

The Power factor of R-L circuit is . What is the value of resistance if inductive 5 reactance is 4 Ω?

(A) 2 Ω	**(B)** 2Ω
(C) 4 Ω	**(D)** 5Ω

Q13. *CUSAT CAT 2023 - Shift 5*

The potential energy of a conservative system is given by PE = ay – by, where y represents the position of the particle and a as well as b are constants. What is the force acting on the system?

(A) b – 2ay	**(B)** ay – b
(C) –by	**(D)** –ay –1

Q14. *CUSAT CAT 2023 - Shift 5*

The powers of two electric bulbs are 100 W and 200 W. Both of them are connected to 220 V main. The ratio of resistance of 100 W and 200 W filaments will be

(A) 1 : 2	**(B)** 2 : 1
(C) 1 : 4	**(D)** 4 : 1

Q15. *CUSAT CAT 2024 - Shift 1*

If the carrier power of a 100 % modulated AM wave is suppressed, the percentage saving in power will be

(A) 50%	**(B)** 100%
(C) 66.66%	**(D)** 75%

Q16. *CUSAT CAT 2024 - Shift 1*

The Wien's displacement law express relation between

(A) Frequency and temperature	**(B)** Temperature and amplitude
(C) Wavelength corresponding to maximum energy and temperature	**(D)** Wavelength and radiating power of black body

Q17. *CUSAT CAT 2024 - Shift 1*

A steel wire of length 2 m is stretched by 2 mm. The cross sectional area of the wire is 2 11 2 4 mm . If the Young's modulus is 2 × 10 N/m , the elastic potential energy stored in the wire in stretched condition is

(A) 0.8 J	**(B)** 1.2 J
(C) 2.4 J	**(D)** 4 J

Q18. *CUSAT CAT 2024 - Shift 1*

The increasing order of penetrating power of α, β and γ radiations of 0.5 MeV energy is

(A) α, β, γ	**(B)** β, γ, α
(C) γ, α, β	**(D)** γ, β, α

Q19. *CUSAT CAT 2024 - Shift 1*

Three equal resistors connected in series across a source of e.m.f. together dissipate 10 watts of power. What would be the power dissipated if the same resistors are connected in parallel across the same source of e.m.f.?

(A) 9 watts	**(B)** 90 watts
(C) 10 watts	**(D)** 100 watts

Q20. *CUSAT CAT 2024 - Shift 1*

A spring of force constant 100 N/m is compressed by 10 cm. What is the potential energy stored in the spring?

(A) 0.5 J	**(B)** 5 J
(C) 10 J	**(D)** 1000 J

Q21. *CUSAT CAT 2024 - Shift 1*

…… should be increased to increase the angular magnification of a simple microscope.

(A) The focal length of the lens	**(B)** Lens aperture
(C) The power of the lens	**(D)** Object size

Q22. *CUSAT CAT 2024 - Shift 2*

A diesel engine pumps 40 kg of water in 1 second. The water comes out vertically upwards with a velocity of 3 m/s. What is the power of the engine in kilo watt?

(A) 12 kW	**(B)** 1.2 kW
(C) 120 kW	**(D)** 1 kW

Q23. *CUSAT CAT 2024 - Shift 2*

Two bodies A and B of equal masses are suspended from two separate massless springs of spring constant k and k respectively. If the bodies oscillate vertically such 1 2 that their maximum velocities are equal, the ratio of the amplitude of A to that of B is k

(A) 1 k 2 k	**(B)** 2 k 1 k
(C) 1 k 2 k	**(D)** 2 k 1

Q24. *CUSAT CAT 2024 - Shift 2*

As an electron makes a transition from an excited state to the ground state of a hydrogen like atom/ion

(A) Kinetic energy decreases, potential energy increases but total energy remains same	**(B)** Kinetic energy and total energy decrease but potential energy increases
(C) Kinetic energy increases but potential energy and total energy decreases	**(D)** Kinetic energy, potential energy and total energy decreases

Q25. *CUSAT CAT 2024 - Shift 2*

Ratio of the angle subtended at the eye by the image to the angle subtended at the eye by the object is called

(A) Magnifying power of a simple microscope	**(B)** Magnifying power of a telescope
(C) Magnifying power of a electron microscope	**(D)** Astigmatism

Q26. *CUSAT CAT 2024 - Shift 2*

1 Joule is equal to

(A) 10 ergs	**(B)** 10 4 ergs
(C) 10 7 ergs	**(D)** 10 9 ergs

Q27. *CUSAT CAT 2024 - Shift 2*

A 250-turn rectangular coil of length 2.1 cm and width 1.25 cm carries a steady current of 85 μA and is subjected to a magnetic field of strength 0.85 T. The work done for rotating the coil by 180° against the torque is

(A) 1.5 pJ	**(B)** 4.55 pJ
(C) 9.5 pJ	**(D)** 2.3 pJ

Q28. *CUSAT CAT 2024 - Shift 2*

An object is thrown vertically upwards with an initial velocity of 20 m/s. Considering the gravitational force acting on the object during its ascent, what is the net work done by the gravitational force?

(A) Zero	**(B)** Negative
(C) Positive	**(D)** Cannot be determined

Q29. *CUSAT CAT 2024 - Shift 2*

The work done against electrostatic force gets stored in the ……. form

(A) Thermal energy	**(B)** Potential energy
(C) Kinetic energy	**(D)** Gravitational energy

Q30. *CUSAT CAT 2024 - Shift 3*

Which of the following graph depicts spring constant k versus length l of the spring correctly?

(A) —	**(B)** —
(C) —	**(D)** —

Q31. *CUSAT CAT 2024 - Shift 3*

The output of a step down transformer is measured to be 24 V when connected to a 12 watt light bulb. The value of the peak current is 1

(A) A 2	**(B)** 2 A
(C) 2A	**(D)** 2 2 A

Q32. *CUSAT CAT 2024 - Shift 3*

A boy pushes a toy box 2.0 m along the floor by means of a force of 10 N directed downward at an angle of 60° to the horizontal. The work done by the boy is

(A) 10 J	**(B)** 12 J
(C) 8 J	**(D)** 6 J

Q33. *CUSAT CAT 2024 - Shift 3*

Which one of the following expressions represents the work-energy theorem?

(A) $W = \Delta E$	**(B)** $W = F \times d$
(C) $W = mgh$	**(D)** $W = P \times t$

Q34. *CUSAT CAT 2024 - Shift 4*

The coefficient of restitution for a perfectly elastic collision is

(A) 1	**(B)** 0
(C) –1	**(D)** ∞

Q35. *CUSAT CAT 2024 - Shift 4*

The potential energy of a test charge will ……. when moved from lower potential point to a higher potential point.

(A) remain the same	**(B)** increase
(C) decrease	**(D)** become zero

Q36. *CUSAT CAT 2024 - Shift 4*

The reason for transmitting A.C. power at high voltages from the power house is

(A) as the rate of transmission is faster	**(B)** as it is more economic due to less power loss
(C) as power cannot be transmitted at low voltages	**(D)** as a precaution against theft of transmission lines

Q37. *CUSAT CAT 2024 - Shift 4*

A force of 15 N acts on an object, causing it to move a distance of 5 m in the direction of the force. What is the work done by the force?

(A) 3 J	**(B)** 5 J
(C) 15 J	**(D)** 75 J

Q38. *CUSAT CAT 2024 - Shift 5*

The average power consumed in a purely inductive circuit is

(A) I × V max max	**(B)** zero
(C) 1 I	**(D)** max V max

Q39. *CUSAT CAT 2024 - Shift 5*

In a circuit, at a given time, 20 C of charge is passed through a battery, whose plates are maintained at a potential difference of 20 V. The work done by the battery is

(A) 1 J	**(B)** 10 J
(C) 20 J	**(D)** 400 J 2

Q40. *CUSAT CAT 2025 - Shift 1*

A man weighing 70 kg carries a 30 kg box to the top of the building of height 20 m. Then the work done by the man is

(A) 980 J	**(B)** 19600 J
(C) 7840 J	**(D)** 20000 J

Q41. *CUSAT CAT 2025 - Shift 1*

When a system is taken from state i to state f along the path iaf, it is found that Q = 50 cal and W = 20 cal. Along the path ibf, if Q = 36 cal, work done along the path ibf is a f i b

(A) 16 cal	**(B)** 66 cal
(C) 14 cal	**(D)** 6 cal

Q42. *CUSAT CAT 2025 - Shift 1*

The ratio of resolving powers of an optical microscope for two wavelengths λ = 4000 Å and λ = 6000 Å is 1 2

(A) 8 : 27	**(B)** 9 : 4
(C) 3 : 2	**(D)** 16 : 81

Q43. *CUSAT CAT 2025 - Shift 1*

When a block of mass M is suspended by a long wire of length L, the length of the wire becomes (L + l). The elastic potential energy stored in the extended wire is

(A) MgL 1	**(B)** Mgl 2 1
(C) MgL 2	**(D)** Mgl

Q44. *CUSAT CAT 2025 - Shift 1*

A body moves in a circular path with constant speed. Which of the following statements is true?

(A) Work is done by centrifugal force	**(B)** Work done by the net force is zero
(C) Kinetic energy of the body changes	**(D)** Velocity of the body remains constant

Q45. *CUSAT CAT 2025 - Shift 1*

A body of mass 0.5 kg travels on straight line with path velocity v = (3x2 + 4) m/s. The net work done by the force during its displacement from x = 0 to x = 2 m is

(A) 64 J	**(B)** 60 J
(C) 120 J	**(D)** 128 J

Q46. *CUSAT CAT 2025 - Shift 2*

In an ac circuit, the voltage applied is E = E sinωt . The resulting current in the 0 π circuit is I = I sinωt– . The power consumption in the circuit is given by 0 2

(A) P = 2E I 0 0 E I	**(B)** P = 0 0 2
(C) = 0 E I 0 0	**(D)** P = 2

Q47. *CUSAT CAT 2025 - Shift 2*

When a long spring is stretched by 2 cm, its potential energy is V. If the spring is stretched by 10 cm, the potential energy in it will be

(A) 10V	**(B)** 25V V
(C) 5	**(D)** 5V

Q48. *CUSAT CAT 2025 - Shift 2*

A diesel engine pumps 40 kg of water in 1 second. The water comes out vertically upwards with a velocity of 3 m/s. What is the power of the engine in kilo watt ?

(A) 12 kW	**(B)** 1.2 kW
(C) 120 kW	**(D)** 1 kW

Q49. *CUSAT CAT 2025 - Shift 2*

The gravitational potential energy of a two-particle system is negative. What does this indicate about the system?

(A) The particles are repelling each other	**(B)** The particles are moving away from each other
(C) The particles are bound to each other	**(D)** None of the above

Q50. *CUSAT CAT 2025 - Shift 2*

A gas expands from an initial volume of 2 m3 to a final volume of 5 m3 at a constant pressure of 100 Pa. Calculate the work done by the gas.

(A) 100 J	**(B)** 200 J
(C) 300 J	**(D)** 400 J

Q51. *CUSAT CAT 2025 - Shift 3*

A position dependent force F = (7 – 2x + 3x2) N, acts on a small body of mass 2 kg and displaces it from x =0 to x = 5 m. Work done is

(A) 35 J	**(B)** 70 J
(C) 135 J	**(D)** 270 J

Q52. *CUSAT CAT 2025 - Shift 3*

The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance

(A) 1.389 H	**(B)** 138.88 H
(C) 0.138 H	**(D)** 13.89 H

Q53. *CUSAT CAT 2025 - Shift 3*

A uniform cable of mass ‘M’ and length ‘L’ is placed on a horizontal surface that is th part is hanging below the edge of the surface. To lift the hanging part of the cable up to the surface, the work done should be

(A) MgL n2	**(B)** MgL 2n2
(C) 2MgL n2	**(D)** nMgL

Q54. *CUSAT CAT 2025 - Shift 4*

A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be

(A) 450	**(B)** 900
(C) 1800	**(D)** 225

Q55. *CUSAT CAT 2025 - Shift 4*

If two identical heaters each rated as 1000 W, 220 V are connected in parallel to 220 V, then the total power consumed is

(A) 200 W	**(B)** 250 W
(C) 2000 W	**(D)** 2500 W

Q56.

CUSAT CAT 2025 - Shift 4

TTThhheee rrreeesssiiisssttttiiivvveee nnneeetttwwwooorrrkkk ssshhhooowwwnnn bbbeeellllooowww iiisss cccooonnnnnneeeccctttteeeddd tttooo aaa DDD...CCC sssooouuurrrcccceee pppfff 111666 VVV... TTThe power ccoonnssuummeedd bbyy tthhee nneettwwoorrkk iiss 44 WWaatttt.. TThhee vvaalluuee ooff R is

(A) 8 Ω	**(B)** 6 Ω
(C) 1 Ω	**(D)** 16 Ω

Q57.

CUSAT CAT 2025 - Shift 4

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase. (take g = 9.8 ms–2)

(A) -62720.0 J	**(B)** -627.2 J
(C) +627.2 J	**(D)** +784.0 J

Q58.

CUSAT CAT 2025 - Shift 4

If two similar springs each of spring constant K are joined in series, the new spring constant and time period (T) would be changed by a factor 1

(A) , 2 2 1	**(B)** , 2 4 1
(C) ,2 2 4 1	**(D)** ,2 2 2

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	A	3	C	4	A	5	A	6	C	7	B	8	A	9	B	10	B
11	D	12	A	13	A	14	B	15	C	16	C	17	A	18	A	19	B	20	A
21	C	22	B	23	D	24	C	25	B	26	C	27	C	28	B	29	B	30	D
31	A	32	A	33	A	34	A	35	B	36	D	37	D	38	B	39	D	40	B
41	D	42	C	43	B	44	B	45	B	46	C	47	B	48	B	49	C	50	C
51	C	52	D	53	B	54	C	55	C	56	A	57	B	58	A				

Rotational Motion

24 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A particle moves in a circle of radius 10.0 cm at a speed that uniformly increases. If the speed changes from 5.0 m/s to 6.0 m/s in 2.0 s, find the angular acceleration. 2

(A) 0.5 rad/s 2	**(B)** 1.0 rad/s 2
(C) 2.5 rad/s 2	**(D)** 5.0 rad/s

Q2. *CUSAT CAT 2023 - Shift 1*

When a planet moves around the Sun

(A) the angular momentum remains conserved	**(B)** the angular speed remains constant
(C) the linear velocity remains constant	**(D)** the linear momentum remains constant

Q3. *CUSAT CAT 2023 - Shift 2*

A constant torque of 31.4 Nm is exerted on a pivoted wheel. If the angular 2 acceleration of wheel is 4π rad/s , then the moment of inertia will be 2

(A) 2.5 kgm 2	**(B)** 4.5 kgm 2
(C) 3.5 kgm 2	**(D)** 5.5 kgm 2

Q4. *CUSAT CAT 2023 - Shift 3*

A pair of forces of equal magnitude but acting in opposite directions with different lines of action is known as a

(A) stretching	**(B)** compression
(C) torque	**(D)** bending

Q5. *CUSAT CAT 2023 - Shift 3*

Analogue of mass in rotational motion is

(A) moment of inertia	**(B)** torque
(C) radius of gyration	**(D)** angular momentum

Q6. *CUSAT CAT 2023 - Shift 4*

A flywheel experiencing a torque τ by an external motor rotates with an angular velocity ω. Then which one of the following is TRUE?

(A) ω is proportional to τ 1	**(B)** ω is proportional to τ
(C) ω is proportional to τ 1	**(D)** ω is proportional to τ

Q7. *CUSAT CAT 2023 - Shift 4*

A dancer on a turn table if suddenly folds her hands, then the speed of the turntable

(A) decreases	**(B)** remain unchanged
(C) increases	**(D)** None of the above

Q8. *CUSAT CAT 2023 - Shift 4*

A current carrying loop is placed in a uniform magnetic field. The torque acting on it does not depend upon

(A) area of the loop	**(B)** value of the current
(C) the magnetic field	**(D)** shape of the loop

Q9. *CUSAT CAT 2023 - Shift 4*

If a solid sphere, disc and hollow cylinder are allowed to roll down an inclined plane without slipping from the same height,

(A) the cylinder will reach the bottom first	**(B)** the disc will reach the bottom first
(C) the sphere will reach the bottom first	**(D)** all will reach the bottom at the same time

Q10. *CUSAT CAT 2024 - Shift 1*

A wheel having moment of inertia 2 kg m about its axis, rotates at 50 rpm, about this axis. Then the torque that can stop the wheel in one minute is π

(A) N-m 18 5π	**(B)** N-m 3 π
(C) N-m 6 2π	**(D)** N-m 3

Q11. *CUSAT CAT 2024 - Shift 1*

Moment of inertia of an object does not depend upon

(A) Mass of the object	**(B)** Mass distribution
(C) Axis of rotation	**(D)** Angular velocity

Q12. *CUSAT CAT 2024 - Shift 2*

A bar magnet of magnetic moment M and moment of inertia I is freely suspended such that the magnetic axial line is in the direction of the magnetic meridian. If the magnetic induction of earth's horizontal field is B and the magnet is displaced by a H very small angle θ, angular acceleration is MB θ

(A) H I IB θ	**(B)** H M Mθ
(C) IB H Iθ	**(D)** MB H

Q13. *CUSAT CAT 2024 - Shift 3*

A solid sphere, hollow sphere and a disc, all having same mass and radius, are placed at the top of an inclined plane and released. The friction coefficients between the objects and inclined plane are same and not sufficient to allow pure rolling. Least time will be taken in reaching the bottom will be by

(A) solid sphere	**(B)** hollow sphere
(C) the disc	**(D)** all will take same time

Q14. *CUSAT CAT 2024 - Shift 3*

What is the condition for rotational equilibrium of a body?

(A) Net force acting on the body is zero	**(B)** Net torque acting on the body is zero
(C) The body must be at rest	**(D)** The body must have constant linear velocity

Q15. *CUSAT CAT 2024 - Shift 4*

Which of the following statements is true regarding a rigid body?

(A) Its shape changes under the action of external forces	**(B)** It cannot rotate about its center of mass
(C) The distance between any two particles remains constant	**(D)** Its center of mass changes position over time

Q16. *CUSAT CAT 2024 - Shift 5*

A circular disc having a moment of inertia 4 kg m 2 about its axis rotates at 40 rpm. The torque required to stop the wheel in 2 minutes is 2

(A) π Nm 45 1	**(B)** π Nm 45 2
(C) π Nm 25 1	**(D)** π Nm 25

Q17. *CUSAT CAT 2025 - Shift 1*

A string is wound around a hollow cylinder of mass 5 kg and radius 0.5 m. If the string is now pulled with a horizontal force of 40 N, and the cylinder is rolling without slipping on a horizontal surface (see the figure) then the angular acceleration of the cylinder will be (neglect the mass and thickness of the string)

(A) 12 rad/s2	**(B)** 16 rad/s2
(C) 10 rad/s2	**(D)** 20 rad/s2

Q18. *CUSAT CAT 2025 - Shift 2*

Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter. The correct statement for this situation is

(A) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane	**(B)** The ring has the greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane
(C) All of them will have same velocity	**(D)** The cylinder has the greatest, and the sphere has the least velocity of the centre of mass at the bottom of inclined plane

Q19. *CUSAT CAT 2025 - Shift 3*

Moment of inertia of a body about a given axis is 1.5 kg m2. Initially the body is at rest. In order to produce rotational kinetic energy of 1200 J, the angular acceleration of 20 rad /s2 must be applied about the axis for a duration of

(A) 2 s	**(B)** 5 s
(C) 2.5 s	**(D)** 3 s

Q20.

CUSAT CAT 2025 - Shift 4

A solid sphere of radius R and mass M is rotating about its diameter. A solid cylinder of same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation KE sphere will be KE cylinder

(A) 3 : 1	**(B)** 2 : 3
(C) 1 : 5	**(D)** 1 : 4

Q21.

CUSAT CAT 2025 - Shift 4

A child stands at the centre of a turntable with two arms outstretched. The turntable is set rotating with an angular speed of 40 rev/min. If he holds his hand back and 2 thereby reduces his moment of inertia to times the initial value, and the turntable 5 rotates without the friction, the angular speed of the child is

(A) 100 rev/min	**(B)** 80 rev/min
(C) 40 rev/min	**(D)** 16 rev/min

Q22.

CUSAT CAT 2025 - Shift 4

A solid sphere and a hollow sphere of equal diameters are raised to the same potential. Then

(A) hollow sphere has more charge	**(B)** both have equal charge
(C) only hollow sphere has charge	**(D)** solid sphere has more charge

Q23.

CUSAT CAT 2025 - Shift 4

Two equal and opposite charges of 2 × 10–10 C are placed at a distance of 1 cm forming a dipole and are placed in an electric field of 2 × 105 N/C. The maximum torque on dipole is

(A) 2 2×10 –6 Nm	**(B)** 4 × 10–9 Nm
(C) 8 × 108 Nm	**(D)** 4 × 10–7 Nm

Q24.

CUSAT CAT 2025 - Shift 4

A disc is rotating with angular velocity ω. If a child sits on it, what is conserved ?

(A) Linear momentum	**(B)** Angular momentum
(C) Kinetic energy	**(D)** Moment of inertia

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	A	4	C	5	A	6	B	7	C	8	D	9	C	10	A
11	D	12	A	13	D	14	B	15	C	16	A	17	B	18	A	19	A	20	C
21	A	22	B	23	D	24	B												

Gravitation

24 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Which of the following is a non-central force?

(A) Electrostatic force	**(B)** Nuclear force
(C) Gravitational force	**(D)** Spring force

Q2. *CUSAT CAT 2023 - Shift 2*

If the gravitational force of attraction between two bodies varies as , the period of 3 r revolution of a planet around the sun would vary as

(A) r 2	**(B)** r
(C) r	**(D)** r 32

Q3. *CUSAT CAT 2023 - Shift 2*

The mass of moon is 7.4 × 10 kg and its radius is 1740 km. The escape velocity from moon is

(A) 1740 km/s	**(B)** 870.4 km/s
(C) 2.38 km/s	**(D)** 1.7 km/s

Q4. *CUSAT CAT 2023 - Shift 2*

There is no atmosphere on the Moon because

(A) it is closer to the Earth	**(B)** it revolves around the Earth
(C) it gets light from the Sun	**(D)** the root mean square velocity of the gas molecules is more than the escape velocity from moon

Q5. *CUSAT CAT 2023 - Shift 3*

A man of weight W = mg is standing on a lift which is moving upward with an a acceleration . If g is the acceleration due to gravity, the apparent weight of the man is 2 a

(A) W1+ g a	**(B)** W1– g a
(C) W1+ 2g a	**(D)** W1– 2g

Q6. *CUSAT CAT 2023 - Shift 4*

Two satellites of masses m and m (m > m) are revolving round the earth in 1 2 1 2 circular orbits of radius r and r (r > r) respectively. Which one of the following is 1 2 1 2 true of their speeds v and v ? 1 2

(A) v = v 1 2	**(B)** v > v 1 2
(C) v < v 1 2 v v	**(D)** 1 = 2 r r 1 2

Q7. *CUSAT CAT 2024 - Shift 1*

It is difficult to cook rice in an open vessel by boiling it at high altitude because of

(A) low boiling point and high pressure	**(B)** high boiling point and low pressure
(C) low boiling point and low pressure	**(D)** high boiling point and high pressure

Q8. *CUSAT CAT 2024 - Shift 1*

Coulomb force is

(A) force between nucleons	**(B)** a gravitational force
(C) an electrostatic force	**(D)** a weak van der Waal's force

Q9. *CUSAT CAT 2024 - Shift 1*

Geostationary satellites are in specified orbits at a height of about

(A) 600 km	**(B)** 36,000 km
(C) 1000 km	**(D)** 100 km

Q10. *CUSAT CAT 2024 - Shift 1*

The time period of a satellite is 5h. If the separation between the earth and the satellite is increased 4 times from the previous value, the new time period will be

(A) 80h	**(B)** 40h
(C) 20h	**(D)** 10h

Q11. *CUSAT CAT 2024 - Shift 1*

What will be the value of acceleration due to gravity on the surface of the earth if the radius of the earth suddenly decreases to 60% of its present value, keeping the mass of the earth unchanged?

(A) 9.81 m/s 2	**(B)** 5.89 m/s 2
(C) 16.35 m/s 2	**(D)** 27.25 m/s 2

Q12. *CUSAT CAT 2024 - Shift 2*

If g is the acceleration due to gravity at a place on the surface of the earth and g' is the acceleration due to gravity at a depth h below the surface of the earth (h < R, the radius of the earth), then h

(A) g'= g– R R	**(B)** g'= g– h
(C) g'= g h	**(D)** g'= g 1– R –5

Q13. *CUSAT CAT 2024 - Shift 2*

Two satellites of mass m, and 10 m, are put in the same orbit around the sun. If T and 1 T T are their time periods, then 1 = 2 T 2

(A) 4	**(B)** 1 1
(C) 2 1	**(D)** 4

Q14. *CUSAT CAT 2024 - Shift 4*

When a pendulum clock keeping correct time is taken to high altitudes, then

(A) It will keep correct time	**(B)** Its length should be decreased to keep correct time
(C) Its length should be increased to keep correct time	**(D)** It cannot keep correct time even if the length is changed

Q15. *CUSAT CAT 2024 - Shift 5*

If the earth stops rotating about its axis, the acceleration due to gravity will remain unchanged at

(A) Equator	**(B)** 45° latitude
(C) 60° latitude	**(D)** Poles

Q16. *CUSAT CAT 2025 - Shift 2*

The radius of a planet is double that of the earth but their average densities are the same. If the escape velocities at the planet and the earth are v and v respectively, p e then

(A) v p =2v e	**(B)** 2v p =v e
(C) v p =v e 1	**(D)** v = v p e 4

Q17. *CUSAT CAT 2025 - Shift 2*

The acceleration due to gravity on the moon's surface is 1.7m/sec2 and acceleration due to gravity on earth is 9.8m/sec2 . If time period of a simple pendulum on the earth is 3.5sec, its time period on the moon is

(A) 7sec	**(B)** 8.4sec
(C) 10.2sec	**(D)** 11.8sec

Q18. *CUSAT CAT 2025 - Shift 2*

Two satellites A and B of masses 200 kg and 400 kg are revolving around the earth at height of 600 km and 1600 km respectively. If T and T are the time periods of A and A B Brespectively then the value of T –T B A [Given: radius of earth = 6400 km, mass of earth = 6 × 1024 kg]

(A) 1.33 × 103 s	**(B)** 3.33 × 102 s
(C) 4.24 × 103 s	**(D)** 4.24 × 102 s

Q19. *CUSAT CAT 2025 - Shift 3*

The acceleration due to gravity on the earth's surface is g. If the diameter of the earth reduces to half of its original value and mass remains constant, then acceleration due to gravity on the surface of earth would be

(A) 2g	**(B)** 4g g
(C) 2 g	**(D)** 4

Q20. *CUSAT CAT 2025 - Shift 3*

Two spheres of masses m and M are situated in air and the gravitational force between them is F. The space around the masses is now filled with a liquid of specific gravity 5. The gravitational force will now be

(A) 3 F	**(B)** F
(C) F 6	**(D)** 5 F

Q21. *CUSAT CAT 2025 - Shift 3*

The time period of satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to 3 times its previous value, then approximate new time period of the satellite will be

(A) 40 hours	**(B)** 36 hours
(C) 30 hours	**(D)** 25 hours

Q22. *CUSAT CAT 2025 - Shift 4*

Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will

(A) keep floating at the same distance between them	**(B)** move towards each other
(C) move away from each other	**(D)** will become stationary

Q23. *CUSAT CAT 2025 - Shift 4*

TThhee aaccccceelleerraattiioonn dduuee ttoo ggrraavviittyy oonn tthhee ppllaanneett A is 9 times the aaccccceelleerraattiioonn dduuee ttoo gravity on the planet BB.. AA mmaann jjuummppss ttoo aa hheeiigghhtt ooff 22 mm oonn tthhee ssuurrffaaccee ooff A. What is tthhee hheeiigghhtt ooff jjuummpp bbyy tthhee ssaammee ppeerrssoonn oonn tthhee ppllaanneett B?

(A) 6 m 2	**(B)** m 3 2
(C) m 9	**(D)** 18 m

Q24. *CUSAT CAT 2025 - Shift 4*

The time period of satellite, revolving above earth's surface at a height equal to R will be (given G = π2m/s2, R = radius of earth)

(A) —	**(B)** √4
(C) √8	**(D)** √32 √2

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	C	4	D	5	C	6	C	7	C	8	C	9	B	10	B
11	D	12	D	13	A	14	B	15	D	16	A	17	B	18	A	19	B	20	B
21	B	22	B	23	D	24	C												

Mechanical Properties of Solids

15 questions

Q1. *CUSAT CAT 2023 - Shift 3*

The stress required to double the length of a wire of Young's modulus Y is

(A) Y	**(B)** 2Y Y
(C) 2 Y	**(D)** 4

Q2. *CUSAT CAT 2023 - Shift 4*

According to Hookes's law of elasticity, if the stress is increased to 4 times its previous value, the ratio of stress to strain

(A) increases by 4 times	**(B)** increases by 2 times
(C) remains unchanged	**(D)** depends on the material property

Q3. *CUSAT CAT 2023 - Shift 4*

A cylinder is stretched by two equal forces applied normal to its cross-sectional area. The restoring force per unit area in this case is called

(A) Poisson ratio	**(B)** Compressive strength
(C) Tensile stress	**(D)** Micro strain 3 2 3t t

Q4. *CUSAT CAT 2023 - Shift 5*

A load of 4.0 kg mass is suspended from a ceiling through a steel wire of radius 2.0 mm. Find the tensile stress developed in the wire when equilibrium is achieved. 6 2

(A) 2.00 × 10 N/m 6 2	**(B)** 3.18 × 10 N/m 6 2
(C) 8.00 × 10 N/m 6 2	**(D)** 16.18 × 10 N/m

Q5. *CUSAT CAT 2023 - Shift 5*

According to Hooke's law of elasticity, if stress is increased, the ratio of stress to strain

(A) becomes zero	**(B)** remains constant
(C) decreases	**(D)** increases

Q6. *CUSAT CAT 2024 - Shift 1*

What is the stress experienced by a solid when a force of 50 N is applied uniformly 2 over a cross-sectional area of 5 m ?

(A) 10 Pa	**(B)** 50 Pa
(C) 100 Pa	**(D)** 250 Pa

Q7. *CUSAT CAT 2024 - Shift 2*

According to Hook's law, if stress is increased, the ratio of stress to strain

(A) decreases	**(B)** increases
(C) remains constant	**(D)** zero

Q8. *CUSAT CAT 2024 - Shift 3*

Two wires 'A' and 'B' are stretched by the same load. The area of cross-section of the wire 'A' is double that of 'B'. Then the stress on 'B' is

(A) equal to that on A	**(B)** twice that on A
(C) half that on A	**(D)** four times that on A

Q9. *CUSAT CAT 2024 - Shift 4*

RReecciipprrooccaall ooff tthhee bbuullkk mmoodduulluuss iiss ccaalllleedd

(A) mmoodduulluuss ooff rriiggiiddiittyy	**(B)** volume stress
(C) volume strain	**(D)** compressibility

Q10. *CUSAT CAT 2024 - Shift 5*

A metal wire is first stretched beyond its elastic limit and then released. It

(A) loses its elastic property completely and it will not contract	**(B)** will contract to its original length
(C) will contract to its length at elastic limit	**(D)** will contract but final length will be greater than original length

Q11. *CUSAT CAT 2025 - Shift 1*

A steel wire having a radius of 2.0 mm, carrying a load of 4 kg, is hanging from ceiling. Given that g = 3.1 πms–2 , what will be the tensile stress that would be developed in the wire ?

(A) 4.8 × 106 Nm–2	**(B)** 5.2 × 106 Nm–2
(C) 6.2 × 106 Nm–2	**(D)** 3.1 × 106 Nm–2

Q12. *CUSAT CAT 2025 - Shift 2*

The rise of a liquid in a capillary tube depends on

(A) Mass of the liquid	**(B)** Radius of the capillary tube
(C) Density of the solid	**(D)** Bulk modulus of liquid

Q13. *CUSAT CAT 2025 - Shift 2*

A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to (Given bulk modulus of metal B = 8 × 1010 Pa)

(A) 0.6	**(B)** 1.67
(C) 5	**(D)** 20

Q14.

CUSAT CAT 2025 - Shift 3

The elastic limit of brass is 379 MPa. What should be the minimum diameter of brass rod if it is to support a 400 N load without exceeding its elastic limit?

(A) 1.16 mm	**(B)** 0.90 mm
(C) 1.36 mm	**(D)** 1.00 mm

Q15.

CUSAT CAT 2025 - Shift 4

The bulk modulus of a spherical object is B. If it is subjected to uniform pressure P, the fractional decrease in radius is P

(A) B B	**(B)** 3P 3P
(C) B P	**(D)** 3B

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	C	3	C	4	B	5	B	6	A	7	C	8	B	9	D	10	D
11	D	12	B	13	B	14	A	15	D										

Mechanical Properties of Fluids

18 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A mercury drop does not spread on a glass plate because the angle of contact between glass and mercury is

(A) less than 90°	**(B)** more than 90°
(C) zero	**(D)** 90°

Q2. *CUSAT CAT 2023 - Shift 2*

Internal energy per mole of gas depends on

(A) viscosity	**(B)** density
(C) temperature	**(D)** thermal conductivity

Q3. *CUSAT CAT 2023 - Shift 2*

Streamline flow is more likely for liquid with

(A) high density and high viscosity	**(B)** low density and low viscosity
(C) high density and low viscosity	**(D)** low density and high viscosity

Q4. *CUSAT CAT 2023 - Shift 4*

The angle of contact at the interface of water-glass is 0°, Ethylalcohol-glass is 0°, mercury-glass is 140° and Methyl iodide-glass is 30°. A glass capillary is put in a trough containing one of these four liquids. It is observed that the meniscus is convex upward. The liquid in the trough is

(A) water	**(B)** ethyl alcohol
(C) mercury	**(D)** methyl iodide

Q5. *CUSAT CAT 2023 - Shift 5*

Which one of the following statements about soap bubble is TRUE?

(A) Excess pressure in a soap bubble is inversely proportional to surface tension	**(B)** Excess pressure in a soap bubble is directly proportional to its radius
(C) Excess pressure in a soap bubble is directly proportional to square of its radius	**(D)** Excess pressure in a soap bubble is directly proportional to surface tension

Q6. *CUSAT CAT 2024 - Shift 1*

Plants get water through the roots by the action of

(A) Capillarity	**(B)** Viscosity
(C) Gravity	**(D)** Elasticity

Q7. *CUSAT CAT 2024 - Shift 2*

A fluid with viscosity η flows through a horizontal pipe of radius r and length L The pressure difference between the two ends of the pipe is P. What is the flow rate of the fluid? 4 πr P

(A) 8ηL 4 πr P	**(B)** 4ηL 4 πr P
(C) 2ηL 3 πr P	**(D)** 8ηL

Q8. *CUSAT CAT 2024 - Shift 2*

Ratio of absolute viscosity to the density of the fluid is called

(A) relative viscosity	**(B)** viscosity index
(C) kinematic viscosity	**(D)** fluidity

Q9. *CUSAT CAT 2024 - Shift 3*

For liquids which do not wet the solids (like Mercury), the angle of contact is

(A) < 90°	**(B)** is equal to 90°
(C) > 90°	**(D)** Zero

Q10. *CUSAT CAT 2024 - Shift 3*

The Reynold's number for fluid flow in a pipe is independent of the

(A) Viscosity of the fluid	**(B)** Length of the pipe
(C) Velocity of the fluid	**(D)** Diameter of the pipe

Q11. *CUSAT CAT 2024 - Shift 4*

The wettability of a surface by a liquid primarily depends on

(A) Surface tension	**(B)** Density
(C) Angle of contact between the surface and liquid	**(D)** Viscosity

Q12. *CUSAT CAT 2024 - Shift 5*

The height up to which water will rise in capillary tube will be

(A) Same at all temperatures	**(B)** Maximum when the temperature of water is 0°C
(C) Maximum when the temperature of water is 4°C	**(D)** Minimum when the temperature of water is 4°C

Q13. *CUSAT CAT 2024 - Shift 5*

The working of a scent sprayer is based on

(A) Archimedes' principle	**(B)** Charles' law
(C) Boyle's law	**(D)** Bernoulli's principle

Q14. *CUSAT CAT 2024 - Shift 5*

Pressure exerted by a liquid depends upon

(A) density	**(B)** viscosity
(C) surface tension	**(D)** buoyancy

Q15. *CUSAT CAT 2024 - Shift 5*

If there were no gravity, which one of the following will not be there for a fluid?

(A) Viscosity	**(B)** Surface tension
(C) Pressure	**(D)** Archimedes upthrust or Buoyant force 4

Q16. *CUSAT CAT 2025 - Shift 1*

A U tube contains water and spirit separated by mercury. The mercury columns in the two arms are in level with 10 cm of water in one arm and 12.5 cm of spirit in the other. Then the relative density of the spirit is

(A) 0.8	**(B)** 0.9
(C) 1	**(D)** 1.12

Q17. *CUSAT CAT 2025 - Shift 1*

A liquid does not wet the solid surface if angle of contact is

(A) equal to 45°	**(B)** equal to 60°
(C) greater than 90°	**(D)** zero

Q18. *CUSAT CAT 2025 - Shift 3*

The wettability of a surface by a liquid depends primarily on

(A) viscosity	**(B)** surface tension
(C) angle of contact between the surface and the liquid	**(D)** density

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	D	4	C	5	D	6	A	7	A	8	C	9	C	10	B
11	C	12	D	13	D	14	A	15	D	16	A	17	C	18	C				

Thermal Properties of Matter

17 questions

Q1. *CUSAT CAT 2023 - Shift 3*

Helium gas is filled in a closed vessel (having negligible thermal expansion coefficient). When it is heated from 300 K to 600 K then average kinetic energy of helium atoms will be

(A) Half	**(B)** unchanged
(C) two times	**(D)** 2 times

Q2. *CUSAT CAT 2023 - Shift 4*

The coefficient of linear expansion of a crystalline substance in the x, y and z –4 –4 –4 directions are 1 × 10 /°C, 2 × 10 /°C and 3 × 10 /°C respectively. The coefficient of cubical expansion of the crystal is equal to

(A) 36 × 10 –4 /°C	**(B)** 6 × 10 –4 /°C
(C) 10 –4 /°C	**(D)** 2 × 10 –4 /°C

Q3. *CUSAT CAT 2023 - Shift 5*

If the temperature of a black body increases from 27°C to 327°C, then the rate of energy radiation increases by 4 327

(A) 27	**(B)** 4
(C) 16	**(D)** 2

Q4. *CUSAT CAT 2024 - Shift 1*

A brass rod has a length of 10 m at 0°C. What is its length at 100°C, if the coefficient of linear expansion of brass is 0.000018 per °C ?

(A) 10 m	**(B)** 10. 018 m
(C) 9.82 m	**(D)** 11.018 m

Q5. *CUSAT CAT 2024 - Shift 1*

Which of the following is wrongly matched?

(A) Barometer-Pressure	**(B)** Lactometer-Milk
(C) Coulomb's law-Charges	**(D)** Humidity-Calorimeter

Q6. *CUSAT CAT 2024 - Shift 2*

The ratio of specific heats (C /C) for monoatomic gas is about p v

(A) 1.5	**(B)** 1.33
(C) 1.4	**(D)** 1.67

Q7. *CUSAT CAT 2024 - Shift 2*

Coefficient of surfacial expansion of a solid is 2 × 10 /°C . Then the coefficient of linear expansion is

(A) 4×10 –5 /°C	**(B)** 3×10 –5 /°C
(C) 2×10 –5 /°C	**(D)** 1×10 –5 /°C

Q8. *CUSAT CAT 2024 - Shift 3*

Water is used to cool radiators of engine, because

(A) of its lower density	**(B)** it is easily available
(C) it is cheap	**(D)** it has high specific heat

Q9. *CUSAT CAT 2024 - Shift 3*

Heat given to a substance for phase change is called

(A) specific heat	**(B)** latent heat
(C) sensible heat	**(D)** radiant heat

Q10. *CUSAT CAT 2024 - Shift 3*

What is the change in volume of a metal rod when its temperature increases by ΔT, given that the coefficient of linear expansion of the metal is α.

(A) $\alpha\Delta T$	**(B)** $2\alpha\Delta T$
(C) $3\alpha\Delta T$	**(D)** $4\alpha\Delta T$

Q11. *CUSAT CAT 2024 - Shift 4*

A black body at a temperature T = 3 K emits radiation of maximum intensity at a wavelength of

(A) 10 7 Å	**(B)** 1 Å
(C) 1 cm	**(D)** 10 –7 Å

Q12. *CUSAT CAT 2024 - Shift 5*

A lead bullet penetrates into a solid object and melts. The initial temperature of the bullet is 27°C and its melting point is 327°C. Latent heat of fusion of 4 lead = 2.5 × 10 J/kg and specific heat capacity of lead is 125 J/kg/K. Assuming that 50% of its kinetic energy was used to heat it, then the initial speed of the bullet is

(A) 200 m/sec	**(B)** 300 m/sec
(C) 400 m/sec	**(D)** 500 m/sec

Q13. *CUSAT CAT 2024 - Shift 5*

A body absorbs all the radiation incident over it is called

(A) perfect black body	**(B)** perfect white body
(C) emitter	**(D)** poor absorber

Q14.

CUSAT CAT 2025 - Shift 1

Coefficient of linear expansion of brass and steel rods are α and α . Lengths of brass 1 2 and steel rods are l and l respectively. If (l – l) is maintained same at all 1 2 2 1 temperatures, which one of the following relations holds good?

(A) =	**(B)** =
(C) =	**(D)** =

Q15.

CUSAT CAT 2025 - Shift 2

The ratio of specific heats =γ in terms of degrees of freedom (f) is given by C v f

(A) 1+ 3 2	**(B)** 1+ f f
(C) 1+ 2 1	**(D)** 1+ f

Q16.

CUSAT CAT 2025 - Shift 4

The intensity of radiation emitted by sun is maximum for the wavelength of 560 nm and that emitted by a star is maximum for the wavelength of 350 nm. If the average temperature of the sun is 2 × 107 K, the temperature of the star is

(A) 3.2 × 107K	**(B)** 4.3 × 106K
(C) 3.2 × 106K	**(D)** 5.4 × 107K

Q17.

CUSAT CAT 2025 - Shift 4

A black body is at a temperature of 300 K. It radiates energy at a rate proportional to

(A) 300	**(B)** (300)2
(C) (300)3	**(D)** (300)4

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	B	3	C	4	B	5	D	6	D	7	D	8	D	9	B	10	C
11	A	12	D	13	A	14	A	15	B	16	A	17	D						

Thermodynamics

24 questions

Q1. *CUSAT CAT 2023 - Shift 2*

The ratio of the slope of isothermal and adiabatic curves is

(A) γ 1	**(B)** γ
(C) 1 3	**(D)** 2

Q2. *CUSAT CAT 2023 - Shift 2*

The pressure of the gas in a constant volume gas thermometer is 80 cm of mercury in melting ice at 1 atm. When the bulb is placed in a liquid, the pressure becomes 160 cm of mercury. The temperature of liquid is

(A) 273.15 K	**(B)** 373.15 K
(C) 546.30 K	**(D)** 636.15 K 2

Q3. *CUSAT CAT 2023 - Shift 3*

In Carnot's engine, efficiency is 50% at hot reservoir temperature T. For efficiency 40%, what will be the temperature of hot reservoir? 2T

(A) 5	**(B)** 6T 6T
(C) 5 5T	**(D)** 6

Q4. *CUSAT CAT 2023 - Shift 3*

When you make ice cubes, the entropy of water

(A) increases	**(B)** decreases
(C) does not change	**(D)** either increase or decrease depending on the process

Q5. *CUSAT CAT 2023 - Shift 4*

An isochoric process is one which takes place at

(A) constant internal energy	**(B)** constant entropy
(C) constant volume	**(D)** constant pressure

Q6. *CUSAT CAT 2023 - Shift 5*

A monoatomic gas at a certain pressure P and volume V is compressed adiabatically 1 1 to one-eighth of its original volume. What is the final pressure of the gas? Given the γ 5 of an ideal monatomic gas = 3

(A) P 1 P	**(B)** 1 8
(C) 64 P 1	**(D)** 32 P 1

Q7. *CUSAT CAT 2023 - Shift 5*

Which law of Thermodynamics states that “Two systems in thermal equilibrium with a third system separately are in thermal equilibrium with each other”?

(A) Zeroth	**(B)** First
(C) Second	**(D)** Third

Q8. *CUSAT CAT 2023 - Shift 5*

If the volume of a gas is doubled at constant pressure, the average translational kinetic energy of its molecules will

(A) be doubled	**(B)** remain the same
(C) increase by a factor	**(D)** increase 4 times

Q9. *CUSAT CAT 2024 - Shift 1*

Physical independence of force is a consequence of

(A) Third law of motion	**(B)** First law of motion
(C) Law of gravitation	**(D)** Second law of motion

Q10. *CUSAT CAT 2024 - Shift 1*

Which of the following does not belong to the thermodynamic state of matter?

(A) Temperature	**(B)** Pressure
(C) Volume	**(D)** Work

Q11. *CUSAT CAT 2024 - Shift 1*

During an adiabatic process, the pressure of a gas is found to be proportional to the Cp cube of its absolute temperature. The ratio for the gas is Cv

(A) 2 3	**(B)** 2 4
(C) 3 5	**(D)** 3

Q12. *CUSAT CAT 2024 - Shift 2*

A Carnot engine is working between steam point and ice point. Then its efficiency will be

(A) 24.9%	**(B)** 25.7%
(C) 26.8%	**(D)** 28.8%

Q13. *CUSAT CAT 2024 - Shift 2*

The temperature of a hypothetical gas increases to 2 times when compressed adiabatically to half the volume. Its equation can be written as

(A) PV = constant 1	**(B)** PV2=constant 3
(C) PV2=constant 5	**(D)** PV2=constant

Q14. *CUSAT CAT 2024 - Shift 2*

The freezer in a refrigerator is located at the top section so that

(A) the entire chamber of the refrigerator is cooled quickly due to convection	**(B)** the motor is not heated
(C) the heat gained from the environment is high	**(D)** the heat gained from the environment is low

Q15. *CUSAT CAT 2024 - Shift 4*

A Carnot engine operates between two temperatures, T and T , with T > T . The 1 2 1 2 efficiency of the engine is given by T

(A) 1– 2 T 1 T	**(B)** T – 2 1 T 1 T
(C) 2 –1 T 1 T	**(D)** 1– 1 T 2

Q16. *CUSAT CAT 2025 - Shift 1*

The heat of 70 calories is required to raise the temperature of 2 mole of an ideal gas at constant pressure from 30°C to 35°C. The amount of heat needed to raise the temperature of the same gas through the same range at constant volume is

(A) 30 cal	**(B)** 40 cal
(C) 45 cal	**(D)** 50 cal

Q17. *CUSAT CAT 2025 - Shift 1*

When sound waves travel in any gaseous medium then at any point of the medium, process is

(A) isothermal	**(B)** isobaric
(C) adiabatic	**(D)** isochoric

Q18. *CUSAT CAT 2025 - Shift 1*

For a monoatomic gas, the adiabatic relation between pressure P and volume V is

(A) PV = constant 5	**(B)** PV3 = constant 7
(C) PV5 = constant 2	**(D)** PV3 = constant

Q19. *CUSAT CAT 2025 - Shift 2*

Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is

(A) adiabatic	**(B)** isochoric
(C) isobaric	**(D)** isothermal

Q20. *CUSAT CAT 2025 - Shift 2*

The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is

(A) 6.25%	**(B)** 20%
(C) 26.8%	**(D)** 12.5%

Q21. *CUSAT CAT 2025 - Shift 3*

A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is (Take 1 cal = 4.2 J)

(A) 2.365 W	**(B)** 23.65 W
(C) 236.5 W	**(D)** 2365 W

Q22. *CUSAT CAT 2025 - Shift 3*

C and C are specific heat at constant pressure and constant volume respectively. It is p v observed that C – C = a for hydrogen gas and C – C = b for nitrogen gas. The p v p v correct relation between a and b is

(A) a = 28b 1	**(B)** a = b 14
(C) a = b	**(D)** a = 14b

Q23. *CUSAT CAT 2025 - Shift 3*

A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be n

(A) n+2 n+2	**(B)** n n
(C) 2n+2 n	**(D)** n–2

Q24. *CUSAT CAT 2025 - Shift 4*

A Carnot engine absorbs 750 J of heat energy from a reservoir at 137°C and rejects 500 J of heat during each cycle, then the temperature of sink is

(A) 0.25°C	**(B)** 0.34°C
(C) 0.44°C	**(D)** 0.54°C

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	D	4	B	5	C	6	D	7	A	8	A	9	B	10	D
11	B	12	C	13	C	14	A	15	A	16	D	17	C	18	B	19	A	20	C
21	C	22	D	23	A	24	B												

Kinetic Theory

3 questions

Q1. *CUSAT CAT 2024 - Shift 2*

The number of degrees of freedom of a triatomic gas is

(A) 1	**(B)** 2
(C) 6	**(D)** 8

Q2. *CUSAT CAT 2024 - Shift 3*

Brownian motion is an evidence for

(A) Kinetic theory of matter	**(B)** Photoelectric effect
(C) Electromagnetic radiation	**(D)** Corpuscular theory of light

Q3. *CUSAT CAT 2025 - Shift 4*

The number of degrees of freedom of a rigid diatomic molecule is

(A) 2	**(B)** 3
(C) 4	**(D)** 5

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	D														

Oscillations

35 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A particle is executing simple harmonic motion from centre with an amplitude A and time period T. Its displacement after 2T will be

(A) A	**(B)** 2A
(C) 4A	**(D)** zero

Q2. *CUSAT CAT 2023 - Shift 1*

An AC voltage source of variable angular frequency ω and fixed amplitude V is o connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω increased,

(A) the bulb glows dimmer	**(B)** the bulb glows brighter
(C) the total impedance of the circuit is unchanged	**(D)** total impedance of the circuit increases 2 –1

Q3. *CUSAT CAT 2023 - Shift 1*

Which one of the following is a simple harmonic motion?

(A) Wave moving through a string fixed at both ends	**(B)** Earth spinning about its own axis
(C) Ball bouncing between two rigid vertical walls	**(D)** Particle moving in a circle with uniform speed

Q4. *CUSAT CAT 2023 - Shift 2*

Which one of the following equations does NOT represent a true simple harmonic motion? (Symbols have usual meaning.)

(A) $x = A \sin(\omega t + \phi)$	**(B)** $x = A \cos(\omega t - \phi)$
(C) $x = A \sin \omega t + B \cos \omega t$	**(D)** $x = A \sin(\omega t + \phi) + B \sin(2\omega t + \phi)$

Q5. *CUSAT CAT 2023 - Shift 2*

If the total energy of the reactants is more than the products of the reaction, heat is released and the reaction is said to be an

(A) exothermic	**(B)** endothermic
(C) elongate	**(D)** amplitude

Q6. *CUSAT CAT 2023 - Shift 2*

If a sound wave enters water from air, then what remains unchanged?

(A) Frequency	**(B)** Amplitude
(C) Velocity	**(D)** Wavelength

Q7. *CUSAT CAT 2023 - Shift 2*

The amplitude of the magnetic field of harmonic electromagnetic wave in vacuum is B = 450 nT. What is the amplitude of electric field part of the wave? 0

(A) 140 N/C	**(B)** 135 N/C
(C) 130 N/C	**(D)** 145 N/C

Q8. *CUSAT CAT 2023 - Shift 3*

When a wave travels from air into glass, there is no change in its

(A) frequency	**(B)** velocity
(C) amplitude	**(D)** wavelength 27 64

Q9. *CUSAT CAT 2023 - Shift 3*

What is the length of a simple pendulum?

(A) 1 m	**(B)** 0.5 m
(C) 0.25 m	**(D)** 0.125 m

Q10. *CUSAT CAT 2023 - Shift 3*

A particle of mass 0.50 kg undergoes a simple harmonic motion under a force F = –(50 N/m)x. If it crosses the centre of oscillation with the speed of 10 m/s, find the amplitude of the motion.

(A) 0.5 m	**(B)** 1 m
(C) 5 m	**(D)** 50 m

Q11. *CUSAT CAT 2023 - Shift 3*

A particle is moving in a circle with uniform speed. Its motion is

(A) not periodic	**(B)** periodic and simple harmonic
(C) periodic but not simple harmonic	**(D)** simple harmonic

Q12. *CUSAT CAT 2023 - Shift 3*

The displacement of a particle in simple harmonic motion is always measured from

(A) extreme position	**(B)** mean position
(C) midpoint of mean and extreme position	**(D)** yield point

Q13. *CUSAT CAT 2023 - Shift 4*

Two simple harmonic motions in the x-y plane are described by x= Acosωt and y= Acos(ωt+φ). The superposition of these two motions is a straight line. The value of ϕ would then be π

(A) 2 3π	**(B)** 2
(C) π π	**(D)** 4

Q14. *CUSAT CAT 2023 - Shift 4*

The phase difference between the displacement and acceleration of a particle executing simple harmonic motion is

(A) Zero π	**(B)** 2
(C) π	**(D)** 2π

Q15. *CUSAT CAT 2023 - Shift 5*

The closest length of the simple pendulum which ticks seconds (seconds pendulum) must be

(A) 1 m	**(B)** 1.5 m
(C) 2 m	**(D)** 10 cm 2

Q16. *CUSAT CAT 2023 - Shift 5*

Which of the following quantities does NOT change periodically for a particle performing SHM?

(A) Velocity	**(B)** Acceleration
(C) Displacement	**(D)** Total energy

Q17. *CUSAT CAT 2023 - Shift 5*

If the amplitude of sound is doubled and frequency reduced to one-fourth, the intensity of sound at the same point will be

(A) increasing by a factor of 2	**(B)** decreasing by a factor of 2
(C) decreasing by a factor of 4	**(D)** unchanged

Q18. *CUSAT CAT 2023 - Shift 5*

A ray of light from air is incident in water. Then, which property of light will not change in water?

(A) Velocity	**(B)** Frequency
(C) Colour	**(D)** Amplitude

Q19. *CUSAT CAT 2024 - Shift 1*

The velocity v and displacement x of a particle executing simple harmonic motion are dv 2 related as v =–ω x. If at x = 0, v=v . The velocity v when the displacement 0 dx becomes x is

(A) v= v2 –ω 2 x 2 0	**(B)** v= v2 +ω 2 x 2 0
(C) v= ω 2 x 2 –v2 0	**(D)** v=v2 +ω 2 x 2 0

Q20. *CUSAT CAT 2024 - Shift 1*

When two waves of the same frequency and amplitude are superimposed in phase, the resulting wave will have

(A) Lower amplitude	**(B)** Higher frequency
(C) Lower frequency	**(D)** Higher amplitude

Q21. *CUSAT CAT 2024 - Shift 1*

Two light sources are said to be coherent when light waves from both the sources have

(A) same speed	**(B)** same amplitude and phase
(C) same wavelength and a constant phase difference	**(D)** same intensity and wavelength

Q22. *CUSAT CAT 2024 - Shift 1*

Audio sine waves of 3 kHz frequency are used to amplitude modulate a carrier signal of 1.5 MHz. Which of the following statements are true?

(A) The side band frequencies are 1506 kHz and 1494 kHz.	**(B)** The bandwidth required for amplitude modulation is 6 kHz.
(C) The bandwidth required for amplitude modulation is 3 MHz.	**(D)** The side band frequencies are 1503 kHz and 1497 kHz.

Q23. *CUSAT CAT 2024 - Shift 1*

In simple harmonic motion, how does the force relate to displacement and its direction?

(A) Force is proportional to displacement, directed away from the centre.	**(B)** Force is inversely proportional to displacement, directed towards the centre
(C) Force is proportional to displacement, directed towards the centre.	**(D)** Force is inversely proportional to displacement, directed away from the centre.

Q24. *CUSAT CAT 2024 - Shift 2*

An amplitude modulated wave varies from 10 V to 4 V. Its percentage modulation is

(A) 36%	**(B)** 42.8%
(C) 54%	**(D)** 68%

Q25. *CUSAT CAT 2024 - Shift 2*

The displacement of a particle in SHM is always measured from

(A) Mean position	**(B)** Extreme position
(C) Mid-point of mean and extreme position	**(D)** Initial position

Q26. *CUSAT CAT 2024 - Shift 2*

When a source of sound moves towards an observer

(A) the frequency of the source is increased	**(B)** the velocity of sound in the medium is increased
(C) the wavelength of the sound in the medium towards the observer is decreased	**(D)** the amplitude of vibration of the particles is increased

Q27. *CUSAT CAT 2024 - Shift 3*

The restoring force of simple harmonic motion is maximum when

(A) at rest	**(B)** crosses mean position
(C) is half way between the mean and extreme position	**(D)** displacement is maximum

Q28. *CUSAT CAT 2024 - Shift 4*

If the length of the simple pendulum is increased, then the time period will

(A) increase	**(B)** decrease
(C) remain constant	**(D)** can't predict

Q29. *CUSAT CAT 2024 - Shift 4*

Match the following: Column A Column B (i) Attenuation (P) mixing of unwanted signal with the informational signal during propagation (ii) Amplification (Q) converts one form of energy into another (iii) Noise (R) the process of increasing the amplitude (iv) Transducer (S) is the loss of strength of a signal during the propagation

(A) (i) - (S), (ii) - (R), (iii) - (P), (iv) - (Q)	**(B)** (i) - (R), (ii) - (S), (iii) - (P), (iv) - (Q)
(C) (i) - (S), (ii) - (R), (iii) - (Q), (iv) - (P)	**(D)** (i) - (P), (ii) - (R), (iii) - (S), (iv) - (Q)

Q30. *CUSAT CAT 2024 - Shift 4*

The frequency of oscillation of a simple pendulum is independent of

(A) Amplitude	**(B)** Acceleration due to gravity
(C) Length of the pendulum	**(D)** Mass of the pendulum bob

Q31. *CUSAT CAT 2024 - Shift 5*

In a plane electromagnetic wave, if the electric field oscillates sinusoidally with a 10 –1 frequency 2 × 10 Hz and amplitude of 48 Vm , then the wavelength of the wave and strength of the oscillating magnetic field are 7

(A) 600 m, 6 × 10 tesla 8 –10	**(B)** 3 × 10 m, 24 × 10 tesla –2 –7
(C) 1.5 × 10 m, 1. 6 × 10 tesla 8 7	**(D)** 3 × 10 m, 1.6 × 10 tesla

Q32. *CUSAT CAT 2024 - Shift 5*

A particle executes simple harmonic motion. The amplitude of oscillation of particle is 2 cm. The displacement of particle in one time period is

(A) 1 cm	**(B)** 2 cm
(C) 4 cm	**(D)** Zero

Q33. *CUSAT CAT 2024 - Shift 5*

A particle undergoes simple harmonic motion of time period T. The time taken by the particle to go directly from the mean position to half the amplitude is T

(A) 2 T	**(B)** 3 T
(C) 6 T	**(D)** 12

Q34.

CUSAT CAT 2025 - Shift 4

A mass m is suspended from a spring of negligible mass and the system oscillates with a frequency f . The frequency of oscillations is f when a mass 16 m is 1 2 f suspended from the same spring. The value of 1 is f 2

(A) 1 :16	**(B)** 1 : 4
(C) 4 : 1	**(D)** 1 : 32

Q35.

CUSAT CAT 2025 - Shift 4

A plane electromagnetic wave has an electric field amplitude E = 300 V/m and 0 propagates in free space. What is the amplitude of the magnetic field (B) associated 0 with the wave?

(A) 0.5 × 10–6 T	**(B)** 1 × 10–6 T
(C) 1.5 × 10–6 T	**(D)** 2 × 10–6 T

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	B	3	A	4	D	5	A	6	A	7	B	8	A	9	A	10	B
11	C	12	B	13	C	14	C	15	A	16	D	17	C	18	B	19	A	20	D
21	C	22	A	23	C	24	B	25	A	26	C	27	D	28	A	29	A	30	D
31	C	32	D	33	D	34	C	35	B										

Waves

15 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A tube closed at one end and filled with air produces when excited; it produces the fundamental note of frequency 512 Hz. If the tube is open at both ends, the fundamental frequency that can be produced is

(A) 128 Hz	**(B)** 256 Hz
(C) 512 Hz	**(D)** 1024 Hz

Q2. *CUSAT CAT 2023 - Shift 2*

Two waves y =0.2sin320t and y =0.2sin314t are propagating along the same 1 2 direction. The number of beats produced per second are π

(A) 3 3	**(B)** π π
(C) 2 2	**(D)** π

Q3. *CUSAT CAT 2023 - Shift 2*

Ultrasonic waves in air produced by a vibrating quartz crystal are

(A) longitudinal waves	**(B)** transverse waves
(C) gravity waves	**(D)** electromagnetic waves

Q4. *CUSAT CAT 2023 - Shift 2*

An open pipe is suddenly closed at one end with the result that the frequency of the third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. Then the fundamental frequency of the open pipe is

(A) 100 Hz	**(B)** 150 Hz
(C) 200 Hz	**(D)** 400 Hz

Q5. *CUSAT CAT 2023 - Shift 2*

A 50 cm long horizontally placed wire of mass 20 g is fixed to a wall at one end and supports a mass of 1.6 kg at the other end hanging downwards with the help of a pulley. The length of the wire between the wall and the pulley is 40 cm. The fundamental frequency of the string between the wall and the pulley is

(A) 15 Hz	**(B)** 25 Hz
(C) 50 Hz	**(D)** 100 Hz

Q6. *CUSAT CAT 2023 - Shift 3*

If the Young's modulus of the material of a rod is 1.5 × 10 n/m and its density is –3 6000 kg m , the time taken by a sound wave to traverse 1 m of the rod will be –4

(A) 2 × 10 s –2	**(B)** 2 × 10 s –4
(C) 10 s –2	**(D)** 10 s

Q7. *CUSAT CAT 2023 - Shift 3*

A transverse wave travels along the z-axis. The particles of the medium must move

(A) along the z-axis	**(B)** along the y-axis
(C) along the x-axis	**(D)** in the x-y plane

Q8. *CUSAT CAT 2023 - Shift 4*

A sound wave of frequency 10 kHz is travelling in air with a speed of 340 m/s. Find the minimum separation between two points where the phase difference is 60°.

(A) 340.00 cm	**(B)** 57.00 cm
(C) 3.40 cm	**(D)** 0.57 cm

Q9. *CUSAT CAT 2023 - Shift 5*

If n ,n and n are the fundamental frequencies of three segments into which a string 1 2 3 is divided, then the original fundamental frequency n of the string is given by 1 1 1 1

(A) = + + n n n n 1 2 3	**(B)** n=n +n +n 1 2 3 1 1 1 1 = + +
(C) n n n n 1 2 3	**(D)** n = n + n + n 1 2 3

Q10. *CUSAT CAT 2023 - Shift 5*

When a stationary wave is formed by superposition, then its frequency is

(A) same as that of the individual waves	**(B)** twice that of the individual waves
(C) half that of the individual waves	**(D)** three fourth of the individual waves

Q11. *CUSAT CAT 2024 - Shift 1*

The fundamental frequency of guitar string of length 90 cm is 124 Hz. To produce a fundamental frequency of 186 Hz it has to be pressed at

(A) 45 cm	**(B)** 60 cm
(C) 75 cm	**(D)** 55 cm

Q12. *CUSAT CAT 2024 - Shift 1*

The sound waves whose frequencies lie below the audible limit are called

(A) Ultrasonic waves	**(B)** Supersonic waves
(C) Inaudible waves	**(D)** Infrasonic waves

Q13. *CUSAT CAT 2024 - Shift 1*

Which of the following effects cannot be observed for sound waves in air?

(A) Interference	**(B)** Refraction
(C) Polarization	**(D)** Diffraction CHEMISTRY UG SHIFT I FINAL ( )

Q14.

CUSAT CAT 2024 - Shift 5

In the transverse waves the direction of vibration in medium of particle is

(A) perpendicular to propagation of wave	**(B)** parallel to propagation
(C) alternate parallel and perpendicular	**(D)** in all three directions

Q15.

CUSAT CAT 2024 - Shift 5

The first overtone frequency of an open organ pipe is equal to the fundamental frequency of a closed organ pipe. What is the length of the closed pipe, if the length of open pipe is 80 cm?

(A) 20 cm	**(B)** 40 cm
(C) 60 cm	**(D)** 80 cm

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	B	3	A	4	C	5	B	6	A	7	D	8	D	9	A	10	A
11	B	12	D	13	C	14	A	15	A										

Electric Charges & Fields

30 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A uniform electric field having a magnitude E and direction along the positive X-axis o exists. If the potential at a point x = 0 is V, then the additional potential at X = +x will be

(A) +x E o	**(B)** –x E o 2
(C) +x E o 2	**(D)** – x E o

Q2. *CUSAT CAT 2023 - Shift 1*

A long thin flat sheet has a uniform surface charge density σ. The magnitude of electric field at a distance r from it is σ

(A) ε 0 σ	**(B)** ε r 0 σ
(C) 2ε 0 σ	**(D)** 2ε r 0

Q3. *CUSAT CAT 2023 - Shift 1*

Choose the CORRECT option from the following.

(A) Gauss's law is valid only for symmetrical charge distributions	**(B)** Gauss's law is valid only for charges placed in vacuum
(C) The electric field calculated by Gauss's law is the field due to the charges inside the Gaussian surface	**(D)** The flux of the electric field through a closed surface due to all the charges is equal to the flux due to the charges enclosed by the surface

Q4. *CUSAT CAT 2023 - Shift 1*

Due to relative motion of the magnet with respect to coil, an emf is induced in the coil in accordance with

(A) Ampere's circuital law	**(B)** Faraday's law
(C) Gauss's law	**(D)** Biot-Savart law

Q5. *CUSAT CAT 2023 - Shift 2*

Two dipoles of charges of magnitude 'e' are placed inside a cube. What will be the total electric flux coming out of the cube? 4e

(A) ε 0 2e	**(B)** ε 0 e
(C) ε 0	**(D)** Zero

Q6. *CUSAT CAT 2023 - Shift 2*

■B dA = 0 is representation of

(A) Gauss's Law for electricity	**(B)** Gauss's Law for magnetism
(C) Faraday's Law	**(D)** Ampere – Maxwell Law

Q7. *CUSAT CAT 2023 - Shift 2*

Tesla is the unit of

(A) electric flux	**(B)** magnetic flux
(C) electric field	**(D)** magnetic induction

Q8. *CUSAT CAT 2023 - Shift 2*

The magnetic field in a travelling electromagnetic wave has a peak value of 35 nT. The peak value of electric field strength is

(A) 6 V/m	**(B)** 8.3 V/m
(C) 10.5 V/m	**(D)** 12.5 V/m

Q9. *CUSAT CAT 2023 - Shift 3*

An electromagnetic wave is propagating in a certain medium with a velocity V in the positive x direction. The instantaneous oscillating electric field of this wave is along the +y direction. Then the direction of the oscillating magnetic field of the wave will be

(A) –y direction	**(B)** +z direction
(C) –z direction	**(D)** –x direction

Q10. *CUSAT CAT 2023 - Shift 3*

The ratio of contributions made by electric field and magnetic field components to the intensity of an electromagnetic wave is, (c being the velocity of light)

(A) 1 : 1	**(B)** c : 1 2
(C) c : 1	**(D)** c: 1

Q11. *CUSAT CAT 2023 - Shift 4*

The average electric field of electromagnetic waves in certain region of free –4 –1 space is 9 × 10 NC . Then the average magnetic field in the same region is of the order of –4

(A) 27 × 10 T –12	**(B)** 3 × 10 T –4
(C) 3 × 10 T 1 –4	**(D)** × 10 T 3

Q12. *CUSAT CAT 2023 - Shift 4*

The magnetic field in a travelling electromagnetic wave has a peak value of 40 nT. The peak value of electric field strength is

(A) 1 V/m	**(B)** 4 V/m
(C) 8 V/m	**(D)** 12 V/m

Q13. *CUSAT CAT 2023 - Shift 5*

An electron of mass of the order of 10 kg carries a charge of the order of –19 10 Coulomb. Then the charge carried by 1 kg of electrons will be of the order of 12

(A) 10 Coulomb –12	**(B)** 10 Coulomb –50
(C) 10 Coulomb 50	**(D)** 10 Coulomb

Q14. *CUSAT CAT 2023 - Shift 5*

Angle between equipotential surface and lines of force of an electric field is

(A) zero	**(B)** 90°
(C) 180°	**(D)** 45° –3

Q15. *CUSAT CAT 2024 - Shift 1*

An inductor may store energy in

(A) its coil	**(B)** its magnetic field
(C) its electric field	**(D)** both in electric and magnetic fields

Q16. *CUSAT CAT 2024 - Shift 1*

Avalanche breakdown is due to

(A) Noise effect	**(B)** High speed electron colliding with the valence electron
(C) Direct breaking of bands due to the electric field	**(D)** Temperature rise

Q17. *CUSAT CAT 2024 - Shift 2*

For which of the following dependences of drift velocity v on electric field E, is d Ohm's law obeyed?

(A) $v \propto E$ d 2	**(B)** $v \propto E$ d
(C) $v \propto E$ d 1	**(D)** $v \propto$ d E

Q18. *CUSAT CAT 2024 - Shift 3*

In the following circuit, what is the total charge on the combination of capacitors? –4

(A) 2 × 10 Coulomb	**(B)** 200 Coulomb 4
(C) 2 × 10 Coulomb	**(D)** 12.5 Coulomb

Q19. *CUSAT CAT 2024 - Shift 3*

A particle of mass 10 kg and charge 5 μC is thrown at a speed of 20 m/sec against 5 –1 a uniform electric field of strength 2 × 10 NC . The distance travelled by particle before coming to rest is

(A) 0.1 m	**(B)** 0.2 m
(C) 0.3 m	**(D)** 0.4 m

Q20. *CUSAT CAT 2024 - Shift 3*

When a point charge is kept at the centre of a metallic insulated spherical shell

(A) Electric field outside the sphere is zero	**(B)** Electric field inside the sphere is zero
(C) Electric potential inside the sphere is zero	**(D)** Net induced charge on the sphere is zero

Q21. *CUSAT CAT 2024 - Shift 4*

When a high energy UV photon beam enters an electric field, it will be

(A) Accelerated	**(B)** Deflected
(C) Retarded	**(D)** Undeflected

Q22. *CUSAT CAT 2024 - Shift 5*

A vertically falling charged liquid droplet of mass 2 × 10 kg is balanced in air by a 5 vertical electric field of magnitude 3.00 × 10 N/C. The charge on the droplet is

(A) 3.00 × 10 –10 C	**(B)** 3.57 × 10 –10 C
(C) 6.67 × 10 –10 C	**(D)** 1.60 × 10 –19 C

Q23. *CUSAT CAT 2025 - Shift 1*

An electric field = 25■+30 ■ N/C exists in a region of space. If the potential at the origin is taken to be zero, then the potential at x = 2 m, y = 2 m is

(A) –110 V	**(B)** –140 V
(C) –120 V	**(D)** –130 V

Q24. *CUSAT CAT 2025 - Shift 1*

Two point charges Q each are placed at a distance d apart. A third point charge q is placed at a distance x from mid-point on the perpendicular bisector. The value of x at which charge q will be experience the maximum coulomb's force is

(A) x=d d	**(B)** x= 2 d
(C) x= 2 d	**(D)** x= 2 2

Q25. *CUSAT CAT 2025 - Shift 2*

The magnetic field near a current carrying conductor is given by

(A) Coulomb's law	**(B)** Lenz's law
(C) Biot-Savart's law	**(D)** Kirchhoff's law

Q26. *CUSAT CAT 2025 - Shift 3*

The electric field of a plane electromagnetic wave is given by E = E (x + y) sin (kx – ωt), its magnetic field will be given by 0 E

(A) 0 (x – y) cos (kx – ωt) C E	**(B)** 0 (–x + y) sin(kz – ωt) C E
(C) 0 (x – y) sin(kz – ωt) C E	**(D)** 0 (x + y) sin(kz - ωt) C

Q27.

CUSAT CAT 2025 - Shift 3

Find the critical angle of a medium for a specific wavelength, if the medium has 4 relative permittivity 3 and relative permeability for this wavelength. 3

(A) 60°	**(B)** 15°
(C) 45°	**(D)** 30°

Q28.

CUSAT CAT 2025 - Shift 4

EEnneerrggyy iinn aa ccuurrrreenntt ccaarrrryyiinngg ccooiill iiss ssttoorreedd iinn tthhee ffoorrmm ooff

(A) electric field	**(B)** magnetic field
(C) ddiieelleeccttrriicc ssttrreennggtthh	**(D)** heat

Q29.

CUSAT CAT 2025 - Shift 4

What prevents protons in a nucleus from collapsing together despite the short-range nuclear attraction?

(A) Pauli exclusion principle	**(B)** Coulomb repulsion
(C) Quantum mechanical zero-point energy	**(D)** Shell model of the nucleus

Q30.

CUSAT CAT 2025 - Shift 4

The electromagnetic waves travel in a medium at a speed of 2.0 × 108 m/s. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be

(A) 2.25	**(B)** 4.25
(C) 6.25	**(D)** 8.25

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	D	4	B	5	D	6	B	7	D	8	C	9	B	10	A
11	B	12	D	13	A	14	B	15	B	16	B	17	A	18	A	19	B	20	D
21	D	22	C	23	A	24	D	25	C	26	B	27	D	28	B	29	B	30	A

Electrostatic Potential & Capacitance

34 questions

Q1. *CUSAT CAT 2023 - Shift 1*

What is the effective capacitance between points X and Y?

(A) 24 µf	**(B)** 18 µf
(C) 12 µf	**(D)** 6 µf

Q2. *CUSAT CAT 2023 - Shift 1*

An inductor of inductance 100 mH is connected in series with a resistance, a variable capacitance and an AC source of frequency 2.0 kHz. What should be the value of the capacitance so that maximum current may be drawn into the circuit?

(A) 6.3 nF	**(B)** 12.6 nF
(C) 43.4 nF	**(D)** 63.0 nF

Q3. *CUSAT CAT 2023 - Shift 1*

The charge on two plates of a parallel plate capacitor of capacitance C are 3Q and –Q, respectively. The potential of the capacitor is [None of the plates of capacitor is grounded] Q

(A) C 3Q	**(B)** C 4Q
(C) C 2Q	**(D)** C

Q4. *CUSAT CAT 2023 - Shift 2*

Three capacitors of capacitances 6µF each are available. The minimum and maximum capacitances, which may be obtained are

(A) 6 µF, 18 µF	**(B)** 3 µF, 12 µF
(C) 2 µF, 12 µF	**(D)** 2 µF, 18 µF

Q5. *CUSAT CAT 2023 - Shift 2*

A parallel plate capacitor has plates of area 200 cm and the separation between the plates 1.00 mm. What potential difference will be developed if a charge of 1.00 nC is given to the capacitor?

(A) 0.17 V	**(B)** 1.00 V
(C) 5.65 V	**(D)** 11.30 V

Q6. *CUSAT CAT 2023 - Shift 3*

A parallel plate capacitor is made by stacking 5 identical equally spaced metallic plates having the same dielectric between the plates. The alternate plates are then connected. If the capacitor formed by two neighbouring plates has a capacitance C, the total capacitance of the combination is

(A) 5 C	**(B)** 4 C C
(C) 5 C	**(D)** 4

Q7. *CUSAT CAT 2023 - Shift 4*

The charge of a parallel plate capacitor is varying as q=q sin2πvt. The plates are 0 very large and close together. Neglecting edge effects, the displacement current through the capacitor is q

(A) Aε 0 q	**(B)** 0 sin2πvt ε 0
(C) 2πvq cos2πvt 0 2πvq	**(D)** 0 cos2πvt ε 0

Q8. *CUSAT CAT 2023 - Shift 5*

Phase difference between voltage and current in a capacitor in an ac circuit is

(A) π π	**(B)** 2
(C) 0 π	**(D)** 3

Q9. *CUSAT CAT 2024 - Shift 1*

Three capacitors C , C and C are connected to a 6 V battery as shown. Then the 1 2 3 charge on C will be 1 C =10μF C =5μF 1 2 6V C =5μF 3

(A) 30 μC	**(B)** 25 μC
(C) 15 μC	**(D)** 60 μC

Q10. *CUSAT CAT 2024 - Shift 1*

The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A is

(A) proportional to the square root of the distance between the plates	**(B)** linearly proportional to the distance between the plates
(C) independent of the distance between the plates	**(D)** inversely proportional to the distance between the plates

Q11. *CUSAT CAT 2024 - Shift 2*

The parallel combination of inductor and capacitor is called

(A) a rectifier circuit	**(B)** a filter circuit
(C) a tank circuit	**(D)** an acceptor circuit

Q12. *CUSAT CAT 2024 - Shift 2*

In a parallel plate capacitor, the capacity increases if

(A) area of the plate is increased	**(B)** distance between the plates increases
(C) area of the plate is decreased	**(D)** dielectric constant decreases

Q13. *CUSAT CAT 2024 - Shift 3*

Two plates of a parallel plate capacitor separated by a layer of air has a capacitance of 15 μF. If a sheet of mica with dielectric constant k = 6 is inserted between the plates, the capacitance will be

(A) 2.5 μF	**(B)** 0.4 μF
(C) 90 μF	**(D)** 9 μF

Q14. *CUSAT CAT 2024 - Shift 3*

A capacitor of capacitance C has charge Q and stored energy is W. If the charge is increased to 2Q, the stored energy will be W

(A) 4 W	**(B)** 2
(C) 2W	**(D)** 4W

Q15. *CUSAT CAT 2024 - Shift 3*

The force between two charges situated in air is F. The force between the same charges if the distance between them is reduced to half and they are situated in a medium having dielectric constant 4 is F

(A) 4	**(B)** 4F
(C) 16F	**(D)** F

Q16. *CUSAT CAT 2024 - Shift 3*

Three capacitors each of capacitance 9 pF are connected in series. Calculate the total capacitance of the combination.

(A) 27 pF	**(B)** 9 pF
(C) 3 pF	**(D)** 6 pF

Q17. *CUSAT CAT 2024 - Shift 5*

Energy stored in the capacitor is 1 2

(A) U = ε E d 0 2 1 2	**(B)** U = ε E 2d 0 2 1 2
(C) U = ε E Ad 0 2 1 2	**(D)** U = ε E 0 2

Q18. *CUSAT CAT 2024 - Shift 5*

A parallel plate capacitor has plates of area 300.00 cm with the separation between them being 2.00 mm. If a charge of 2 nC is given to the capacitor, what will be the potential difference that is developed between the plates?

(A) 5.42 V	**(B)** 15.04 V
(C) 20.12 V (D 75.00 V CHEMISTRY UG (SHIFT V FINAL)	**(D)** —

Q19. *CUSAT CAT 2025 - Shift 1*

A 4 μF capacitor is charged to 400 V and then its plates are joined through a resistance of 1 kΩ. The heat produced in the resistance is

(A) 0.16 J	**(B)** 1.28 J
(C) 0.64 J	**(D)** 0.32 J

Q20. *CUSAT CAT 2025 - Shift 1*

A series of LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance R = 3 kΩ, an inductor of inductive reactance X =250 π Ω and an unknown capacitor. L The value of capacitance to maximize the average power should be (take π2 = 10)

(A) 400 µF	**(B)** 4 µF
(C) 40 µF	**(D)** 25 µF

Q21. *CUSAT CAT 2025 - Shift 1*

Two identical thin metal plates has charge q and q respectively such that q ■ q . The 1 2 1 2 plates were brought close to each other to form a parallel plate capacitor of capacitance C . The potential difference between them is (q +q)

(A) 1 2 C (q –q)	**(B)** 1 2 C (q –q)
(C) 1 2 2C 2(q –q)	**(D)** 1 2 C

Q22. *CUSAT CAT 2025 - Shift 2*

Two capacitors have a capacitance of 5µF when connected in parallel and 1.2µF when connected in series. Then the capacitances are

(A) 2µF and 3µF	**(B)** 3µF and 5µF
(C) 2µF and 4µF	**(D)** 1µF and 2µF

Q23. *CUSAT CAT 2025 - Shift 2*

In the capacitor of capacitance C, charge Q and energy W is stored. If charge is increased upto 2Q, the energy stored will be W

(A) 4 W	**(B)** 2
(C) 2W	**(D)** 4W

Q24. *CUSAT CAT 2025 - Shift 2*

Sphere A with radius a and sphere B with radius b are at same electric potential. Then the ratio of the surface charge densities of A and B is a

(A) b b	**(B)** a 2 a
(C) 2 b 2 b	**(D)** 2 a

Q25. *CUSAT CAT 2025 - Shift 2*

In an LCR circuit, the capacitance is changed from C to 2C. For the resonant frequency to remain unchanged, the inductance should be changed from L to

(A) 4L	**(B)** 2L L
(C) 2 L	**(D)** 4

Q26.

CUSAT CAT 2025 - Shift 3

A parallel plate condenser with oil between the plates (dielectric constant of oil k = 2) has a capacitance C. If the oil is removed, then capacitance of the capacitor becomes

(A) 2C C	**(B)** 2
(C) 2C C	**(D)** 2

Q27.

CUSAT CAT 2025 - Shift 3

A parallel-plate capacitor initially has a capacitance of 10 μF and is charged to 100 V. If the plate separation is reduced by half while maintaining the same charge on the capacitor, what is the resulting capacitance?

(A) 5 μF	**(B)** 10 μF
(C) 20 μF	**(D)** 40 μF

Q28.

CUSAT CAT 2025 - Shift 3

A capacitor of capacitance C = 1 μF is suddenly connected to a battery of 100 volt through a resistance R = 100 Ω. The time taken for the capacitor to be charged to get 50 V is Takeln2 = 0.69

(A) 1.44 × 10–4 S	**(B)** 3.33 × 10–4 S
(C) 0.69 × 10–4 S	**(D)** 0.30 × 10–4 S

Q29.

CUSAT CAT 2025 - Shift 3

In an LC oscillator, if values of inductance and capacitance become twice and eight times, respectively, then the resonant frequency of oscillator becomes ‘x’ times its initial resonance frequency ω . The value of ‘x’ is 0 1

(A) 4	**(B)** 4
(C) 16 1	**(D)** 6

Q30.

CUSAT CAT 2025 - Shift 4

A capacitor of capacitance C has reactance X . If capacitance and frequency become c double, then the capacitive reactance will be

(A) 2X c	**(B)** 4X c X c
(C) 2 X c	**(D)** 4

Q31.

CUSAT CAT 2025 - Shift 4

An alternating voltage E (in V) = 200 2 sin (100 t) is applied to a 1 μ F capacitor through an AC ammeter. Then the ammeter will read a current of

(A) 10 mA	**(B)** 20 mA
(C) 40 mA	**(D)** 80 mA

Q32.

CUSAT CAT 2025 - Shift 4

To establish an instantaneous current of 2 A through a 1 μF capacitor; the potential difference across the capacitor plates should be changed at the rate of

(A) 2 × 10 4 V/s	**(B)** 4 × 10 6 V/s
(C) 2 × 10 6 V/s	**(D)** 4 × 10 4 V/s

Q33.

CUSAT CAT 2025 - Shift 4

A 90 pF capacitor is connected to a 12 V battery. How many electrons are transferred from one plate to another?

(A) 1.1 × 10 9	**(B)** 6.7 × 10 9
(C) 4 × 10 19	**(D)** 5 × 10 19

Q34.

CUSAT CAT 2025 - Shift 4

A capacitor of 2 μμFFF iiisss ccchhhaaarrrgggeeeddd aaasss ssshhhooowwwnnn iiinnn ttthhheee fffiiiggguuurrreee... WWWhhheeennn ttthhheee ssswwwiiitttccchhh S is turned to ppoossiittiioonn 22,, tthhee ppeerrcceennttaaggee ooff iittss ssttoorreedd eenneerrggyy ddiisssiippaatteedd iiss

(A) 20%	**(B)** 75%
(C) 80%	**(D)** 0%

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	D	3	D	4	D	5	C	6	B	7	C	8	B	9	A	10	A
11	C	12	A	13	C	14	D	15	D	16	C	17	C	18	B	19	D	20	C
21	C	22	A	23	D	24	B	25	C	26	D	27	C	28	C	29	A	30	D
31	B	32	C	33	B	34	C												

Current Electricity

58 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A wire of resistance 8 Ω is bent into a circle. The resistance between the ends of a diameter of the circle is 1

(A) Ω 4 1	**(B)** Ω 8
(C) 2 Ω	**(D)** 16 Ω

Q2. *CUSAT CAT 2023 - Shift 1*

A silver wire has a resistance of 2.1 Ω at 27.5°C, and a resistance of 2.7 Ω at 100°C. What is the temperature coefficient of resistivity of silver? –1

(A) 0.0059°C –1	**(B)** 0.0039°C –1
(C) 0.0129°C –1	**(D)** 0.0030°C

Q3. *CUSAT CAT 2023 - Shift 1*

An induced e.m.f. is produced when a magnet is plunged into a coil. The strength of the induced e.m.f. is independent of

(A) the strength of the magnet	**(B)** number of turns of coil
(C) the resistivity of the wire of the coil	**(D)** speed with which the magnet is moved

Q4. *CUSAT CAT 2023 - Shift 2*

A compass needle suffers a deflection when placed near a wire carrying an electric current. This was observed by

(A) Hans Christian Oersted	**(B)** Hendrik Antoon Lorentz
(C) Gustav Robert Kirchhoff	**(D)** Georg Simon Ohm

Q5. *CUSAT CAT 2023 - Shift 2*

To convert a galvanometer into an ammeter, we should connect

(A) a low resistance in series	**(B)** a low resistance in parallel
(C) a high resistance in series	**(D)** a high resistance in parallel

Q6. *CUSAT CAT 2023 - Shift 2*

What is the voltage gain in a common emitter amplifier where input resistance is 3 Ω and load resistance 24 Ω, $\beta = 0.6$?

(A) 8.4	**(B)** 4.8
(C) 2.4	**(D)** 480

Q7. *CUSAT CAT 2023 - Shift 2*

Resistance of the metallic conductors

(A) increases with rise in temperature	**(B)** decreases with rise in temperature
(C) remains unchanged with change in temperature	**(D)** becomes zero at very high temperature CHEMISTRY (UG) – SHIFT II (FINAL)

Q8. *CUSAT CAT 2023 - Shift 3*

A platinum resistance thermometer records the temperature as 0°C and 100°C when the resistance of the platinum wire is 80 Ω and 90 Ω. Find the temperature read by the thermometer when the resistance of the platinum wire is 86 Ω.

(A) 10°C	**(B)** 20°C
(C) 30°C	**(D)** 60°C

Q9. *CUSAT CAT 2023 - Shift 3*

A common emitter transistor amplifier has a current gain of 50. If the input resistance and the load resistance of the amplifier circuit are 500 Ω and 4 kΩ respectively, the voltage gain of the amplifier is

(A) 100	**(B)** 200
(C) 300	**(D)** 400

Q10. *CUSAT CAT 2023 - Shift 3*

A resistor develops 400 J of thermal energy in 10 s when a current of 2 A is passed through it. Find its resistance.

(A) 10 Ω	**(B)** 20 Ω
(C) 200 Ω	**(D)** 400 Ω

Q11. *CUSAT CAT 2023 - Shift 3*

A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 20 mV is added to the base-emitter voltage, the base current changes by 20 μA and the collector current changes by 2 mA. If the load resistance is 5 kΩ, find the β factor.

(A) 1	**(B)** 10
(C) 100	**(D)** 1000

Q12. *CUSAT CAT 2023 - Shift 3*

A battery of emf E and negligible internal resistance is connected to a resistor R. Taking the heating effect of current and its further effects on circuit into the considerations, indicate which of the plots shown in figure best represents the rate of production of thermal energy in the resistor. P A B C D t

(A) A	**(B)** B
(C) C	**(D)** D

Q13. *CUSAT CAT 2023 - Shift 4*

The colour code on a resistor is red, red, black. The value of its resistance is

(A) (22±2.2) Ω	**(B)** 22 Ω
(C) (22±0.44) Ω	**(D)** (22±4.4) Ω

Q14. *CUSAT CAT 2023 - Shift 4*

A milliammeter of resistance 40 Ω has a range of 0-30 mA. What will be the resistance used in series to convert it into a voltmeter of range 0-15 V?

(A) 300 Ω	**(B)** 360 Ω
(C) 400 Ω	**(D)** 460 Ω

Q15. *CUSAT CAT 2023 - Shift 4*

The potential difference across 8 Ω resistance is 48 V as shown in the figure. The value of potential difference across X and Y points will be

(A) 160 V	**(B)** 128 V
(C) 80 V	**(D)** 62 V

Q16. *CUSAT CAT 2023 - Shift 4*

A battery of emf 2 V and internal resistance 0.5 Ω is connected across a resistance of 9.5 Ω. How many electrons cross through a cross-section of the resistance in 1 second? 18

(A) 1.25 × 10 18	**(B)** 3.25 × 10 15
(C) 1.25 × 10 15	**(D)** 3.25 × 10

Q17. *CUSAT CAT 2023 - Shift 4*

What will be the reading of the voltmeter across the resistance and ammeter in the circuit shown in the figure? R=100ΩΩΩΩ V V V 100 V 100 V A 220V, 50Hz

(A) 300 V, 2A	**(B)** 800 V, 2A
(C) 100 V, 2A	**(D)** 220 V, 2.2A

Q18. *CUSAT CAT 2023 - Shift 5*

The smallest resistance that can be obtained by combining n resistors, each of resistance R is

(A) nR R	**(B)** n
(C) n 2 R R	**(D)** 2 n 1

Q19. *CUSAT CAT 2023 - Shift 5*

A wire of resistance R is cut into n equal parts. These parts are then connected in parallel. The equivalent resistance of the combination will be

(A) nR R	**(B)** n n
(C) R R	**(D)** 2 n

Q20. *CUSAT CAT 2023 - Shift 5*

The resistivity of certain metals or alloys drops to zero when they are cooled below a certain temperature, this phenomenon is known as

(A) Conductivity	**(B)** Partial conductivity
(C) Superconductivity	**(D)** Non-conductivity

Q21. *CUSAT CAT 2024 - Shift 1*

Eight identical cells each of emf E and internal resistance r are connected in series. If the polarity of two cells are reversed, then the total internal resistance of the circuit will be

(A) 8r	**(B)** 6r
(C) 4r	**(D)** 2r

Q22. *CUSAT CAT 2024 - Shift 1*

An AC voltage is applied to a circuit consisting of a resistor and inductor in series. If the resistance and inductive reactance of both are 3 Ω, what is the phase difference between the applied voltage and current in the circuit? π

(A) 6 π	**(B)** 4 π
(C) 2	**(D)** Zero

Q23. *CUSAT CAT 2024 - Shift 1*

Platinum and silicon are heated up to 250°C and after that cooled. In the process of cooling

(A) resistance of platinum will increase and that of silicon will decrease	**(B)** resistance of silicon will increase and that of platinum will decrease
(C) resistance of both will increase	**(D)** resistance of both will decrease

Q24. *CUSAT CAT 2024 - Shift 1*

In the given circuit, the current through the battery is D3 5Ω D1 10Ω D2 5Ω 20Ω 10V

(A) 1 A	**(B)** 1.5 A
(C) 2 A	**(D)** 10 A

Q25. *CUSAT CAT 2024 - Shift 2*

Two resistances R = 10.0 Ω ± 0.1 Ω and R = 5.0 ± 0.1 Ω are connected in parallel. 1 2 Then % of error in the resultant value of R is

(A) 4.3%	**(B)** 1.393%
(C) 3.83%	**(D)** 0.0133%

Q26. *CUSAT CAT 2024 - Shift 2*

If an ammeter is used in the place of a voltmeter, then we must connect the ammeter with

(A) a low resistance in parallel	**(B)** a high resistance in parallel
(C) a high resistance in series	**(D)** a low resistance in series

Q27. *CUSAT CAT 2024 - Shift 3*

When a cell is undergoing charging process

(A) There is no voltage drop in its internal resistance	**(B)** Its terminal potential is less than its emf
(C) Its terminal potential is more than its emf	**(D)** Its terminal potential is zero

Q28. *CUSAT CAT 2024 - Shift 3*

In LCR circuit, if resistance increases, quality factor

(A) increases finitely	**(B)** decreases finitely
(C) remains constant	**(D)** None of the above

Q29. *CUSAT CAT 2024 - Shift 3*

If the galvanometer current is 10 mA, the resistance of the galvanometer is 40 Ω and shunt of 2 Ω is connected to the galvanometer, the maximum current which can be measured by this ammeter is

(A) 0.21 A	**(B)** 2.1 A
(C) 210 A	**(D)** 21 A –3

Q30. *CUSAT CAT 2024 - Shift 3*

Which of the following statement is true?

(A) A Van de Graff generator produces large voltage and less current	**(B)** A Van de Graff generator produces large resistance and less voltage
(C) A Van de Graff generator produces large current and large resistance	**(D)** A Van de Graff generator produces large current and less voltage

Q31. *CUSAT CAT 2024 - Shift 4*

Pieces of copper and germanium are cooled from room temperature to 80 K. The resistance of

(A) Copper increases and germanium increases	**(B)** Copper increases and germanium decreases
(C) Copper decreases and germanium decreases	**(D)** Copper decreases and germanium increases

Q32. *CUSAT CAT 2024 - Shift 4*

The fuse is made from a wire of

(A) low resistance and high melting point	**(B)** high resistance and high melting point
(C) high resistance and low melting point	**(D)** low resistance and low melting point

Q33. *CUSAT CAT 2024 - Shift 4*

A meter bridge is set up as shown, to determine an unknown resistance ‘X’ using a standard 100 ohm resistor. The galvanometer shows null point when the tapping-key is at 52 cm mark. The end-corrections are 3 cm and 2 cm respectively for ends A and B. The determined value of ‘X’ is

(A) 100 ohm	**(B)** 106 ohm
(C) 108 ohm	**(D)** 110 ohm

Q34. *CUSAT CAT 2024 - Shift 4*

In a coil of resistance 100 Ω, a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is

(A) 200 Wb	**(B)** 225 Wb
(C) 250 Wb	**(D)** 275 Wb

Q35. *CUSAT CAT 2024 - Shift 4*

Two resistors of resistance 5 ohms and 15 ohms are connected in series. If the voltage drop across 5 ohms resistor is 2.5 V, the current across 15 ohms resistor is

(A) 0.5 A	**(B)** 1 A
(C) 5 A	**(D)** insufficient data

Q36. *CUSAT CAT 2024 - Shift 4*

The coils in resistance boxes are made from doubled insulated wire to nullify the effect of

(A) heating	**(B)** magnetism
(C) pressure	**(D)** self induced e.m.f.

Q37. *CUSAT CAT 2024 - Shift 5*

No current flows between two charged bodies when connected, if they have same

(A) capacity	**(B)** potential
(C) resistance	**(D)** charges

Q38. *CUSAT CAT 2024 - Shift 5*

What happens during the regulation action of a Zener diode?

(A) The current and voltage across the Zener diode remains fixed	**(B)** The Zener resistance is constant
(C) The resistance offered by the Zener diode changes	**(D)** The current through the series resistance (R) does not change s

Q39. *CUSAT CAT 2024 - Shift 5*

What is immaterial for an electric fuse wire?

(A) Specific resistance	**(B)** Radius
(C) Length	**(D)** Current flowing through it

Q40. *CUSAT CAT 2024 - Shift 5*

The temperature coefficient of resistance is positive for

(A) Carbon	**(B)** Germanium
(C) Electrolyte	**(D)** Aluminium

Q41. *CUSAT CAT 2024 - Shift 5*

A certain carbon resistor has following colour codes: I band – Yellow; II band – Violet; III band – Orange; IV band – Gold Then

(A) the value of the resistance is 4.7 KΩ ±5% tolerance	**(B)** the value of the resistance is 47 KΩ ±5% tolerance
(C) the value of the resistance is 47 KΩ ±10% tolerance	**(D)** the value of the resistance is 4.7 KΩ ±10% tolerance

Q42. *CUSAT CAT 2025 - Shift 1*

In a potentiometer arrangement, a cell of emf 1.5 V gives a balance point at 27 cm length of wire. If the cell is replaced by another cell and balance point shifts to 54 cm, the emf of the second cell is

(A) 3 V	**(B)** 1.5 V
(C) 0.75 V	**(D)** 2.25 V

Q43. *CUSAT CAT 2025 - Shift 1*

The specific resistance of a conductor increases with

(A) increase in temperature	**(B)** increase in cross sectional area
(C) decrease in length	**(D)** decrease in cross sectional area

Q44. *CUSAT CAT 2025 - Shift 1*

A cell of emf 90 V is connected across series combination of two resistors each of 100 Ω resistance. A voltmeter of resistance 400 Ω is used to measure the potential difference across each resistor. The reading of the voltmeter will be

(A) 45 V	**(B)** 40 V
(C) 80 V	**(D)** 90 V

Q45. *CUSAT CAT 2025 - Shift 2*

At room temperature (27°C), the resistance of a heating element is 100Ω. At temperature T°C, the resistance is found to be 117Ω. Given that temperature coefficient of the resistor material is 1.7 × 10–4/°C,then the temperature T is

(A) 1027°C	**(B)** 820°C
(C) 527°C	**(D)** 1232°C

Q46. *CUSAT CAT 2025 - Shift 2*

Si and Cu are cooled from 300K to a temperature of 60K. Then resistivity

(A) for Si increases and for Cu decreases	**(B)** for Cu increases and for Si decreases
(C) decreases for both Si and Cu	**(D)** increases for both Si and Cu

Q47. *CUSAT CAT 2025 - Shift 2*

A strip of Copper and another of Germanium are cooled to a temperature of 80 K. The resistance of

(A) each of these decreases	**(B)** each of these increases
(C) Copper strip decreases and that of Germanium increases	**(D)** Copper strip increases and that of Germanium decreases

Q48. *CUSAT CAT 2025 - Shift 2*

A wire of 1 Ω has a length of 1 m. It is stretched till its length increases by 25%. The percentage change in resistance to the nearest integer is

(A) 56%	**(B)** 25%
(C) 12.5%	**(D)** 76%

Q49. *CUSAT CAT 2025 - Shift 2*

A 12 V battery connected to a coil of resistance 6 Ω through a switch, drives a constant current in the circuit. The switch is opened in 1 ms.Theemf induced across the coil is 20 V. The inductance of the coil is

(A) 5 mH	**(B)** 12 mH
(C) 8 mH	**(D)** 10 mH

Q50. *CUSAT CAT 2025 - Shift 3*

The electrical conductivity of the material of a conductor of length 3 m, area of cross section 0.02 mm2 having a resistance of 2 Ω is

(A) 75 × 106/ Ω-m	**(B)** 85 × 105/Ω-m
(C) 75 × 105/ Ω-m	**(D)** 85 ×106/ Ω-m

Q51. *CUSAT CAT 2025 - Shift 3*

A carbon resistance is having a following coding: green, orange, black, gold. The resistance of the resistor is

(A) 53 × 100 ± 5%	**(B)** 53 × 101 ± 5%
(C) 53 × 100 ± 10%	**(D)** 53 × 101 ± 10%

Q52. *CUSAT CAT 2025 - Shift 3*

Which of the following statement is FALSE?

(A) Wheatstone bridge is most sensitive when all the four resistances are of the same order of magnitude	**(B)** In a balanced Wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed
(C) A rheostat can be used as a potential divider	**(D)** Kirchhoff's second law represents energy conservation

Q53. *CUSAT CAT 2025 - Shift 3*

The magnetic dipole moment of a current-carrying circular coil is doubled by

(A) Doubling the radius of the loop	**(B)** Doubling the current through the loop
(C) Doubling the number of turns	**(D)** Doubling the current through the loop and the number of turns

Q54. *CUSAT CAT 2025 - Shift 3*

An NPN transistor is used in common emitter configuration as an amplifier with 1 kΩ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector and 15 μA change in the base current of the amplifier. The input resistance and voltage gain are

(A) 0.033 kΩ, 1.5	**(B)** 0.67 kΩ, 200
(C) 0.33 kΩ, 300	**(D)** 0.67 kΩ, 300

Q55. *CUSAT CAT 2025 - Shift 3*

A potentiometer wire PQ of 1 m length is connected to a standard cell E . Another 1 cell E of emf 1.02 V is connected with a resistance ‘r’ and switch S (as shown in 2 figure). With switch S open, the null position is obtained at a distance of 49 cm from Q. The potential gradient in the potentiometer wire is

(A) 0.02 V/cm	**(B)** 0.04 V/cm
(C) 0.01 V/cm	**(D)** 0.03 V/cm

Q56. *CUSAT CAT 2025 - Shift 3*

In a series, LCR Resonance circuit, if we change the resistance only, from a lower to higher value then

(A) the bandwidth of resonance circuit will increase	**(B)** the resonance frequency will increase
(C) the quality factor will increase	**(D)** the quality factor and the residence frequency will remain constant

Q57. *CUSAT CAT 2025 - Shift 4*

The free electrons of a copper wire of cross sectional area 10–6m2 acquire a drift velocity of 10–4 m/sec when a certain potential difference is applied across the wire. If the density of free electrons in copper is 8.5 × 1028 electrons/m3, then current flowing in the wire is

(A) 1.36 A	**(B)** 2.45 A
(C) 3.65 A	**(D)** 13.6 A

Q58. *CUSAT CAT 2025 - Shift 4*

What is the main purpose of doping a semiconductor material?

(A) Increase its resistance	**(B)** Enhance its conductivity
(C) Reduce its melting point	**(D)** Increase its hardness

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	B	3	C	4	A	5	B	6	B	7	A	8	D	9	D	10	A
11	C	12	D	13	B	14	D	15	A	16	A	17	D	18	B	19	D	20	C
21	A	22	B	23	B	24	B	25	A	26	C	27	C	28	B	29	A	30	A
31	D	32	D	33	D	34	C	35	A	36	D	37	B	38	C	39	C	40	D
41	B	42	A	43	A	44	B	45	A	46	A	47	C	48	A	49	D	50	A
51	A	52	B	53	D	54	D	55	A	56	A	57	A	58	B				

Magnetic Effects of Current

12 questions

Q1. *CUSAT CAT 2023 - Shift 2*

An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?

(A) The electron will be accelerated along the axis	**(B)** The electron path will be circular about the axis
(C) The electron will experience a force at 45° to the axis and hence execute a helical path	**(D)** The electron will continue to move with uniform velocity along the axis of the solenoid

Q2. *CUSAT CAT 2023 - Shift 2*

The polarity of induced emf is given by

(A) Ampere's circuital law	**(B)** Biot-Savart law
(C) Lenz's law	**(D)** Fleming's right hand rule

Q3. *CUSAT CAT 2023 - Shift 3*

If in a moving coil galvanometer, a current i produces a deflection θ, then

(A) i isproportionaltotanθ	**(B)** i isproportionaltoθ
(C) i isproportionaltoθ 2	**(D)** i isproportionalto θ

Q4. *CUSAT CAT 2023 - Shift 4*

The television uses the …….. to generate magnetic fields needed

(A) solenoid	**(B)** toroid
(C) both solenoid and toroid	**(D)** solar cell

Q5. *CUSAT CAT 2024 - Shift 3*

The polarity of induced emf is given by

(A) Ampere's circuital law	**(B)** Biot-Savart law
(C) Lenz's law	**(D)** Fleming's right hand rule

Q6. *CUSAT CAT 2024 - Shift 3*

Which one of the following rules is used to identify the direction of the current induced in a wire moving in a magnetic field?

(A) Fleming's left-hand rule	**(B)** Ampere's rule
(C) Fleming's right-hand rule	**(D)** Biot-Savart law

Q7. *CUSAT CAT 2024 - Shift 3*

…… states that an emf is induced whenever there is a change in the magnetic field linked with electric circuits.

(A) Lenz's Law	**(B)** Faraday's Law of Electromagnetic Induction
(C) Ohm's Law	**(D)** Biot-Savart Law

Q8. *CUSAT CAT 2025 - Shift 1*

A moving coil galvanometer has a coil with 175 turns and area 1 cm2. It uses a torsion band of torsion constant 10–6 N-m/rad. The coil is placed in a magnetic field B parallel to its plane. The coil deflect by 1° for a current of 1 mA. The value of B (in Tesla) is approximately

(A) 10–3	**(B)** 10–1
(C) 10–4	**(D)** 10–2

Q9. *CUSAT CAT 2025 - Shift 2*

The number of turns of the coil of a moving coil galvanometer is increased in order to increase current sensitivity by 50%. The percentage change in voltage sensitivity of the galvanometer will be

(A) 100%	**(B)** 50%
(C) 75%	**(D)** 0%

Q10. *CUSAT CAT 2025 - Shift 3*

When two parallel wires carry currents in the same direction

(A) they attract each other	**(B)** they repel each other
(C) magnetic forces on two wires are perpendicular to each other	**(D)** they do not experience any magnetic force

Q11. *CUSAT CAT 2025 - Shift 4*

A solenoid coil of 300 turns/m is carrying a current of 5 A. The length of the solenoid is 0.5 m and has a radius of 1 cm. The magnitude of the magnetic field inside the solenoid is

(A) 1.9 × 10–3 T	**(B)** 4.5 × 10–3 T
(C) 2.3 × 10–3 T	**(D)** 3.2 × 10–3 T

Q12. *CUSAT CAT 2025 - Shift 4*

A solenoid of inductance = 2H is connected to a battery. If the current changes uniformly at the rate of 5 A/s, what is the EMF induced in the solenoid?

(A) 2 V	**(B)** 5 V
(C) 10 V	**(D)** 20 V

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key

1	D	2	C	3	B	4	A	5	C	6	A	7	B	8	A	9	D	10	A
11	A	12	C																

Magnetism & Matter

12 questions

Q1. *CUSAT CAT 2023 - Shift 2*

The ratio of magnetic field at the centre of a current carrying circular coil to its magnetic moment is r. If the current and the radius both are halved, then the new ratio becomes r

(A) 2	**(B)** 2r r
(C) 8	**(D)** 8r

Q2. *CUSAT CAT 2023 - Shift 3*

The magnetic susceptibility of a paramagnetic material at –73°C is 0.0075. Its value at –173°C will be

(A) 0.0150	**(B)** 0.0075
(C) 0.0045	**(D)** 0.0030

Q3. *CUSAT CAT 2023 - Shift 4*

Two circular current carrying coils of radii 2 cm and 3 cm are equivalent to magnetic dipoles having equal magnetic moments. The ratio of currents through the coils are 4

(A) 9 9	**(B)** 4 2
(C) 3 3	**(D)** 4

Q4. *CUSAT CAT 2023 - Shift 5*

The magnetic susceptibility of a certain paramagnetic substance at –73°C is 6 × 10 . Its susceptibility at –173°C will be –2

(A) 1.2 × 10 –3	**(B)** 1.8 × 10 –3
(C) 3 × 10 –3	**(D)** 4.5 × 10

Q5. *CUSAT CAT 2023 - Shift 5*

Which among the following materials display higher magnetic susceptibility?

(A) Ferromagnetic material	**(B)** Paramagnetic material
(C) Diamagnetic material	**(D)** None of the above

Q6. *CUSAT CAT 2024 - Shift 2*

Which of the following is best method to reduce eddy currents?

(A) Laminating core	**(B)** Using thick wires
(C) Reducing hysteresis loss	**(D)** Using thin wires

Q7. *CUSAT CAT 2024 - Shift 2*

At magnetic poles the angle of dip is

(A) 45°	**(B)** 90°
(C) 30°	**(D)** 5°

Q8. *CUSAT CAT 2024 - Shift 3*

Adiabatic demagnetization of a paramagnetic salt usually results in

(A) drop in temperature	**(B)** drop in magnetic susceptibility
(C) rise in temperature	**(D)** critical magnetic filed

Q9. *CUSAT CAT 2024 - Shift 4*

Materials with susceptibility χ in the range $-1 \leq \chi < 0$ are called

(A) Ferromagnetic	**(B)** Paramagnetic
(C) Antiferromagnetic	**(D)** Diamagnetic

Q10. *CUSAT CAT 2024 - Shift 5*

Which magnetic pole does not exist?

(A) Dipole	**(B)** Quadrupole
(C) Octupole	**(D)** Monopole

Q11. *CUSAT CAT 2025 - Shift 2*

If a magnetic material is having magnetic susceptibility (χ) = – 1, then the relative magnetic permeability (μ r) and type of magnetic material are

(A) 0, diamagnetic	**(B)** 2, ferromagnetic
(C) 1, paramagnetic	**(D)** –1, diamagnetic

Q12. *CUSAT CAT 2025 - Shift 2*

Nickel shows ferromagnetic property at room temperature. If the temperature is increased beyond Curie temperature, then it will show

(A) Paramagnetism	**(B)** anti-ferromagnetism
(C) no magnetic property	**(D)** diamagnetism

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	B	4	A	5	A	6	A	7	B	8	A	9	D	10	D
11	A	12	A																

Electromagnetic Induction

13 questions

Q1. *CUSAT CAT 2023 - Shift 1*

When a bulk piece of conductor is subjected to changing magnetic flux, induced currents are produced in them. However, their flow patterns resemble swirling in water. These currents are called

(A) Saturation current	**(B)** Eddy current
(C) Leak current	**(D)** Dark current

Q2. *CUSAT CAT 2023 - Shift 1*

Faraday's laws of electromagnetic induction are consequence of conservation of

(A) energy	**(B)** energy and magnetic field
(C) charge	**(D)** magnetic field

Q3. *CUSAT CAT 2023 - Shift 1*

In an induction coil the current increases from zero to 6 amp in 0.3 s by which induced emf of 30 V is produced. The value of coefficient of self induction of coil will be

(A) 1 Henry	**(B)** 1.5 Henry
(C) 2 Henry	**(D)** 3 Henry

Q4. *CUSAT CAT 2023 - Shift 3*

The magnetic flux (in Webers) linked with a coil is given by the equation $2\phi = 3t + 4t + 9$. Then the magnitude of the induced emf at time t = 2s will be

(A) 2 V	**(B)** 4 V
(C) 8 V	**(D)** 16 V

Q5. *CUSAT CAT 2023 - Shift 3*

Two pure inductors each of self inductance L are connected in series, the net inductance is

(A) L	**(B)** 2 L L
(C) 2 L	**(D)** 4 –7

Q6. *CUSAT CAT 2023 - Shift 4*

The self inductance of a coil is L. Keeping the length and area same, the number of turns in the coil is increased to four times. The self inductance of the coil will be 1

(A) L 4	**(B)** L
(C) 4L	**(D)** 16L

Q7. *CUSAT CAT 2023 - Shift 5*

In a coil, when the current is changed from 10.0 A in one direction to 10.0 A in the opposite direction in 0.5 s, the average induced emf is 0.4 V. The self inductance of the coil is

(A) 10.0 mH	**(B)** 5.0 mH
(C) 2.0 mH	**(D)** 0.5 mH

Q8. *CUSAT CAT 2024 - Shift 1*

If number of turns in primary and secondary coils is increased to two times each, the mutual inductance

(A) becomes 4 times	**(B)** becomes 2 times
(C) reduces to 0.5 times	**(D)** remains unchanged

Q9. *CUSAT CAT 2024 - Shift 3*

Determine the self-inductance of a coil, which has a magnetic flux of 50 milliwebers that is produced when a current of 5 A flows through it

(A) 1 × 10 –2 Wb	**(B)** 1 × 10 –3 Wb
(C) 100 Wb	**(D)** 1 × 10 3 Wb

Q10. *CUSAT CAT 2024 - Shift 5*

Two coils of self inductances L and L are placed close together, if mutual 1 2 inductance between them is M, then M is proportional to …….., where the proportionality constant is known as …….. L

(A) 1 , coefficient of coupling L 2	**(B)** L L , coefficient of coupling 1 2 2
(C) (L 1 L 2) , coefficient of coupling	**(D)** LL , coupling coefficient 1 2

Q11. *CUSAT CAT 2025 - Shift 1*

The core of transformer is laminated to

(A) prevent it from moisture	**(B)** prevent it from noise
(C) prevent it from heat	**(D)** reduce the loss of energy

Q12. *CUSAT CAT 2025 - Shift 3*

The equation φ = 3t2 + 4t + 9 represents the magnetic flux (in Weber) linked with a coil. Then the magnitude of the emf induced after t = 2s will be

(A) 2 V	**(B)** 4 V
(C) 8 V	**(D)** 16 V

Q13. *CUSAT CAT 2025 - Shift 4*

Suppose the number of turns in a coil be tripled, the value of magnetic flux linked with it

(A) remains unchanged 1	**(B)** becomes 3
(C) is tripled	**(D)** None of the above

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	A	3	B	4	D	5	B	6	D	7	A	8	A	9	A	10	D
11	D	12	D	13	C														

Alternating Current

11 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A ccaappaacciittoorr wwoorrkkss aass aa cchhaarrggee ssttoorriinngg ddeevviiccee iinn

(A) AC circuits	**(B)** DC circuits
(C) bbootthh AACC aanndd DDCC cciirrccuuiittss	**(D)** NNoonnee ooff tthhee aabboovvee

Q2. *CUSAT CAT 2023 - Shift 2*

AA sstteepp ddoowwnn ttrraannssffoorrmmeerr hhaass aa 1100 V sseeccoonnddaarryy ttaappppiinngg.. IIff tthhee pprriimmaarryy aatt 222200 V draws a current of 5 A,, aanndd tthhee sseeccoonnddaarryy ddrraawwss aa ccuurrrreenntt ooff 8888 A,, wwhhaatt iiss tthhee eeffffiicciieennccyy ooff the transformer?

(A) 8.8%	**(B)** 88%
(C) 80%	**(D)** 8% –10

Q3. *CUSAT CAT 2023 - Shift 3*

The frequency of LC circuit is 1

(A) LC 2π 1	**(B)** 2πLC 1 L
(C) 2π C 1 1	**(D)** 2π LC

Q4. *CUSAT CAT 2024 - Shift 2*

An inductor of inductance 30 mH is connected to a 220 V, 100 Hz AC source. It offers an inductive reactance of

(A) 10.58 Ω	**(B)** 12.64 Ω
(C) 18.85 Ω	**(D)** 22.67 Ω

Q5. *CUSAT CAT 2024 - Shift 2*

In a step-down transformer with 20 turns in the secondary coil and 100 turns in the primary coil, if the voltage applied to the primary coil is 120 V, the voltage output from the secondary coil is

(A) 24 V	**(B)** 12 V
(C) 6 V	**(D)** 18 V

Q6. *CUSAT CAT 2024 - Shift 3*

A parallel LC circuit is operated at a frequency ω below its resonance frequency ω . o Then its reactance will be

(A) resistive	**(B)** capacitative
(C) inductive	**(D)** zero

Q7. *CUSAT CAT 2024 - Shift 3*

The graph between inductive reactance and frequency is

(A) parabola	**(B)** straight line
(C) hyperbola	**(D)** an arc of a circle

Q8. *CUSAT CAT 2024 - Shift 4*

If the frequency of the AC is made four times of its initial value, then the inductive reactance will

(A) be 4 times	**(B)** be 2 times
(C) remain same	**(D)** be half

Q9. *CUSAT CAT 2024 - Shift 5*

In a circuit, if the alternating current is represented by I = 4 cos (ωt +ϕ), then the r.m.s value of current is

(A) 2	**(B)** 4 2
(C) 2 2	**(D)** 1/ 2

Q10. *CUSAT CAT 2025 - Shift 1*

An AC circuit has R = 100 Ω, C = 2 μF and L= 80 mH, connected in series. The quality factor of the circuit is

(A) 0.5	**(B)** 2
(C) 20	**(D)** 400

Q11. *CUSAT CAT 2025 - Shift 2*

For an RLC circuit driven with voltage of amplitude v and frequency ω = The m 0 . quality factor, Q is given by √

(A) ω	**(B)** ωL
(C) ωR	**(D)** R ω

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	D	4	C	5	A	6	C	7	B	8	A	9	C	10	B
11	B																		

Electromagnetic Waves

22 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The waves used by artificial satellites in communication is

(A) Infrared rays	**(B)** Radio waves
(C) X rays	**(D)** Micro waves

Q2. *CUSAT CAT 2023 - Shift 2*

Consider the following two statements (I) and (II) and identify the correct choice. (I) The characteristic X-ray spectrum depends on the nature of the material of the target. (II) The short wavelength limit of the continuous X-ray spectrum varies inversely on the potential difference applied to the X-ray tube.

(A) (I) is true and (II) is false	**(B)** (I) is false and (II) is true
(C) Both (I) and (II) are true	**(D)** Both (I) and (II) false

Q3. *CUSAT CAT 2023 - Shift 3*

In the hydrogen spectrum, Paschen and Brackett series are

(A) infrared region	**(B)** UV region
(C) visible region	**(D)** microwave region

Q4. *CUSAT CAT 2023 - Shift 3*

Which wavelength of the radiation from sun is used for conversion into electrical energy?

(A) Radio waves	**(B)** Infrared waves
(C) Visible light	**(D)** Micro waves

Q5. *CUSAT CAT 2023 - Shift 4*

Which one of the following cannot be used for detection of X-Rays?

(A) Photographic film	**(B)** Photocells
(C) Geiger tubes	**(D)** Ionisation chamber

Q6. *CUSAT CAT 2023 - Shift 5*

Match List - I (Electromagnetic wave type) with List - II (its association/application) and select the correct match option from the choices given below: List - I List - II (a) Infrared waves (i) To treat muscular strain (b) Radio waves (ii) For broadcasting (c) X-rays (iii) To detect fracture of bones (d) Ultraviolet rays (iv) Absorbed by the ozone layer of the atmosphere

(A) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)	**(B)** (a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)
(C) (a) - (iii), (b)- (ii), (c) - (i), (d) - (iv)	**(D)** (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

Q7. *CUSAT CAT 2024 - Shift 1*

Maximum diffraction takes place in a given slit for

(A) ultraviolet rays	**(B)** infrared rays
(C) radio waves	**(D)** gamma rays

Q8. *CUSAT CAT 2024 - Shift 1*

The green house effect is due to trapping of …….. in the atmosphere.

(A) Ultraviolet rays	**(B)** X-rays
(C) Infrared rays	**(D)** Radiowaves

Q9. *CUSAT CAT 2024 - Shift 2*

Speed of electromagnetic waves in vacuum is equal to 1

(A) με 0 0	**(B)** με 0 0
(C) με 0 0 1	**(D)** με 0 0

Q10. *CUSAT CAT 2024 - Shift 2*

The frequency of X-rays, ϒ-rays and ultraviolet rays are respectively a, b and c. Then

(A) a > b, b < c	**(B)** a < b, b > c
(C) a < b, b < c	**(D)** a > b, b > c

Q11. *CUSAT CAT 2024 - Shift 3*

Vibrational motion of the molecules occurs at frequencies in the

(A) ultraviolet region	**(B)** radio frequency region
(C) infra red region	**(D)** gamma ray region

Q12. *CUSAT CAT 2024 - Shift 3*

Communication between the ground station and an artificial satellite occurs normally at

(A) radio frequencies	**(B)** microwave frequencies
(C) infrared frequencies	**(D)** ultraviolet frequencies

Q13. *CUSAT CAT 2024 - Shift 3*

The wavelength of a source transmitting electromagnetic waves of frequency 6 8.2 × 10 Hz is

(A) 36.6 m	**(B)** 18.8 m
(C) 42.8 m	**(D)** 58.0 m

Q14. *CUSAT CAT 2024 - Shift 4*

Pick out the correct increasing order of energy of electromagnetic waves from the following

(A) E < E < E < E < E gamma ultraviolet visible infrared micro	**(B)** E < E < E < E < E micro infrared ultraviolet visible gamma
(C) E < E < E < E < E micro infrared visible gamma ultraviolet	**(D)** E < E < E < E < E micro infrared visible ultraviolet gamma

Q15. *CUSAT CAT 2024 - Shift 4*

Which of the following is a correct statement?

(A) Beta rays are the same as cathode rays	**(B)** Gamma rays are high energy neutrons
(C) Alpha particles are singly ionized helium atoms	**(D)** Protons and neutrons have the same mass

Q16. *CUSAT CAT 2024 - Shift 5*

The structural determination of the solids can be obtained by using

(A) Ultraviolet rays	**(B)** Cosmic rays
(C) Gamma rays	**(D)** X-rays

Q17. *CUSAT CAT 2024 - Shift 5*

Which of the following electromagnetic radiation has maximum penetrating power?

(A) Radio waves	**(B)** Microwaves
(C) Ultraviolet rays	**(D)** Gamma rays

Q18. *CUSAT CAT 2024 - Shift 5*

The Kα X-ray of molybdenum has wavelength 71 pm. If the energy of a molybdenum atom with a K electron knocked out is 23.32 keV, what will be the energy of this atom when an L electron is knocked out?

(A) 5.82 eV	**(B)** 6.52 eV
(C) 8.22 eV	**(D)** 4.32 eV

Q19. *CUSAT CAT 2025 - Shift 1*

Which is the correct ascending order of wavelengths ?

(A) $\lambda < \lambda < \lambda < \lambda$ visible X-ray gamma-ray microwave	**(B)** $\lambda < \lambda < \lambda < \lambda$ gamma-ray X-ray visible microwave
(C) $\lambda < \lambda < \lambda < \lambda$ X-ray gamma-ray visible microwave	**(D)** $\lambda < \lambda < \lambda < \lambda$ microwave visible gamma-ray X-ray

Q20. *CUSAT CAT 2025 - Shift 3*

The most penetrating radiation out of the following is

(A) β- rays	**(B)** γ-rays
(C) X-rays	**(D)** α-rays

Q21.

CUSAT CAT 2025 - Shift 3

The 21 cm radio wave emitted by hydrogen in interstellar space is due to the interaction called the hyperfine interaction in atomic hydrogen. The energy of the emitted wave is nearly

(A) 10–17 J	**(B)** 1 J
(C) 7 × 10–6 J	**(D)** 10–24 J

Q22.

CUSAT CAT 2025 - Shift 3

An electron having de-Broglie wavelength λ is incident on a target in an X- ray tube. Cut-off wavelength of emitted X-ray is

(A) 0	**(B)** 2 λ ■ λ
(C) 2 ■ ■	**(D)** —

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	C	3	A	4	B	5	B	6	D	7	C	8	C	9	A	10	B
11	C	12	B	13	A	14	D	15	A	16	D	17	D	18	A	19	D	20	B
21	D	22	C																

Ray Optics

24 questions

Q1. *CUSAT CAT 2023 - Shift 1*

An equiconvex lens has a focal length f. If the lens is cut along the line perpendicular to the principal axis and passing through the pole, what will be the focal length of any half part? f

(A) 2	**(B)** 2f
(C) f f	**(D)** 4

Q2. *CUSAT CAT 2023 - Shift 1*

A glass prism of μ = 1.5 is immersed in water as shown in the figure. A beam of light incident normally on the face ab is internally reflected from the face ad so as to 4 incident normally on face bd. Given that refractive index of water is . Then the 3 value of θ is 8 –1

(A) θ>sin 9 2 –1	**(B)** θ>sin 3 2 –1
(C) θ<sin 3 4 –1	**(D)** θ<sin 3

Q3. *CUSAT CAT 2023 - Shift 2*

The monochromatic light is refracted from air into glass of refractive index μ. The ratio of the wavelengths of the incident and refractive waves is

(A) 1 : 1	**(B)** μ : 1
(C) 1 : μ 2	**(D)** μ : 1

Q4. *CUSAT CAT 2023 - Shift 3*

A double convex lens of focal length 20 cm with same radius of curvature for both the faces needs to manufactured. If the refractive index of the glass is 1.55, the radius of curvature required will be

(A) 10 cm	**(B)** 17.5 cm
(C) –17.5 cm	**(D)** 22 cm 4

Q5. *CUSAT CAT 2023 - Shift 3*

An achromatic combination of lenses is formed by joining

(A) 2 convex lenses	**(B)** 2 concave lenses
(C) 1 convex and 1 concave lens	**(D)** 1 convex lens and a plane mirror

Q6. *CUSAT CAT 2023 - Shift 4*

The time taken by light to pass through a glass slab of thickness 5 mm and refractive index 1.5 is –13

(A) 2.5 × 10 s –13	**(B)** 3 × 10 s –11
(C) 2.5 × 10 s –11	**(D)** 3 × 10 s

Q7. *CUSAT CAT 2023 - Shift 4*

If Young's double slit apparatus is immersed in a liquid of refractive index μ, the fringe width β' β

(A) reduces to μ	**(B)** increases to μβ
(C) remains constant	**(D)** increases to (μ + 1)β

Q8. *CUSAT CAT 2023 - Shift 4*

Mechanism of light transmission in optical fibers are based on

(A) total internal reflection	**(B)** scattering
(C) refraction	**(D)** absorption

Q9. *CUSAT CAT 2023 - Shift 4*

The refractive index of the water is 1.33. What will be the speed of light in water? 8 –1

(A) 3 × 10 ms 8 –1	**(B)** 2.25 × 10 ms 8 –1
(C) 4 × 10 ms 8 –1	**(D)** 1.33 × 10 ms

Q10. *CUSAT CAT 2023 - Shift 4*

The magnification of an image by a convex lens is positive only when the object is placed

(A) at its focus F	**(B)** between F and 2F
(C) at 2F	**(D)** between F and optical centre

Q11. *CUSAT CAT 2023 - Shift 5*

The limit of resolution of a microscope is given by the formula (λ is wavelength, n is refractive index and θ is semi vertical angle) λ

(A) 2sinθ 1.22λ	**(B)** 2sinθ λ
(C) 2nsinθ 1.22λ	**(D)** sinθ

Q12. *CUSAT CAT 2023 - Shift 5*

Sonar technology works based on the principle of

(A) Photoelectric effect	**(B)** Total internal reflection of light
(C) Magnetic confinement of plasma	**(D)** Reflection of ultrasonic waves

Q13. *CUSAT CAT 2024 - Shift 3*

The speed of light in a certain glass is 1.91 × 10 m/s. What is the refractive index of the glass?

(A) 1.57	**(B)** 0.64
(C) 1.09	**(D)** 4.90

Q14. *CUSAT CAT 2024 - Shift 5*

Sparkling of diamond is due to

(A) reflection	**(B)** diffraction
(C) total internal reflection	**(D)** the high refractive index of diamond

Q15. *CUSAT CAT 2024 - Shift 5*

Which of the following principle is used in optical fibres?

(A) Diffraction	**(B)** Scattering
(C) Total internal reflection	**(D)** Refraction

Q16. *CUSAT CAT 2024 - Shift 5*

When the angle of incidence on a certain material is 60°, the reflected light is completely polarized. The refractive index of the material is then

(A) 1.732	**(B)** 0.866
(C) 0.577	**(D)** 1.5

Q17. *CUSAT CAT 2024 - Shift 5*

A monochromatic beam of light is incident on a glass block at an angle of 60° with respect to the surface normal. The reflected beam is observed to be completely polarised. The critical angle for the glass block is

(A) 15.77°	**(B)** 25.34°
(C) 35.27°	**(D)** 55.71°

Q18. *CUSAT CAT 2025 - Shift 1*

The radii of curvature of a double convex lens are 15 cm and 30 cm and its refractive index is 1.5 then its focal length is

(A) 20 cm	**(B)** 30 cm
(C) 15 cm	**(D)** 12 cm

Q19. *CUSAT CAT 2025 - Shift 2*

How does the focal length of a convex lens change if monochromatic red light is used instead of violet light?

(A) Focal length is increased	**(B)** Focal length is decreased
(C) Focal length remains same	**(D)** Focal length does not depend on colour of light used

Q20. *CUSAT CAT 2025 - Shift 2*

Two objects A and B are placed at 15 cm and 25 cm from the pole in front of a concave mirror having radius of curvature 40 cm . The distance between images formed by the mirror is

(A) 40 cm	**(B)** 60 cm
(C) 160 cm	**(D)** 100 cm

Q21. *CUSAT CAT 2025 - Shift 3*

When light is incident normally on an air-glass interface, what percentage of light is reflected back, if the refractive index of the glass is 1.5?

(A) 80%	**(B)** 50%
(C) 10%	**(D)** 4%

Q22. *CUSAT CAT 2025 - Shift 4*

A substance has a critical angle of 45° for yellow light. Then its refractive index is

(A) 2 1	**(B)** 2 3
(C) 2 1	**(D)** 2

Q23. *CUSAT CAT 2025 - Shift 4*

A signal is transmitted through an optical fiber with a refractive index of n = 1. If the length of the fiber is 10 km, what is the time taken for the signal to travel through it?

(A) 33.3 μs	**(B)** 51.3 μs
(C) 63.3 μs	**(D)** 70 μs

Q24. *CUSAT CAT 2025 - Shift 4*

A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index is poured inside the bucket, the microscope have to raised by 30 cm to focus the object again. The height of the liquid in the bucket is

(A) 75 cm	**(B)** 50 cm
(C) 18 cm	**(D)** 12 cm

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	A	3	B	4	D	5	C	6	C	7	A	8	A	9	B	10	D
11	C	12	D	13	A	14	C	15	C	16	A	17	C	18	A	19	A	20	C
21	D	22	A	23	A	24	A												

Wave Optics

8 questions

Q1.

CUSAT CAT 2024 - Shift 3

In a double slit experiment, when a thin film of thickness t having refractive index μ is inserted in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. If λ is the wavelength of the light used, the value of t is λ

(A) μ–1 λ	**(B)** 2μ–1 2λ
(C) μ–1 λ	**(D)** 2(μ–1)

Q2.

CUSAT CAT 2024 - Shift 4

Circularly polarized light is passed through a quarter wave plate. Then the general state of polarization state of the outgoing radiation is

(A) Unpolarized	**(B)** Elliptical polarized
(C) Circular polarized	**(D)** Linearly polarized

Q3.

CUSAT CAT 2024 - Shift 4

When a drop of oil is spread on a water surface, it displays beautiful colours in daylight because of

(A) dispersion of light	**(B)** interference of light
(C) reflection of light	**(D)** scattering of light

Q4.

CUSAT CAT 2024 - Shift 4

AAA llliiiggghhhttt bbbeeeaaammm ooofff wwwaaavvveeellleeennnggggtttthhh 666000000 nnnmmm fffrrrooommm aaa dddiiisssttaaannnttt sssoooouuurrrcccceee fffaaallllllsss on a single slit of 1 mmmmmm wwwiiidddeee aaannnddd tttthhheee rrreeesssuuullltttiiinnnggg dddiiiffffffrrraaaccctttiiiooonnn pppaaatttttteeerrrnnn iiisss ooobbbssseeerrrvvveeeddd ooonnn aaa ssscccrrreeeeeennn 222 mmm aaawwwaaayyy... TTThhheee dddiiisssttaaannnccceee bbbeeetttwwweeeeennn tttthheee fffiiirrrssttt dddaaarrrkkk fffrrriiinnnggeeesss ooonnn eeeiiittthheeerrr sssiiidddeee ooofff tttthheee cccceennnttttrrraaalll bbbrrriiiggghhhttt fffrrriiinnnggeee iiisss

(A) 2.4 mm	**(B)** 1.2 mm
(C) 1.2 cm	**(D)** 2.4 cm

Q5.

CUSAT CAT 2024 - Shift 5

If Young's double slit experiment is performed in water

(A) The fringe width decreases	**(B)** The fringe width increases
(C) The fringe width is unchanged	**(D)** There will be no fringes observed

Q6. *CUSAT CAT 2025 - Shift 1*

In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen, when light of wavelength 600 nm is used. If the wavelength of the light is changed to 400 nm, the number of fringes observed in the same segment of the screen is given by

(A) 12	**(B)** 18
(C) 24	**(D)** 30

Q7. *CUSAT CAT 2025 - Shift 1*

Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of waves is in the ratio

(A) 4 : 1	**(B)** 25 : 9
(C) 16 : 9	**(D)** 5 : 3

Q8. *CUSAT CAT 2025 - Shift 4*

Which of the following phenomena cannot be explained by the wave theory of light?

(A) Diffraction	**(B)** Interference
(C) Polarization	**(D)** Photoelectric effect

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	D	3	B	4	A	5	A	6	B	7	B	8	D				

Dual Nature of Radiation

16 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Which of the following is WRONGLY matched?

(A) Raman effect – Scattering of light	**(B)** Thomson effect – Thermoelectricity
(C) Hall effect – Work function	**(D)** Photoelectric effect – Quantum nature of light

Q2. *CUSAT CAT 2023 - Shift 2*

Ultraviolet light of wavelength 280 nm is used in an experiment on photoelectric effect with lithium cathode whose work function is 2.5 eV. Find the maximum kinetic energy of the photoelectrons

(A) 1.9 eV	**(B)** 2.5 eV
(C) 4.4 eV	**(D)** 7.5 eV

Q3. *CUSAT CAT 2023 - Shift 5*

The threshold wavelength for a metal whose work function W is λ. What is the 2W threshold wavelength of a wave function whose work function is ? 3 2

(A) λ 3 3	**(B)** λ 4 1
(C) λ 3 3	**(D)** λ 2

Q4. *CUSAT CAT 2023 - Shift 5*

Einstein's work on the photoelectric effect provided support for the equation

(A) E =mc 2	**(B)** E =hν –Rhc
(C) E = 2 n 1 2	**(D)** KE = mv 2

Q5. *CUSAT CAT 2024 - Shift 1*

Louis de-Broglie was credited for his work on

(A) Theory of relativity	**(B)** Matter waves
(C) Electromagnetic theory	**(D)** Photoelectric effect

Q6. *CUSAT CAT 2024 - Shift 2*

What is the de Broglie wavelength of a ball of mass 150 g moving at a speed of 50 m/s?

(A) 8.8 × 10 –34 m	**(B)** –30 8.8 × 10 m
(C) –25 8.8 × 10 m	**(D)** –35 8.8 × 10 m

Q7. *CUSAT CAT 2024 - Shift 3*

Consider a beam of electrons (each electron with energy E) incident on a metal 0 surface kept in an evacuated chamber. Then,

(A) no electrons will be emitted as only photons are emitted by the metal	**(B)** electrons can be emitted but all with an energy E 0
(C) electrons can be emitted with any energy, with a maximum of E – ■ (■ is the 0 work function)	**(D)** electron can be emitted with any energy, with a maximum of E 0

Q8. *CUSAT CAT 2024 - Shift 4*

In photoelectric effect, the photocurrent

(A) depends both on intensity and frequency of the incident light	**(B)** does not depend on the frequency of the incident light but depends on the intensity of the incident light.
(C) decreases with increase in frequency of incident light.	**(D)** increases with increase in frequency of incident light.

Q9. *CUSAT CAT 2024 - Shift 4*

Planck's constant explains

(A) the Quantum nature of light	**(B)** the Particle nature of light
(C) the Wave nature of light	**(D)** both the Quantum nature and Wave nature of light

Q10. *CUSAT CAT 2024 - Shift 5*

Kinetic energy and stopping potential are zero, when the frequency of incident radiation is

(A) equal to the threshold frequency	**(B)** very minimum
(C) lower than the threshold frequency	**(D)** very high

Q11. *CUSAT CAT 2024 - Shift 5*

Work function of three metals A, B and C are 4.7 eV, 4.5 eV and 3.5 eV respectively. If a light of wavelength 3310 A° is incident on the metals, then photoelectrons are emitted by

(A) All the metals	**(B)** C alone
(C) A and B alone	**(D)** No metals

Q12. *CUSAT CAT 2025 - Shift 1*

An electron and a proton are associated with the same de Broglie wavelength. Then the ratio of their kinetic energy is

(A) 1: 1836 2	**(B)** 1: 1836
(C) 1: 1836 1	**(D)** 1: 1836

Q13.

CUSAT CAT 2025 - Shift 2

Which of the following is wrongly matched?

(A) Raman effect- Scattering of light	**(B)** Thomson effect- Thermoelectricity
(C) Hall effect- Work function	**(D)** Photoelectric effect- Quantum nature of light

Q14.

CUSAT CAT 2025 - Shift 3

The cathode of a photoelectric cell is changed such that the work function changes from W toW (W >W . If the current before and after changes are I and I , all 1 2 1 2) 1 2 other conditions remaining unchanged, then (assuminghν>W) 2

(A) I = I 1 2	**(B)** I < I 1 2
(C) I > I 1 2	**(D)** None of the above

Q15.

CUSAT CAT 2025 - Shift 3

The de-Broglie wavelength of a molecule in a gas at room temperature (300 K) is λ . 1 If the temperature of the gas is increased to 600 K, then the de Broglie wavelength of the same gas molecule becomes 1

(A) λ 1 2	**(B)** 2 λ 1 1
(C) λ 1 2	**(D)** 2λ 1

Q16.

CUSAT CAT 2025 - Shift 4

The work function of a metal is 4.5 eV. When monochromatic light with a wavelength of 200 nm strikes the metal surface, what is the maximum kinetic energy of the emitted photoelectrons?

(A) 1.1 eV	**(B)** 1.7 eV
(C) 2.1 eV	**(D)** 2.7 eV

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	D	4	B	5	B	6	D	7	D	8	B	9	D	10	A
11	B	12	A	13	C	14	A	15	A	16	B								

Atoms

11 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Rydberg constant, R is –7 –1

(A) 1.097 × 10 m 7 –1	**(B)** 1.097 × 10 m 9 –1
(C) 1.97 × 10 m 11 –1	**(D)** 1.097 × 10 m

Q2. *CUSAT CAT 2023 - Shift 1*

In which of the following series, does the 121.5 nm line of the spectrum of the hydrogen atom lies?

(A) Lyman series	**(B)** Balmer series
(C) Paschen series	**(D)** Brackett series

Q3. *CUSAT CAT 2023 - Shift 2*

Consider the spectral line resulting from the transition n = 2 → n = 1 in the atoms and ions given below. The shortest wavelength is produced by

(A) hydrogen atom	**(B)** deuterium atom
(C) singly ionized helium	**(D)** doubly ionized lithium

Q4. *CUSAT CAT 2023 - Shift 4*

The ground state energy of hydrogen atom is

(A) 13.6 eV	**(B)** 27.2 eV
(C) –13.6 eV	**(D)** infinity

Q5. *CUSAT CAT 2023 - Shift 4*

When an electron jumps from the fourth orbit to the second orbit, one gets

(A) second line of Paschen series	**(B)** second line of Lyman series
(C) second line of Balmer series	**(D)** first line of Pfund series

Q6. *CUSAT CAT 2023 - Shift 5*

In hydrogen atom, an electron makes a transition from orbit n = 4 to n = 2. The wave number of the emitted radiation will be (R = Rydberg's constant) 16

(A) 3R 2R	**(B)** 16 3R
(C) 16 4R	**(D)** 16

Q7.

CUSAT CAT 2024 - Shift 3

When an electron jumps from the fourth orbit to the second orbit one gets the

(A) Second line of Paschen series	**(B)** Second line of Balmer series
(C) First line of Pfund series	**(D)** Second line of Lyman series

Q8.

CUSAT CAT 2025 - Shift 1

The energy of hydrogen atom in the nth orbit is E , then the energy in the nth orbit of n single ionised helium atom is E n

(A) 2	**(B)** 2E n
(C) 4E n E n	**(D)** 4

Q9.

CUSAT CAT 2025 - Shift 1

An electron makes a transition from orbit n = 4 to orbit n = 2 of a hydrogen atom. The wave number of the emitted radiation is (R = Rydberg constant) 16

(A) 3R 3R	**(B)** 16 2R
(C) 16 4R	**(D)** 16

Q10.

CUSAT CAT 2025 - Shift 1

Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be

(A) 2	**(B)** 3
(C) 5	**(D)** 6

Q11.

CUSAT CAT 2025 - Shift 1

TTThhheee rrraaatttiiiooo ooofff wwwaaavvveeellleeennnggttthhhsss ooofff ttthhheee lllaaassssttt llliiinnneee ooofff BBBaaalllmmmeeerrr sssseeerrriiieeesss aaannnddd ttthhheee lllaaassssttt llliiinnneee ooofff Lyman series is

(A) 2	**(B)** 1
(C) 4	**(D)** 0.5

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	A	3	D	4	C	5	C	6	C	7	B	8	C	9	B	10	D
11	C																		

Nuclei

17 questions

Q1. *CUSAT CAT 2023 - Shift 2*

A nucleus (atomic number 100 and mass number 200) emits four -particles and the resultant nucleus emits five -particles. The atomic and mass number of the final nucleus are

(A) 191, 92	**(B)** 186, 87
(C) 186, 97	**(D)** 191, 97

Q2. *CUSAT CAT 2023 - Shift 3*

In the following nuclear reaction 7 4 p + Li → 2 He 3 2 if the binding energy per nucleon of the Li and He nuclei are 5.6 and 7.06 MeV, estimate what must be the energy of the proton?

(A) 39.2 MeV	**(B)** 28.24 MeV
(C) 17.28 MeV	**(D)** 1.46 MeV

Q3. *CUSAT CAT 2023 - Shift 3*

If the radius of nucleus of Al is 3.6 Fermi, the approximate nuclear radius of Cu in Fermi is

(A) 1.2	**(B)** 2.4
(C) 3.6	**(D)** 4.8

Q4. *CUSAT CAT 2023 - Shift 3*

Emission of β rays in radioactive decay results in the change of

(A) mass but not in charge	**(B)** charge but not in mass
(C) both mass and charge	**(D)** either mass or charge

Q5. *CUSAT CAT 2023 - Shift 5*

A nucleus X emits a α-particle. The resultant nucleus emits a β particle. The z respective atomic and mass numbers of the final nucleus will be

(A) z – 3, A – 4	**(B)** z – 1, A – 4
(C) z– 2, A – 4	**(D)** z, A – 2

Q6. *CUSAT CAT 2023 - Shift 5*

The energy gap is much more in silicon than in germanium because

(A) it has less number of electrons	**(B)** it has high atomic mass number
(C) its crystal has much stronger bonds called ionic bonds	**(D)** its valence electrons are more tightly bound to their parent nuclei

Q7. *CUSAT CAT 2024 - Shift 1*

Nuclei with same atomic number and mass number but existing in different energy states is known as

(A) Isotones	**(B)** Isomers
(C) Isobars	**(D)** Isotopes

Q8. *CUSAT CAT 2024 - Shift 3*

The decay constant (λ) and the half-life (T) of a radioactive isotope are related by the formula,

(A) λ = 1/log 2 T e	**(B)** λ = log 2 / T e
(C) λ = 1/log 2 e	**(D)** λ = 2/T

Q9. *CUSAT CAT 2024 - Shift 4*

Two isotopes of an element have

(A) the same number of nucleons in their nuclei	**(B)** the same number of protons as well as neutrons
(C) the same number of protons but a different number of neutrons	**(D)** different number of protons but the same number of neutrons

Q10. *CUSAT CAT 2024 - Shift 5*

Mirror nuclei are nuclei having

(A) same mass number but with different atomic number	**(B)** same mass number but with proton number and neutron number interchanged
(C) equal number of neutrons	**(D)** mirror images

Q11. *CUSAT CAT 2025 - Shift 1*

The K line from molybdenum (atomic number = 42) has a wavelength of 0.7078Å. α The wavelength of K line of zinc (atomic number = 30) will be α

(A) 1 Å	**(B)** 1.3872 Å
(C) 0.3541 Å	**(D)** 0.5 Å

Q12. *CUSAT CAT 2025 - Shift 1*

A radioactive nucleus undergoes a series of decay process represented by the scheme below: A (α–decay) → A (β–decay) → A (α–decay) → A (γ–decay) → A 1 2 3 4 If the mass number and atomic number of A are 180 and 72 respectively, then what are these numbers for A ? 4

(A) 172 and 69	**(B)** 174 and 70
(C) 176 and 69	**(D)** 176 and 70

Q13. *CUSAT CAT 2025 - Shift 2*

The half-life of radioactive sample is 4 days. After 16 days how much quantity of matter remain undecayed? 1

(A) 16 1	**(B)** 32 1
(C) 4 1	**(D)** 2

Q14.

CUSAT CAT 2025 - Shift 2

The number of beta particles emitted by a radioactive substance is twice the number of alpha particles emitted by it. The resulting daughter is an

(A) isobar of parent	**(B)** isomer of parent
(C) isotone of parent	**(D)** isotope of parent 236 117 117

Q15.

CUSAT CAT 2025 - Shift 2

In a fission reaction, U → X + Y +n+n the binding energy per nucleon of 92 236 X and Y is 8.5 MeV whereas of U is 7.6 MeV. The total energy liberated will be about

(A) 2000 MeV	**(B)** 200 MeV
(C) 2 MeV	**(D)** 1 keV

Q16.

CUSAT CAT 2025 - Shift 3

What is the ratio of Bohr magneton to the nuclear magneton? m p

(A) m e	**(B)** 1 2 m p
(C) 2 m e	**(D)** —

Q17.

CUSAT CAT 2025 - Shift 4

The β particle is emitted from the nucleus of mass number A with velocity v, then the recoil speed of nucleus will be

(A) –	**(B)** 4 +4
(C) —	**(D)** 4 –4

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	D	4	B	5	A	6	D	7	B	8	B	9	C	10	B
11	C	12	A	13	A	14	D	15	B	16	A	17	A						

Semiconductor Electronics

31 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The V-I Characteristics of the diode lie in the st nd

(A) 1 and 2 quadrant st rd	**(B)** 1 and 3 quadrant st th
(C) 1 and 4 quadrant st	**(D)** only in the 1 quadrant

Q2. *CUSAT CAT 2023 - Shift 2*

If a pn diode is reverse biased then the resistance measured by an ohmmeter will be

(A) zero	**(B)** low
(C) high	**(D)** infinite

Q3. *CUSAT CAT 2023 - Shift 2*

The color of an LED can be changed by

(A) increasing the charge carriers	**(B)** varying the doping level of the semiconductors
(C) increasing applied voltage	**(D)** by using different bandgap semiconductors

Q4. *CUSAT CAT 2023 - Shift 3*

A reverse bias PN junction has

(A) very narrow depletion layer	**(B)** almost no current
(C) very low resistance	**(D)** large current flow

Q5. *CUSAT CAT 2023 - Shift 4*

The collector current in a transistor is 4 mA and the base current is 200 μA. The current gain for the common base configuration is

(A) 0.95	**(B)** 0.90
(C) 0.85	**(D)** 0.80

Q6. *CUSAT CAT 2023 - Shift 4*

Zener diode is used as a

(A) amplifier	**(B)** voltage regulator
(C) rectifier	**(D)** oscillator

Q7. *CUSAT CAT 2023 - Shift 4*

At absolute temperature, an intrinsic semiconductor has

(A) a few free electrons	**(B)** many holes
(C) many free electrons	**(D)** no holes or free electrons

Q8. *CUSAT CAT 2023 - Shift 5*

The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than the 2480 nm is incident on it. The band gap (in eV) for the semiconductor is

(A) 0.9	**(B)** 0.7
(C) 0.5	**(D)** 1.1

Q9. *CUSAT CAT 2023 - Shift 5*

In semiconductor the forbidden energy gap lies

(A) just below the conduction band	**(B)** just above the conduction band
(C) either above or below the conduction band	**(D)** between the valence band and conduction band

Q10. *CUSAT CAT 2024 - Shift 1*

Which one of the following represents a forward biased junction diode?

(A) —	**(B)** —
(C) —	**(D)** —

Q11. *CUSAT CAT 2024 - Shift 1*

Zener diode is used for

(A) Rectification	**(B)** Stabilization
(C) Amplification	**(D)** Modulation

Q12. *CUSAT CAT 2024 - Shift 1*

The current amplification factor α of a common base transistor and the current amplification factor β of a common emitter transistor are not related by β

(A) $\alpha= 1+\beta$ 1 1	**(B)** – =1 α β α
(C) $\beta= 1-\alpha$ α	**(D)** $\beta= 1+\alpha$

Q13. *CUSAT CAT 2024 - Shift 2*

In a p-type semiconductor, the majority and minority charge carriers are, respectively,

(A) Protons and Neutrons	**(B)** Electrons and Holes
(C) Electrons and Protons	**(D)** Holes and Electrons

Q14. *CUSAT CAT 2024 - Shift 3*

If the output of a semiconductor transistor to be proportional to the input signal, the transistor has to be operated in the

(A) Plateau region	**(B)** Cut-off region
(C) Active region	**(D)** Saturation region

Q15. *CUSAT CAT 2024 - Shift 3*

Doping of germanium with arsenic will result in

(A) n-type semiconductor	**(B)** p-type semiconductor
(C) Insulator	**(D)** Intrinsic semiconductor

Q16. *CUSAT CAT 2024 - Shift 3*

Which one of the following statements is not true?

(A) Resistance of an intrinsic semiconductor decreases with increase in temperature.	**(B)** Doping of pure Si with trivalent impurities gives p-type semiconductor.
(C) The majority carriers in n-type semiconductor are holes.	**(D)** A p-n junction can act as semiconductor diode.

Q17. *CUSAT CAT 2024 - Shift 3*

Bipolar junction transistor is a

(A) current controlled current device	**(B)** current controlled voltage device
(C) voltage controlled current device	**(D)** voltage controlled voltage device 8

Q18. *CUSAT CAT 2024 - Shift 4*

In a good conductor, the energy gap between the conduction band and the valence band is

(A) Infinity	**(B)** Wide
(C) Narrow	**(D)** Zero

Q19. *CUSAT CAT 2024 - Shift 4*

A photodiode is essentially a

(A) P-type semiconductor	**(B)** N-type semiconductor
(C) P-N junction	**(D)** Metal-Insulator-Semiconductor

Q20. *CUSAT CAT 2024 - Shift 4*

When n-type semiconductor is heated

(A) number of electrons increases while number of holes decreases	**(B)** number of holes increases while number of electrons decreases
(C) number electrons and holes remains the same	**(D)** number of electrons and holes increases equally –1 –2

Q21. *CUSAT CAT 2024 - Shift 5*

In a p-type semiconductor, there is

(A) excess of one electron	**(B)** a missing of atom
(C) a donor level	**(D)** absence of one electron

Q22. *CUSAT CAT 2024 - Shift 5*

To get output 1 at R for the given logic gate circuit, the input values must be

(A) X = 0, Y = 1	**(B)** X = 1, Y = 1
(C) X = 1, Y = 0	**(D)** X = 0, Y = 0

Q23. *CUSAT CAT 2024 - Shift 5*

Fermi level in the case of intrinsic semiconductor lies

(A) close to the conduction band	**(B)** close to the valence band
(C) in the middle of the forbidden energy gap	**(D)** above the conduction band

Q24. *CUSAT CAT 2024 - Shift 5*

In case of a common emitter transistor amplifier, the ratio of the collector current to the emitter current Ic /Ie is 0.96. The current gain of the amplifier is

(A) 6	**(B)** 48
(C) 24	**(D)** 12

Q25. *CUSAT CAT 2025 - Shift 1*

In forward bias the width of depletion layer

(A) decreases with increase in V	**(B)** increases with increase in V
(C) independent of V	**(D)** is zero

Q26. *CUSAT CAT 2025 - Shift 1*

The given combination of logic gates represents the following gate

(A) OR	**(B)** XOR
(C) NAND	**(D)** NOR

Q27. *CUSAT CAT 2025 - Shift 1*

The current voltage relation of diode is given by I = (– 1) mA, where the applied voltage V is in volts and the temperature T is in degree Kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300 K, what will be the error in the value of current on mA ?

(A) 0.5 mA	**(B)** 0.05 mA
(C) 0.2 mA	**(D)** 0.02 mA

Q28. *CUSAT CAT 2025 - Shift 2*

In a common base mode of a transistor, the collector current is 5.488mA for an emitter current of 5.6mA. The value of the base current amplification factor (β) will be

(A) 49	**(B)** 50
(C) 51	**(D)** 48

Q29.

CUSAT CAT 2025 - Shift 2

The colour of the light emitted by the light emitting diode (LED) depends on

(A) the magnitude of the current injected into the diode junction	**(B)** semiconductor material used in the making of LED
(C) doping concentration on either side of the diode junction	**(D)** forward bias voltage

Q30.

CUSAT CAT 2025 - Shift 2

If a semiconductor photodiode can detect a phonon with a maximum wavelength of 400 nm, then its band gap energy is Plank's constant h = 6.63 × 10–34 J s Speed constant c = 3 × 108 m/s

(A) 2.0 eV	**(B)** 1.5 eV
(C) 3.1 eV	**(D)** 1.1 eV

Q31.

CUSAT CAT 2025 - Shift 3

Which one of the following statements is NOT correct?

(A) A diode does not obey Ohm's law.	**(B)** A p-n junction diode symbol shows an arrow identifying the direction of current (forward) flow
(C) An ideal diode is an open switch	**(D)** An ideal diode is an ideal one way conductor

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	D	4	B	5	A	6	B	7	D	8	C	9	D	10	A
11	B	12	D	13	D	14	C	15	A	16	C	17	A	18	D	19	C	20	D
21	D	22	C	23	C	24	C	25	A	26	A	27	C	28	A	29	B	30	C
31	C																		

Physics – Miscellaneous

382 questions

Q1.

CUSAT CAT 2023 - Shift 1

WWaatteerr iiss uusseedd aass aa ccoooollaanntt iinn eennggiinneess bbeeccaauussee

(A) wwaatteerr iiss aa bbaadd ccoonndduuccttoorr ooff hheeaatt	**(B)** wwaatteerr hhaass llooww ddeennssiittyy
(C) wwaatteerr hhaass hhiigghh ssppeecciiffiicc hheeaatt	**(D)** wwaatteerr iiss ppoollaarr ssoollvveenntt

Q2.

CUSAT CAT 2023 - Shift 1

In the following P-VVV dddiiiaaagggrrraaammm fffooorrr aaa gggaaasss wwwhhhiiiccchhh ooonnneee ooofff ttthhheee fffooollllllooowwwiiinnnggg iiisss CORRECT?

(A) TTThhheee ttteeemmmpppeeerrraaatttuuurrreee ooofff ttthhheee gggaaasss wwwiiiiiiiii iiinnnccccrrreeeaaassseee aaasss iiittt gggoooeeesss fffrrrooommm AAA tttooo BBB	**(B)** TTThhheee ttteeemmmpppeeerrraaatttuuurrreee ooofff ttthhheee gggaaasss wwwiiiiiiiii iiinnnccccrrreeeaaassseee aaasss iiittt gggoooeeesss fffrrrooommm BBB tttooo CCC
(C) TThhee tteemmppeerraattuurree ooff tthhee ggaass rreemmaaiinn ccoonnssttaanntt dduurriinngg tthheessee cchhaannggeess	**(D)** TThhee tteemmppeerraattuurree ooff tthhee ggaass ddeeccrreeaassee aass iitt ggooeess ffrroomm DD ttoo AA

Q3.

CUSAT CAT 2023 - Shift 1

AAA ssswwwiiimmmmmmeeerrr iiisss ssswwwiiimmmmmmiiinnnggg 111000 mmm bbbeeelllooowww ttthhheee wwwaaattteeerrr sssuuurrrfffaaaccceee ooofff aaa lllaaakkkeee... HHHeee ttthhheeennn eeexxxpppeeerrriiieeennnccceeesss a pressure of

(A) 2 atm	**(B)** 5 atm
(C) 1 atm	**(D)** 7 atm

Q4.

CUSAT CAT 2023 - Shift 1

When the mass of the electron becomes equal to three times its rest mass, its speed will become (c = velocity of light) 2 2

(A) c 3 2	**(B)** c 3 1
(C) c 3 1	**(D)** c 4

Q5.

CUSAT CAT 2023 - Shift 1

In the cciirrccuuiitt ggiivveenn bbeellooww,, tthhee vvaalluuee ooff tthhee ccuurrrreenntt iiss

(A) Zero –2	**(B)** 10 A 2
(C) 10 A –3	**(D)** 10 A

Q6.

CUSAT CAT 2023 - Shift 1

IIInnn YYYoooouuunnnggg'''sss dddoooouuubbbllleee ssslllliiittt eeexxxpppeeerrriiimmmeeennnttt,,, tttwwwooo wwwaaavvveeesss iiinnnttteeerrrfffeeerrreee tttooo pppprrrooodddduuuccceee aaannn iiinnnttteeerrrfffeeerrreeennnccceee ppaattttteerrnn.. TThhee tthhiirrdd mmiinniimmaa ooff tthhee ppaattttteerrnn hhaavvee aa

(A) PPhhaassee ddiiffffeerreennccee ooff 3π 5π	**(B)** PPhhaassee ddiiffffeerreennccee ooff 2
(C) PPaatthh ddiiffffeerreennccee ooff 33λ 5λ	**(D)** PPaatthh ddiiffffeerreennccee ooff 2

Q7.

CUSAT CAT 2023 - Shift 1

If ω iiss tthhee ffrreeqquueennccyy ooff aa ccaarrrriieerr wwaavvee aanndd ω is the frequency ooff tthhee mmoodduullaattiioonn c m ssiiggnnaall,, tthheenn tthhee aammpplliittuuddee mmoodduullaatteedd wwaavvee wwiillll hhaavvee ffrreeqquueenncciieess

(A) ω and ω c m	**(B)** ω ,ω +ω aanndd ω –ω c c m c m
(C) ω and ωωωω c mm cc	**(D)** ω and ω ·ω c cc mm

Q8.

CUSAT CAT 2023 - Shift 1

For a head-oonn ccoollllliissiioonn ooff α- ppaarrttiicclleess wwiitthh aa ggoolldd nnuuccllleeuuss,, tthhee iimmppaacctt ppaarraammeetteerr iiss

(A) zero –6	**(B)** ooff tthhee oorrddeerr ooff 1100 m –10
(C) of tthhee oorrddeerr ooff 1100 m –14	**(D)** ooff tthhee oorrddeerr ooff 1100 m

Q9.

CUSAT CAT 2023 - Shift 1

A particle covers half of the circle of radius r. The displacement and distance of the particle are, respectively

(A) πr,r	**(B)** 2r,πr πr
(C) ,2r 2	**(D)** 2πr,0

Q10.

CUSAT CAT 2023 - Shift 1

When 40 g of water at 10°C is mixed with 80 g of water at 100°C. The resultant temperature is

(A) 55°C	**(B)** 60°C
(C) 65°C	**(D)** 70°C

Q11.

CUSAT CAT 2023 - Shift 1

Which of the following particles can be added to the nucleus without changing its chemical properties?

(A) Electrons	**(B)** Neutrons
(C) Protons	**(D)** -particles

Q12.

CUSAT CAT 2023 - Shift 1

Which one of the following is a non-polar molecule?

(A) HCl	**(B)** CO
(C) H O 2	**(D)** O 2

Q13. *CUSAT CAT 2023 - Shift 1*

In 1929, Nobel Prize in Physics was awarded to …… for his discovery of the wave nature of electrons.

(A) Schrodinger	**(B)** Hamiltonian
(C) Debye	**(D)** de Broglie

Q14. *CUSAT CAT 2023 - Shift 1*

The mass (kg) of observable Universe is about 55

(A) 10 41	**(B)** 10 30
(C) 10 25	**(D)** 10

Q15. *CUSAT CAT 2023 - Shift 1*

On an average, a human heart is found to beat 75 times in a minute. Then its period is

(A) 0.4 s	**(B)** 0.8 s
(C) 1.2 s	**(D)** 2.4 s

Q16. *CUSAT CAT 2023 - Shift 1*

A ray of light travelling in the direction 1(iˆ+ 3iˆ) is incident on a plane mirror. 2 After reflection, it travels along the direction 1(iˆ– 3iˆ) . The angle of incidence is 2

(A) 30°	**(B)** 60°
(C) 45°	**(D)** 75°

Q17. *CUSAT CAT 2023 - Shift 1*

Two springs of force constant 1200 N/m and 2400 N/m respectively are stretched with a same force. Their potential energies will be in the ratio of

(A) 4 : 1	**(B)** 1 : 2
(C) 1 : 4	**(D)** 2 : 1

Q18. *CUSAT CAT 2023 - Shift 1*

Water in a bucket is whirled in a vertical circle with a string attached to it. The Water does not fall down even when the bucket is inverted at the top of its path. We conclude that in this position 2 mv

(A) mg = r 2 mv	**(B)** mg is greater than r 2 mv
(C) mg is not greater than r 2 mv	**(D)** mg is not less than r

Q19. *CUSAT CAT 2023 - Shift 1*

For photoelectric emission from certain metal, the cut off frequency is ν. If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be (m is the mass of the electron) hν

(A) 2m hν	**(B)** m 2hν
(C) m hν	**(D)** 2 m

Q20. *CUSAT CAT 2023 - Shift 1*

A carbon resistor of (47 ±4.7) kΩ is to be marked with rings of different colours for its identification. The colour code sequence will be

(A) Violet-Yellow-Orange-Silver	**(B)** Green-Orange-Violet-Gold
(C) Yellow-Green-Violet-Gold	**(D)** Yellow-Violet-Orange-Silver

Q21. *CUSAT CAT 2023 - Shift 1*

A toy boat containing pieces of iron is floating in a dish of water. If one of the pieces of iron is removed from the boat and placed in the water outside the boat, then

(A) the level of water in the dish will rise	**(B)** the level of water in dish will fall
(C) the level of water in the dish will remain unchanged	**(D)** the toy boat will sink

Q22. *CUSAT CAT 2023 - Shift 1*

The truth table given below is for which gate? A B C 0 0 1 0 1 1 1 0 1 1 1 0

(A) XOR	**(B)** OR
(C) AND	**(D)** NAND

Q23. *CUSAT CAT 2023 - Shift 1*

A source of light is placed at a distance of 1 m from the photocell and the cut-off potential is found to be V . If the lamp is moved to a distance of 2 m then the cut-off 0 potential will become

(A) 2 V 0 V	**(B)** 0 2 V
(C) 0 4	**(D)** V 0

Q24. *CUSAT CAT 2023 - Shift 1*

An arrow of mass 20 g, moving at 150 m/s penetrates 5 cm into a log and then comes to rest. Assuming that the force exerted by the log is uniform, the magnitude of the force is

(A) 450 N	**(B)** 4500 N
(C) 22500 N	**(D)** 2250000 N

Q25. *CUSAT CAT 2023 - Shift 1*

A wheel of perimeter 220 cm rolls on a level road at a speed of 9 km/h. How many revolutions does the wheel make per second? 25

(A) rev/s 22 9	**(B)** rev/s 220 90000
(C) rev/s 22 9×2π	**(D)** rev/s 220 3

Q26. *CUSAT CAT 2023 - Shift 1*

The volume of water decreases from its initial volume of 1000 cm , under a 5 2 6 2 pressure change from 10 N/m to 10 N/m . If the compressibility of water –11 –1 2 is 50 × 10 N m , the decrease in volume of water is 3

(A) 0.45 cm 3	**(B)** 2.22 cm 3
(C) 8.42 cm 3	**(D)** 50.22 cm

Q27. *CUSAT CAT 2023 - Shift 1*

An object is seen through a simple microscope of focal length 12 cm. Find the angular magnification produced if the image is formed at the near point of the eye which is 25 cm away from it.

(A) 2.08	**(B)** 3.08
(C) 6.16	**(D)** 9.24 –6 –6

Q28. *CUSAT CAT 2023 - Shift 1*

Two particles A and B having charges 8.0 × 10 C and –2.0 × 10 C respectively are held fixed with a separation of 20 cm. At what distance from B should a third charged particle be placed so that it does not experience a net electric force?

(A) 2 cm	**(B)** 20 cm
(C) 40 cm	**(D)** 80 cm

Q29. *CUSAT CAT 2023 - Shift 1*

When a body is taken from the equator to the poles, its weight

(A) remains the same	**(B)** decreases
(C) increases	**(D)** increases at north pole and decreases at south pole

Q30. *CUSAT CAT 2023 - Shift 1*

When a body is stationary

(A) there is no force acting on it	**(B)** the forces acting on it are not in contact with it
(C) the combination of the forces acting on it balance each other	**(D)** the body is in vacuum

Q31. *CUSAT CAT 2023 - Shift 1*

Force F applied on a body moves it through a distance S along F . Energy spent is

(A) F×S F	**(B)** S
(C) F×S 2 F	**(D)** 2 S

Q32. *CUSAT CAT 2023 - Shift 1*

What is the surface energy of an air bubble inside a soap solution? 2

(A) 4πr T 2	**(B)** 8πr T 2
(C) 2πr T 2	**(D)** πr T

Q33.

CUSAT CAT 2023 - Shift 1

Temperature can be expressed as a derived quantity in terms of

(A) length and mass	**(B)** mass and time
(C) length, mass and time	**(D)** length and time

Q34.

CUSAT CAT 2023 - Shift 2

For cooking pppuuurrrpppooossseee,,, iiittt iiisss dddeeesssiiirrraaabbbllleee ttthhhaaattt ttthhheee mmmaaattteeerrriiiaaalll ooofff ttthhheee uuuttteeennnsssiiilll ssshhhooouuullldd hhhaaavvveee

(A) llooww ccoonndduuccttiivviittyy aanndd llooww ssppeecciiffiicc hheeaatt	**(B)** hhiigghh ccoonndduuccttiivviittyy aanndd llooww ssppeecciiffiicc hheeaatt
(C) llooww ccoonndduuccttiivviittyy aanndd hhiigghh ssppeecciiffiicc hheeaatt	**(D)** hhiigghh ccoonndduuccttiivviittyy aanndd hhiigghh ssppeecciiffiicc hheeaatt

Q35.

CUSAT CAT 2023 - Shift 2

The P-VVV ppllloootttsss fffooorrr tttwwwooo gggaaassseeesss (((111 aaannnddd 222))) ddduuurrriiinnggg aaadddiiiaaabbbaaatttiiiccc ppprrrooocceeesssssseeesss aaarrreee ssshhhooowwwnnn bbbeeellllooowww... Then

(A) 11 aanndd 22 bbootthh sshhoouulldd ccoorrrreessppoonndd ttoo aa ddiiaattoommiicc ggaass	**(B)** 111 cccooorrrrrreeessspppooonnnddsss tttooo dddiiiaaatttooommmiiiccc aaannnddd 222 cccooorrrrrreeessspppooonnnddsss tttooo mmmooonnnooaaatttooommmiiiccc gggaaasss
(C) 11 aanndd 22 bbootthh sshhoouulldd ccoorrrreessppoonndd ttoo aa monoatomic gas	**(D)** 111 ssshhhooouuullldd cccooorrrrrreeessspppooonnnddd tttooo mmmooonnnooaaatttooommmiiiccc aaannnddd 222 ssshhhooouuullldd cccooorrrrrreeessspppooonnnddd tttooo dddiiiaaatttooommmiiiccc gggaaasss

Q36.

CUSAT CAT 2023 - Shift 2

TThhee aavveerraaggee ddeepptthh ooff IInnddiiaann OOcceeaann iiss aabbooutt 33000000 m.. IIff tthhee bbuullkk mmoodduulluuss ooff oocceeaann 9 –22 ∆V water is 2 × 10 Nm , the fractional compression ooff wwaatteerr aatt tthhee bbootttoomm ooff tthhee V –2 oocceeaann wwiillll bbee ((aassssuummee g = 9.81 ms)

(A) 1.36%	**(B)** 1.47%
(C) 1.8%	**(D)** 2%

Q37.

CUSAT CAT 2023 - Shift 2

IIInnn ttthhheee fffooollllllooowwwiiinnnggg ccciiirrrcccuuuiiittt,,, ttthhheee cccaaapppaaaccciiitttaaannnccceee bbbeeetttwwweeeeeennn ttthhheee pppoooiiinnnttttsss AAA aaannnddd BBB wwwiiiiiiiii bbbeee

(A) 3 μF	**(B)** 12 μF
(C) 4 μF	**(D)** 1 μF

Q38.

CUSAT CAT 2023 - Shift 2

Which one ooofff ttthhheee fffooollllllooowwwiiinnnggg iiisss mmmooossssttt sssuuuiiiitttaaabbbllleee fffooorrr mmmaaakkkiiinnnggg pppeeerrrmmmaaannneeennnttt mmmaaagggnnneeettt???

(A) Steel	**(B)** Soft iron
(C) Copper	**(D)** Nickel

Q39.

CUSAT CAT 2023 - Shift 2

TThhee BBoooolleeaann eexxpprreessssiioonn ffoorr tthhee cciirrccuuiitt bbeelloow iiss

(A) Y = A·B + C ( )	**(B)** Y = A· B+C ( )
(C) Y = A· B+C	**(D)** Y = A·(B+C)

Q40.

CUSAT CAT 2023 - Shift 2

WWWhhhiiiccchhh ooonnneee ooofff ttthhheee fffooollllllooowwwiiinnnggg uuuttttiiiillliiizzzeeesss ttthhheee ppphhheeennnooommmeeennnooonnn ooofff tttoootttaaalll iiinnnttteeerrrnnnaaalll rrreeeffffllleeeccctttiiiooonnn???

(A) FFoorrmmaattiioonn ooff mmiirraaggee	**(B)** FFoorrmmaattiioonn ooff rraaiinnbbooww
(C) LLiigghhtt pprrooppaaggaattiioonn iinn ooppttiiccaall ffiibbeerrss	**(D)** All of the aabboovvee

Q41.

CUSAT CAT 2023 - Shift 2

Three stars A, B and CC have surface temperature of T , T and T rreesspeeccttiivveellyy.. IIff A A B C appears bluish, B aappppeeaarrss rreeddddiisshh aanndd C appears yellowish, then

(A) T > T > T A C B	**(B)** T > T > T A B C
(C) T > T > T B C A	**(D)** T > T > T C B A

Q42.

CUSAT CAT 2023 - Shift 2

A stone of mmmaaassssss 222000000 ggg iiisss tttiiieeeddd tttooo ttthhheee eeennnddd ooofff aaa ssstttrrriiinnnggg ooofff lleeennnggttthhh 000...555 mmm... IIttt iiisss wwwhhhiiirrrlleeeddd iiinnn aaa cciirrccllee wwiitthh aa ffrreeqquueennccyy ooff 00..55 HHzz.. WWhhaatt iiss tthhee tteennssiioonn iinn tthhee ssttrriinngg?? 2 π

(A) N 2 2 π	**(B)** N 4 2 π
(C) N 5 2 π	**(D)** N 10 1

Q43.

CUSAT CAT 2023 - Shift 2

The shift in wavelength due to Doppler's effect is 0.2 Å for a star producing wavelength of 3000 Å. The velocity of recession of the star will be

(A) 5 km/s	**(B)** 10 km/s
(C) 20 km/s	**(D)** 30 km/s

Q44. *CUSAT CAT 2023 - Shift 2*

If the light is polarised by reflection, then the angle between reflected and refracted light is

(A) 36°	**(B)** 45°
(C) 90°	**(D)** 180°

Q45. *CUSAT CAT 2023 - Shift 2*

In Pfund series, the value of principal quantum number ‘n’ starts from

(A) n = 2, 3, 4, ...	**(B)** n = 4, 5, 6, ...
(C) n = 5, 6, 7, ...	**(D)** n = 6, 7, 8, ...

Q46. *CUSAT CAT 2023 - Shift 2*

The atomic mass of chlorine atom is

(A) 17.23 u	**(B)** 35.47 u
(C) 71.47 u	**(D)** 3.547 u

Q47. *CUSAT CAT 2023 - Shift 2*

A point source of light is placed at a distance of 2f from a converging lens of focal length f. The intensity on the other side of the lens is maximum at a distance

(A) f	**(B)** 2f
(C) more than 2f	**(D)** between f and 2f

Q48. *CUSAT CAT 2023 - Shift 2*

If the water falls from a dam into a turbine wheel 19.6 m below, then the velocity of the water at the turbine is

(A) 9.8 m/sec	**(B)** 32.6 m/sec
(C) 58.8 m/sec	**(D)** 19.6 m/sec

Q49. *CUSAT CAT 2023 - Shift 2*

If the velocity of a particle is v= At+Bt where A and B are constants. Then the distance travelled by it between 1 s and 2 s is 3

(A) A+4B 2	**(B)** 3A+7B 3 7
(C) A+ B 2 3 1 1	**(D)** A+ B 2 3

Q50. *CUSAT CAT 2023 - Shift 2*

The slits in Young’s double slit experiment are having equal width and the source is placed symmetrically with respect to the slits. The intensity at the central fringe is I . o If one of the slit is closed, the intensity at this point will be

(A) I o I o	**(B)** 4 I o
(C) 2	**(D)** 4I o

Q51. *CUSAT CAT 2023 - Shift 2*

If a stationary charge is put inside a magnetic field, the charge will

(A) move in a straight line	**(B)** move in circle
(C) remain stationary	**(D)** move in a helix

Q52. *CUSAT CAT 2023 - Shift 2*

Two plane mirrors are inclined at an angle of 60°. An object is placed symmetrically between the mirrors. The total number of images formed by the two mirrors is

(A) 4	**(B)** 6
(C) 5	**(D)** 7

Q53. *CUSAT CAT 2023 - Shift 2*

Two vectors having equal magnitudes Y make an angle α with each other. The magnitude of the resultant is

(A) Y	**(B)** 2Y α
(C) 2Y cos 2	**(D)** 2Y cos (α) g

Q54. *CUSAT CAT 2023 - Shift 2*

A cart of weight 72 kg, and a man of weight 65 kg move towards each other on a smooth horizontal surface. The velocity of the cart and the man are 3 km/h and 6 km/h respectively. When the man reaches the cart, he jumps into it. Then the speed of the cart carrying the man is

(A) 4.42 km/h	**(B)** 3.23 km/h
(C) 1.87 km/h	**(D)** 1.27 km/h

Q55. *CUSAT CAT 2023 - Shift 2*

A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be

(A) doubled	**(B)** four times
(C) one-fourth	**(D)** halved

Q56. *CUSAT CAT 2023 - Shift 2*

In an n-type silicon, which of the following statement is true?

(A) Electrons are the majority carriers and trivalent atoms are the dopants	**(B)** Electrons are the minority carriers and pentavalent atoms are the dopants
(C) Holes are the minority carriers and pentavalent atoms are the dopants	**(D)** Holes are the majority carriers and trivalent atoms are the dopants

Q57. *CUSAT CAT 2023 - Shift 2*

The motion of holes and free electrons due to thermal agitation is called

(A) diffusion	**(B)** drift
(C) translation	**(D)** conduction

Q58. *CUSAT CAT 2023 - Shift 2*

Consider a system of two identical particles. One of the particles is at permanent rest and the other has acceleration a. The centre of mass has acceleration

(A) Zero a	**(B)** 2
(C) a	**(D)** 2a

Q59. *CUSAT CAT 2023 - Shift 3*

If V and V represent the velocity of sound in moist air and dry air at the same m d temperature, which one of the following is TRUE?

(A) V = V m d	**(B)** V > V m d
(C) V < V m d	**(D)** V = 2V m d

Q60. *CUSAT CAT 2023 - Shift 3*

On which one of the following scales of temperature, the temperature is never negative?

(A) Kelvin	**(B)** Fahrenheit
(C) Celcius	**(D)** Reaumur

Q61. *CUSAT CAT 2023 - Shift 3*

A real gas behaves as an ideal gas at

(A) very low pressure and high temperature	**(B)** high pressure and low temperature
(C) high pressure and high temperature	**(D)** low pressure and low temperature

Q62. *CUSAT CAT 2023 - Shift 3*

Which one of the following is the defining equation for uniform circular motion? (Symbols have usual meaning.)

(A) V = ω r ω	**(B)** V = r
(C) V = ω 2 r r	**(D)** V = ω

Q63. *CUSAT CAT 2023 - Shift 3*

The centripetal force acting on a coin weighing 0.1 kg place 0.1 m from the centre of a disc rotating at 600 rpm is 2 π

(A) N 2	**(B)** 2π 2 N 2 π
(C) N 4	**(D)** 4π 2 N

Q64. *CUSAT CAT 2023 - Shift 3*

The earth's magnetic field is approximately 5

(A) 2 × 10 T –5	**(B)** 2 × 10 T –5
(C) 4 × 10 T –5	**(D)** 8 × 10 T

Q65. *CUSAT CAT 2023 - Shift 3*

The frequency of rotation of water molecules is about

(A) 2.45 GHz	**(B)** 2.45 MHz
(C) 2.45 Hz	**(D)** 0.245 Hz

Q66. *CUSAT CAT 2023 - Shift 3*

1 curie is equal to 10

(A) 3.7 × 10 Bq 8	**(B)** 2.7 × 10 Bq 6
(C) 1.7 × 10 Bq –10	**(D)** 3.7 × 10 Bq

Q67. *CUSAT CAT 2023 - Shift 3*

A rubber ball is dropped from a height of 5 m on a plane. On bouncing it rises to 1.8 m. The ball loses its velocity on bouncing by a factor of 3

(A) 5 2	**(B)** 5 16
(C) 25 9	**(D)** 25

Q68. *CUSAT CAT 2023 - Shift 3*

The pulleys and strings shown below are smooth and of negligible mass. For the system to remain in equilibrium, the angle of θ should be

(A) 30°	**(B)** 45°
(C) 90°	**(D)** 60°

Q69. *CUSAT CAT 2023 - Shift 3*

Which of the following methods can be used to measure the speed of light in laboratory?

(A) Roemer method	**(B)** Fizeau method
(C) Foucault method	**(D)** Michelson method

Q70. *CUSAT CAT 2023 - Shift 3*

As the wavelength is increased from violet to red, the luminosity of a source

(A) continuously increases	**(B)** continuously decreases
(C) decreases then increases	**(D)** increases then decreases

Q71. *CUSAT CAT 2023 - Shift 3*

In the electric network shown, when no current flows through the 4 Ω resistor in the arm EB the potential difference between the points A and D will be

(A) 6 V	**(B)** 3 V
(C) 5 V	**(D)** 4 V

Q72.

CUSAT CAT 2023 - Shift 3

Two balls each of radius R, equal mass and density are placed in contact, then the force of gravitation between them is proportional to 1

(A) F ∝ 2 R	**(B)** F ∝ R
(C) F ∝ R 4 1	**(D)** F ∝ R

Q73.

CUSAT CAT 2023 - Shift 3

A hollow metallic sphere of radius 10 cm is charged such that the potential of its surface becomes 70 V. The potential at the centre of the sphere is

(A) 100 V	**(B)** 35 V
(C) 70 V	**(D)** 7 V

Q74.

CUSAT CAT 2023 - Shift 3

A steel ball weighing 1 Kg is dropped from the Leaning tower of Pisa. It starts form 2 rest and falls freely. The position after 1 sec is (g is 9.8 m/s)

(A) 4.9 m	**(B)** –9.8 m
(C) –4.9 m	**(D)** 19.6 m

Q75.

CUSAT CAT 2023 - Shift 3

If the horizontal and vertical components of the earth's magnetic field are equal at a certain place, then the angle of dip at that place will be

(A) 90°	**(B)** 60°
(C) 45°	**(D)** 0°

Q76.

CUSAT CAT 2023 - Shift 3

AA mmaatteerriiaall wwiitthh oovveerrllaappppiinngg ccoonnddduuccttiioonn aanndd vvaallaannccee bbaannddss wwiillll bbee

(A) an insulator	**(B)** aa sseemmiiccoonnddduuccttoorr
(C) a metal	**(D)** aa ssuuppeerrccoonnddduuccttoorr

Q77.

CUSAT CAT 2023 - Shift 3

FFFooorrr ttthhheee ppphhhoootttoooeeellleeeccctttrrriiiccc eeeffffffeeeccctttt,,, ttthhheee mmmaaaxxxiiimmmuuummm kkkiiinnneeetttiiiccc eeennneeerrrgggyyy KKKEEE ooofff ttthhheee eeemmmiiitttttteeeddd pphhoottooeelleeccttrroonnss iiss ppllooottteedd aaggaaiinnsstt tthhee ffrreeqquueennccyy ν ooff tthhee iinncciiddeenntt pphhoottoonnss aass sshhoowwnn iinn tthhee ffiigguurree bbeellooww.. TThhee sslloopee ooff tthhee ccuurrvvee ggiivveess

(A) charge ooff tthhee eelleeccttrroonn	**(B)** work ffuunnccttiioonn ooff tthhee mmeettaall
(C) Plank's ccoonnssttaanntt	**(D)** ratio of the PPllaannkk''ss constant to electronic charge

Q78.

CUSAT CAT 2023 - Shift 3

AAA pppaaarrrrtttiiicccllleee ssstttaaarrrtttsss wwwiiittthhh aaannn iiinnniiitttiiiaaalll vvveeellloooccciiitttyyy 222...555000 mmm///sss aaallloooonnnggg ttthhheee pppooosssiiitttiiivvveee x direction and 2 iitt aaccccceelleerraatteess uunniiffoorrmmllyy aatt tthhee rraattee 00..5500 mm//ss .. TThhee ddiissttaannccee ttrraavveelleedd bbyy iitt iinn tthhee ffiirrsstt 2 seconds is

(A) 2.0 m	**(B)** 4.0 m
(C) 6.0 m	**(D)** 8.0 m

Q79.

CUSAT CAT 2023 - Shift 3

A toy cccaaarrr ooofff 111 kkkggg mmmaaassssss mmmooovvveeesss wwwiiittthhh ccciiirrrcccuuulllaaarrr ssspppeeeeeeddd iiinnn aaa hhhooorrriiizzzooonnnttttaaalll ccciiirrrcccuuulllaaarrr gggrrrooooovvveee,,, wwwiiittthhh vveerrttiiccaall ssiiddee wwaalllss,, ooff rraaddiiuuss 2255 ccmm.. TThhee ttooyy ccaarr ttaakkeess 22..00 sseeccoonnddss ttoo ccoommpplleettee oonnee rroouunndd.. TThhee nnoorrmmaall ccoonnttaacctt ffoorrccee bbyy tthhee ssiiddee wwaalllll ooff tthhee ggrrooovvee iiss

(A) 0.025 N	**(B)** 0.247 N
(C) 2.47 N	**(D)** 24.7 N

Q80.

CUSAT CAT 2023 - Shift 3

The displacement of a particle of a string carrying a travelling wave is y = (3.0 cm) sin 6.28(0.50 x – 50 t), where x is in centimetre and t is in second. The speed of the wave is

(A) 100 m/s	**(B)** 10 m/s
(C) 100 cm/s	**(D)** 10 cm/s 3

Q81.

CUSAT CAT 2023 - Shift 3

A vessel of volume 2000 cm contains 0.1 mole of oxygen and 0.2 mole of carbon dioxide. If the temperature of the mixture is 300 K, find its pressure. 5

(A) 1.25 × 10 Pa 5	**(B)** 2.50 × 10 Pa 5
(C) 3.75 × 10 Pa 5	**(D)** 7.50 × 10 Pa

Q82.

CUSAT CAT 2023 - Shift 3

Regarding diffraction and interference phenomena consider the following statements: (i) In diffraction phenomena, the interfering beams originate from a continuous distribution of sources and interference phenomena, the interfering beams originate from a discrete number of sources. (ii) In the far-field (or Fraunhofer) diffraction as the viewing screen is moved relative to the aperture, the size of the diffraction scales uniformly, but the shape of the diffraction pattern does not change. (iii) In the near field (or Fresnel) diffraction both the shape and size of the diffraction pattern depend on the distance between the aperture and the screen. As the screen is moved away from the aperture, the image of the aperture passes through the forms predicted in turn by geometrical optics, near-field diffraction and far-field diffraction

(A) (i) only is correct	**(B)** (i), (ii) only are correct
(C) (ii), (iii) only are correct	**(D)** (i), (ii), (iii) are correct

Q83. *CUSAT CAT 2023 - Shift 3*

A positive charge enters in a magnetic field and travels opposite the field and it experiences

(A) an upward force	**(B)** a downward force
(C) an accelerated force	**(D)** no force

Q84. *CUSAT CAT 2023 - Shift 3*

What is the force between two charged spheres having charges of 2 × 10 C and –7 3 × 10 C placed 30 cm apart in air? 4

(A) 6 × 10 N –3	**(B)** 6 × 10 N –4
(C) 4 × 10 N –3	**(D)** 4 × 10 N

Q85. *CUSAT CAT 2023 - Shift 3*

Can an ideal gas be liquefied?

(A) Yes	**(B)** No
(C) Can be liquefied at low pressure	**(D)** Can be liquefied at high temperature

Q86. *CUSAT CAT 2023 - Shift 4*

IInn aann aaddiiaabbaattiicc eexxppaannssiioonn ooff aa ggaass,, tthhee ggaass ddooeess 2255 J ooff wwoorrkk.. TThheenn tthhee difference in the iinntteerrnnaall eenneerrggyy iinn tthhee pprroocceessss iiss

(A) 25 J	**(B)** –25 J
(C) Zero	**(D)** 50 J

Q87. *CUSAT CAT 2023 - Shift 4*

TThhee eeqquuaattiioonn ooff ssttaattee ffoorr 88 gg ooff OO molecule will be 2

(A) PV = 8RT RT	**(B)** PV = 4
(C) PV = RT RT	**(D)** PV = 2

Q88. *CUSAT CAT 2023 - Shift 4*

A ball of mass 5 kg iiss ttaakkeenn ttoo aa hheeiigghhtt ooff 33 m aabboovvee tthhee ggrroouunndd.. TThheenn tthhee ppootteennttiiaall eenneerrggyy ssttoorreedd bbyy tthhee bbaallll iiss aabboouutt

(A) 121.2 J	**(B)** 147. 15 J
(C) 227.31 J	**(D)** 182. 21 J

Q89.

CUSAT CAT 2023 - Shift 4

AA rreeccttaanngguullaarr lloooopp ccaarrrryyiinngg aa ccuurrrreenntt i iiss ppllaacceedd nneeaarr aa lloonngg ssttrraaiigghhtt wwiirree ccaarrrryyiinngg aa steady current I sssuuuccchhh ttthhhaaattt ttthhheee wwwiiirrreee iiisss pppaaarrraaallllleeelll tttooo ooonnneee ooofff ttthhheee sssiiidddeee ooofff the loop and is in ppllaannee ooff tthhee lloooopp,, aass sshhoowwnn iinn tthhee ffiigguurree bbeellooww.. The loop then the

(A) rroottaatteess aabbooutt aann aaxxiiss ppaarraallllleell ttoo tthhee lloonngg wwiirree	**(B)** mmoovveess aawwaayy ffrroomm tthhee wwiirree
(C) mmoovveess ttoowwaarrddss tthhee wwiirree	**(D)** rreemmaaiinnss ssttaattiioonnaarryy

Q90.

CUSAT CAT 2023 - Shift 4

An unstable heavy nucleus at rest breaks into two daughter nuclei which move away with velocities in the ratio 9 : 27. The ratio of the radii of the nuclei (assuming them to be spherical) is

(A) 9 : 27	**(B)** 2 : 3
(C) 4 : 9	**(D)** 3 : 2

Q91.

CUSAT CAT 2023 - Shift 4

AAA ssstttooonnneee dddrrrooopppppppeeeddd fffrrrooommm ttthhheee tttoooppp ooofff aaa tttooowwweeerrr hhhiiitttsss ttthhheee gggrrrooouuunnnddd aaafffttteeerrr 666 sss... HHHooowww mmmuuuccchhh tttiiimmmeee ddooeess iitt ttaakkee ttoo ccoovveerr tthhee ffiirrsstt oonnee-tthhiirrdd ooff tthhee ddiissttaannccee ffrroomm tthhee ttoopp ooff tthhee ttoowweerr??

(A) 4 s	**(B)** 2 3s
(C) 3s	**(D)** 2 s

Q92.

CUSAT CAT 2023 - Shift 4

AAA cccuuubbbeee ooofff iiiccceee iiisss fffllloooaaatttiiinnnggg iiinnn wwwaaattteeerrr cccooonnnttttaaaiiinnneeeddd iiinnn aaa vvveeessssssseeelll... WWWhhheeennn ttthhheee iiiccceee mmmeeellltttsss,,, ttthhheee lleeevvveeelll of water in the vessel

(A) falls	**(B)** rises
(C) rreemmaaiinnss uunncchhaannggeedd	**(D)** ffaalllss aatt ffiirrsstt aanndd tthheenn rriisseess

Q93.

CUSAT CAT 2023 - Shift 4

AA 2200 kkgg bbllooockk iiss ssuussppeenddeedd bbyy ttwwoo lliigghhtt spring balances aass sshhoowwnn iinn tthhee ffiigguurree. What will bbee rreeaaddiinngg ooff bbootthh ssccaalleess??

(A) 20 kg, 20 kg	**(B)** 10 kg, 10 kg
(C) 0 kg, 20 kg	**(D)** 20 kg, 0 kg

Q94.

CUSAT CAT 2023 - Shift 4

Two circular loops A and B of radii R and NR rreessppeeccttiivveellyy aarree mmaaddee ffrroomm aa uunniiffoorrmm wwiirree.. TThhee mmoommeenntt ooff iinneerrttiiaa ooff B aabbooutt iittss aaxxiiss iiss 33 ttiimmeess tthhee mmoommeenntt ooff iinneerrttiiaa ooff A aabbooutt iittss aaxxiiss.. TThhee vvaalluuee ooff N is

(A) 3 –13	**(B)** 3 13
(C) 3 23	**(D)** 3 –23

Q95. *CUSAT CAT 2023 - Shift 4*

In Young's double slit experiment, the central point on the screen is

(A) dark	**(B)** bright
(C) first bright and then dark	**(D)** first dark and then bright

Q96. *CUSAT CAT 2023 - Shift 4*

The root mean square speed of the molecule of an enclosed gas is v. If the pressure is doubled keeping the temperature constant, the root mean square speed will be

(A) v	**(B)** 2v v
(C) 2	**(D)** 2v

Q97. *CUSAT CAT 2023 - Shift 4*

Two uniform aluminium rods A and B of length l and 2l and radius r and 2r respectively are heated to same temperature. The ratio of the increase in the length of A to that of B is

(A) 1 : 2	**(B)** 2 : 1
(C) 1 : 1	**(D)** 1 : 4

Q98. *CUSAT CAT 2023 - Shift 4*

A microscope …… objects and produces their larger image

(A) magnifies	**(B)** resolves
(C) capture	**(D)** reduce

Q99. *CUSAT CAT 2023 - Shift 4*

β-decay emits

(A) electrons or positrons	**(B)** proton
(C) phonon	**(D)** hole

Q100. *CUSAT CAT 2023 - Shift 4*

A particle starts rotating from rest according to θ= – , then the angular 20 3 acceleration at the end of 5 seconds is

(A) 7.92 rad/sec 2	**(B)** 3.83 rad/sec
(C) 5.48 rad/sec 2	**(D)** 8.23 rad/sec

Q101. *CUSAT CAT 2023 - Shift 4*

How much water a pump of 2 kW can raise in one minute to a height of 10 m? 2 (Take g = 9.8m/s)

(A) 1000 litres	**(B)** 100 litres
(C) 1200 litres	**(D)** 2000 litres R

Q102. *CUSAT CAT 2023 - Shift 4*

Longitudinal waves CANNOT

(A) have a unique wavelength	**(B)** have a unique wave velocity
(C) transmit energy	**(D)** be polarized

Q103. *CUSAT CAT 2023 - Shift 4*

A charge q is placed at the centre of the line joining two equal positive charges Q. The system of the three charges will be in equilibrium, if q is equal to Q

(A) – 2 Q	**(B)** – 4 Q
(C) + 2 Q	**(D)** + 4 2

Q104. *CUSAT CAT 2023 - Shift 4*

Two identical particles move towards each other with velocities 2v and v respectively. The velocity of their centre of mass will be

(A) v v	**(B)** 3 v
(C) 2	**(D)** 2v

Q105. *CUSAT CAT 2023 - Shift 4*

In a carnival ride, a passenger travel in a merry-go-around at constant speed in a circle of radius 5 m and he took 4 sec to complete. The acceleration is 2

(A) 12 m/s 2	**(B)** 49 m/s 2
(C) 3 m/s 2	**(D)** 8 m/s

Q106. *CUSAT CAT 2023 - Shift 4*

A ball is dropped from a balloon going up with a uniform velocity of 5.0 m/s. If the balloon was 50 m high when the ball was dropped, find its height when the ball hits the ground.

(A) 50.0 m	**(B)** 63.3 m
(C) 68.5 m	**(D)** 74.4 m

Q107. *CUSAT CAT 2023 - Shift 4*

The density of air near earth's surface is 1.3 kg/m and the atmospheric pressure is 5 2 1 × 10 N/m . If the atmosphere had uniform density, same as that observed at the surface of the earth, what would be the height of the atmosphere to exert the same pressure? 5

(A) 10 m	**(B)** 8848 m
(C) 7692 m	**(D)** 4231 m

Q108. *CUSAT CAT 2023 - Shift 4*

A ball starts from rest and falls on ground vertically from a height of 20 m. The distance traveled by the ball during the last 0.3 s is

(A) 2.35 m	**(B)** 4.85 m
(C) 5.65 m	**(D)** 7.25 m

Q109. *CUSAT CAT 2023 - Shift 4*

When current in a coil changes from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self-inductance of the coil is

(A) 1.67 H	**(B)** 6 H
(C) 3 H	**(D)** 0.67 H

Q110. *CUSAT CAT 2023 - Shift 4*

What is the charge on sphere with diameter of 2.4 m and has a surface charge density 2 of 80 μC/m ? –3

(A) 1.45 × 10 C –3	**(B)** 2.35 × 10 C –3
(C) 1.35 × 10 C –5	**(D)** 1.45 × 10 C

Q111. *CUSAT CAT 2023 - Shift 4*

A sine wave is travelling in a medium. The minimum distance between the two particles always having same speed is λ

(A) 4 λ	**(B)** 3 λ
(C) 2	**(D)** λ

Q112. *CUSAT CAT 2023 - Shift 4*

When a copper ball is heated, the largest percentage increase will occur in its

(A) diameter	**(B)** area
(C) volume	**(D)** density

Q113. *CUSAT CAT 2023 - Shift 4*

Optical fibers for communication use are mostly fabricated from

(A) Plastic	**(B)** Silica or multicomponent glass
(C) Ceramics	**(D)** Copper

Q114. *CUSAT CAT 2023 - Shift 4*

In a metallic conductor, electric current is generally due to the movement of

(A) Ions	**(B)** Phonons
(C) Electrons	**(D)** Protons CHEMISTRY (UG) – SHIFT IV (FINAL)

Q115.

CUSAT CAT 2023 - Shift 5

A force of 1 KN is applied to a wire of cross section 10 µm results in an increase in its length by 0.1%. The Young's modulus of the wire will be 12 –2

(A) 10 Nm –2	**(B)** 10 GNm
(C) 10 11 Nm –2 6 –2	**(D)** 10 MNm

Q116.

CUSAT CAT 2023 - Shift 5

A hole is drilled on a copper sheet. The diameter of the hole at room temperature is 4.25 cm. The diameter of the hole when the sheet is heated to 70°C will be

(A) 4.25 cm only	**(B)** more than 4.25 cm
(C) less than 4.25 cm	**(D)** data insufficient to predict

Q117.

CUSAT CAT 2023 - Shift 5

If V is the root mean square value of velocity, V is the average velocity and rms av V velocity of the molecules of a gas, which one of the following is true? mp

(A) V = V ≠ V rms av mp	**(B)** V < V < V rms av mp
(C) V > V > V rms av mp	**(D)** V ≠ V = V rms av mp –31

Q118.

CUSAT CAT 2023 - Shift 5

An electric iron box of rating 1000 W was used for 5 hours per day for 20 days. The electrical energy utilized is then

(A) 200 kWh	**(B)** 120 kWh
(C) 100 kWh	**(D)** 500 kWh

Q119.

CUSAT CAT 2023 - Shift 5

The ffoollllloowwiinngg ffiigguurree sshhoowwss tthhee ddiisspplaacceemmeenntt ooff aa ppaarrttiicllee aalloonngg tthhee x-axis as a function of time. TTThhheee fffoorrrcceee aaacccttiiinnggg ooonnn ttthhheee pppaaarrrttiiicclleee iiisss zzzeeerrrooo iiinnn ttthhheee rrreeeggiiioonnnsss

(A) AB, DE	**(B)** BC, CD
(C) AB, CD	**(D)** BC, DE

Q120.

CUSAT CAT 2023 - Shift 5

AA ssqquuaarree ffrraammee ooff eeaacchh ssiiddee L iiss ddiipppeedd iinn aa ssooaapp ssoolluuttiioonn aanndd ttaakkeenn oouutt.. IIff T is sssuuurrrfffaaaccceee ttteeennnsssiiiooonnn ooofff sssoooaaappp sssoollluuuttiiiooonnn,,, ttthhheee fffoorrrcceee aaacccttiiinnggg ooonnn ttthhheee fffiiilllmmm fffoorrrmmmeeeddd iiisss

(A) TL	**(B)** 2TL
(C) 4TL	**(D)** 8TL

Q121. *CUSAT CAT 2023 - Shift 5*

AAnn uunnppoollaarriisseedd bbeeaamm ooff iinntteennssiittyy I iiss iinncciiddeenntt oonn aa ppaaiirr ooff NNiiccooll pprriissmmss mmaakkiinngg aann o angle of 60° wwwiiittthhh eeeaaaccchhh ooottthhheeerrr... TTThhheee iiinnnttteeennnsssiiitttyyy ooofff lllliiiggghhhttt eeemmmeeerrrgggiiinnnggg fffrrrooommm ttthhheee pppaaaiiirrr iiisss

(A) l o l o	**(B)** 2 l o
(C) 4 l o	**(D)** 8

Q122. *CUSAT CAT 2023 - Shift 5*

Pressure of an ideal gas is doubled keeping the temperature constant. The kinetic energy of the molecules

(A) becomes double	**(B)** becomes half
(C) remains the same	**(D)** may increase or decrease depending on the nature of gas

Q123. *CUSAT CAT 2023 - Shift 5*

How many NAND gates are used in an OR gate?

(A) 3	**(B)** 2
(C) 5	**(D)** 4

Q124. *CUSAT CAT 2023 - Shift 5*

The value of Bohr magneton is 24 2

(A) 9.27 × 10 Am –24 2	**(B)** 9.27 × 10 Am –34 2
(C) 6.27 × 10 Am 34 2	**(D)** 6.27 × 10 Am

Q125. *CUSAT CAT 2023 - Shift 5*

Photo cells are used to measure the …… of light.

(A) intensity	**(B)** wavelength
(C) wave-number	**(D)** speed

Q126. *CUSAT CAT 2023 - Shift 5*

The existence of discrete energy levels in an atom was directly verified in 1914 by

(A) James Franck and Gustav Hertz	**(B)** Stern Gerlach
(C) Heisenberg	**(D)** Langevin

Q127. *CUSAT CAT 2023 - Shift 5*

The electromagnetic force between charged particles arise due to the exchange of particles called

(A) Gravitons	**(B)** Vector Bosons
(C) Electrons	**(D)** Photons

Q128. *CUSAT CAT 2023 - Shift 5*

A ball is kicked at an angle of 30° with the vertical. If the horizontal component of its velocity is 19.6 m/sec then the maximum height ball reached is

(A) 58.8 m	**(B)** 68.8 m
(C) 48.8 m	**(D)** 72.3 m

Q129. *CUSAT CAT 2023 - Shift 5*

A ball is dropped from a spacecraft revolving around the earth at a height of 120 km. What will happen to the ball?

(A) It will continue to move with the same speed along the original orbit of the spacecraft	**(B)** It will move with the same speed, tangentially to the spacecraft
(C) It will fall down to the earth gradually	**(D)** It will go very far in the space

Q130. *CUSAT CAT 2023 - Shift 5*

The sum of all electromagnetic forces between different particles of a system of charged particles is zero.

(A) only if all the particles are positively charged	**(B)** only if all the particles are negatively charged
(C) only if half the particles are positively charged and half are negatively charged	**(D)** irrespective of the signs of the charges

Q131. *CUSAT CAT 2023 - Shift 5*

Water flows in a horizontal tube as shown in figure. The pressure of water changes by 2 2 2 600 N/m between A and B where the areas of cross section are 30 cm and 15 cm respectively. Then the rate of flow of water through the tube is 3 –1

(A) 2300 cm s 3 –1	**(B)** 1890 cm s 3 –1
(C) 3200 cm s 3 –1	**(D)** 1000 cm s

Q132. *CUSAT CAT 2023 - Shift 5*

Consider the following two statements. (I) Line spectra contain information about atoms (II) Band spectra contain information about molecules

(A) Both (I) and (II) are wrong	**(B)** (I) is correct but (II) is wrong
(C) (II) is correct but (I) is wrong	**(D)** Both (I) and (II) are correct

Q133. *CUSAT CAT 2023 - Shift 5*

When deuterium and helium are subjected to an accelerating field simultaneously then

(A) both acquire same energy	**(B)** deuterium accelerates faster
(C) helium accelerates faster	**(D)** neither of them accelerated

Q134. *CUSAT CAT 2023 - Shift 5*

If the polarizing angle for air-glass interface is 56.3°, the angle of refraction in glass is

(A) 45°	**(B)** 123.7°
(C) 90°	**(D)** 33.7°

Q135. *CUSAT CAT 2023 - Shift 5*

The maximum intensity in case of interferences of n identical coherent waves each of intensity I is o

(A) nI o 2	**(B)** n I o
(C) n(n+1)I o	**(D)** (2n–1)I o

Q136. *CUSAT CAT 2023 - Shift 5*

A proton has kinetic energy E = 100 eV which is equal to that of a photon. The λ wavelength of photon is λ and that of proton is λ. The ratio of 2 is proportional to 2 1 λ 1

(A) E 2	**(B)** E 12
(C) E –1	**(D)** E –12

Q137. *CUSAT CAT 2023 - Shift 5*

A cricket ball of mass 250 g collides with a bat with velocity 10 ms and returns with the same velocity with in 0.01 second. The force acting on the bat is

(A) 25 N	**(B)** 50 N
(C) 250 N	**(D)** 500 N

Q138. *CUSAT CAT 2023 - Shift 5*

Zener breakdown will occur if impurity level is

(A) low	**(B)** high
(C) less in n side	**(D)** less in p side

Q139. *CUSAT CAT 2023 - Shift 5*

The force on a particle of mass 10 g is (10 i + 5 j) N. If it starts from rest, what would be its position at time t = 5s?

(A) (12500 i + 6250 j) m	**(B)** (1250 i + 625 j) m
(C) (12.5 i + 62.5 j) m	**(D)** (12.5 i – 62.5 j) m

Q140. *CUSAT CAT 2023 - Shift 5*

A uniform sphere of mass 2000 g rolls without slipping on a plane surface so that its centre moves at a speed of 2.00 cm/s. Find its kinetic energy. –4

(A) 5.6 × 10 J –1	**(B)** 5.6 × 10 J –4
(C) 8 × 10 J –1	**(D)** 8 × 10 J

Q141. *CUSAT CAT 2023 - Shift 5*

In a Young's double slit experiment, the separation between the slits is 0.10 mm, the wavelength of light used is 600 nm and the interference pattern is observed on a screen 1.0 m away. Find the separation between the successive bright fringes

(A) 6.00 cm	**(B)** 6.00 mm
(C) 30.00 cm	**(D)** 3.00 cm

Q142. *CUSAT CAT 2023 - Shift 5*

The light from the sun is found to have a maximum intensity near the wavelength of 470 nm. Assuming that the surface of a sun emits as a blackbody, calculate the temperature of the surface of the sun.

(A) 470 K	**(B)** 840 K
(C) 4128 K	**(D)** 6128 K 5

Q143. *CUSAT CAT 2023 - Shift 5*

A beam of protons with a velocity of 4 × 10 m/s enters a uniform magnetic field of 0.3 T. The velocity makes an angle of 60° with the magnetic field. Find the radius of the helical path taken by the proton beam.

(A) 0.3 cm	**(B)** 1.2 cm
(C) 2.4 cm	**(D)** 9.6 cm

Q144. *CUSAT CAT 2023 - Shift 5*

A player in a ground throws a cricket ball with a speed of 12.0 m/s at an angle of 45° with the horizontal. At what distance will it hit the ground?

(A) 6.6 m	**(B)** 8.3 m
(C) 11.2 m	**(D)** 14.4 m

Q145. *CUSAT CAT 2023 - Shift 5*

The radii of first three Bohr orbits is in the ratio

(A) 1 : 2 : 3	**(B)** 2 : 4 : 6
(C) 1 : 4 : 9	**(D)** 1 : 3 : 5

Q146. *CUSAT CAT 2023 - Shift 5*

The Earth revolves around the Sun in an elliptical orbit. Its speed

(A) goes on decreasing continuously	**(B)** is greatest when it is closest to the Sun
(C) is constant at the all the points on the orbit	**(D)** is greatest when it is farthest from the Sun

Q147. *CUSAT CAT 2023 - Shift 5*

For the resultant of two vectors to be maximum. What must be the angle between them?

(A) 0°	**(B)** 60°
(C) 90°	**(D)** 180°

Q148. *CUSAT CAT 2024 - Shift 1*

A body of weight w is suspended from the ceiling of a room by a chain of weight w . 1 2 The ceiling pulls the chain by a force

(A) w 1	**(B)** w 2 w +w
(C) 1 2 2	**(D)** w +w 1 2

Q149.

CUSAT CAT 2024 - Shift 1

A tank containing liquid has an orifice on its vertical side. If the center of the orifice is 19.6 m below the water surface level in the tank, what is the velocity of discharge of the water?

(A) 19.6 m/s	**(B)** 20 m/s
(C) 15 m/s	**(D)** 9.8 m/s

Q150.

CUSAT CAT 2024 - Shift 1

The kinetic energy of photoelectrons depends on

(A) the intensity of the radiation incident on the photoelectric material	**(B)** the frequency of the radiation
(C) the time for which the radiation is incident	**(D)** the nature of the emitting surface

Q151.

CUSAT CAT 2024 - Shift 1

Two points are located at a distance of 10 m and 15 m from the source of oscillation. The period of the oscillation is 0.05 s and the velocity of the wave is 300 m/s. Then the phase difference between the oscillations at the two points is π

(A) 6	**(B)** π π
(C) 3 2π	**(D)** 3

Q152.

CUSAT CAT 2024 - Shift 1

A magnet drops down a long vertical copper tube. Its velocity as it falls down the tube

(A) increases	**(B)** remains constant
(C) decreases	**(D)** first increases and then decreases

Q153.

CUSAT CAT 2024 - Shift 1

Which is the correct diagram of a half wave rectifier?

(A) —	**(B)** —
(C) —	**(D)** —

Q154.

CUSAT CAT 2024 - Shift 1

Direction: In the given question a statement of assertion and a corresponding statement of reason is given Assertion: When a light wave travels from a rarer to denser medium, its speed decreases. The decrease in speed imply a reduction in energy carried by the light wave. Reason: The energy of a wave is inversely proportional to velocity of wave Choose the correct alternative from given below.

(A) Both assertion and reason are true and reason is the is the correct explanation of assertion	**(B)** Both assertion and reason are true but reason is not the correct explanation of assertion
(C) Assertion is true but reason is false	**(D)** Both assertion and reason are false

Q155. *CUSAT CAT 2024 - Shift 1*

Direction of current induced in a wire moving in a magnetic field is found using

(A) Fleming's left hand rule	**(B)** Fleming's right hand rule
(C) Ampere's rule	**(D)** Right hand clasp rule

Q156. *CUSAT CAT 2024 - Shift 1*

Heat required to raise the temperature of one gram of water through 1°C is

(A) 0.1 cal	**(B)** 0.1 kcal
(C) 0.001 kcal	**(D)** 1 kcal

Q157. *CUSAT CAT 2024 - Shift 1*

When the cells are connected in parallel, then

(A) the current capacity increases	**(B)** the current capacity decreases
(C) the emf increases	**(D)** the emf decreases

Q158. *CUSAT CAT 2024 - Shift 1*

The primary origin of magnetism lies in

(A) Intrinsic spin of electrons	**(B)** Polar and non polar nature of molecules
(C) Pauli exclusion principle	**(D)** Extrinsic spin of electrons

Q159. *CUSAT CAT 2024 - Shift 1*

If the force applied to a body is doubled and the mass is cut in half, what would be the accelerations' ratio?

(A) 1 : 2	**(B)** 2 : 1
(C) 1 : 4	**(D)** 4 : 1

Q160. *CUSAT CAT 2024 - Shift 1*

Which of the following statement is wrong?

(A) Clouds appear white for most of the day because the sun radiates all visible wavelengths of light	**(B)** The rays have to travel a shorter distance than usual at sunrise and sunset
(C) Danger signals are red because red color gets scattered the least	**(D)** The sky is blue because blue light is scattered more than other colors since it travels as shorter waves

Q161. *CUSAT CAT 2024 - Shift 1*

What should be done to obtain an OR gate from a NAND gate?

(A) We need only 3 NAND gates	**(B)** We need two NOT gates obtained from NAND gates and one NAND gate
(C) We need 3 NOT gates obtained from NAND gates and 3 NAND gates	**(D)** We need 2 NAND gates and 4 AND gates obtained from NAND gates

Q162. *CUSAT CAT 2024 - Shift 1*

A quasi-static process is

(A) a slow process	**(B)** an infinitely slow process
(C) a fast process	**(D)** an infinitely fast process

Q163. *CUSAT CAT 2024 - Shift 1*

A mass of 1.0 kg is attached to a spring of stiffness constant 16 N/m. Find its natural frequency.

(A) 6.40 Hz	**(B)** 64 Hz
(C) 6400 Hz	**(D)** 0.64 Hz –7

Q164. *CUSAT CAT 2024 - Shift 1*

Calculate the energy of an electron of wavelength 4.35 × 10 m –19

(A) 4.56 × 10 Cal –19	**(B)** 4.56 × 10 J –19
(C) 4.56 × 10 KJ –19	**(D)** 4.56 × 10 Kcal

Q165. *CUSAT CAT 2024 - Shift 1*

When making side-by-side comparisons of glass fragments for color analysis, it is essential that the

(A) Fragments are of equal and sufficient size	**(B)** Fragments are visually identical
(C) Fragments are as small as possible	**(D)** Fragment sizes are not relevant

Q166. *CUSAT CAT 2024 - Shift 1*

A spring of force constant k is stretched a certain distance. It takes twice as much work to stretch a second spring by half this distance. The force constant of the second spring is

(A) k	**(B)** 2k
(C) 4k	**(D)** 8k

Q167. *CUSAT CAT 2024 - Shift 2*

Which one of the following is not a green house gas?

(A) Water vapour	**(B)** Carbon dioxide
(C) Methane	**(D)** Nitrogen

Q168. *CUSAT CAT 2024 - Shift 2*

A closed organ pipe has a frequency ‘n’. If its length is doubled, its frequency nearly becomes

(A) Halved	**(B)** Doubled
(C) Three times	**(D)** Four times

Q169. *CUSAT CAT 2024 - Shift 2*

Two slits are made one millimeter apart and the screen is placed one meter away. What is the fringe separation when the blue-green light of wavelength 500 nm is used?

(A) 0.05 mm	**(B)** 0.01 mm
(C) 0.5 mm	**(D)** 0.1 mm

Q170. *CUSAT CAT 2024 - Shift 2*

What is the ratio of output frequency of a full wave rectifier to half wave rectifier fed with a input signal of frequency 50 Hz?

(A) 1 : 2	**(B)** 1 : 4
(C) 2 : 1	**(D)** 4 : 1

Q171. *CUSAT CAT 2024 - Shift 2*

Which of the following is fundamental quantity?

(A) Velocity	**(B)** Time
(C) Volume	**(D)** Force

Q172. *CUSAT CAT 2024 - Shift 2*

If earth shrinks to half its radius without changing its mass then the length of the day will be

(A) 6 hours	**(B)** 12 hours
(C) 8 hours	**(D)** 24 hours

Q173. *CUSAT CAT 2024 - Shift 2*

$\sigma 1$ and $\sigma 2$ are the electrical conductivities of Ge and Na respectively. If these substances are heated then

(A) $\sigma 1$ and $\sigma 2$ increases	**(B)** $\sigma 1$ increases and $\sigma 2$ decreases
(C) $\sigma 1$ decreases and $\sigma 2$ increases	**(D)** $\sigma 1$ and $\sigma 2$ decreases

Q174. *CUSAT CAT 2024 - Shift 2*

Which one of the following phenomena is not explained by Huygen's construction of wavefront?

(A) Refraction	**(B)** Reflection
(C) Diffraction	**(D)** Origin of spectra

Q175. *CUSAT CAT 2024 - Shift 2*

If the angle between two nonzero vectors A and B is 120°, its resultant C

(A) must be equal to A–B	**(B)** must be greater than A–B
(C) may be equal to A–B	**(D)** must be less than A–B

Q176. *CUSAT CAT 2024 - Shift 2*

Two mercury drops of same radius merge to form a bigger drop. Then the released surface energy is 2

(A) 1.65 πr S 2	**(B)** 3.3 πr S 3
(C) 2.56 πr S 2	**(D)** 2.56 πr S

Q177. *CUSAT CAT 2024 - Shift 2*

Two radioactive materials A and B have decay constants 10λ and λ, respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of 1 A to that of B will be after a time e 1

(A) 9λ 11	**(B)** 10λ 1
(C) 10λ 1	**(D)** 11λ

Q178. *CUSAT CAT 2024 - Shift 2*

Two rods of same length and area of cross section and Young's modulii Y and Y are 1 2 joined end to end. The equivalent Young's modulus for the composite rod is Y +Y

(A) 1 2 2 Y +Y	**(B)** 1 2 YY 1 2 2YY
(C) 1 2 Y +Y 1 2	**(D)** Y +Y 1 2

Q179. *CUSAT CAT 2024 - Shift 2*

A prism can produce a minimum deviation δ in a light beam. If three such prisms are combined, the minimum deviation that can be produced in this beam is

(A) 0 δ	**(B)** 3
(C) δ	**(D)** 3δ

Q180. *CUSAT CAT 2024 - Shift 2*

Light year is used to measure the

(A) Time	**(B)** Distance
(C) Energy	**(D)** Mass

Q181. *CUSAT CAT 2024 - Shift 2*

Convection does not take place in

(A) Boiling of water	**(B)** Sea and land breeze
(C) Heating air around a furnace	**(D)** Warming of glass of the bulb due to filament

Q182. *CUSAT CAT 2024 - Shift 2*

A stone tied to a string is rotated in a circle with the other end of the string as the centre. If the string breaks the stone will

(A) move towards the centre	**(B)** move away from the centre
(C) move along the tangent	**(D)** stop

Q183. *CUSAT CAT 2024 - Shift 2*

If the diameter of the objective of a telescope is doubled, the intensity of the image is increased to

(A) Two times	**(B)** Four times
(C) Eight times	**(D)** Sixteen times

Q184. *CUSAT CAT 2024 - Shift 2*

The time taken for a radio signal to travel 150 km from the transmitter to receiving antenna is –4

(A) 5 × 10 s –3	**(B)** 4.5 × 10 s –8
(C) 5 × 10 s –6	**(D)** 4.5 × 10 s

Q185. *CUSAT CAT 2024 - Shift 2*

What is the number of molecules in 2.24 L of SO at STP? 2

(A) 6.023×10 23	**(B)** 6.023×10 22
(C) 6.023×10 20	**(D)** 6.023×10 2

Q186. *CUSAT CAT 2024 - Shift 2*

In the common emitter amplifier, the phase difference between the input voltage and output voltage signal across the collector and emitter is

(A) 0 π	**(B)** 2
(C) π π	**(D)** 4

Q187. *CUSAT CAT 2024 - Shift 2*

A rocket of mass 1000 kg is initially at rest in space. The rocket engines are turned on, and a constant thrust of 5000 N is exerted for 20 seconds. What is the final velocity of the rocket?

(A) 50 m/s	**(B)** 100 m/s
(C) 150 m/s	**(D)** 200 m/s

Q188. *CUSAT CAT 2024 - Shift 2*

What is the speed of a wave with a frequency of 50 Hz and a wavelength of 2 m?

(A) 25 m/s	**(B)** 50 m/s
(C) 100 m/s	**(D)** 200 m/s

Q189. *CUSAT CAT 2024 - Shift 2*

Inversion temperature of a gas is the temperature above which

(A) the gas shows cooling effect while passing through a narrow orifice	**(B)** the temperature of the gas remains unaltered while passing through narrow orifice
(C) the gas shows heating effect while passing through a narrow orifice	**(D)** the gas can be liquefied CHEMISTRY UG (SHIFT II - FINAL)

Q190. *CUSAT CAT 2024 - Shift 3*

The figure below shows the displacement-time graph of a moving particle on the x-axis. Then which one of the following is CORRECT?

(A) The particle is continuously going in the positive x-direction	**(B)** The particle is at rest
(C) The velocity increases up to a time t and then becomes constant o	**(D)** The particle moves at a constant velocity up to a time t and then stops o

Q191. *CUSAT CAT 2024 - Shift 3*

According to Bohr principle, the relation between principal quantum number n and the Radius (r) of the orbit is 1

(A) $r \propto n$	**(B)** $r \propto n$
(C) $r \propto n$ 2 1	**(D)** $r \propto 2n$

Q192. *CUSAT CAT 2024 - Shift 3*

An α – particle with a velocity v bombards a heavy stationary nucleus. Then the distance of closest approach depends on the velocity of the α – particle as 1

(A) v 1	**(B)** v2
(C) v2	**(D)** v

Q193. *CUSAT CAT 2024 - Shift 3*

A thunder clap is heard 5.5 seconds later the lightning flash. The distance of the flash is (Given velocity of sound = 330 m/s)

(A) 1760 m	**(B)** 1540 m
(C) 1870 m	**(D)** 1815 m

Q194. *CUSAT CAT 2024 - Shift 3*

Which one of the following is not an optical defect?

(A) Optical rotation	**(B)** Chromatic aberration
(C) Coma	**(D)** Astigmatism

Q195. *CUSAT CAT 2024 - Shift 3*

Velocity – time curve for a body projected vertically upwards is

(A) Parabola	**(B)** Hyperbola
(C) Straight line	**(D)** Ellipse

Q196. *CUSAT CAT 2024 - Shift 3*

The velocity of light is maximum in

(A) Glass	**(B)** Vacuum
(C) Diamond	**(D)** Water

Q197. *CUSAT CAT 2024 - Shift 3*

A metal cube is placed in empty vessel. When the water is filled in the vessel so that cube is completely immersed in water, the force on the bottom of the vessel in contact with cube

(A) will increase	**(B)** will decrease
(C) will remain same	**(D)** will become zero

Q198. *CUSAT CAT 2024 - Shift 3*

The main difference between the electric lines of force and magnetic lines of force is

(A) the electric lines of force are closed curves whereas magnetic lines of force are not	**(B)** the electric lines of force are in the form of open curves whereas the magnetic lines are closed curves
(C) the electric lines of force tend to contract lengthwise whereas the magnetic lines of force do not	**(D)** the electric lines of force are continuous and magnetic lines of force are discrete

Q199. *CUSAT CAT 2024 - Shift 3*

The electron configuration for the noble gas atom is 2 2 6

(A) 1s 2s 2p 2 2 5	**(B)** 1s 2s 2p 2 2 6 2 6 2
(C) 1s 2s 2p 3s 3p 4p 2 2	**(D)** 1s 2s

Q200. *CUSAT CAT 2024 - Shift 3*

The truth table given below is for …….. gate A B C 0 0 1 0 1 1 1 0 1 1 1 0

(A) XOR	**(B)** OR
(C) AND	**(D)** NAND

Q201. *CUSAT CAT 2024 - Shift 3*

A wire can sustain the weight of 20 kg before breaking. If the wire is cut into two equal parts, each part can sustain weight of

(A) 10 kg	**(B)** 5 kg
(C) 20 kg	**(D)** 40 kg

Q202. *CUSAT CAT 2024 - Shift 3*

A 20 N metal block is suspended by a spring balance. A beaker containing some water is placed on a weighing machine which reads 40 N. The spring balance is now lowered so that the block gets immersed in the water. The spring balance now reads

(A) 60 N	**(B)** 44 N
(C) 36 N	**(D)** 20 N

Q203. *CUSAT CAT 2024 - Shift 3*

The electric charge is quantized. This was established by

(A) J. J. Thomson	**(B)** William Crookes
(C) Wilhelm Roentgen	**(D)** R.A. Millikan

Q204. *CUSAT CAT 2024 - Shift 3*

An ice cube is suspended in vacuum in a gravity free hall. As the ice melts it

(A) will retain its cubical shape	**(B)** will change its shape to spherical
(C) will fall down on the floor of the hall	**(D)** will fly up

Q205. *CUSAT CAT 2024 - Shift 3*

Distance -Time graph of accelerated motion is a

(A) Straight line	**(B)** Parabola
(C) Ellipse	**(D)** Hyperbola

Q206. *CUSAT CAT 2024 - Shift 3*

The measured weight of a person sitting in a lift accelerating upwards is

(A) less than actual weight	**(B)** equal to actual weight
(C) more than actual weight	**(D)** zero

Q207. *CUSAT CAT 2024 - Shift 3*

A 1 kΩ resistor has 20 mA of current flowing through it. The applied voltage is then

(A) 0.05 V	**(B)** 1 V
(C) 15 V	**(D)** 20 V

Q208. *CUSAT CAT 2024 - Shift 3*

Order of magnitude of magnetic field on the earth's surface is 8

(A) 10 Tesla	**(B)** 1 Tesla –2
(C) 10 Tesla	**(D)** 10 –5 Tesla

Q209. *CUSAT CAT 2024 - Shift 3*

What is the relation between the internal energy and heat supplied in the process 1 and 2 shown in the diagram? Both paths start at A and end at B.

(A) U > U , Q > Q 1 2 1 2	**(B)** U < U , Q > Q 1 2 1 2
(C) U =U , Q = Q 1 2 1 2	**(D)** U = U , Q > Q 1 2 2 2

Q210. *CUSAT CAT 2024 - Shift 3*

Which one of the following is a scalar quantity?

(A) Velocity	**(B)** Acceleration
(C) Distance	**(D)** Force

Q211. *CUSAT CAT 2024 - Shift 3*

In which one of the following states of matter do the particles possess the highest kinetic energy?

(A) Gas	**(B)** Liquid
(C) Solid	**(D)** Plasma

Q212. *CUSAT CAT 2024 - Shift 3*

An electron microscope is used to look at an atom of 0.1 nm diameter. If the desired resolution is 0.005 nm, the minimum energy of the electron should be

(A) 0.957 eV 4	**(B)** 6 × 10 eV 7
(C) 1 × 10 eV	**(D)** 1.24 keV CHEMISTRY UG (SHIFT III - FINAL)

Q213. *CUSAT CAT 2024 - Shift 4*

Vectors A and B have same magnitude. In addition, the magnitude of their resultant is also equal to the magnitude of either of them. Then A and B are at an angle

(A) 120o	**(B)** 60o
(C) 90o	**(D)** 45o

Q214. *CUSAT CAT 2024 - Shift 4*

A stone is released from an elevator going up with acceleration a. The acceleration of the stone after the release is

(A) a upwards	**(B)** (g – a) upwards
(C) (g – a) downwards	**(D)** g downwards

Q215. *CUSAT CAT 2024 - Shift 4*

A jar has a mixture of Hydrogen and Oxygen gases in the ratio 1 : 5. The ratio of mean kinetic energies of the molecules of the Hydrogen and Oxygen gas is

(A) 1 : 16	**(B)** 5 : 1
(C) 1 : 1	**(D)** 1 : 5

Q216. *CUSAT CAT 2024 - Shift 4*

A fraction f ooofff aaa rrraaadddiiiooooaaaccctttiiivvveee sssaaammmppplllleee dddeeeccccaaayyysss iiinnn ooonnneee mmmeeeaaannn llliiifffeee aaannnddd aaa fffrrrraaaccctttiiiooonnn f decays 1 2 in one half life. Then

(A) f > f 1 2	**(B)** f < f 1 2
(C) f = f 1 2	**(D)** rreellaattiivvee ffrraaccttiioonn ooff f and f ddeeppeennddss oonn iinniittiiaall aammoouunntt ooff tthhee ssaammppllee 1 2

Q217.

CUSAT CAT 2024 - Shift 4

FFoorr sstteeaaddyy ffllooww ooff aa ffllluuiidd,, tthhe Raynold's number must be

(A) less than 2000	**(B)** ggrreeaatteerr tthhaann 22000000 bbuutt lleessss tthhaann 33000000
(C) negative	**(D)** ggrreeaatteerr tthhaann 33000000

Q218.

CUSAT CAT 2024 - Shift 4

TTThhheee fffiiiiggguuurrreee bbbeeelllooowww ssshhhooowwwsss aaa ssspppphhheeerrriiicccaaalll GGGaaauuussssssiiiaaannn sssuuurrrfffaaaccceee aaannnddd aaa ccchhhaaarrrgggeee dddiiissstttrrriiibbbuuutttiiioonnn... TTThhheee eelleeccttrriicc ffiieelldd aatt aannyy ppooiinntt oovveerr tthhee ssuurrface will be due to the charge

(A) + q alone 3	**(B)** +q and +q 1 3
(C) +q , +q and –qq 1 3 2	**(D)** +q and –q 1 2

Q219.

CUSAT CAT 2024 - Shift 4

If E and B rrreeepppprrreeessseeennnttt ttthhheee eeellleeecccttttrrriiiccc aaannnddd mmmaaagggnnneeetttiiiccc fffiiieeellldddd cccooommmpppooonnneeennnttt ooofff ttthhheee eeellleeecccttttrrrooommmaaagggnnneeetttiiiccc wwaavveess,, tthheenn tthhee ddiirreeccttiioonn ooff pprrooppaaggaattiioonn ooff tthe wave will be

(A) parallel to E ddiirreeccttiioonn	**(B)** parallel to B ddiirreeccttiioonn
(C) in aa ddiirreeccttiioonn ggiivveenn bbyy E × B	**(D)** in aa ddiirreeccttiioonn ggiivveenn bbyy B × E

Q220.

CUSAT CAT 2024 - Shift 4

If the particles listed below all have same kinetic energy, which one of them have the shortest de-Broglie wavelength?

(A) Electron	**(B)** Proton
(C) Deuteron	**(D)** α-particle

Q221.

CUSAT CAT 2024 - Shift 4

Three stars A, B and C have temperatures T , T and T A appears bluish, B appears A B C. reddish and C appears yellowish. Then we can conclude that

(A) T > T > T A C B	**(B)** T > T > T A B C
(C) T > T > T B C A	**(D)** T > T > T A B A

Q222.

CUSAT CAT 2024 - Shift 4

An AND gate with inverters on each of its input is equivalent to

(A) NAND gate	**(B)** NOR gate
(C) XOR gate	**(D)** OR gate

Q223. *CUSAT CAT 2024 - Shift 4*

For which of the following does the centre of mass lie outside the body?

(A) A pencil	**(B)** A shot put
(C) A dice	**(D)** A bangle

Q224. *CUSAT CAT 2024 - Shift 4*

At what rate the earth should rotate so that the weight of a body at the equator of the earth becomes zero?

(A) 17 time the present speed	**(B)** 8 times the present speed
(C) 170 times the present speed	**(D)** 80 times the present speed

Q225. *CUSAT CAT 2024 - Shift 4*

For hydrogen gas Cp – C = a and for oxygen gas Cp – C = b, so the relation v v between a and b is given by

(A) a = 16b	**(B)** 4a = b
(C) a = 4b	**(D)** a = b

Q226. *CUSAT CAT 2024 - Shift 4*

A particle is moving with speed v=b x along positive x axis. If the particle is at origin when t = 0 sec, the speed of the particle at t = τ is

(A) b 2 τ 2 b τ	**(B)** 2 2 b τ
(C) 2 2 b τ	**(D)** 4

Q227. *CUSAT CAT 2024 - Shift 4*

An astronomical telescope having an objective of focal length 150 cm and eyepiece of focal length 5 cm, is focused to see an object 10 km from the objective. If the final image is formed at infinity, the angular magnification produced by the telescope is

(A) –30	**(B)** 750
(C) –0.033	**(D)** –750

Q228. *CUSAT CAT 2024 - Shift 4*

A transmitting antenna of height 20 m and the receiving antenna of height h are separated by a distance of 40 km for satisfactory communication in line-of-sight mode. If the radius of earth is 6400 km, the value of h is

(A) 40 m	**(B)** 45 m
(C) 30 m	**(D)** 25 m

Q229. *CUSAT CAT 2024 - Shift 4*

To double the covering range of a TV transmission tower, its height should be multiplied by 1

(A) 2	**(B)** 2
(C) 4	**(D)** 2

Q230. *CUSAT CAT 2024 - Shift 4*

In the combination of the following gates, the output Y can be written in terms of inputs A and B as

(A) A.B	**(B)** A.B+ A.B
(C) A.B+ A.B	**(D)** A+B

Q231. *CUSAT CAT 2024 - Shift 4*

The correct plot of the magnitude of magnetic field Bvs distance r from the centre of the wire, if the radius of the wire is R B

(A) 0 R r B	**(B)** 0 R r B
(C) 0 R r B	**(D)** 0 R r

Q232. *CUSAT CAT 2024 - Shift 4*

An oscillator is nothing but an amplifier with

(A) larger gain	**(B)** no feed back
(C) positive feed back	**(D)** negative feed back

Q233. *CUSAT CAT 2024 - Shift 4*

A free neutron decays to a proton but a free proton does not decay to a neutron. This is because

(A) Neutron is a composite particle made of a proton and an electron where as proton is a fundamental particle	**(B)** Neutron is a uncharged particle where as proton is a charged particle
(C) Neutron has a larger rest mass than the proton	**(D)** Weak forces can operate in a neutron but not in a proton

Q234. *CUSAT CAT 2024 - Shift 4*

The motion of a car on a straight road can be called as

(A) Rectilinear	**(B)** Circular
(C) Periodic	**(D)** Harmonic

Q235. *CUSAT CAT 2024 - Shift 4*

Beta rays emitted by radioactive material are

(A) neutral particles	**(B)** electrons orbiting around the nucleus
(C) electromagnetic radiation	**(D)** charged particles emitted by the nucleus

Q236. *CUSAT CAT 2024 - Shift 4*

In a common base amplifier the phase difference between input and output voltage is

(A) 45°	**(B)** 0
(C) 90°	**(D)** 180°

Q237.

CUSAT CAT 2024 - Shift 4

A monkey is sitting at a spring balance that reads 60 kilograms. What effect will the monkey having jumped off the balance have on the reading?

(A) Decrease	**(B)** Increase
(C) No change in reading	**(D)** First increase and then becomes zero

Q238.

CUSAT CAT 2024 - Shift 4

TThhrreeee oobbjjeeccttss ooff mmaassss 11 kkgg eeaacchh aarree ppllaacceedd aatt ccoorrnneerrss ooff aa rriigghhtt-aanngglleedd ttrriiaanngglllee AAOOBB as ssshhhhooowwwnnn iiinnn ttthhheee fffiiiggguuurrreee... CCCooonnnsssiiiidddeeerrr ‘‘‘OOO’’’ aaasss ttthhheee ooorrriiiggiiinnn aaannnddd OOOAAA === OOOBBB === 111 mmm... DDDeeettteeerrrmmmiiinnneee ttthhheee ppoossiittiioonn ooff tthhee cceennttrree ooff mmaassss ooff tthhiiss ssyysstteemm 1

(A) (i – j) m 3 2	**(B)** (i + j) m 3 1
(C) (i + j) m 3	**(D)** (i – j) m

Q239.

CUSAT CAT 2024 - Shift 4

IIInnn aaa ttthhheeerrrmmmoooddyyynnnaaammmiiiccc ppprrrooocccceeessssss,,, ppprrreeessssssuuurrreee ooofff aaa fffiiixxxeeeddd mmmaaassssss ooofff ttthhheee gggaaasss iiisss ccchhhaaannnggeeeddd iiinnn sssuuuccchhh aaa mmmaaannnnnneeerrr ttthhhaaattt ttthhheee gggaaasss mmmooollleeecccuuullleeesss gggiiivvveee ooouuuttt 333000 JJJ ooofff hhheeeaaattt... AAAlllsssooo,,, 111000 JJJ ooofff wwwooorrrkkk iiisss dddooonnneee ooonnn ttthhheee ggaass.. IIff tthhee iinntteerrnnaall eenneerrggyy ooff tthhee ggaass wwaass 4400 JJ,, wwhhaatt wwiillll bbee tthhee ffiinnaall iinntteerrnnaall eenneerrggyy ??

(A) 20 J	**(B)** –20 J
(C) 80 J	**(D)** zero

Q240.

CUSAT CAT 2024 - Shift 4

TTThhheee fffiiiggguuurrreee bbbeeelllooowww ssshhhooowwwsss ttthhheee eeellleeeccctttrrriiiccc fffiiieeellldddd llliiinnneeesss ddduuueee tttooo tttwwwooo pppooosssiiittttiiivvveee ccchhhaaarrrgggeeesss... TTThhheee magnitudes E , E aanndd EE ooff tthhee eelleeccttrriicc ffiieelldss aatt ppooiinntt AA,, BB aanndd CC rreessppeeccttiivveellyy aarree A B C related as

(A) E > E > E A B C	**(B)** E > E > E B A C
(C) E = E > E A C B	**(D)** E > E = E A B C

Q241.

CUSAT CAT 2024 - Shift 4

A particle is initially moving with a velocity of 10 m/s. If it undergoes a constant 2 acceleration of 2 m/s , what is its velocity after 4 seconds?

(A) 12 m/s	**(B)** 14 m/s
(C) 16 m/s	**(D)** 18 m/s

Q242.

CUSAT CAT 2024 - Shift 4

A force of 20 N is applied to an object of mass 5 kg. What is the acceleration of the object? 2

(A) 2 m/s 2	**(B)** 4 m/s 2
(C) 10 m/s 2	**(D)** 100 m/s

Q243.

CUSAT CAT 2024 - Shift 4

Ideeennnttttiiiffffyyy ttthhheee cccooorrrrrreeeccccttt eeeqqquuuaaatttiiiooonnn bbbaaassseeeddd ooonnn ttthhheee ttthhhrrreeeeee eeeqqquuuaaatttiiiooonnnsss ooofff mmmoooottttiiiooonnn... 2 ut at

(A) s = + 2 2 ut 2	**(B)** s = +at 2 2 2ut at
(C) s = + 2 2 2 at	**(D)** s =2ut+ 2

Q244.

CUSAT CAT 2024 - Shift 4

TThhee pprrooppeerrttyy ooff bbrreeaakkiinngg ooff mmaatteerriiaall wwiitthh lliittttllee ppeerrmmaanneenntt ddiissttoorrttiioonn iiss ccaalllleedd

(A) Ductility	**(B)** Malleability
(C) Rigidity	**(D)** Brittleness

Q245.

CUSAT CAT 2024 - Shift 4

WWhhiicchh oonnee ooff tthhee ffoollllloowwiinngg statements hhoollddss ggoooodd wwhheenn tthhee lliigghhtt iiss rreeffrraacctteedd into a medium?

(A) BBootthh ffrreeqquueennccyy aanndd wwaavveelleennggtthh ooff tthhee lliigghhtt iinnccrreeaassee	**(B)** Both waveelleennggtthh aanndd ffrreeqquueennccyy ddeeccrreeaassee
(C) TTThhheee wwwaaavvveeellleeennnggggtttthhh dddeeecccrrreeeaaassseeesss bbbuuuttt ttthhheee fffrrreeeqqquuueeennnccccyyy rrreeemmmaaaiiinnnsss cccooonnnsssttttaaannnttt	**(D)** TTThhheee wwwaaavvveeellleeennnggggtttthhh iiinnnccccrrreeeaaassseeesss bbbuuuttt ttthhheee fffrrreeeqqquuueeennnccccyyy rrreeemmmaaaiiinnnsss uuunnnccchhhaaannnggggeeeddd

Q246.

CUSAT CAT 2024 - Shift 4

CCCaaalllcccuuulllaaattteee ttthhheee cccuuurrrrrreeennnttt fffllllooowww ttthhhrrrooouuuggghhh ttthhheee ZZZeeennneeerrr dddiiiooodddeee bbbaaassseeeddd ooonnn ttthhheee ccciiirrrcccuuuiiittt bbbeeellllooowww

(A) 6 mA	**(B)** 0 mA
(C) 7 mA	**(D)** 5.3 mA

Q247.

CUSAT CAT 2024 - Shift 4

For turbulent flow, the velocity at the centre is …… times the mean velocity.

(A) 1.2	**(B)** 2.2
(C) 2.0	**(D)** 1.5

Q248. *CUSAT CAT 2024 - Shift 4*

Negative feedback in an amplifier

(A) reduces the voltage gain	**(B)** increases the voltage gain
(C) does not affect the voltage gain	**(D)** converts the amplifier into an oscillator

Q249. *CUSAT CAT 2024 - Shift 4*

The colour of light that travels with maximum speed in glass is

(A) red	**(B)** blue
(C) violet	**(D)** green CHEMISTRY UG (SHIFT IV - FINAL)

Q250. *CUSAT CAT 2024 - Shift 5*

When the source move away from the observer, the frequency as measured by the observer will be

(A) Larger	**(B)** Smaller
(C) Infinite	**(D)** Zero

Q251. *CUSAT CAT 2024 - Shift 5*

What will be the energy of the photon of red color compared with that of violet color?

(A) Equal	**(B)** Larger
(C) Smaller	**(D)** Cannot be compared

Q252. *CUSAT CAT 2024 - Shift 5*

When a system is at equilibrium, the free energy of the reaction is

(A) positive	**(B)** negative
(C) zero	**(D)** can be negative or positive

Q253. *CUSAT CAT 2024 - Shift 5*

If × = × , then the angle between A and B is

(A) π π	**(B)** 3 π
(C) 2 π	**(D)** 4

Q254. *CUSAT CAT 2024 - Shift 5*

The lines of forces due to the Earth's horizontal magnetic field are

(A) elliptical	**(B)** curved lines
(C) parallel and straight	**(D)** concentric circles

Q255. *CUSAT CAT 2024 - Shift 5*

de Broglie proposed that electron has

(A) dual characters of wave and particle nature	**(B)** only particle nature
(C) only wave nature	**(D)** either wave or particle nature

Q256. *CUSAT CAT 2024 - Shift 5*

The Speed of Light in Vacuum is μ

(A) 0 ε 0	**(B)** με 0 0 1
(C) με 0 0 1	**(D)** με 0 0

Q257. *CUSAT CAT 2024 - Shift 5*

A student sets a source at a distance 10 cm away from a double slit which are separated by 5 cm apart and keeps the micrometer eye piece at 60 cm from the double slit φ, but unable to observe the interference fringes. You must advice him to

(A) increase the distance of the source	**(B)** decrease the distance between the double slit
(C) replace the light with white light source	**(D)** replace the micrometer eyepiece with telescope

Q258. *CUSAT CAT 2024 - Shift 5*

The molecular weights of oxygen and hydrogen are 32 and 2 respectively. The root mean square velocities of oxygen and hydrogen at NTP are in the ratio

(A) 4 : 1	**(B)** 1 : 16
(C) 16 : 1	**(D)** 1 : 4

Q259. *CUSAT CAT 2024 - Shift 5*

Magnetic domain formation is the necessary feature of

(A) diamagnetism	**(B)** ferromagnetism
(C) paramagnetism	**(D)** superconductivity

Q260. *CUSAT CAT 2024 - Shift 5*

Neutrino is

(A) chargeless, fermion of negligible mass	**(B)** massless, chargeless 1
(C) massless, fermion of charge e 3	**(D)** chargeless, spinless boson

Q261. *CUSAT CAT 2024 - Shift 5*

Which law states that the pressure exerted by a gaseous mixture is equal to the sum of partial pressure of each component present in the mixture?

(A) Graham's law	**(B)** Boyle's law
(C) Dalton's law	**(D)** Charles' law

Q262. *CUSAT CAT 2024 - Shift 5*

If the intensity of sound is increased three times of its initial intensity, by how many decibels does the sound level increase?

(A) Remains same	**(B)** Tripled
(C) 4.77 dB	**(D)** 6 dB

Q263. *CUSAT CAT 2024 - Shift 5*

An athlete completes one round of a circular track of radius R in 20 seconds. What will be his displacement at the end of 2 minutes 20 seconds?

(A) Zero	**(B)** 2R
(C) 2πR	**(D)** 7πR

Q264. *CUSAT CAT 2024 - Shift 5*

The position of a particle varies with time t as x = pt – qt . The acceleration of the particle will be 2

(A) 2pt–3qt	**(B)** p /q
(C) 2p–6qt	**(D)** zero

Q265. *CUSAT CAT 2024 - Shift 5*

A 600 W electric heater is designed to operate from 240 V lines. Then it would draw a current of

(A) 0.4 A	**(B)** 2.5 A
(C) 5 A	**(D)** 15 A

Q266. *CUSAT CAT 2024 - Shift 5*

Bonding in Germanium crystal is

(A) Ionic	**(B)** van der Waals' type
(C) Metallic	**(D)** Covalent 235

Q267. *CUSAT CAT 2024 - Shift 5*

The percentage of U present in natural Uranium is

(A) 100%	**(B)** 0.714%
(C) 99.28%	**(D)** 0.006%

Q268. *CUSAT CAT 2024 - Shift 5*

The image formed by the objective of a compound microscope is

(A) virtual and enlarged	**(B)** virtual and diminished
(C) real and diminished	**(D)** real and enlarged

Q269. *CUSAT CAT 2024 - Shift 5*

The decimal equivalent of the binary number 1111 is

(A) 15	**(B)** 4
(C) 14	**(D)** 1

Q270. *CUSAT CAT 2024 - Shift 5*

In Young's double slit experiment the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth fringe is measured to be 1.2 cm. Determine the wavelength of light used in this experiment.

(A) 650 nm	**(B)** 400 nm
(C) 500 nm	**(D)** 600 nm

Q271. *CUSAT CAT 2024 - Shift 5*

A U-shaped tube has areas of cross-sections 2 cm 2 and 20 cm 2 at its two ends. It is filled with water, and a force of 10 N is applied in the thinner arm using a piston. For the water to remain in equilibrium the force to be applied in the thicker arm is

(A) 20 N	**(B)** 40 N
(C) 100 N	**(D)** 200 N

Q272. *CUSAT CAT 2024 - Shift 5*

An ambulance moves away from an observer at a speed of 20.00 m/s. The siren in the ambulance is detected by the observer and is found to be at a frequency of 2200.00 Hz. Considering the observer to be immobile and speed of sound to be 320.00 m/s, what is the original frequency of the siren?

(A) 2200.00 Hz	**(B)** 2337.44 Hz
(C) 2557.35 Hz	**(D)** 3201.23 Hz

Q273. *CUSAT CAT 2024 - Shift 5*

The beam from a laser source is incident on a double slit with the distance between them being 430 micron. A screen is placed at a distance of 2.00 m from the double slit to observe the fringes. The first bright fringe is observed at 2.2 mm from the central fringe. The wavelength of the laser source is

(A) 355.05 nm	**(B)** 423.24 nm
(C) 468.09 nm	**(D)** 532.01 nm

Q274. *CUSAT CAT 2024 - Shift 5*

A sealed glass container has a volume of 3 liters. It contains 0.5 mole of oxygen and 0.3 mole of nitrogen. If the contents of the container are at 300 K, then the pressure in the chamber is 5

(A) 2.50 × 10 Pa 5	**(B)** 4.16 × 10 Pa 5
(C) 6.65 × 10 Pa 5	**(D)** 8.32 × 10 Pa –5

Q275. *CUSAT CAT 2025 - Shift 1*

If the mass of a gas molecule be m, then the root mean square speed of the gas molecule at temperature T will be 3kT

(A) m 2kT	**(B)** m 8kT
(C) mπ m	**(D)** 3kT

Q276. *CUSAT CAT 2025 - Shift 1*

When a vibrating tuning fork is placed vertically on a table, then the vibrations in the table board will be

(A) free vibrations	**(B)** forced vibrations
(C) resonant vibrations	**(D)** maintained vibrations

Q277. *CUSAT CAT 2025 - Shift 1*

The slope of the time-speed graph gives

(A) velocity of the body	**(B)** distance travelled by the body
(C) acceleration of the body	**(D)** direction of motion

Q278. *CUSAT CAT 2025 - Shift 1*

The value of the current I in the circuit is I A 30Ω 2V 30Ω B 30Ω C

(A) 0.1 A	**(B)** 1 A
(C) 2 A	**(D)** 0.3 A

Q279. *CUSAT CAT 2025 - Shift 1*

A charged particle is moving along a magnetic field line. The magnetic force on the particle is

(A) along its velocity	**(B)** opposite to its velocity
(C) perpendicular to its velocity	**(D)** zero

Q280. *CUSAT CAT 2025 - Shift 1*

How much work is required to carry a 6 μC charge from the negative to the positive terminal of a 9 V battery?

(A) 54 × 10–3 J	**(B)** 54 × 10–6 J
(C) 54 × 10–9 J	**(D)** 54 × 10–12 J

Q281. *CUSAT CAT 2025 - Shift 1*

The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is

(A) 2.5%	**(B)** 3.5%
(C) 4.5%	**(D)** 6%

Q282.

CUSAT CAT 2025 - Shift 1

Two points are located at a distance of 10 m and 15 m from the source of oscillation. The period of oscillation is 0.05 s and the velocity of the wave is 300 m/s. What is the phase difference between the oscillations of two points?

(A) π π	**(B)** 6 π
(C) 3 2π	**(D)** 3

Q283.

CUSAT CAT 2025 - Shift 1

TThhee rreessiissttaannccee ooff aa wwiirree iiss R. It is bent at the middle by 180° aanndd bbootthh tthhee eennddss aarree tttwwwiiisssttteeeddd tttoooggeeettthhheeerrr tttooo mmmaaakkkeee aaa ssshhhooorrrtteeerrr wwwiiirrreee... TTThhheeennn ttthhheee rrreeesssiiisssttaaannnccceee ooofff ttthhheee nnneeewww wwwiiirrreee iiisss

(A) 2 R R	**(B)** 2 R
(C) 4 R	**(D)** 8

Q284.

CUSAT CAT 2025 - Shift 1

TThhee mmaaxxiimmuumm ppoowweerr ddiissssiippaatteedd iinn aann eexxtteerrnnaall rreessiissttaannccee R,, wwhheenn ccoonnnneeccteedd ttoo aa cceellll of emf E aanndd iinntteerrnnaall rreessiissttaannccee r will be 2 E

(A) r 2 E	**(B)** 2r 2 E
(C) 3r 2 E	**(D)** 4r

Q285.

CUSAT CAT 2025 - Shift 1

WWWhhhaaattt iiisss ttthhheee mmmaaagggnnneeetttiiiccc eeennneeerrrgggyyy ssstttooorrreeeddd iiinnn ttthhheee cccoooiiilll ooofff ttthhheee ccciiirrrcccuuuiiittt bbbeeelllooowww???

(A) Zero	**(B)** Infinite
(C) 25 J	**(D)** 50 J

Q286.

CUSAT CAT 2025 - Shift 1

A circular coil carrying a certain current produces a magnetic field B at its centre. o The coil is now rewound so as to have 3 turns and the same current is passed through it. The new magnetic field at the centre is

(A) 3B o B o	**(B)** 3
(C) 9B o B o	**(D)** 9

Q287.

CUSAT CAT 2025 - Shift 1

In any fission process, ratio of mass of daughter nucleus to mass of parent nucleus is

(A) less than 1	**(B)** greater than1
(C) equal to 1	**(D)** depends on the mass of parent nucleus

Q288.

CUSAT CAT 2025 - Shift 1

A person standing on the floor of an elevator drops a coin. The coin reaches the floor in time t if the elevator is at rest and in time t if the elevator is moving uniformly. 1 2 Then which of the following option is correct?

(A) t < t or t > t depending upon whether the lift is going up or down 1 2 1 2	**(B)** t < t 1 2
(C) t > t 1 2	**(D)** t = t 1 2

Q289.

CUSAT CAT 2025 - Shift 1

TTThhheee cccooollloorrr cccooodddeee ooofff aaa rrreeesssiiisssttttaaannnccceee iiisss gggiiivvveeennn bbbeeellloooowww... TTThhheee vvvaaallluuueeesss ooofff rrreeesssiiisssttttaaannnccceee aaannnddd tttooollleeerrraaannnccceee respectively, are

(A) 47 kΩ, 10%	**(B)** 4.7 kΩ, 5%
(C) 470 Ω, 5%	**(D)** 470 kΩ, 5%

Q290.

CUSAT CAT 2025 - Shift 1

LLiigghhtt wwiittthh aann aavveerraaggee fflluuxx ooff 2200 WW//ccmm2 falls on a non-rreefflleeccttiinngg ssuurrffaaccee aatt nnoorrmmaall iinnccciiddeennccee hhaavviinngg ssuurrffaaccee aarreeaa 2200 ccmm2.. TThhee eenneerrggyy rreecceeiivveedd bbyy tthhee ssuurrffaaccee dduurriinngg ttiimmee span of 1 minutes is

(A) 16 × 103 J	**(B)** 24 × 103 J
(C) 48 × 103 J	**(D)** 12 × 103 J

Q291.

CUSAT CAT 2025 - Shift 1

OOOuuuttt ooofff tttthhheee fffooollllllooowwwiiinnnggg ooopppttttiiiooonnnsss wwwhhhiiiccchhh ooonnneee cccaaannn bbbeee uuussseeeddd tttooo pppprrrooodddduuuccceee aaa pppprrrooopppaaagggaaatttiiinnnggg eelleeccttrroommaaggnneettiicc wwaavvee??

(A) AA ssttaattiioonnaarryy cchhaarrggee	**(B)** AA cchhaarrggeelleessss ppaarrttiiccllee
(C) AAnn aaccccceelleerraattiinngg cchhaarrggee	**(D)** AA cchhaarrggee mmoovviinngg aatt ccoonnssttaanntt vveelloocciittyy

Q292.

CUSAT CAT 2025 - Shift 1

FFoorr wwhhiicchh oonnee ooff tthhee ffoollllloowwiinngg,, BBoohhrr mmooddeell iiss nnoott vvaalliidd??

(A) SSiinnggllyy iioonniisseedd hheelliiuumm aattoomm ((HHee+)	**(B)** Deuteron atom
(C) SSiinnggllyy iioonniisseedd nneeoonn aattoomm ((NNee+)	**(D)** HHyyddrrooggeenn aattoomm

Q293.

CUSAT CAT 2025 - Shift 1

A bicycle starts from rest and accelerates at 1.5 m/s2 for 4 seconds. Find the total distance covered by the bicycle.

(A) 6 m	**(B)** 8 m
(C) 10 m	**(D)** 12 m

Q294. *CUSAT CAT 2025 - Shift 1*

A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength 980 Å. The radius of the atom in the excited state, in terms of Bohr radius a will be (h = 12500 eV-Å) 0, c

(A) 9 a 0	**(B)** 25 a 0
(C) 4 a 0	**(D)** 16 a 0

Q295. *CUSAT CAT 2025 - Shift 1*

Which one of the following will be the output of the given circuit ?

(A) NOR Gate	**(B)** NAND Gate
(C) AND Gate	**(D)** XOR Gate

Q296. *CUSAT CAT 2025 - Shift 1*

Ratio of thermal energy released in two resistor R and 3R connected in parallel in an electric circuit is

(A) 3 : 1	**(B)** 1 : 1
(C) 1 : 3	**(D)** 1 : 27

Q297. *CUSAT CAT 2025 - Shift 1*

A hollow spherical shell of outer radius R floats just submerged under the water surface. The inner radius of the shell is r. If the specific gravity of the shell material is 27 with respect to water, the value of r is 8 4

(A) R 9 8	**(B)** R 9 1
(C) R 3 2	**(D)** R 3 π

Q298. *CUSAT CAT 2025 - Shift 1*

In the wave equation, Y = 0.5 sin (400t – x) the velocity of wave will be

(A) 200 m/s	**(B)** 200 2 m/s
(C) 400 m/s	**(D)** 400 2 m/s

Q299. *CUSAT CAT 2025 - Shift 2*

The radius of the earth is approximately 6000km, the weight of the object 6000km above the surface of the earth is

(A) 4g	**(B)** g g
(C) 2 g	**(D)** 4

Q300. *CUSAT CAT 2025 - Shift 2*

An electric current passes through a long straight wire. At a distance of 5cm from the wire, the magnetic field is B. The magnetic field at 20cm from the straight wire would be B

(A) 6 B	**(B)** 4 B
(C) 3 B	**(D)** 2 C p

Q301. *CUSAT CAT 2025 - Shift 2*

What is the wavelength of a television station which transmits vision on 500 MHz?

(A) 0.6m	**(B)** 0.8m
(C) 1m	**(D)** 1.1m

Q302. *CUSAT CAT 2025 - Shift 2*

A source of sound of frequency 600Hz is placed inside water. The speed of the sound in water is 1500m/sec and in air is 300m/sec. The frequency of sound detected in air is

(A) 200Hz	**(B)** 600Hz
(C) 750Hz	**(D)** 3000Hz

Q303. *CUSAT CAT 2025 - Shift 2*

An ideal gas is filled in a closed rigid and thermally insulated container. A coil of 100Ω resistor carrying current 1A for 5minutes supplies heat to the gas. The change in internal energy of the gas is

(A) 10kJ	**(B)** 20kJ
(C) 30kJ	**(D)** 0 kJ

Q304. *CUSAT CAT 2025 - Shift 2*

A positive charge is moving upward in a magnetic field directed towards north. The particle will be deflected towards

(A) east	**(B)** west
(C) north	**(D)** south

Q305. *CUSAT CAT 2025 - Shift 2*

An electron of mass m and charge e is accelerated from rest through a potential difference V in vacuum. Its final velocity will be

(A) 2	**(B)** —
(C) —	**(D)** 2

Q306. *CUSAT CAT 2025 - Shift 2*

The speed of light in the medium is

(A) maximum on the axis of the beam	**(B)** minimum on the axis of the beam
(C) the same everywhere in the beam	**(D)** directly proportional to the intensity I

Q307. *CUSAT CAT 2025 - Shift 2*

The centripetal force in a circular motion is given by the relation 2 mv

(A) r 3 mv	**(B)** r 3 mv
(C) 3 r mv	**(D)** r

Q308. *CUSAT CAT 2025 - Shift 2*

Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is

(A) 5 m 10	**(B)** m 3 20
(C) m 3	**(D)** 10 m

Q309. *CUSAT CAT 2025 - Shift 2*

An electron enters a higher potential region V from a lower potential region V . Then 2 1 what happens to its velocity?

(A) Its velocity decreases	**(B)** Its velocity increases
(C) There is no change in magnitude of the velocity but its direction changes	**(D)** Its velocity reduces to zero

Q310. *CUSAT CAT 2025 - Shift 2*

In the circuit given below, the value of the current is

(A) Zero	**(B)** 10–2 A
(C) 10–3 A	**(D)** 102 A

Q311. *CUSAT CAT 2025 - Shift 2*

An explosion breaks a rock into three parts in a horizontal plane. Two of them go off at right angles to each other. The first part of mass 1 kg moves with a speed of 12 ms–1 and the second part of mass 2 kg moves with 8 ms–1 speed. If the third part flies off with 4 ms–1 speed, then its mass is

(A) 3 kg	**(B)** 5 kg
(C) 7 kg	**(D)** 17 kg

Q312. *CUSAT CAT 2025 - Shift 2*

A child is swinging a swing. Minimum and maximum heights of swing from the earth's surface are 0.75 m and 2 m respectively. The maximum velocity of this swing is

(A) 5 m/s	**(B)** 10 m/s
(C) 15 m/s	**(D)** 20 m/s

Q313. *CUSAT CAT 2025 - Shift 2*

A magnetic field of 2 × 10–2 T acts at right angles to a coil of area 100 cm2, with 50 turns. The average emf induced in the coil is 0.1 V, when it is removed from the field in t second. The value of t is

(A) 10 s	**(B)** 0.1 s
(C) 0.01 s	**(D)** 1 s

Q314. *CUSAT CAT 2025 - Shift 2*

The Brewster's angle i for an interface should be b

(A) 30° <i < 45° b	**(B)** 45° <i < 90° b
(C) i = 90° b	**(D)** 0° <i < 30° b

Q315. *CUSAT CAT 2025 - Shift 2*

A uniform rod of length 2 m is pivoted at one end. If the rod is in equilibrium under its own weight, what is the distance of its center of mass from the pivot point?

(A) 1.0 m	**(B)** 0.5 m
(C) 0.75 m	**(D)** 0.25 m

Q316. *CUSAT CAT 2025 - Shift 2*

In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of 7.5 × 10–12 m, the minimum electron energy required is close to

(A) 100 KeV	**(B)** 500 KeV
(C) 25 KeV	**(D)** 1 KeV

Q317. *CUSAT CAT 2025 - Shift 2*

An excited He+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n, corresponding to its initial excited state is (for photon of wavelength λ, energy E =). λ ( )

(A) n = 5	**(B)** n = 4
(C) n = 6	**(D)** n = 7

Q318. *CUSAT CAT 2025 - Shift 2*

The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for license what broadcast frequency will you allot ?

(A) 2750 kHz	**(B)** 2000 kHz
(C) 2250 kHz	**(D)** 2900 kHz

Q319. *CUSAT CAT 2025 - Shift 2*

A proton and an αparticle (with their masses in the ratio of 1 : 4 and charges in the ratio of 1: 2) are accelerated from rest through a potential difference V. If a uniform magnetic field

(A) —	**(B)** is set up perpendicular to their velocities, the ratio of their radii r : r of the circular paths described by them will be p α
(C) 1 : 3	**(D)** 1 : √3

Q320. *CUSAT CAT 2025 - Shift 2*

A child of mass 5 kg is going to round a merry-go-round that makes 1 rotation in 3.14 s. The radius of the merry-go-round is 2 m. The centrifugal force on the child will be

(A) 80 N	**(B)** 50 N
(C) 100 N	**(D)** 40 N

Q321.

CUSAT CAT 2025 - Shift 2

TThhee rroooott mmeeaann ssqquuaarree ssppeeeedd ooff mmoolleeccuulleess ooff nniittrrooggeenn ggaass aatt 2277°°CC iiss aappppprrooxxiimmaatteellyy ((ggiivveenn mmaassss ooff aa nniittrrooggeenn mmoolleeccuullee == 44..66××1100–26 kkgg aanndd ttaakkee BBoollttzzmmaannnn ccoonnssttaanntt K = 1.4 × 10–13 JK–1)) B

(A) 523 m/s	**(B)** 1260 m/s
(C) 91 m/s	**(D)** 27.4 m/s

Q322.

CUSAT CAT 2025 - Shift 2

AA ggrraanniittee rroodd ooff 6600 ccmm lleennggtthh iiss ccllaammppeedd aatt iittss mmiiddddllee ppooiinntt aanndd iiss sseett iinnttoo llaattiittuuddiinnaall vviibbrraattiioonnss.. TThhee ddeennssiittyy ooff ggrraanniittee iiss 22..77 ×× 11003 aanndd iittss YYoouunngg''ss mmoodduulluuss iiss 9.27 × 1010 PPPaaa... WWWhhhaaattt wwwiiiiiiiii bbbeee ttthhheee fffuuunnnddddaaammmeeennnttttaaalll fffrrreeeqqquuueeennnccccyyy ooofff ttthhheee lllaaatttiiiittttuuuddddiiinnnaaalll vvviiibbbrrraaatttiiioooonnnsss???

(A) 7.5 kHz	**(B)** 5 kHz
(C) 2.5 kHz	**(D)** 10 kHz

Q323.

CUSAT CAT 2025 - Shift 2

AAA rrroooddd ooofff lleeennnggggtttthhh 555000 cccmmm iiisss pppiiivvvooottteeeddd aaattt ooonnneee eeennnddd... IIIttt iiisss rrraaaiiissseeeddd sssuuuccchhh ttthhhaaattt iiittt mmmaaakkkeeesss aaannn aaannnggllleee ooofff 333000°°° fffrrrooommm ttthhheee hhhooorrriiizzzooonnnttttaaalll aaasss ssshhhooowwwnnn aaannnddd rrreeellleeeaaassseeeddd fffrrrooommm rrreeesssttt... IIIttttsss aaannnggguuulllaaarrr ssspppeeeeeeddd wwwhhheeennn iiittt ppaassssseess tthhrroouugghh tthhee hhoorriizzoonnttaall ((iinn rraadd ss–1) will be (g = 10 m/s–2)

(A) 30 30	**(B)** 2 30
(C) 2 20	**(D)** 2

Q324.

CUSAT CAT 2025 - Shift 2

If the size of an atom (~ 1Å) is enlarged to the size of the earth (~ 107) what would be the size of the nucleus (10–15 m)?

(A) 10 m	**(B)** 100 m
(C) 1000 m	**(D)** 1 m

Q325.

CUSAT CAT 2025 - Shift 3

The distance between the centres of carbon and oxygen atoms in CO molecule is 0.12 nm. The distance of the centre of the mass of the molecule from the carbon atom is

(A) 0.03 nm	**(B)** 0.05 nm
(C) 0.06 nm	**(D)** 0.07 nm

Q326. *CUSAT CAT 2025 - Shift 3*

An ideal gas undergoes a quasi static reversible process in which its molar heat capacity C remains constant. If during this the relation of pressure P and volume V is given by PV n = constant, then n is given by C–C p

(A) n= C–C v C –C p	**(B)** n= C–C v C–C
(C) n= v C–C p C p	**(D)** n= C v

Q327. *CUSAT CAT 2025 - Shift 3*

The displacement equation of a progressive wave is y = 3sin (2000t – 5x), where the distance x and y are in metre and time in second the speed of the wave is

(A) 200 m/sec	**(B)** 400 m/sec
(C) 600 m/sec	**(D)** 800 m/sec

Q328. *CUSAT CAT 2025 - Shift 3*

A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will be

(A) doubled	**(B)** four times
(C) one fourth	**(D)** one third

Q329. *CUSAT CAT 2025 - Shift 3*

A boy standing on a stationary lift (open from above) throws a ball upwards with the maximum initial speed he can, equal to 49 m/sec. How much time does the ball take to return to his hands?

(A) 5 secs	**(B)** 10 secs
(C) 15 secs	**(D)** 30 secs

Q330. *CUSAT CAT 2025 - Shift 3*

The half life of radium is 1600 years. The fraction of a sample of radium that would remain after 6400 years 1

(A) 4 1	**(B)** 2 1
(C) 8 1	**(D)** 16

Q331. *CUSAT CAT 2025 - Shift 3*

The energy of a photon whose wavelength 6840 Å is

(A) 1.81 eV	**(B)** 3.6 eV
(C) 12.1 eV	**(D)** 13.2 eV

Q332. *CUSAT CAT 2025 - Shift 3*

In Thomson's experiment to measure e/m of electrons, the electric and magnetic fields are

(A) in the same direction	**(B)** in the opposite direction
(C) at an angle of 45° with each other	**(D)** perpendicular to each other

Q333. *CUSAT CAT 2025 - Shift 3*

Two thin lenses, one of focal length 60 cm and the other of focal length –20 cm are put in contact, the combined focal length is

(A) 15 cm	**(B)** –15 cm
(C) –30 cm	**(D)** 30 cm

Q334. *CUSAT CAT 2025 - Shift 3*

Within the limit of elasticity, which of the following graphs obey Hooke's law?

(A) —	**(B)** —
(C) —	**(D)** —

Q335. *CUSAT CAT 2025 - Shift 3*

A long, straight wire of radius R carries a current distributed uniformly over its cross- section. The magnitude of the magnetic field is

(A) maximum at the axis of the wire	**(B)** minimum at the axis of the wire
(C) maximum at the surface of the wire	**(D)** minimum at the surface of the wire

Q336. *CUSAT CAT 2025 - Shift 3*

An image of an object is formed by a lens. If half of the lens is covered with black paper, how will the image change?

(A) Size of the image will be reduced to one half	**(B)** Brightness of the image will be reduced
(C) The image will disappear	**(D)** An inverted image will be formed

Q337. *CUSAT CAT 2025 - Shift 3*

In Young's double slit experiment, when two light waves interfere to produce a interference pattern of maxima and minima, at the third minima, the two waves have

(A) phase difference of 3π 5π	**(B)** phase difference of 2
(C) path difference of 3λ 5λ	**(D)** path difference of 2

Q338. *CUSAT CAT 2025 - Shift 3*

To get output 1for the following circuit, the correct choice for the input is

(A) A = 1, B = 0, C = 0	**(B)** A = 1, B = 1, C = 0
(C) A = 0, B = 0, C = 1	**(D)** A = 0, B = 1, C = 1

Q339. *CUSAT CAT 2025 - Shift 3*

The combination of gates as shown in the figure forms the

(A) AND gate	**(B)** OR gate
(C) NOR gate	**(D)** NOT gate

Q340.

CUSAT CAT 2025 - Shift 3

In a carbon monoxide molecule, the carbon and the oxygen atoms are separated by a distance 1.12 × 10–10 m. The distance of the centre of mass from the carbon atom is

(A) 0.64 × 10–10 m	**(B)** 0.56 × 10–10 m
(C) 0.51 × 10–10 m	**(D)** 0.48 × 10–10 m

Q341.

CUSAT CAT 2025 - Shift 3

A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C. Its density is (R = 83 Jmol–1K–1)

(A) 0.2 kg/m3	**(B)** 0.1 kg/m3
(C) 0.02 kg/m3	**(D)** 0.5 kg/m3

Q342.

CUSAT CAT 2025 - Shift 3

Three containers of the same volume contain three different gases. The masses of the molecules are m ,m and m and the number of molecules in their respective 1 2 3 containers are N ,N and N . The gas pressure in the containers are p , p and p 1 2 3 1 2 3 respectively. All the gases are now mixed and put in one of these containers. The pressure p of the mixture will be ( )

(A) p< p + p + p 1 2 3 p + p + p	**(B)** p = 1 2 3 3
(C) p = p + p + p 1 2 3 ( )	**(D)** p > p + p + p 1 2 3

Q343.

CUSAT CAT 2025 - Shift 3

For hydrogen gas C –C =a and for oxygen gas C –C =b, so the relation p v p v between a and b is given by

(A) a = 16b	**(B)** 16b = a
(C) a = 4b	**(D)** a = b

Q344.

CUSAT CAT 2025 - Shift 3

An n-p-n tttrrraaannnsssiiissstttoorrr iiisss cccooonnnnnneeeccctttteeeddd iiinnn cccooommmmmmooonnn eeemmmiiitttttteeerrr cccooonnnfffiiiggguuurrraaatttiiiooonnn (((ssseeeeee fffiiiggguuurrreee))) iiinnn wwhhiicchh ccoollllleeccttoorr vvoollttaaggee ddrroopp aaccrroossss llooaadd rreessiissttaanncccee ((880000 ΩΩ)) ccoonnnneeecctteedd ttoo tthhee ccoollllleeccttoorr cciirrccuuiitt iiss 00..88 VV.. TThhee ccoollllleeccttoorr ccuurrrrent is

(A) 2 mA	**(B)** 0.1 mA
(C) 1 mA	**(D)** 0.2 mA

Q345.

CUSAT CAT 2025 - Shift 3

TTThhheee vvvooollltttaaagggeee gggaaaiiinnn ooofff aaannn aaammmpppllliiiffffiiieeerrr wwwiiittthhh 999%%% nnneeegggaaatttiiivvveee fffeeeeeedddbbbaaaccckkk iiisss 111000... TTThhheee vvvooollltttaaagggeee gggaaaiiinnn wwiitthhoouutt ffeeeeddbbaacckk wwiillll bbee

(A) 90	**(B)** 10
(C) 1.25	**(D)** 100

Q346. *CUSAT CAT 2025 - Shift 3*

TThhee ffiigguurree sshhoowwss aa llooggiicc cciirrccuuiitt wwiitthh ttwwoo iinnpputs A and B aanndd tthhee oouuttppuutt C. The vvoollttaaggee wwaavvee ffoorrmmss aaccrroossss A, B and C aarree aass ggiivveenn.. TThhee llooggiicc cciirrccuuiitt ggaattee iiss

(A) OR gate	**(B)** NOR gate
(C) AND gate	**(D)** NAND gate

Q347. *CUSAT CAT 2025 - Shift 3*

A steady current of 5A flows through a copper wire for 10 s. Calculate the total charge that passes through the wire during this time.

(A) 2 C	**(B)** 5 C
(C) 15 C	**(D)** 50 C

Q348. *CUSAT CAT 2025 - Shift 3*

The modulation index for an A.M. wave having maximum and minimum peak to peak voltages of 14 mV and 6 mV respectively is

(A) 1.4	**(B)** 0.4
(C) 0.2	**(D)** 0.6

Q349. *CUSAT CAT 2025 - Shift 3*

A beam of protons with speed 4 × 105 ms–1 enters a uniform magnetic field of 0.3 T at an angle of 60o to the magnetic field. Then the pitch of the resulting helical path of protons is close to [Mass of the proton =1.67 × 10–27 kg, charge of the proton =1.69×10–19 C]

(A) 12 cm	**(B)** 4 cm
(C) 5 cm	**(D)** 2 cm

Q350. *CUSAT CAT 2025 - Shift 3*

A fly wheel is accelerated uniformly from the rest and rotates through 5 rad in the first second. The angle rotated by fly wheel in the next second, will be

(A) 7.5 rad	**(B)** 15 rad
(C) 20 rad	**(D)** 30 rad

Q351. *CUSAT CAT 2025 - Shift 3*

105 Fermi is equal to

(A) 1 meter	**(B)** 100 micron
(C) 1 Angstrom	**(D)** 1 mm

Q352. *CUSAT CAT 2025 - Shift 4*

Two particles of the same mass are revolving with the same speed on circular paths of radii r and r . The ratio of their centripetal forces will be 1 2 r

(A) 1 r 2 r	**(B)** 2 r 1 2 r
(C) 1 r 2 2 r	**(D)** 2 r 1

Q353. *CUSAT CAT 2025 - Shift 4*

Two springs of force constant k and k are attached to mass m as shown. The time 1 2 period of the oscillation is 1 m

(A) 2π k m(k +k)	**(B)** 2π 1 2 k +k 1 2 m
(C) 2π k +k 1 2 1 m(k +k)	**(D)** 1 2 2π k +k 1 2 π

Q354. *CUSAT CAT 2025 - Shift 4*

In a circuit the current lags behind the voltage by a phase difference of . The circuit 2 contains which of the following?

(A) only R	**(B)** only L
(C) only C	**(D)** R and C

Q355. *CUSAT CAT 2025 - Shift 4*

Four metal conductors having different shapes: (i) a sphere, (ii) a cylinder (iii) a pear (iv) a lightning conductor are mounted on an insulating stands and charged. The one which is best suited to retain the charges for a longer time is

(A) sphere	**(B)** cylinder
(C) pear	**(D)** lightning conductor

Q356. *CUSAT CAT 2025 - Shift 4*

The emf E (in volts) of a certain thermocouple is found to vary with 2 θ θ (in °C) according to the equation E =20θ– , where θ is the temperature of the 20 hot function, the cold function being kept at 0°C. Then, the neutral temperature of the thermocouple is

(A) 300°C	**(B)** 100°C
(C) 340°C	**(D)** 200°C

Q357. *CUSAT CAT 2025 - Shift 4*

Magnetic field at the centre of a current carrying circular loop having 1 A current and number of turns one (radius of the loop is 1 m) will be

(A) 2	**(B)** 2μ o
(C) 4	**(D)** 4μ o

Q358. *CUSAT CAT 2025 - Shift 4*

The existence of positively charged nucleus in an atom was first established by

(A) Bohr's theory of hydrogen atom	**(B)** Positive rays analysis
(C) α-particle scattering experiment	**(D)** Thomson's model of atom

Q359. *CUSAT CAT 2025 - Shift 4*

A photocell is illuminated by a small bright source placed 1 m away. When the source of light is placed 0.5 m away, the number of electrons emitted by photo cathode would

(A) decrease by a factor of 4	**(B)** increase by a factor of 4
(C) decrease by a factor of 2	**(D)** increase by a factor of 2

Q360. *CUSAT CAT 2025 - Shift 4*

Which of the phenomena is not common to sound and light waves?

(A) Interference	**(B)** Diffraction
(C) Coherence	**(D)** Polarisation

Q361. *CUSAT CAT 2025 - Shift 4*

Three vessels A, B and C of different shapes contain a water up to the same height as shown in the figure. P , P and P be the pressures exerted by the water at the A B C bottom of the vessels A, B and C respectively. Then

(A) P > P > P A B C	**(B)** P > P > P B C A
(C) P > P > P C B A	**(D)** P = P = P A B C

Q362. *CUSAT CAT 2025 - Shift 4*

The length of a metallic rod is 5 m at 0°C and becomes 5.01 m, on heating up to 100°C. The linear expansion of the metal will be

(A) 2.33 × 10–5/ C ■	**(B)** 6.0 × 10–5/ C ■
(C) 4.0 × 10–5/ C ■	**(D)** 2.0 × 10–5/ C ■

Q363. *CUSAT CAT 2025 - Shift 4*

A small angle prism (μ = 1.62) gives a deviation of 4.8°. The angle of prism is

(A) 5°	**(B)** 6.36°
(C) 3°	**(D)** 7.74°

Q364. *CUSAT CAT 2025 - Shift 4*

A proton, a neutron, an electron and an α -particle have same energy. Then their de- Broglie wavelengths compare as

(A) λ = λ > λ > λ p n e α	**(B)** λ < λ = λ < λ α p n e
(C) λ < λ = λ > λ e p n α	**(D)** λ = λ > λ > λ e p n α

Q365. *CUSAT CAT 2025 - Shift 4*

According to wave theory, time required for photo emission is

(A) less than 10s	**(B)** around 10 –9 s
(C) around 1s	**(D)** around few hours

Q366.

CUSAT CAT 2025 - Shift 4

Range of frequencies allotted for commercial FM radio broadcast is

(A) 88 to 108 MHz	**(B)** 88 to 108 KHz
(C) 8 to 88 MHz	**(D)** 88 to 108 GHz

Q367.

CUSAT CAT 2025 - Shift 4

For an AM wave, the maximum voltage was found to be 10V and minimum voltage was 4V. The modulation index of the wave is

(A) 0.33	**(B)** 0.43
(C) 0.56	**(D)** 0.64

Q368.

CUSAT CAT 2025 - Shift 4

In aa gguuiittaarr,, ttwwoo ssttrriinnggss A and B mmaaddee ooff ssaammee mmaatteerriiaall aarree sslliigghhttllyy oouutt ooff ttuunnee aanndd pprroodduuccee bbeeaattss ooff ffrreeqquueennccyy 66 HHzz.. WWhheenn tteennssiioonn iinn B iiss sslliigghhttllyy ddeeccrreeaasseedd,, tthhee bbeeaatt ffrreeqquueennccyy iinnccrreeaasseess ttoo 77 HHzz.. IIff tthhee ffrreeqquueennccyy ooff A iiss 553300 HHzz,, tthhee oorriiggiinnaall ffrreeqquueennccyy ooff B will be

(A) 524 Hz	**(B)** 536 Hz
(C) 537 Hz	**(D)** 523 Hz

Q369.

CUSAT CAT 2025 - Shift 4

TThhee vvaarriiaattiioonn ooff eelleeccttrroossttaattiicc ppootteennttiiaall wwiitthh rraaddiiaall ddiiissttaannccee r ffrroomm tthhee cceennttrree ooff aa ppoossiittiivveellyy cchhaarrggeedd mmeettaallllliicc tthhiinn sshheellll ooff rraaddiiuuss R iiss ggiivveenn bbyy tthhee ggrraapphh

(A) —	**(B)** —
(C) —	**(D)** —

Q370.

CUSAT CAT 2025 - Shift 4

AAA ssstteeeppp dddooowwwnnn tttrrraaannnsssfffooorrrmmmeeerrr cccooonnnnnneeeccctteeeddd tttooo aaannn AAACCC mmmaaaiiinnnsss sssuuuppppplllyyy ooofff 222222000 VVV iiisss mmmaaadddeee tttooo ooopppeeerrraaattteee aaattt 111111 VVV,,, 444444 WWW lllaaammmppp... IIIgggnnnooorrriiinnnggg pppooowwweeerrr lllooossssseeesss iiinnn ttthhheee tttrrraaannnsssfffooorrrmmmeeerrr,,, wwwhhhaaattt iiisss ttthhheee ccuurrrreenntt iinn tthhee pprriimmaarryy circuit?

(A) 0.2 A	**(B)** 0.4 A
(C) 2 A	**(D)** 4 A

Q371.

CUSAT CAT 2025 - Shift 4

AAAnnn eeellleeeccctttrrrooonnn iiisss aaaccccccceeellleeerrraaattteeeddd fffrrrooommm rrreeessttt ttthhhrrrooouuuggghhh aaa pppoooottteeennnttiiiaaalll dddiiiffffffeeerrreeennnccceee ooofff V volt. If the de-BBrrooglliiee wwaavveelleennggtthh ooff tthhee eelleeccttrroonn iiss 11..222277 ××1100–2 nnmm,, tthhee ppootteennttiiaall ddiiffffeerreennccee iiss

(A) 102 V	**(B)** 103 V
(C) 104 V	**(D)** 10 V

Q372.

CUSAT CAT 2025 - Shift 4

TTThhheee sssppeeeccctttrrruuummm ooobbbtttaaaiiinnneeeddd fffrrrooommm aaa sssoooddddiiiuuummm vvvaaapppooouuurrr lllaaammmppp iiisss aaannn eeexxxaaammmpppllleee ooofff

(A) band spectrum	**(B)** ccoonnttiinnuuoouuss ssppeeccttrruumm
(C) eemmiissssiioonn ssppeeccttrruumm	**(D)** aabbssoorrppttiioonn ssppeeccttrruumm

Q373.

CUSAT CAT 2025 - Shift 4

An electron accelerated through a potential difference V has a de-Broglie wavelength 1 of λ. When the potential is changed to V , its de-Broglie wavelength increases by 2 V 50%. The value of 1 is equal to V 2

(A) 3 9	**(B)** 4 3
(C) 2	**(D)** 4

Q374.

CUSAT CAT 2025 - Shift 4

Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number to nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be

(A) 1 : 16	**(B)** 4 : 1
(C) 1 : 4	**(D)** 5 : 4

Q375.

CUSAT CAT 2025 - Shift 4

If an emitter current is changed by 4 mA, the collector current changes by 3.5 mA. The value of β will be

(A) 7	**(B)** 0.5
(C) 0.875	**(D)** 3.5

Q376.

CUSAT CAT 2025 - Shift 4

AA ggaallvvaannoommeetteerr hhaavviinngg aa ccooiill rreessiissttaannccee 110000 Ω gives a full- ssccaallee ddeeffllleeccttiioonn wwhheenn aa cccuuurrrrrreeennnttt ooofff mmmAAA iiisss pppaaassssseeeddd ttthhhrrrooouuuggghhh iiittt... WWWhhhaaattt iiisss ttthhheee vvvaaallluuueee ooofff ttthhheee rrreeesssiiissstance which can ccoonnvveerrtt tthhiiss ggaallvvaannoommeetteerr iinnttoo aa vvoollttmmeetteerr ggiivviinngg ffuullll-ssccaallee ddeeffllleeccttiioonn ffoorr aa ppootteennttiiaall difference of 10 V ?

(A) 9.9 kΩ	**(B)** 8.9 kΩ
(C) 7.9 kΩ	**(D)** 10 kΩ

Q377.

CUSAT CAT 2025 - Shift 4

AA mmaaggnneettiicc nneeeeddllee ooff mmaaggnneettiicc mmoommeenntt 66..77 ×× 1100–2 Am2 aanndd mmoommeenntt ooff iinneerrttiiaa 7.5×10–6 kg m2 iiisss pppeeerrrfffooorrrmmmiiinnnggg sssiiimmmpppllleee hhhaaarrrmmmooonnniiiccc ooossscccciiillllllaaatttiiiooonnnsss iiinnn aaa mmmaaagggnnneeetttiiiccc fffield of 00..0011 TT.. TTiimmee ttaakkeenn ffoorr 1100 ccoommpplleettee aassssoocciiaattiioonnss iiss

(A) 6.98 s	**(B)** 8.76 s
(C) 6.65 s	**(D)** 8.89 s

Q378.

CUSAT CAT 2025 - Shift 4

The AC current is given by I = I sin ωt + cos ωt. A hot wire ammeter will give a 1 reading

(A) – 2	**(B)** + 2
(C) + √2	**(D)** + 2√2

Q379.

CUSAT CAT 2025 - Shift 4

A hydraulic automobile lift is designed to lift vehicles of mass 5000 kg. The area of cross section of the cylinder carrying the Load is 250 cm2. The maximum pressure the smaller piston would have to bear is [Assuming g = 10 m/s2]

(A) 2 × 105 Pa	**(B)** 200 × 106 Pa
(C) 20 × 106 Pa	**(D)** 2 × 106 Pa

Q380.

CUSAT CAT 2025 - Shift 4

A certain charge Q is divided into two parts q and (Q – q). How should the charges Q and q be divided so that q and (Q – q) placed at a certain distance apart experience maximum electrostatic repulsion ? q

(A) Q = 2	**(B)** Q = 2q
(C) Q = 4q	**(D)** Q = 3q

Q381.

CUSAT CAT 2025 - Shift 4

A body sliding down on a smooth inclined plane, slides th of its distance in 2 s. The 4 time required for sliding the complete plane is

(A) 4 s	**(B)** 2 s
(C) 5 s	**(D)** 3 s

Q382.

CUSAT CAT 2025 - Shift 4

The ratio of magnitude of displacement to distance for a moving object is always

(A) equal to 1	**(B)** greater than 1
(C) less than 1	**(D)** less than or equal to 1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	A	4	A	5	B	6	D	7	B	8	A	9	B	10	D

11	B	12	D	13	D	14	A	15	B	16	A	17	D	18	C	19	C	20	D
21	B	22	D	23	D	24	B	25	A	26	A	27	B	28	B	29	C	30	C
31	A	32	B	33	C	34	B	35	B	36	C	37	C	38	A	39	D	40	D
41	A	42	D	43	C	44	C	45	D	46	B	47	B	48	D	49	C	50	B
51	C	52	C	53	C	54	D	55	A	56	C	57	A	58	B	59	B	60	A
61	A	62	A	63	D	64	C	65	A	66	A	67	B	68	B	69	C	70	D
71	C	72	C	73	C	74	C	75	C	76	C	77	C	78	C	79	C	80	C
81	C	82	C	83	D	84	B	85	B	86	B	87	B	88	B	89	C	90	D
91	B	92	C	93	A	94	B	95	B	96	A	97	A	98	A	99	A	100	B
101	C	102	D	103	B	104	C	105	A	106	C	107	C	108	C	109	A	110	A
111	C	112	C	113	B	114	C	115	C	116	B	117	C	118	C	119	C	120	D
121	C	122	C	123	A	124	B	125	A	126	A	127	D	128	A	129	A	130	D
131	B	132	D	133	D	134	D	135	B	136	D	137	D	138	B	139	A	140	A
141	B	142	D	143	B	144	D	145	C	146	B	147	A	148	D	149	A	150	B
151	D	152	B	153	B	154	D	155	B	156	C	157	A	158	A	159	C	160	B
161	B	162	B	163	D	164	B	165	A	166	D	167	D	168	A	169	C	170	C
171	B	172	A	173	B	174	D	175	D	176	A	177	A	178	C	179	C	180	B
181	D	182	C	183	B	184	A	185	B	186	C	187	B	188	C	189	C	190	D
191	C	192	B	193	D	194	A	195	C	196	B	197	B	198	B	199	A	200	D
201	C	202	B	203	D	204	B	205	B	206	C	207	D	208	D	209	D	210	C
211	D	212	B	213	A	214	D	215	C	216	A	217	A	218	C	219	C	220	D
221	A	222	B	223	D	224	A	225	D	226	B	227	A	228	B	229	C	230	B
231	B	232	C	233	C	234	A	235	D	236	B	237	D	238	C	239	A	240	C
241	D	242	B	243	C	244	D	245	D	246	C	247	A	248	A	249	A	250	B
251	C	252	C	253	A	254	B	255	A	256	C	257	B	258	D	259	B	260	A
261	C	262	C	263	A	264	C	265	B	266	D	267	B	268	D	269	A	270	D
271	C	272	B	273	C	274	C	275	A	276	B	277	C	278	A	279	D	280	B
281	C	282	D	283	C	284	D	285	C	286	C	287	A	288	D	289	C	290	B
291	C	292	C	293	D	294	D	295	B	296	A	297	B	298	C	299	D	300	B
301	A	302	B	303	C	304	B	305	A	306	B	307	A	308	C	309	B	310	B
311	B	312	A	313	B	314	B	315	A	316	C	317	A	318	B	319	A	320	D
321	A	322	B	323	A	324	B	325	D	326	A	327	B	328	A	329	B	330	D

331	A	332	D	333	C	334	C	335	A	336	B	337	D	338	D	339	A	340	A
341	A	342	C	343	D	344	C	345	D	346	A	347	D	348	B	349	B	350	B
351	C	352	B	353	C	354	B	355	A	356	D	357	A	358	C	359	B	360	D
361	D	362	D	363	D	364	B	365	D	366	A	367	B	368	A	369	B	370	A
371	C	372	C	373	B	374	D	375	A	376	A	377	C	378	B	379	D	380	B
381	A	382	D																

CHEMISTRY

NCERT Chapter-Wise Compilation

CUSAT CAT 2023 · 2024 · 2025 — All 14 Shifts

Some Basic Concepts

14 questions

Q1. *CUSAT CAT 2023 - Shift 2*

How many distinct internal alkynes share the molecular formula C H ? 6 10

(A) 4	**(B)** 3
(C) 2	**(D)** 1

Q2. *CUSAT CAT 2023 - Shift 2*

Pick the statement that is NOT true for Freon refrigerant R-22.

(A) It causes ozone depletion	**(B)** Its molecular formula is CCl F 2 2
(C) It is prepared from CHCl 3	**(D)** It is used for preparing tetrafluoroethene (C F) 2 4

Q3. *CUSAT CAT 2023 - Shift 3*

Which of the following is the correct set with reference to molecular formula, hybridisation of central atom and shape of the molecule? 2

(A) CO , sp , bent 2 2	**(B)** H O , sp , bent 2 2
(C) BeCl , sp , linear 2 3	**(D)** H O, sp , bent 2

Q4. *CUSAT CAT 2023 - Shift 4*

What is the weight of CuO, that will react with 2.6 L of hydrogen at NTP (molar mass of CuO is 79.5 g)?

(A) 79.5 g	**(B)** 2.2 g
(C) 9.23 g	**(D)** 11.2 g

Q5. *CUSAT CAT 2023 - Shift 5*

Which of the following units of concentration is temperature dependent?

(A) Mole fraction	**(B)** Molality
(C) Normality	**(D)** Weight percentage

Q6. *CUSAT CAT 2023 - Shift 5*

The total pressure exerted by a number of non reacting gases is equal to the sum of the partial pressures of the gases under the same conditions is known as

(A) Boyle's law	**(B)** Charle's law
(C) Avogadro's law	**(D)** Dalton's law PV

Q7. *CUSAT CAT 2023 - Shift 5*

How much volume of CO will be obtained by thermal decomposition of 25 g of 2 calcium carbonate (molar mass of calcium carbonate = 100 g) ?

(A) 1.0 L	**(B)** 5.6 L
(C) 11.2 L	**(D)** 22.4 L

Q8. *CUSAT CAT 2024 - Shift 1*

The number of entities or particles together in mole concept is known as

(A) Avogadro's number	**(B)** Boltzmann constant
(C) Universal gas constant	**(D)** Reynold's number

Q9. *CUSAT CAT 2024 - Shift 2*

There are two different isomeric compounds with molecular formula C H O. One of 2 6 these isomers has a much higher boiling point than the other. Explain why?

(A) Due to difference in their ability to form strong intermolecular hydrogen bonds	**(B)** Due to large difference in their dipole moments
(C) Due to difference in the number of carbon-carbon bonds in the two isomers	**(D)** Due to difference in carbon oxygen bonds in the two isomers

Q10. *CUSAT CAT 2024 - Shift 3*

The rates of diffusion of gases are inversely proportional to square root of their densities. This statement refers to

(A) Daltons law	**(B)** Avogadros law
(C) Boyles law	**(D)** Grahams law

Q11. *CUSAT CAT 2024 - Shift 4*

3 M 1.5 L of KOH solution is mixed with another 1 M 650 mL KOH solution. What is the molarity of the resultant solution?

(A) 2.00 M	**(B)** 2.40 M
(C) 2.66 M	**(D)** 1.50 M

Q12. *CUSAT CAT 2024 - Shift 5*

The percent composition of an organic compound was estimated as: 72% C, 12% H. The molar mass is 100 g/mole. What is the molecular formula of this compound?

(A) C H O 5 8 2	**(B)** C H 7 16
(C) C H O 6 12	**(D)** C H O 4 4 3

Q13. *CUSAT CAT 2025 - Shift 1*

Which of the following is dependent on temperature?

(A) Mole fraction	**(B)** Molality
(C) Weight percentage	**(D)** Molarity

Q14.

CUSAT CAT 2025 - Shift 1

Henry's law constant for molality of methane ib benzene at 298 K is 4.27 × 105 mm Hg. The mole fraction of methane in benzene at 298 K under 760 mm Hg is

(A) 1.78 × 10–3	**(B)** 0.174
(C) 0.114	**(D)** 0.281

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	D	4	C	5	B	6	D	7	B	8	A	9	A	10	D
11	B	12	C	13	D	14	A												

Structure of Atom

24 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The electronic configuration with the highest first ionization energy is 2 1

(A) [Ne]3s 3p 2 2	**(B)** [Ne]3s 3p 2 3
(C) [Ne]3s 3p 2 4	**(D)** [Ne]3s 3p 2–

Q2. *CUSAT CAT 2023 - Shift 1*

Which of the following represents the general electronic configuration of an element belonging to the p-block of the periodic table? 0 0 2 0-6

(A) (n – 2)f (n – 1)d ns np 0 1-10 2 1-6	**(B)** (n – 2)f (n – 1)d ns np 0 0 2 1-6
(C) (n – 2)f (n – 1)d ns np 1-14 1-10 2 1-6	**(D)** (n – 2)f (n – 1)d ns np

Q3. *CUSAT CAT 2023 - Shift 2*

How many electrons are associated with the quantum number n = 4?

(A) 32	**(B)** 64
(C) 16	**(D)** 8

Q4. *CUSAT CAT 2023 - Shift 3*

The region at which the probability density function reduces to zero is called as

(A) Density region	**(B)** Nodal surfaces
(C) Orientation surfaces	**(D)** Wave function 2 3

Q5. *CUSAT CAT 2023 - Shift 3*

The correct electronic configuration and spin only magnetic moment of Gd are 7

(A) [Xe] 4f and 7.9BM 7	**(B)** [Xe] 4f and 8.9BM 6 1
(C) [Xe] 4f 5d and 7.9BM 7	**(D)** [Xe] 5f and 7.9BM

Q6. *CUSAT CAT 2023 - Shift 3*

Orbital angular momentum depends on the quantum number/s

(A) l	**(B)** n and l
(C) n and m	**(D)** m and s

Q7. *CUSAT CAT 2023 - Shift 4*

Electronic configuration of a transition element X in +3 oxidation state is [Ar]3d . What is its atomic number?

(A) 25	**(B)** 26
(C) 27	**(D)** 24

Q8. *CUSAT CAT 2023 - Shift 5*

Which statement about the four quantum numbers that describe electrons in atoms is INCORRECT?

(A) n = principal quantum number, n = 1, 2, 3, ...	**(B)** l = subsidiary (or azimuthal) quantum number, l = 1, 2, 3, ... , (n + 1)
(C) m = magnetic quantum number, m = (–l), ... , 0, ... , (+l) l l	**(D)** m = spin quantum number, m = +1/2 or –1/2 s s

Q9. *CUSAT CAT 2023 - Shift 5*

For the element with the electronic configuration [Xe]4f 5d 6s 6p , which is most stable oxidation state?

(A) +3	**(B)** +4
(C) +2	**(D)** +1

Q10. *CUSAT CAT 2024 - Shift 1*

For a value l = 2, then the minimum value of principle quantum number for d-orbital is

(A) 2	**(B)** 3
(C) 1	**(D)** 0

Q11. *CUSAT CAT 2024 - Shift 1*

Which of the following statements describes a correct step in the formation of an ionic bond between sodium and chlorine?

(A) Removing an electron from sodium will provide energy for the bond formation	**(B)** Adding an electron to chlorine will require energy for bond formation
(C) The Coulombic potential between the ions will release energy	**(D)** Overcoming the Pauli repulsion from the overlap of wave functions of core electrons will release energy

Q12. *CUSAT CAT 2024 - Shift 3*

The four quantum numbers for the electron in the outer most orbital of potassium are

(A) n = 3, l = 0, m = 1, s = +½	**(B)** n = 4, l = 0, m = 0, s = +½
(C) n = 2, l = 0, m = 1, s = +½	**(D)** n = 4, l = 0, m = 1, s = +½

Q13. *CUSAT CAT 2024 - Shift 3*

What is the Bohr radius of the electron of hydrogen atom?

(A) 0.24 Å	**(B)** 0.48 Å
(C) 0.53 Å	**(D)** 1.06 Å

Q14. *CUSAT CAT 2024 - Shift 3*

Which of the following set of quantum numbers cannot represent an electron?

(A) 1,1,1, ±½	**(B)** 1,0,0, ±½
(C) 1,0,0, +½	**(D)** 2,0,0, +½ –1

Q15. *CUSAT CAT 2024 - Shift 4*

The correct ground statement electronic configuration of chromium atom is

(A) [Ar] 3d 5 4s 1	**(B)** [Ar] 3d 5 4s 2
(C) [Ar] 3d 6 4s 0	**(D)** [Ar] 3d 4 4s 1 4p 1

Q16. *CUSAT CAT 2024 - Shift 5*

Principal, azimuthal and magnetic quantum numbers are respectively related to

(A) size, shape and orientation	**(B)** shape, size and orientation
(C) size, orientation and shape	**(D)** None of the above

Q17. *CUSAT CAT 2025 - Shift 1*

Ψ2=0 represents

(A) A node	**(B)** An orbital
(C) Angular wave function	**(D)** Wave function

Q18. *CUSAT CAT 2025 - Shift 1*

Which combination of quantum numbers n, l, m and s for the electron in the atom does not provide permissible solution of wave equation?

(A) 3,2,-2,1/2	**(B)** 3,3,1,-1/2
(C) 3, 2, 1, ½	**(D)** 3,1,1,1/2

Q19. *CUSAT CAT 2025 - Shift 1*

Pd has exceptional electronic configuration of 4d105s0.It belongs to

(A) Period 4, group 11	**(B)** Period 5, group 10
(C) Period 6, group 9	**(D)** Period 3, group 16

Q20. *CUSAT CAT 2025 - Shift 2*

Which of the following electronic configuration has associated magnetic moment of 4.89 B.M.?

(A) [Ar] 3d5	**(B)** [Ar] 3d6
(C) [Ar] 3d7	**(D)** [Ar] 3d8

Q21. *CUSAT CAT 2025 - Shift 3*

What is the correct set of four quantum numbers for the valence electron of an atom with Z = 37 ? 1

(A) 5,0,1,+ 2 1	**(B)** 5,1,0,– 2 1
(C) 5,0,0,+ 2 1	**(D)** 5,1,1,– 2

Q22.

CUSAT CAT 2025 - Shift 3

The wave function ( ) of 2s is given by = # 2– % &'%■ ! . Find the √ ! ! value of r where radial node is formed

(A) 0	**(B)**) *
(C) 2) *	**(D)** 3) *

Q23.

CUSAT CAT 2025 - Shift 3

Which of the following represents isoelectronic species?

(A) Li+, Na+, Mg2+, Ca2+	**(B)** Ba2+, Sr2+, K+, Ca2+
(C) O2■, F■, S2■, Ca2+	**(D)** Ar, K+, Cl■, Sc3+ 121 Which among the atomic orbitals 6s, 7s, 7p, 8p and 8d have 5 radial nodes?

Q24.

CUSAT CAT 2025 - Shift 4

Which of the following explains the sequence of filling the electrons in different shells?

(A) Aufbau principle	**(B)** Octet rule
(C) Hund's rule	**(D)** All of the above

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	B	4	B	5	A	6	A	7	B	8	B	9	D	10	B
11	C	12	B	13	C	14	A	15	A	16	A	17	A	18	B	19	B	20	B
21	C	22	C	23	D	24	A												

Periodic Classification

12 questions

Q1. *CUSAT CAT 2023 - Shift 2*

Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar?

(A) Ca < S < Ba < Se < Ar	**(B)** S < Se < Ca < Ba < Ar
(C) Ba < Ca < Se < S < Ar	**(D)** Ca < Ba < S < Se <Ar

Q2. *CUSAT CAT 2023 - Shift 3*

What is the most important factor which makes Li strong reducing agent?

(A) Sublimation energy	**(B)** Ionization energy
(C) Hydration energy	**(D)** Electron gain enthalpy

Q3. *CUSAT CAT 2023 - Shift 4*

The elements x, y and z are present in one period of the periodic table. Chemically their oxides are acidic, amphoteric and basic respectively. Arrange the elements in order of increasing atomic number.

(A) x, y, z	**(B)** z, y, x
(C) y, z, x	**(D)** y, x, z

Q4. *CUSAT CAT 2023 - Shift 5*

Stable ionic solid MX is formed with

(A) metal having high electron affinity	**(B)** metal having high ionisation potential
(C) X having high electron affinity	**(D)** metal having low ionisation potential and X having high electron affinity

Q5. *CUSAT CAT 2023 - Shift 5*

With respect to periodic properties the correct statement is

(A) Electron affinity order is: F > O > Cl	**(B)** First ionisation energy order is: Al > Mg > K
(C) Atomic radius order is: N > P > As	**(D)** Ionic radius order is: K + > Ca 2+ > Mg 2+ MATHEMATICS UG SHIFT V - (FINAL) 2 –1

Q6. *CUSAT CAT 2024 - Shift 1*

How many elements are there in the periodic table?

(A) 116	**(B)** 114
(C) 118	**(D)** 120

Q7. *CUSAT CAT 2024 - Shift 4*

Which one exhibits anomalous behavior in terms of its chemical reactivity, particularly forming stable compounds with a higher oxidation state than expected, and what is the primary reason for this anomaly?

(A) Nitrogen (N); Small atomic size and high electronegativity	**(B)** Phosphorus (P); Ability to exhibit various oxidation states 5+
(C) Arsenic (As); Exceptionally stable As oxidation state	**(D)** Bismuth (Bi); High atomic mass and low electronegativity

Q8. *CUSAT CAT 2024 - Shift 4*

Be and Al show diagonal relationship hence both have

(A) Same degree of electronegativity	**(B)** Approximately same charge /radius ratio
(C) Amphoteric nature of oxides	**(D)** All of the above

Q9. *CUSAT CAT 2024 - Shift 5*

Among the alkali earth metals, which one forms an insoluble sulfide, and what specific factor contributes to this behavior?

(A) Beryllium (Be); Small size and high lattice energy	**(B)** Magnesium (Mg); Low ionization energy
(C) Calcium (Ca); Exceptionally stable Ca 2+ ions	**(D)** Barium (Ba); Increased atomic size

Q10. *CUSAT CAT 2024 - Shift 5*

Ionic bonds will be formed more easily between the elements having comparatively

(A) low ionization enthalpy and high electron affinity	**(B)** high ionization enthalpy and high electron affinity
(C) low ionization enthalpy and low electron affinity	**(D)** high ionization enthalpy and low electron affinity

Q11. *CUSAT CAT 2025 - Shift 3*

Identify the correct order of electronegativity

(A) Br > O > P > C	**(B)** C > P > O > Br
(C) O > Br > C > P	**(D)** P > C > O > Br

Q12. *CUSAT CAT 2025 - Shift 3*

Which of the following transition metal has the highest third ionization enthalpy?

(A) Cr	**(B)** Mn
(C) V	**(D)** Fe

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	B	4	D	5	D	6	C	7	C	8	D	9	A	10	A
11	C	12	B																

Chemical Bonding & Structure

30 questions

Q1. *CUSAT CAT 2023 - Shift 1*

What is the hybridization of carbons in C60 (buckminsterfullerene)? Identify the number of pentagons and hexagons in C60. 2

(A) Both sp and sp . 12 pentagons, 20 hexagons	**(B)** sp only. 12 pentagons, 20 hexagons 2 3
(C) sp, sp and sp . 12 pentagons, 24 hexagons 2	**(D)** sp only. 12 pentagons, 20 hexagons

Q2. *CUSAT CAT 2023 - Shift 1*

Pick the closest change in the HCH bond angle in ethene when it reacts with bromine to give 1,2-dibromoethane

(A) 30°	**(B)** 60°
(C) 10°	**(D)** 0°

Q3. *CUSAT CAT 2023 - Shift 1*

The eclipsed and staggered confirmations of n-butane differ drastically in

(A) C -C bond distance 2 3	**(B)** angle between C -C -C and C -C -C planes 1 2 3 2 3 4
(C) C -C -C bondangle 1 2 3	**(D)** H-C-H bond angles

Q4. *CUSAT CAT 2023 - Shift 2*

The molecule which has zero dipole moment is

(A) CH Cl 2 2	**(B)** BF 3
(C) NF 3	**(D)** ClO 2 0

Q5. *CUSAT CAT 2023 - Shift 2*

Hybridization and geometry of carbon in ethene are 3

(A) sp and Planar 2	**(B)** sp and Planar
(C) sp and Linear 2	**(D)** sp and Tetrahedral

Q6. *CUSAT CAT 2023 - Shift 2*

Hybridization of [Fe(CN)] 6 3 2

(A) sp d 3	**(B)** sp d 2 3
(C) d sp 3	**(D)** dsp MATHEMATICS UG (SHIFT – II FINAL)

Q7. *CUSAT CAT 2023 - Shift 3*

In the following the correct bond order sequence is 2– + –

(A) O > O > O > O 2 2 2 2 + – 2–	**(B)** O > O > O > O 2 2 2 2 + – 2–
(C) O > O > O > O 2 2 2 2 – 2– +	**(D)** O > O > O > O 2 2 2 2

Q8. *CUSAT CAT 2023 - Shift 4*

Which of the following has the least bond angles?

(A) NH 3	**(B)** PH 3
(C) AsH 3	**(D)** SbH 3

Q9. *CUSAT CAT 2023 - Shift 4*

According to VSEPR theory

(A) bp-bp > lp-bp > lp-lp	**(B)** bp-lp > bp-bp > lp-lp
(C) lp-lp > lp-bp > bp-bp	**(D)** lp-lp < lp-bp < bp-bp where bp is bond pair and lp is lone pair.

Q10. *CUSAT CAT 2023 - Shift 5*

A simple Allene is a compound with the following chemical structure H C C CH 2 2 * What is the state of hybridization of the starred carbon (middle carbon)? 3

(A) sp 2	**(B)** sp
(C) alternates between sp 2 and sp	**(D)** sp

Q11. *CUSAT CAT 2024 - Shift 1*

C-C-C Bond angle in 1-propyne is

(A) 109.5°	**(B)** 120°
(C) 180°	**(D)** 104.3°

Q12. *CUSAT CAT 2024 - Shift 2*

Nitration of phenol is much faster than nitration of benzene due to ……. of OH group

(A) +I effect	**(B)** +M effect
(C) –I effect	**(D)** Hydrogen bonding with the incoming nitronium ion electrophile

Q13. *CUSAT CAT 2024 - Shift 2*

What type of bond, a hydrogen atom need have for hydrogen bonding to occur?

(A) An ionic bond	**(B)** A polar covalent bond
(C) A non-polar covalent bond	**(D)** An alkaline bond

Q14. *CUSAT CAT 2024 - Shift 2*

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R Assertion A : The H–O–H bond angle in water molecule is 104.5°. Reason R : The lone pair - lone pair repulsion of electrons is higher than the bond pair - bond pair repulsion. Choose the correct alternative from given below.

(A) A is false but R is true	**(B)** Both A and R are true, but R is not the correct explanation of A
(C) A is true but R is false	**(D)** Both A and R are true, and R is the correct explanation of A

Q15. *CUSAT CAT 2024 - Shift 2*

The incorrect statement regarding the structure of C60 is

(A) The six-membered rings are fused to both six and five-membered rings	**(B)** Each carbon atom forms three sigma bonds
(C) The five-membered rings are fused only to six-membered rings	**(D)** It contains 12 six-membered rings and 24 five-membered rings

Q16. *CUSAT CAT 2024 - Shift 2*

The bond order and magnetic behaviour of O ion are 2

(A) 1.5 and paramagnetic	**(B)** 1.5 and diamagnetic
(C) 2 and diamagnetic	**(D)** 1 and paramagnetic

Q17. *CUSAT CAT 2024 - Shift 2*

According to molecular orbital theory, the species among the following that does not exist is +

(A) He 2 –	**(B)** He 2
(C) Be 2 2–	**(D)** O 2

Q18. *CUSAT CAT 2024 - Shift 3*

What is the approximate change in the HCH bond angle in ethene when it reacts with chlorine to form 1,2-dichloroethane?

(A) 30°	**(B)** 15°
(C) 11°	**(D)** 0°

Q19. *CUSAT CAT 2024 - Shift 4*

The polar molecular solids are held by

(A) Hydrogen bonding	**(B)** London forces
(C) Dipole - Dipole interaction	**(D)** Ionic interaction

Q20. *CUSAT CAT 2024 - Shift 4*

Geometry and hybridization of carbon in benzene are

(A) Angular and sp3	**(B)** Planar and sp2
(C) Linear and sp	**(D)** Tetrahedral and sp3

Q21. *CUSAT CAT 2024 - Shift 4*

Simple aliphatic alcohols have higher boiling point than isomeric ethers due to difference in

(A) Intermolecular hydrogen bonding	**(B)** Intramolecular hydrogen bonding
(C) van der Waals forces of attraction	**(D)** Polarizability

Q22. *CUSAT CAT 2025 - Shift 1*

The hybridization of N in (C H) N and its CNC bond angle are: 2 5 3

(A) sp2, >109.5°	**(B)** sp2, <109.5°
(C) sp3, >109.5°	**(D)** sp3,<109.5°

Q23. *CUSAT CAT 2025 - Shift 1*

Which of the following chemical species pairs has ions with the same central atom hybridization?

(A) NO –and NH – 2 2	**(B)** NO – and NO – 2 3
(C) NH + and NO – 4 3	**(D)** SCN– and NH – 2

Q24. *CUSAT CAT 2025 - Shift 2*

Which one of the following does not have sp2 hybridized carbon?

(A) Acetonitrile	**(B)** Acetic acid
(C) Acetone	**(D)** Acetamide

Q25. *CUSAT CAT 2025 - Shift 3*

When the hybridization state of carbon atom changes from sp3 to sp2 and finally to sp, the angle between the hybridized orbitals

(A) decreases gradually	**(B)** decreases when hybridization state changes from sp3 to sp2 and increases when hybridization state changes from sp2 to sp
(C) remains unchanged	**(D)** increases progressively

Q26. *CUSAT CAT 2025 - Shift 3*

Pick the correct statement about allene (propa-1,2-diene): H H C C C H H

(A) only sp2 hybridized carbons are present in allene	**(B)** both sp2 and sp hybridized carbons are present in allene
(C) all four hydrogens in allene lie in the same plane	**(D)** C-C-C bond angle in allene is about 120 ± 5 degrees

Q27.

CUSAT CAT 2025 - Shift 3

Formal charge of nitrogen atom in nitric acid (HNO) and nitrous acid (HNO) is 3 2

(A) +1 and 0 respectively	**(B)** 0 and +1 respectively
(C) +1 and -1 respectively	**(D)** +1 and +1 respectively

Q28.

CUSAT CAT 2025 - Shift 4

Formate ion is resonance stabilized. One of the two contributing structures (canonical forms) of formate anion is shown below. Consider other contributing structure and answer the following question on C-O bonds marked A and B. Which C–O bond is longer?

(A) A	**(B)** B
(C) Both A and B are of equal length	**(D)** Depends on the nature of the counter ion

Q29.

CUSAT CAT 2025 - Shift 4

The hybridization and geometry in SF is 4

(A) sp3 - Tetrahedral	**(B)** dsp2 - Square planar
(C) sp3d - See-saw	**(D)** sp3d2 - Octahedral

Q30.

CUSAT CAT 2025 - Shift 4

The bond order of O + is 2

(A) 1	**(B)** 1.5
(C) 2	**(D)** 2.5

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	C	3	B	4	B	5	B	6	C	7	C	8	D	9	C	10	D
11	C	12	B	13	B	14	D	15	D	16	A	17	C	18	C	19	C	20	B
21	A	22	D	23	B	24	A	25	D	26	B	27	A	28	B	29	C	30	D

States of Matter

1 questions

Q1. *CUSAT CAT 2023 - Shift 5*

is known as nRT

(A) Pressure factor	**(B)** Temperature factor
(C) Volume factor	**(D)** Compressibility factor

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D																		

Thermodynamics (Chem)

9 questions

Q1.

CUSAT CAT 2023 - Shift 3

For a spontaneous reaction, the Gibbs free energy change (ΔG), the equilibrium o constant (K) and E will be respectively cell

(A) –ve, > 1, +ve	**(B)** +ve, > 1, –ve
(C) –ve, < 1, –ve	**(D)** –ve, > 1, –ve

Q2.

CUSAT CAT 2024 - Shift 3

If the standard enthalpy and entropy change of a reaction are positive and negative, respectively, the correct statement is

(A) the reaction is non-spontaneous at all temperatures	**(B)** the reaction is spontaneous at all temperatures
(C) the reaction is non-spontaneous only at relatively very high temperatures	**(D)** the reaction is spontaneous only at relatively very high temperatures

Q3.

CUSAT CAT 2024 - Shift 5

One mole of an ideal gas is expanded from 10 L to 20 L at 298 K isothermally and reversibly. The correct statement describing the entropy change for the above process is

(A) Δ = 2 1	**(B)** Δ = 2
(C) Δ = 2 1	**(D)** Δ = 2

Q4.

CUSAT CAT 2025 - Shift 1

From the following thermochemical data provided,the enthalpy of formation of CH (g) from graphite and H (g) is ……. 4 2 CH (g) + 2 O (g) → CO (g) + 2 H O(l); ΔH = -890.3 kJ/mol ……(i) 4 2 2 2 C(graphite) + O (g) → CO (g); ΔH = -393.5 kJ/mol ……..(ii) 2 2 H (g) + ½ O (g) → H O(l); ΔH = -285.8 kJ/mol ……(iii) 2 2 2

(A) –75 kJ/mol	**(B)** +75 kJ/mol
(C) –175 kJ/mol	**(D)** +175 kJ/mol

Q5.

CUSAT CAT 2025 - Shift 1

The heat of combustion of liquid toluene (C H) at 298 K and 1 atm pressure is 7 8 –3910.3 kJ.mol–1. C H (g) + 9O (g) → 7(CO) + 4 H O (ΔH = –3910.3 kJ.mol–1) 7 8 2 2 2 How much heat energy will be evolved when 46 g of toluene is burnt in an open container? (1 Cal = 4.1 J)

(A) 106 kCal	**(B)** 238 kCal
(C) 477 kCal	**(D)** 954 kCal

Q6. *CUSAT CAT 2025 - Shift 3*

Yellow and Red phosphorous are two important allotropes of Phosphorous. Heat of combustion of red and yellow phosphorous are –8.78 kJ and – 9.91 kJ respectively. The heat of transition of yellow to Red phosphorous is

(A) –18.69 kJ	**(B)** +1.13 kJ
(C) +18.69 kJ	**(D)** –1.13 kJ

Q7. *CUSAT CAT 2025 - Shift 3*

Which of the following have highest bond dissociation energy?

(A) F 2	**(B)** Cl 2
(C) Br 2	**(D)** I 2

Q8. *CUSAT CAT 2025 - Shift 4*

In a thermochemical process, 500 J of heat is absorbed by a system and 250 J of work is done by the system. The change in internal energy during the entire process is ……..

(A) +750 J	**(B)** +250 J
(C) -250 J	**(D)** -750 J

Q9. *CUSAT CAT 2025 - Shift 4*

Spontaneous adsorption of gas molecules on solid surfaces is an exothermic process because of ……

(A) Increase in enthalpy	**(B)** Increase in entropy
(C) Decrease in entropy	**(D)** Decrease in enthalpy

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	A	3	C	4	A	5	C	6	D	7	B	8	B	9	C		

Equilibrium

7 questions

Q1. *CUSAT CAT 2023 - Shift 1*

For 10 M CH COOH solution if K is 10 then find out α (degree of dissociation) 3 a –6

(A) 10 –5	**(B)** 10 –3
(C) 10 –2	**(D)** 10

Q2. *CUSAT CAT 2023 - Shift 4*

What is the degree of dissociation of 0.1 M acetic acid? (K = 10) a –3

(A) 10 –2	**(B)** 10 –1
(C) 10	**(D)** 1.0

Q3. *CUSAT CAT 2023 - Shift 4*

Pick the statement that is TRUE for E2 eliminations.

(A) Follows unimolecular kinetics	**(B)** Follows bimolecular kinetics
(C) Involves free radical intermediates	**(D)** It is catalyzed by weak acids

Q4. *CUSAT CAT 2024 - Shift 5*

A weak acid has a dissociation constant of 1 × 10 . The equilibrium constant for its reaction with a strong base is –4

(A) 1.0 × 10 –10	**(B)** 1.0 × 10 10
(C) 1.0 × 10 14	**(D)** 1.0 × 10

Q5. *CUSAT CAT 2025 - Shift 1*

Consider the endothermic forward reaction of gas-phase water splitting as shown below, 2H O (g) ■ 2H (g) + O (g) 2 2 2 Which of the following step will decrease the amount of water in the equilibrium condition?

(A) Adding more O into the reaction system 2	**(B)** Introduction of inert He-atmosphere for the reaction
(C) Decreasing the total volume of the container	**(D)** Increasing temperature at constant pressure

Q6. *CUSAT CAT 2025 - Shift 2*

A chemical reaction having higher equilibrium constant suggests

(A) Rate of forward reaction is faster	**(B)** Rate of backward reaction is faster
(C) Rate of both backward and forward reactions are equal	**(D)** None of the above

Q7.

CUSAT CAT 2025 - Shift 3

A buffer solution was prepared by mixing equal concentration of acid (ionization constant Ka) and a salt. The pH of the buffer is

(A) 14 – pKa	**(B)** pKa + 1
(C) pKa	**(D)** pKa – 14

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	B	3	B	4	C	5	D	6	A	7	C						

Redox Reactions

17 questions

Q1. *CUSAT CAT 2023 - Shift 2*

Which of the following does NOT show +4 oxidation state?

(A) Dy	**(B)** Ce
(C) Eu	**(D)** Tb

Q2. *CUSAT CAT 2023 - Shift 3*

Which of the following arrangements does NOT represent the correct order of the property stated against it? 2+ 2+ 2+ 2+

(A) V < Cr < Mn < Fe : paramagnetic behaviour 2+ 2+ 2+	**(B)** Ni < Co < Fe2+ < Mn : ionic size 3+ 3+ 3+ 3+
(C) Co < Fe < Cr < Sc : stability in aqueous solution	**(D)** Sc < Ti < Cr < Mn : number of oxidation states

Q3. *CUSAT CAT 2023 - Shift 3*

Which one of the following characteristics of the transition metals is associated with higher catalytic activity?

(A) High enthalpy of atomisation	**(B)** Paramagnetic behaviour
(C) Colour of hydrate ions	**(D)** Variable oxidation states

Q4. *CUSAT CAT 2023 - Shift 3*

The oxidation number of cobalt in K[Co(CO)] is 4

(A) +1	**(B)** +3
(C) –1	**(D)** –3 +3

Q5. *CUSAT CAT 2023 - Shift 5*

Oxidation state of Fe in cytochrome is/are

(A) +2 and +4 only	**(B)** +3 and +4 only
(C) +2, +3, and +4 in catalytic intermediates	**(D)** +4 only

Q6. *CUSAT CAT 2023 - Shift 5*

The actinides which exhibit the +7 oxidation state are

(A) Pu, Am	**(B)** U, Np
(C) Am, Cm	**(D)** Np, Pu 3+ 3+ 3+ 3+

Q7. *CUSAT CAT 2023 - Shift 5*

Maximum oxidation state is presented by

(A) CrO Cl and MnO 2 2 4	**(B)** MnO 2 3– 3–
(C) [Fe(CN)] and [Co(CN)] 6 6	**(D)** MnO

Q8. *CUSAT CAT 2024 - Shift 2*

The oxidation states of X in H X O , H X O and H X O , respectively, are 4 2 7 4 2 5 4 2 6

(A) 7, 5 and 6	**(B)** 5, 4 and 3
(C) 5, 3 and 4	**(D)** 6, 4 and 5 x

Q9. *CUSAT CAT 2024 - Shift 3*

Which of the following does not show variable oxidation state?

(A) Bromine	**(B)** Fluorine
(C) Iodine	**(D)** Chlorine

Q10. *CUSAT CAT 2024 - Shift 4*

The common oxidation state of the elements of the lanthanide series is

(A) +1	**(B)** +3
(C) +4	**(D)** +6

Q11. *CUSAT CAT 2024 - Shift 5*

Among the following lanthanides, which element is known for exhibiting an unusual oxidation state, +4, in addition to its more common oxidation states, and what is the specific compound where this unusual oxidation state is observed?

(A) Europium (Eu); Exhibits +4 oxidation state in europium (IV) oxide	**(B)** Cerium (Ce); Shows +4 oxidation state in cerium (IV) fluoride
(C) Praseodymium (Pr); Displays +4 oxidation state in praseodymium(IV) sulfide	**(D)** Gadolinium (Gd); Demonstrates +4 oxidation state in gadolinium(IV) chloride

Q12. *CUSAT CAT 2025 - Shift 1*

Oxidation of cyclohexene yields a dicarboxylic acid G which is a monomer for a polyamide polymer. Identify the oxidizing agent.

(A) OsO ,NaOH 4	**(B)** O , Me S 3 2
(C) O , H O 3 2 2	**(D)** O , NaBH 3 4

Q13. *CUSAT CAT 2025 - Shift 2*

The E0 3+ 2+values for Cr, Mn, Fe and Co are –0.41, + 1.57, + 0.77 and +1.97 V M /M respectively. For which one of the metals, the change in oxidation state from +2 to +3 is easiest?

(A) Cr	**(B)** Mn
(C) Fe	**(D)** Co

Q14.

CUSAT CAT 2025 - Shift 2

S and O both are non-metals in the oxygen family, but their melting points are largely different O = 90K and S = 718K this is because

(A) S is larger in size than O S exhibit more oxidation state +2, +4, +6 whereas O exhibit –2 and +2	**(B)** oxidation state
(C) O exist as diatomic molecule whereas S exhibit polyatomic molecule (S) 8	**(D)** S has more allotropes than O

Q15.

CUSAT CAT 2025 - Shift 3

What is the change of oxidation state of Cr in potassium dichromate as a strong oxidising agent in acidic solution?

(A) +3 to +1	**(B)** +6 to 0
(C) +6 to +3	**(D)** +3 to +2

Q16.

CUSAT CAT 2025 - Shift 4

Which of the following shows +7 oxidation state?

(A) Co	**(B)** Cr
(C) V	**(D)** Mn

Q17.

CUSAT CAT 2025 - Shift 4

In the reaction 2[K Fe(CN)] +2 KOH + H O → 2[K Fe(CN)] +2H O +O , H O 3 6 2 2 4 6 2 2 2 2 acts as

(A) Oxidizing agent	**(B)** Reducing agent
(C) Bleaching agent	**(D)** Both oxidizing agent and bleaching agent

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	D	4	C	5	C	6	D	7	A	8	C	9	B	10	B
11	B	12	C	13	A	14	C	15	C	16	D	17	B						

Hydrogen & Its Compounds

3 questions

Q1. *CUSAT CAT 2023 - Shift 2*

The number of protons, electrons and neutrons in a molecule of heavy water are respectively

(A) 8,10,11	**(B)** 10,10,10
(C) 10,11,10	**(D)** 11,10,10

Q2. *CUSAT CAT 2023 - Shift 5*

The maximum possible number of hydrogen bonds that can be formed by a water molecule is

(A) 2	**(B)** 3
(C) 4	**(D)** 1

Q3. *CUSAT CAT 2024 - Shift 3*

From among the following statements on Friedel-Crafts reactions, pick the wrong statement

(A) Mechanistically, Friedel-Crafts reaction is an aromatic electrophilic substitution reaction	**(B)** Presence of electron withdrawing groups such as nitro groups enhances the rate of Friedel-Crafts reaction
(C) In Friedel-Crafts alkylation reaction, AlCl is used in catalytic amounts 3 whereas in Friedel-Crafts acylation reaction AlCl and acid chloride are used 3 in a 1:1 molar ratio	**(D)** Both acid chlorides and acid anhydrides can be used for Friedel crafts acylation reaction

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	B														

s-Block Elements

10 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Which of the following alkali metals does react with water least vigorously?

(A) Li	**(B)** Na
(C) K	**(D)** Cs

Q2. *CUSAT CAT 2023 - Shift 5*

Permanent hardness of water is due to the presence of

(A) sulphates of Na and K	**(B)** sulphates of Ca and Mg
(C) hydrogen carbonates of Ca and Mg	**(D)** carbonates of alkali metals in water 14 10 2 1

Q3. *CUSAT CAT 2024 - Shift 1*

Which of the following formulas for magnesium compounds is correct?

(A) MgO 2	**(B)** MgS 2
(C) MgF 2	**(D)** Mg O 2

Q4. *CUSAT CAT 2024 - Shift 4*

Among the alkali metals (Group 1) and alkaline earth metals (Group 2), which individual element in each group tends to deviate from the general trend in ionization energy, and what specific factor or anomaly contributes to this deviation? 1. Lithium (Li) in Group 1; Anomalous electron configuration. 2. Cesium (Cs) in Group 1; Increased atomic size and electronic configuration. 3. Beryllium (Be) in Group 2; Unique electron configuration and higher effective nuclear charge 4. Strontium (Sr) in Group 2; Screening effect and increased atomic size.

(A) 1, 2	**(B)** 1, 3
(C) 2, 3	**(D)** 2, 4

Q5. *CUSAT CAT 2024 - Shift 4*

Which of the following statements is incorrect with respect to Beryllium?

(A) It is rendered passive by nitric acid	**(B)** It forms Be C 2
(C) Its salt rarely hydrolyze	**(D)** Its hydride is electron deficient and polymeric

Q6. *CUSAT CAT 2024 - Shift 5*

Solubility of the alkaline earth metal sulphates in water decreases in the sequence

(A) Ba > Mg > Sr > Ca	**(B)** Mg >Ca > Sr > Ba
(C) Ca > Sr > Ba > Mg	**(D)** Sr >Ca > Mg > Ba

Q7. *CUSAT CAT 2024 - Shift 5*

The elements with atomic number 35, 53 and 85 are

(A) Noble gas	**(B)** Halogens
(C) Alkali metals	**(D)** Alkaline earth metals

Q8. *CUSAT CAT 2025 - Shift 1*

Which of the following is also known as Schiff base?

(A) imine	**(B)** cyanohydrin
(C) lithium hydroxide	**(D)** sodamide

Q9. *CUSAT CAT 2025 - Shift 1*

Which of the following alkaline earth metal sulphates has hydration enthalpy higher than the lattice energy?

(A) CaSO 4	**(B)** BeSO 4
(C) BaSO 4	**(D)** SrSO 4

Q10. *CUSAT CAT 2025 - Shift 2*

Which of the following statements about lithium compounds is untrue?

(A) The hydroxides, carbonates, nitrate, decompose, to give oxides on heating	**(B)** It is the most electronegative among alkali metals
(C) The hydrogen carbonates cannot be isolate as stable solid	**(D)** It forms a peroxide but not superoxide

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	B	3	C	4	C	5	C	6	B	7	B	8	A	9	B	10	D

p-Block (Groups 13 & 14)

17 questions

Q1. *CUSAT CAT 2023 - Shift 1*

In graphite and diamond, the percentage of p-character in their hybrid orbitals respectively are

(A) 33 and 25	**(B)** 67 and 75
(C) 50 and 75	**(D)** 33 and 75

Q2. *CUSAT CAT 2023 - Shift 1*

Which of the following has the maximum C – C bond length?

(A) Graphite	**(B)** Naphthalene
(C) C 60	**(D)** Diamond

Q3. *CUSAT CAT 2023 - Shift 2*

Strong heating of an aqueous solution of aluminium chloride to dryness will give

(A) Al(OH)Cl 2	**(B)** Al O 2 3
(C) Al Cl 2 6	**(D)** AlCl 3

Q4. *CUSAT CAT 2023 - Shift 2*

Match the ores (Column A) with the metals (Column B): (Column A) (Column B) (I) Siderite (a) Zinc (II) Kaolinite (b) Copper (III) Malachite (c) Iron (IV) Calamine (d) Aluminium

(A) (I) –(a); (II) – (b); (III) –(c); (IV) – (d)	**(B)** (I) –(c); (II) – (d); (III) –(b); (IV) – (a)
(C) (I) –(c); (II) – (d); (III) –(a); (IV) – (b)	**(D)** (I) –(b); (II) – (c); (III) –(d); (IV) – (a)

Q5. *CUSAT CAT 2023 - Shift 4*

The WRONG statement among the following is

(A) Nitrogen atom and nitride ion have same atomic number	**(B)** Aluminium atom and its ion have same mass number
(C) Iron atom and ferrous ion have same electron configuration	**(D)** Nuclear charge is same in both chlorine atom, chloride ion

Q6. *CUSAT CAT 2023 - Shift 5*

To get n-type of semiconductor, Germanium should be doped with

(A) Gallium	**(B)** Arsenic
(C) Aluminium	**(D)** Boron

Q7. *CUSAT CAT 2024 - Shift 1*

Which element is in group 2 and period 3 of the periodic table?

(A) Magnesium	**(B)** Aluminium
(C) Boron	**(D)** Calcium

Q8. *CUSAT CAT 2024 - Shift 2*

Strongest back donation of an electron pair from halide to boron is expected in

(A) BCl 3	**(B)** BF 3
(C) BBr 3	**(D)** BI 3

Q9. *CUSAT CAT 2024 - Shift 3*

Permutit is chemically

(A) Hydrated sodium aluminium silicate	**(B)** Sodium hexaphosphate
(C) Sodium bicarbonate	**(D)** Calcium hydroxide

Q10. *CUSAT CAT 2024 - Shift 4*

In the extraction processes of zinc, aluminium, copper, and iron; which metal is predominantly extracted using the electrolytic method?

(A) Zinc; High melting point of the metal	**(B)** Aluminium; Low energy consumption and high purity
(C) Copper; Efficient separation of impurities	**(D)** Iron; Cost-effectiveness and simplicity

Q11. *CUSAT CAT 2024 - Shift 4*

Among the allotropes of carbon, which one exhibits a unique property of superconductivity at relatively high temperatures compared to other known superconductors?

(A) Graphite	**(B)** Diamond
(C) Amorphous carbon	**(D)** Fullerene (specifically C60)

Q12. *CUSAT CAT 2024 - Shift 4*

Boron nitride has the structure of the type

(A) Graphite	**(B)** Diamond
(C) NaCl	**(D)** Both Graphite and Diamond

Q13. *CUSAT CAT 2025 - Shift 1*

Which of the statement is incorrect?

(A) Aluminium reacts with excess NaOH to give Al(OH) 3	**(B)** NaHCO on heating gives Na CO 3 2 3
(C) Pure sodium metal dissolves in liquid ammonia to give blue solution	**(D)** NaOH reacts with glass to give sodium silicate

Q14. *CUSAT CAT 2025 - Shift 2*

Metallic magnesium is prepared by

(A) Displacement of magnesium by iron from magnesium sulphate solution	**(B)** Electrolysis of an aqueous solution of Mg(NO) 3
(C) Electrolysis of molten MgCl 2	**(D)** Reduction of MgO by aluminium

Q15. *CUSAT CAT 2025 - Shift 2*

Which one of the following anions is present in the chain structure of silicates?

(A) SiO 4– 4	**(B)** (Si O)6– 2 7
(C) (SiO)2– 3	**(D)** (Si O)2– 2 5

Q16. *CUSAT CAT 2025 - Shift 2*

In borax bead test which compound is formed?

(A) Orthoborate	**(B)** Metaborate
(C) Double oxide	**(D)** Tetraborate

Q17. *CUSAT CAT 2025 - Shift 3*

Why boric acid is solid and BF is gas at room temperature? 3

(A) Strong ionic bond in boric acid	**(B)** Strong covalent bond in BF 3
(C) Strong van der Waal's interaction in boric acid	**(D)** Strong hydrogen bond in boric acid

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	D	3	B	4	B	5	C	6	B	7	A	8	B	9	A	10	B
11	D	12	A	13	A	14	C	15	D	16	B	17	D						

Organic Chemistry Basics

17 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Compounds ‘A’ and ‘C’ in the following reaction are

(A) identical	**(B)** positional isomers
(C) functional isomers	**(D)** optical isomers

Q2. *CUSAT CAT 2023 - Shift 2*

What does a full “curved” arrow ( ) signify while writing reaction mechanisms?

(A) That two structures are enantiomers	**(B)** Represents movement of two electrons in the direction of the arrow
(C) Represents movement of an atom or a group in the direction of the arrow	**(D)** Resonating structures

Q3. *CUSAT CAT 2023 - Shift 2*

Glucose and fructose are

(A) optical isomers	**(B)** tautomers
(C) functional isomers	**(D)** chain isomers

Q4. *CUSAT CAT 2023 - Shift 5*

Which among the following carbanions is most stable? CH 2

(A) benzyl carbanion CH 2	**(B)** 4-methylbenzyl carbanion CH 3 CH 2
(C) 4-methoxybenzyl carbanion OCH 3 CH 2	**(D)** 4-nitrobenzyl carbanion NO 2

Q5. *CUSAT CAT 2024 - Shift 1*

Types of isomerism found in butene is/are

(A) Position isomerism only	**(B)** Both position and geometrical isomerism
(C) Geometrical isomerism only	**(D)** Both Position and optical isomerism

Q6. *CUSAT CAT 2024 - Shift 2*

Reaction of silver salts of carboxylic acids with halogen to form organic halides with one carbon less (Hunsdiecker reaction) proceeds through

(A) Carbocation intermediate	**(B)** Carbanion intermediate
(C) Radical intermediate	**(D)** Carbene intermediate

Q7. *CUSAT CAT 2024 - Shift 2*

The number of geometric and optical isomers of the complex [Co(en)]Cl are 3 3

(A) 0 and 0	**(B)** 0 and 2
(C) 1 and 0	**(D)** 1 and 1

Q8. *CUSAT CAT 2024 - Shift 3*

Which among the following would act as an electrophile?

(A) BH 3	**(B)** NH 3
(C) H O 2	**(D)** H S 2

Q9. *CUSAT CAT 2024 - Shift 3*

Addition of bromine to trans-but-2-ene to give meso-2,3-dibromobutane proceeds through

(A) Carbocation intermediate	**(B)** Carbanion intermediate
(C) Bromonium ion intermediate	**(D)** Free radical intermediate

Q10. *CUSAT CAT 2024 - Shift 4*

Identify functional groups present in the following molecule O NH O N O O HO

(A) Acid, Amide, Amine, Ester, Aromatic ring	**(B)** Alcohol, Ether, Amide, Amine, Aromatic ring
(C) Acid, Nitro, Aromatic ring, Amide	**(D)** Acid, Ester

Q11. *CUSAT CAT 2024 - Shift 4*

The allene shown here exhibits Me Me C H H

(A) optical isomerism	**(B)** geometrical isomerism
(C) both optical and geometrical isomerism	**(D)** neither optical nor geometrical isomerism

Q12. *CUSAT CAT 2024 - Shift 4*

Which among the following set of species have a carbon atom with a vacant orbital?

(A) carbocation and triplet carbene	**(B)** carbocation and singlet carbene
(C) free radical and triplet carbene	**(D)** carbanion and singlet carbene

Q13. *CUSAT CAT 2024 - Shift 5*

SN substitution proceeds through 2

(A) carbocation intermediate	**(B)** carbanion intermediate
(C) free radical intermediate	**(D)** no intermediate, only a transition state

Q14.

CUSAT CAT 2025 - Shift 2

To which of the following molecules will ammonia (a nucleophile) add in a 1,4- fashion?

(A) trans-but-2-ene	**(B)** butenone
(C) 4-penten-2-one	**(D)** cyclopentanone

Q15.

CUSAT CAT 2025 - Shift 2

Listed below are one each of olefin, carbanion, carbocation and carbon centered radical. Stability of one among the following species cannot be explained on the basis of hyperconjugation. Identify it.

(A) —	**(B)** —
(C) —	**(D)** —

Q16.

CUSAT CAT 2025 - Shift 2

Major product formed in the catalytic hydrogenation of the following optically active eneyne compound in the presence of poisoned palladium catalyst is

(A) an optically active compound	**(B)** an optically inactive compound
(C) a racemic mixture of two optical isomers	**(D)** a mixture of several optically active compounds

Q17.

CUSAT CAT 2025 - Shift 4

Arrange following carbanions in decreasing stability order. (i) (CH) C- 3 3 (ii) CH - 3 (iii) Benzyl anion (C H CH -) 6 5 2 (iv) CH COCH - 3 2

(A) (ii) > (iii) > (iv) > (i)	**(B)** (iv) > (iii) > (ii) > (i)
(C) (iii) > (iv) > (ii) > (i)	**(D)** (ii) > (iii) > (i) > (iv)

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	C	4	D	5	B	6	C	7	B	8	A	9	C	10	A
11	A	12	B	13	D	14	B	15	B	16	B	17	B						

Hydrocarbons

19 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Which among the following name reaction yields alkanes?

(A) Wurtz-Fittig reaction	**(B)** Wurtz reaction
(C) Friedel-Crafts reaction	**(D)** Vilsmeyer-Haack reaction

Q2. *CUSAT CAT 2023 - Shift 2*

An alkene on ozonolysis followed by hydrolysis gave acetaldehyde as the only product. This alkene can exhibit

(A) optical isomerism	**(B)** geometrical isomerism
(C) both optical and geometrical isomerism	**(D)** neither optical nor geometrical isomerism

Q3. *CUSAT CAT 2023 - Shift 2*

Major product formed in the reaction of tolune with N-Bromosuccinimide (NBS) in presence of benzoyl peroxide is

(A) a mixture of 2- and 4-bromotoluene	**(B)** benzyl bromide
(C) 3-bromotoluene	**(D)** 4-bromotoluene 2– 3– – 2+ + 3+

Q4. *CUSAT CAT 2023 - Shift 3*

In the following reaction sequence, the major product ‘Z’ is

(A) toluene	**(B)** biphenyl
(C) 4-chlorotoluene	**(D)** N-methylaniline

Q5. *CUSAT CAT 2023 - Shift 4*

Which among the following food additives is effective in controlling rancidity of unsaturated edible oils?

(A) Butylated hydroxytoluene (BHT)	**(B)** Monosodium glutamate (MSG)
(C) Sodium nitrate	**(D)** Caramel

Q6. *CUSAT CAT 2023 - Shift 5*

In the following sequence B and C are H 2 O Red hot iron tube AlCl 3 CaC 2 A B C CH 3 Cl

(A) Benzene and acetylene respectively	**(B)** Toluene and benzene respectively
(C) Benzene and toluene respectively	**(D)** Toluene and acetylene respectively

Q7. *CUSAT CAT 2024 - Shift 1*

Which of the following solvents forms explosive peroxides when they come in contact with air and light?

(A) Toluene	**(B)** Chloroform
(C) Hexane	**(D)** Diethyl ether

Q8. *CUSAT CAT 2024 - Shift 1*

The reaction of toluene with bromine in the presence of light yields

(A) o-bromotoluene	**(B)** benzyl bromide
(C) m-bromotoluene	**(D)** p-bromotoluene

Q9. *CUSAT CAT 2024 - Shift 2*

How to accomplish the following transformation?

(A) Anhydrous AlCl catalyzed rearrangement 3	**(B)** Markovnikov addition of HBr followed by base catalyzed E1 elimination of HBr
(C) Anti-Markovnikov addition of HBr followed by base catalyzed E2 elimination of HBr	**(D)** Base catalysed equilibration

Q10. *CUSAT CAT 2024 - Shift 2*

Picric acid is

(A) Strong acid that can protonate electron rich aromatic compounds	**(B)** Monomer used in the preparation of polyester PET
(C) 2,4,6-trinitrobenzoic acid	**(D)** 2,4,6-trinitrophenol

Q11. *CUSAT CAT 2024 - Shift 3*

IUPAC name of the alkyne that can be converted into 3-ethylpentane upon hydrogenation is

(A) 3-ethylpent-1-yne	**(B)** 3-ethylpent-4-yne
(C) 3,3-diethylprop-1-yne	**(D)** 1,1-diethylprop-3-yne

Q12. *CUSAT CAT 2024 - Shift 4*

Hydroboration-oxidation is a method to prepare alcohols from olefins. The product alcohol is an anti-Markovnikov product. Predict the product formed from the following hydroboration-oxidation reaction. CH 3 B H 2 6 ? H O , NaOH 2 2

(A) —	**(B)** —
(C) —	**(D)** —

Q13. *CUSAT CAT 2024 - Shift 4*

What type of reaction is the addition of HBr to an alkene in the absence of light and radical initiators?

(A) Electrophilic addition	**(B)** Nucleophilic addition
(C) Concerted addition	**(D)** None of the above

Q14. *CUSAT CAT 2024 - Shift 4*

Among the following compounds, under identical conditions, which will dissolve fastest in conc. H SO ? 2 4

(A) Benzene	**(B)** Nitrobenzene
(C) Toluene	**(D)** Chlorobenzene

Q15. *CUSAT CAT 2024 - Shift 5*

An alkyne on treatment with dilute H SO and HgSO gave a mixture of two isomeric 2 4 4 ketones (C H O). Identify the structure of the alkyne. 5 10

(A) —	**(B)** —
(C) —	**(D)** —

Q16. *CUSAT CAT 2024 - Shift 5*

Iodoalkane is obtained by the reaction of the corresponding bromoalkane or chloroalkane with NaI in dry acetone. The reaction is known as

(A) Hunsdiecker reaction	**(B)** Wurtz reaction
(C) Finkelstein reaction	**(D)** Iodoform reaction

Q17. *CUSAT CAT 2025 - Shift 1*

Identify the product formed in each case - when an alkyne C H reacts with: 3 4 (i) Hg(OAc) /H O; NaBH , 2 2 4 (ii) BH ; H O /OH- 3 2 2

(A) (i) acetone (ii) propionaldehyde	**(B)** (i) propionaldehyde (ii) acetone
(C) (i) ethanol (ii) propionaldehyde	**(D)** (i) ethanol (ii) acetone

Q18. *CUSAT CAT 2025 - Shift 2*

Most reactive compound among the following towards sulphonation is

(A) toluene	**(B)** chlorobenzene
(C) nitrobenzene	**(D)** thiophene

Q19. *CUSAT CAT 2025 - Shift 2*

Ozonolysis of an organic compound 'A' gave a 1:1 mixture of propanone and propanal. Identify 'A' from the following compounds

(A) hex-1-ene	**(B)** hex-2-ene
(C) 2-methylpent-2-ene	**(D)** 2-methylpent-1-ene

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	B	4	A	5	A	6	C	7	D	8	B	9	C	10	D
11	A	12	A	13	A	14	C	15	B	16	C	17	A	18	D	19	C		

Solid State

20 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A unit cell has the following characteristics, a ≠ b ≠c; α = γ = 90°, β ≠ 90°. The unit cell belongs to the crystal system

(A) Orthorhombic	**(B)** Rhombohedral
(C) Monoclinic	**(D)** Triclinic

Q2. *CUSAT CAT 2023 - Shift 1*

Ferrous oxide has a cubic structure, with the edge length of 5.0 Å. If the density of the –3 2+ 2– solid is 3.84 g cm , the number of Fe and O ions present per unit cell is 23 (N = 6.02 × 10) A 2+ 2–

(A) 2Fe and 2O 2+ 2–	**(B)** Fe and O 2+ 2–
(C) 4Fe and 4O 2+ 2–	**(D)** 3Fe and 4O

Q3. *CUSAT CAT 2023 - Shift 2*

A compound X has face centred cubic structure its density is 3.4 g cm . The length of the unit cell is (molecular weight of X is 99 g/mol)

(A) 5.78 Å	**(B)** 6.78 Å
(C) 7.78 Å	**(D)** 8.783 Å

Q4. *CUSAT CAT 2023 - Shift 3*

The number of atoms in 100 g of fcc crystal with density d = 10 g cm and cell edge equal to 100 pm is 25

(A) 4 × 10 25	**(B)** 3 × 10 25
(C) 2 × 10 25	**(D)** 1 × 10

Q5. *CUSAT CAT 2023 - Shift 4*

The density of a metal which crystallises in bcc lattice with unit cell edge length 300 pm –1 and molar mass 50 g mol will be –3

(A) 10 g cm –3	**(B)** 14.2 g cm –3
(C) 6.15 g cm –3	**(D)** 9.32 g cm

Q6. *CUSAT CAT 2023 - Shift 5*

Silver halides generally show

(A) Schottky defect	**(B)** Frenkel defect
(C) Both Frenkel and Schottky defects	**(D)** Cation excess defect

Q7. *CUSAT CAT 2023 - Shift 5*

Coordination number of FCC crystal is

(A) 4	**(B)** 8
(C) 12	**(D)** 16 –6

Q8. *CUSAT CAT 2024 - Shift 3*

NaCl crystals generally shows …….

(A) Schottky defect	**(B)** Frenkel defect
(C) Metal excess defect	**(D)** Both Schottky defect and Frenkel defect

Q9. *CUSAT CAT 2024 - Shift 3*

A given metal crystallizes out with a cubic structure having edge length of 361 pm. If there are four metal atoms in one unit cell, what is the radius of one atom?

(A) 108 pm	**(B)** 40 pm
(C) 127 pm	**(D)** 80 pm

Q10. *CUSAT CAT 2024 - Shift 4*

When constituent particles are present only on the corner positions of a unit cell, it is said to be

(A) face centered unit cell	**(B)** primitive unit cell
(C) body centered unit cell	**(D)** end-centered unit cell

Q11. *CUSAT CAT 2024 - Shift 5*

Schottky defect is an example of

(A) point defect	**(B)** metal deficiency
(C) metal excess	**(D)** electron defect

Q12. *CUSAT CAT 2025 - Shift 1*

What is the effect of Frenkel defect on the density of ionic solids?

(A) The density of the crystal increases	**(B)** The density of the crystal decreases
(C) The density of the crystal remains unchanged	**(D)** There is no relationship between density of a crystal and defect present in it

Q13. *CUSAT CAT 2025 - Shift 2*

P, F, B represents Primitive, Face - centered, Body centered cubic structures respectively. The crystal structure types of CsCl and NaCl are

(A) B, F	**(B)** F, C
(C) P, F	**(D)** F, P

Q14. *CUSAT CAT 2025 - Shift 3*

In Zinc blende structure

(A) zinc ions occupy half of the tetrahedral sites	**(B)** each Zn2– ion is surrounded by six sulphide ions
(C) each S2– ion is surrounded by six Zn2+ ions	**(D)** it has fcc structure

Q15. *CUSAT CAT 2025 - Shift 3*

A metal crystallizes into two cubic phases, body centered cubic and face centered cubic whose unit cell length are 2.5 and 2.0 Aº respectively. The ratio of densities of body centered cubic and face centered cubic is

(A) 0.24	**(B)** 0.25
(C) 0.26	**(D)** 0.23

Q16. *CUSAT CAT 2025 - Shift 4*

An element with atomic mass 100 has a bcc structure and edge length 400 pm. The density of element is

(A) 10.37 g cm–3	**(B)** 5.19 g cm–3
(C) 7.29 g cm–3	**(D)** 2.14 g cm–3

Q17. *CUSAT CAT 2025 - Shift 4*

An element possesses cubic close packing structure. Calculate the radius (r) of the atom in the unit cell. (The edge length of unit cell is a = 252 nm)

(A) 89.36 nm	**(B)** 126 nm
(C) 152 nm	**(D)** 109.1 nm

Q18. *CUSAT CAT 2025 - Shift 4*

How many atoms are present in hexagonal close-packed unit cell?

(A) 4	**(B)** 2
(C) 8	**(D)** 6

Q19. *CUSAT CAT 2025 - Shift 4*

A compound having molecular mass of 99 has face-centred cubic structure. Its ` density is 3.4g cm-3. The length of unit cell in Å (in Angstroms) is

(A) 6.783	**(B)** 5.783
(C) 7.783	**(D)** 8.683

Q20. *CUSAT CAT 2025 - Shift 4*

Appearance of colour in solid alkali metal halides is generally due to

(A) F-centres	**(B)** Frenkel defects
(C) Schottky defects	**(D)** Interstitial positions

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	A	4	A	5	C	6	C	7	C	8	A	9	C	10	B
11	A	12	C	13	C	14	C	15	B	16	A	17	A	18	D	19	B	20	A

Solutions

4 questions

Q1.

CUSAT CAT 2023 - Shift 4

Find out the WRONG statement.

(A) A non-volatile solute dissolved in water lowers vapor pressure	**(B)** A non-volatile solute dissolved in water raises boiling point
(C) A non-volatile solute dissolved in water raises density	**(D)** A non-volatile solute dissolved in water lowers osmotic pressure

Q2.

CUSAT CAT 2023 - Shift 4

Calculate the osmotic pressure associated with 50.0 g of an enzyme of molecular weight 98,000 g/mol dissolved in water to give 2600 mL of solution at 30.0°C.

(A) 3.71 torr	**(B)** 1.68 torr
(C) 1.96 torr	**(D)** 2.48 torr

Q3.

CUSAT CAT 2023 - Shift 4

An azeotropic solution of two liquids has boiling point lower than either when it

(A) shows a negative deviation from Raoult's law	**(B)** shows no deviation from Raoult's law
(C) shows positive deviation from Raoult's law	**(D)** is saturated

Q4.

CUSAT CAT 2024 - Shift 3

Choose the correct relation for osmotic pressure, temperature and concentration C

(A) $\Pi \propto$ RT	**(B)** $\Pi \propto$CRT 1
(C) $\Pi \propto$ CRT T	**(D)** $\Pi \propto$ CR

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	C	4	B												

Electrochemistry

11 questions

Q1. *CUSAT CAT 2023 - Shift 2*

Limiting molar conductivity of NH OH (i.e., ■ (NH OH)) is equal to 4 m 4 0 0 0

(A) ■ (NH Cl) + ■ (NaCl) – ■ (NaOH) m 4 m m 0 0 0	**(B)** ■ (NaOH) + ■ (NaCl) – ■ (NH Cl) m m m 4 0 0 0
(C) ■ (NH OH) + ■ (NH Cl) – ■ (HCl) m 4 m 4 m 0 0 0	**(D)** ■ (NH Cl) + ■ (NaOH) – ■ NaCl) m 4 m m (

Q2. *CUSAT CAT 2023 - Shift 3*

Electrode potentials are reported with reference to

(A) standard hydrogen electrode	**(B)** normal calomel electrode
(C) glass electrode	**(D)** silver-silver chloride electrode

Q3. *CUSAT CAT 2023 - Shift 3*

An electrochemical cell can behave like an electrolytic cell when

(A) E = 0 cell	**(B)** E > E cell ext
(C) E > E ext cell	**(D)** E = E cell ext

Q4. *CUSAT CAT 2023 - Shift 4*

The standard reduction potentials of three metallic ions X, Y, Z are 0.52 V, –3.03 V, –1.18 V respectively. The order of reducing power of the corresponding metals is

(A) Y > Z > X	**(B)** X > Y > Z
(C) Z > Y > X	**(D)** Z > X > Y

Q5. *CUSAT CAT 2024 - Shift 3*

For the Daniel cell Zn | Zn 2+ || Cu 2+ | Cu | with E° cell = 1.1 V the application of opposite potential greater than 1.1 V results in 2+

(A) oxidation of Cu and reduction of Zn	**(B)** increase in cell potential
(C) increase in current	**(D)** decrease in cell potential

Q6. *CUSAT CAT 2024 - Shift 5*

Which of the following metal cannot be obtained by electrolysis of an aqueous solution?

(A) Copper	**(B)** Magnesium
(C) Calcium	**(D)** Chromium

Q7. *CUSAT CAT 2025 - Shift 1*

The standard reduction potentials of Cr O 2-/Cr3+ and Cr3+/Cr are 1.33 and 0.74 V 2 7 respectively. The reduction potential of Cr O 2-/Cr is 2 7

(A) 2.07 V	**(B)** 1.77 V
(C) 1.33 V	**(D)** 0.59 V

Q8. *CUSAT CAT 2025 - Shift 1*

The molar conductivities at infinite dilution for the strong electrolytes NH Cl, 4 NaOH and NaCl are respectively 149.7 Ω–1cm2mol–1, 248.1 Ω–1cm2mol–1 and 126.5 Ω–1cm2mol–1. The molar conductivity of NH OH at infinite dilution is 4

(A) 271.3 Ω–1cm2mol–1	**(B)** 224.9 Ω–1cm2mol–1
(C) 177.8 Ω–1cm2mol–1	**(D)** 28.1 Ω–1cm2mol–1

Q9. *CUSAT CAT 2025 - Shift 2*

The sodium is made by the electrolysis of a molten mixture of about 40% NaCl and 60% CaCl because 2

(A) Ca2+ can displace Na from NaCl	**(B)** This mixture has a lower melting point than NaCl
(C) CaCl helps in conduction of electricity 2	**(D)** Ca2+ can reduce NaCl to Na

Q10. *CUSAT CAT 2025 - Shift 4*

The mass of Cu deposited on cathode during electrolysis when 268 milli-amperes (268 mA) of current was passed through a Faradaic cell containing a Cu cathode in saturated CuSO4 solution for 1 hour is …….. (atomic weight of Cu is 63.546 u)

(A) 415 mg	**(B)** 315 mg
(C) 215 mg	**(D)** 115 mg

Q11. *CUSAT CAT 2025 - Shift 4*

The standard reduction potentials of X, Y, Z metals are 0.52, –3.03, –1.18 respectively. The order of reducing power of the corresponding metals is

(A) Y > Z > X	**(B)** X > Y > Z
(C) Z > Y > X	**(D)** Z > X > Y

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	C	4	A	5	A	6	C	7	D	8	A	9	B	10	B
11	B																		

Chemical Kinetics

27 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The rate constant of a zero-order reaction is 0.04 M sec . The concentration of the reactant remaining after 25 sec is 0.5 M. The initial concentration of the reactant is

(A) 0.5 M	**(B)** 1.25 M
(C) 0.125 M	**(D)** 1.5 M

Q2. *CUSAT CAT 2023 - Shift 2*

3A→2B then the rate of reaction dt 3 d[A]

(A) – 2 dt 2 d[A]	**(B)** – 3 dt 1 d[A]
(C) – 3 dt d[A]	**(D)** 2 dt

Q3. *CUSAT CAT 2023 - Shift 3*

For a first order reaction, t is 138.6 sec. Its specific rate constant (in sec) is 0.75 –2

(A) 10 –4	**(B)** 10 –5
(C) 10 –6	**(D)** 10 3

Q4. *CUSAT CAT 2023 - Shift 3*

If the rate of reaction is equal to the rate constant, the order of the reaction is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q5. *CUSAT CAT 2023 - Shift 4*

The half-life period of a first order reaction which takes 40 min for 30% decomposition is

(A) 77.7 min	**(B)** 52.5 min
(C) 22.7 min	**(D)** 46.2 min 2–

Q6. *CUSAT CAT 2023 - Shift 4*

The role of a catalyst is to lower the

(A) Gibbs energy of reaction	**(B)** enthalpy of reaction
(C) activation energy of reaction	**(D)** equilibrium constant 2+ –

Q7. *CUSAT CAT 2023 - Shift 5*

Addition of catalyst to a reaction at a particular temperature lowers

(A) the value of equilibrium constant	**(B)** the value of activation energy for the forward reaction
(C) the value of free energy change of reaction	**(D)** the value of frequency factor in Arrhenius equation

Q8. *CUSAT CAT 2024 - Shift 1*

Arrhenius equation is given by

(A) k = Ae –E a R	**(B)** A=ke –E a RT
(C) k = Ae –E a RT	**(D)** A=ke –E a R –1

Q9. *CUSAT CAT 2024 - Shift 1*

The rate constant of a zero-order reaction is 0.04 M sec . The concentration of the reactant remaining after 25 sec is 0.5 M. The initial concentration of the reactant is

(A) 0.5 M	**(B)** 1.25 M
(C) 0.125 M	**(D)** 1.5 M

Q10. *CUSAT CAT 2024 - Shift 1*

If the first order reaction involves gaseous reactant and gaseous products, the units of its rate are

(A) atm	**(B)** atm sec
(C) –1 atm sec	**(D)** atm 2 sec 2

Q11. *CUSAT CAT 2024 - Shift 1*

The half-life of a first order reaction

(A) depends on the reactant concentration raised to the first power	**(B)** is inversely proportional to the square of the reactant concentration
(C) is inversely proportional to the reactant concentration	**(D)** is independent of the reactant concentration

Q12. *CUSAT CAT 2024 - Shift 1*

Select the correct statement. (a) Physical adsorption is reversible while the chemical adsorption is irreversible (b) High activation energy is involved in physical adsorption (c) Physical adsorption is specific while chemical adsorption is nonspecific (d) High activation energy is involved in chemical adsorption

(A) a and d	**(B)** b and c
(C) b and d	**(D)** a and c

Q13. *CUSAT CAT 2024 - Shift 1*

For a hypothetical reaction: 2A + B → Product The rate constant, k is equal to 5.6 × 10 –6 Lmol –1 s –1, the order of the reaction is

(A) One	**(B)** Two
(C) Three	**(D)** Fractional

Q14. *CUSAT CAT 2024 - Shift 2*

According to Arrhenius theory, the activation energy (Ea) for a reaction with rate constant k at a temperature T is …….. (R is gas constant) =

(A) = –	**(B)** =
(C) —	**(D)** = –

Q15. *CUSAT CAT 2024 - Shift 2*

Arrhenius equation is k = A e–Ea RT . Choose the correct statement.

(A) A depends on temperature	**(B)** Ea depends on temperature
(C) Both Ea and A are independent of temperature	**(D)** Both Ea and A depend on temperature

Q16. *CUSAT CAT 2024 - Shift 3*

The rate constant for a first order reaction is 60 s . The time taken to reduce the 1 th initial concentration of the reactant to its value will be 16

(A) 0.00462 s	**(B)** 0.462 s
(C) 0.0462 s	**(D)** 4.63 s

Q17. *CUSAT CAT 2024 - Shift 3*

Which of the following statement is not true?

(A) A reaction having fraction order must be a complex reaction	**(B)** Order of reaction is determined from experimental data
(C) Molecularity of a reaction may be equal to three	**(D)** Order of the reaction is always positive integers

Q18. *CUSAT CAT 2024 - Shift 4*

The unit of the rate and rate constant are the same for a reaction of order

(A) 0	**(B)** 1 1
(C) 2	**(D)** 2

Q19. *CUSAT CAT 2024 - Shift 4*

The activation energy for a chemical reaction depends upon

(A) temperature	**(B)** nature of reactants
(C) concentration of reactants	**(D)** collision frequency

Q20. *CUSAT CAT 2024 - Shift 5*

For a first order reaction, t is 138.6 sec. Its specific rate constant (in sec) is 0.75

(A) 10 –2	**(B)** 10 –4
(C) 10 –5	**(D)** 10 –6 2

Q21. *CUSAT CAT 2024 - Shift 5*

If the rate law for a reaction is, rate = k[A][B] , the unit for k is –1 –1

(A) mol L s –1	**(B)** mol s 2 –2 –1
(C) L mol s	**(D)** mol 2 L –2 s –1 x

Q22. *CUSAT CAT 2025 - Shift 1*

The decomposition of carbon disulfide, (CS), to carbonmonosulfide (CS), and sulfur 2 is first order with k = 2.8 × 10–7 s–1 at 1000°C CS →CS + S 2 The half-life of this reaction at 1000°C is

(A) 5.0 × 107 s	**(B)** 2.5 × 106 s
(C) 3.8 × 105 s	**(D)** 6.1 × 104 s

Q23. *CUSAT CAT 2025 - Shift 1*

For a chemical reaction A → B, it is found that the rate of the reaction quadruples when the concentration of A is doubled. The rate expression for the reaction is, rate = k [A]n where the value of n is

(A) 1	**(B)** 2
(C) 0	**(D)** 3

Q24. *CUSAT CAT 2025 - Shift 1*

If 75% of a first order reaction was completed in 60 minutes, 50% of the same reaction under the same conditions would be completed in

(A) 20 minutes	**(B)** 30 minutes
(C) 35 minutes	**(D)** 75 minutes

Q25. *CUSAT CAT 2025 - Shift 1*

Radioactive disintegration is an example of

(A) zero order reaction	**(B)** first order reaction
(C) second order reaction	**(D)** third order reaction

Q26. *CUSAT CAT 2025 - Shift 2*

The chemical reactions of high molecularity are rare because

(A) Activation energy for many body collisions is very high	**(B)** Many body collisions are energetically less favorable
(C) Many body collisions have a low probability	**(D)** All of the above

Q27.

CUSAT CAT 2025 - Shift 3

What is the activation energy for a reaction if its rate doubles when the temperature is raised from 200 K to 400 K? (R = 8.314 JK–1mol–1)

(A) 234.65 kJ mol–1	**(B)** 434.65 kJ mol–1
(C) 229.1 J mol–1	**(D)** 334.65 J mol–1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	B	3	A	4	A	5	A	6	C	7	B	8	C	9	D	10	C
11	D	12	A	13	B	14	A	15	C	16	C	17	D	18	A	19	B	20	A
21	C	22	B	23	B	24	B	25	B	26	C	27	C						

Surface Chemistry

14 questions

Q1. *CUSAT CAT 2023 - Shift 2*

In physical adsorption, the forces of attraction are,

(A) ionic	**(B)** covalent
(C) Van der Waal's	**(D)** H-bonding

Q2. *CUSAT CAT 2023 - Shift 5*

If the dispersed phase is a liquid and the dispersion medium is solid, the colloid is known as

(A) foam	**(B)** sol
(C) emulsion	**(D)** gel

Q3. *CUSAT CAT 2024 - Shift 1*

The Langmuir adsorption isotherm is given by θ= , where p is the pressure of 1+k p the adsorbate gas. The Langmuir adsorption isotherm for diatomic gas A undergoing 2 dissociative adsorption is k p

(A) θ= 1+k p 2k p	**(B)** θ= 1+2k p 2 ( ) k p
(C) θ= 2 ( ) 1+ k p 12 ( ) k p	**(D)** θ= 12 ( ) 1+ k p

Q4. *CUSAT CAT 2024 - Shift 1*

Which among the following statement is correct with respect to adsorption of gases on a solid? n I. The extent of adsorption is equal to kp according to Freundlich isotherm II. The extent of adsorption is equal to kp 1n according to Freundlich isotherm 1+bp III. The extent of adsorption is equal to according to Langmuir isotherm ap bp IV. The extent of adsorption is equal to according to Langmuir isotherm 1+bp

(A) I and III	**(B)** I and IV
(C) II and III	**(D)** II and IV

Q5. *CUSAT CAT 2024 - Shift 1*

Which of the following involves coagulation? I. Peptization II. Formation of delta regions III. Treatment of drinking water by potash alum IV. Clotting of blood by the use of ferric chloride

(A) I and IV	**(B)** II, III and IV
(C) I, III and IV	**(D)** I, II and III

Q6. *CUSAT CAT 2024 - Shift 3*

In physical adsorption, the forces of attraction are

(A) ionic	**(B)** covalent
(C) Vander Waal's	**(D)** H-bonding

Q7. *CUSAT CAT 2024 - Shift 4*

The example of emulsion is

(A) Soap	**(B)** Butter
(C) Cheese	**(D)** Starch

Q8. *CUSAT CAT 2024 - Shift 4*

The adsorption isotherm is defined as the dependence of

(A) surface coverage on the temperature at fixed pressure	**(B)** surface coverage on the pressure at fixed temperature
(C) surface coverage on the oxidation stable of the surface material	**(D)** rate of surface reaction at a fixed temperature

Q9. *CUSAT CAT 2024 - Shift 4*

The gas which is commonly employed in the study of adsorption is

(A) O 2	**(B)** Cl 2
(C) CO 2	**(D)** N 2

Q10. *CUSAT CAT 2024 - Shift 4*

One of the methods of purifying a colloidal solution is

(A) Coagulation	**(B)** Peptization
(C) Dialysis	**(D)** Electro-osmosis

Q11. *CUSAT CAT 2024 - Shift 5*

For a linear plot of log versus log p in a Freundlich adsorption isotherm, the m correct statement is (k and n are constants). 1

(A) Both k and appear in the slope term n 1	**(B)** appears as the intercept n 1
(C) Only appears as the slope n 1	**(D)** log appears as the intercept n

Q12. *CUSAT CAT 2024 - Shift 5*

Multilayer adsorption is assumed in

(A) Langmuir equation	**(B)** Freundlich equation
(C) BET equation	**(D)** Both Langmuir equation and BET equation

Q13. *CUSAT CAT 2024 - Shift 5*

The principle involved in column chromatography over silica gel is

(A) Absorption	**(B)** Partition
(C) Adsorption	**(D)** Size exclusion

Q14.

CUSAT CAT 2025 - Shift 3

The exterior surface of a soap micelle in water is:

(A) hydrophobic	**(B)** hydrophilic
(C) amphiphilic	**(D)** lipophilic

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	D	3	D	4	D	5	B	6	C	7	B	8	B	9	D	10	D
11	C	12	C	13	C	14	B												

p-Block (Groups 15–18)

9 questions

Q1. *CUSAT CAT 2023 - Shift 4*

The characteristics of the blue solution of sodium in liquid ammonia is /are (a) diamagnetic (b) paramagnetic (c) reducing in nature (d) conducts electricity

(A) (a) only	**(B)** (b), (c) and (d) only
(C) (b) and (c) only	**(D)** (a), (c) and (d) only

Q2. *CUSAT CAT 2024 - Shift 1*

Chemical produced by addition reaction between benzene and chlorine is

(A) DDT	**(B)** BHC
(C) Chlorobenzene	**(D)** Chloroprene

Q3. *CUSAT CAT 2024 - Shift 4*

Stratospheric ozone layer depletion is primarily initiated by

(A) Chlorine and bromine atoms	**(B)** Hydroxyl radical
(C) NO 2	**(D)** SO 2

Q4. *CUSAT CAT 2024 - Shift 4*

Which among the following is not a chlorine containing compound?

(A) DDT	**(B)** ABS Rubber
(C) Neoprene	**(D)** PVC

Q5. *CUSAT CAT 2025 - Shift 1*

With a density of 1.80 g ml-1, concentrated aqueous sulfuric acid is 98% H SO by 2 4 mass. Acid volume needed to produce one liter of 0.1 M H SO is 2 4

(A) 5.55 ml	**(B)** 11.10 ml
(C) 16.65 ml	**(D)** 22.20 ml

Q6. *CUSAT CAT 2025 - Shift 3*

Which of the following does not give a white precipitate with ammoniacal AgNO 3 solution?

(A) propyne	**(B)** but-1-yne
(C) but-2-yne	**(D)** ethyne

Q7. *CUSAT CAT 2025 - Shift 3*

Which among the following reactions is best suited to convert benzyl bromide to benzylamine with minimal side product formation?

(A) Reaction with liquid ammonia	**(B)** Reaction with KCN followed by reduction with LiAlH 4
(C) Reaction with potassium phthalimide followed by hydrolysis with strong base	**(D)** Reaction with sodium nitrite (NaNO)followed by reduction with LiAlH 2 4

Q8. *CUSAT CAT 2025 - Shift 4*

Which of the following condition favors formation of ammonia in Haber Process?

(A) High Temperature and High Pressure	**(B)** Low Temperature and Low Pressure
(C) Low Temperature and High Pressure	**(D)** High Temperature and Low Pressure

Q9. *CUSAT CAT 2025 - Shift 4*

Which of the following solutions shows positive deviation from Raoult's law?

(A) Acetone + Aniline	**(B)** Acetone + Ethanol
(C) Water + Nitric acid	**(D)** Chloroform + Benzene

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	A	4	B	5	A	6	C	7	C	8	C	9	B		

d & f Block Elements

4 questions

Q1. *CUSAT CAT 2024 - Shift 5*

Propionaldehyde is reduced in presence of sodium borohydride to compound P which is treated first with thionyl chloride and the resultant compound Q is treated with magnesium in dry ether to form R. R can react with propionaldehyde to form S. What product is formed when S is oxidized with potassium dichromate?

(A) —	**(B)** —
(C) —	**(D)** —

Q2. *CUSAT CAT 2024 - Shift 5*

Which of the following elements and its specific factor which can form superconducting compounds at relatively higher temperatures in comparison to other d-block elements?

(A) Chromium (Cr); Due to its exceptional hardness	**(B)** Vanadium (V); Exhibits variable oxidation states
(C) Yttrium (Y); Forms oxygen deficient compounds	**(D)** Manganese (Mn); Displays diverse magnetic properties

Q3. *CUSAT CAT 2024 - Shift 5*

Among the lanthanides, which element is employed in certain medical imaging applications due to its ability to emit X-rays when exposed to high-energy photons, and what is the specific medical imaging technique where this element is utilized?

(A) Gadolinium (Gd); Used in magnetic resonance imaging (MRI)	**(B)** Samarium (Sm); Employed in positron emission tomography (PET)
(C) Lutetium (Lu); Utilized in single-photon emission computed tomography (SPECT)	**(D)** Terbium (Tb); Applied in X-ray fluorescence imaging

Q4. *CUSAT CAT 2025 - Shift 1*

Which one of them is expected to have highest third ionization energy?

(A) Vanadium	**(B)** Chromium
(C) Manganese	**(D)** Iron

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	D	4	C												

Coordination Compounds

13 questions

Q1. *CUSAT CAT 2023 - Shift 1*

What is common between Heme and Chlorophyl?

(A) Both are red in colour	**(B)** Both have magnesium in them
(C) Both are proteins	**(D)** Both are co-ordination complexes of porphyrinic ligands

Q2. *CUSAT CAT 2023 - Shift 1*

If A, B, C and D are four different ligands, how many geometrical isomers will be 2+ formed for square planar [PtABCD] ?

(A) 3	**(B)** 2
(C) 1	**(D)** 4

Q3. *CUSAT CAT 2023 - Shift 2*

The crystal field stabilization energy (CFSE) for an octahedral complex; [CoCl] is 6 –1 2– 18000 cm . The CFSE for tetrahedral [CoCl] will be 4 4–

(A) greater than that of [CoCl] 6 –1	**(B)** 15000 cm 4–
(C) equal to that of [CoCl] 6 –1	**(D)** 8000 cm 3–

Q4. *CUSAT CAT 2023 - Shift 3*

Crystal field stabilization energy (CFSE) of a high spin octahedral iron(III) complex (atomic number of iron = 26)

(A) –20 Dq	**(B)** 0
(C) –6 Dq	**(D)** –4 Dq

Q5. *CUSAT CAT 2024 - Shift 1*

The number of ligands in the outer coordination sphere of CoCl .6NH , CoCl .5NH 3 3 3 3 and CoCl .4NH are 3 3

(A) 1, 2 and 3	**(B)** 2, 2 and 3
(C) 3, 2 and 1	**(D)** 3, 2 and 2

Q6. *CUSAT CAT 2024 - Shift 5*

Which of the following is hexadentate ligand?

(A) Acetylacetonate	**(B)** 1,10-phenanthroline
(C) Ethylene diamine	**(D)** Ethylene diamine tetraacetate

Q7. *CUSAT CAT 2025 - Shift 2*

Which of the following complex ion is not expected to absorb visible light?

(A) [Ni(CN)]2- 4	**(B)** [Cr(NH)]3+ 6
(C) [Fe(H O)]2+ 2 6	**(D)** [Ni(H O)]2+ 2 6

Q8. *CUSAT CAT 2025 - Shift 2*

The number of unpaired electrons in the complex ion [CoF]3-is 6

(A) 1	**(B)** 6
(C) 4	**(D)** 2

Q9. *CUSAT CAT 2025 - Shift 4*

Which of the following complex ions shows geometrical isomerism?

(A) [Cr(H O) Cl]+ 2 4 2	**(B)** [Pt(NH) Cl]2■ 3 3
(C) [Co(NH)]3■ 3 6	**(D)** [Co(CN) (NC)]3■ 5

Q10. *CUSAT CAT 2025 - Shift 4*

Which is the reason for the purple colour of KMnO ? 4

(A) Ligand to metal charge transfer	**(B)** Metal to ligand charge transfer
(C) d-d transition	**(D)** p-d transition

Q11. *CUSAT CAT 2025 - Shift 4*

Which of the following is a hexadentate ligand?

(A) Ethylene diamine	**(B)** Ethylene diamine tetraacetic acid
(C) 1,10-phenanthroline	**(D)** Acetyl acetonato

Q12. *CUSAT CAT 2025 - Shift 4*

Which of the following is a π-acid ligand?

(A) NH 3	**(B)** CO
(C) F■	**(D)** H O 2

Q13. *CUSAT CAT 2025 - Shift 4*

The ligand system present in Vitamin B is 12

(A) Porphyrine	**(B)** Corrin
(C) Crown ether	**(D)** Chlorin

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key

1	D	2	A	3	D	4	B	5	C	6	D	7	A	8	C	9	A	10	A
11	B	12	B	13	B														

Haloalkanes & Haloarenes

7 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Among carbon tetrachloride, chlorobenzene and benzyl chloride which will give a white precipitate with alcoholic silver nitrate?

(A) Carbon tetrachloride and chlorobenzene	**(B)** Carbon tetrachloride and benzyl chloride
(C) Benzyl chloride only	**(D)** All the three halides

Q2. *CUSAT CAT 2023 - Shift 3*

Which among the following methods is NOT suitable for preparing phenol under mild conditions?

(A) Treatment of chlorobenzene with NaOH in water	**(B)** Hydrolysis of phenyl acetate
(C) Treatment of methoxybenzene (anisole) with 57% HI	**(D)** Treatment of cumene hydroperoxide with dil. HCl

Q3. *CUSAT CAT 2023 - Shift 3*

Reaction of CH -CH -CH -Cl with KCN in acetonitrile to give CH -CH -CH -CN 3 2 2 3 2 2 proceeds predominantly by

(A) Nucleophilic Substitution bimolecular	**(B)** Nucleophilic Substitution unimolecular
(C) Radical Substitution bimolecular	**(D)** Radical Substitution unimolecular

Q4. *CUSAT CAT 2023 - Shift 4*

The reaction between sodium alkoxide and alkyl halide to give the corresponding ether is called

(A) Wurtz reaction	**(B)** Kolbe's reaction
(C) Williamson's synthesis	**(D)** Zeisel reaction

Q5. *CUSAT CAT 2024 - Shift 2*

Benzene diazonium chloride can be converted to chlorobenzene by treating with

(A) PCl 5	**(B)** Cl , FeCl 2 3
(C) CuCl 2	**(D)** CuCl

Q6. *CUSAT CAT 2024 - Shift 3*

Arrange the following compounds according to the ease of nucleophilic substitution of Cl by hydroxyl anion to give the corresponding alcohols. Cl Cl Cl Cl I II III IV

(A) IV > I > III > II	**(B)** I > IV > II > III
(C) II > III > I > IV	**(D)** III > II > IV > I

Q7.

CUSAT CAT 2025 - Shift 3

Which of the following halides can undergo nucleophilic substitution under mild conditions? (i) 3-chloroprop-1-ene (ii) 1-chloropropane (iii) Benzyl chloride (iv) Chloroethene (Vinyl Chloride)

(A) (i), (ii), and (iii)	**(B)** (i) and (ii)
(C) (ii) and (iv)	**(D)** (i) and (iii)

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	A	4	C	5	D	6	A	7	A						

Alcohols, Phenols & Ethers

17 questions

Q1. *CUSAT CAT 2023 - Shift 2*

Ethylmagnesium Iodide on treatment with water gives

(A) Ethanol	**(B)** Ethene
(C) Ethane	**(D)** n-Butanol

Q2. *CUSAT CAT 2023 - Shift 2*

Phenol reacts with bromine in water at room temperature to give

(A) 4-bromophenol without the evolution of HBr	**(B)** a mixture of 2- and 4- bromophenols without the evolution of HBr
(C) the corresponding addition product 1,2,3,4,5,6-hexabromocyclohexanol	**(D)** 2,4,6-tribromophenol with the evolution of HBr

Q3. *CUSAT CAT 2023 - Shift 3*

In the following sequence of reactions, the product ‘D’ is

(A) Ethanol	**(B)** Ethyne
(C) Ethylamine	**(D)** Ethene

Q4. *CUSAT CAT 2023 - Shift 4*

Methyl isocyanate (CH –N=C=O), the chemical responsible for Bhopal gas tragedy is 3 an intermediate involved in

(A) reaction between methylamine and carbon dioxide	**(B)** reaction of acetamide (CH CONH) with Br in presence of NaOH 3 2 2
(C) reaction between methylamine and formaldehyde	**(D)** reaction of ammonium formate (HCO NH) with methanol 2 4

Q5. *CUSAT CAT 2023 - Shift 4*

Benzene diazonium chloride on treatment with phenol in the presence of NaOH gives

(A) diphenyl ether	**(B)** 4-hydroxybiphenyl
(C) an azo dye	**(D)** polyaniline, a conducting polymer

Q6. *CUSAT CAT 2023 - Shift 5*

Following reaction is an example for O peracid O O

(A) Baeyer-Villiger oxidation	**(B)** Epoxidation
(C) Fischer esterification	**(D)** Claisen reaction

Q7. *CUSAT CAT 2023 - Shift 5*

Which among the following methods is best suited for the selective preparation of methyl tert-butyl ether?

(A) Acid catalyzed addition of methanol to propyne Williamson ether synthesis involving tert-butyl bromide and sodium salt of	**(B)** methanol
(C) Williamson ether synthesis involving sodium salt of tert-butanol and methyl iodide	**(D)** Selective methylation of dimethyl ether O H O

Q8. *CUSAT CAT 2024 - Shift 1*

Ethanol can be converted to ethanal by

(A) treatment with chromic acid	**(B)** treatment with alkaline KMnO 4
(C) treatment with pyridinium chlorochromate	**(D)** treatment with OsO 4

Q9. *CUSAT CAT 2024 - Shift 1*

Hinsberg test is used to differentiate between

(A) Primary, Secondary and Tertiary alcohols	**(B)** Phenols and Aliphatic alcohols
(C) Aromatic amines and Phenols	**(D)** Primary, Secondary and Tertiary amines

Q10. *CUSAT CAT 2024 - Shift 3*

Phenol gives a violet colour with

(A) Bromine water	**(B)** Neutral FeCl solution 3
(C) Fehling solution	**(D)** Diazomethane

Q11. *CUSAT CAT 2024 - Shift 4*

Compound M (C H Br) was treated with alcoholic KOH and the product formed was 4 9 ozonolyzed and subsequently reduced to yield ethanol as the only product. What is the structure of compound M?

(A) —	**(B)** —
(C) —	**(D)** —

Q12. *CUSAT CAT 2024 - Shift 5*

Among the following, the amino acid which has phenolic side chain is?

(A) Aspartic Acid	**(B)** Asparagine
(C) Leucine	**(D)** Tyrosine

Q13. *CUSAT CAT 2025 - Shift 2*

Arrange (i) 3- nitrophenol, (ii) 3-chlorophenol, (iii) 4-nitrophenol and (iv) phenol in decreasing order of acidity

(A) (i) > (ii) > (iii) > (iv)	**(B)** (iii) >(i) > (ii) > (iv)
(C) (iv) >(i) > (ii) > (iii)	**(D)** (iv) > (iii) > (ii) > (i)

Q14.

CUSAT CAT 2025 - Shift 2

Phenol fails to

(A) release carbon dioxide from aqueous sodium bicarbonate solution	**(B)** decolorize bromine water
(C) give a violet color with neutral FeCl solution 3	**(D)** dissolve in cold dilute sodium hydroxide solution

Q15.

CUSAT CAT 2025 - Shift 3

An example of a polar protic solvent is:

(A) Acetonitrile	**(B)** Methanol
(C) Acetone	**(D)** Dimethyl formamide

Q16.

CUSAT CAT 2025 - Shift 4

Major product formed when 2-bromo-2-methylpentane is treated with NaOEt in ethanol is (Note: under these conditions Zaitsev elimination predominates):

(A) 2-methylpent-1-ene	**(B)** 2-methylpent-2-ene
(C) (E)-4-methylpent-2-ene	**(D)** (Z)-4-methylpent-2-ene

Q17.

CUSAT CAT 2025 - Shift 4

Cyclohexanol (I), 2,4,6-trichlorophenol (II), 2,4,6-trinitrophenol (III) and 2,4,6- trimethoxyphenol (IV) are given. In these, the order of decreasing acidic character will be: OH OH OH OH Cl Cl ON NO MeO OMe 2 2 Cl NO OMe 2 I II III IV

(A) III > II > IV > I	**(B)** II > III > I > IV
(C) II > III > IV > I	**(D)** III > IV > II > I

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	D	3	B	4	B	5	C	6	A	7	C	8	C	9	D	10	B
11	B	12	D	13	B	14	A	15	B	16	C	17	A						

Aldehydes, Ketones & Acids

28 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Schiff's bases are prepared by

(A) condensation of aldehydes and ketones with hydrazine	**(B)** condensation of aldehydes and ketones with secondary amines
(C) condensation of aldehydes and ketones with primary amines	**(D)** condensation of aldehydes and hydroxylamine

Q2. *CUSAT CAT 2023 - Shift 2*

Triphenyl carbinol can be prepared in one pot by the reaction of two equivalents of phenylmagnesium bromide with Ph Ph C OH Ph triphenyl carbinol

(A) C H COOH (benzoic acid) 6 5	**(B)** C H CN (benzonitrile) 6 5
(C) PhCOOCH (methyl benzoate) 3	**(D)** PhCHO (benzaldehyde)

Q3. *CUSAT CAT 2023 - Shift 2*

Base catalyst condensation between two molecules of acetaldehyde to give 3-hydroxybutanal is an example for

(A) Stobbe condensation	**(B)** Benzoin condensation
(C) Aldol condensation	**(D)** Perkin reaction

Q4. *CUSAT CAT 2023 - Shift 3*

What is the main chemical component of oil of wintergreen?

(A) Methyl salicylate	**(B)** Menthol
(C) Benzaldehyde	**(D)** Camphor

Q5. *CUSAT CAT 2023 - Shift 3*

Which among the following statements is incorrect about Hunsdiecker reaction?

(A) It proceeds through a carbocation intermediate	**(B)** Silver salts of carboxylic acids are used in this reaction
(C) One equivalent of CO is liberated in this reaction 2	**(D)** It is not suitable for the generation of alkyl fluoride

Q6. *CUSAT CAT 2023 - Shift 3*

A 1 : 1 mixture of benzaldehyde and formaldehyde on heating with concentrated aq NaOH solution gives

(A) sodium benzoate and methyl alcohol	**(B)** methyl benzoate
(C) disodium salt of phthalic acid	**(D)** benzyl alcohol and sodium formate

Q7. *CUSAT CAT 2023 - Shift 4*

Which among the following is an example for Diels-Alder reaction?

(A) Reaction of benzene with acetyl chloride in the presence of anhydrous AlCl to 3 give acetophenone	**(B)** Reaction between 1,3-butadiene and ethene to give cyclohexene
(C) Anhydrous AlCl catalyzed rearrangement of phenyl acetate to give a mixture 3 of 2- and 4-hydroxyacetophenones	**(D)** Acid catalyzed rearrangement of 1,2-diols to give ketones

Q8. *CUSAT CAT 2023 - Shift 4*

Which among following compounds, fails to undergo Cannizzaro reaction?

(A) Benzaldehyde	**(B)** Trichloroacetaldehyde
(C) 2,2-dimethylpropanal	**(D)** Formaldehyde

Q9. *CUSAT CAT 2023 - Shift 4*

2-Hydroxybenzaldehyde can be prepared by treating Phenol with

(A) Formaldehyde in the presence of anhydrous aluminium chloride	**(B)** Carbon dioxide under pressure in sodium hydroxide solution
(C) Chloroform and concentrated sodium hydroxide	**(D)** Carbon monoxide and 5N HCl

Q10. *CUSAT CAT 2023 - Shift 4*

The chemical reaction between aldehydes and Borche's reagent (2,4-dinitrophenylhydrazine) to give an orange coloured precipitate is an example of

(A) Condensation reaction	**(B)** Addition reaction
(C) Substitution reaction	**(D)** Polymerization reaction

Q11. *CUSAT CAT 2023 - Shift 4*

Formation of phenol-formaldehyde resin involves the reaction between

(A) three equivalents of formaldehyde with one equivalent of phenol	**(B)** one equivalent of formaldehyde with three equivalents of phenol
(C) one equivalent of formaldehyde with one equivalent of phenol	**(D)** six equivalents of formaldehyde with one equivalent of phenol

Q12. *CUSAT CAT 2023 - Shift 5*

Identify the product 'Z' in the following sequence of reactions

(A) Aspirin	**(B)** Salicylaldehyde
(C) Benzoic acid	**(D)** Salicylic acid

Q13. *CUSAT CAT 2023 - Shift 5*

An organic compound that readily undergoes Cannizaro reaction but does not give a positive test with Fehling's solution is

(A) methanal	**(B)** 2,2-dimethylpropanal
(C) benzaldehyde	**(D)** acetone

Q14. *CUSAT CAT 2024 - Shift 1*

Product/products formed in the reaction of benzaldehyde (PhCHO) with sodium hydroxide followed by acidification is/are

(A) PhCO H + HCHO 2	**(B)** PhCH(OH)CO H 2
(C) PhCH OH + HCHO 2	**(D)** PhCH OH + PhCO H 2 2

Q15. *CUSAT CAT 2024 - Shift 1*

How many moles of formic acid and formaldehyde are formed when 1 mole of glucose (C H O) is oxidized using excess HIO ? 6 12 6 4

(A) 2 and 4 respectively	**(B)** 3 and 3 respectively
(C) 4 and 2 respectively	**(D)** 5 and 1 respectively

Q16. *CUSAT CAT 2024 - Shift 3*

Pick the wrong statement regarding Clemmensen reduction and Wolff-Kishner reduction

(A) Clemmensen reduction is more suitable for acid sensitive compounds while Wolff-Kishner reduction is more suitable for base sensitive compounds	**(B)** Zinc amalgam is used as a reagent in Clemmensen reduction while hydrazine is used as a reagent in Wolff-Kishner reduction.
(C) Both are useful to reduce ketones and aldehydes to the corresponding hydrocarbons	**(D)** Both reactions are successful with aromatic and aliphatic aldehydes and ketones

Q17. *CUSAT CAT 2024 - Shift 4*

Chemically coconut oil is a/an

(A) Alcohol	**(B)** Acid
(C) Ester	**(D)** Ketone

Q18. *CUSAT CAT 2024 - Shift 4*

Primary alcohols having two additional carbons can be prepared from Grignard reagents by the reaction with

(A) ethylene oxide	**(B)** acetaldehyde
(C) Acetylene	**(D)** Two equivalents of Formaldehyde

Q19. *CUSAT CAT 2024 - Shift 5*

Which of the following is a polyamide?

(A) Polyurethane	**(B)** Nylon 6
(C) Polyaniline	**(D)** Melamine formaldehyde resin

Q20. *CUSAT CAT 2025 - Shift 1*

Correct IUPAC nomenclature for the following is

(A) 2-butan-(2-cyclohexanone) ketone	**(B)** 2-(3-oxobutyl)cyclohexanone
(C) 2-(2-oxobutyl)cyclohexanone	**(D)** 2-oxobutan-2-cyclohexanone

Q21. *CUSAT CAT 2025 - Shift 1*

A monomer of polyurethane is

(A) diisocyanate	**(B)** diisocyanide
(C) diamine	**(D)** dicarboxylic acid

Q22. *CUSAT CAT 2025 - Shift 1*

Cyclic trimer of formaldehyde is

(A) paraformaldehyde	**(B)** paraldehyde
(C) 1,3,5-trioxane	**(D)** bakelite

Q23. *CUSAT CAT 2025 - Shift 1*

Under normal conditions, which of the following reactions is unsuccessful with benzaldehyde?

(A) Oxidation with Fehling's reagent	**(B)** Cannizaro reaction
(C) Aldol condensation	**(D)** Benzoin condensation

Q24. *CUSAT CAT 2025 - Shift 2*

In Wolff-Kishner reduction, in the first step ketones (or aldehydes) are converted to the corresponding

(A) imine	**(B)** hydrazone
(C) semicarbazone	**(D)** cyanohydrin

Q25. *CUSAT CAT 2025 - Shift 2*

Which reagent can be used to efficiently convert carboxylic acid to acid chloride derivative?

(A) SOCl 2	**(B)** Cl in the presence of light 2
(C) Interhalogen compound bromine monochloride (Br-Cl) in the presence of peroxides	**(D)** Sodium hypochlorite (NaOCl)

Q26.

CUSAT CAT 2025 - Shift 3

Benzoic acid is treated with thionyl chloride to give compound Q which is converted to benzaldehyde as shown below. What reagent is required to convert Q to benzaldehyde? PhCOOH → Q → PhCHO

(A) NH Cl 4	**(B)** H /Pd-C 2
(C) H , Pd/BaSO 2 4	**(D)** O , Pt 2

Q27.

CUSAT CAT 2025 - Shift 4

Which among the following carboxylic acid derivatives is least reactive towards dilute NaOH solution in water?

(A) Amide	**(B)** Ester
(C) Acid chloride	**(D)** Anhydride

Q28.

CUSAT CAT 2025 - Shift 4

What is common for both Cannizzaro reaction and Meerwein-Pondroff-Verley reduction?

(A) Aluminium triisopropoxide is used in both reactions	**(B)** Cannot be performed successfully with aromatic aldehydes
(C) Both reactions give a 1:1 mixture of the corresponding alcohol and carboxylic acid from aliphatic aldehydes	**(D)** Both reactions involve hydride (H–) transfer in a key step.

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	C	4	A	5	A	6	D	7	B	8	B	9	C	10	A
11	A	12	D	13	C	14	D	15	D	16	A	17	B	18	A	19	B	20	B
21	A	22	C	23	A	24	B	25	A	26	C	27	A	28	D				

Amines

7 questions

Q1. *CUSAT CAT 2023 - Shift 1*

An amino acid having secondary amine component is

(A) Alanine	**(B)** Tryptophan
(C) Asparagine	**(D)** Proline

Q2. *CUSAT CAT 2023 - Shift 1*

Diethyl oxalate forms a solid oxamide derivative with

(A) both primary and secondary amines	**(B)** both secondary and tertiary amines
(C) primary amines only	**(D)** aromatic secondary amines only

Q3. *CUSAT CAT 2023 - Shift 2*

Which among the following compounds is NOT formed in the positive test for nitrogen in Lassaigne's test for aniline?

(A) Fe [Fe(CN)] 4 6 3	**(B)** Na [Fe(CN)] 3 6
(C) Fe(CN) 3	**(D)** Na [Fe(CN) NOS] 3 5

Q4. *CUSAT CAT 2023 - Shift 3*

Aniline is selectively converted to 4-bromoaniline in high yields by

(A) treatment with bromine water	**(B)** conversion to acetanilide by treatment with acetic anhydride followed by bromination and hydrolysis
(C) treatment with bromine in the presence of a halogen carrier	**(D)** treatment with bromine in CCl in the presence of light 4

Q5. *CUSAT CAT 2023 - Shift 4*

Choose the weakest base among the following

(A) 4-chloroaniline	**(B)** 4-methylaniline
(C) 4-methoxyaniline	**(D)** 4-nitroaniline

Q6. *CUSAT CAT 2023 - Shift 5*

The foul smelling compound formed by the action of alcoholic caustic potash on chloroform and aniline is

(A) Phenyl isocyanide (C H NC) 6 5	**(B)** Benzotrichloride (C H CCl) 6 5 3
(C) Benonitrile (C H CN) 6 5	**(D)** Phenyl isocyanate (C H NCO) 6 5

Q7.

CUSAT CAT 2025 - Shift 2

Aniline is successively treated with i) excess bromine ii) NaNO , HCl iii) H PO . 2 3 2 Major product formed is

(A) 2,4,6-tribromoaniline	**(B)** 2,4-dibromoaniline
(C) 1,3,5-tribromobenzene	**(D)** 2,4-dibromobenzene

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	C	3	A	4	B	5	D	6	A	7	C						

Biomolecules

29 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Most crucial function of insulin in human body is to

(A) regulate blood sugar level	**(B)** regulate kidney function
(C) regulate hormone balance in body	**(D)** regulate hydrolysis of starch

Q2. *CUSAT CAT 2023 - Shift 2*

What is the function of heme in the human body?

(A) It helps transport oxygen in the blood	**(B)** It helps to digest food in the stomach
(C) It is a structural component of cell membranes	**(D)** It is a cofactor in enzymes involved in DNA replication

Q3. *CUSAT CAT 2023 - Shift 2*

Which among the following essential amino acids has sulphur containing side chain?

(A) Glycine	**(B)** Cysteine
(C) Methionine	**(D)** Tyrosine

Q4. *CUSAT CAT 2023 - Shift 3*

Among the following, the amino acid which does not have aromatic/heteroaromatic ring residue is

(A) Tyrosine	**(B)** Asparagine
(C) Tryptophan	**(D)** Histidine

Q5. *CUSAT CAT 2023 - Shift 3*

Maltose is a disaccharide made up of

(A) α-D-glucose only	**(B)** one each of α- and β-D-glucose units
(C) β-D-glucose and D-fructose	**(D)** two D-fructose units only

Q6. *CUSAT CAT 2023 - Shift 4*

Which among the following natural amino acids is achiral?

(A) Threonine	**(B)** Tyrosine
(C) Alanine	**(D)** Glycine

Q7. *CUSAT CAT 2023 - Shift 4*

Among vitamins A, C, and E, which is/are considered as dietary antioxidant/s?

(A) Vitamin C only	**(B)** Vitamins A and C only
(C) Vitamins C and E only	**(D)** Vitamins A, C and E

Q8. *CUSAT CAT 2023 - Shift 5*

Nucleic acids are biopolymers containing

(A) phosphite linkers	**(B)** hexoses
(C) amino acids	**(D)** purine and pyrimidine bases +

Q9. *CUSAT CAT 2023 - Shift 5*

Which among the following Vitamins helps to release energy from food?

(A) K	**(B)** A
(C) B	**(D)** D

Q10. *CUSAT CAT 2024 - Shift 1*

Function of adrenaline hormone is

(A) to make heart beat faster and lungs breathe more efficiently	**(B)** to control metabolism of glucose
(C) to reduce fat burning	**(D)** to control calcium metabolism

Q11. *CUSAT CAT 2024 - Shift 1*

An example of a sulphur containing amino acid present in human hair is

(A) Lysine	**(B)** Valine
(C) Serine	**(D)** Cysteine

Q12. *CUSAT CAT 2024 - Shift 2*

Which is the reserve carbohydrate in animals?

(A) Starch	**(B)** Pectin
(C) Glycogen	**(D)** Inulin

Q13. *CUSAT CAT 2024 - Shift 3*

A Natural amino acid having benzyl side chain is

(A) Alanine	**(B)** Valine
(C) Asparagine	**(D)** Phenylalanine

Q14. *CUSAT CAT 2024 - Shift 3*

Molisch's test is used to check the presence of

(A) Carbohydrates	**(B)** Steroids
(C) Alkaloids	**(D)** Amino acids

Q15. *CUSAT CAT 2024 - Shift 4*

Which among the elements given below is present in the hormone thyroxine?

(A) Cl	**(B)** I
(C) Ca	**(D)** Mg

Q16. *CUSAT CAT 2024 - Shift 4*

Name the following amino acid NH 2 CO H 2 N H

(A) Glycine	**(B)** Aspartic Acid
(C) Tryptophan	**(D)** Tyrosine

Q17. *CUSAT CAT 2024 - Shift 5*

An epimer of glucose is (remember, All Altruists Gladly Make Gum In Gallon Tanks):

(A) Mannose	**(B)** Fructose
(C) Allose	**(D)** Tallose

Q18. *CUSAT CAT 2024 - Shift 5*

Coagulation of blood is promoted by Vitamin

(A) B	**(B)** C
(C) D	**(D)** K

Q19. *CUSAT CAT 2024 - Shift 5*

Which among the following is a β-lactam antibiotic?

(A) Chloramphenicol	**(B)** Penicillin
(C) Vitamin K	**(D)** Griesofulvin

Q20. *CUSAT CAT 2024 - Shift 5*

The metal atoms present in Haemoglobin and Vitamin B respectively is 12

(A) Fe, Mg	**(B)** Fe, Co
(C) Co, Fe	**(D)** Co, Mg

Q21. *CUSAT CAT 2025 - Shift 1*

Hydrolysis of sucrose yields

(A) glucose + maltose	**(B)** glucose + fructose
(C) glucose + galactose	**(D)** maltose + fructose

Q22. *CUSAT CAT 2025 - Shift 1*

The deficiency of which of the following vitamins causes scurvy?

(A) Vitamin A	**(B)** Vitamin B
(C) Vitamin C	**(D)** Vitamin D

Q23. *CUSAT CAT 2025 - Shift 2*

Which among the following amino acids has an aromatic ring residue?

(A) Tryptophan	**(B)** Valine
(C) Isoleucine	**(D)** Aspartic Acid

Q24. *CUSAT CAT 2025 - Shift 2*

Among the following saccharides, identify the one that does not form silver mirror with Tollen's reagent

(A) Lactose	**(B)** Maltose
(C) Galactose	**(D)** Sucrose

Q25. *CUSAT CAT 2025 - Shift 3*

Which among the following amino acids does not have a primary amino (NH) 2 group?

(A) Serine	**(B)** Proline
(C) Tyrosine	**(D)** Lysine

Q26. *CUSAT CAT 2025 - Shift 3*

The term invert sugar refers to an equimolar mixture of:

(A) D-glucose and D-galactose	**(B)** D-glucose and D-fructose
(C) D-glucose and D-mannose	**(D)** D-glucose and D-ribose

Q27. *CUSAT CAT 2025 - Shift 4*

Which of the following amino acid has a –CH OH residue? 2

(A) Tyrosine	**(B)** Serine
(C) Threonine	**(D)** Lysine

Q28. *CUSAT CAT 2025 - Shift 4*

Preservative used widely in processed food is:

(A) Saccharin	**(B)** Menthol
(C) Corn starch	**(D)** Sodium benzoate

Q29.

CUSAT CAT 2025 - Shift 4

Which among the following vitamins is considered as a steroid?

(A) A	**(B)** E
(C) D	**(D)** K

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	A	3	C	4	B	5	A	6	D	7	D	8	D	9	C	10	A
11	D	12	C	13	D	14	A	15	B	16	C	17	A	18	D	19	B	20	B
21	B	22	C	23	A	24	D	25	B	26	B	27	B	28	D	29	C		

Polymers

10 questions

Q1. *CUSAT CAT 2023 - Shift 3*

Dynel, polymer used for hair wigs, is a

(A) copolymer of butadiene and styrene	**(B)** copolymer of vinyl chloride and acrylonitrile
(C) polyamide resin	**(D)** cross linked polystyrene

Q2. *CUSAT CAT 2023 - Shift 4*

Atropisomerism in stereoisomerism originating due to

(A) restricted rotation around a single bond	**(B)** four groups attached to carbon are different
(C) helical nature of polymers	**(D)** sheet like geometry of polymers

Q3. *CUSAT CAT 2023 - Shift 4*

Styrene-butadiene rubber finds application in

(A) Styrofoam	**(B)** Bubble gums
(C) Flame retardant coatings	**(D)** High-performance elastomer for space application

Q4. *CUSAT CAT 2023 - Shift 5*

Teflon coated cookware is losing popularity because

(A) Teflon is very expensive	**(B)** Teflon is highly carcinogenic
(C) PFOA used in creating Teflon coating is not good for human health	**(D)** Bakelite is a now emerging as a more economical alternative

Q5. *CUSAT CAT 2024 - Shift 1*

Nylon 6,6 is prepared by reacting

(A) Hexamethylenediamine with adipic acid by condensation reaction	**(B)** Hexamethylenediamine with succinic acid by addition reaction
(C) Hexamethylenediamine with glutaric acid by elimination reaction	**(D)** Ring opening polymerization of ε-caprolactam

Q6. *CUSAT CAT 2024 - Shift 2*

Nucleic acids are biopolymers. The monomer units present in them

(A) Pyrimidine and purine bases	**(B)** Sugars
(C) Nucleosides	**(D)** Nucleotides

Q7. *CUSAT CAT 2024 - Shift 4*

Nitric oxide is paramagnetic in

(A) Gaseous state	**(B)** Liquid state
(C) Solid state	**(D)** Polymeric state

Q8. *CUSAT CAT 2025 - Shift 2*

Which of the following is not an addition polymer?

(A) Teflon	**(B)** Plexiglass
(C) Orlon	**(D)** Kevlar

Q9. *CUSAT CAT 2025 - Shift 4*

Styrene (PhCH=CH) is not used as a monomer in the industrial production of which 2 among the following polymers?

(A) ABS	**(B)** PPS
(C) SAN	**(D)** SBR

Q10. *CUSAT CAT 2025 - Shift 4*

The Catalyst used for stereoregular polymerization of propene is:

(A) Ziegler-Natta catalyst	**(B)** Wilkinson's catalyst
(C) Lindlar catalyst	**(D)** Zeise's salt complex

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	A	3	B	4	C	5	A	6	D	7	A	8	D	9	B	10	A

Chemistry in Everyday Life

2 questions

Q1. *CUSAT CAT 2024 - Shift 4*

Kjeldahl's method is a

(A) Quantitative method for determining nitrogen in an organic compound	**(B)** Qualitative method for detecting the presence of nitrogen and halogens in an organic compound
(C) Quantitative method for determining sulphur in pesticides	**(D)** Method for determining iodine number of oils and fats

Q2. *CUSAT CAT 2025 - Shift 2*

Pyrethrins are well-known as

(A) Anti-oxidants	**(B)** Insecticides
(C) Flavouring agent	**(D)** Drugs

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	B																

Chemistry – Miscellaneous

457 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A radioactive source has a half-life of 40 s, how long will it take for of the source 8 to decay?

(A) 127 s	**(B)** 57 s
(C) 147 s	**(D)** 560 s

Q2. *CUSAT CAT 2023 - Shift 1*

Which of the following two gases can be cooled from room temperature by the Joule-Thomson effect?

(A) Hydrogen and Oxygen	**(B)** Helium and Nitrogen
(C) Helium and Hydrogen	**(D)** Nitrogen and Oxygen

Q3. *CUSAT CAT 2023 - Shift 1*

The increase in internal energy of the system is 100 J when 300 J heat is supplied to it. What is the amount of work done by the system?

(A) 100 J	**(B)** 200 J
(C) 300 J	**(D)** 400 J

Q4. *CUSAT CAT 2023 - Shift 1*

A container of volume 5.0 L is divided into two compartments of equal size. In the left compartment there is nitrogen at 1.0 atmosphere pressure and 25°C, in the right compartment there is hydrogen at the same temperature and pressure. What will happen when the partition is removed?

(A) The entropy increases, and the free energy decreases	**(B)** The entropy decreases, and the free energy decreases
(C) The entropy increases, and the free energy increases	**(D)** The entropy decreases, and the free energy increases

Q5. *CUSAT CAT 2023 - Shift 1*

The hydrogen electrode is dipped in a solution of pH 3 at 25°C, the potential of the 2.303RT cell would be (the value of =0.059V) F

(A) +0.177 V	**(B)** –0.177 V
(C) +0.087 V	**(D)** +0.059 V

Q6. *CUSAT CAT 2023 - Shift 1*

To protect iron against corrosion the most durable metal plating on it is

(A) Tin plating	**(B)** Copper plating
(C) Zinc plating	**(D)** Nickel plating –1

Q7. *CUSAT CAT 2023 - Shift 1*

The pH of the blood is maintained by buffer system given by

(A) NaCl and HCl	**(B)** NH Cl and NH OH 4 4
(C) Sodium citrate and Citric acid	**(D)** HCO – and H CO 3 2 3

Q8. *CUSAT CAT 2023 - Shift 1*

The Clausius-Clayperon equation helps to calculate

(A) Latent heat of vaporization	**(B)** Melting point of the solvent
(C) Heat of neutralization	**(D)** Molecular weight of solute

Q9. *CUSAT CAT 2023 - Shift 1*

Electrical conductivity of an electrolyte depends upon the

(A) number of molecules in the electrolytes	**(B)** number of ions present in the electrolytes
(C) number of molecules present in the solvent	**(D)** number of charges present in the solution + 2+ 3+ 4+

Q10. *CUSAT CAT 2023 - Shift 1*

Addition of Ag , Pb , Fe and Si causes coagulation of negatively charged colloidal sol. Then which of the following is true? 3+ 4+ 2+ +

(A) Fe > Si > Pb > Ag + 2+ 3+ 4+	**(B)** Ag < Pb < Fe < Si + 2+ 3+ 4+
(C) Ag > Pb > Fe > Si + 2+ 3+ 4+	**(D)** Ag = Pb = Fe = Si

Q11. *CUSAT CAT 2023 - Shift 1*

The root mean square velocity (u), average velocity (p) and most probable velocity (q) of a molecule are related to each other. Then which of the following is true?

(A) q > p > u	**(B)** p > q > u
(C) u > p > q	**(D)** u = p = q

Q12. *CUSAT CAT 2023 - Shift 1*

A plant cell shrinks when it is kept in

(A) Hypertonic solution	**(B)** Hypotonic solution
(C) Water	**(D)** Isotonic solution with cell sap

Q13. *CUSAT CAT 2023 - Shift 1*

One Einstein of energy is 2.859 5 –1 E = ×10 cal mol

(A) λ 2.859 –5 –1 E = ×10 k cal mol	**(B)** λ 2.859 5 –1 E = ×10 J mol
(C) λ 2.859 –5 –1 E = ×10 kJ mol	**(D)** λ

Q14. *CUSAT CAT 2023 - Shift 1*

The correct unit of Van der Waal's constant 'a' is 2 –2

(A) atm mol L –2 2	**(B)** atm mol L –1 –2 2
(C) atm mol L 2 2	**(D)** atm mol L

Q15. *CUSAT CAT 2023 - Shift 1*

If travelling at same speeds, which of the following matter waves have the shortest wave length?

(A) Electron	**(B)** Alpha particle
(C) Neutron	**(D)** Proton

Q16. *CUSAT CAT 2023 - Shift 1*

In H -O fuel cell, the reaction occurring at cathode is 2 2 – –

(A) 2H O + O + 4e → 4OH 2 2	**(B)** 2H + O → 2H O 2 2 2 –
(C) H + OH → H O 2 –	**(D)** H + e → ½ H 2 –5

Q17. *CUSAT CAT 2023 - Shift 1*

Number of isomers possible for C2H2Cl2 is/are

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q18. *CUSAT CAT 2023 - Shift 1*

Identify X in the following chemical reaction O Br 2 /KOH X CH 3 OH NH OCH NH 3 2 O

(A) —	**(B)** —
(C) —	**(D)** —

Q19. *CUSAT CAT 2023 - Shift 1*

Turmeric is a good source for

(A) Curcumin	**(B)** Nicotine
(C) Terramycin	**(D)** Anabasine

Q20. *CUSAT CAT 2023 - Shift 1*

2-Methylbutane on reacting with I in presence of sunlight gives mainly 2

(A) 2-iodo-2-methylbutane	**(B)** unchanged 2-methylbutane
(C) 1-iodo-2-methylbutane	**(D)** 1-iodo-3-methylbutane

Q21. *CUSAT CAT 2023 - Shift 1*

In the reaction below, A and B are respectively

(A) CH COOH, Cl CCOOH 3 3	**(B)** CH CH OH, CH CH Cl 3 2 3 2
(C) CH CHO, Cl CCHO 3 3	**(D)** CH COCH , Cl CCOCH 3 3 3 3

Q22. *CUSAT CAT 2023 - Shift 1*

Acetic anhydride is a controlled substance in India since it is

(A) highly toxic	**(B)** used in the synthesis of illegal substances like heroin
(C) extremely expensive	**(D)** highly explosive

Q23. *CUSAT CAT 2023 - Shift 1*

Iodoform (CHI) on heating with Ag powder forms 3

(A) Ethyne	**(B)** Ethene
(C) Ethane	**(D)** Methanoic acid

Q24. *CUSAT CAT 2023 - Shift 1*

One mole of methoxybenzene (anisole, C H OCH) on heating with conc. HI gives 6 5 3

(A) One mole of CH OH and one mole of C H I 3 6 5	**(B)** One mole of CH I and one mole of C H OH 3 6 5
(C) One mole of CH I and one mole of C H I 3 6 5	**(D)** 4-Iodoanisole as the only product

Q25. *CUSAT CAT 2023 - Shift 1*

In the following reaction sequence The end product

(A) —	**(B)** —
(C) is	**(D)** Ethane

Q26. *CUSAT CAT 2023 - Shift 1*

The group having isoelectronic species is 2– – + 2+

(A) O , F , Na , Mg – – +	**(B)** O , F , Na, Mg 2– – 2+
(C) O , F , Na, Mg – – + 2+	**(D)** O , F , Na , Mg

Q27. *CUSAT CAT 2023 - Shift 1*

The number and type of bonds in C ion in CaC are 2 2

(A) One σ bond and one ■ – bond	**(B)** One σ bond and two ■ – bonds
(C) Two σ bonds and two ■ – bonds	**(D)** Two σ bonds and one ■ – bond

Q28. *CUSAT CAT 2023 - Shift 1*

Which of the following is an amphoteric hydroxide?

(A) Be(OH) 2	**(B)** Ca(OH) 2
(C) Mg(OH) 2	**(D)** Sr(OH) 2

Q29. *CUSAT CAT 2023 - Shift 1*

0.5 moles of gas A and x moles of gas B exert pressure of 200 Pa in a container of 3 –1 –1 volume 10 m at 1000 K. Given R is the gas constant in JK mol , x is 2

(A) 4+ 2	**(B)** 4– 4+
(C) 2 4–	**(D)** 2

Q30. *CUSAT CAT 2023 - Shift 1*

In basic medium, H O exhibits which of the following reactions? 2 2 2+ 4+ – (a) Mn → Mn (b) I → I (c) PbS → PbSO 2 4 Choose the most appropriate answer from the options given below.

(A) (a) and (c)	**(B)** (a) and (b)
(C) (a) only	**(D)** (b) only

Q31. *CUSAT CAT 2023 - Shift 1*

The number of P-O bonds in P O is 4 6

(A) 9	**(B)** 6
(C) 12	**(D)** 18

Q32. *CUSAT CAT 2023 - Shift 1*

Which of the following has the highest crystal field stabilization energy? 3–

(A) [Fe(OH)] 5 3–	**(B)** [Fe(Cl)] 6 3–
(C) [Fe(CN)] 6 3+	**(D)** [Fe(H O)] 2 6 3+

Q33. *CUSAT CAT 2023 - Shift 1*

Magnetic moment of Gd ion (Z = 64) is

(A) 3.62 BM	**(B)** 9.72 BM
(C) 7.9 BM	**(D)** 10.60 BM

Q34. *CUSAT CAT 2023 - Shift 1*

The coordination number of Th in K [Th(C O) (H O)] is 4 2 4 4 2 2

(A) 14	**(B)** 6
(C) 8	**(D)** 10

Q35. *CUSAT CAT 2023 - Shift 1*

How many number of molecules and atoms are present in 2.24 L of a diatomic gas at STP? 22 22

(A) 3.0 × 10 ; 6.0 × 10 23 22	**(B)** 6.02 × 10 ; 15.0 × 10 22 22
(C) 6.0 × 10 ; 12.0 × 10 22 22	**(D)** 15.0 × 10 ; 7.5 × 10

Q36. *CUSAT CAT 2023 - Shift 1*

Which of the following series of lines are the only lines in the hydrogen spectrum, which appear in the near infrared region?

(A) Lyman	**(B)** Balmer
(C) Paschen	**(D)** Brackett

Q37. *CUSAT CAT 2023 - Shift 1*

Number of angular nodes for 4f orbital

(A) 4	**(B)** 3
(C) 2	**(D)** 1

Q38. *CUSAT CAT 2023 - Shift 1*

0.2 mol AgCl is obtained, when 0.1 mol of the complex, MCl (NH) ,is treated with 3 3 5 excess of AgNO . The complex will be 3

(A) 1 : 3 electrolyte	**(B)** 1 : 2 electrolyte
(C) 1 : 1 electrolyte	**(D)** 3 : 1 electrolyte

Q39. *CUSAT CAT 2023 - Shift 1*

The most stable radioactive isotope of hydrogen is

(A) deuterium	**(B)** hydronium
(C) protium	**(D)** tritium MATHEMATICS UG (SHIFT – I FINAL) 2 2

Q40. *CUSAT CAT 2023 - Shift 2*

Which of the following sets of conditions makes a process spontaneous at all temperatures?

(A) $\Delta H = 0; \Delta S > 0$	**(B)** $\Delta H = 0; \Delta S < 0$
(C) $\Delta H < 0; \Delta S > 0$	**(D)** $\Delta H < 0; \Delta S < 0$

Q41. *CUSAT CAT 2023 - Shift 2*

The alkali hydrolysis of an ester represented by – RCOOR' + OH RCOO– + R'OH This reaction is,

(A) bimolecular and second-order	**(B)** bimolecular but first-order
(C) bimolecular but not second-order	**(D)** second-order but not bimolecular

Q42. *CUSAT CAT 2023 - Shift 2*

Which of the following can be used to measure pH?

(A) Glass electrode	**(B)** Hydrogen electrode
(C) Quinhydrone electrode	**(D)** All of the above

Q43. *CUSAT CAT 2023 - Shift 2*

Kinetic energy of a single molecule is given by the expression

(A) RT	**(B)** RT/N
(C) kT	**(D)** 1.5RT/N

Q44. *CUSAT CAT 2023 - Shift 2*

In a reversible isothermal process change in internal energy

(A) $\Delta U = 0$	**(B)** $\Delta U > 0$
(C) $\Delta U < 0$	**(D)** $\Delta U = P\Delta V$

Q45. *CUSAT CAT 2023 - Shift 2*

Heat of formation of sulphur dioxide, $\Delta H = -297$ kJ. The heat liberated on burning 8 g of sulphur in oxygen is

(A) 297 kJ	**(B)** 223 kJ
(C) 148.5 kJ	**(D)** 74.25 kJ

Q46. *CUSAT CAT 2023 - Shift 2*

In a second order reaction $2A \rightarrow$ products, if the concentration of A is doubled, t of 1/2 the reaction

(A) doubled	**(B)** quadrupled
(C) halved	**(D)** unchanged

Q47. *CUSAT CAT 2023 - Shift 2*

The ionic strength (μ) for 0.1 M BaCl is given by 2

(A) 0.3 M	**(B)** 0.1 M
(C) 0.2 M	**(D)** 0.6 M

Q48. *CUSAT CAT 2023 - Shift 2*

If 60 calories are added to a system and system does work of 20 calories on the surroundings, the change in internal energy of system?

(A) 20 calories	**(B)** 50 calories
(C) 40 calories	**(D)** 30 calories –1

Q49. *CUSAT CAT 2023 - Shift 2*

For certain reaction R→P, the value of ΔH and ΔS are 60 KJ and 80 JK respectively. The temperature at which ΔG = 0 is

(A) 477°C	**(B)** 750°C
(C) 800°C	**(D)** 450°C

Q50. *CUSAT CAT 2023 - Shift 2*

Root mean square velocity of molecule A is four times greater than that of molecule B at same temperature. The ratio of molecular weight of A to B (M : M) A B

(A) 1 : 4	**(B)** 2 : 1
(C) 4 : 1	**(D)** 1 : 16

Q51. *CUSAT CAT 2023 - Shift 2*

If a spoon of copper metal is placed in a solution of ferrous sulphate, then

(A) Cu will precipitate out	**(B)** Iron will precipitate
(C) Cu and Fe will precipitate	**(D)** no reaction will take place

Q52. *CUSAT CAT 2023 - Shift 2*

The vapour pressure of two liquids P and Q are 80 and 60 Torr respectively. The total vapour pressure of the solution obtained by mixing three moles of P and two moles of Q would be

(A) 68 Torr	**(B)** 72 Torr
(C) 140 Torr	**(D)** 20 Torr d[B]

Q53. *CUSAT CAT 2023 - Shift 2*

Unit of third order reaction 2 –2 –1

(A) L mol t –1 –1 –1	**(B)** mol L t –1 –1
(C) mol L S –1 –1	**(D)** mol S –3

Q54. *CUSAT CAT 2023 - Shift 2*

The orbital angular momentum of an electron in 2s orbital is

(A) zero	**(B)** one
(C) two	**(D)** three

Q55. *CUSAT CAT 2023 - Shift 2*

Following reaction involves CH 3 CH 3 CH 3 HCl CH 3 CH CH 2 CH 3 C C CH 3 CH Cl H 3

(A) Anti Markovnikov addition and rearrangement	**(B)** Markovnikov addition and rearrangement
(C) Rearrangement followed by Markovnikov addition	**(D)** Rearrangement followed by anti Markovnikov addition

Q56. *CUSAT CAT 2023 - Shift 2*

Which among the following reaction sequences would result in retention of configuration at the reaction centre?

(A) S 1 followed by E1 N	**(B)** S 2 followed by E2 N
(C) S 2 followed by another S 2 N N	**(D)** S 2 followed by S 1 N N

Q57. *CUSAT CAT 2023 - Shift 2*

In the below sequence of reactions, ‘D’ is

(A) —	**(B)** —
(C) —	**(D)** —

Q58. *CUSAT CAT 2023 - Shift 2*

In nucleic acids the phosphate diester bond links

(A) two sugar units at the 5’, 3’ positions	**(B)** two nitrogen bases together
(C) one sugar unit with one nitrogen base	**(D)** two nucleic acid strands together

Q59. *CUSAT CAT 2023 - Shift 2*

When injured, body generates prostaglandins as a warning signal. Prostaglandins stimulate inflammation in the tissue and cause sensation of pain. Which among the following drugs can inhibit the generation of prostaglandins?

(A) Meprobamate	**(B)** Dimetap
(C) Aspirin	**(D)** Salvarsan

Q60. *CUSAT CAT 2023 - Shift 2*

Taj Mahal is being slowly disfigured and discoloured. This is primarily due to

(A) acid rain	**(B)** deposition of soot from stubble burning
(C) increased exposure to UV light due to ozone depletion	**(D)** global warming

Q61. *CUSAT CAT 2023 - Shift 2*

The correct order of the ionic radii of O , N , F , Mg , Na and Al is

(A) N 3– < O 2– < F – < Na + < Mg 2+ < Al 3+ 3+ + 2+ 2– – 3–	**(B)** Al < Na < Mg < O < F < N 3+ 2+ + – 2– 3–
(C) Al < Mg < Na < F < O < N 3– – 2– 2+ + 3+	**(D)** N < F < O < Mg < Na < Al

Q62. *CUSAT CAT 2023 - Shift 2*

Among the following molecules/ions: 2– 2– 2– C 2 , N 2 , O 2 , O 2 , which one is diamagnetic and exhibiting the highest bond length?

(A) O 2	**(B)** N 2– 2
(C) C 2– 2	**(D)** O 2– 2

Q63. *CUSAT CAT 2023 - Shift 2*

Which of the following has unpaired electron(s)?

(A) N 2 –	**(B)** O 2 2+
(C) O 2 2–	**(D)** O 2

Q64. *CUSAT CAT 2023 - Shift 2*

Which of the following metals is used in photoelectric cell?

(A) Na	**(B)** Li
(C) Rb	**(D)** Cs

Q65. *CUSAT CAT 2023 - Shift 2*

The alloy used in the construction of aircrafts is

(A) Mg-Al	**(B)** Mg-Zn
(C) Mg-Sn	**(D)** Mg-Mn

Q66. *CUSAT CAT 2023 - Shift 2*

Which one of the following substances has the highest proton affinity?

(A) H S 2	**(B)** NH 3
(C) PH 3	**(D)** H O 2

Q67. *CUSAT CAT 2023 - Shift 2*

Which of the following has maximum number of lone pairs associated with Xe?

(A) XeF 4	**(B)** XeF 6
(C) XeF 2	**(D)** XeO 3

Q68. *CUSAT CAT 2023 - Shift 2*

Which one of the following is paramagnetic? 3+

(A) [Fe(H O)] 2 6 2–	**(B)** [Ni(CN)] 4 3–
(C) [Co(CN)] 6	**(D)** [Ni(CO)] 4

Q69. *CUSAT CAT 2023 - Shift 2*

The compound used in the treatment of lead poisoning is

(A) D-penicillamine	**(B)** Desferrioxime B
(C) Cis-Platin	**(D)** EDTA

Q70. *CUSAT CAT 2023 - Shift 2*

Which of the following gases will have least volume, if 5 g of each gas is taken at the same temperature and pressure?

(A) N 2	**(B)** O 2
(C) HCl	**(D)** SO 2

Q71. *CUSAT CAT 2023 - Shift 2*

What will be the standard molar volume of Ne, if its density is 0.8999 g/L at STP?

(A) 11.2 L	**(B)** 5.6 L
(C) 2.8 L	**(D)** 22.4 L

Q72. *CUSAT CAT 2023 - Shift 2*

Salvarsan, which is used for the syphilis treatment, contains

(A) phosphorus	**(B)** arsenic
(C) antimony	**(D)** iron 4–

Q73. *CUSAT CAT 2023 - Shift 3*

A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to convert carbon monoxide to carbon dioxide. If ΔH if the enthalpy change and ΔU is the change in internal energy, then

(A) $\Delta H > \Delta U$	**(B)** $\Delta H < \Delta U$
(C) $\Delta H = \Delta U$	**(D)** $\Delta U = \Delta H = 0$

Q74. *CUSAT CAT 2023 - Shift 3*

The decay of a radioactive element exhibits the characteristics of a reaction of

(A) Zero order	**(B)** First order
(C) Second order	**(D)** Fractional order

Q75. *CUSAT CAT 2023 - Shift 3*

Two platinum electrodes were immersed in a solution of copper sulphate and electric current was passed till copper sulphate is completely electrolysed. The resultant solution contains

(A) Platinum sulphate	**(B)** Copper hydroxide
(C) Only water	**(D)** Dilute sulphuric acid

Q76. *CUSAT CAT 2023 - Shift 3*

A pressure cooker reduces cooking time for food because

(A) heat is more easily distributed in the cooking space	**(B)** the higher pressure inside the cooker crushes the food material
(C) boiling point of water involved in cooking is increased	**(D)** cooking involves chemical changes helped by a rise in temperature

Q77. *CUSAT CAT 2023 - Shift 3*

The molar conductances of NaCl, HCl and CH COONa at infinite dilution are 3 –1 2 –1 126.45, 426.16 and 91 Ohm cm mol , respectively. The molar conductance of acetic acid at infinite dilution is –1 2 –1

(A) 590.71 ohm cm mol –1 2 –1	**(B)** 698.28 ohm cm mol –1 2 –1
(C) 217.45 ohm cm mol –1 2 –1	**(D)** 390.71 ohm cm mol

Q78. *CUSAT CAT 2023 - Shift 3*

In salt bridge, KCl is used because

(A) KCl is present in the calomel electrode + –	**(B)** K and Cl ions have same transport number + –
(C) K and Cl ions are isoelectronic	**(D)** KCl is a strong electrolyte

Q79. *CUSAT CAT 2023 - Shift 3*

E o =0.80 V and E o =-0.25 V. The EMF of the cell Ni-Ag is + 2+ Ag Ag Ni Ni

(A) +0.21 V	**(B)** +1.05 V
(C) –2.10 V	**(D)** –1.05 V

Q80. *CUSAT CAT 2023 - Shift 3*

40 mg of pure sodium hydroxide is dissolved in 10 litre of distilled water. The pH of the solution is

(A) 9.0	**(B)** 10.0
(C) 11.0	**(D)** 12.0

Q81. *CUSAT CAT 2023 - Shift 3*

Out of Cu, Ag, Fe and Zn, the metal which can displace all others from salt solution is

(A) Ag	**(B)** Cu
(C) Zn	**(D)** Fe

Q82. *CUSAT CAT 2023 - Shift 3*

The value of diffusion coefficient of methane at a given temperature is half of a gas X. The molecule weight of X is

(A) 4	**(B)** 32
(C) 8	**(D)** 64

Q83. *CUSAT CAT 2023 - Shift 3*

A current of 9.65 A flowing for 10 min deposits 3.0 g of a metal. The equivalent of the metal is

(A) 10	**(B)** 30
(C) 50	**(D)** 96.5

Q84. *CUSAT CAT 2023 - Shift 3*

The vapour pressure of pure liquid solvent A is 0.80 atm. When non-volatile substance B is added to the solvent its vapour pressure drops to 0.40 atm. What is the mole fraction of B in solution?

(A) 0.50	**(B)** 0.28
(C) 0.75	**(D)** 0.40

Q85. *CUSAT CAT 2023 - Shift 3*

What is the molar solubility, s, of Ba (PO) in terms of K ? 3 4 2 sp 1/2

(A) s = K sp 1/5	**(B)** s = K sp 1/5
(C) s = [K /27] sp 1/5	**(D)** s = [K /108] sp

Q86. *CUSAT CAT 2023 - Shift 3*

Identify the compound having sp, sp and sp hybridized carbons.

(A) 2,3-pentadiene	**(B)** 1,3-butadiene
(C) 2-heptyne	**(D)** phenylacetylene

Q87. *CUSAT CAT 2023 - Shift 3*

Identify X and Y in the following chemical steps OH OH Conc H 2 SO 4 X Br 2 Y H 2 O/H+ Br SO AcOH 3 OH OH

(A) X = SO 3 H Y = SO 3 H Br OH OH Br	**(B)** X = Y = SO 3 H SO 3 H OH OH
(C) X = Y = Br SO 3 H SO 3 H OH OH	**(D)** X = Y = Br SO 3 H SO 3 H

Q88. *CUSAT CAT 2023 - Shift 3*

Lindlar Catalyst is

(A) NaBH 4	**(B)** NH NH 2 2
(C) HCl/ZnCl 2	**(D)** Pd/BaSO poisoned with Quinoline 4

Q89. *CUSAT CAT 2023 - Shift 3*

Thionyl chloride mediated rearrangement of benzophenone oxime to benzanilide is an example for OH O N thionyl chloride C Ph C Ph N Ph Ph H

(A) Hofmann rearrangement	**(B)** Curtius rearrangement
(C) Schmidt rearrangement	**(D)** Beckmann rearrangement

Q90. *CUSAT CAT 2023 - Shift 3*

IUPAC name for the following compound is

(A) 2-methoxy-4-ethoxy-3-pentanone	**(B)** 2-ethoxy-4-methoxy-3-pentanone
(C) 2-ethoxy-3-methoxy-3-pentanone	**(D)** None of the above

Q91. *CUSAT CAT 2023 - Shift 3*

Which of the following is Lucas Reagent?

(A) Ammonical silver nitrate	**(B)** Br /CCl 2 4
(C) Anhy. ZnCl /conc. HCl 2	**(D)** Alk. KMnO 4

Q92. *CUSAT CAT 2023 - Shift 3*

The following reaction is called

(A) Wurtz reaction	**(B)** Kolbe's reaction
(C) Reimer-Tiemann reaction	**(D)** Schotten-Baumann reaction

Q93. *CUSAT CAT 2023 - Shift 3*

Oxides of nitrogen and sulphur, formed by burning fossil fuels, combine with oxygen in the presence of sun light to form

(A) dioxins	**(B)** thiourea
(C) smog	**(D)** ammonium sulphate nano particles

Q94. *CUSAT CAT 2023 - Shift 3*

Carbocation intermediates are involved in

(A) S N 1 and S N 2 substitution reaction	**(B)** E1 and E2 elimination reaction
(C) S 2 substitution and E2 elimination N	**(D)** S 1 substitution and E1 elimination N

Q95. *CUSAT CAT 2023 - Shift 3*

Which is the correct order of second ionization potential of C, N, O, and F in the following?

(A) O > N > F > C	**(B)** O > F > N > C
(C) F > O > N > C	**(D)** C > N > O > F

Q96. *CUSAT CAT 2023 - Shift 3*

Which of the following ions has the lowest ionic conductivity in aqueous solution? +

(A) Na +	**(B)** Rb +
(C) Li +	**(D)** K

Q97. *CUSAT CAT 2023 - Shift 3*

The element that shows greater ability to form pπ-pπ multiple bonds, is

(A) Sn	**(B)** C
(C) Ge	**(D)** Si

Q98. *CUSAT CAT 2023 - Shift 3*

The form of iron obtained from blast furnace is

(A) Steel	**(B)** Cast iron
(C) Pig iron	**(D)** Wrought iron

Q99. *CUSAT CAT 2023 - Shift 3*

The element that can be refined by distillation is

(A) Nickel	**(B)** Zinc
(C) Tin	**(D)** Gallium

Q100. *CUSAT CAT 2023 - Shift 3*

XeF on partial hydrolysis with water produces a compound ‘x’. The same compound 6 ‘x’ is formed when XeF reacts with silica. The compound ‘x’ is 6

(A) XeF 2	**(B)** XeF 4
(C) XeOF 4	**(D)** XeO 3

Q101. *CUSAT CAT 2023 - Shift 3*

The correct order of atomic radii is

(A) Nd > Dy > Ce > Yb	**(B)** Nd > Ce > Dy > Yb
(C) Ce > Dy > Yb > Nd	**(D)** Ce > Nd > Dy > Yb

Q102. *CUSAT CAT 2023 - Shift 3*

Photon of which light has maximum energy?

(A) Red	**(B)** Blue
(C) Violet	**(D)** Green

Q103. *CUSAT CAT 2023 - Shift 3*

The covalent energy is maximised in

(A) Heteronuclear molecule of AB type + –	**(B)** Heteronuclear molecule of A B type
(C) Homonuclear diatomic molecule	**(D)** None of the above

Q104. *CUSAT CAT 2023 - Shift 3*

How many moles of O can be produced during electrolytic decomposition of 90 g of 2 water?

(A) 1.25 moles	**(B)** 2.5 moles
(C) 5.0 moles	**(D)** 3.5 moles

Q105. *CUSAT CAT 2023 - Shift 3*

Which of the following complex of M (atomic number 26) will be most stable?

(A) [M(CO)] 5	**(B)** [M(CO)] 4 –––––
(C) [M(CO)] 5	**(D)** [M(CO)] 6 MATHEMATICS UG (SHIFT – III FINAL)

Q106. *CUSAT CAT 2023 - Shift 4*

The most suitable coagulating agent for ferric hydroxide sol is

(A) potassium ferricyanide	**(B)** potassium chloride
(C) potassium oxalate	**(D)** potassium sulphate

Q107. *CUSAT CAT 2023 - Shift 4*

The relation between K and K of a reversible reaction is p c Δn

(A) K = K (RT) c p Δn	**(B)** K = K (RT) p c
(C) K = K c p	**(D)** K = 1/K p c

Q108. *CUSAT CAT 2023 - Shift 4*

For a monoatomic gas, the specific heat ratio is

(A) 0	**(B)** 1.40
(C) 1.67	**(D)** 1.33 –10

Q109. *CUSAT CAT 2023 - Shift 4*

What is the solubility of AgCl in water if K = 1.6 × 10 ? sp –5

(A) 1.6 × 10 –10	**(B)** 3.2 × 10 –5
(C) 1.3 × 10 –10	**(D)** 1.6 × 10

Q110. *CUSAT CAT 2023 - Shift 4*

How many coulombs of electricity is required to reduce 1 mole of Cr O in acidic 2 7 medium?

(A) 4	**(B)** 6
(C) 2	**(D)** 1

Q111. *CUSAT CAT 2023 - Shift 4*

van’t Hoff factor of 0.005M aqueous solution of KCl is 1.95. The degree of ionization is

(A) 0.94	**(B)** 0.96
(C) 0.95	**(D)** 0.59 –5

Q112. *CUSAT CAT 2023 - Shift 4*

In a zero-order reaction for every 10°C rise of temperature the rate is doubled. If the temperature is increased from 10°C to 100°C the rate of the reaction will be

(A) 256 times	**(B)** 512 times
(C) 64 times	**(D)** 128 times

Q113. *CUSAT CAT 2023 - Shift 4*

For a second-order reaction, what is the unit of the rate of the reaction? –1

(A) s –1 –1	**(B)** mol L s –1 –1
(C) mol L s –2 2 –1	**(D)** mol L s

Q114. *CUSAT CAT 2023 - Shift 4*

The equivalent conductance of Ba and Cl are respectively 127 and 76 ––––1 ––––1 ––––1 ohm cm eq at infinite dilution. The equivalent conductance of BaCl at infinite dilution will be 2

(A) 139.5	**(B)** 203
(C) 279	**(D)** 101.5 2+ 2+

Q115. *CUSAT CAT 2023 - Shift 4*

Consider the cell reaction Cd(s)| Cd (1.0 M) || Cu (1.0 M) | Cu(s). The voltage can be lowered by 2+ 2+

(A) increasing the concentration of Cd and Cu 2+	**(B)** increasing only the concentration of Cd to 2.0 M 2+
(C) increasing only the concentration of Cu to 2.0 M	**(D)** 2+ increasing the concentration of Cd 2.0 M and decreasing the concentration of 2+ Cu 0.1 M

Q116. *CUSAT CAT 2023 - Shift 4*

The value for Henry's constant for helium, hydrogen, nitrogen and oxygen at 293 K are 144.97 kbar, 69.16 kbar, 76.48 kbar and 34.86 kbar respectively. Which of the gas will be having maximum solubility?

(A) Hydrogen	**(B)** Oxygen
(C) Helium	**(D)** Nitrogen

Q117. *CUSAT CAT 2023 - Shift 4*

Fe O (magnetite) is an example of 3 4

(A) normal spinel structure	**(B)** inverse spinel structure
(C) fluorite structure	**(D)** antifluorite structure

Q118. *CUSAT CAT 2023 - Shift 4*

Which among the following pigments is predominantly present in photosystem II of land plants?

(A) Chlorophyll a	**(B)** Chlorophyll b
(C) β-carotene	**(D)** Xanthophyll

Q119. *CUSAT CAT 2023 - Shift 4*

β-D-Glucose is represented as: OH

(A) O HO OH HO OH OH OH	**(B)** O OH HO OH OH OH O
(C) HO HO OH OH O	**(D)** HO HO OH OH

Q120. *CUSAT CAT 2023 - Shift 4*

Mono nitration of which among the following compounds is expected to yield the corresponding meta nitro derivative as the major product?

(A) C H -CH 6 5 3	**(B)** C H -CH Cl 6 5 2
(C) C H -CHCl 6 5 2	**(D)** C H -CCl 6 5 3

Q121. *CUSAT CAT 2023 - Shift 4*

Free radical intermediates are involved in

(A) Hydroboration-oxidation of propene to give propan-1-ol	**(B)** Hunsdiecker Reaction
(C) Hell Volhard Zelinsky Reaction	**(D)** Chlorination of benzene in the presence of Cl and a halogen carrier 2

Q122. *CUSAT CAT 2023 - Shift 4*

Which among the following is an example of Sandmeyer reaction? CuCl/HCl

(A) C H N +Cl- C H Cl 6 5 2 6 5 H O/Δ	**(B)** C H N +Cl- 2 C H OH 6 5 2 6 5 HBF
(C) C H N +Cl- 4 C H F 6 5 2 6 5 KI/warm	**(D)** C H N +Cl- C H I 6 5 2 6 5

Q123. *CUSAT CAT 2023 - Shift 4*

If the magnetic moment of a dioxygen species is 1.73 B.M, it may be – +

(A) O or O 2 2 +	**(B)** O or O 2 2 –
(C) O or O 2 2 – +	**(D)** O , O or O 2 2 2

Q124. *CUSAT CAT 2023 - Shift 4*

Which of the following has highest solubility in water?

(A) KF	**(B)** LiF
(C) NaF	**(D)** RbF

Q125. *CUSAT CAT 2023 - Shift 4*

The decreasing order of catenation is

(A) C > Sn > Si > Ge	**(B)** C > Si > Ge > Sn
(C) Si > Sn > C > Ge	**(D)** Ge > Sn > Si > C

Q126. *CUSAT CAT 2023 - Shift 4*

The electrolytes usually used in the electroplating of gold and silver, respectively are – –

(A) [Au(CN)] and [Ag(CN)] 2 2 – –	**(B)** [Au(CN)] and[Ag(Cl)] 2 2 – –
(C) [Au(OH)] and[Ag(OH)] 4 2 – –	**(D)** [Au(NH)] and[Ag(CN)] 3 2 2

Q127. *CUSAT CAT 2023 - Shift 4*

Trigonal bipyramidal geometry is shown by

(A) XeO F 2 2	**(B)** XeO F 3 2
(C) FXeOSO F 2 2–	**(D)** [XeF] 8

Q128. *CUSAT CAT 2023 - Shift 4*

Which of the following ions has the maximum magnetic moment? 2+

(A) Mn 2+	**(B)** Fe 2+
(C) Ti 2+	**(D)** Cr

Q129. *CUSAT CAT 2023 - Shift 4*

A double bond resulting in O molecule is from 2

(A) two bonding electrons	**(B)** four π-bonding electrons
(C) two π-bonding and two π-anti bonding electrons	**(D)** two bonding, four π-bonding and two π-anti bonding electrons

Q130. *CUSAT CAT 2023 - Shift 4*

Which of the following is most stable in aqueous solution? 2+

(A) Mn 3+	**(B)** Cr 3+
(C) V 3+	**(D)** Ti

Q131. *CUSAT CAT 2023 - Shift 4*

The complex that absorbs light of shortest wavelength is 3–

(A) [CoF] 6 +3	**(B)** [Co(H O)] 2 6 +3
(C) [Co(NH)] 3 6 3–	**(D)** [Co(ox)] 3 (ox = oxalate ion)

Q132. *CUSAT CAT 2023 - Shift 4*

In the manufacture of sulphuric acid by contact process Tyndall box is used to

(A) convert SO and SO 2 3	**(B)** test the presence of dust particles
(C) filter dust particles	**(D)** remove impurities 5

Q133. *CUSAT CAT 2023 - Shift 4*

The third line of Balmer series in the emission spectrum of hydrogen atom is due to the transition from

(A) fifth Bohr orbit to the second Bohr orbit	**(B)** fourth Bohr orbit to the first Bohr orbit
(C) sixth Bohr orbit to the third Bohr orbit	**(D)** eight Bohr orbit to the fourth Bohr orbit

Q134. *CUSAT CAT 2023 - Shift 4*

How many orbitals have the values n = 4, l = 2 and ml = –2?

(A) 1 orbital	**(B)** 2 orbitals
(C) 3 orbitals	**(D)** 4 orbitals

Q135. *CUSAT CAT 2023 - Shift 4*

For a complex, the order of energy of d orbitals, d = d > d = d = d . The z2 x2-y2 xy xz yz complex has

(A) octahedral structure	**(B)** square planar structure
(C) tetrahedral structure	**(D)** square pyramidal structure MATHEMATICS UG SHIFT IV - (FINAL)

Q136. *CUSAT CAT 2023 - Shift 5*

If the mean free path of atoms is doubled then the pressure of the gas will become P

(A) 4 P	**(B)** 2 P
(C) 8	**(D)** P

Q137. *CUSAT CAT 2023 - Shift 5*

In what manner will increase of pressure affect the following equation? C(s) + H O → CO(g) + H (g) 2 2

(A) Shift in the reverse direction	**(B)** Shift in the forward direction
(C) Increase in the yield of Hydrogen	**(D)** No effect

Q138. *CUSAT CAT 2023 - Shift 5*

What is the frequency of light having a wavelength of 4.50 × 10 cm? –12 –1

(A) 2.84 × 10 s 4 –1	**(B)** 2.10 × 10 s 14 –1
(C) 4.29 × 10 s 15 –1	**(D)** 6.67 × 10 s

Q139. *CUSAT CAT 2023 - Shift 5*

Which of the following statement is NOT true regarding the effect of increasing temperature on the distribution of molecular motion in a gas?

(A) Most probable speed increases	**(B)** Distribution curve become broader
(C) The fraction of molecule with higher velocity increases	**(D)** Distribution is independent of temperature

Q140. *CUSAT CAT 2023 - Shift 5*

In which of the following thermodynamics process there is no flow of heat between the system and the surroundings?

(A) Isobaric	**(B)** Isochoric
(C) Adiabatic	**(D)** Isothermal

Q141. *CUSAT CAT 2023 - Shift 5*

For an ideally dilute binary solution Raoult's law is valid for

(A) solute only	**(B)** solvent only
(C) both of the solute and the solvent	**(D)** none of the solute and the solvent

Q142. *CUSAT CAT 2023 - Shift 5*

Stability of dispersion is associated with magnitude of

(A) asymmetry effect	**(B)** Coulomb repulsion
(C) zeta potential	**(D)** Yukawa potential

Q143. *CUSAT CAT 2023 - Shift 5*

Which one is a correct relation? ∂P ∂V ∂T

(A) ≥0 ∂V ∂T ∂P T P V ∂P ∂V ∂T	**(B)** =1 ∂V ∂T ∂P T P V ∂P ∂V ∂T
(C) =–1 ∂V ∂T ∂P T P V ∂P ∂V ∂T	**(D)** ≤0 ∂V ∂T ∂P T P V

Q144. *CUSAT CAT 2023 - Shift 5*

Which of the following will increase the voltage of the cell with the cell reaction? + 2+ Sn(s) + 2Ag (aq)→Sn (aq) + 2Ag(s)

(A) Increase in concentration of both Sn 2+ and Ag +	**(B)** Increase in amount of Sn metal
(C) Increase in concentration of Sn 2+ +	**(D)** Increase in concentration of Ag ions

Q145. *CUSAT CAT 2023 - Shift 5*

The quantity of electricity required to deposit 1.15 g of sodium from molten NaCl [given atomic weight : Na = 23, Cl = 35.5] is

(A) 1 F	**(B)** 0.5 F
(C) 0.05 F	**(D)** 1.5 F –1 –1

Q146. *CUSAT CAT 2023 - Shift 5*

The specific conductance of a 0.1 M KCl solution at 23°C is 0.012 ohm cm . The resistance of cell containing the solution at the same temperature was found to be 55 ohm. The cell constant is –1

(A) 6.6 cm –1	**(B)** 0.66 cm –1
(C) 0.066 cm –1	**(D)** 0.12 cm

Q147. *CUSAT CAT 2023 - Shift 5*

Nitration of t-butylbenzene gives

(A) 2-nitro derivative exclusively	**(B)** 3-nitro derivative predominantly
(C) a 2:1 mixture of 2- and 4-nitro derivatives	**(D)** 4-nitro derivative almost exclusively

Q148. *CUSAT CAT 2023 - Shift 5*

Adipic acid can be written as HOOC COOH n where n equals to

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q149. *CUSAT CAT 2023 - Shift 5*

Which of the following, in aqueous solutions of equal concentration, has the lowest pH?

(A) ClCH CO H 2 2	**(B)** CH CO H 3 2
(C) CF CO H 3 2	**(D)** C H OH 2 5

Q150. *CUSAT CAT 2023 - Shift 5*

Which statement about the intermediate [C H NO] , formed during the 6 6 2 mononitration of benzene, is correct?

(A) It retains aromaticity 3	**(B)** It contains only one sp hybridized carbon atom
(C) It is antiaromatic in nature	**(D)** It is nonaromatic, but planar in nature

Q151. *CUSAT CAT 2023 - Shift 5*

In order to distinguish between C H NH and C H NH , which of the following 6 13 2 6 5 2 reagents is useful?

(A) Hinsberg reagent	**(B)** Bromine water
(C) Solubility test in dilute acid	**(D)** Lucas reagent

Q152. *CUSAT CAT 2023 - Shift 5*

Which among the following name reactions will give a hydrocarbon as the end product?

(A) Hunsdiecker reaction	**(B)** Meerwein-Verley-Ponndorf reduction
(C) Wurtz-Fittig reaction	**(D)** Reformatzky reaction

Q153. *CUSAT CAT 2023 - Shift 5*

The reaction given below is a typical example for Lindlar Catalyst C H COCl + H C H CHO + HCl 6 5 2 6 5

(A) Meerwein-Verley-Ponndorf reduction	**(B)** Clemmenson reduction
(C) Rosenmund reduction	**(D)** Wolff-Kishner reduction

Q154. *CUSAT CAT 2023 - Shift 5*

Selective method to convert pent-1-ene to pentan-1-ol is

(A) Acid catalyzed addition of water	**(B)** Oxymercuration-demercuration
(C) Hydroboration-oxidation	**(D)** Addition of HBr followed by treatment with NaOH in water

Q155. *CUSAT CAT 2023 - Shift 5*

'A' + H O→NaOH; 'A'2 →'B'2→NaOH + O 2 400°C at 25°C 2 'B' is used for oxygenating in submarine. 'A' and 'B' are respectively

(A) Na O and Na O 2 2 2	**(B)** Na O and Na O 2 2 2
(C) Na O and O 2 2 2	**(D)** NaO and O 2

Q156. *CUSAT CAT 2023 - Shift 5*

The formation of atomic hydrogen from molecular hydrogen is favoured under what conditions of temperature and pressure?

(A) Low temperature and low pressure	**(B)** High temperature and low pressure
(C) Low temperature and high pressure	**(D)** High temperature and high pressure

Q157. *CUSAT CAT 2023 - Shift 5*

In a sealed nickel vessel, Xe and F taken in 1 : 20 volume ratio were heated to 2 400°C. The product obtained is

(A) XeF	**(B)** XeF 2
(C) XeF 4	**(D)** XeF 6

Q158. *CUSAT CAT 2023 - Shift 5*

The number of P-O-P bonds in cyclic metaphosphoric acid is

(A) zero	**(B)** two
(C) three	**(D)** four

Q159. *CUSAT CAT 2023 - Shift 5*

Which is the correct increasing order of lone pair of electrons on the central atom?

(A) IF < XeF < ClF < IF 7 2 3 5	**(B)** IF < IF < ClF < XeF 7 5 3 2
(C) IF < ClF < XeF < IF 7 3 2 5	**(D)** IF < XeF < IF < ClF 7 2 5 3

Q160. *CUSAT CAT 2023 - Shift 5*

The correct order of ionic radii of Y , La , Eu and Lu is 3+ 3+ 3+ 3+

(A) Y < Lu < Eu < La 3+ 3+ 3+ 3+	**(B)** Y < La < Eu < Lu 3+ 3+ 3+ 3+
(C) Lu < Eu < La < Y 3+ 3+ 3+ 3+	**(D)** La < Eu < Lu < Y

Q161. *CUSAT CAT 2023 - Shift 5*

The number of radial nodes for the 4f orbital

(A) 3	**(B)** 4
(C) 0	**(D)** 1

Q162. *CUSAT CAT 2023 - Shift 5*

Equanil is

(A) artificial sweetener	**(B)** tranquilizer
(C) antihistamine	**(D)** antifertility drug

Q163. *CUSAT CAT 2023 - Shift 5*

What type of orbital is designated by n = 4, l = 2, ml = –2?

(A) 4p orbital	**(B)** 4s orbital
(C) 4d orbital	**(D)** 4f orbital

Q164. *CUSAT CAT 2023 - Shift 5*

Balmer series of lines in the hydrogen spectrum are observed in

(A) UV region	**(B)** visible region
(C) infrared region	**(D)** far infrared region

Q165. *CUSAT CAT 2023 - Shift 5*

Which of the following metal hydroxide is least basic?

(A) Ca(OH) 2	**(B)** Mg(OH) 2
(C) Sr(OH) 2	**(D)** Ba(OH) 2

Q166. *CUSAT CAT 2023 - Shift 5*

A surface ejects electrons when it is hit by green light; but does not eject electron when it is hit by yellow light. Will electrons be ejected if the surface is hit by red light?

(A) Yes	**(B)** No
(C) Yes if the red beam is quite intense	**(D)** Yes if the red beam continues to fall upon

Q167. *CUSAT CAT 2024 - Shift 1*

Which of the following sets of conditions makes a process spontaneous at all temperatures?

(A) $\Delta H = 0; \Delta S > 0$	**(B)** $\Delta H = 0; \Delta S < 0$
(C) $\Delta H < 0; \Delta S > 0$	**(D)** $\Delta H < 0; \Delta S < 0$

Q168. *CUSAT CAT 2024 - Shift 1*

The increase in internal energy of the system is 100 J when 300 J heat is supplied to it. What is the amount of work done by the system?

(A) –200 J	**(B)** +200 J
(C) –300 J	**(D)** –400 J

Q169. *CUSAT CAT 2024 - Shift 1*

The hydrogen ion concentration of a solution with pH value 3.69 is –4

(A) 2.042 × 10 M –2	**(B)** 3.69 × 10 M –4
(C) 4.31 × 10 M	**(D)** 0.369 M

Q170. *CUSAT CAT 2024 - Shift 1*

The correct statement between free energy change in a reaction and the corresponding equilibrium constant K is c

(A) ΔG = RT ln K c	**(B)** –ΔG = RT ln K c
(C) ΔG° = RT ln K c	**(D)** –ΔGo = RT ln K c

Q171. *CUSAT CAT 2024 - Shift 1*

The molecular weight of NaCl, assuming 100% dissociation in solution, as determined by elevation of boiling point method is

(A) 58.5	**(B)** 29.25
(C) > 58.5	**(D)** Zero

Q172. *CUSAT CAT 2024 - Shift 1*

The molar conductances of NaCl, HCl and CH COONa at infinite dilution are 126.45, 3 –1 2 –1 426.16 and 91 Ohm cm mol respectively. The molar conductance of acetic acid at infinite dilution is –1 2 –1

(A) 590.71 ohm cm mol –1 2 –1	**(B)** 698.28 ohm cm mol –1 2 –1
(C) 252.9 ohm cm mol	**(D)** 390.71 ohm –1 cm 2 mol –1

Q173. *CUSAT CAT 2024 - Shift 1*

The hydrogen electrode is dipped in a solution of pH = 3 at 25°C, the potential of the 2.303RT cell would be (the value of = 0.059 V) F

(A) +0.177 V	**(B)** –0.177 V
(C) +0.087 V	**(D)** +0.059 V

Q174. *CUSAT CAT 2024 - Shift 1*

Viscosity of gases will ……. with increase in temperature

(A) increase	**(B)** not change
(C) decrease	**(D)** increase or decrease k p

Q175. *CUSAT CAT 2024 - Shift 1*

The s-orbital of an atom is

(A) Independent of angles	**(B)** Dependent of angles
(C) Dependent on only sinθ	**(D)** Dependent on sinθand cosθ

Q176. *CUSAT CAT 2024 - Shift 1*

Give the correct IUPAC nomenclature for the following compound.

(A) 1-Bromo-4-chloro-1-ethyl-2-methylpentane	**(B)** 3-Bromo-6-chloro-4-methylheptane
(C) 2-Chloro-4 (bromopropyl)-pentane	**(D)** 5-Bromo-2-chloro-4-methylheptane

Q177. *CUSAT CAT 2024 - Shift 1*

Arrange the following compounds based on the order of number of hyperconjugative structures possible.

(A) I > II > III > IV	**(B)** I > IV > II > III
(C) IV > I > II > III	**(D)** IV > III > I > II

Q178. *CUSAT CAT 2024 - Shift 1*

A major pollutant in photochemical smog is

(A) CO 2	**(B)** H CO 2 3
(C) CH 4	**(D)** Peroxyacetyl nitrate

Q179. *CUSAT CAT 2024 - Shift 1*

Which among the following substituted benzyl carbocations is the most stable?

(A) HO +CH 2	**(B)** O 2 N +CH 2
(C) H 3 C +CH 2	**(D)** Cl +CH 2

Q180. *CUSAT CAT 2024 - Shift 1*

In aromatic electrophilic substitution reaction, cyano (C N) group present in benzene ring is

(A) Deactivating and meta directing	**(B)** Activating and ortho, para directing
(C) Deactivating and ortho, para directing	**(D)** Activating and meta directing

Q181. *CUSAT CAT 2024 - Shift 1*

Which among the following compounds will give a positive iodoform test?

(A) Iodomethane	**(B)** 2-Iodopropane
(C) Iodobenzene	**(D)** Butan-2-ol

Q182. *CUSAT CAT 2024 - Shift 1*

α-D-Glucose is represented as OH O HO

(A) HO OH OH OH OH O	**(B)** OH HO OH OH O
(C) OH HO HO OH OH OH O HO	**(D)** HO OH

Q183. *CUSAT CAT 2024 - Shift 1*

Butane-2,3-diol has two chiral carbon. One of the stereoisomers of butane-2,3-diol exhibited specific rotation of 13.0°. Other stereoisomers will exhibit optical rotation

(A) -13.0, +26.0, -26.0 degrees	**(B)** 0, 0, 0 degrees
(C) -13.0, +6.5, -6.5 degrees	**(D)** 0 and -13.0 degrees

Q184. *CUSAT CAT 2024 - Shift 1*

What is the number of available coordination sites in ethylene diamine tetraacetate?

(A) 2	**(B)** 4
(C) 6	**(D)** 8

Q185. *CUSAT CAT 2024 - Shift 1*

The number of ions that can react with AgNO in PtCl .2NH are 3 2 3

(A) 0	**(B)** 1
(C) 2	**(D)** 3 4– 3–

Q186. *CUSAT CAT 2024 - Shift 1*

Predict the stability of the complexes [Fe(CN)] and [Fe(CN)] according to 6 6 EAN rule

(A) stable, stable	**(B)** stable, unstable
(C) unstable, stable	**(D)** unstable, unstable

Q187. *CUSAT CAT 2024 - Shift 1*

The IUPAC nomenclature of [Pt(NH)BrCl(CH NH)] is 3 3 2

(A) methyl amine chloro bromo ammine platinum(II)	**(B)** bromo chloro ammine methyl amine platinum(II)
(C) amine methyl amine chloro bromo platinum(II)	**(D)** ammine bromo chloro methyl amine platinum(II) 4– 3+

Q188. *CUSAT CAT 2024 - Shift 1*

Which among the complexes [Co(CN)] , [Cr(NH)] exhibit Jahn-Teller distortion? 6 3 6

(A) 4– 3+ [Co(CN)] , [Cr(NH)] 6 3 6	**(B)** 4– [Co(CN)] 6
(C) [Cr(NH)] 3+ 3 6	**(D)** None of the above

Q189. *CUSAT CAT 2024 - Shift 1*

What is the chemical element for the symbol Rn?

(A) Rutherfordium	**(B)** Rhenium
(C) Radon	**(D)** Radium

Q190. *CUSAT CAT 2024 - Shift 1*

Which of the following pairs of elements belong to the same group?

(A) H and He	**(B)** Li and Be
(C) C and Pb	**(D)** Ga and Ge

Q191. *CUSAT CAT 2024 - Shift 1*

State the number of valence electrons in a sulphur atom

(A) 16	**(B)** 12
(C) 10	**(D)** 6

Q192. *CUSAT CAT 2024 - Shift 1*

Based on the values for the first through fifth ionization energies in the table, what is the most likely identity of Element A and B? IE (kJ/mol) IE (kJ/mol) IE (kJ/mol) IE (kJ/mol) IE (kJ/mol) 1 2 3 4 5 Element A 498 4560 6910 9540 13400 Element B 787 1575 3220 4350 16100

(A) Ca and S	**(B)** Na and Si
(C) K and P	**(D)** Ne and Al

Q193. *CUSAT CAT 2024 - Shift 1*

How many covalent bonds can carbon form?

(A) 0	**(B)** 2
(C) 3	**(D)** 4

Q194. *CUSAT CAT 2024 - Shift 1*

Which of the following statements explains why silicon dioxide has a high melting point?

(A) It has a giant ionic structure with strong electrostatic attraction between ions	**(B)** It has a giant covalent structure with strong covalent bonds between atoms
(C) It has a simple molecular structure with weak forces between molecules	**(D)** It has a giant metallic structure with a strong attraction between positive ions and the sea of electrons

Q195. *CUSAT CAT 2024 - Shift 1*

Which of the following does not contain ionic bonds?

(A) Sulphur dioxide	**(B)** Sodium oxide
(C) Silver oxide	**(D)** Strontium oxide

Q196.

CUSAT CAT 2024 - Shift 1

Which of the following gives the best explanation for why a substance does not conduct electricity?

(A) The bonding in the substance is not ionic	**(B)** The bonding model in the substance does not have free electrons
(C) The bonding model does not have ions which are free to move or free electrons	**(D)** The bonding model does not have ions or free electrons

Q197.

CUSAT CAT 2024 - Shift 1

What is the electron pair geometry and molecular geometry of ICl ? 3

(A) trigonal-planar and T-shaped	**(B)** trigonal-bipyramidal and trigonal-planar
(C) trigonal-bipyramidal and linear	**(D)** trigonal-bipyramidal and T-shaped

Q198.

CUSAT CAT 2024 - Shift 1

Which of the following was not one of the assumptions of Bohr's model for the hydrogen atom?

(A) The energy of the hydrogen atom can only change when a photon of energy equal to the energy difference between two allowed energy levels is emitted or absorbed	**(B)** The energy of the hydrogen atom does not change while the electron moves around the nucleus
(C) A hydrogen atom has certain allowed energy levels	**(D)** It is not possible to know simultaneously the exact position and momentum of the electron as it moves around the nucleus MATHEMATICS UG - SHIFT I (FINAL) 43 1 6

Q199.

CUSAT CAT 2024 - Shift 2

The heat of formation of sulphur dioxide is ΔH = –297 kJ. The heat liberated on f burning 8 g of sulphur in oxygen is

(A) –148.5 kJ	**(B)** –74.25 kJ
(C) 148.5 kJ	**(D)** 74.25 kJ

Q200.

CUSAT CAT 2024 - Shift 2

The efficiency of a heat engine is given by W T –T

(A) = 2 1 q T 2 2 W T –T	**(B)** = 1 2 q T 2 2 W T –T
(C) = 2 1 q T 2 1 W T –T	**(D)** = 1 2 q T 2 1

Q201.

CUSAT CAT 2024 - Shift 2

The pH of the blood is maintained by buffer system given by

(A) NaCl and HCl	**(B)** NH Cl and NH OH 4 4
(C) Sodium citrate and Citric acid –	**(D)** HCO and H CO 3 2 3

Q202. *CUSAT CAT 2024 - Shift 2*

In which of the following reactions, increase in the volume at constant temperature does not affect the number of moles at equilibrium?

(A) 2NH → N + 3H 3 2 2	**(B)** C (g) + ½ O (g) → CO (g) 2
(C) H (g) + O (g) → H O (g) 2 2 2 2	**(D)** None of the above

Q203. *CUSAT CAT 2024 - Shift 2*

An azeotropic solution of two liquids has the boiling point lower than either of them, when it

(A) shows negative deviation from Raoult's law	**(B)** shows no deviation from Raoul's law
(C) shows positive deviation from Raoult's law	**(D)** contains non-volatile solute

Q204. *CUSAT CAT 2024 - Shift 2*

In a H -O fuel cell, combustion of hydrogen occurs to 2 2

(A) Remove adsorbed oxygen from electrode surface	**(B)** Create potential difference between the two electrodes
(C) Produce high purity water	**(D)** Generate heat

Q205. *CUSAT CAT 2024 - Shift 2*

Out of Cu, Ag, Fe and Zn, the metal which can displace all others from salt solution is

(A) Ag	**(B)** Cu
(C) Zn	**(D)** Fe

Q206. *CUSAT CAT 2024 - Shift 2*

The presence of double layer in colloids accounts for

(A) kinetic properties	**(B)** electrokinetic properties
(C) optical properties	**(D)** tyndal effect

Q207. *CUSAT CAT 2024 - Shift 2*

Which of the following is an example of isobar? 14 15

(A) C and C 6 6 14 14	**(B)** C and N 6 7 15 14
(C) N and C 7 7 6 7	**(D)** C and C 14 14

Q208. *CUSAT CAT 2024 - Shift 2*

Among the given equations, the correct one representing a closed system is

(A) Δ = –	**(B)** Δ =
(C) Δ = –	**(D)** Δ = Δ+Δ

Q209. *CUSAT CAT 2024 - Shift 2*

If in a nearly saturated solution, the dissolution process is endothermic, then the solubility should

(A) decrease with rise in temperature	**(B)** increase with rise in temperature
(C) increase with reduce in temperature	**(D)** decrease with reduce in temperature

Q210. *CUSAT CAT 2024 - Shift 2*

A (g) + 2B(g)■C(g) + 2D(s), the K for the given reaction will be 2 c 2 [C][D]

(A) K c = 2 [A][B] 2 [C] K =	**(B)** c 2 [A][B] 2 2 [A][B]
(C) K = 2 c 2 [C][D] 2 [A][B]	**(D)** K = 2 c [C]

Q211. *CUSAT CAT 2024 - Shift 2*

If the initial concentration of the reactants in a reaction increases then the equilibrium constant

(A) also increases	**(B)** decreases
(C) remain constant	**(D)** may increase or decrease

Q212. *CUSAT CAT 2024 - Shift 2*

The enthalpy of unit mass for any system is

(A) H = U + pV + S	**(B)** H = U + pV – S
(C) H = U + pV	**(D)** H = U – pV + S 2

Q213. *CUSAT CAT 2024 - Shift 2*

Within an atom, the probability density ψ is always

(A) zero	**(B)** negative
(C) ±1	**(D)** positive

Q214. *CUSAT CAT 2024 - Shift 2*

According to Ostwald's dilution law k

(A) α= a C 1 k 2	**(B)** α= a C k
(C) α= a V k a	**(D)** α= V

Q215. *CUSAT CAT 2024 - Shift 2*

The fuel cell reaction generates electricity directly from 2H + O →2H O. What 2 2 2 potential should the battery generate (under standard conditions)?

(A) 0.40	**(B)** 0.43
(C) 0.68	**(D)** 1.23

Q216. *CUSAT CAT 2024 - Shift 2*

For a second order reaction, the time required for half change is

(A) Directly proportional to the initial concentration	**(B)** Independent of the initial concentration
(C) Inversely proportional to the initial concentration	**(D)** Inversely proportional to the square of the initial concentration

Q217. *CUSAT CAT 2024 - Shift 2*

Overall 3D structure of a simple protein is called

(A) Quaternary structure	**(B)** Tertiary structure
(C) Secondary structure	**(D)** Primary structure

Q218. *CUSAT CAT 2024 - Shift 2*

The reactive intermediate involved when HBr is added to propene in the presence of peroxide is

(A) —	**(B)** —
(C) —	**(D)** —

Q219. *CUSAT CAT 2024 - Shift 2*

Reagents used in Wolff-Kishner reduction of cyclohexanone to give cyclohexane is/are

(A) LiAlH 4	**(B)** Na, liq NH 3
(C) Zn, HCl	**(D)** NH NH , NaOH 2 2

Q220. *CUSAT CAT 2024 - Shift 2*

Which among the following is alleged to have been used to poison a famous historic figure?

(A) Mg	**(B)** Si
(C) Al	**(D)** As

Q221. *CUSAT CAT 2024 - Shift 2*

AZT is the azido derivative of a DNA nucleoside based anti-AIDS drug. The nitrogen base present in AZT is not present in RNA. Which among the following nucleic acid base does AZT contain?

(A) Uracil	**(B)** Cytosine
(C) Thymine	**(D)** Guanine

Q222. *CUSAT CAT 2024 - Shift 2*

Trans addition of Br to Compound X gave meso-2,3-dibromobutane. What is the 2 identity of compound X?

(A) 1-butene	**(B)** cis-but-2-ene
(C) trans-but-2-ene	**(D)** 1,3-butadiene HOOC COOH n

Q223. *CUSAT CAT 2024 - Shift 2*

Glutaric acid can be written as with n equals to (remember, Oh, My Such Good Apples)

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q224. *CUSAT CAT 2024 - Shift 2*

Which of the following, in aqueous solutions of equal concentration, has the lowest pH?

(A) ClCH CO H 2 2	**(B)** CH CO H 3 2
(C) Cl CHCO H 2 2	**(D)** C H OH 2 5

Q225. *CUSAT CAT 2024 - Shift 2*

Shown below is Cholesterol, which is the most abundant compound from steroids in animals. (Note: Thick solid wedge indicates a bond coming above the plane, while broken wedge indicates a bond below the plane) How many chiral carbons are present in Cholesterol?

(A) 10	**(B)** 8
(C) 6	**(D)** 4

Q226. *CUSAT CAT 2024 - Shift 2*

According to rules for drawing Fischer projection, the groups on the horizontal line are

(A) Above the plane of the paper	**(B)** Below the plane of the paper
(C) In the plane of paper	**(D)** Group appearing on the right side is above and group appearing on the left is below the plane of the paper

Q227. *CUSAT CAT 2024 - Shift 2*

Aspirin is an acetylation product of

(A) Aspartic acid	**(B)** o-Hydroxybenzoic acid
(C) Asparagine	**(D)** m-Hydroxybenzoic acid

Q228. *CUSAT CAT 2024 - Shift 2*

Resonance energy of benzene is

(A) 152 kJ/mol	**(B)** 36 kJ/mol
(C) 36 eV/mol	**(D)** 36 cal/mol

Q229. *CUSAT CAT 2024 - Shift 2*

The number of unpaired electrons in the complexes [Cu(en) (H O)]Cl and K [CrI] are 2 2 2 2 3 6

(A) 0 and 1	**(B)** 1 and 2
(C) 1 and 3	**(D)** 2 and 3

Q230. *CUSAT CAT 2024 - Shift 2*

Which of the following pairs does not represent an acid and its conjugate base?

(A) HNO and NO ¯ 3 3	**(B)** H SO and HSO ¯ 2 4 4 2
(C) H SO and SO ¯ 2 4 4 2	**(D)** HSO ¯ and SO ¯ 4 4

Q231. *CUSAT CAT 2024 - Shift 2*

Which is the most acidic compound?

(A) CH 4	**(B)** NH 3
(C) H O 2	**(D)** H S 2

Q232. *CUSAT CAT 2024 - Shift 2*

Which of the following compounds cannot act as a Lewis base?

(A) NF 3	**(B)** PCl 5
(C) SF 4	**(D)** ClF 3

Q233. *CUSAT CAT 2024 - Shift 2*

The interaction energy of London forces between two particles is proportional to r , where r is the distance between the particles. The value of x is

(A) 3	**(B)** –3
(C) –6	**(D)** 6 –

Q234. *CUSAT CAT 2024 - Shift 2*

Which among the following is NOT isostructural?

(A) SO 3 –	**(B)** NO 3
(C) PCl 3 2–	**(D)** CO 3

Q235. *CUSAT CAT 2024 - Shift 2*

The correct statement about B H is 2 6

(A) Terminal B–H bonds have less p-character when compared to bridging bonds	**(B)** The two B–H–B bonds are not of same length
(C) All B–H–B angles are of 120°	**(D)** Its fragment, BH , behaves as a Lewis base 3

Q236. *CUSAT CAT 2024 - Shift 2*

Which among the following species has unequal bond lengths? –

(A) BF 4	**(B)** XeF 4
(C) SF 4	**(D)** SiF 4

Q237. *CUSAT CAT 2024 - Shift 2*

Identify the ion having the 'spin only' magnetic moment equal to 4.9 BM

(A) Ni 2+	**(B)** Cr 2+
(C) V 2+	**(D)** Mn 2+

Q238. *CUSAT CAT 2024 - Shift 2*

The number of bonds between the two Mn in Mn (CO) is 2 10

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q239. *CUSAT CAT 2024 - Shift 2*

Select the correct option 2-

(A) [Ni(CO)] and [NiCl] both are diamagnetic 4 4 2-	**(B)** [Ni(CO)] is paramagnetic and [NiCl] is diamagnetic 4 4 2-
(C) [Ni(CO)] is diamagnetic and [NiCl] is paramagnetic 4 4 2-	**(D)** [Ni(CO)] and [NiCl] both are paramagnetic 4 4

Q240. *CUSAT CAT 2024 - Shift 2*

The Cr compounds that is not involved in the reaction sequence of chromyl chloride test for confirmation of Cl^{-} ion is

(A) K Cr O 2 2 7	**(B)** CrO Cl 2 2
(C) Cr O 2 3	**(D)** CrO 3 MATHEMATICS UG – SHIFT II (FINAL) 3 2 1 ω ω 3

Q241. *CUSAT CAT 2024 - Shift 3*

A mixture of 2 moles of carbon monoxide and one mole of oxygen in a closed vessel is ignited to convert carbon monoxide to carbon dioxide. If ΔH is the enthalpy change and ΔE is the change in internal energy, then

(A) $\Delta H > \Delta E$	**(B)** $\Delta H < \Delta E$
(C) $\Delta H = \Delta E$	**(D)** The relationship depends on the volume of the vessel

Q242. *CUSAT CAT 2024 - Shift 3*

The Clausius-Clayperon equation helps to calculate

(A) Latent heat of vaporization	**(B)** Melting point of the solvent
(C) Heat of neutralization	**(D)** Molecular weight of solute

Q243. *CUSAT CAT 2024 - Shift 3*

40 mg of pure sodium hydroxide is dissolved in 10 litre of distilled water. The pH of the solution is

(A) 9.0	**(B)** 10.0
(C) 11.0	**(D)** 12.0

Q244. *CUSAT CAT 2024 - Shift 3*

When 10 mL of 0.2 M of H SO is added to 90 mL of H O, the concentration of 2 4 2 H SO is. 2 4

(A) 0.02 N	**(B)** 0.04 M
(C) 0.04 N	**(D)** 0.002 M

Q245. *CUSAT CAT 2024 - Shift 3*

A 0.1 molal solution has boiling point of 100.052°C, then the molal elevation constant of water is –1

(A) 5.2 Km –1	**(B)** 0.52 Km –1
(C) 0.052 Km –1	**(D)** 52 Km

Q246. *CUSAT CAT 2024 - Shift 3*

E° =0.80V and E° =–0.25V. The EMF of the cell Ni-Ag is Ag+ Ag Ni+ Ni

(A) +0.21 V	**(B)** +1.05 V
(C) –2.10 V	**(D)** –1.05 V

Q247. *CUSAT CAT 2024 - Shift 3*

The decay of a radioactive element exhibits the characteristics of a reaction of

(A) Zero order	**(B)** First order
(C) Second order	**(D)** Fractional order –1

Q248. *CUSAT CAT 2024 - Shift 3*

For reaction: 4A + B→2C+ 2D Which of the following statement is incorrect? th 1

(A) Rate of disappearance of B is of rate of disappearance of A 4 th 1	**(B)** Rate of formation of C is of rate of consumption of A 2 th 1
(C) Rate of appearance of D is of rate of disappearance of B 2	**(D)** Rate of appearance of C and D are equal

Q249. *CUSAT CAT 2024 - Shift 3*

Which factor would increase the rate of chemical reaction? I. Increasing the temperature II. Removing products as they are formed III. Adding a catalyst

(A) I and II	**(B)** II and III
(C) I, II and III	**(D)** I only

Q250. *CUSAT CAT 2024 - Shift 3*

The thermodynamical potential enthalpy H = U + pV , then which of the following relation hold true? ∂H ∂H

(A) T =– and V = ∂S ∂P v S ∂H ∂H	**(B)** T = and V =– ∂S ∂P P S ∂H ∂H
(C) T =– and V = ∂P ∂P S v ∂H ∂H	**(D)** V = and T = ∂P ∂S S P

Q251. *CUSAT CAT 2024 - Shift 3*

In the electrochemical reaction: 2Fe 3+ + Zn → Zn 2+ + 2Fe 2+ Increasing the concentration of Fe2+

(A) Increase cell emf	**(B)** Increase the current flow
(C) Decrease the cell emf	**(D)** Will not alter cell emf

Q252. *CUSAT CAT 2024 - Shift 3*

Graph between concentration x of the product and time of the reaction A→B is of the following type d[A] Hence, graph and time will be of the dt

(A) —	**(B)** —
(C) —	**(D)** —

Q253. *CUSAT CAT 2024 - Shift 3*

Which of the following reagents will react with n-butane to give isolable product/s?

(A) Alkaline KMnO at room temperature 4	**(B)** Aqueous KOH solution
(C) Iodine in presence of light	**(D)** Bromine in presence of light

Q254. *CUSAT CAT 2024 - Shift 3*

Which among the following is least toxic to humans?

(A) Mustard gas	**(B)** Chloropicrin
(C) Phosgene	**(D)** Menthol

Q255. *CUSAT CAT 2024 - Shift 3*

Benzene is nitrated using conc. HNO and conc.H SO . Which among the following 3 2 4 ion is involved in the nitration step? –

(A) Nitrite (NO) 2 +	**(B)** Nitrosonium (NO) +
(C) Nitronium (NO) 2	**(D)** Nitrate (NO) – 3

Q256. *CUSAT CAT 2024 - Shift 3*

1°, 2°, 3° alcohols can be distinguished by

(A) Victor-Meyer test	**(B)** Hinsberg test
(C) Molisch's test	**(D)** Biuret test

Q257. *CUSAT CAT 2024 - Shift 3*

Theoretically BOD (biological oxygen demand) value of clean water should not be above

(A) 10 ppm	**(B)** 5 ppm
(C) 3 ppm	**(D)** 20 ppm

Q258. *CUSAT CAT 2024 - Shift 3*

Keratin, the protein present in hair contains cysteine residues. “Hair styling” is possible due to ease of formation of

(A) S-S bond	**(B)** P-P bond
(C) C-F bond	**(D)** N-F bond

Q259. *CUSAT CAT 2024 - Shift 3*

Mobile Phase and Retention Factor are the terms used in which among the following Purification and Separation Techniques?

(A) Resolution of racemic mixtures by forming chiral inclusion complexes	**(B)** Column Chromatography
(C) Steam Distillation	**(D)** Solvent-Solvent extraction

Q260. *CUSAT CAT 2024 - Shift 3*

How many planes of symmetry does cis-1,3-Dimethylcyclobutane have ?

(A) Zero	**(B)** One
(C) Two	**(D)** Three

Q261. *CUSAT CAT 2024 - Shift 3*

Which among the following molecules gives garlic its distinctive odour?

(A) Allicin	**(B)** Nicotine
(C) Quinine	**(D)** Ornithine

Q262. *CUSAT CAT 2024 - Shift 3*

The common name of 3-Bromopropene is

(A) Propylidene bromide	**(B)** Vinyl bromide
(C) tert-Butyl bromide	**(D)** Allyl bromide

Q263. *CUSAT CAT 2024 - Shift 3*

The orange colour of K Cr O and purple colour of KMnO is due to 2 2 7 4

(A) d-d transitions and charge transfer transitions	**(B)** d-d transitions in both
(C) charge transfer transition in both	**(D)** charge transfer transitions and d-d transitions

Q264. *CUSAT CAT 2024 - Shift 3*

The number of moles of oxalate ions oxidized by one mole of MnO $^{-}$ ion is 4 1

(A) 5	**(B)** 5 2
(C) 5 5	**(D)** 2

Q265. *CUSAT CAT 2024 - Shift 3*

The number of radial node(s) for 3p orbital is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q266. *CUSAT CAT 2024 - Shift 3*

Which of the following statements exactly represents Dalton's atomic theory?

(A) Atoms are indivisible and indestructible	**(B)** Atoms of different elements can combine in simple whole-number ratios to form compounds
(C) All atoms of a given element are identical in mass and properties	**(D)** Atoms can be created or destroyed in chemical reactions

Q267. *CUSAT CAT 2024 - Shift 3*

What is the position uncertainty in locating an electron with speed of 25 ms having uncertainty of 0.1%? –5

(A) $\Delta x = 3.317 \times 10 -3$	**(B)** $\Delta x = 2.317 \times 10 -5$
(C) $\Delta x = 1.220 \times 10 -4$	**(D)** $\Delta x = 3.323 \times 10$

Q268. *CUSAT CAT 2024 - Shift 3*

Which of the following anion has dπ-pπ bond? 2–

(A) CO 3 3–	**(B)** PO 4 –
(C) NO 3 –	**(D)** NO 2

Q269. *CUSAT CAT 2024 - Shift 3*

In BrF molecule, the lone pair of electrons occupy equatorial positions to minimize 3

(A) lone pair-bond pair repulsion only	**(B)** bond pair-bond pair repulsion only
(C) lone pair-lone pair repulsion and lone pair-bond pair repulsion	**(D)** lone pair-lone pair repulsion only

Q270. *CUSAT CAT 2024 - Shift 3*

The hybrid orbital of ……. has only 20% d character.

(A) CCl 4	**(B)** SF 6
(C) SF 4	**(D)** Cl O 2

Q271. *CUSAT CAT 2024 - Shift 3*

Exactly same number of σ and π bonds is present in

(A) CH (CN) 2 2 –	**(B)** HCO 3
(C) XeO 4	**(D)** (CN) 2

Q272. *CUSAT CAT 2024 - Shift 3*

Which one of the following species cannot be identified in normal temperature and pressure conditions? 2+

(A) Be	**(B)** Be 2
(C) B 2	**(D)** Li 2

Q273. *CUSAT CAT 2024 - Shift 3*

Which of the following halides has the highest cation to anion size ratio?

(A) CsI	**(B)** CsF
(C) LiF	**(D)** NaF

Q274. *CUSAT CAT 2024 - Shift 3*

Arrange O, S, F and Cl in the order of their increasing electron gain enthalpy.

(A) Cl < F < O < S	**(B)** O < S < F < Cl
(C) F < S < O < Cl	**(D)** S < O < Cl < F

Q275. *CUSAT CAT 2024 - Shift 3*

Find out the number of valence electrons for the element X. The successive ionization energies of X is given below. IE : 250 kJ mol –1 1 IE : 820 kJ mol –1 2 IE : 1100 kJ mol –1 3 IE : 1400 kJ mol –1 4

(A) 3	**(B)** 4
(C) 2	**(D)** 1

Q276. *CUSAT CAT 2024 - Shift 3*

Hydrolith reacts with water to produce

(A) H 2	**(B)** H O 2 2
(C) NaH	**(D)** N 2 MATHEMATICS UG – SHIFT III (FINAL) n

Q277. *CUSAT CAT 2024 - Shift 4*

Plants and living beings are the examples of

(A) Isolated system	**(B)** Adiabatic system
(C) Open system	**(D)** Closed system

Q278. *CUSAT CAT 2024 - Shift 4*

Which of the following statement is false?

(A) Temperature is a state function	**(B)** Work is a state function
(C) Change in the state is completely defined when the initial and final states are specified	**(D)** Work appears at the boundary of the system

Q279. *CUSAT CAT 2024 - Shift 4*

The compound that is not a Lewis acid is

(A) BF 3	**(B)** AlCl 3
(C) ZnCl 2	**(D)** NI 3 + 2+ 3+ 4+

Q280. *CUSAT CAT 2024 - Shift 4*

Addition of Ag , Pb , Fe and Si causes coagulation of negatively charged colloidal sol in the following order

(A) Fe 3+ > Si 4+ > Pb 2+ > Ag +	**(B)** Ag + < Pb 2+ < Fe 3+ < Si4+
(C) Ag + > Pb 2+ > Fe 3+ > Si 4+	**(D)** Ag + = Pb 2+ = Fe 3+ = Si 4+

Q281. *CUSAT CAT 2024 - Shift 4*

When a solution of a weak electrolyte is diluted

(A) specific conductance increases	**(B)** specific conductance increases but equivalent conductance decreases
(C) both specific conductance and equivalent conductance decreases	**(D)** specific conductance decreases and equivalent conductance increases

Q282. *CUSAT CAT 2024 - Shift 4*

Which of the following is not a reference electrode?

(A) Saturated calomel electrode	**(B)** Platinum electrode
(C) Silver-Silver chloride electrode	**(D)** Standard hydrogen electrode

Q283. *CUSAT CAT 2024 - Shift 4*

Match Column I with Column II and select the correct answer using the codes given below. Column I Column II a. Isothermal process 1. No heat exchange b. Adiabatic process 2. Constant temperature c. Isochoric process 3. Constant pressure d. Joule Thomson 4. Constant volume process 5. Constant enthalpy

(A) a-2, b-1, c-4, d-5	**(B)** a-1, b-2, c-3, d-5
(C) a-2, b-1, c-4, d-3	**(D)** a-1, b-2, c-4, d-3

Q284. *CUSAT CAT 2024 - Shift 4*

For a zero order reaction where the concentration of the reactants do not affect the rate of the reaction, the plot of concentration of reactant vs time will be

(A) linearly decreasing	**(B)** linearly increasing
(C) exponentially decreasing	**(D)** exponentially increasing

Q285. *CUSAT CAT 2024 - Shift 4*

Which of the following decreases with increase in concentration?

(A) Conductance	**(B)** Specific conductance
(C) Cell constant	**(D)** Molar conductance

Q286. *CUSAT CAT 2024 - Shift 4*

A solution will boil when its vapor pressure equals

(A) standard state pressure of solvent	**(B)** enthalpy of vaporization
(C) external pressure	**(D)** enthalpy of fusion

Q287. *CUSAT CAT 2024 - Shift 4*

Calculate the P H at the equivalence point when a solution of 0.01 M CH COOH is 3 treated with a solution of 0.01 M NaOH. pKa of CH COOH is 4.74. 3

(A) 2.11	**(B)** 4.22
(C) 6.22	**(D)** 8.22

Q288. *CUSAT CAT 2024 - Shift 4*

Which among the following compounds will give positive iodoform and Tollen's silver mirror tests?

(A) —	**(B)** —
(C) —	**(D)** —

Q289. *CUSAT CAT 2024 - Shift 4*

Predict the major product formed in the reaction given below

(A) —	**(B)** —
(C) —	**(D)** —

Q290. *CUSAT CAT 2024 - Shift 4*

An additive used in LPG cylinders to aid gas leak detection is:

(A) diphenyl sulphide	**(B)** 4-Methyl-4-sulfanylpentan-2-one
(C) methyl isothiocyanate	**(D)** Ethanethiol

Q291. *CUSAT CAT 2024 - Shift 4*

Triphenyl carbinol may be prepared by the reaction of 2 equivalents of phenylmagnesium chloride with

(A) CH CHO 3	**(B)** HCHO
(C) Methyl benzoate (PhCO CH) 2 3	**(D)** PhCHO

Q292. *CUSAT CAT 2024 - Shift 4*

Which one shows a unique reactivity with hydrogen, forming a stable compound under normal conditions?

(A) Sodium (Na); Forms stable compound NaH	**(B)** Potassium (K); Forms stable compound KH
(C) Rubidium (Rb); Forms stable compound RbH	**(D)** Cesium (Cs); Forms stable compound CsH

Q293. *CUSAT CAT 2024 - Shift 4*

Which of the metal gives NH NO with very dilute HNO ? 4 3 3

(A) Fe	**(B)** Ti
(C) Cu	**(D)** Hg

Q294. *CUSAT CAT 2024 - Shift 4*

AlF is soluble in HF only in the presence of KF. It is due to the formation of 3

(A) K [AlF H] 3 3 3	**(B)** K [AlF] 3 6
(C) AlH 3	**(D)** K[AlF H] 3

Q295. *CUSAT CAT 2024 - Shift 4*

Identify the following which cannot produce diborane.

(A) Na B O + HCl 2 4 7	**(B)** NaBH + I 4 2
(C) BF + LiAlH 3 4	**(D)** BF + NaH 3

Q296. *CUSAT CAT 2024 - Shift 4*

Select the incorrect statement.

(A) Ag + forms complexes, Na + doesn't	**(B)** NaCl is water soluble AgCl is insoluble
(C) Na CO is thermally stable while Ag CO decompose into Ag, CO and O 2 3 2 3 2 2	**(D)** NaCl and AgCl both give colour in flame when ignited

Q297. *CUSAT CAT 2024 - Shift 4*

The composition of mica is

(A) NaAlSiO .3H O 4 2	**(B)** K O .3Al O .6SiO .2H O 2 3 2 3 2 2
(C) K HAl (SiO) 2 4 3	**(D)** Na.K.SiO .10H O 4 2

Q298. *CUSAT CAT 2024 - Shift 4*

In solid state PCl exists as 5

(A) PCl 3	**(B)** PCl + 4
(C) PCl – 6	**(D)** PCl + and PCl – 4 6

Q299. *CUSAT CAT 2024 - Shift 4*

The acid strength of oxo-acids of halogen is in order

(A) HOI > HOBr > HOCl > HOF	**(B)** HOF > HOCl > HOBr > HOI
(C) HOCl > HOBr > HOI > HOF	**(D)** HOI > HOF > HOBr > HOCl

Q300. *CUSAT CAT 2024 - Shift 4*

End product of the hydrolysis of XeF is 6

(A) XeO 3 –	**(B)** XeO 3
(C) XeOF 4	**(D)** XeO F 2 2

Q301. *CUSAT CAT 2024 - Shift 4*

Which shows the maximum magnetic moment among the bivalent ions of the first transition series? +2

(A) Mn +2	**(B)** Fe +2
(C) Co +2	**(D)** Ni MATHEMATICS UG – SHIFT IV (FINAL) 1 1 1

Q302. *CUSAT CAT 2024 - Shift 5*

In a reversible isothermal process, the change in internal energy

(A) $\Delta E = 0$	**(B)** $\Delta E > 0$
(C) $\Delta E < 0$	**(D)** $\Delta E \geq 0$

Q303. *CUSAT CAT 2024 - Shift 5*

Which of the following two gases can be cooled from room temperature by the Joule-Thomson effect?

(A) Hydrogen and oxygen	**(B)** Helium and nitrogen
(C) Helium and hydrogen	**(D)** Nitrogen and oxygen –4

Q304. *CUSAT CAT 2024 - Shift 5*

A plant cell shrinks when it is kept in

(A) Hypertonic solution	**(B)** Hypotonic solution
(C) Water	**(D)** Isotonic solution with cell sap

Q305. *CUSAT CAT 2024 - Shift 5*

Two platinum electrodes were immersed in a solution of copper sulphate and electric current was passed till the solution becomes colourless. The colourless solution contains

(A) Platinum sulphate	**(B)** Copper hydroxide
(C) Only water	**(D)** Dilute sulphuric acid

Q306. *CUSAT CAT 2024 - Shift 5*

To protect iron against corrosion, the most durable metal plating on it is

(A) Tin plating	**(B)** Copper plating
(C) Zinc plating	**(D)** Nickel plating

Q307. *CUSAT CAT 2024 - Shift 5*

The intensive property listed below is

(A) Internal energy	**(B)** Volume
(C) Mole fraction	**(D)** Mass

Q308. *CUSAT CAT 2024 - Shift 5*

When a system releases 10 kJ of energy to the surroundings as work, the internal energy of the system

(A) decreases by 10 kJ	**(B)** increases by 5 Kj
(C) reduced 20 kJ	**(D)** remains constant

Q309. *CUSAT CAT 2024 - Shift 5*

Quantitative relation between pressure and solubility of a gas in a solvent is given by …….. law.

(A) Le Chatelier	**(B)** Dalton
(C) Henry	**(D)** Raoult

Q310. *CUSAT CAT 2024 - Shift 5*

Which of the following relation is true for Rhombohedral crystal system?

(A) $\alpha=\beta=\gamma=90°$	**(B)** $\alpha\neq\beta=\gamma=90°$
(C) $\alpha=\beta=90°\gamma=120°$	**(D)** $\alpha=\beta=\gamma\neq90°$

Q311. *CUSAT CAT 2024 - Shift 5*

In Ziegler Natta catalyst the commonly used catalyst system is

(A) TiCl , Al(C H) 4 2 5 3	**(B)** (η 5 -cp) TiCl , Al(OEt) 2 2 3
(C) VO (acac) Al (CH) 2 2 3 6	**(D)** TiCl , BF 4 3

Q312. *CUSAT CAT 2024 - Shift 5*

In an isothermal expansion of an ideal gas, the internal energy of molecules

(A) may increase or decrease	**(B)** does not change
(C) increase	**(D)** decrease

Q313. *CUSAT CAT 2024 - Shift 5*

The combined form of first and second law of thermodynamics is given by (p = Pressure, V = Volume, T = Temperature, U = Internal energy, S = Entropy, Q = Quantity of heat)

(A) TdS =dU + pdV	**(B)** dQ=TdS+ pdV
(C) dU =TdS+dQ	**(D)** TdS =dU – pdV

Q314. *CUSAT CAT 2024 - Shift 5*

In which case rate of diffusion is highest if all are present in the same container at the same temperature?

(A) 4g H 2	**(B)** 32g O 2
(C) 22g CO 2	**(D)** 56g N 2

Q315. *CUSAT CAT 2024 - Shift 5*

Maleic acid (shown below) is a geometrical isomer of fumaric acid. Which of the statements given below is true? (pK is a measure of acidity) a

(A) Both have same pK (1 st and 2 nd) a	**(B)** 1 st pK of maleic acid < 1 st pK of fumaric acid a a
(C) 2 nd pK of maleic acid < 2 nd pK of fumaric acid a a	**(D)** 1 st pK of maleic and fumaric acids are same, a nd nd while 2 pK of maleic acid < 2 pK of fumaric acid a a

Q316. *CUSAT CAT 2024 - Shift 5*

A more environmentally friendly alternative to using CFC for refrigeration is:

(A) Tetrafluoroethylene	**(B)** Hydrofluorocarbons
(C) Bromoflurocarbons	**(D)** Tetrachloroethylene

Q317. *CUSAT CAT 2024 - Shift 5*

Cyclohexene can be converted to adipic acid [CO H(CH) CO H] using 2 2 4 2

(A) O , NaBH 3 4	**(B)** O , Me S 3 2
(C) O , H O 3 2	**(D)** O , H O 3 2 2

Q318. *CUSAT CAT 2024 - Shift 5*

In which of the following compounds π-delocalization is NOT possible

(A) 1,3-Butadiene	**(B)** 1,2-Butadiene
(C) 1,3,5-Hexatriene	**(D)** Benzene

Q319. *CUSAT CAT 2024 - Shift 5*

The following reaction is best accomplished by H CO H CO 3 3 ? +

(A) Treating with strong base	**(B)** Neat heating at 50 °C
(C) Treating with strong acids like conc. H SO 2 4	**(D)** Using free radical initiators

Q320. *CUSAT CAT 2024 - Shift 5*

Explain why 2,5-dimethyl-1,4-dioxane (shown below) is not chiral. O O

(A) It has an axis of symmetry	**(B)** It has a plane of symmetry
(C) It has no chiral carbons	**(D)** It has a centre of symmetry

Q321. *CUSAT CAT 2024 - Shift 5*

Oximes of benzophenones rearranges on reacting with SOCl to form the 2 corresponding Benzanilides. This rearrangement reaction is an example for

(A) Hofmann Rearrangment	**(B)** Curtius Rearrangement
(C) Wagner-Meerwein Rearrangement	**(D)** Beckmann Rearrangement

Q322. *CUSAT CAT 2024 - Shift 5*

An organic compound which has the ability to exhibit optical activity is

(A) 2–Methylbutane	**(B)** 3–Methylpentane
(C) Methylcyclopentane	**(D)** 3–Methylhexane

Q323. *CUSAT CAT 2024 - Shift 5*

Williamson's ether synthesis works most efficiently in

(A) acidic medium	**(B)** basic medium
(C) neutral conditions	**(D)** neat heating above 50 °C

Q324. *CUSAT CAT 2024 - Shift 5*

Calculate the electron gain enthalpy of sodium ion, whose first ionisation potential is 5.1 eV.

(A) –2.55 eV	**(B)** 5.1 eV
(C) 10.2 eV	**(D)** 2.55 eV

Q325. *CUSAT CAT 2024 - Shift 5*

Calomel on reaction with NH OH gives 4

(A) Hg O 2	**(B)** HgO
(C) HgNH Cl 2	**(D)** NH Hg Cl 2 2

Q326. *CUSAT CAT 2024 - Shift 5*

Which among the following catalysts and the unique property made them to act both as homogeneous and heterogeneous catalyst?

(A) Platinum (Pt); Due to its high surface area	**(B)** Iron (Fe); Exhibits redox activity
(C) Palladium (Pd); Forms stable complexes in solution	**(D)** Zeolites; Possess well-defined porous structures

Q327. *CUSAT CAT 2024 - Shift 5*

The number of ions per molecule of the complexes CoCl .5NH in aqueous solution will 3 3 be

(A) 2	**(B)** 3
(C) 4	**(D)** 9

Q328. *CUSAT CAT 2024 - Shift 5*

The crystal field splitting energy for octahedral (Δ) and tetrahedral (Δ) complexes is o t related as

(A) Δ = 4/9 Δ t o	**(B)** Δ = 1/2 Δ t o
(C) Δ = 1/2 Δ o t	**(D)** Δ = 4/9 Δ o t

Q329. *CUSAT CAT 2024 - Shift 5*

The complexes [Co(NH) NO]Cl and [Co(NH) ONO]Cl are examples of 3 5 2 2 3 5 2

(A) Linkage isomers	**(B)** Ionisation isomers
(C) Geometrical isomers	**(D)** Co-ordination isomers

Q330. *CUSAT CAT 2024 - Shift 5*

The IUPAC name of the compound K [Fe (CN)] is 4 6

(A) Potassium hexacyanoferrate(II)	**(B)** Potassium hexacyanoferrate(III)
(C) Hexacyano potassiumferrate(II)	**(D)** Hexacyano potassiumferrate(III)

Q331. *CUSAT CAT 2024 - Shift 5*

To which block do the elements with atomic number 56 belongs?

(A) p	**(B)** f
(C) d	**(D)** s

Q332. *CUSAT CAT 2024 - Shift 5*

Which state of matter has the highest kinetic energy?

(A) Solid	**(B)** Liquid
(C) Gas	**(D)** Plasma MATHEMATICS UG (SHIFT V FINAL)

Q333. *CUSAT CAT 2025 - Shift 1*

For a general chemical reaction aA + bB■cC + dD, If the equilibrium constant is K. Then equilibrium constant for the reaction 2aA + 2bB ■2cC + 2dD will be

(A) 2K	**(B)** K
(C) K2	**(D)** K/2

Q334. *CUSAT CAT 2025 - Shift 1*

The formation of carbon monoxide from coal is shown by the equation C(s)+H O(g)■CO(g)+H (g) 2 2 The wrong statement about the chemical equilibria of the reaction is I As concentration of water increases, quantity of Hydrogen will increase II Addition of CO will decrease the amount of Hydrogen III By increasing the pressure, the reaction proceeds in forward direction IV Quantity of CO remains unchanged by the addition of coal into the system

(A) I	**(B)** II
(C) III	**(D)** IV

Q335. *CUSAT CAT 2025 - Shift 1*

The mole fraction of 10 moles of Glucose in water solution is 0.50. The volume of water taken for dissolving glucose is

(A) 9 mL	**(B)** 18 mL
(C) 90 mL	**(D)** 180 mL

Q336. *CUSAT CAT 2025 - Shift 1*

Rate of chemical reactions in solution do not depend to a great extend upon

(A) Pressure	**(B)** Temperature
(C) Concentration	**(D)** Catalyst

Q337. *CUSAT CAT 2025 - Shift 1*

The position of both an electron and helium atom is known within 1.0 nm. The momentum of the electron is known within 5.0 × 10–26kg ms–1. The minimum uncertainty in the measurement of the momentum of the helium atom is

(A) 7.0 × 10–26kg ms–1	**(B)** 5.0 × 10–26kg ms–1
(C) 8.0 × 10–26kg ms–1	**(D)** 6.0 × 10–26kg ms–1

Q338. *CUSAT CAT 2025 - Shift 1*

The solubility products of Al(OH) and Zn(OH) are 8.5 × 10–23 and 1.8 × 10–14 3 2 respectively. If NH OH is added to a solution containing Al3+ and Zn2+ ions, then 4 substance precipitated first is

(A) Al(OH) 3	**(B)** Zn(OH) 2
(C) Both together	**(D)** No precipitate is formed

Q339. *CUSAT CAT 2025 - Shift 1*

The e.m.f. of the cell Zn/Zn2+ (0.01 M) || Fe2+ (0.001 M) Fe at 298 K is 0.2905 volt. Then the value of equilibrium constant for the cell reaction is

(A) 10	**(B)** 1010 1
(C) 10 1	**(D)** 10 10

Q340. *CUSAT CAT 2025 - Shift 1*

Most acidic proton among the highlighted protons of HOCH CH CH C≡CH is 2 2 2

(A) C≡CH	**(B)** HOCH 2
(C) HOCH 2	**(D)** CH C≡CH 2

Q341. *CUSAT CAT 2025 - Shift 1*

Identify the Hoffmann elimination product formed when 2-bromopentane reacts with sodium tert-butoxide in tert-butanol.

(A) —	**(B)** —
(C) —	**(D)** —

Q342. *CUSAT CAT 2025 - Shift 1*

What products among below are formed by reaction of buta-1,3-diene with HCl (1 equivalent)?

(A) i only	**(B)** i and iv
(C) i and ii	**(D)** i and iii

Q343. *CUSAT CAT 2025 - Shift 1*

Acetophenone reacts with iodine - KI solution and aqueous NaOH to give yellow solid P, which can be removed by filtration. The filtrate when acidified with HCl produces a white precipitate of compound Q. Identify P and Q.

(A) PhCHO and PhCO H 2	**(B)** CHI and PhCHO 3
(C) CHI and PhCO H 3 2	**(D)** CH I and PhCOCO H 3 2

Q344. *CUSAT CAT 2025 - Shift 1*

Which among MgBr , (CH) P, CH CH OH and (CH) NH can act as both Lewis 2 3 3 3 2 3 2 acid and Lewis base?

(A) MgBr 2	**(B)** (CH) NH 3 2
(C) CH CH OH 3 2	**(D)** (CH) P 3 3

Q345. *CUSAT CAT 2025 - Shift 1*

An aminoacid containing sulphur atom is:

(A) Valine	**(B)** Serine
(C) Lysine	**(D)** Cysteine

Q346. *CUSAT CAT 2025 - Shift 1*

Carbonyl carbon in which of the following compounds are in the same oxidation level? CH COOH, CH CHO, CH COCl, CH CONH 3 3 3 3 2

(A) CH COOH and CH CHO 3 3	**(B)** CH COOH, CH CHO and CH COCl 3 3 3
(C) All four	**(D)** CH COOH, CH COCl and CH CONH 3 3 3 2

Q347. *CUSAT CAT 2025 - Shift 1*

Which among following compounds test negative with Tollens reagent? O

(A) OH CHO	**(B)** O
(C) H O	**(D)** OH

Q348. *CUSAT CAT 2025 - Shift 1*

1-Bromo-2-methylpropane is treated sequentially with i) Mg, Et O ii) CO , iii) H O+. 2 2 3 What is the product formed?

(A) 2-methylpropanoic acid	**(B)** 2-methylbutanoic acid
(C) 3-methylpropanoic acid	**(D)** 3-methylbutanoic acid

Q349. *CUSAT CAT 2025 - Shift 1*

Benzene reacting with CH COCl + anhydrous AlCl to give acetophenone is an 3 3 example for

(A) Aromatic nuceleophilic substitution	**(B)** Lewis acid catalyzed addition
(C) Hunsdiecker reaction	**(D)** Aromatic electrophilic substitution

Q350. *CUSAT CAT 2025 - Shift 1*

An element X has the following isotopic composition of 200X :90%199X : 8% 202X : 2% . X's weighted average atomic mass is most similar to

(A) 201 amu	**(B)** 202 amu
(C) 199 amu	**(D)** 200 amu

Q351. *CUSAT CAT 2025 - Shift 1*

If an electron's energy in the ground state of a hydrogen atom is -x eV, then energy in the second excited state of He+ is

(A) -x eV	**(B)** -4/9 x eV
(C) +2 eV	**(D)** -9/4 x eV

Q352. *CUSAT CAT 2025 - Shift 1*

Which of the following is not true for covalent molecules?

(A) They may exhibit isomerism	**(B)** They have low melting and boiling points
(C) They show ionic reactions	**(D)** They show molecular reactions

Q353. *CUSAT CAT 2025 - Shift 1*

Which of the following species contains equal number of σ- and π- bonds?

(A) CH (CN) 2 2	**(B)** HCO – 3
(C) XeO 4	**(D)** (CN) 2

Q354. *CUSAT CAT 2025 - Shift 1*

Which of the following pairs of ions are isoelectronic and isostructural?

(A) ClO –, SO 2– 3 3	**(B)** CO 2–, SO 2– 3 3
(C) ClO –, CO 2– 3 3	**(D)** SO 2–, NO – 3 3

Q355. *CUSAT CAT 2025 - Shift 1*

What percent of a sample of nitrogen must be allowed to escape if its temperature, pressure and volume are to be changed from 220°C, 3 atmand 1.65 litre to 110■, 0.7 atm and 1.00 litre respectively?

(A) 81.8%	**(B)** 71.8%
(C) 76.8%	**(D)** 86.8%

Q356. *CUSAT CAT 2025 - Shift 1*

At room temperature, which of the following gaseous mixtures defies Dalton's law of partial pressure?

(A) NO and O 2 2	**(B)** NH and HCl 3
(C) CO and CO 2	**(D)** SO and SO 2 3

Q357. *CUSAT CAT 2025 - Shift 1*

The partial pressure of hydrogen in a flask containing 2g H and 32g SO is 2 2

(A) 1/16th of total pressure	**(B)** 1/9th of total pressure
(C) 2/3rd of total pressure	**(D)** 1/8th of total pressure

Q358. *CUSAT CAT 2025 - Shift 1*

The mixture of three gases X, Y, and Z is enclosed in closed vessel at constant temperature. Molecular weight of X is highest and that of Y is the least. When equilibrium is established the

(A) Gas X will be more at bottom	**(B)** Gas Y will be more at top
(C) Gases X, Y and Z homogeneously present	**(D)** Gas Y will be more at bottom

Q359. *CUSAT CAT 2025 - Shift 1*

Which one the following pairs of substances on reaction will not evolve H gas? 2

(A) copper and HCl	**(B)** iron and steam
(C) iron and sulphuric acid	**(D)** sodium and ethyl alcohol

Q360. *CUSAT CAT 2025 - Shift 1*

Hydrogen can be used to form helium at

(A) High temperature and high pressure	**(B)** High temperature and low pressure
(C) Low temperature and high pressure	**(D)** Low temperature and low pressure

Q361. *CUSAT CAT 2025 - Shift 1*

Which of these produces oxide when heated vigorously?

(A) LiNO 3	**(B)** NaNO 3
(C) KNO 3	**(D)** RbNO 3

Q362. *CUSAT CAT 2025 - Shift 1*

Which of the following oxides is not expected to react with NaOH?

(A) CaO	**(B)** SiO 2
(C) BeO	**(D)** B O 2 3

Q363.

CUSAT CAT 2025 - Shift 2

Which of the following fuel has the highest calorific value?

(A) Coal	**(B)** Gasoline
(C) Kerosine	**(D)** Hydrogen

Q364.

CUSAT CAT 2025 - Shift 2

Palmitic acid (Molecular weight = 256.42 g/mol) is one of the major sources of energy in our diet as shown in the chemical reaction provided below C H O (s)+23O (g)→16CO (g)+16H O(l) 16 32 2 2 2 2 The energy released by the combustion of 1 g of palmitic acid is……. kJ/g (ΔH0 (CO) = -393.5 kJ/mol; ΔH0 (H O) = -285.8 kJ/mol; ΔH0 (C16H32O2) = f 2 f 2 f –891.5kJ/mol; ΔH0 (O2) = 0 kJ/mol) f

(A) – 19 kJ/g	**(B)** – 29 kJ/g
(C) – 39 kJ/g	**(D)** – 49 kJ/g

Q365.

CUSAT CAT 2025 - Shift 2

For a general chemical equilibria AX (g) AX (g) + ½ X (g), the equilibrium 3 2 2 constant is K ; The equilibrium constant for■ the reaction 2AX (g) + X (g) 2 AX c 2 2 3 (g) will be …….. ■

(A) 2K c	**(B)** K /2 c
(C) K 2 c	**(D)** 1/K 2 c

Q366.

CUSAT CAT 2025 - Shift 2

For the Haber Process, N (g) + 3H (g) 2NH (g), the value of K at 300 oCis 9.60. 2 2 3 c The value of K at same temperature will■ be p (The value of R, is 0.08206 L-atm. mol-1.K-1)

(A) 4.34 x 10-3	**(B)** 2.17 x 104
(C) 1.6 x 10-2	**(D)** 5.8 x 103

Q367.

CUSAT CAT 2025 - Shift 2

The standard EMF for the cell reaction Zn + Cu2+→ Cu + Zn2+ is 1.10 V at 25 oC. The EMF of the cell when 0.1M Zn2+ and 0.1M Cu2+ solutions are used at 25 oC is

(A) 1.10 V	**(B)** 0.55 V
(C) 0.275 V	**(D)** 0.11 V

Q368.

CUSAT CAT 2025 - Shift 2

The equilibrium constant for a cell reaction, Cu(s) + 2Ag+(aq) → Cu2+(aq) + 2Ag (s) is 4× 1016. E0(cell) for the cell reaction is

(A) 0.63 V	**(B)** 0.49 V
(C) 0.77 V	**(D)** 0.89 V

Q369. *CUSAT CAT 2025 - Shift 2*

The shape of orbital with the value of l = 2, m = 0 is

(A) Spherical	**(B)** Double dumb-bell
(C) Trigonal planar	**(D)** Square planar

Q370. *CUSAT CAT 2025 - Shift 2*

The vapour pressure of water at 300 K in a closed container is 0.4 atm. If the volume of container is doubled, its vapour pressure at 300 K will be

(A) 0.8 atm	**(B)** 0.2 atm
(C) 0.4 atm	**(D)** 0.6 atm

Q371. *CUSAT CAT 2025 - Shift 2*

Internal energy does not include

(A) Rotational energy	**(B)** Vibrational energy
(C) Nuclear energy	**(D)** Gravitational energy pull

Q372. *CUSAT CAT 2025 - Shift 2*

Precipitation takes place when

(A) Ionic product < Ksp	**(B)** Ionic product > Ksp
(C) [Ionic product]2 = Ksp	**(D)** [Ionic product]2 > Ksp

Q373. *CUSAT CAT 2025 - Shift 2*

In the reaction 2A + B → A B, if the concentration of A is doubled and that of B is 2 halved, then the rate of the reaction will

(A) increase 2 times	**(B)** increase 4 times
(C) decrease 2 times	**(D)** remain the same

Q374. *CUSAT CAT 2025 - Shift 2*

Hess' law is applicable for the determination of heat of

(A) Transition	**(B)** Formation
(C) Reaction	**(D)** All of the above

Q375. *CUSAT CAT 2025 - Shift 2*

Which of the following is an example of a fractional order reaction?

(A) NH NO → N + 2H O 4 2 2 2	**(B)** NO + O → NO + O 3 2 2
(C) 2NO + Br → 2NOBr 2	**(D)** CH CHO → CH + CO 3 4

Q376. *CUSAT CAT 2025 - Shift 2*

Glucose of 18 g was dissolved in 170 g of water. The vapour pressure of water for this aqueous solution at 37°C is [Given : vapour pressure of water as solvent is 760 torr]

(A) 652 torr	**(B)** 752 torr
(C) 756 torr	**(D)** 854 torr

Q377. *CUSAT CAT 2025 - Shift 2*

A mixed oxide has ccp arrangement in which the cations 'X' occupy 1/3rd of octahedral voids and the cations 'Y' occupy 1/3rd of tetrahedral voids. The formula of oxide is

(A) X Y O 2 3 2	**(B)** XY O 3
(C) X YO 2 3	**(D)** XY O 3

Q378. *CUSAT CAT 2025 - Shift 2*

Calculate the energy associated with the second orbital of Li2+. What is the radius of this orbit?

(A) –3.27 × 10–18 J, 0.0793 nm	**(B)** –4.905 × 10–18 J, 0.0705 nm
(C) –9.81 × 10–18 J, 0.1190 nm	**(D)** 1.635 × 10–18 J, 0.0352 nm

Q379. *CUSAT CAT 2025 - Shift 2*

is named as

(A) 1,1-dimethyl-3-cycloheptene	**(B)** 4,4-dimethyl-cycloheptene
(C) 3,3-dimethyl-cycloheptene	**(D)** 6,6-dimethyl-cycloheptene

Q380. *CUSAT CAT 2025 - Shift 2*

In E2 elimination, some compounds follow Hofmann's rule which means

(A) the double bond goes to the most substituted position	**(B)** newly formed double bond will be in conjugation with double bond already present in the molecule
(C) when strong bases are used double bond goes to the less substituted carbon, and when weak bases are used double bond goes to the less substituted carbon	**(D)** the double bond goes mainly towards the least substituted carbon

Q381. *CUSAT CAT 2025 - Shift 2*

When pent-2-yne is treated with dil. H SO and HgSO , major product formed is 2 4 4

(A) pentan-2-ol	**(B)** pentan-3-ol
(C) pentan-3-one	**(D)** pentane-2,3-diol

Q382. *CUSAT CAT 2025 - Shift 2*

Which among the following fails to give a positive test for nitrogen in Lassaigne's test?

(A) RNA	**(B)** Nicotine
(C) Insulin	**(D)** Cholesterol

Q383. *CUSAT CAT 2025 - Shift 2*

Selective reduction of PhCH=CHCO Et to PhCH=CHCH OH is performed using 2 2

(A) Zn/Hg, HCl	**(B)** H /Pd-C 2
(C) NH NH , NaOH 2 2	**(D)** LiAlH in dry ether 4

Q384. *CUSAT CAT 2025 - Shift 2*

Upon heating sodium hydrogen carbonate, the products formed are

(A) Na O + CO + H O 2 2 2	**(B)** Na CO + CO 2 3 2
(C) Na CO + CO + H O 2 3 2 2	**(D)** NaCO + H O 3 2

Q385. *CUSAT CAT 2025 - Shift 2*

Of the following, which one isn't a reducing oxide?

(A) SO 2	**(B)** SeO 2
(C) TeO	**(D)** SO 3

Q386. *CUSAT CAT 2025 - Shift 2*

The compound insoluble in aqueous NH is 3

(A) AgI	**(B)** AgCl
(C) ZnSO 4	**(D)** HgCl 2

Q387. *CUSAT CAT 2025 - Shift 2*

Carbon and silicon belong to (IV) group. The maximum coordination number of carbon in commonly occurring compounds is 4, whereas that of silicon is 6. This is due to

(A) Availability of low lying d-orbitals in Silicon	**(B)** Large size of Silicon
(C) More electropositive nature of Silicon	**(D)** None of the above

Q388. *CUSAT CAT 2025 - Shift 2*

Nitrogen forms N , but phosphorus does not form P , However, it forms P reason is 2 2 4,

(A) Triple bond present between phosphorus atom	**(B)** pπ-pπ bonding is weak
(C) pπ-pπ bonding is strong	**(D)** Multiple bonds form easily

Q389. *CUSAT CAT 2025 - Shift 2*

Transuranic elements start with

(A) Np	**(B)** Cm
(C) Pu	**(D)** U

Q390. *CUSAT CAT 2025 - Shift 2*

The number of moles of KMnO required to reduce one mole of KI in alkaline 4 medium is

(A) 1/4	**(B)** 5
(C) 1	**(D)** 2

Q391. *CUSAT CAT 2025 - Shift 2*

Which of the following is diamagnetic complex?

(A) [Fe(H O)]3+ 2 6	**(B)** [Fe(H O)]2+ 2 6
(C) [Fe(CN)]3- 6	**(D)** [Fe(CN)]4- 6

Q392. *CUSAT CAT 2025 - Shift 2*

A magnetic moment of 1.73 BM will be shown by which one among the following?

(A) [Ni(CN)]2- 4	**(B)** TiCl 4
(C) [CoCl]4- 6	**(D)** [Cu(NH)]2+ 3

Q393. *CUSAT CAT 2025 - Shift 2*

The colorless species are

(A) Cu Cl and [NiCl]2- 2 2 4	**(B)** [TiF]2- and Cu Cl 6 2 2
(C) [CoF]3-and [NiCl]2- 6 4	**(D)** [TiF]2-and [CoF]3- 6 6

Q394. *CUSAT CAT 2025 - Shift 2*

What is the number of electrons in all the completely filled subshells of n = 4 ?

(A) 30	**(B)** 32
(C) 34	**(D)** 36

Q395. *CUSAT CAT 2025 - Shift 3*

According to the first law of thermodynamics, for an isolated system, ΔU =

(A) Negative	**(B)** Zero
(C) Positive	**(D)** The value of ΔU cannot be predicted by using first law

Q396. *CUSAT CAT 2025 - Shift 3*

The ionization of water (H O(l) H+(aq) + OH-(aq)) has an equilibrium constant of 2 ■ 1 x 10-14 at 298 K. The ΔG0 of the reaction is ……..kJ/mol

(A) – 79.8 kJ/mol	**(B)** +79.8 kJ/mol
(C) +39.9 kJ/mol	**(D)** – 39.9 kJ/mol

Q397. *CUSAT CAT 2025 - Shift 3*

pOH of a 1 × 10–9 M aqueous solution of hydronium ion in room temperature will be

(A) 9	**(B)** 7.9
(C) 5	**(D)** 10.9

Q398. *CUSAT CAT 2025 - Shift 3*

A solution containing equimolar concentration of SnCl , NiCl , MgCl , AgNO , 2 2 2 3 AuCl is being electrolyzed by using inert electrodes. The standard electrode 3 potentials (in Volts) for each metal ions is Sn2+/Sn = –0.14V, Ni2+/Ni = –0.25 V, Mg2+/Mg= –2.36 V, Ag+/Ag = 0.80 V and Au3+/Au = 1.40V With an increasing voltage, the order of metal deposition on the cathode will be

(A) Sn, Ni, Ag, Au, Mg	**(B)** Au, Ag, Sn, Ni, Mg
(C) Mg, Ni, Sn, Ag, Au	**(D)** Mg, Au, Ag, Ni, Sn

Q399. *CUSAT CAT 2025 - Shift 3*

The amount of O dissolved in one liter of water at 25°C under a partial pressure of 2 oxygen 0.75 atm is ……. (Henry's law constant k = 1.4 × 10–3 mol/L/atm) H

(A) 33.6 mg	**(B)** 3.36 mg
(C) 33.6 g	**(D)** 3.36 g

Q400. *CUSAT CAT 2025 - Shift 3*

…….. gets deposited on both anodic and cathodic plates of a discharged lead-acid battery

(A) Pb	**(B)** PbO 2
(C) Pb O 2 4	**(D)** PbSO 4

Q401. *CUSAT CAT 2025 - Shift 3*

Which of the following expression gives the de Broglie relationship? h

(A) λ= mp h	**(B)** p = mv
(C) ε=mc 2 h	**(D)** λ= mv

Q402. *CUSAT CAT 2025 - Shift 3*

The enthalpies of all elements in their standard states are

(A) Unity	**(B)** Zero
(C) < 0	**(D)** Different for each element

Q403. *CUSAT CAT 2025 - Shift 3*

Half-life of 10 g of radioactive substance is 10 days. The half-life of 20 g is

(A) 10 days	**(B)** 20 days
(C) 40 days	**(D)** 15 days

Q404. *CUSAT CAT 2025 - Shift 3*

Ksp for AgCl (mol wt 143.5) is 17 × 10–11 at 298 K. The molar solubility of AgCl at 298 K is

(A) 18.7 × 10–5	**(B)** 2.3 × 10–3
(C) 9.90 × 10–3	**(D)** 1.3 × 10–5

Q405. *CUSAT CAT 2025 - Shift 3*

DDT on exposure to water decomposes. Half-life = 10 years. How much time will it take for its 90% decomposition?

(A) 50 years	**(B)** 33.3 years
(C) 70 years	**(D)** 80 years

Q406. *CUSAT CAT 2025 - Shift 3*

Which of the following is used as an electrolyte in an H -O fuel cell? 2 2

(A) KOH	**(B)** NH OH 4
(C) Fe(OH) 2	**(D)** Cu(OH) 2

Q407. *CUSAT CAT 2025 - Shift 3*

If Zn2+/Zn electrode is diluted 100 times, then the change in emf is

(A) increase of 59 mV	**(B)** decrease of 59 mV
(C) increase of 29.5 mV	**(D)** decrease of 29.5 mV

Q408. *CUSAT CAT 2025 - Shift 3*

A + 2B → C + D. If –d[A]/dt = 5 × 10–4 mol L–1 s–1, then –d[B]/dt is

(A) 2.5 × 10–4 mol L–1 s–1	**(B)** 5.0 × 10–4 mol L–1 s–1
(C) 2.5 × 10–3 mol L–1 s–1	**(D)** 1.0 × 10–3 mol L–1 s–1

Q409. *CUSAT CAT 2025 - Shift 3*

The vapour pressure of A and B at 27°C are 60 mm Hg and 40 mm Hg respectively. If A and B are mixed such that the mole fraction of A in the mixture is 0.8, then the mole of B in the vapour phase is Z × 10–1. The value of Z is

(A) 1.0	**(B)** 1.2
(C) 1.4	**(D)** 1.6

Q410. *CUSAT CAT 2025 - Shift 3*

IUPAC name of (CH) CHCH=CHCH is 3 2 3

(A) 2-methylpent-3-ene	**(B)** 4-methylpent-2-ene
(C) 1,1-dimethylbut-2-ene	**(D)** 4,4,-dimethylbut-2-ene

Q411. *CUSAT CAT 2025 - Shift 3*

Major product formed when 2,2-dibromobutane is heated at 200 °C in presence of molten KOH is (note: reaction is carried out under thermodynamically controlled conditions)

(A) 1-bromo-but-1-yne	**(B)** 1-bromo-but-2-yne
(C) but-1-yne	**(D)** but-2-yne

Q412. *CUSAT CAT 2025 - Shift 3*

Pick the major product formed in the following reaction sequence.

(A) —	**(B)** —
(C) —	**(D)** —

Q413. *CUSAT CAT 2025 - Shift 3*

Which of the following reactions will not result in the formation of new carbon-carbon bonds?

(A) Reimer-Tieman reaction	**(B)** Cannizaro reaction
(C) Wurtz reaction	**(D)** Friedel-Crafts acylation

Q414. *CUSAT CAT 2025 - Shift 3*

SN1 substitution on a chiral carbon of an optically active compound proceeds with:

(A) retention in configuration	**(B)** inversion in configuration
(C) racemization	**(D)** inversion at higher temperature and retention at lower temperature

Q415. *CUSAT CAT 2025 - Shift 3*

Additive that was used as an anti-knock agent in petrol that has now been banned due to environmental and health concerns is:

(A) Ethyl acetate	**(B)** Lead tetrachloride
(C) Lead acetate	**(D)** Tetraethyllead [(C H) Pb] 2 5 4

Q416. *CUSAT CAT 2025 - Shift 3*

Which among the following name reactions is best suited to convert acetic acid to bromoacetic acid? CH CO H → Br-CH CO H 3 2 2 2

(A) Hell-Volhard-Zelinsky reaction	**(B)** Hunsdiecker reaction
(C) Reformatsky raction	**(D)** Hoffmann bromamide reaction

Q417. *CUSAT CAT 2025 - Shift 3*

Which one of the following bases is not present in DNA?

(A) Adenine	**(B)** Guanine
(C) Cytosine	**(D)** Uracil

Q418. *CUSAT CAT 2025 - Shift 3*

Major product formed when hex-3-yne reacts with Na/liq. NH is: 3

(A) cis-hex-3-ene	**(B)** trans-hex-3-ene
(C) 3-aminohexane	**(D)** 3,4-diaminohexane

Q419. *CUSAT CAT 2025 - Shift 3*

What is the number of radial nodes for 3p orbital ?

(A) 1	**(B)** 2
(C) 3	**(D)** 4 "

Q420. *CUSAT CAT 2025 - Shift 3*

Which transition in the hydrogen atom spectrum would have the same wavelength as the Balmer type transition from n = 4 to n = 2 of He+ spectrum

(A) n = 2 to n = 1	**(B)** n = 3 to n = 1
(C) n = 2 to n = 4	**(D)** n = 3 to n = 4

Q421. *CUSAT CAT 2025 - Shift 3*

Statement I: 5f electrons can participate in bonding to a far greater extent than 4f electrons Statement II: 5f orbitals are not as buried as 4f orbitals Choose the correct option

(A) Both the Statements are true	**(B)** Statement I is true but Statement II is false
(C) Statement I is false and Statement II is true	**(D)** Both the Statements are false

Q422. *CUSAT CAT 2025 - Shift 3*

Which among the following is the most basic oxide?

(A) Ga O 2 3	**(B)** Al O 2 3
(C) B O 2 3	**(D)** In O 2 3

Q423. *CUSAT CAT 2025 - Shift 3*

What is the spin only magnetic moment value of [Pt(NH) Cl(NH CH)]Cl? 3 2 2 3

(A) 0	**(B)** 2.6
(C) 3.4	**(D)** 4.9

Q424. *CUSAT CAT 2025 - Shift 3*

Which of the following has square pyramidal shape?

(A) [Ni(CN)]2■ 4	**(B)** PCl 5
(C) BrF 5	**(D)** PF 5

Q425. *CUSAT CAT 2025 - Shift 3*

What is the geometry of B in BF and BF NMe ? 3 3 3

(A) Trigonal planar and tetrahedral	**(B)** Trigonal planar and pentagonal bipyramidal
(C) Tetrahedral and pentagonal bipyramidal	**(D)** Tetrahedral and trigonal planar

Q426. *CUSAT CAT 2025 - Shift 3*

Identify the correct statement about B H . 2 6

(A) All B■H■B angles are 120°	**(B)** The two B■H■B bonds have different length
(C) Terminal B■H bonds have less p-character than the bridging B■H bonds	**(D)** Its fragment BH3 is a Lewis base

Q427. *CUSAT CAT 2025 - Shift 3*

How many geometrical isomers are possible for the square planar complex [Pd(py)(Cl)(Br)(NH)], py = pyridine? 3

(A) 2	**(B)** 4
(C) 5	**(D)** 3

Q428. *CUSAT CAT 2025 - Shift 3*

Why is cuprous ion colourless while cupric ion is coloured?

(A) Both have half-filled p- and d-orbitals	**(B)** Cuprous ion has incomplete d-orbital and cupric ion has a complete d-orbital
(C) Both have unpaired electrons in the d-orbitals	**(D)** Cuprous ion has complete d-orbital and cupric ion has an incomplete d-orbital

Q429. *CUSAT CAT 2025 - Shift 3*

Identify the correct geometry and magnetic property of [Ni(CO)] complex. 4

(A) Square planar and diamagnetic	**(B)** Tetrahedral and diamagnetic
(C) Tetrahedral and paramagnetic	**(D)** Square planar and paramagnetic

Q430. *CUSAT CAT 2025 - Shift 4*

The exhaust of a car shows a carbonmonoxide (CO) emission of 400 ppm. The concentration of CO per Liter of exhaust is ……

(A) 0.004 mL	**(B)** 0.04 mL
(C) 0.4 mL	**(D)** 4.0 mL

Q431. *CUSAT CAT 2025 - Shift 4*

During an experiment on observing depression in freezing point of an organic solvent (5g) by the addition of an non-volatile, non-electrolytic miscible solute (weight = 0.5 g), depression of freezing point 5 oC was observed. Molal depression constant of the pure solvent is 7.0 K. Kg.mol-1. The molecular weight of the solute is ……g/mol

(A) 280	**(B)** 210
(C) 140	**(D)** 70

Q432. *CUSAT CAT 2025 - Shift 4*

Which of the following graph represents integrated rate equation for a second order reaction? [A is the reactant] The image cannot be displayed. Your computer may not have enough memory to open the image, or the image may have been corrupted. Restart your computer, and then open the file again. If the red x still appears, you may have to delete the image and then insert it again.

(A) (I)	**(B)** (II)
(C) (III)	**(D)** None of the above

Q433. *CUSAT CAT 2025 - Shift 4*

Radioactive decay follows …….. order kinetics

(A) Zero	**(B)** First
(C) Second	**(D)** Third

Q434. *CUSAT CAT 2025 - Shift 4*

For the reaction H (g) + I (g) → 2HI(g) the change in enthalpy (ΔH) will be 2 2

(A) = ΔU	**(B)** > ΔU
(C) < ΔU	**(D)** zero

Q435. *CUSAT CAT 2025 - Shift 4*

When a catalyst is added to a reversible reaction in equilibrium state the value of the equilibrium constant

(A) increases	**(B)** decreases
(C) does not change	**(D)** becomes zero

Q436. *CUSAT CAT 2025 - Shift 4*

How many gram of CaC O will dissolve in one litre of saturated solution? Ksp of 2 4 CaC O is 2.5 × 10–9 mol–2 and its molecular weight is 128 2 4

(A) 0.0064 g	**(B)** 0.0128 g
(C) 0.0032 g	**(D)** 0.0640 g

Q437. *CUSAT CAT 2025 - Shift 4*

During the decomposition of H O to give dioxygen, 48 g O is formed per minute at 2 2 2 certain point of time. The rate of formation of water at this point is

(A) 0.75 mol min–1	**(B)** 1.5 mol min–1
(C) 2.25 mol min–1	**(D)** 3.0 mol min–1

Q438. *CUSAT CAT 2025 - Shift 4*

For which one of the following equilibria, K = K ? p c

(A) 2SO (g)+O (g) 2SO (g) 2 2 3	**(B)** C H (g) C H■(g) + H (g) 2 6 2 4 2
(C) 2HI(g) ■ H (g) + I (g) 2 2	**(D)** N (g) + ■ 3H (g) 2NH (g) 2 2 3 ■

Q439. *CUSAT CAT 2025 - Shift 4*

Which of the following relationship is correct?

(A) 1 poise = 1 g cm–1s–1	**(B)** 1 poise = 10–2 kg m–1s–1
(C) 1 poise = 10–1 kg m–1s–2	**(D)** 1 poise = 1 g cm–1s–2

Q440. *CUSAT CAT 2025 - Shift 4*

An ideal gas is allowed to expand from 1 l to 10 l at 300 K against constant external pressure of 1 atm. The work done is

(A) 6 kJ	**(B)** 6 l atm.
(C) 9 l atm.	**(D)** 9 kJ

Q441. *CUSAT CAT 2025 - Shift 4*

The IUPAC name of the following compound is:

(A) (E)-1-cyclohexyl-2-hexene	**(B)** (E)-2-cyclohexyl-2-hexene
(C) (Z)-1-cyclohexyl-1-hexene	**(D)** (Z)-2-cyclohexyl-2-hexene

Q442. *CUSAT CAT 2025 - Shift 4*

Which sets of reagents would selectively give the expected product?

(A) Hg(CF CO) 3 2 2	**(B)** Hg(OAc) in THF/H O, NaBH 2 2 4
(C) BH , NaOH/H O 3 2 2	**(D)** BH , H O /HO-, NaH, CH I 3 2 2 3

Q443. *CUSAT CAT 2025 - Shift 4*

Which among the following organic halide/halides on reaction with sodium metal (Wurtz coupling reaction) will give n-hexane as the only product?

(A) 1:1 mixture of chloroethane and 1-chlororobutane	**(B)** 1:1 mixture of bromomethane and 1-bromopentane
(C) 1-bromopropane only	**(D)** 1-bromopropane and propene

Q444. *CUSAT CAT 2025 - Shift 4*

Which of the following does not give a yellow precipitate with I +NaOH? 2

(A) CH CH OH 3 2	**(B)** CH OH 3
(C) CH CHO 3	**(D)** CH COCH 3 3

Q445. *CUSAT CAT 2025 - Shift 4*

Pick the correct statement on C-H bond distance in ethane, ethene and ethyne.

(A) C-H bond distance is equal in all the three compounds	**(B)** C-H bond distance in ethene is shorter than that in ethane but longer than that in ethyne
(C) C-H bond distance is shortest in ethyne and highest in ethane	**(D)** C-H bond distance is shortest in ethane and highest in ethyne

Q446. *CUSAT CAT 2025 - Shift 4*

Peptide bonds are formed by what type of linkages?

(A) ester	**(B)** amide
(C) urethane	**(D)** disulfide

Q447. *CUSAT CAT 2025 - Shift 4*

The compound formed as a result of oxidation of ethylbenzene with alkaline KMnO 4 followed by acidification is:

(A) benzyl alcohol	**(B)** benzophenone
(C) acetophenone	**(D)** benzoic acid

Q448. *CUSAT CAT 2025 - Shift 4*

Name of following Nobel Prize winning reaction.

(A) Claisen Condensation	**(B)** Diel's Alder reaction
(C) Dieckmann cyclisation	**(D)** Michael addition reaction

Q449. *CUSAT CAT 2025 - Shift 4*

RNA is different from DNA because RNA contains

(A) Ribose sugar and thymine	**(B)** Ribose sugar and uracil
(C) Deoxyribose sugar and thymine	**(D)** Deoxyribose sugar and uracil

Q450. *CUSAT CAT 2025 - Shift 4*

Identify the product in the nuclear reaction Mg24 + He4 = n1 + …….? 12 2 0

(A) Al27 13	**(B)** Si27 14
(C) Al28 13	**(D)** Mg25 12

Q451. *CUSAT CAT 2025 - Shift 4*

What is the n/p ratio of a stable nuclides?

(A) n/p =1	**(B)** n/p = 2
(C) n/p > 1	**(D)** n/p < 1

Q452. *CUSAT CAT 2025 - Shift 4*

Which of the following statements is not true?

(A) Si, Ga can be purified by zone refining	**(B)** Zone refining is carried out in an inert atmosphere
(C) Zinc can be refined by distillation	**(D)** Principle of zone refining is different from fractional crystallization

Q453. *CUSAT CAT 2025 - Shift 4*

The essential element for nitrogen fixation is

(A) Zn	**(B)** Cu
(C) Co	**(D)** Mo

Q454. *CUSAT CAT 2025 - Shift 4*

Which of the following has highest magnetic moment?

(A) Fe2+	**(B)** Cr2+
(C) Mn2+	**(D)** Co2+

Q455. *CUSAT CAT 2025 - Shift 4*

Volume of O liberated from 0.96g of H O at STP is 2 2 2

(A) 316.2 ml	**(B)** 224. 6 ml
(C) 613.2 ml	**(D)** 449.2 ml

Q456. *CUSAT CAT 2025 - Shift 4*

Shape of CIO – ion is 4

(A) Tetrahedral	**(B)** Square planar
(C) Square Pyramidal	**(D)** Trigonal bipyramidal

Q457. *CUSAT CAT 2025 - Shift 4*

Bell metal is an alloy of

(A) Sn and Pb	**(B)** Cu and Sn
(C) Sn and Sb	**(D)** Cu and Pb

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	D	3	B	4	A	5	B	6	C	7	D	8	A	9	D	10	B
11	C	12	A	13	B	14	B	15	B	16	A	17	C	18	A	19	A	20	B
21	A	22	B	23	A	24	B	25	B	26	A	27	B	28	A	29	D	30	B
31	C	32	C	33	C	34	D	35	C	36	C	37	B	38	B	39	D	40	C
41	A	42	D	43	D	44	A	45	D	46	C	47	A	48	C	49	A	50	D
51	D	52	B	53	A	54	A	55	B	56	C	57	B	58	A	59	C	60	A
61	C	62	D	63	B	64	D	65	A	66	D	67	C	68	A	69	D	70	D
71	D	72	B	73	B	74	B	75	D	76	C	77	D	78	B	79	B	80	B
81	C	82	A	83	C	84	A	85	D	86	A	87	B	88	D	89	D	90	D
91	C	92	D	93	C	94	D	95	B	96	C	97	B	98	C	99	B	100	C
101	D	102	C	103	C	104	B	105	A	106	A	107	B	108	C	109	C	110	B

111	C	112	B	113	B	114	A	115	D	116	B	117	B	118	B	119	A	120	D
121	B	122	A	123	A	124	D	125	B	126	A	127	B	128	A	129	D	130	B
131	C	132	B	133	A	134	A	135	A	136	B	137	A	138	D	139	D	140	C
141	B	142	C	143	C	144	D	145	C	146	B	147	D	148	D	149	C	150	B
151	B	152	C	153	C	154	C	155	B	156	B	157	D	158	C	159	B	160	A
161	C	162	B	163	C	164	B	165	B	166	B	167	C	168	B	169	A	170	D
171	B	172	D	173	B	174	A	175	A	176	D	177	B	178	D	179	A	180	A
181	D	182	A	183	D	184	C	185	A	186	B	187	D	188	B	189	C	190	C
191	D	192	B	193	D	194	B	195	A	196	C	197	D	198	D	199	D	200	A
201	D	202	D	203	C	204	B	205	C	206	B	207	B	208	A	209	B	210	B
211	C	212	C	213	D	214	B	215	D	216	C	217	B	218	A	219	D	220	D
221	C	222	C	223	C	224	C	225	B	226	A	227	B	228	A	229	C	230	C
231	D	232	B	233	C	234	C	235	A	236	C	237	B	238	B	239	C	240	C
241	B	242	A	243	B	244	C	245	B	246	B	247	B	248	C	249	C	250	D
251	C	252	B	253	D	254	D	255	C	256	A	257	B	258	A	259	B	260	C
261	A	262	D	263	C	264	D	265	B	266	B	267	B	268	B	269	C	270	C
271	C	272	B	273	B	274	B	275	D	276	A	277	C	278	B	279	D	280	B
281	D	282	B	283	A	284	A	285	D	286	C	287	D	288	C	289	C	290	D
291	C	292	C	293	A	294	B	295	A	296	D	297	B	298	D	299	A	300	A
301	A	302	A	303	D	304	A	305	D	306	C	307	C	308	A	309	C	310	D
311	A	312	B	313	A	314	A	315	B	316	B	317	D	318	B	319	C	320	D
321	D	322	D	323	B	324	B	325	C	326	C	327	B	328	A	329	A	330	A
331	D	332	D	333	C	334	C	335	D	336	A	337	B	338	A	339	B	340	C
341	B	342	B	343	C	344	C	345	D	346	D	347	D	348	D	349	D	350	D
351	B	352	C	353	C	354	A	355	A	356	B	357	C	358	C	359	A	360	A
361	A	362	A	363	D	364	C	365	D	366	A	367	A	368	B	369	B	370	C
371	D	372	B	373	A	374	D	375	D	376	B	377	D	378	B	379	B	380	D
381	C	382	D	383	D	384	C	385	D	386	A	387	A	388	B	389	A	390	D
391	D	392	D	393	B	394	B	395	B	396	B	397	C	398	B	399	A	400	D

401	D	402	B	403	A	404	D	405	B	406	A	407	B	408	D	409	C	410	B
411	D	412	D	413	B	414	C	415	D	416	A	417	D	418	B	419	A	420	A
421	A	422	D	423	A	424	C	425	A	426	C	427	D	428	D	429	B	430	C
431	C	432	A	433	B	434	A	435	C	436	A	437	D	438	A	439	C	440	C
441	B	442	D	443	C	444	B	445	C	446	B	447	D	448	B	449	B	450	B
451	A	452	D	453	D	454	C	455	A	456	A	457	B						

MATHEMATICS

NCERT Chapter-Wise Compilation

CUSAT CAT 2023 · 2024 · 2025 — All 14 Shifts

Sets, Relations & Functions

41 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The function f(x)= x +mx+n,where m and n real constants, describes

(A) a one-to-one mapping	**(B)** an onto mapping
(C) a non one-to-one but an onto mapping	**(D)** a mapping which is neither one-to-one nor onto 2

Q2. *CUSAT CAT 2023 - Shift 1*

With reference to a universal set, the relation of inclusion of a subset in another, is

(A) a symmetric relation	**(B)** an equivalence relation
(C) a reflexive relation	**(D)** not a transitive relation 1 1 1 α β γ 1 1 1 3 2

Q3. *CUSAT CAT 2023 - Shift 2*

Let ~ be the binary relation defined on ■ defined by a~b if and only if a–b+ 3 is an irrational number. Then ~ is

(A) an equivalence relation	**(B)** transitive only
(C) symmetric only	**(D)** reflexive only

Q4. *CUSAT CAT 2023 - Shift 2*

Let R = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on the set A = {3, 6, 9, 12}. Then the relation R is

(A) reflexive and transitive only	**(B)** an equivalence relation
(C) reflexive only	**(D)** reflexive and symmetric only 4 4

Q5. *CUSAT CAT 2023 - Shift 2*

Let R be a relation on the set L of lines defined by l Rl if l is perpendicular to l . 1 2 1 2 Then the relation R is

(A) reflexive and symmetric	**(B)** symmetric and transitive
(C) an equivalence relation	**(D)** symmetric

Q6. *CUSAT CAT 2023 - Shift 4*

Let ' f ' and 'g' be two bijective functions defined on a set A and such that f(x)=2x+1 and (g f)(x)=3x+2. Then g(x) is 3 5

(A) x– 2 2 3 1	**(B)** x+ 2 2 3
(C) 2x– 2	**(D)** 5x+3

Q7. *CUSAT CAT 2023 - Shift 4*

Two finite sets have m and n elements respectively. The total number of subsets of first set is 56 more than the total number of subsets of the second set. The values of m and n respectively are

(A) 7, 6	**(B)** 6, 3
(C) 5, 1	**(D)** 8, 7 3 –1

Q8. *CUSAT CAT 2023 - Shift 4*

Let X = { x:x=n 3 +2n+1,n∈■ } and Y = { x:x=3n 2 +7,n∈■ } . Then X ∩Y is a subset of

(A) X ={x:x=3n+5,n∈■}	**(B)** X = { x:x=n 2 +n+1,n∈■ }
(C) X ={x:x=7n–1,n∈■}	**(D)** None of the above

Q9. *CUSAT CAT 2023 - Shift 5*

The set onto which the derivative of the function f(x)= x(logx–1)maps the ray [1, ∞) is

(A) [1, ∞)	**(B)** (0, ∞)
(C) [0, ∞)	**(D)** (1, ∞) 3 5

Q10. *CUSAT CAT 2023 - Shift 5*

The number of subsets of the set A = a ,a ,...,a which contain even number of 1 2 n elements is

(A) 2 n–1	**(B)** 2 n –1
(C) 2 n–2	**(D)** 2 n

Q11. *CUSAT CAT 2023 - Shift 5*

Let S be the set of integers. For a, b ∈ S , define aRb iff a–b < 1. Then

(A) R is not reflexive	**(B)** R is not symmetric
(C) R is an equivalence relation	**(D)** R is not an equivalence relation

Q12. *CUSAT CAT 2023 - Shift 5*

Let S be a non-empty subset of R. Consider the following statement: P : There is a rational number x∈S such that x > 0. Then the negation of P is

(A) Every rational number x∈S satisfies x≤0	**(B)** x∈S and x≤0⇒ x is not a rational number
(C) There is a rational number x∈S such that x≤0	**(D)** There is no rational number x∈S such that x≤0 68 68

Q13. *CUSAT CAT 2024 - Shift 1*

The number of subsets of the set X = x ,x ,...,x which contain even 1 2 n number of elements is n

(A) 2 n	**(B)** 2 – 1 n – 2
(C) 2 n –1	**(D)** 2

Q14. *CUSAT CAT 2024 - Shift 1*

Let f :(–1,1)→ B be the function defined by f(x)=sin and such that it 1+x 2 is a one-one and onto function. Then B is π π

(A) – , 2 2 π π	**(B)** – , 2 2 π
(C) 0, 2 π	**(D)** 0, 2

Q15. *CUSAT CAT 2024 - Shift 2*

The number of onto mappings from the set A = {1, 2, . . . , 100} to set B = {1, 2} is 100

(A) 2 100	**(B)** 2 – 2 99
(C) 2 – 2 99	**(D)** 2 2

Q16. *CUSAT CAT 2024 - Shift 2*

Let * be the binary operation defined on R by a*b= , for every a,b∈R, then 4 the operation * is

(A) Commutative and Associative	**(B)** Commutative but not Associative
(C) Associative but not Commutative	**(D)** Neither Associative nor Commutative 1+ 3 8+4 3

Q17. *CUSAT CAT 2024 - Shift 3*

An onto function f : {1, 2, 3} → {4, 5, 6}

(A) may not be one-one	**(B)** always one-one
(C) never one-one	**(D)** one-one if f is identity function

Q18. *CUSAT CAT 2024 - Shift 3*

Let R be a relation over the set N × N defined by (a, b) R (c, d) ⇔ a + d = b + c. Then R is

(A) reflexive only	**(B)** symmetric only
(C) transitive only	**(D)** an equivalence relation

Q19. *CUSAT CAT 2024 - Shift 4*

Let f :Z →Zdefined by f(x)= x . Then

(A) f is neither one-one nor onto	**(B)** f is onto
(C) f is both one-one and onto	**(D)** f is one-one

Q20. *CUSAT CAT 2024 - Shift 4*

Let f, g : ■→■ be defined by f (x) = [x] and g(x)= . Then 4

(A) f is not one-one and g is onto	**(B)** f is onto and g is not one-one
(C) neither f nor g is one-one	**(D)** both f and g is onto

Q21.

CUSAT CAT 2024 - Shift 4

One function is selected from all the functions F : S → S, where S = {a, b, c, d, e}. The probability that it is an onto function is

(A) 5 81	**(B)** 5 162
(C) 5 324	**(D)** 7 324 2 π tan α

Q22.

CUSAT CAT 2024 - Shift 4

Let R be the relation over the set of integers defined by (x, y) ∈ R ⇔ |x – y| ≤ 1 . Then R is

(A) reflexive and transitive	**(B)** reflexive and symmetric
(C) symmetric and not transitive	**(D)** an equivalence relation

Q23.

CUSAT CAT 2024 - Shift 5

A binary relation ρ, defined in R by xρy if 1+xy >0 for x,y∈R, is

(A) reflexive and symmetric	**(B)** an equivalence relation
(C) symmetric and transitive	**(D)** only transitive

Q24.

CUSAT CAT 2024 - Shift 5

Let f :R→R, be defined by f(x)=(x+1)2 –1, x≥1. Then f –1(x)

(A) does not exists as f is not one-to-one	**(B)** does not exists as f is not onto
(C) =–1– x+1	**(D)** =–1+ x+1 d 1–cosx

Q25.

CUSAT CAT 2025 - Shift 1

Let f :[0,∞)→■ be defined by f(x)= x. Then

(A) f is one-one	**(B)** f is onto
(C) f is both one-one and onto	**(D)** f is neither one-one nor onto –1

Q26.

CUSAT CAT 2025 - Shift 1

The number of subsets of {2,3,4,5,6,7,8,9} that contains at least one composite number is

(A) 16	**(B)** 32
(C) 64	**(D)** 240

Q27.

CUSAT CAT 2025 - Shift 1

Which of the following is TRUE ?

(A) Division is a closed binary operation in ■	**(B)** Division is a closed binary operation in ■
(C) Division is a closed binary operation in ■\{0}	**(D)** Division is a closed binary operation in■

Q28. *CUSAT CAT 2025 - Shift 1*

Let f :■ →■ be a mapping defined by f(x)= x 3 +4. Then f is

(A) bijective	**(B)** surjective
(C) injective	**(D)** not bijective

Q29. *CUSAT CAT 2025 - Shift 1*

Let a function f :N → N be defined by f(n)=n–1, n=3,7,11,15,..., then f is n+1 ,n=1,5,9,13,... 2

(A) one-one and onto	**(B)** one-one but not onto
(C) onto but not one-one	**(D)** neither one-one nor onto

Q30. *CUSAT CAT 2025 - Shift 2*

The function f :■ →■, defined by f(x)=sinx+cosx, is

(A) one-one	**(B)** onto
(C) one-one and onto	**(D)** neither one-one nor onto –1

Q31. *CUSAT CAT 2025 - Shift 2*

An onto function f :{1,2,3,4,5}→{ 1{,12,2,3,3,6,6,7,7}}is

(A) one-to-one in some cases	**(B)** always one-to-one
(C) not one-to-one in some cases	**(D)** one-to-one if f is the identity

Q32. *CUSAT CAT 2025 - Shift 2*

Two finite sets X and Y have p and q elements respectively. If the total number of subsets of X is 56 more than the total number of subsets of Y. Then the value of q is

(A) 2	**(B)** 3
(C) 6	**(D)** 28

Q33. *CUSAT CAT 2025 - Shift 3*

The number of one-one functions from A to A , where A={1,2,3}, is

(A) 1	**(B)** 2
(C) 4	**(D)** 6

Q34. *CUSAT CAT 2025 - Shift 3*

The mapping f :[–2,2]→■, defined by f(x)= x x , is

(A) bijective	**(B)** not one-one but onto
(C) not onto	**(D)** not a function

Q35. *CUSAT CAT 2025 - Shift 3*

The mapping f : , →■, defined by f(x)= tanx, is 2 2

(A) one-one	**(B)** onto
(C) bijective	**(D)** not a function –1 –1

Q36. *CUSAT CAT 2025 - Shift 3*

Let f :X → X be a mapping such that f (f(x))= x for all x∈ X. Then

(A) f is one-to-one and onto	**(B)** f is one-to-one but not onto
(C) f is onto but not one-to-one	**(D)** f is neither one-to-one nor onto –2x

Q37. *CUSAT CAT 2025 - Shift 4*

Let f :[–1,1]→B be defined by f(x)= x 2 . Then the set B for which f is onto, is

(A) (0, 1)	**(B)** [–1,1]
(C) (–1,1]	**(D)** [0,1] 5 4

Q38. *CUSAT CAT 2025 - Shift 4*

Which of the following function is onto?

(A) f :■→■ defined by f(n)=n+2 1	**(B)** f :■–{0}→■ defined by f(x)= x
(C) f :[0,∞)→■ defined by f(x)= x	**(D)** f :■∪{0}→■defined by f(n)=n+1

Q39. *CUSAT CAT 2025 - Shift 4*

LetR be the relation over the set all real numbers as x and y are related under R if xy≥0. Then R is

(A) reflexive and transitive	**(B)** reflexive and symmetric
(C) symmetric and transitive	**(D)** an equivalence relation

Q40. *CUSAT CAT 2025 - Shift 4*

The number of onto mappings from the set X ={1,2,...,11} to the set Y ={1,2} is

(A) 11 2 –2	**(B)** 11 2
(C) 10 2 –2	**(D)** 22 4

Q41. *CUSAT CAT 2025 - Shift 4*

In a set of (2n + 1) elements, the number of subsets which contain at most n elements is

(A) 2 n	**(B)** 2 n+1
(C) 2 n–1	**(D)** 2 2n 1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key

1	D	2	C	3	D	4	A	5	D	6	B	7	B	8	C	9	C	10	A
11	C	12	A	13	D	14	A	15	B	16	B	17	B	18	D	19	A	20	A
21	C	22	C	23	A	24	D	25	A	26	D	27	C	28	A	29	A	30	D
31	B	32	B	33	D	34	C	35	C	36	A	37	D	38	D	39	D	40	A
41	D																		

Trigonometric Functions

356 questions

Q1. *CUSAT CAT 2023 - Shift 1*

If A and B are square matrices of same order and A is non-singular, then, for a n (–1) positive integer n, A BA is

(A) A –n B n A n	**(B)** A n B n A –n
(C) A –1 B n A (–1)	**(D)** n A BA 4 0 x

Q2. *CUSAT CAT 2023 - Shift 1*

cos + cos + cos = 7 7 7 1

(A) 2 1	**(B)** 4 1
(C) – 2	**(D)** 1 1

Q3. *CUSAT CAT 2023 - Shift 1*

Let a= for all real x. Then 5cosx+12sinx 1

(A) a≥ 13	**(B)** a=1 2
(C) a=– 13 2	**(D)** a= 13

Q4. *CUSAT CAT 2023 - Shift 1*

Let λ and λ be two values lying in [0, 2π] for which tanλ=θ. Then 1 2 λ λ tan 1 · tan 2 = 2 2

(A) 0	**(B)** –1
(C) 2	**(D)** 1

Q5. *CUSAT CAT 2023 - Shift 1*

If tanA=2tanB+cotB, then 2tan(A–B)=

(A) 2tanB	**(B)** 2cotB
(C) cotB	**(D)** tanB 3 15

Q6. *CUSAT CAT 2023 - Shift 1*

If the sines of two angles of a triangle are equal to and , then the cosine of the 5 17 third angle is 11

(A) 85 13	**(B)** 85 23
(C) 85 19	**(D)** 85 1–x 1 –1 –1

Q7. *CUSAT CAT 2023 - Shift 1*

If tan = tan x, then the value of x is 1+x 2 1

(A) 2 1	**(B)** 3
(C) 3	**(D)** 2

Q8. *CUSAT CAT 2023 - Shift 1*

Let y = cosx + sinx . Then at x= is dx 3

(A) 1	**(B)** 0 1– 3
(C) 2 3–1	**(D)** 2 3 2 3 x –2 x +1

Q9. *CUSAT CAT 2023 - Shift 1*

The minimum value of f(x)=asecx–btanx, a>b>0, is

(A) a 2 +b 2	**(B)** a 2 –b 2
(C) a 2 +b 2	**(D)** a 2 –b 2 2x2–2x+1 2

Q10. *CUSAT CAT 2023 - Shift 1*

The minimum value of e sin x is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q11. *CUSAT CAT 2023 - Shift 1*

If x –ax+b=0and x – px+q=0 have a root in common and the second equation has equal roots, then

(A) b+q=ap	**(B)** p+q=a
(C) a+ p=q	**(D)** 2(b+q)= pa

Q12. *CUSAT CAT 2023 - Shift 1*

If f(x)=sin x+cos x+1, then the range of f(x) is 3

(A) , 2 2 3	**(B)** 1, 2
(C) [1, 2]	**(D)** [2, 3]

Q13. *CUSAT CAT 2023 - Shift 1*

The fifth term of a GP is 2. Then the product of first 9 terms is

(A) 128	**(B)** 64
(C) 256	**(D)** 512 1–cos2x

Q14. *CUSAT CAT 2023 - Shift 1*

The ordinary differential equation corresponding to the general solution y= Asin2t+Bcos2t is 2 d y dy

(A) + +4y =0 dt 2 dt 2 d y	**(B)** +4y =0 2 dt 2 d y
(C) +8y =0 2 dt 2 d y	**(D)** –4y =0 2 dt 1–3p 1+4p 1+ p

Q15. *CUSAT CAT 2023 - Shift 1*

The imaginary part of (z – 1)(cos α – i sin α) + (z – 1) –1(cos α + i sin α) is zero if

(A) |z – 1| = 2	**(B)** arg (z – 1) = 2α
(C) arg (z – 1) = α	**(D)** |z| = 1

Q16. *CUSAT CAT 2023 - Shift 1*

The solution set for 2+ 2 + 2– 2 =2.2 is

(A) {0}	**(B)** {0, 2}
(C) {2}	**(D)** [0, 2] 3 a x sinx

Q17. *CUSAT CAT 2023 - Shift 1*

The value of is equal to 1+sinθ π θ

(A) tan – 3 2 π θ	**(B)** tan – 4 2 θ π
(C) tan – 3 4 θ π	**(D)** tan – 4 4

Q18. *CUSAT CAT 2023 - Shift 1*

The equation of the circle passing through (1, –3) and the points common to two 2 2 2 2 circles x + y –6x+8y–16=0 and x + y +4x–2y–8=0 is

(A) x 2 + y 2 –4x+6y+24=0	**(B)** 2x 2 +2y 2 +3x+ y–20=0
(C) 3x 2 +3y 2 –5x+7y–19=0	**(D)** 3x 2 +3y 2 +5x–7y–19=0 2n 1

Q19. *CUSAT CAT 2023 - Shift 1*

Let f(x) be a continuous function defined for 1≤ x≤3. If f(x) takes rational values for all x and f(2)=10, then f(1.5) is

(A) 0	**(B)** 10
(C) not defined	**(D)** a constant 4 1 x

Q20. *CUSAT CAT 2023 - Shift 1*

If α+β=γ and tan γ = 22 , ‘ a ’ is the arithmetic and ‘ b ’ is the geometric mean 3 a respectively of tan α and tan β, then the value of is equal to (2)3 1–b

(A) 1331	**(B)** 1320
(C) 1330	**(D)** 1335 3 2

Q21. *CUSAT CAT 2023 - Shift 1*

If the lines joining the origin to the intersection of the line y = mx + 2 and the curve 2 2 x + y =1 are at right angles, then

(A) m 2 =1	**(B)** m 2 =3
(C) m 2 =7	**(D)** 2m 2 =1

Q22. *CUSAT CAT 2023 - Shift 1*

If the numerical value of cos +tan is , then 5 3 b

(A) a+b=23	**(B)** a+b=0
(C) 3b=a+2	**(D)** 2a=3b

Q23. *CUSAT CAT 2023 - Shift 1*

If A and B are acute positive angles satisfying the equations 3sin A+2sin B =1 and 3sin2A–2sin2B=0, then A+2B is equal to π

(A) 4 π	**(B)** 2 3π
(C) 4 2π	**(D)** 3 3π 816

Q24. *CUSAT CAT 2023 - Shift 1*

If sin –1 x+sin –1 y+sin –1 z = then the value of 3000(x+ y+z) – is 2 x 2 + y 2 +z 2 equal to

(A) 8724	**(B)** 8728
(C) 8772	**(D)** 8767

Q25. *CUSAT CAT 2023 - Shift 1*

Let P, Q, R be three points on a parabola y 2 =4ax whose ordinates are in geometrical progression, then the tangents at P and R meet on

(A) the line through Q parallel to x-axis	**(B)** the line through Q parallel to y-axis
(C) the line joining Q and the vertex	**(D)** the line joining Q and the focus

Q26. *CUSAT CAT 2023 - Shift 2*

The value of cosA·cos(60°–A)cos(60°+ A) is 1

(A) cos3A 2 cos3A	**(B)** 4
(C) cos3A	**(D)** 1 7 π x

Q27. *CUSAT CAT 2023 - Shift 2*

If sinx+cosx= where x∈ 0, , then tan is equal to 2 4 2 3+ 7

(A) 2 3– 7	**(B)** 2 4– 7
(C) 3 7 –2	**(D)** 3 π π π

Q28. *CUSAT CAT 2023 - Shift 2*

The maximum value of sinx+ +cosx+ in the interval 0, is attained at 6 6 2 π

(A) 12 π	**(B)** 6 π
(C) 3 π	**(D)** 2

Q29.

CUSAT CAT 2023 - Shift 2

The number of values of $\theta \in [0,5\pi]$ satisfying the equation $3\cos 2\theta - 10\cos\theta + 7 = 0$ is

(A) 1	**(B)** 2
(C) 6	**(D)** 8 A B 2

Q30.

CUSAT CAT 2023 - Shift 2

Let ABC be a triangle with tan and tan are roots of bx –5x+1=0. Then the 2 2 triangle is

(A) equilateral	**(B)** isosceles
(C) right angled	**(D)** a triangle with the largest angle 70°

Q31.

CUSAT CAT 2023 - Shift 2

The lines of regression of y on x and x on y are respectively y = x and 4x– y–3=0 and the second moment about the origin of x is 2. The variance of y is

(A) 9	**(B)** 8
(C) 5	**(D)** 4 2 2 2 2

Q32.

CUSAT CAT 2023 - Shift 2

The ranges of k for which the circles x + y =4 and x + y –4kx+9=0 have two common tangents, are 13

(A) k < 8 13	**(B)** 1<k < 8 13 13
(C) k > or k <– 8 8 13 13	**(D)** k∈– , 8 8 π

Q33.

CUSAT CAT 2023 - Shift 2

Let f(x)=sinx, g(x)=2x and h(x)=cosx. If $\phi(x)=(g\ fh)(x)$, then ϕ'' = 4

(A) 2	**(B)** –2
(C) –4	**(D)** 4 x –x 3 –3

Q34.

CUSAT CAT 2023 - Shift 2

lim = x→0 secx–cosx

(A) 0	**(B)** ∞
(C) 1	**(D)** –1

Q35.

CUSAT CAT 2023 - Shift 2

If ∫log (1+x 2) dx= xlog (1+x 2) –k+c, where c is a constant, then the value of k is

(A) 2x+2tan –1 x	**(B)** –2x+2tan –1 x
(C) 2x–2tan –1 x	**(D)** –2x–2tan –1 x 1 4π sin t

Q36.

CUSAT CAT 2023 - Shift 2

Let I =∫ sint dt . Then ∫ 2 dt = 1+t 4π+2–t 0 4π–2

(A) 2I	**(B)** –2I
(C) I	**(D)** –I (2) x 1+ y dy

Q37.

CUSAT CAT 2023 - Shift 2

∫ is equal to 6 x x –16 3 1 x –1

(A) sec +c 3 4 3 x –1	**(B)** cosh +c 4 3 1 x –1
(C) sec +c 12 4 3 x –1	**(D)** sec +c 4 dy

Q38.

CUSAT CAT 2023 - Shift 2

The equation of the common tangent touching the circle (x–3) + y =9 and the 2 parabola y =4x above the x-axis is

(A) 3y =3x+1	**(B)** 3y =–(x+3)
(C) 3y = x+3	**(D)** 3y =–(3x+1)

Q39.

CUSAT CAT 2023 - Shift 2

The points on the curve y 2 =4ax+a+sin at which the tangent is parallel to a x-axis, lie on

(A) a straight line	**(B)** a circle
(C) an ellipse	**(D)** a parabola 2

Q40.

CUSAT CAT 2023 - Shift 2

The total number of solutions of sin x + cos x = sin x.cos x in [0, 2π] is equal to

(A) 2	**(B)** 4
(C) 6	**(D)** 8

Q41.

CUSAT CAT 2023 - Shift 2

The angle between the straight line r = (2i +3j +k) +t(iˆ– ˆj+k ˆ) and the plane 2x– y+z =5 is 2 –1

(A) sin 3 3 2 3 –1	**(B)** sin 3 3 –1
(C) cos 2 3 2 –1	**(D)** cos 3

Q42.

CUSAT CAT 2023 - Shift 2

Two n bit binary strings, S and S are chosen randomly with uniform probability. 1 2 The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d

(A) n 2 nC d	**(B)** d 2 nC
(C) d n 2 1	**(D)** d 2 cos(2πz)

Q43. *CUSAT CAT 2023 - Shift 2*

The angle between the line = = and the plane 3x+ y+z =7 is 2 3 6 7 11 –1

(A) sin 15 15 –1	**(B)** sin 7 11 7 11 –1
(C) cos 15 15 –1	**(D)** cos 7 11

Q44. *CUSAT CAT 2023 - Shift 2*

The value of the integral ∫ (sinx + cosx) dx is a

(A) aπ	**(B)** 2aπ aπ
(C) 2	**(D)** independent of a

Q45. *CUSAT CAT 2023 - Shift 2*

Consider the equation y– y =m (x–x) . In this equation, if m and x are fixed and 1 1 1 different lines are drawn for different values of y , then 1

(A) the lines will pass through a single point	**(B)** there will be one possible line only
(C) there will be a set of parallel lines	**(D)** there are at least two lines

Q46. *CUSAT CAT 2023 - Shift 2*

If the standard deviation of a set of observations is 4 and if each observation is divided by 4, the standard deviation of the new set of observations will be

(A) 4	**(B)** 3
(C) 2	**(D)** 1

Q47. *CUSAT CAT 2023 - Shift 2*

The period of the function f(x)= tanx is

(A) π	**(B)** 2π π
(C) 2 π	**(D)** 4

Q48. *CUSAT CAT 2023 - Shift 2*

If common chord of the circle C with centre at (2, 1) and of radius r and the circle 2 2 x + y –2x–6y+6=0is a diameter of the second circle, then value of r is

(A) 3	**(B)** 2 3
(C) 2	**(D)** 1 15

Q49. *CUSAT CAT 2023 - Shift 2*

Maximum value of 2997 sin x + 3996 cos x is equal to

(A) 4998	**(B)** 4932
(C) 4900	**(D)** 4995 2 π tan x

Q50. *CUSAT CAT 2023 - Shift 2*

If $\alpha \in 0$, , then the expression x 2 +x + is always greater than or equal to 2 2 x +x

(A) 2tanα	**(B)** 2
(C) 1	**(D)** sec 2 α 2003

Q51. *CUSAT CAT 2023 - Shift 2*

The diagonal of a parallelogram PQRS are along the lines x + 3y = 4 and 6x – 2y = 7, then PQRS must be a

(A) rectangle	**(B)** square
(C) cyclic quadrilateral	**(D)** rhombus 2 2

Q52. *CUSAT CAT 2023 - Shift 2*

P(sinθ,cosθ) and Q(cosθ,sinθ)are two points whose mid point is at the origin. R(sin2θ,cosθ) is a point on the plane whose distance from the origin is 2

(A) 3 3	**(B)** 2 3
(C) 2 1	**(D)** 2

Q53. *CUSAT CAT 2023 - Shift 3*

Let f(x)= where {.} and [.] respectively denote the fractional part and sin{x} greatest integer. Then range of f is

(A) real numbers	**(B)** negative integers
(C) natural numbers	**(D)** rationals

Q54. *CUSAT CAT 2023 - Shift 3*

Two die are thrown simultaneously to get the coordinates of a point on x– y plane. Then the probability that this point lies inside or on the region bounded by x + y ≤3 is 2

(A) 14 3	**(B)** 14 1
(C) 12 4	**(D)** 14

Q55. *CUSAT CAT 2023 - Shift 3*

If sinθ=n(sin(θ+2α)), then tan(θ+α) is equal to

(A) ntanα 1–n	**(B)** tanα 1+n
(C) tanα 1+n	**(D)** tanα 1–n π 2π (n–1)π 2 2 2

Q56. *CUSAT CAT 2023 - Shift 3*

If S =cos +cos +...+cos , then S is equal to n n n n(n+1)

(A) 2 1	**(B)** (n–1) 2 1
(C) (n–2) 2 n	**(D)** 2 9

Q57. *CUSAT CAT 2023 - Shift 3*

If cosecA+cot A= , then tanA is 2 77

(A) 36 36	**(B)** 77 18
(C) 77 77	**(D)** 72 2

Q58. *CUSAT CAT 2023 - Shift 3*

If tan x+secx–a =0 has atleast one solution, then the complete set of values of 'a' is

(A) (–∞, 2)	**(B)** (–1,1)
(C) [–1,1]	**(D)** [–1, ∞) 4 2

Q59. *CUSAT CAT 2023 - Shift 3*

The values of 'λ', for which the equation cos x–(λ+2)cos x–(λ+3)=0 possesses a solution, lies in

(A) [–3, –1]	**(B)** [–3, –2]
(C) [0, 2]	**(D)** (0, 3) π –1 –1

Q60. *CUSAT CAT 2023 - Shift 3*

Let sin x–cos x= . Then x is 6 1

(A) 2 3	**(B)** 2 1
(C) – 2 3	**(D)** 4

Q61. *CUSAT CAT 2023 - Shift 3*

The sum of 10 terms is 12 and the sum of their squares is 17. Then the standard deviation will be 3

(A) – 5 6	**(B)** 5 3
(C) 5 4	**(D)** 5 2 2

Q62. *CUSAT CAT 2023 - Shift 3*

If the circle x + y +4x+22y+c=0 bisects the circumference of the circle 2 2 x + y –2x+8y–d =0, then c+d is equal to

(A) 60	**(B)** 40
(C) 80	**(D)** 50 π 3 3

Q63. *CUSAT CAT 2023 - Shift 3*

The second order derivative of asin t with respect to acos t at t = is 4 4 2

(A) 3a 2	**(B)** 3a 4
(C) 3a 4	**(D)** a 2 πx πx

Q64. *CUSAT CAT 2023 - Shift 3*

The period of the function f(x)=cos –sin is n! (n+1)!

(A) 2n!	**(B)** n!
(C) 2(n+1)!	**(D)** (n+1)! sin(cosx)–cosx , x≠ π 2 2

Q65. *CUSAT CAT 2023 - Shift 3*

∫ (x 3 +xcosx+tan 5 x+1) dx= –π2

(A) 0	**(B)** 2
(C) π	**(D)** 1

Q66. *CUSAT CAT 2023 - Shift 3*

If y = f(t)sint+ f '(t)cost and x= f(t)cost– f '(t)sint, then + is dt dt equal to 2

(A) [f(t)– f ''(t)] 2	**(B)** [f(t)+ f ''(t)] 2
(C) [f(t)+ f '(t)] 2	**(D)** [f(t)– f '(t)] sinnx

Q67. *CUSAT CAT 2023 - Shift 3*

The value of integer n for which the function f(x)= has 4π as its period is x sin n

(A) 2	**(B)** 3
(C) 4	**(D)** 5 3 2

Q68. *CUSAT CAT 2023 - Shift 3*

If f(x) = 2x + 9x + λx + 20 is a decreasing function of x in the largest possible interval (–2,–1), then λ is equal to

(A) 6	**(B)** –6
(C) 12	**(D)** –12

Q69. *CUSAT CAT 2023 - Shift 3*

The value of sintan – 3 +cos is 2

(A) ∞	**(B)** –1
(C) 0	**(D)** 1

Q70. *CUSAT CAT 2023 - Shift 3*

Consider the circle z–5–5i =2 in the complex plane (x,y) with z = x+iy. Then the minimum distance from the origin to the circle is

(A) 5 2–2	**(B)** 5 2
(C) 34	**(D)** 54 3 2

Q71. *CUSAT CAT 2023 - Shift 3*

If a straight line through C – 8, 8 making an angle of 135° with the x-axis and cuts the circle x=5cosθ, y =5sinθ at points A and B, then the length of AB is

(A) 3	**(B)** 7
(C) 10	**(D)** 5

Q72. *CUSAT CAT 2023 - Shift 3*

The function f(x)=sec log x+ 1+x is

(A) even	**(B)** odd
(C) constant	**(D)** periodic 2 3n

Q73. *CUSAT CAT 2023 - Shift 3*

If a 2 –2acosx+1=674 and tan =7 then the integral value of a is 2

(A) 25	**(B)** 49
(C) 67	**(D)** 74 1

Q74. *CUSAT CAT 2023 - Shift 3*

If the point (3, 4) lies on the locus of the point of intersection of the lines xcosα+ ysinα=a and xsinα– ycosα=b,(α is a variable), and the point (a,b) 4 4 lies on the line 3x – 4y = 0, then 9a +16b +34 is equal to

(A) 3634	**(B)** 3684
(C) 3845	**(D)** 3874 2

Q75. *CUSAT CAT 2023 - Shift 3*

If x = 9 is the chord of contact of the hyperbola x 2 – y2 =9, then the equation of the corresponding pair of tangents is

(A) 9x 2 –8y 2 +18x–9=0	**(B)** 9x 2 –8y 2 –18x+9=0
(C) 9x 2 –8y 2 –18x–9=0	**(D)** 9x 2 –8y 2 +18x+9=0 2 2

Q76. *CUSAT CAT 2023 - Shift 3*

The lines joining the origin to the points of intersection of x + y +2gx+c=0 and 2 2 x + y +2fy–c=0 are at right angles. Then

(A) g 2 + f 2 =c	**(B)** g 2 – f 2 =c
(C) g 2 – f 2 =2c	**(D)** g 2 + f 2 =c 2

Q77. *CUSAT CAT 2023 - Shift 4*

If in a triangle ABC, tan A + tan B + tan C = 3 3, then the triangle is

(A) right-angled but not isosceles	**(B)** right-angled and isosceles
(C) equilateral	**(D)** isosceles but not right-angled

Q78. *CUSAT CAT 2023 - Shift 4*

If e = x , then is dx logx

(A) 2 (logx–1) logx–1	**(B)** 2 (logx+1) logx–2
(C) 2 (logx–1)	**(D)** 1 x2 e –cosx

Q79. *CUSAT CAT 2023 - Shift 4*

The value of lim is equal to x→0 x 2 1

(A) 2 3	**(B)** 2 2
(C) 3	**(D)** 2 cosx

Q80. *CUSAT CAT 2023 - Shift 4*

lim (cosx) is π x→ 2

(A) 1	**(B)** 0 1
(C) e 2	**(D)** e

Q81. *CUSAT CAT 2023 - Shift 4*

The locus of the point of intersection of the lines xsinθ+(1–cosθ)y =asinθ and xsinθ–(1+cosθ)y+asinθ=0 is

(A) x 2 – y 2 =a 2	**(B)** x 2 + y 2 =a 2
(C) y 2 =ax	**(D)** y 2 –x 2 =a 3 e2 2 x 1 e

Q82. *CUSAT CAT 2023 - Shift 4*

The equation of a common tangent to the curves y =8x and xy =–1 is

(A) 3y =9x+2	**(B)** y =2x+1
(C) 2y = x+8	**(D)** y = x+2 3

Q83. *CUSAT CAT 2023 - Shift 4*

If ∫ =kπ then the value of k is (2)(2) x +4 x +9 0 1

(A) 60 1	**(B)** 80 1
(C) 40 1	**(D)** 20 sin5x , x≠0

Q84. *CUSAT CAT 2023 - Shift 4*

The number of words that can be formed by using the letters of the word MATHEMATICS that start as well as end with T is

(A) 80720	**(B)** 90720
(C) 20860	**(D)** 37258

Q85. *CUSAT CAT 2023 - Shift 4*

If the acute angle between the lines x+ky+3=0 and 2x+ y–7=0 is tan , 4 then ‘ k ’ is

(A) 1	**(B)** 2
(C) 3	**(D)** 4 (2) log 1+x

Q86. *CUSAT CAT 2023 - Shift 4*

The complex number z = x + iy which satisfy the equation =1 lie on z+5i

(A) the x-axis	**(B)** the straight line y = 5
(C) a circle passing through the origin	**(D)** the line x= y

Q87. *CUSAT CAT 2023 - Shift 4*

The equation of the horizontal tangent to the graph of the function y =e +e is

(A) y =–2	**(B)** y =–1
(C) y =2	**(D)** y =1

Q88. *CUSAT CAT 2023 - Shift 4*

The differential equation of all circles passing through the origin and having their centres on the x-axis is dy 2 2

(A) x = y +xy dx dy 2 2	**(B)** x = y +3xy dx dy 2 2
(C) y = x +2xy dx dy 2 2	**(D)** y = x –2xy dx

Q89. *CUSAT CAT 2023 - Shift 4*

If a line OP through the origin O makes angles α,45°and60°with x, y and z axis respectively, then the direction cosines of OP are 1 1 1

(A) , , 2 2 2 1 1 1	**(B)** , , 2 2 2 1 1 1
(C) , , 2 2 2	**(D)** 1,1,1

Q90. *CUSAT CAT 2023 - Shift 4*

The three numbers are chosen from 1 to 30. Then the probability that they are not consecutive is 142

(A) 145 144	**(B)** 145 143
(C) 145 1	**(D)** 145 x

Q91. *CUSAT CAT 2023 - Shift 4*

Let f(x)= lim . Then f is discontinuous at n→∞1+(2sinx) 2n

(A) π π	**(B)** 3 π
(C) 4 π	**(D)** 6

Q92. *CUSAT CAT 2023 - Shift 4*

The probability of drawing a diamond card in each of the two consecutive draws from a well shuffled pack of cards, if the card drawn is not replaced after the first draw, is 4

(A) 17 13	**(B)** 17 1
(C) 17	**(D)** 0 x+ y x x

Q93. *CUSAT CAT 2023 - Shift 4*

If ∫t f(t)dt =sinx–xcosx– for all x∈R–{0} , then the value of f is 2 6 e

(A) 0	**(B)** 1 1
(C) – 2 3	**(D)** 2 1 1 1 π 1 1 1

Q94. *CUSAT CAT 2023 - Shift 4*

If a+ib= i , then (a,b)= k=1

(A) (0, 1)	**(B)** (0, 0)
(C) (0, –1)	**(D)** (1, 1) 2 sin A 1+cosA sin A

Q95. *CUSAT CAT 2023 - Shift 4*

The value of 1– + – , is 1+cosA sin A 1–cosA

(A) cos A	**(B)** 0
(C) 1	**(D)** sin A n 1+i

Q96. *CUSAT CAT 2023 - Shift 4*

If =–1, n∈N, then the least value of n is 1–i

(A) 1	**(B)** 2
(C) 3	**(D)** 4 xtan2x–2xtanx

Q97. *CUSAT CAT 2023 - Shift 4*

The value of lim is x→0 (1–cos2x) 2

(A) 2	**(B)** –2 1
(C) 2 –1	**(D)** 2

Q98. *CUSAT CAT 2023 - Shift 4*

The complex numbers z = x+iy which satisfy the equation =1,lie on z+5i

(A) the x-axis	**(B)** the straight line y = 5
(C) a circle passing through origin	**(D)** Re(z)>3 dy

Q99. *CUSAT CAT 2023 - Shift 4*

A solution of the equation x = y(log y–logx+1) is dx cx

(A) y = xe 2 y	**(B)** =cx x
(C) y 2 =cxlogx	**(D)** log y =cx 2+3isinθ

Q100. *CUSAT CAT 2023 - Shift 4*

A value of θ for which z = is purely imaginary, is 1–2isinθ π

(A) 3 π	**(B)** 6 3 –1
(C) sin 4 1 –1	**(D)** sin 3 n 1 ∑ r

Q101. *CUSAT CAT 2023 - Shift 4*

The least value of cosec x+25sec x is

(A) 26	**(B)** 28
(C) 36	**(D)** 0 2

Q102. *CUSAT CAT 2023 - Shift 5*

The value of log (ax) where a is a constant is equal to e dx

(A) 1	**(B)** log ax e 1
(C) a	**(D)** log (ax)+1 e n(n+1)x [ ] [ ] [ ] [ ]

Q103. *CUSAT CAT 2023 - Shift 5*

A point P (other than origin) on y =4x –2x such that the tangent at P passes through the origin is

(A) (1, 3)	**(B)** (1, 2)
(C) (–1, 2)	**(D)** (2, –32)

Q104. *CUSAT CAT 2023 - Shift 5*

The differential equation representing a curve passing through (1, 2) is = . dx x Then the equation of the curve is

(A) y =2x 2	**(B)** y = x 2 +1
(C) y =2x	**(D)** x=2y–3

Q105. *CUSAT CAT 2023 - Shift 5*

Let f(x)= , x≠0. Then the value of f(0) so that f secx–cosx is continuous at x = 0 is

(A) 1	**(B)** 0
(C) 2	**(D)** –1

Q106.

CUSAT CAT 2023 - Shift 5

The equation of the line passing through (3, 3) and making an angle of 60° with the positive direction of the x-axis is

(A) x– 3y+3–3 3 = 0	**(B)** x+ 3y+3–3 3 =0
(C) 3x– y+3–3 3 =0	**(D)** 3x– y–3+3 3 =0

Q107.

CUSAT CAT 2023 - Shift 5

The area of the figure bounded by the lines x=0, x= , f(x)=sinx, g(x)=cosx is 2 ( )

(A) 2 2–1	**(B)** 3–1 ( )
(C) 2 3–1 ( )	**(D)** 2 2+1

Q108.

CUSAT CAT 2023 - Shift 5

The point on the y-axis whose perpendicular distance from the line 4x – 3y – 12 = 0 is

(A) (0, 1)	**(B)** (0, 3)
(C) (0, –9)	**(D)** Both

Q109.

CUSAT CAT 2023 - Shift 5

A set of direction cosines of the normal to the plane 6x – 3y – 2z = 10 is 6 –3 –2

(A) , , 7 7 7	**(B)** (6,–3,–2) 6 –3 –2
(C) , , 49 49 49 –6 3 2	**(D)** , , 49 49 49

Q110.

CUSAT CAT 2023 - Shift 5

Six dice are thrown simultaneously. Then the probability that all of them show the different faces is 1

(A) 65 6!	**(B)** 66 1
(C) 6! 5!	**(D)** 66

Q111.

CUSAT CAT 2023 - Shift 5

Two tangents are drawn from the origin to a circle with centre at (2,–1) . If the equation of one of the tangents is 3x + y = 0, the equation of the other tangent is

(A) 3x – y = 0	**(B)** x + 3y = 0
(C) x – 3y = 0	**(D)** x + 2y = 0 1 1 5

Q112.

CUSAT CAT 2023 - Shift 5

If ∫ f(x)dx=–2cos x +c, then f(x) is equal to

(A) sin x sin x	**(B)** x
(C) 2cos x	**(D)** cos x 2

Q113.

CUSAT CAT 2023 - Shift 5

Let x= 1+i 7 and y =cos +isin . Let Δ= 1 x+ y y . Then Δ is 3 4 4 1 x x+ y equal to

(A) – 7	**(B)** 7
(C) i	**(D)** –1

Q114.

CUSAT CAT 2023 - Shift 5

If z =1 and z ≠ ±1, then all the values of lie on 2 1–z

(A) a line not passing through the origin	**(B)** z =2
(C) the x-axis	**(D)** the y-axis sinx

Q115.

CUSAT CAT 2023 - Shift 5

If 7sin θ+3cos θ=4 and 0≤θ≤ , then the value of tanθ is 2 3

(A) 7 2	**(B)** 7 1
(C) 3 1	**(D)** 7 2

Q116.

CUSAT CAT 2023 - Shift 5

If cosθ+secθ=2, then the value of cos θ+sec θ equals

(A) 1	**(B)** 2
(C) 3	**(D)** 68

Q117.

CUSAT CAT 2023 - Shift 5

The domain of the function f(x)=sin log is 3 2

(A) [–1, 9]	**(B)** [1, 9]
(C) [–9, 1]	**(D)** [3, 9] 3

Q118.

CUSAT CAT 2023 - Shift 5

Which of the following functions is not 1-1?

(A) f :■→■ defined by f(x)=2x+5	**(B)** f :[0,π]→[–1,1] defined by f(x)=cosx –π π
(C) f : , →[1,7] defined by f(x)=3sinx+4 2 2	**(D)** f :■→[–1,] defined by f(x)=sinx 1

Q119.

CUSAT CAT 2023 - Shift 5

If A is a set with n elements, then the cardinality of {(x,y,z):x,y,z∈ A,x≠ y,y ≠ z,z ≠ x} is

(A) n 3	**(B)** n(n–1) 2
(C) n 2 (n–2)	**(D)** n 3 –3n 2 +2n 2 cos (x) cosxsinx –sinx π 2

Q120.

CUSAT CAT 2023 - Shift 5

Let Δ(x)= cosxsinx sin 2 x cosx . Then ∫[Δ(x)+Δ'(x)]dx equals sinx –cosx 0 0 π

(A) 3 π	**(B)** 2
(C) 2π 3π	**(D)** 2

Q121.

CUSAT CAT 2024 - Shift 1

cos4θ=

(A) 8cos 4 θ–8cos 2 θ+1	**(B)** 8cos 4 θ+8cos 2 θ–1
(C) 4cos 4 θ–8cos 2 θ+1	**(D)** 8cos 4 θ+4cos 2 θ–1 dy y x

Q122.

CUSAT CAT 2024 - Shift 1

If (cosx) =(sin y) , then = dx log(sin y)+ ytanx

(A) log(cosx)–xcot y log(cosx)–xcot y	**(B)** log(sin y)+ ytanx log(sin y)– ytanx
(C) log(cosx)+xcot y log(cosx)+xcot y	**(D)** log(sin y)– ytanx π 2

Q123.

CUSAT CAT 2024 - Shift 1

Let f(x)=sinx and g(x)= tanx. Then is not g

(A) continuous at x= 1 2	**(B)** continuous at x=0 π
(C) defined at x= 2	**(D)** defined at x=1 3 1 27 27 3

Q124.

CUSAT CAT 2024 - Shift 1

If cot =θ, then sinθ is 5 5

(A) 26 1	**(B)** 26 5
(C) 26 1	**(D)** 26 1

Q125.

CUSAT CAT 2024 - Shift 1

Which of the following statement is false? 1 –1

(A) If z =r(cosθ+isinθ), then z = (cosθ–isinθ) r	**(B)** Given any complex number cosθ+isinθ and any integer n n, (cosθ+isinθ) =cosnθ+insinθ
(C) Given any complex number cosθ+isinθ and any real x x, (cosθ+isinθ) =cosxθ+isinxθ 1 1 θ+2kπ θ+2kπ	**(D)** zn =rn cos +cos , k =0,1,2,3,... n n 1 cosθ 1

Q126.

CUSAT CAT 2024 - Shift 1

Let –cosθ 1 cosθ , where 0≤θ≤2π. Then –1 –cosθ 1

(A) det	**(B)** det
(C) det	**(D)** det

Q127.

CUSAT CAT 2024 - Shift 1

Inverse of cosine function exists if the domain of cosine is

(A) ■ –π π	**(B)** , 2 2
(C) [0,2π]	**(D)** [0,π] x y

Q128.

CUSAT CAT 2024 - Shift 1

The expression tanilog reduces to a+ib ab

(A) 2 2 a +b 2ab	**(B)** 2 2 a –b ab
(C) 2 2 a –b 2ab	**(D)** 2 2 a +b

Q129.

CUSAT CAT 2024 - Shift 1

The total number of flags with three horizontal strips, in order that can be formed using 2 identical red, 2 identical green and 2 identical white strips, is equal to

(A) 4!	**(B)** 2.(4!)
(C) 3.(4!)	**(D)** 4(4!) 0 1 0

Q130.

CUSAT CAT 2024 - Shift 1

Given the matrix A = 0 0 1 . If constants p, q, r satisfy 1 2 –1 3 2 A = pA + qA + rI, then

(A) p = 2, q = 2, r = –1	**(B)** p = 1, q = 2, r = –1
(C) p = –1, q = 2, r = 1	**(D)** p = –2, q = 2, r = 1 1 0 0 1 0 0 3 2

Q131.

CUSAT CAT 2024 - Shift 1

If the function f(x)=cos(logx), then the value of 2 (2) (2) 1 x (2 2) f x f y – f + f x y is 2 y 2

(A) –2	**(B)** –1
(C) 1	**(D)** 0 1+x

Q132.

CUSAT CAT 2024 - Shift 1

The value of sin A sin(60° – A) sin(60° + A) is equal to 1

(A) sin3A 4 1	**(B)** sin3A 2 1
(C) sin2A 4 1	**(D)** sin3A 3 0, x<2 (3+2x)

Q133.

CUSAT CAT 2024 - Shift 1

For 0 ≤ x ≤ 2π, the number of real solutions of the equation [sin x] = [1 + sin x] + [1 – cos x] is

(A) 0	**(B)** 1
(C) 2	**(D)** 4 2 2 2 a +b +c

Q134.

CUSAT CAT 2024 - Shift 1

In any ΔABC, a(bcosC–ccosB) is equal to

(A) b 2 +c 2	**(B)** b 2 –c 2 1 1
(C) + b c 1 1	**(D)** – 2 2 b c 2 2 x + y

Q135.

CUSAT CAT 2024 - Shift 1

If sinθ= , then 2 2 x – y

(A) x≠ y	**(B)** x≠ y, x,y∈■
(C) x=0 or y =0	**(D)** x≠ yand x≠0,y ≠0 ∂z x y z

Q136.

CUSAT CAT 2024 - Shift 1

If x y z =c, then is equal to ∂x 1+logx

(A) 1+logz 1+logx	**(B)** – 1+logz 1+logz
(C) 1+logx 1–logz	**(D)** 1+logx sinxcosx

Q137.

CUSAT CAT 2024 - Shift 1

∫ dx is equal to 4 1–sin x 1 –1(2)

(A) sin sin x +c 2 1 –1(2)	**(B)** cos sin x +c 2 1 –1(2)
(C) tan sin x +c 2 1 –1(2)	**(D)** tan 2sin x +c 2

Q138.

CUSAT CAT 2024 - Shift 1

Solution of the differential equation x dy – y dx = 0 represents

(A) parabola whose vertex is at origin	**(B)** circle whose center is a origin
(C) a rectangular hyperbola	**(D)** straight line passing through origin 12 9 4

Q139.

CUSAT CAT 2024 - Shift 1

The circles x + y –10x + 16 = 0 and x + y = a intersect at two distinct points, if

(A) a < 2	**(B)** 2 < a < 8
(C) a > 8	**(D)** a = 2

Q140.

CUSAT CAT 2024 - Shift 1

The complex number sinx+icos2x and cosx–isin2x are conjugates to each other for

(A) x=nπ 1	**(B)** x=n+ π 2
(C) x=0	**(D)** no values of x

Q141. *CUSAT CAT 2024 - Shift 1*

For the curve y =7x–x , if x increases at the rate of 4 units/sec. then at x=2, the slope of the curve is changing at

(A) 72 units/sec	**(B)** –72 units/sec
(C) 48 units/sec	**(D)** –48 units/sec

Q142. *CUSAT CAT 2024 - Shift 1*

For x∈[–1,1], the number of solutions of the equation sin x=2tan x is equal to

(A) 2	**(B)** 3
(C) 1	**(D)** 4 sinx –sinx

Q143. *CUSAT CAT 2024 - Shift 1*

If θ is the acute angle between any two diagonals of a cube, then 1

(A) cosθ= 3 1	**(B)** sinθ= 3 1
(C) cosθ= 3 1	**(D)** sinθ= 3 6( )

Q144. *CUSAT CAT 2024 - Shift 2*

The series + + + +... is 1 2 3 4

(A) divergent	**(B)** convergent
(C) oscillatory	**(D)** constant x–sinx

Q145. *CUSAT CAT 2024 - Shift 2*

A missile fixed from ground level rises x meters vertically upwards in t seconds and 25 2 x=100t– t . The maximum height reached is 2

(A) 100 m	**(B)** 300 m
(C) 150 m	**(D)** 200 m

Q146. *CUSAT CAT 2024 - Shift 2*

∫sin 6 θcos 4 θdθ= 3π

(A) 512 π	**(B)** 512 5π
(C) 512 7π	**(D)** 512

Q147. *CUSAT CAT 2024 - Shift 2*

If the first 2 moments of a distribution about the value 5 are 2 and 20, then the standard deviation of the distribution is

(A) 2	**(B)** 3
(C) 1	**(D)** 4

Q148. *CUSAT CAT 2024 - Shift 2*

The area bounded by the curve y = sin x between x = 0 and x = π is

(A) 2 sq.unit	**(B)** 4 sq.unit
(C) 8 sq.unit	**(D)** 16 sq.unit 3

Q149. *CUSAT CAT 2024 - Shift 2*

The equation of the tangent to the curve y = x at the point (1, 1) is

(A) y = 3x	**(B)** y = 3x – 1
(C) y = 3x – 3	**(D)** y = 3x – 2

Q150. *CUSAT CAT 2024 - Shift 2*

The differentiation of log(cosx)is

(A) cotx –cosx	**(B)** sinx
(C) –cot 2 x	**(D)** –tan 2 x

Q151. *CUSAT CAT 2024 - Shift 2*

Which of the functions have maxima? x

(A) f(x)=e 1	**(B)** f(x)= x
(C) f(x)= tanx	**(D)** f(x)=sinx

Q152. *CUSAT CAT 2024 - Shift 2*

Which of the following statement is false about inverse sine function?

(A) Domain is [–1, 1] –π π	**(B)** Range is , 2 2
(C) An even function	**(D)** An odd function 1 2 –1 –1

Q153. *CUSAT CAT 2024 - Shift 2*

tan +tan = 4 9 3 –1

(A) cos 5 2 –1	**(B)** cos 5 2 –1
(C) cos 11 4 –1	**(D)** cos 5 3 t 2

Q154. *CUSAT CAT 2024 - Shift 2*

The distance covered by the particle is given by s(t)= –t +3. After starts 3 moving, the velocity and acceleration are zero respectively at

(A) t = 1, t = 2	**(B)** t = 0, t = 3
(C) t = 2, t = 1	**(D)** t = 0, t = 1 dy

Q155. *CUSAT CAT 2024 - Shift 2*

The general solution of differential equation log = x+ y is dx

(A) e x +e y =C	**(B)** e x +e –y =C
(C) e –x +e y =C	**(D)** e –x +e –y =C

Q156. *CUSAT CAT 2024 - Shift 2*

In a multiple choice examination with four multiple options (out of which, only one is correct), for each of five questions, the probability that a candidate would get four correct answers by just guessing is 15

(A) 256 3	**(B)** 1024 405
(C) 1024 15	**(D)** 1024

Q157. *CUSAT CAT 2024 - Shift 2*

The cost function of certain commodity is c(x) = 30x + 250 and the price of the commodity is fixed at `55. Then the break even point is

(A) 5	**(B)** 10
(C) 15	**(D)** 20

Q158. *CUSAT CAT 2024 - Shift 2*

Which of the following is true?

(A) sinθ+icosθ=i(cosθ–isinθ) –n	**(B)** (cosθ–isinθ) =cosnθ–isinnθ –n
(C) (cosθ+isinθ) =–cosnθ–isinnθ 1 cosnθ n	**(D)** z + = ,where z =cosθ+isinθ n z 2 2 3

Q159. *CUSAT CAT 2024 - Shift 2*

The product of three consecutive positive integers is 8 times their sum. Then the sum of their squares is

(A) 50	**(B)** 77
(C) 110	**(D)** 194

Q160. *CUSAT CAT 2024 - Shift 2*

The sum of the series cosx– cos x+ cos x– cos x+... 2 3 4 x

(A) log2+2log cos 2 x	**(B)** 2log2–2log cos 2 x
(C) log2+log cos 2 x	**(D)** 2log2+2log cos 2 3 3 3 2 3 4

Q161. *CUSAT CAT 2024 - Shift 2*

If A is a matrix such that A + A + 2I = 0, then

(A) A is non-singular	**(B)** A ≠0
(C) A is symmetric 1 –1	**(D)** A =– (A+I) 2

Q162. *CUSAT CAT 2024 - Shift 2*

If sin α, sin β and cos α are in Geometric Progression, then the roots of the equation 2 x + 2x cot β + 1 = 0 are always

(A) equal	**(B)** real
(C) imaginary	**(D)** greater than 1

Q163. *CUSAT CAT 2024 - Shift 2*

An example for a rational number is

(A) sin 15°	**(B)** cos 15°
(C) sin 15° cos 15°	**(D)** sin 15° cos 75° 4 4

Q164. *CUSAT CAT 2024 - Shift 2*

The solution of the equation 4sin x+cos x=1 is

(A) x=2nπ	**(B)** x=nπ+1
(C) x=(n+2)π 2 –1	**(D)** x=2nπ or nπ±sin 5 π

Q165. *CUSAT CAT 2024 - Shift 2*

tan cos +tan is equal to 5 3 13

(A) 6 13	**(B)** – 6 17
(C) 6 17	**(D)** – 6

Q166. *CUSAT CAT 2024 - Shift 2*

The lines x + 2y + 3 = 0, x + 2y – 7 = 0 and 2x – y – 4 = 0 are the sides of a square. Equation of a remaining side of the square can be

(A) 2x – y – 14 = 0	**(B)** 2x – y + 8 = 0
(C) 2x – y – 10 = 0	**(D)** 2x – y – 6 = 0 x2 e –cosx

Q167. *CUSAT CAT 2024 - Shift 2*

lim is equal to 2 x→0 x 1

(A) 2 2	**(B)** 3 3
(C) 2	**(D)** 2 π 2 cosx–sinx

Q168. *CUSAT CAT 2024 - Shift 2*

∫ is equal to 1+cosxsinx 0

(A) 0 π	**(B)** 2 π
(C) 4 π	**(D)** 6 2 dy dy

Q169.

CUSAT CAT 2024 - Shift 2

The number of solutions of the equation sin e =3 +3 is

(A) 0	**(B)** 2
(C) 1	**(D)** Infinitely many 2

Q170.

CUSAT CAT 2024 - Shift 2

The tangent to the parabola y = 16x, which is perpendicular to a line y – 3x – 1 = 0 is

(A) x – y + 36 = 0	**(B)** 3y – x – 36 = 0
(C) x + y – 36 = 0	**(D)** 3y + x + 36 = 0 dy 3 x

Q171.

CUSAT CAT 2024 - Shift 2

The solution of the differential equationx =2y+x e , where y = 0 when x = 1, is dx 2(x)

(A) y = x e –e 3(x)	**(B)** y = x e–e 2(x)
(C) y = x e–e	**(D)** tanx=(secx+c)y

Q172.

CUSAT CAT 2024 - Shift 2

The equation of a tangent to the parabola x =8y, which makes an angle θ with the positive direction of x-axis, is:

(A) y = ytanθ+2cotθ	**(B)** y = ytanθ–2cotθ
(C) x= ycotθ+2tanθ	**(D)** x= ycotθ–2tanθ

Q173.

CUSAT CAT 2024 - Shift 2

tan –1 +sec –1 is equal to 3+ 3 6+3 3 π

(A) 4 π	**(B)** 2 π
(C) 6 π	**(D)** 3 2

Q174.

CUSAT CAT 2024 - Shift 2

∫x 51(tan –1 x+cot –1 x) dx equals to 52

(A) x (tan –1 x–cot –1 x) +c 52 52 πx π	**(B)** + +c 104 2 52
(C) x (tan –1 x+cot –1 x) +c 52 52 πx π	**(D)** + +c 52 2

Q175.

CUSAT CAT 2024 - Shift 3

If x=2cost–cos2t and y =2sint–sin2t, then at t = , = 2 dx

(A) 1	**(B)** 2
(C) –1	**(D)** –2 3 2

Q176.

CUSAT CAT 2024 - Shift 3

A particle moves along a straight line according to the law s =t –6t +9t+3. The velocity at the instant when its acceleration is zero is

(A) –3 units/sec	**(B)** +3 units/sec
(C) –6 units/sec	**(D)** +6 units/sec 2

Q177.

CUSAT CAT 2024 - Shift 3

The point on the curve y = x at which the tangent makes an angle of 45° with the x- axis, is 1 1

(A) , 2 2 1 1	**(B)** , 2 3 1 1
(C) , 2 4 1 1	**(D)** , 4 2

Q178.

CUSAT CAT 2024 - Shift 3

∫sin7xcos5xdx= 1

(A) cos7xsin5x+c 35 1cos12x cos2x	**(B)** – + +c 2 12 2 1cos6x
(C) – +cosx +c 2 6 1cos12x cos2x	**(D)** + +c 2 12 2

Q179.

CUSAT CAT 2024 - Shift 3

The curves x – y =a and xy =c

(A) touch each other	**(B)** intersect each other
(C) cut orthogonally	**(D)** parallel to each other 2

Q180.

CUSAT CAT 2024 - Shift 3

The rate of change of area of a circle per second with respect to its radius at r = cm is π 2

(A) 20 m /s 2	**(B)** 20π cm /s 2
(C) 0.02 m /s 2	**(D)** 0.02 cm /s

Q181.

CUSAT CAT 2024 - Shift 3

Angle between sine and cosine curve at the point of intersection in the interval 0, is 2 π

(A) 4 –1( )	**(B)** tan 2 –1( )
(C) tan 2 2	**(D)** 0

Q182.

CUSAT CAT 2024 - Shift 3

The amplitude of complex number is 3+2i 5 –1

(A) tan 13 12 –1	**(B)** tan 13 π
(C) 4 5 –1	**(D)** tan 12

Q183. *CUSAT CAT 2024 - Shift 3*

Let a, b, c, d, e be five consecutive terms in an arithmetic sequence and suppose a + b + c + d + e = 30. Which of the following can be found?

(A) a	**(B)** b
(C) c	**(D)** d 5 4 3 2

Q184. *CUSAT CAT 2024 - Shift 3*

A deck of cards has red and black cards. The probability of a randomly chosen card 1 being red is . When four black card are added to the deck, the probability of 3 1 choosing red becomes . Then the number of cards in the deck originally is 4

(A) 9	**(B)** 12
(C) 16	**(D)** 20 3 3 3 3 3 3

Q185. *CUSAT CAT 2024 - Shift 3*

The set of values of x for which the inequalities x + 6x – 27 > 0 and x – 3x – 4 < 0 hold simultaneously is

(A) x > 3	**(B)** x < 4
(C) 3 < x < 4 7	**(D)** x= 2

Q186. *CUSAT CAT 2024 - Shift 3*

The product of the four values of cos +isin is 3 3

(A) 1	**(B)** –1
(C) 2	**(D)** –2

Q187. *CUSAT CAT 2024 - Shift 3*

The maximum number of points into which 4 circles and 4 straight lines intersect is

(A) 26	**(B)** 36
(C) 50	**(D)** 72

Q188. *CUSAT CAT 2024 - Shift 3*

If 0< x< ,then the sum of the series tanx– tan x+ tan x+...+∞ is 4 3 5

(A) x	**(B)** logx
(C) 2x	**(D)** x + logx

Q189. *CUSAT CAT 2024 - Shift 3*

Let a, b, c be three positive real numbers. The value of the determinant a + b 2 c c bc + 2a c 2c is b+ ac bc c ( )

(A) c 2b–c b ( )	**(B)** b 2c– y b ( )
(C) a 2b–c b ( )	**(D)** c 2b+c b sinx cosx

Q190. *CUSAT CAT 2024 - Shift 3*

If $\alpha + \beta =$ and $\beta + \gamma = \alpha$, then $\tan \alpha$ is equal to 2

(A) 2(tan β + tan γ)	**(B)** tan β + tan γ
(C) 2 tan β + tan γ	**(D)** tan β + 2 tan γ

Q191. *CUSAT CAT 2024 - Shift 3*

If in a ΔABC, the values of cot A, cot B, cot C are in Arithmetic Progression, then tanA is equal to 4

(A) 3 3	**(B)** 4 8
(C) 15 6	**(D)** 15 B C

Q192. *CUSAT CAT 2024 - Shift 3*

In a ΔABC, 1–tan tan is equal to 2 2 2a

(A) a+b+c 2b	**(B)** a+b+c 2c
(C) a+b+c 2	**(D)** a+b+c π –1 –1 –1

Q193. *CUSAT CAT 2024 - Shift 3*

If cot x+cot y+cot z = , then x+ y+z is equal to 2 1 1 1

(A) + + x y z	**(B)** xyz
(C) xy+ yz+zx	**(D)** 2

Q194. *CUSAT CAT 2024 - Shift 3*

The two points on the line x + y = 4 that lie at a unit distance from the line 4x + 3y = 10 are

(A) (–3, 1), (7, 11)	**(B)** (3, 1), (–7, 11)
(C) (3, 1), (7, 11)	**(D)** (3, 2), (–7, 12) x –x e +e –2cosx–4

Q195. *CUSAT CAT 2024 - Shift 3*

For all z∈Con the curve C : z =4, let the locus of the point z+ be the curve C . 1 2 z Then the curves

(A) C and C intersect at 4 points. 1 2	**(B)** C lies inside C 1 2
(C) C and C intersect at 2 points 1 2	**(D)** C lies inside C 2 1 2 p q

Q196. *CUSAT CAT 2024 - Shift 3*

If sin x– + – +...= , where x <2, then the value of x is 2 4 8 6 2

(A) 3 2	**(B)** – 3 3
(C) – 2 3	**(D)** 2 3 1

Q197. *CUSAT CAT 2024 - Shift 4*

If u = n +1– n –1, then the series u is n n

(A) Divergent	**(B)** Convergent
(C) Constant	**(D)** Oscillatory dy

Q198. *CUSAT CAT 2024 - Shift 4*

If x=sint cos2t and y =cost cos2t then = dx

(A) tan3t	**(B)** – tan3t
(C) tan2t	**(D)** – tan2t dx

Q199. *CUSAT CAT 2024 - Shift 4*

The system of the equations x – 4y + 7z =14, 3x + 8y – 2z = 13 and 7x – 8y + 26z = 5 has

(A) one solution	**(B)** finite number of solutions
(C) infinitely many solutions	**(D)** no solution sin2x

Q200. *CUSAT CAT 2024 - Shift 4*

sin –sin = 3 2 1

(A) 2 1	**(B)** 3 1
(C) 4	**(D)** 1 4

Q201. *CUSAT CAT 2024 - Shift 4*

Which of the following is true?

(A) Range of log function is the set of all real numbers	**(B)** The point (0, 1) is always on the graph of log function
(C) log function is ever increasing	**(D)** None of the above

Q202. *CUSAT CAT 2024 - Shift 4*

The function f(x)= tanx is π π

(A) increasing in – , 2 2 π	**(B)** decreasing in0, 2 π
(C) decreasing in– ,0 2	**(D)** increasing in(0,π)

Q203. *CUSAT CAT 2024 - Shift 4*

The directional cosines of z axis are

(A) 0, 1, 0	**(B)** 1, 1, 0
(C) 0, 1, 1	**(D)** 0, 0, 1

Q204.

CUSAT CAT 2024 - Shift 4

Suppose sina+sinb= , cosa+cosb=1. Then cos(a–b)is 3 5

(A) –1 3 1	**(B)** 3 1
(C) 2 2	**(D)** 3 x–1 y–3 z–4

Q205.

CUSAT CAT 2024 - Shift 4

If y =e then is dx y

(A) 1– y	**(B)** e x+y
(C) 2e x+y	**(D)** e y sinx

Q206.

CUSAT CAT 2024 - Shift 4

The smallest value of 3sinθ+17is

(A) 14	**(B)** 15
(C) 16	**(D)** 17

Q207.

CUSAT CAT 2024 - Shift 4

A traffic light runs repeatedly through the following cycle: green for 30 seconds, then yellow for 3 seconds and then red for 30 seconds. A man picks a random three-second time interval to watch the light. Then the probability that the colour changes while he is watching is 1

(A) 63 1	**(B)** 21 1
(C) 10 1	**(D)** 7 ∞ ∑ 2n

Q208.

CUSAT CAT 2024 - Shift 4

If cos θ=5, then the value of cos2θ is n=0 1

(A) 5 2	**(B)** 5 3
(C) 5 4	**(D)** 5

Q209.

CUSAT CAT 2024 - Shift 4

If a matrix A is both symmetric and skew symmetric, then A is a

(A) diagonal matrix	**(B)** zero matrix
(C) square matrix	**(D)** rectangular matrix a+b

Q210.

CUSAT CAT 2024 - Shift 4

Let a∈0, . Then the expression x 2 +x + is always greater than or 2 2 x +x equal to

(A) sec 2 α	**(B)** 2tanα
(C) 2	**(D)** 1

Q211. *CUSAT CAT 2024 - Shift 4*

If t =cos +isin , then the value of is 3 3 2 n

(A) (–1) n (–1)	**(B)** 3n 2 1
(C) 3n 2 n	**(D)** (–1) + 1 r2

Q212. *CUSAT CAT 2024 - Shift 4*

The values of x satisfying log cosx +log sinx =2 when x∈ 0, is sinx cosx 2 π

(A) 4 π	**(B)** 3 π
(C) 0, 4 π π	**(D)** , 4 2 2 3 4 n n + 10

Q213. *CUSAT CAT 2024 - Shift 4*

The number of triangles which can be formed by using the vertices of a regular polygon of t + 3 sides is 680. Then t is equal to

(A) 8	**(B)** 11
(C) 14	**(D)** 17 18 1

Q214. *CUSAT CAT 2024 - Shift 4*

If the k term in the expansion of – is a constant multiple of x , 3 x 2 then k is equal to

(A) 2	**(B)** 3
(C) 4	**(D)** 5 x–1 5x 7 2 3 2

Q215. *CUSAT CAT 2024 - Shift 4*

If cos(x – y), cos x and cos(x + y) are in Harmonic Progression, then cos x. sec 2 is equal to

(A) ±1	**(B)** ±2
(C) ± 3	**(D)** ± 2 2

Q216. *CUSAT CAT 2024 - Shift 4*

If sin x + a sin x + 1 = 0 has no real solutions, then

(A) | a | ≥ 2	**(B)** | a | ≥ 1
(C) | a | < 2	**(D)** | a | > 2 2 2

Q217. *CUSAT CAT 2024 - Shift 4*

The area of the ∆ is a – (b – c) . Then

(A) a 2 , b 2 , c 2 are in Arithmetic Progression	**(B)** a, b, c are in Arithmetic Progression
(C) cos A, cos B, cos C are in Arithmetic Progression	**(D)** sin A, sin B, sin C are in Arithmetic Progression

Q218. *CUSAT CAT 2024 - Shift 4*

The equation 3sinx+cosx=4 has

(A) only one solution	**(B)** two solutions
(C) infinitely many solutions	**(D)** no solution a b π –1 –1

Q219. *CUSAT CAT 2024 - Shift 4*

If tan +tan = , then x is equal to x x 2

(A) ab	**(B)** 2ab
(C) 2ab	**(D)** ab dy y x

Q220. *CUSAT CAT 2024 - Shift 4*

If f(x)=2x3+9x 2 +λx+20 is a decreasing function of x in the largest possible interval (–2, –1), then λ is equal to

(A) 12	**(B)** –12
(C) 6	**(D)** –6 2

Q221. *CUSAT CAT 2024 - Shift 4*

If centroid of tetrahedron OABC, where A,B,C are given by (a,1,2), (2,b,3) and (3,1,c) respectively be (1,2,–1), then distance of P(a,b,c) from origin is equal to

(A) 107	**(B)** 114
(C) 109	**(D)** 118 1 1

Q222. *CUSAT CAT 2024 - Shift 4*

The maximum and minimum values of f(x)=3sin x+4cos x is

(A) {–4, –3}	**(B)** {7, 3}
(C) {4, –3}	**(D)** {4, 3} x e –1

Q223. *CUSAT CAT 2024 - Shift 4*

The value of lim is equal to x→0 1–cosx 1

(A) 2	**(B)** – 2
(C) 2	**(D)** Does not exist

Q224. *CUSAT CAT 2024 - Shift 4*

If the solution curve f(x,y)=0 of the differential equation dy (1+log x) –xlog x=e y , x>0, passes through the point (1, 0), then arbitrary e e dx constant c in the solution is equal to

(A) –2	**(B)** –1
(C) 1	**(D)** 0

Q225. *CUSAT CAT 2024 - Shift 5*

The value of 2+ 2+ 2+2cos8θ where 0<θ< , is equal to 8

(A) cosθ	**(B)** 2cosθ
(C) 2sinθ	**(D)** –2cosθ ( )

Q226. *CUSAT CAT 2024 - Shift 5*

If f(x)=cosx sinx+ sin2 x+sin2θ , where θ is a constant, then the maximum value of f(x) is

(A) 1–cos 2 θ	**(B)** 1+sin 2 θ
(C) cosθ	**(D)** sinθ

Q227. *CUSAT CAT 2024 - Shift 5*

If f(x)=sin6 x+cos6 x, then the range of f(x) is 5

(A) 1, 4 1 3	**(B)** , 4 4 3 5
(C) , 4 4 1	**(D)** ,1 4 1 1 z

Q228. *CUSAT CAT 2024 - Shift 5*

Let x=cos55°, y =cos65° and z =cos175°. Then + + = x y xy

(A) –1	**(B)** 1
(C) 0	**(D)** ∞

Q229. *CUSAT CAT 2024 - Shift 5*

If A,B and C are angles of a triangle such that angle A is obtuse, then tanB·tanC will be less than 1

(A) 3 3	**(B)** 2
(C) 1	**(D)** 2 a b π

Q230. *CUSAT CAT 2024 - Shift 5*

If tan–1 +tan–1 = , then x is equal to x x 2

(A) a2b	**(B)** 2ab
(C) 2ab	**(D)** ab x2 –1

Q231. *CUSAT CAT 2024 - Shift 5*

The value of k in order that f(x)=sinx–cosx–kx+l decreases for all real values, is given by

(A) k <1	**(B)** k ≥1
(C) k < 2	**(D)** k ≥ 2 3

Q232. *CUSAT CAT 2024 - Shift 5*

A curve passing through (2,3) and satisfying ∫t y(t)dt = x2y(x), (x>0) is 0 x2 y2

(A) + =1 8 18	**(B)** xy =C
(C) x2 + y2 =13 9	**(D)** y2 = x 2

Q233. *CUSAT CAT 2024 - Shift 5*

If sinα,cosα are the roots of the equation ax2 +bx+c=0, c≠0, then

(A) a2 +b2 +2ac=0	**(B)** b2 –a2 –2ac=0
(C) a2 –b2 –2ac=0	**(D)** a2 –b2 +2ac=0

Q234. *CUSAT CAT 2024 - Shift 5*

A common tangent to 9x –16y =144 and x + y =9 is 3x 15

(A) y = + 7 7 2 15	**(B)** y =3 x+ 7 7 3
(C) y =2 x+15 7 7 2	**(D)** y =3 x+15 7 7

Q235. *CUSAT CAT 2024 - Shift 5*

If cosec x, 2cotx, 2cosecx, where 0< x< , are in Geometric Progression, then 2 2 x is equal to

(A) 2π	**(B)** π π
(C) 2 π	**(D)** 3

Q236. *CUSAT CAT 2024 - Shift 5*

The sum of the focal distances from any point on the ellipse 9x +16y =144is

(A) 32	**(B)** 18
(C) 16	**(D)** 8

Q237. *CUSAT CAT 2024 - Shift 5*

For all real numbers of x, the value of cos x + sin x lies in the interval

(A) (1, 2)	**(B)** [1, 2] 3
(C) ,1 4 5	**(D)** ,2 4 n 20 ∑

Q238. *CUSAT CAT 2024 - Shift 5*

The value of lim xtan –tan equals x→∞ x+2 x+2

(A) 1	**(B)** –1 1
(C) – 2 1	**(D)** 2 x x –x

Q239. *CUSAT CAT 2024 - Shift 5*

The number of real roots of sin(e) = 5 + 5 is

(A) 0	**(B)** 1
(C) 2	**(D)** infinitely many 2

Q240. *CUSAT CAT 2024 - Shift 5*

The singularities of the analytic function f(z)= are at 2 z +1

(A) i and –i	**(B)** 1 and –i
(C) 1 and i	**(D)** 1 and –1

Q241. *CUSAT CAT 2024 - Shift 5*

If the graph of the function f(x) has a unique tangent at the point (a,0), then (1+6f(x)) lim log is x→a 3f(x)

(A) 1	**(B)** 0
(C) 2 1	**(D)** 3

Q242. *CUSAT CAT 2024 - Shift 5*

If the equation sin4 x+asin4 x+1=0 has a solution, then a lies in

(A) (–∞,–2)	**(B)** (–∞,2)
(C) (2,∞)	**(D)** (–2,∞)

Q243. *CUSAT CAT 2024 - Shift 5*

If the sum of the distance of a point to two perpendicular lines in a plane is 1, then its locus is

(A) square	**(B)** circle
(C) straight line	**(D)** two intersecting lines x2 y2

Q244. *CUSAT CAT 2024 - Shift 5*

The least value of cosec2 θ+36sec2 θ is

(A) 36	**(B)** 40
(C) 49	**(D)** 72

Q245. *CUSAT CAT 2024 - Shift 5*

The number of divisors of 9600 including 1 and 9600 are

(A) 60	**(B)** 58
(C) 48	**(D)** 46 sin2x cos2x

Q246.

CUSAT CAT 2024 - Shift 5

There are two bags one of which contains 3 black and 4 white balls while the other contains 4 black and 3 white balls. A die is cast. If the face 1 or 3 turns up with, a ball is taken from the first bag and if any other face turn up with, a ball is chosen from the second bag. The probability of choosing a black ball is 13

(A) 21 11	**(B)** 31 11
(C) 21 16	**(D)** 21

Q247.

CUSAT CAT 2024 - Shift 5

The value of log(tan1) · log(tan2) ··· log(tan89)=

(A) 0	**(B)** –1
(C) 2	**(D)** –2

Q248.

CUSAT CAT 2024 - Shift 5

Equation of the tangent to the circle with centre (2, –1) is 3x + y = 0 . The square of the length of the tangent to the circle from the point (23, 17) is

(A) 1354	**(B)** 1452
(C) 1320	**(D)** 1525 2

Q249.

CUSAT CAT 2024 - Shift 5

= 2+ 2+ 2+2cos4x x

(A) sec 2	**(B)** secx
(C) cosecx	**(D)** 1 1+x2 π

Q250.

CUSAT CAT 2024 - Shift 5

The number of solutions of the equation sin–1 = (sec(x–1)) is 2x 2

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q251.

CUSAT CAT 2024 - Shift 5

If the system of linear equations x+ y+z =6, x+2y+3z =14 and 2x+5y+λz =μ, (λ,μ∈■) has a unique solution, then

(A) λ≠8	**(B)** λ=8,μ≠36
(C) λ=8,μ=36	**(D)** λ≠8,μ=36 1 3cosx 1

Q252.

CUSAT CAT 2024 - Shift 5

If f(x)= sinx 1 3cosx , the maximum value of f(x) is 1 sinx 1

(A) 10	**(B)** 11
(C) 6	**(D)** 7

Q253. *CUSAT CAT 2025 - Shift 1*

The number of asymptotes of tan x is

(A) 1	**(B)** 2
(C) 3	**(D)** infinite

Q254. *CUSAT CAT 2025 - Shift 1*

sinx°= dxπ 2

(A) cosx° π π	**(B)** cosx° 180 π
(C) cosx° 90 1	**(D)** cosx° 90 3

Q255. *CUSAT CAT 2025 - Shift 1*

Let S ={x∈■|tanx= x}. Then S is

(A) infinite	**(B)** finite
(C) 1	**(D)** 0 1

Q256. *CUSAT CAT 2025 - Shift 1*

The number of strings which can be formed using the letters of the word ‘MATRIX’ either starts with A or ends with R, is

(A) 100	**(B)** 216
(C) 432	**(D)** 720

Q257. *CUSAT CAT 2025 - Shift 1*

A straight line through the point (2, 2) intersects the lines 3x+ y =0 and 3x– y =0 at the points A and B. The equation of the line AB so that the∆OAB is equilateral, is

(A) x–2=0	**(B)** y–2=0
(C) x+ y–4=0	**(D)** x+ y+4=0

Q258. *CUSAT CAT 2025 - Shift 1*

The slopes of the lines represented by x 2 +6hxy+3y 2 =0 are two tangents to a circle. Then the radius of the circle is 1

(A) ± 2 2	**(B)** ± 3
(C) ±1 3	**(D)** ± 2

Q259. *CUSAT CAT 2025 - Shift 1*

The probability that an even number appears on all when throwing three dice simultaneously, is 1

(A) 8 5	**(B)** 8 2
(C) 9 4	**(D)** 13

Q260.

CUSAT CAT 2025 - Shift 1

An example of a rational number is

(A) sin 15°	**(B)** cos 15°
(C) sin 15° cos 15°	**(D)** sin 15° cos 75° 2 2

Q261.

CUSAT CAT 2025 - Shift 1

The quadratic equation whose roots are sin 18° and cos 36°, is

(A) 16x 2 –12x+1=0	**(B)** 16x 2 +12x+1=0
(C) 16x 2 –12x–1=0	**(D)** 16x 2 +10x+1=0 π 2 2

Q262.

CUSAT CAT 2025 - Shift 1

In a triangle ABC, if ∠B= , then cos A+cos C is equal to 2

(A) –2	**(B)** 1
(C) 2	**(D)** 0

Q263.

CUSAT CAT 2025 - Shift 1

The two points on the line x + y = 4 that lie at a unit distance from the line 4x + 3y = 10 are

(A) (–3,1),(7,11)	**(B)** (3,1),(–7,11)
(C) (3,1),(7,11)	**(D)** (3,1),(–7,–11) 2

Q264.

CUSAT CAT 2025 - Shift 1

The shortest distance between the line y–x=1 and the curve x= y , is 3 2

(A) 8 2 3	**(B)** 8 3 2
(C) 5 3	**(D)** 4 ∂z x y z

Q265.

CUSAT CAT 2025 - Shift 1

If x y z =c, then is equal to ∂x 1+logx

(A) 1+logz 1+logx	**(B)** – 1+logz 1+logz
(C) 1+logx 1–logz	**(D)** 1–logx x –x e +e +2cosx–4

Q266.

CUSAT CAT 2025 - Shift 1

In the interval (–3,3), the function f(x)= + , x≠0 is 3 x

(A) increasing	**(B)** decreasing
(C) neither increasing nor decreasing	**(D)** partly increasing and partly decreasing π 2 cosx–sinx

Q267.

CUSAT CAT 2025 - Shift 1

∫ is equal to 1+cosxsinx 0

(A) 0 π	**(B)** 2 π
(C) 4 π	**(D)** 6 dy 3 x

Q268.

CUSAT CAT 2025 - Shift 1

The solution of the differential equation x =2y+x e , where y =0 when x=1 is dx 2(x)

(A) y = x e –e 3(x)	**(B)** y = x e–e 3(x)
(C) x= y e–e	**(D)** tanx=(secx+c)y

Q269.

CUSAT CAT 2025 - Shift 1

If z is a point on the argand plane such that z–1 =1, then is equal to z

(A) tan(argz)	**(B)** cot(argz)
(C) itan(argz)	**(D)** 2

Q270.

CUSAT CAT 2025 - Shift 1

The number of solutions of the equations sin = x –2 3x+4 is 2 3

(A) 0	**(B)** 1
(C) 2	**(D)** more than 2 (n+2)!–(n+1)!

Q271.

CUSAT CAT 2025 - Shift 1

The value of tan –1sincos –1 is 3 π

(A) 4 π	**(B)** 2 π
(C) 3 π	**(D)** 6 2 4

Q272.

CUSAT CAT 2025 - Shift 1

The integral ∫sec3xcosec3x dx= –1

(A) –3tan 3 x+c –4 3	**(B)** – tan 3 x+c 4 –1
(C) 3cot 3 x+c –1	**(D)** 3tan 3 x+c

Q273.

CUSAT CAT 2025 - Shift 1

The distance between the straight lines 2x+ y+4=0 and 2x+ y+8=0 is 4

(A) 5 3	**(B)** 5 9
(C) 5 3	**(D)** 2

Q274.

CUSAT CAT 2025 - Shift 1

The domain of the function cos (2x–1) is

(A) [–2,0]	**(B)** [0,2]
(C) [1,0]	**(D)** [0,1] x 1

Q275.

CUSAT CAT 2025 - Shift 1

The order of the differential equation whose general solution is 2 2 y =c cos2x+c cos x+c sin x+c , is 1 2 3 4

(A) 1	**(B)** 2
(C) 3	**(D)** 4 x+3 (–4–2x) dx

Q276.

CUSAT CAT 2025 - Shift 2

The value of y =sec (–2) is π

(A) 3 2π	**(B)** 3 5π
(C) 3 7π	**(D)** 3 cosθ sinθ π 2025

Q277.

CUSAT CAT 2025 - Shift 2

If A= , then the matrix A at θ= is –sinθ cosθ 3 0 0

(A) 0 0 1 0	**(B)** 0 1 –1 0
(C) 0 1 –1 0	**(D)** 0 –1 1 –3 2

Q278.

CUSAT CAT 2025 - Shift 2

The value of x for which the matrix A= 4 –1 x is singular, is 3 5 2 31

(A) 7 32	**(B)** 7 33
(C) 7 34	**(D)** 7

Q279.

CUSAT CAT 2025 - Shift 2

The number of points at which sinx attains its minimum in ■, is

(A) 1	**(B)** 2
(C) 0	**(D)** infinitely many

Q280.

CUSAT CAT 2025 - Shift 2

The value of cosec(–1080°) is

(A) 2 1	**(B)** 2
(C) 0	**(D)** ∞

Q281.

CUSAT CAT 2025 - Shift 2

cosec cosec x = x holds if x belongs to

(A) ■ –π π	**(B)** , 2 2
(C) ■–(–1,1)	**(D)** [–1,1]

Q282.

CUSAT CAT 2025 - Shift 2

cos = 6 n=0 2

(A) 2– 3 3	**(B)** 2
(C) 2	**(D)** ∞ cosx –sinx T

Q283.

CUSAT CAT 2025 - Shift 2

Consider A= . Let S be set of all real x satisfying A·A = I . Then sinx cosx 2 S is equal to

(A) {0}	**(B)** {2π}
(C) ■	**(D)** ■

Q284.

CUSAT CAT 2025 - Shift 2

If 3x–6y+10=0and 12y–6x+16=0 are two tangents to a circle, then the radius of the circle is 3

(A) 2 5 3	**(B)** 5 3
(C) 4 5 3	**(D)** 2

Q285.

CUSAT CAT 2025 - Shift 2

If the matrix –3 0 4 is not invertible, then –1 1 2

(A) 10	**(B)** –10
(C) –4	**(D)** 4 2 2 cos α cosαsinα cos β cosβsinβ

Q286.

CUSAT CAT 2025 - Shift 2

If A= and B = are two matrices cosαsinα sin 2 α cosβsinα sin 2 β such that AB is a null matrix, then α–βis

(A) 0 π	**(B)** an odd multiple of 2 π
(C) an even multiple of 2	**(D)** α=β 1 1 1 2 3 4

Q287.

CUSAT CAT 2025 - Shift 2

The sum of the series cosx– cos x+ cos x– cos x+... is 2 3 4 x

(A) log2+log cos 2 x	**(B)** log2–2log cos 2 x
(C) log2–log cos 2 x	**(D)** log2+2log cos 2

Q288. *CUSAT CAT 2025 - Shift 2*

Seven balls are drawn simultaneously from a bag containing 5 white and 6 green balls. The probability of drawing 3 white and 4 green balls is 7

(A) 11 C 7 5 6 C + C	**(B)** 3 4 11 C 7 5 6 C + C
(C) 3 2 11 C 7 6 5 C + C	**(D)** 3 4 11 C 7

Q289. *CUSAT CAT 2025 - Shift 2*

The maximum value of 5cosx+5sinx+5 is

(A) 5	**(B)** 10
(C) 10+ 2	**(D)** 5(1+ 2)

Q290. *CUSAT CAT 2025 - Shift 2*

The equation 4 sin x + 5 cos x = 7 has

(A) only one solution	**(B)** two solutions
(C) infinitely many solutions	**(D)** no solution 3

Q291. *CUSAT CAT 2025 - Shift 2*

In a triangle ABC, b= 3,c=2 and sinB= , then C is equal to 4

(A) 30°	**(B)** 60°
(C) 90°	**(D)** 120°

Q292. *CUSAT CAT 2025 - Shift 2*

If αand β are solutions of sin x+asinx+b =0 as well that of 2 cos x+ccosx+d =0, thenα+β is equal to 2bd

(A) 2 2 b +d 2 2 a +c	**(B)** 2ac 2 2 b +d
(C) 2bd 2ac	**(D)** 2 2 a +c

Q293. *CUSAT CAT 2025 - Shift 2*

If f(x)=a+b x +c x , where a,b,c are real constants, then f(x) is differentiable at x=0 if

(A) a =0	**(B)** b=0
(C) c=0	**(D)** a =b=c

Q294. *CUSAT CAT 2025 - Shift 2*

If f(x)=cosxcos2xcos4xcos8x, then f ' is 4

(A) –1	**(B)** 2
(C) 2	**(D)** 2 2 2n 2n ∑ –1 ∑

Q295. *CUSAT CAT 2025 - Shift 2*

If cos x =0 then x is i i i=1 i=1

(A) n	**(B)** 2n n(n+1)
(C) 2 n(n–1)	**(D)** 2

Q296. *CUSAT CAT 2025 - Shift 2*

If a = 7+ 7+ 7+... having n radical signs, then by methods of mathematical n induction which is true

(A) a >7,∀n≥1 n	**(B)** a <2,∀n≥1 n
(C) a <7,∀n≥1 n	**(D)** a >13,∀n≥1 n sinx –sinx

Q297. *CUSAT CAT 2025 - Shift 2*

The value of cos 3 cos +sin 3 sin is 8 8 8 8 1

(A) 2 1	**(B)** 2 2 1
(C) 2 1	**(D)** 4 4 1+i 3

Q298. *CUSAT CAT 2025 - Shift 2*

tanx is not defined when x is

(A) 0 nπ	**(B)** 2 (2n+1)π
(C) 2	**(D)** nπ

Q299. *CUSAT CAT 2025 - Shift 2*

If the slope of a line is , then the slope of the line perpendicular to that line is 3 3

(A) – 2 3	**(B)** 2 2
(C) 3 2	**(D)** – 3 2 sin x

Q300. *CUSAT CAT 2025 - Shift 2*

The principal value of cos cos is 6 13π

(A) 6 π	**(B)** 2 π
(C) 3 π	**(D)** 6 π

Q301. *CUSAT CAT 2025 - Shift 2*

The direction cosines of the y-axis are respectively

(A) 1,0,0	**(B)** 1,1,0
(C) 0,1,0	**(D)** 0,0,1 2

Q302. *CUSAT CAT 2025 - Shift 2*

The multiplicative inverse of the given complex number z =12–5i is

(A) –12+5i 12 5	**(B)** + i 169 169 12 5
(C) + i 13 13	**(D)** 12+5i 1 –1 tan x

Q303. *CUSAT CAT 2025 - Shift 2*

If C be the centroid of the triangle having vertices (3,–1), (1,3) and (2,4). Let A be the point of intersection of the lines x+3y–1=0 and 3x– y+1=0, then the line passing through the points C and A also pass through the point

(A) (–9, –7)	**(B)** (–9, –6)
(C) (7, 6)	**(D)** (9, 7) y+ey+ey+...∞ dy

Q304. *CUSAT CAT 2025 - Shift 3*

If the directional cosines of a straight line are 0, 0, 1, then the equation of the straight line is

(A) x=0, y =0	**(B)** y =0, z =0
(C) x=0, z =0	**(D)** x+ y+ z =0 2

Q305. *CUSAT CAT 2025 - Shift 3*

If the system of equations x+ay =0, az+ y =0, ax+z =0 has infinite number of solutions, then the value of a is

(A) 0	**(B)** –1
(C) 1	**(D)** z sinx

Q306. *CUSAT CAT 2025 - Shift 3*

Let S be the set of points at which the function f :■ →■ defined by f(x)= cosx is not differentiable. Then S is equal to

(A) 0	**(B)** 1
(C) 2	**(D)** infinite

Q307. *CUSAT CAT 2025 - Shift 3*

The sum of three numbers is 96. The first number is 6 times the third number, and the third number is 40 less than the second number. Then the absolute value of the difference between the first and second numbers is

(A) 2	**(B)** 3
(C) 4	**(D)** 5 5

Q308. *CUSAT CAT 2025 - Shift 3*

If sina+sinb= and cosa+cosb=1, then cos(a–b) is equal to 3 1

(A) 3 2	**(B)** 3 3
(C) 4	**(D)** 1 13 5

Q309. *CUSAT CAT 2025 - Shift 3*

The number of constant functions from a set containing m elements to a set containing n elements is

(A) mn	**(B)** m
(C) n	**(D)** m + n –π π

Q310. *CUSAT CAT 2025 - Shift 3*

The value of cos x+sin x is

(A) 0	**(B)** 1 π
(C) 2	**(D)** π

Q311. *CUSAT CAT 2025 - Shift 3*

Which of the following function has maxima?

(A) f(x)=e 2x , x∈■ 1	**(B)** f(x)= , x∈■ 2 x
(C) f(x)=cot(x) , x∈■	**(D)** f(x)=cos(x) , x∈■

Q312. *CUSAT CAT 2025 - Shift 3*

If the point P(4,5) is shifted by a distance 2 2 units parallel to the line y = x, then the coordinates of P in new position is

(A) (6,7)	**(B)** ( ) 4+ 2,5+ 2
(C) ( ) 4+ 2,5– 2	**(D)** (5,4)

Q313. *CUSAT CAT 2025 - Shift 3*

The angle between the tangents drawn from the point (1, 4) to the parabola y =4x is π

(A) 6 π	**(B)** 4 π
(C) 3 π	**(D)** 2

Q314. *CUSAT CAT 2025 - Shift 3*

If a=cos +isin , then the value of is 3 3 2 n

(A) (–1) n (–1)	**(B)** 3n 2 1
(C) 3n 2 n	**(D)** (–1) +1

Q315. *CUSAT CAT 2025 - Shift 3*

The number of solutions of the equation log sinx=1–log cosx, in∈ [–2π,2π], 0.5 0.5 is

(A) 1	**(B)** 2
(C) 3	**(D)** 4 2 4 3

Q316. *CUSAT CAT 2025 - Shift 3*

The number of triangles which can be formed by using the vertices of a regular polygon of n + 3 sides is 220. Then n is equal to

(A) 8	**(B)** 9
(C) 10	**(D)** 11 18 1

Q317. *CUSAT CAT 2025 - Shift 3*

If g(x)=cos(logx), then g(x)g(y)– g +g(xy) is equal to 2 y

(A) 1 1	**(B)** 2
(C) –2	**(D)** 0

Q318. *CUSAT CAT 2025 - Shift 3*

An urn contains 3 red and 5 blue balls. The probability that two balls are drawn in which the second ball drawn is blue without replacement is 5

(A) 16 5	**(B)** 56 5
(C) 8 20	**(D)** 56

Q319. *CUSAT CAT 2025 - Shift 3*

The number of integral values of k for which the equation 7 cos x + 5 sin x = 2k + 1 has a solution is

(A) 4	**(B)** 8
(C) 10	**(D)** 12 B C

Q320. *CUSAT CAT 2025 - Shift 3*

In a triangle, sides are a, b, c and angles are A, B, C. If b + c = 3a, then cot cot 2 2 is equal to

(A) 3	**(B)** 1
(C) 4	**(D)** 2

Q321. *CUSAT CAT 2025 - Shift 3*

The four straight lines given by 12x +7xy–12y =0 and –12x 2 +7xy–12y 2 –x+7y–1=0 lie along the sides of a

(A) square	**(B)** parallelogram
(C) rectangle	**(D)** rhombus π dy –1 –1

Q322. *CUSAT CAT 2025 - Shift 3*

If sin x+sin y = , then is equal to 2 dx x

(A) y x	**(B)** – y y
(C) x y	**(D)** – x 3x–9 e +7

Q323. *CUSAT CAT 2025 - Shift 3*

The value of lim is equal to x→3cos(3–x) 3

(A) 7	**(B)** ∞
(C) 3	**(D)** 0

Q324. *CUSAT CAT 2025 - Shift 3*

If f(x)=2x 3 +9x 2 +λx+20 is a decreasing function of x in the interval (–2,–1), then λ is equal to

(A) 12	**(B)** –12
(C) 6	**(D)** –6

Q325. *CUSAT CAT 2025 - Shift 3*

The solution of the differential equation xdy– ydx =0 represents a

(A) parabola whose vertex is at origin	**(B)** circle whose center is at origin
(C) rectangular hyperbola	**(D)** straight line passing through origin

Q326. *CUSAT CAT 2025 - Shift 3*

The complex numbers sinx+icos2x and cosx–isin2x are conjugate to each other for

(A) x=nπ 1	**(B)** x=n+ π 2
(C) x=0	**(D)** no value of x

Q327. *CUSAT CAT 2025 - Shift 3*

The sum of n terms of the series 1 2 1 2 1 –1 –1 –1 –1 –1 tan +tan +tan +tan +tan +... is 2 5 8 25 18

(A) tan –1 (–3) 1 –1	**(B)** cot 3 1 –1
(C) tan 3 2 –1	**(D)** cot 3

Q328. *CUSAT CAT 2025 - Shift 3*

The number of ways of choosing 10 balls from infinite white, red, blue and green balls is

(A) 70	**(B)** 84
(C) 286	**(D)** 86 2 2

Q329. *CUSAT CAT 2025 - Shift 3*

The tangents to the hyperbola x – y =3 are parallel to the straight line 2x+ y+8=0 at the points

(A) (2, 1) or (1, 2)	**(B)** (2,–1) or (–2, 1)
(C) (–1,–2)	**(D)** (–2,–1)

Q330. *CUSAT CAT 2025 - Shift 3*

A plane passes through the point (1, –2, 3) and is parallel to the plane 2x–2y+z =0. The distance of the point (–1, 2, 0) from the plane is

(A) 2	**(B)** 3
(C) 4	**(D)** 5

Q331. *CUSAT CAT 2025 - Shift 3*

Nine digit numbers are formed using 1, 2, 3,…. ,9 without repetition. The probability that a number so formed is divisible by 4 is 2

(A) 7 3	**(B)** 8 1
(C) 9 2	**(D)** 9 30 –1+i 3

Q332. *CUSAT CAT 2025 - Shift 3*

A point on the straight line, 3x + 5y = 15 which is equidistant from the coordinate axes will lie only in

(A) 4th quadrant	**(B)** 1st quadrant
(C) 1st and 2nd quadrant	**(D)** 1st , 2nd and 4th quadrant 2 2

Q333. *CUSAT CAT 2025 - Shift 3*

If cosx= then cos2x is 2

(A) 0	**(B)** –1 1
(C) 2	**(D)** 1

Q334. *CUSAT CAT 2025 - Shift 3*

The number of tangents that can be drawn from (1,2) to the circle x + y =5 is

(A) 0	**(B)** 1
(C) 2	**(D)** more than 2 2 x –1

Q335. *CUSAT CAT 2025 - Shift 3*

The value of sin logi +cos logi is

(A) –1	**(B)** 1
(C) 0	**(D)** –i 2n 1–i

Q336. *CUSAT CAT 2025 - Shift 4*

The minimum value of cosec x is

(A) 0 π	**(B)** 2 π
(C) – 2	**(D)** π 2025 2024 0

Q337. *CUSAT CAT 2025 - Shift 4*

The area bounded by the curve cosx and the x-axis in 0, is 2

(A) 0 1	**(B)** 2
(C) 1	**(D)** 2 19–x

Q338. *CUSAT CAT 2025 - Shift 4*

How many five digit even numbers can be formed using the digits 3, 4, 5, 6, 7 in all possible ways?

(A) 120	**(B)** 240
(C) 10	**(D)** 24 4 3 2

Q339. *CUSAT CAT 2025 - Shift 4*

The number of ‘x’ satisfying cos2x>sin is 2

(A) 0	**(B)** 1
(C) 2	**(D)** ∞

Q340. *CUSAT CAT 2025 - Shift 4*

With respect to multiplication, the set {0,1,–1} does not form a group, since it fails to satisfy

(A) associativity	**(B)** existence of inverse
(C) existence of identity	**(D)** closure law { n }

Q341. *CUSAT CAT 2025 - Shift 4*

cosilog is equal to a+ib

(A) ab 2 2 a –b	**(B)** 2 2 a +b 2 2 a –b
(C) 2ab 2ab	**(D)** 2 2 a +b 3

Q342. *CUSAT CAT 2025 - Shift 4*

The product of all values of (cosα+isinα)5is

(A) 1	**(B)** cosα+isinα
(C) cos3α+isin3α	**(D)** cos5α+isin5α 1 1

Q343. *CUSAT CAT 2025 - Shift 4*

The sides PQ, QR, RP of a triangle PQR have 3,4,5 interior points on them. Total number of triangles that can be formed using these points as vertices is equal to

(A) 135	**(B)** 145
(C) 178	**(D)** 205 2 3 x 2

Q344. *CUSAT CAT 2025 - Shift 4*

If p=cos55°,q=cos65°and r =cos175°, then the value of + + is p q pq

(A) 0	**(B)** –1
(C) 1	**(D)** 2

Q345. *CUSAT CAT 2025 - Shift 4*

Given that the sides of a triangle ABC are in AP. If c < b < a, then cosA is equal to 3c–4b

(A) 2b 3c–4b	**(B)** 2c 4c–3b
(C) 2b 4c–3b	**(D)** 2c 2 2

Q346. *CUSAT CAT 2025 - Shift 4*

The number of points on the circle 2x +2y –5x=0which are at distance 2 from the point (1, –3) is

(A) 2	**(B)** 0
(C) 1	**(D)** 3 1+x dy –1

Q347. *CUSAT CAT 2025 - Shift 4*

If sec =a,then is 1– y dx y–1

(A) x+1 y+1	**(B)** x–1 x–1
(C) y–1 x–1	**(D)** y+1 tanx

Q348. *CUSAT CAT 2025 - Shift 4*

lim log (sinx) is x→0 e

(A) 1	**(B)** –1
(C) 0	**(D)** ∞

Q349. *CUSAT CAT 2025 - Shift 4*

Angle of intersection of the curves r =sinθ+cosθ and r =2sinθ is equal to π

(A) 2 π	**(B)** 3 π
(C) 4 π	**(D)** 6 1

Q350. *CUSAT CAT 2025 - Shift 4*

The solution of the differential equationcocossxxddyy = y(sinx– y)dx, 0< x< is 2

(A) secx= tanx+c	**(B)** ysecx= tanx+c
(C) y(tanx+c)=secx	**(D)** tanx=(secx+c)y

Q351. *CUSAT CAT 2025 - Shift 4*

If 3 =a+ib, then (a–2) 2 +b 2 is equal to 2+cosθ+isinθ

(A) 0	**(B)** 1
(C) –1	**(D)** 2

Q352. *CUSAT CAT 2025 - Shift 4*

In a polygon no three diagonals are concurrent. If the total number of points of intersection of diagonals interior to the polygon be 70, then the number of diagonals of the polygon is

(A) 20	**(B)** 10
(C) 28	**(D)** 8

Q353. *CUSAT CAT 2025 - Shift 4*

The shortest distance between the line y–x=1 and the curve x= y is 2 3

(A) 8 3 2	**(B)** 5 3
(C) 4 3 2	**(D)** 8 2 f(x) –9

Q354. *CUSAT CAT 2025 - Shift 4*

If x=a(cost+tsint) and y = a(sint–tcost), then = dx

(A) tant	**(B)** cott
(C) –tant	**(D)** –cott

Q355. *CUSAT CAT 2025 - Shift 4*

The number of tangents that can be drawn from (1,2) to x 2 + y 2 =5 is

(A) 0	**(B)** 1
(C) 2	**(D)** infinitely many –1 –1 –1

Q356.

CUSAT CAT 2025 - Shift 4

If cos x+cos y+cos z=3π; then the value of xy+ yz+zx is

(A) 3	**(B)** 1
(C) –1	**(D)** 0 ANSWER BOOK SI SI SI SI SI SI SI SI No Key No Key No Key No Key No Key No Key No Key No Key 1 A 31 C 61 D 91 B 121 A 151 A 181 A 211 B 2 C 32 B 62 A 92 A 122 B 152 B 182 A 212 B 3 D 33 D 63 A 93 A 123 D 153 A 183 C 213 B 4 C 34 B 64 C 94 C 124 B 154 D 184 C 214 B 5 B 35 D 65 B 95 C 125 D 155 D 185 A 215 A 6 A 36 D 66 A 96 B 126 D 156 C 186 C 216 C 7 D 37 A 67 B 97 B 127 B 157 A 187 C 217 D 8 A 38 B 68 B 98 C 128 C 158 D 188 D 218 B 9 C 39 B 69 C 99 D 129 C 159 B 189 B 219 D 10 D 40 B 70 D 100 C 130 A 160 A 190 C 220 D 11 C 41 D 71 A 101 B 131 B 161 B 191 A 221 B 12 B 42 A 72 B 102 A 132 A 162 D 192 A 222 C 13 A 43 B 73 B 103 B 133 A 163 B 193 D 223 A 14 B 44 C 74 A 104 C 134 B 164 C 194 B 224 B 15 A 45 B 75 D 105 B 135 D 165 C 195 B 225 A 16 C 46 A 76 B 106 B 136 C 166 A 196 D 17 D 47 C 77 C 107 D 137 C 167 C 197 C 18 A 48 C 78 C 108 A 138 A 168 B 198 A 19 A 49 C 79 C 109 B 139 D 169 D 199 D 20 A 50 B 80 C 110 C 140 C 170 D 200 D 21 A 51 D 81 B 111 D 141 D 171 A 201 D 22 C 52 B 82 A 112 D 142 D 172 A 202 D 23 B 53 B 83 B 113 A 143 A 173 D 203 B 24 D 54 B 84 A 114 B 144 C 174 A 204 A 25 C 55 A 85 A 115 B 145 C 175 B 205 D 26 D 56 A 86 C 116 A 146 B 176 C 206 A 27 D 57 A 87 A 117 A 147 D 177 A 207 A 28 D 58 B 88 D 118 B 148 A 178 B 208 B 29 B 59 D 89 A 119 D 149 B 179 D 209 D 30 B 60 A 90 B 120 B 150 D 180 A 210 D

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	A	4	B	5	C	6	B	7	B	8	D	9	D	10	B
11	D	12	A	13	D	14	B	15	C	16	A	17	B	18	B	19	B	20	A
21	C	22	A	23	B	24	B	25	B	26	C	27	D	28	A	29	D	30	C
31	D	32	C	33	C	34	C	35	B	36	D	37	C	38	C	39	D	40	A
41	B	42	C	43	B	44	D	45	B	46	D	47	A	48	A	49	D	50	A
51	D	52	C	53	C	54	C	55	D	56	C	57	B	58	D	59	B	60	B
61	C	62	D	63	A	64	C	65	C	66	B	67	A	68	C	69	D	70	A
71	C	72	A	73	A	74	A	75	B	76	C	77	C	78	C	79	B	80	A
81	B	82	D	83	A	84	B	85	B	86	A	87	C	88	C	89	C	90	B
91	D	92	C	93	C	94	A	95	A	96	B	97	C	98	A	99	A	100	D
101	C	102	D	103	B	104	A	105	A	106	C	107	A	108	D	109	A	110	B

111	C	112	B	113	C	114	D	115	C	116	B	117	B	118	D	119	D	120	B
121	A	122	A	123	C	124	A	125	C	126	D	127	D	128	B	129	A	130	C
131	D	132	A	133	A	134	B	135	C	136	B	137	A	138	D	139	B	140	D
141	D	142	B	143	A	144	A	145	D	146	A	147	D	148	A	149	D	150	B
151	D	152	C	153	B	154	C	155	B	156	D	157	B	158	A	159	B	160	A
161	D	162	B	163	C	164	D	165	C	166	A	167	C	168	A	169	A	170	D
171	A	172	C	173	D	174	B	175	C	176	A	177	D	178	B	179	C	180	C
181	C	182	D	183	C	184	B	185	C	186	A	187	C	188	A	189	A	190	D
191	C	192	A	193	B	194	B	195	A	196	A	197	B	198	B	199	D	200	A
201	B	202	A	203	D	204	B	205	A	206	A	207	D	208	C	209	B	210	B
211	B	212	A	213	C	214	B	215	D	216	C	217	A	218	D	219	A	220	A
221	D	222	D	223	D	224	D	225	B	226	B	227	D	228	C	229	C	230	D
231	D	232	B	233	D	234	B	235	D	236	D	237	C	238	D	239	A	240	A
241	C	242	A	243	A	244	C	245	C	246	C	247	A	248	D	249	A	250	B
251	A	252	A	253	B	254	D	255	D	256	B	257	B	258	B	259	A	260	C
261	A	262	B	263	B	264	A	265	B	266	B	267	A	268	A	269	C	270	B
271	D	272	A	273	A	274	D	275	D	276	B	277	C	278	D	279	D	280	A
281	C	282	A	283	D	284	A	285	C	286	B	287	D	288	C	289	D	290	D
291	B	292	D	293	B	294	C	295	B	296	C	297	B	298	C	299	A	300	D
301	C	302	B	303	B	304	A	305	B	306	D	307	D	308	A	309	C	310	C
311	D	312	A	313	C	314	B	315	B	316	B	317	D	318	C	319	B	320	D
321	A	322	B	323	B	324	A	325	D	326	D	327	B	328	C	329	B	330	D
331	D	332	C	333	B	334	B	335	A	336	C	337	C	338	B	339	A	340	B
341	B	342	C	343	D	344	B	345	D	346	A	347	A	348	C	349	C	350	C
351	B	352	A	353	D	354	A	355	B	356	A								

Complex Numbers

26 questions

Q1. *CUSAT CAT 2023 - Shift 1*

If w is a non-real cube root of unity, then the value of (2) (2) (2) 1·(2–w) 2–w +2·(3–w) 3–w +···+(n–1)(n–w) n–w is

(A) n 3 n 2(n–1) 2	**(B)** –n+1 4 2 n(n+1)
(C) –n 2 2 n(n–1)	**(D)** 4 4 5 6

Q2. *CUSAT CAT 2023 - Shift 1*

The number of complex numbers z satisfying Re z =0 and z = 3 is

(A) 0	**(B)** 2
(C) 3	**(D)** 4

Q3. *CUSAT CAT 2023 - Shift 2*

Assume that z satisfies the condition that the imaginary part of is –1. Then the iz+1 locus of z is

(A) a point	**(B)** a circle
(C) a pair of straight lines	**(D)** an ellipse

Q4. *CUSAT CAT 2023 - Shift 2*

If P represents z = x + iy in the argand plane and |z – 1| + |z + 1| = 4, then the locus of P is 2 2

(A) x + y = 2 2 2	**(B)** x + y = 1 2 2
(C) x + y = 4	**(D)** x + y = 2 2 2 2

Q5. *CUSAT CAT 2023 - Shift 2*

If z = 1+i 3 , find the value of 3Im z z n 5 4

(A) 1536	**(B)** 1436
(C) 1578	**(D)** 1565

Q6. *CUSAT CAT 2023 - Shift 3*

The value of 4+5– +i –3 +i is equal to 2 2 2 2

(A) 1–i 3	**(B)** –1+i 3
(C) 4 3i	**(D)** –i 3 2

Q7. *CUSAT CAT 2023 - Shift 3*

The complex number z satisfying the equations z –4= z–i – z–5i =0 is

(A) 3–i	**(B)** 2 3
(C) –2 3–2i	**(D)** 0

Q8. *CUSAT CAT 2023 - Shift 3*

The centre of a regular hexagon is at the point z = i. If one of the vertices is at 2 + i, then the adjacent vertices of 2 + i are at the points

(A) 1±2i	**(B)** i+1± 3
(C) 2+i+(1± 3)	**(D)** 1+i(1± 3) 2 2 dx dy

Q9. *CUSAT CAT 2023 - Shift 4*

The number of complex numbers z such that z < and a z =1 r 3 r=1 where a <2 is r

(A) 0	**(B)** 1
(C) 4	**(D)** infinite 1 1 x– x+

Q10. *CUSAT CAT 2023 - Shift 4*

If z =3, the area of the triangle whose sides are z,ωz and z+ωz (here ω is a complex cube root of unity) is 9 3

(A) 4 3 3	**(B)** 2 5
(C) 2 8 3	**(D)** 3

Q11. *CUSAT CAT 2023 - Shift 5*

If ω(≠1) is a complex cube root of unity, the least value of n∈■ for which n n (2) (4) 1+ω = 1+ω is

(A) 6	**(B)** 5
(C) 3	**(D)** 2 ∞ 2 x π

Q12. *CUSAT CAT 2024 - Shift 1*

(1+i) =

(A) 512	**(B)** 512i
(C) 1 + i	**(D)** 1 – i

Q13. *CUSAT CAT 2024 - Shift 1*

A function f is defined by f(z)=iz . The number of values of z that satisfy both |z| = 5 and f (z) = z is

(A) 0	**(B)** 1
(C) 2	**(D)** 4

Q14. *CUSAT CAT 2024 - Shift 1*

There are 24 different complex numbers z such that z = 1 . Then the number of z 6 such that z is real is

(A) 0	**(B)** 4
(C) 6	**(D)** 12 x e

Q15. *CUSAT CAT 2024 - Shift 4*

For a complex number u = a + ib and v = c + id , define u * v = ac + ibd. Suppose z is 2 a complex number such that z * z = z + 40 . Then |z| is

(A) 5	**(B)** 5
(C) 10	**(D)** 5 2

Q16. *CUSAT CAT 2024 - Shift 4*

A polynomial equation in x of degree n always has

(A) n distinct roots	**(B)** n real roots
(C) n complex roots	**(D)** at most one root 3–2x

Q17. *CUSAT CAT 2024 - Shift 4*

For complex numbers z ,z , let z =12 and z –3–4i =5. Then the minimum 1 2 1 2 value of z –z is 1 2

(A) 0	**(B)** 2
(C) 7	**(D)** 17 8x 6x 4x 2x

Q18. *CUSAT CAT 2024 - Shift 5*

For any complex number z, the minimum value of z + z–2i is

(A) 2	**(B)** 1
(C) 0	**(D)** 3

Q19. *CUSAT CAT 2024 - Shift 5*

Let ω be the cube root of unity. The number of complex numbers satisfying the system of equations z3 +ω7=0 and z5 +ω 11 =1is

(A) 2	**(B)** 1
(C) 0	**(D)** 4

Q20. *CUSAT CAT 2025 - Shift 1*

If a =10, b =2 and a·b =12, then the value of a×b is

(A) 5	**(B)** 10
(C) 14	**(D)** 16 4 1+i

Q21. *CUSAT CAT 2025 - Shift 1*

If ω is a cube root of unity, then the value of (1–ω 8)(1–ω 4)(1–ω 2) (1–ω) is

(A) 1	**(B)** 9
(C) 0	**(D)** 6 dx

Q22.

CUSAT CAT 2025 - Shift 2

If ω≠1 is the complex cube root of unity and matrix H = , then H is 0 ω equal to

(A) 0	**(B)** –H
(C) 2 H	**(D)** H 1

Q23.

CUSAT CAT 2025 - Shift 3

The number of complex numbers that satisfy the equation z = z is

(A) 2	**(B)** 3
(C) 5	**(D)** 7 1

Q24.

CUSAT CAT 2025 - Shift 3

The smallest positive integer for which =–1 where i= –1 is 1+i

(A) 1	**(B)** 2
(C) 3	**(D)** 4 dx

Q25.

CUSAT CAT 2025 - Shift 4

The value of 1+ω–2ω , where ω is the cube root of unity is

(A) –81	**(B)** 81ω 2
(C) –81ω 2	**(D)** 81 5π

Q26.

CUSAT CAT 2025 - Shift 4

The polar form of the complex number–1+i is 5π

(A) 2, 4 3π	**(B)** 2, 4 π
(C) – 2, 4 π	**(D)** 2, 4

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	D	3	B	4	B	5	A	6	C	7	C	8	D	9	A	10	A
11	C	12	B	13	C	14	D	15	D	16	C	17	B	18	A	19	A	20	D
21	B	22	D	23	D	24	A	25	B	26	B								

Permutations & Combinations

17 questions

Q1. *CUSAT CAT 2023 - Shift 1*

A box contains 2 white balls, 3 black balls and 4 red balls. The number of ways in which 3 balls can be drawn from the box if at least one black ball is to be included in the draw is

(A) 32	**(B)** 64
(C) 96	**(D)** 128 2n+1 2n–1

Q2. *CUSAT CAT 2023 - Shift 3*

If the median of 21 observations is 40 and if the observations greater than the median are increased by 5, then the median of the new data will be

(A) 45	**(B)** 40 50
(C) 40+ 21 50	**(D)** 40– 21

Q3. *CUSAT CAT 2023 - Shift 3*

The number of ways in which the number 94864 can be resolved as a product of two factors

(A) 22	**(B)** 24
(C) 23	**(D)** 26

Q4. *CUSAT CAT 2023 - Shift 3*

In a group of 8 girls, two girls are sisters. The number of ways in which the girls can sit so that two sisters are not sitting together, is

(A) 48200	**(B)** 14106
(C) 28300	**(D)** 30240 x

Q5. *CUSAT CAT 2023 - Shift 5*

The number of ways in which six '+' and four '–' signs can be arranged in a line so that no two '–' signs occur together is

(A) 30	**(B)** 35
(C) 6!5!	**(D)** 10!

Q6. *CUSAT CAT 2024 - Shift 1*

In a group of 9 boys, three are brothers. The number of ways in which the boys can sit so that three brothers are not sitting together is

(A) 332640	**(B)** 352800
(C) 347760	**(D)** 364640 4 n

Q7. *CUSAT CAT 2024 - Shift 1*

Eight people are to be transported from city A to city B in three cars of different makes. If each car can accommodate at most three persons, then the number of ways, in which they can be transported, is

(A) 1120	**(B)** 560
(C) 3360	**(D)** 1680

Q8. *CUSAT CAT 2024 - Shift 3*

The number of ways to give away 20 apples to 3 boys, each boy receiving at least 4 apples, is 10

(A) C 8	**(B)** 90 22
(C) C 20 10	**(D)** C 12 2 –a ab ac

Q9. *CUSAT CAT 2024 - Shift 4*

From 4 gentlemen and 6 ladies, a committee of five is to be selected. The number of ways, in which the committee can be formed so that gentlemen are in majority, is

(A) 60	**(B)** 66
(C) 90	**(D)** 140 10 th x 2 4

Q10. *CUSAT CAT 2024 - Shift 5*

In a group of eight girls, two girls are sisters. The number of ways in which the girls can sit in a row so that two sisters are not sitting together is

(A) 4820	**(B)** 1410
(C) 2830	**(D)** 30240

Q11. *CUSAT CAT 2025 - Shift 1*

In a cricket squad of 15 members, only 5 can bowl. From this, a team of 11 members with 4 bowlers has to be selected. The probability for this selection is 2

(A) 455 6	**(B)** 1365 40
(C) 91 54	**(D)** 91

Q12. *CUSAT CAT 2025 - Shift 1*

The number of ways of arranging the letters of the name EULER is

(A) 24	**(B)** 10
(C) 120	**(D)** 60

Q13. *CUSAT CAT 2025 - Shift 2*

There are 7 bulbs in a room. Then the number of ways in which the room can be illuminated is

(A) 127	**(B)** 128
(C) 63	**(D)** 64 (–1)

Q14.

CUSAT CAT 2025 - Shift 3

An urn contains 3 red pens, 4 green pens and 6 yellow pens. The number of ways of drawing 4 pens from the urn if at least one red pen is to be included in the draw is

(A) 500	**(B)** 505
(C) 510	**(D)** 520

Q15.

CUSAT CAT 2025 - Shift 4

The number of arrangements of alphabets $\alpha,\alpha,\alpha,\alpha,\beta,\gamma,\delta$ such that no two α's are adjacent is

(A) 3!	**(B)** 4!
(C) 5!	**(D)** 6!

Q16.

CUSAT CAT 2025 - Shift 4

12 seats in a row are to be occupied by 4 boys the number of possible arrangement of there should be at least two empty seats between any two boys is

(A) 4!×6C 4 ( )	**(B)** 12C –5C ×3! 4 3 12P ×4!
(C) 4 3! 8P ×4!	**(D)** 4 2!

Q17.

CUSAT CAT 2025 - Shift 4

The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

(A) is increased by 2	**(B)** is decreased by 2
(C) is two times the original median	**(D)** remains the same as the that of the original set 1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	B	3	C	4	D	5	B	6	A	7	D	8	A	9	B	10	D
11	D	12	D	13	A	14	B	15	A	16	A	17	D						

Binomial Theorem

18 questions

Q1. *CUSAT CAT 2023 - Shift 1*

If coefficient of x n in (1+x) 101(1–x+x 2)100 is non-zero, then n cannot be of the form

(A) 3t + 1	**(B)** 3t
(C) 3t + 2	**(D)** 6t + 1

Q2. *CUSAT CAT 2023 - Shift 2*

The sum of the rational terms in the expansion of 2+ 3 is

(A) 9	**(B)** 31
(C) 32	**(D)** 41 2 2 2 x y z

Q3. *CUSAT CAT 2023 - Shift 2*

The ratio of the coefficient of x to the term independent of x in the expansion of 15 2 2 x + is x

(A) 1 : 8	**(B)** 1 : 12
(C) 1 : 16	**(D)** 1 : 32

Q4. *CUSAT CAT 2023 - Shift 3*

The largest term in the expansion of (3+2x) , where x= is 5

(A) 5 th	**(B)** 6 th
(C) 8 th	**(D)** 9 th

Q5. *CUSAT CAT 2024 - Shift 1*

The coefficient of x in the quadratic equation ax + bx + c = 0 was wrongly taken as 17 in place of 13 and its roots were found to be –2 and –15 . The actual roots of the equation are

(A) –2 and 15	**(B)** –3 and –10
(C) –4 and –9	**(D)** –5 and –6 2x + 1 2 x + 3 x

Q6. *CUSAT CAT 2024 - Shift 1*

The coefficient of x in (1+x) +(1+x) +...+(1+x) is 16

(A) C 4 16	**(B)** C 5 16
(C) C 6 16	**(D)** C 7 { }

Q7. *CUSAT CAT 2024 - Shift 1*

The coefficient of n in the expansion of log (1+x) (1–x) is equal to 1

(A) 30 1	**(B)** 15 1
(C) 10 1	**(D)** 45

Q8. *CUSAT CAT 2024 - Shift 1*

The coefficient of x in the expansion of 1+x x +2+ is x 2 30

(A) C 10 30	**(B)** C 25
(C) 1	**(D)** 0

Q9. *CUSAT CAT 2024 - Shift 2*

If the coefficient of the p term in the expansion of (2x + 3) is 1440, then p is

(A) 2	**(B)** 3
(C) 4	**(D)** 5 n 10 20 p ∑

Q10. *CUSAT CAT 2024 - Shift 2*

The number of integral terms in the expansion of 52 +54 is equal to

(A) 170	**(B)** 171
(C) 136	**(D)** 137 2

Q11. *CUSAT CAT 2024 - Shift 3*

If c ,c ,c ,...,c denote the binomial coefficients in the expansion of (1+x) , then 0 1 2 n the value of c +2c +3c +...+nc is 1 2 3 n

(A) n.2 n	**(B)** n.2 n–1
(C) n.2 n+1	**(D)** 2n 1 1

Q12. *CUSAT CAT 2024 - Shift 4*

The middle term in the expansion of x– is x 18

(A) C 9 18	**(B)** – C 9 18
(C) C 10 18	**(D)** – C 10

Q13. *CUSAT CAT 2024 - Shift 5*

The coefficient of xk in the expansion of is e–x 1–k–k2

(A) k! k2 +1	**(B)** k! 1–k
(C) k! 1	**(D)** k!

Q14. *CUSAT CAT 2025 - Shift 1*

If the coefficient of x and x in the expansion of (4+kx) are equal, then the value of k is

(A) 2	**(B)** 4
(C) 7	**(D)** 1 x+2y+z =4

Q15.

CUSAT CAT 2025 - Shift 1

The sum of the last ten coefficients in the expansion of (1+x) 19 when expanded in ascending powers of x, is

(A) 18 2	**(B)** 19 2
(C) 17 2	**(D)** 20 2

Q16.

CUSAT CAT 2025 - Shift 1

If α and β are the coefficients of x and x respectively in the expansion of ( )6 ( )6 2 2 x+ x –1 + x– x –1 , then

(A) α+β=60	**(B)** α+β=–30
(C) α–β=60	**(D)** α–β=–132 2 2 (2)

Q17.

CUSAT CAT 2025 - Shift 2

The coefficient of x in (1+x) +(1+x) +...+(1+x) is 16

(A) C 9 16 6	**(B)** C – C 5 5 16
(C) C –1 6 16 6	**(D)** C – C 6 5 λ 1 6

Q18.

CUSAT CAT 2025 - Shift 3

The middle term in the expansion of x– is x 18

(A) C 9 18	**(B)** – C 9 18
(C) C 10 18	**(D)** – C 10 a b 1 0

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	D	3	D	4	B	5	B	6	D	7	B	8	B	9	A	10	B
11	B	12	B	13	A	14	B	15	A	16	D	17	A	18	B				

Sequences & Series

25 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Sum to n terms of the series + + +... 1·2·3 2·3·4 3·4·5 5 (2n+5)

(A) + 4 2(n+1)(n+2) 1 (2n+5)	**(B)** – 4 2(n+1)(n+2) 5 (2n+5)
(C) – 4 2(n+1)(n+2) 1 (2n+5)	**(D)** + 4 2(n+1)(n+2) 4 7 10 167. Sum to infinity of the series 1+ + + +... is 5 5 2 5 3 16

Q2. *CUSAT CAT 2023 - Shift 2*

Given that a, b, c are in Arithmetic Progression and a ,b , c <1. 2 2 Let p = 1 + a + a + …+ ∞, q = 1 + b + b + . . .+ ∞ and 2 r = 1 + c + c + . . . + ∞. Then p, q, r are in

(A) AP	**(B)** GP
(C) HP	**(D)** equal 2 2

Q3. *CUSAT CAT 2023 - Shift 2*

Let a, b, c be distinct non-native numbers. If the vectors aiˆ+aˆj+ck ˆ , iˆ+k ˆ and ciˆ+cˆj+bk ˆ lie in a plane, then c is

(A) the harmonic mean of a and b	**(B)** equal to 0
(C) the arithmetic mean of a and b	**(D)** the geometric mean of a and b

Q4. *CUSAT CAT 2023 - Shift 3*

A student read common difference of an A.P. as –3 instead of 3 and obtained the sum of first 10 terms as –30. Then, the actual sum of the first 10 terms is equal to

(A) 120	**(B)** 240
(C) 180	**(D)** 300 2 2

Q5. *CUSAT CAT 2023 - Shift 4*

If , and are in Arithmetic progression, then the value of n can be 4 5 6

(A) 6	**(B)** 7
(C) 8	**(D)** 9 x 2 ; 0< x<2 f(11)– f(–11)

Q6. *CUSAT CAT 2023 - Shift 4*

Let a ,a ,...a be in Arithmetic Progression and h ,h ,...,h be in harmonic 1 2 10 1 2 10 Progression. If a =h =2 and a =h =3 then a h is 1 1 10 10 4 7

(A) 2	**(B)** 3
(C) 5	**(D)** 6

Q7. *CUSAT CAT 2023 - Shift 4*

Consider an infinite geometric series with the first term a and the common ratio r < 1. 3 If its sum is 4 and the second term is , then 4 4 3

(A) a = , r = 7 7 3	**(B)** a =2, r = 8 3 1
(C) a = , r = 2 2 1	**(D)** a =3, r = 4

Q8. *CUSAT CAT 2023 - Shift 4*

The interior angles of a polygon are in Arithmetic Progression. If the smallest angle is 100° and the common difference is 4°, then the number of sides is

(A) 5	**(B)** 7
(C) 36	**(D)** 44

Q9. *CUSAT CAT 2023 - Shift 4*

If a, b, c are in Arithmetic Progression, p, q, r are in Harmonic Progression and p r ap, bq, cr are in Geometric Progression, then + is equal to r p a c

(A) + c a a c	**(B)** – c a b q
(C) + q b b a	**(D)** – q p

Q10. *CUSAT CAT 2023 - Shift 5*

If x > 1, y > 1, z > 1 are in Geometric Progression, then , , are in 1+lnx 1+ln y 1+lnz

(A) Arithmetic Progression	**(B)** Geometric Progression
(C) Harmonic Progression	**(D)** both Arithmetic Progression and Harmonic Progression

Q11. *CUSAT CAT 2023 - Shift 5*

If three positive real numbers a, b, c are in Arithmetic Progression such that abc = 4, then the minimum possible value of b is 3

(A) 22 2	**(B)** 23 1
(C) 23 5	**(D)** 22

Q12. *CUSAT CAT 2024 - Shift 1*

Let a = 0 and a , a , a , . . . , a be real numbers such that |a| = |a + 1| for all i . 1 1 2 3 n i i–1 Then the arithmetic mean of the numbers a , a , a , . . . , a has a value t where 1 2 3 n 1

(A) t ≤ 2 1	**(B)** t <– 2 1
(C) t ≥– 2 1	**(D)** t > 2

Q13. *CUSAT CAT 2024 - Shift 1*

If the fourth, seventh and tenth terms of a Geometric Progression are p, q and r respectively, then 2 2 2

(A) p = q + r 2	**(B)** p = qr 2
(C) q = pr 2 2 2	**(D)** r = p + q 3 2

Q14. *CUSAT CAT 2024 - Shift 2*

Assume that m, n, p are in Harmonic Progression. Then , , are in n+ p p+m m+n

(A) Harmonic Progression	**(B)** Geometric Progression
(C) Arithmetic Progression	**(D)** Standard Progression 2 3

Q15. *CUSAT CAT 2024 - Shift 3*

If a, b, c, d are non-zero real numbers such that 2 2 2 2 2 2 2 (a + b + c)(b + c + d) ≤ (ab + bc + cd) , then a, b, c, d are in

(A) Arithmetic progression (AP)	**(B)** Geometric Progression (GP)
(C) Harmonic progression (HP)	**(D)** Neither AP nor GP nor HP

Q16. *CUSAT CAT 2024 - Shift 3*

The number of terms common to two Arithmetic Progressions 3, 7, 11, . . . , 407 and 2, 9, 16, . . . , 709 are

(A) 11	**(B)** 14
(C) 21	**(D)** 28 1 1 1 1

Q17. *CUSAT CAT 2024 - Shift 3*

If + = + and b≠a+c, then a,b,c are in b–a b–c a c

(A) Harmonic Progression	**(B)** Geometric Progression
(C) Arithmetic Progression	**(D)** Constant Progression 4 1

Q18. *CUSAT CAT 2024 - Shift 3*

Let N be the sum of the numbers appeared when two fair dice are rolled and let the k probability that N –2, 3N,N +2 are in geometric progression be . Then, the 36 value of k is

(A) 2	**(B)** 3
(C) 4	**(D)** 6

Q19. *CUSAT CAT 2024 - Shift 4*

Assume that a, b, c are three distinct real numbers and they are in Geometric Progression. If a + b + c = xb , then

(A) x < –1 or x > 3	**(B)** x ≤ –3 or x ≥ 2
(C) x < –4 or x > 3	**(D)** x ≤ –1 or x ≥ 3 π

Q20. *CUSAT CAT 2024 - Shift 5*

If S denotes the sum of n terms of an A.P., then S –3S +3S –S is equal to n n+3 n+2 n+1 n

(A) 2	**(B)** 1 1
(C) 2	**(D)** 0

Q21.

CUSAT CAT 2025 - Shift 1

The first three terms of a geometric progression are 2, 2 and 2. Then the next term is

(A) 9 2	**(B)** 8 2
(C) 7 2	**(D)** 1

Q22.

CUSAT CAT 2025 - Shift 2

Sum of n terms of the series 3+ 27 + 75+ 147 +... is n(n+1)(2n+1)

(A) 2 3 3n(n+1)	**(B)** 2 (n+1)
(C) 3 n(n–1)(2n–1)	**(D)** 2 3 P(n)

Q23.

CUSAT CAT 2025 - Shift 2

The sum of series + + +...+∞ equals 1! 2! 3!

(A) e e	**(B)** 2 3e
(C) 2 e	**(D)** 1+ 2 2

Q24.

CUSAT CAT 2025 - Shift 4

The 5th and 6th term of an arithmetic progression are 20 and 17. Then which term of the arithmetic progression lies between –1127 and –1125?

(A) 375	**(B)** 376
(C) 377	**(D)** 378 x –1

Q25.

CUSAT CAT 2025 - Shift 4

If the general term of an A.P. is3n, then the common difference is

(A) 2	**(B)** 3
(C) 5	**(D)** 6

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	C	3	D	4	B	5	B	6	D	7	D	8	A	9	A	10	C
11	B	12	C	13	C	14	A	15	B	16	B	17	A	18	B	19	D	20	D
21	D	22	A	23	C	24	D	25	B										

Straight Lines & Conic Sections

34 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse 2 2 x + 4y = 16 is 1 2 2

(A) (x–1) +(y–0) = 3 7 2 2	**(B)** (x–1) +(y–0) = 3 11 2 2
(C) (x–1) +(y–0) = 3 13 2 2	**(D)** (x–1) +(y–0) = 3

Q2. *CUSAT CAT 2023 - Shift 1*

If the imaginary part of is – 4, then the locus of the point representing z in the iz+1 complex plane is

(A) a straight line	**(B)** a parabola
(C) an ellipse	**(D)** a circle

Q3. *CUSAT CAT 2023 - Shift 2*

The differential equation = represents dx y (1+x 2)

(A) straight line	**(B)** hyperbola
(C) circle	**(D)** ellipse 6–x x

Q4. *CUSAT CAT 2023 - Shift 2*

The solution of the differential equation x+ y =0 is a family of dx

(A) straight lines	**(B)** circles
(C) ellipses	**(D)** parabolas 3z

Q5. *CUSAT CAT 2023 - Shift 2*

The length of the straight line x – 3y – 1 = 0 intercepted by the hyperbola 2 2 x – 4y = 1 is 6

(A) 10 5 6	**(B)** 10 7
(C) 10 6	**(D)** 5 2 2

Q6. *CUSAT CAT 2023 - Shift 2*

The locus of Re(z+1)= z–1 is

(A) a straight line	**(B)** a circle
(C) a parabola	**(D)** an ellipse 360 1 ∑

Q7. *CUSAT CAT 2023 - Shift 3*

An ellipse has OB as semi-minor axis, F and F' its foci and the angle FBF' is a right angle. Then, the eccentricity of the ellipse is 1

(A) 3 1	**(B)** 4 1
(C) 2 1	**(D)** 2 e4 2

Q8. *CUSAT CAT 2023 - Shift 3*

If the length of the semi major axis of an ellipse is 68 and the eccentricity is then 2 the area of the rectangle formed by joining the vertices of the latera recta of the ellipse is equal to

(A) 6846	**(B)** 6936
(C) 6676	**(D)** 7244 3 3 2 2

Q9. *CUSAT CAT 2023 - Shift 4*

If the line y =3x+λ touches the hyperbola 9x2 –5y2 =45, then the value of λ 2 is

(A) 45	**(B)** 36
(C) 6	**(D)** 15 1 1 1 1

Q10. *CUSAT CAT 2023 - Shift 4*

The equation of the directrix of the parabola 4y +12x–12y+39=0 is 7

(A) x = – 4 3	**(B)** y = 4
(C) x – y = 0	**(D)** 2x – 3y = 0

Q11. *CUSAT CAT 2023 - Shift 4*

The locus of the centre of a circle which touches the circle z–z =a externally is 1

(A) an ellipse	**(B)** a hyperbola
(C) a circle	**(D)** a pair of straight lines f(x)

Q12. *CUSAT CAT 2023 - Shift 4*

Equation of the directrix of the parabola y +4y+4x+2=0 is

(A) x=–1	**(B)** x=1 –3
(C) x= 2 3	**(D)** x= 2 x 2 x π

Q13. *CUSAT CAT 2023 - Shift 5*

The latus rectum of an ellipse is equal to half of its minor axis. Its eccentricity is 1

(A) 2 3	**(B)** 2 3
(C) 2 1	**(D)** 2 { }

Q14. *CUSAT CAT 2024 - Shift 2*

If the line y = x + a passes through the vertex of the parabola y = x + a , then the number of possible values of 'a' is

(A) 0	**(B)** 1
(C) 2	**(D)** ∞

Q15. *CUSAT CAT 2024 - Shift 3*

The focus of the parabola y – 4y – 8x + 2 = 0 is

(A) (1, 4)	**(B)** (1, 0)
(C) (–1, 2)	**(D)** (3, 2) dy x x x

Q16. *CUSAT CAT 2024 - Shift 3*

A curve through (1, 0) and satisfying the differential equation (1 + y) dx – xy dy = 0 represents

(A) a circle	**(B)** a parabola
(C) an ellipse	**(D)** a hyperbola

Q17. *CUSAT CAT 2024 - Shift 3*

The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is 4

(A) 3 2	**(B)** 3
(C) 3 4	**(D)** 3

Q18. *CUSAT CAT 2024 - Shift 4*

Volume of the solid formed by revolving the region bounded by the parabola y = x , x-axis, x = 0, x = 1 about x-axis is π

(A) 5 π	**(B)** 4 π
(C) 3 π	**(D)** 2 2

Q19. *CUSAT CAT 2024 - Shift 4*

The equation z =a + , r >0 represents (z–a)

(A) an ellipse	**(B)** a parabola
(C) a circle	**(D)** a straight line through the point a 1 1 1

Q20. *CUSAT CAT 2024 - Shift 5*

The equation to the line touching both the parabolas y2 =4x and x2 =–32y is

(A) x–2y+3=0	**(B)** x–4y+2=0
(C) x–2y+4=0	**(D)** x–4y+3=0 1 2 n dy

Q21. *CUSAT CAT 2024 - Shift 5*

The vertex of the parabola whose parametric equations are x=t2 +1,y =t+1 t ∈ ■ is

(A) (1, 2)	**(B)** (1,–2)
(C) (1, 1)	**(D)** (–1, 1) 4 2

Q22. *CUSAT CAT 2024 - Shift 5*

Equation + =1 represents an ellipse if k k–1

(A) 0 < k < 1	**(B)** k > 1
(C) k < 0 or k > 1	**(D)** 1 < k < 2

Q23. *CUSAT CAT 2024 - Shift 5*

If a =b,c∈■ and b2 >c, then zz +az+bz +c=0 represents

(A) a circle	**(B)** a parabola
(C) a straight line	**(D)** a hyperbola

Q24. *CUSAT CAT 2025 - Shift 1*

The solution of the differential equation 2x – y=3 represents dx

(A) circles	**(B)** straight lines
(C) ellipses	**(D)** parabolas

Q25. *CUSAT CAT 2025 - Shift 1*

The equation x + y =α where α is constant represents a family of dy

(A) circles	**(B)** parabolas
(C) hyperbolas	**(D)** exponential curves

Q26. *CUSAT CAT 2025 - Shift 2*

Three normals are drawn to the parabola y 2 = x through the point (a, 0). Then 1

(A) a = 2 1	**(B)** a = 4 1
(C) a > 2 1	**(D)** a< 4

Q27. *CUSAT CAT 2025 - Shift 2*

The equation z+3–2i = z+2i–3 represents

(A) a pair of straight lines	**(B)** two points
(C) a circle	**(D)** an ellipse z–i

Q28. *CUSAT CAT 2025 - Shift 2*

A curve through (1, 0) and satisfying the differential equation 1+ y dx–xydy =0 represents a/an

(A) circle	**(B)** parabola
(C) ellipse	**(D)** hyperbola

Q29.

CUSAT CAT 2025 - Shift 2

If the foci of the ellipse + =1 and the hyperbola – = coincide, then 16 b 2 144 81 25 2 the value of b is

(A) 9	**(B)** 1
(C) 5	**(D)** 7 ω 0 70

Q30.

CUSAT CAT 2025 - Shift 3

Locus of the point z satisfying the equation iz–1+ z–i =2 is

(A) an ellipse	**(B)** a parabola
(C) a circle	**(D)** a straight line 1

Q31.

CUSAT CAT 2025 - Shift 3

The slope of the line touching both the parabolas y =4x and x =–32y is 1

(A) 8 2	**(B)** 3 1
(C) 2 3	**(D)** 2 –4 –1 (2016 2015 2014)

Q32.

CUSAT CAT 2025 - Shift 4

The solution of the differential equation 2x – y =5 represents dx

(A) circles	**(B)** straight lines
(C) ellipses	**(D)** parabolas

Q33.

CUSAT CAT 2025 - Shift 4

The equation x +4y –4xy+4+x–2y =1 represents

(A) a straight line	**(B)** an ellipse
(C) a circle	**(D)** a parabola 2 2

Q34.

CUSAT CAT 2025 - Shift 4

If the ellipse + y =1 meets the ellipse x + =1 in four distinct points and 4 a 2 a = b2– 5b + 7, then b does not lie in

(A) [4, 5]	**(B)** (–∞, 2) ∪ (3,∞)
(C) (–∞, 0)	**(D)** [2, 3]

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	D	3	B	4	B	5	A	6	C	7	D	8	B	9	B	10	A
11	B	12	D	13	C	14	C	15	D	16	D	17	B	18	A	19	C	20	C
21	C	22	B	23	A	24	D	25	A	26	B	27	A	28	D	29	D	30	D
31	C	32	D	33	A	34	D												

Limits & Derivatives

2 questions

Q1. *CUSAT CAT 2023 - Shift 1*

lim x→0 2x

(A) exists and equal to 1	**(B)** exists and equal to –1
(C) exists	**(D)** does not exist 1 2 –0.1 a –1

Q2. *CUSAT CAT 2025 - Shift 4*

One of possible values of lim x→1 x–1

(A) –1	**(B)** 0 1
(C) 2	**(D)** 1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	C																

Matrices & Determinants

32 questions

Q1. *CUSAT CAT 2023 - Shift 2*

If A and B are two square matrices such that B=–A BA, then (A+B) is equal to

(A) 0	**(B)** A 2 +B 2
(C) A 2 +2AB+B 2	**(D)** A+B

Q2. *CUSAT CAT 2023 - Shift 3*

The determinant –b c bα+c =0, if aα+b bα+c 0

(A) b 3 =ac	**(B)** ab=c 3
(C) x–α is a factor of bx 2 +2ax+c	**(D)** x–α is a factor of ax 2 +2bx+c 2 4 6

Q3. *CUSAT CAT 2023 - Shift 3*

The maximum number of different elements required to form a symmetric matrix of order 12 is

(A) 72	**(B)** 78
(C) 75	**(D)** 70

Q4. *CUSAT CAT 2023 - Shift 3*

If A and B are two distinct matrices such that A = B and A B = B A, 3 3 then det (A + B) is equal to

(A) 1	**(B)** 2
(C) 0	**(D)** –1

Q5. *CUSAT CAT 2023 - Shift 3*

Suppose a matrix A satisfies A – 5A + 7I = 0 . If A = aA + bI , then a =

(A) 1265	**(B)** 5299
(C) 1259	**(D)** 5432 1 50

Q6. *CUSAT CAT 2023 - Shift 4*

If A is a matrix of order 3, then det (kA) is 3

(A) k det	**(B)** 3 det
(C) det	**(D)** 3k det

Q7. *CUSAT CAT 2023 - Shift 4*

If the rank of the matrix 0 λ –1 is 2, then λ is –1 0 λ

(A) 2	**(B)** 3
(C) 1	**(D)** any real number 3 1 8 n

Q8. *CUSAT CAT 2023 - Shift 4*

If A is a square matrix of order n, then | adj A | is 2

(A) A n	**(B)** A n–1
(C) A	**(D)** A n n n

Q9. *CUSAT CAT 2023 - Shift 5*

The rank of the diagonal matrix 0 is 4 0

(A) 1	**(B)** 3
(C) 5	**(D)** 2 1 –4 20

Q10. *CUSAT CAT 2023 - Shift 5*

If A and B are two skew symmetric matrices of order n, then

(A) AB is a skew symmetric matrix	**(B)** AB is a symmetric matrix
(C) AB is a symmetric matrix if A and B commute	**(D)** AB is a singular matrix

Q11. *CUSAT CAT 2023 - Shift 5*

If ω is a complex cube root of unity, the matrix A=ω 2 ω 1 is a ω 1 ω 2

(A) singular matrix	**(B)** non-singular matrix
(C) skew symmetric matrix	**(D)** symmetric matrix sinx –sinx

Q12. *CUSAT CAT 2024 - Shift 1*

The rank of the matrix 2 3 4 is 3 5 7

(A) 3	**(B)** 2
(C) 1	**(D)** 0

Q13. *CUSAT CAT 2024 - Shift 2*

Let A be non singular square matrix of order 5 × 5 . Then |adj A| is equal to

(A) |A|	**(B)** |A| 2
(C) |A| 3	**(D)** |A| 4

Q14. *CUSAT CAT 2024 - Shift 2*

Let M = :x∈■–{0}. If X,E∈M such that X ·E = X for all X ∈M, x x where · is matrix multiplication. Then E is 1 1

(A) 1 1 1 0	**(B)** 0 1 1 1
(C) 2 2 1 1 2 2	**(D)** 0 1 1 0

Q15. *CUSAT CAT 2024 - Shift 2*

The matrix A= satisfies the equation 1 2 2

(A) A – I = 2A 2	**(B)** A – 2A + I = 0 2
(C) A – 2I = 0 2	**(D)** A – 4A + I = 0 1

Q16. *CUSAT CAT 2024 - Shift 4*

Number of all possible matrices of order 3 × 3 with each entry as 1 or 2 is

(A) 27	**(B)** 256
(C) 81	**(D)** 512

Q17. *CUSAT CAT 2024 - Shift 5*

Let B be a matrix such that B = B and let A = I – B. Then 2

(A) A = A 2	**(B)** A = I
(C) AB ≠ 0	**(D)** BA ≠ 0

Q18. *CUSAT CAT 2024 - Shift 5*

The rank of a 3 × 3 matrix Z = XY, found by multiplying a non-zero column matrix X of size 3 × 1 and a non-zero row matrix Y of size 1 × 3, is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q19. *CUSAT CAT 2024 - Shift 5*

If A is symmetric as well as skew symmetric matrix, then A is

(A) diagonal	**(B)** null
(C) upper triangular	**(D)** lower triangular

Q20. *CUSAT CAT 2024 - Shift 5*

If A is a 3 × 3 matrix such that A'A= I and A =1, then the value of A–I is

(A) 1	**(B)** 2
(C) 0	**(D)** 3

Q21. *CUSAT CAT 2025 - Shift 1*

The number of possible orders of a matrix with 10 elements is

(A) 1	**(B)** 2
(C) 3	**(D)** 4 x–1 x x+2

Q22.

CUSAT CAT 2025 - Shift 1

Let M denote set of all matrices of form , x∈■\{0}. Then the identity x x element of M under the matrix multiplication is 1 0

(A) 0 1 1 1	**(B)** 1 1 1 1 2 2
(C) 1 1 2 2 1 1	**(D)** 0 1

Q23.

CUSAT CAT 2025 - Shift 2

In 4×4 matrix, sum of each row, column and both the main diagonal is α, then the sum of four diagonal corner elements

(A) α	**(B)** may not be α
(C) its never equal to α	**(D)** α 2

Q24.

CUSAT CAT 2025 - Shift 3

If a non zero matrix A of order 5 × 5 satisfies the equation A –5A=0, then the matrix A is 5 5

(A) A= 5 5 5 0	**(B)** A= 0 5 5 0
(C) A= 0 0 0 5	**(D)** A= 5 0

Q25.

CUSAT CAT 2025 - Shift 3

Let M be the set of all n×n matrices such that each matrix is both symmetric and skew symmetric. Then M is

(A) n	**(B)** 1
(C) n 2	**(D)** ∞

Q26.

CUSAT CAT 2025 - Shift 3

A set of matrices of same order under matrix multiplication can never be

(A) a monoid	**(B)** a non abelian group
(C) a semi group	**(D)** an abelian group 3n 4π 4π 1+a

Q27.

CUSAT CAT 2025 - Shift 3

If the matrix commutes with the matrix , then c d 1 1

(A) a =0, b=c	**(B)** b=0, c=d
(C) b=0, d =a	**(D)** d =0, a =b –x x 2

Q28.

CUSAT CAT 2025 - Shift 3

The values of x for which the matrix 2 x –x will be invertible are x –2 –x

(A) –2≤ x≤2	**(B)** x≠2 and x≠–2
(C) x≥2	**(D)** x≤–2 { 2 }

Q29.

CUSAT CAT 2025 - Shift 3

Which of the following is correct?

(A) Skew symmetric matrix of even order is always singular	**(B)** Skew symmetric matrix of odd order is always non-singular
(C) Skew symmetric matrix of odd order is always singular	**(D)** None of the above b

Q30.

CUSAT CAT 2025 - Shift 3

If A= , the determinant of the matrix A –2A –A is 3 1

(A) –175	**(B)** 2014
(C) 2016	**(D)** –25

Q31.

CUSAT CAT 2025 - Shift 3

If the matrix A is both symmetric and skew-symmetric, then

(A) A is a diagonal matrix	**(B)** A is a zero matrix
(C) A is a square matrix	**(D)** None of the above 2

Q32.

CUSAT CAT 2025 - Shift 4

If A – A+I =0, then the inverse of the matrix A is

(A) A+I	**(B)** A
(C) A–I	**(D)** I –A

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	D	3	B	4	C	5	A	6	A	7	C	8	C	9	B	10	C
11	A	12	A	13	D	14	C	15	D	16	D	17	A	18	B	19	B	20	C
21	D	22	C	23	A	24	B	25	B	26	D	27	C	28	B	29	C	30	D
31	B	32	D																

Continuity & Differentiability

6 questions

Q1. *CUSAT CAT 2023 - Shift 3*

If f(x)= (π–2x) is continuous at x= π then k = 2 π k, x= 2

(A) 0 1	**(B)** 2
(C) 1	**(D)** –1

Q2. *CUSAT CAT 2023 - Shift 3*

If 4x+3 y = y, then y as a function of x is

(A) not continuous at x = 0	**(B)** not defined for all real x dy 1
(C) = for x<0 dx 2	**(D)** differentiable at x=0

Q3. *CUSAT CAT 2023 - Shift 4*

The value of the k which makes f(x)= x continuous at x=0 is k, x=0

(A) 1	**(B)** 2
(C) 5	**(D)** 6

Q4. *CUSAT CAT 2023 - Shift 4*

Let f be defined on [–5, 5] as f(x)= . Then –x if (x) is irrational

(A) f(x)is continuous at every x, except x = 0	**(B)** f(x)is discontinuous at every x, except x = 0
(C) f(x)is continuous everywhere	**(D)** f(x)is discontinuous everywhere

Q5. *CUSAT CAT 2024 - Shift 4*

The function f(x)= , when x≠0; f(x)=1 when x=0, is x

(A) continuous at x=0	**(B)** not continuous at x=0
(C) both continuous and differentiable atx=0	**(D)** continuous but not differentiable atx=0 1+x –1

Q6. *CUSAT CAT 2025 - Shift 2*

Let f and g be continuous functions at a point x . Then which of the following is 0 NOT necessarily true?

(A) f +g is continuous at x 0	**(B)** f –g is continuous at x 0
(C) f ·g is continuous at x 0 f	**(D)** is continuous at x 0 g

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	C	3	C	4	B	5	B	6	D								

Applications of Derivatives

24 questions

Q1. *CUSAT CAT 2023 - Shift 1*

The minimum value of P=bcx+cay+abz when xyz =abc, is

(A) 3abc	**(B)** 6abc
(C) abc	**(D)** 4abc 2 2

Q2. *CUSAT CAT 2023 - Shift 1*

The maximum value of S =5+4 +4 +... is equal to 3 3

(A) 30	**(B)** 40
(C) 55	**(D)** 60

Q3. *CUSAT CAT 2023 - Shift 1*

The number of solutions of the equation a f(x) +g(x)=0, a>0, g(x)≠0 and has 1 minimum value of , is 2

(A) one	**(B)** two
(C) zero	**(D)** infinitely many 1

Q4. *CUSAT CAT 2023 - Shift 1*

The maximum value of (7–x) (2+x) , when x lies between –2 and 7, is

(A) (4 4 5 5)6	**(B)** (4 4 5 4)9
(C) (4 4 5 5)9	**(D)** (4 5 5 4)9

Q5. *CUSAT CAT 2023 - Shift 2*

If n is an integer between 0 and 21, then the minimum value of n!(21–n)! is

(A) 9!21!	**(B)** 10!11!
(C) 20!	**(D)** 21! 10 (5)

Q6. *CUSAT CAT 2023 - Shift 3*

On the interval [0,1], the function x 25 ·(1–x) 75 takes its maximum value at the point 1

(A) 4	**(B)** 0 1
(C) 2 1	**(D)** 3 x2–x x2–x

Q7. *CUSAT CAT 2023 - Shift 3*

If a >0 for i =1,2,...,n and a a ...a =1, then the minimum value of i 1 2 n (1+a)(1+a) ...(1+a) is 1 2 n

(A) 2 n 2	**(B)** 2 n
(C) 2 2n	**(D)** 1

Q8. *CUSAT CAT 2023 - Shift 3*

If a, b, c are positive numbers in A.P. such that their product is 64, then the minimum value of b is

(A) 6	**(B)** 4
(C) 2	**(D)** 1 6 7 8 9 10 31

Q9. *CUSAT CAT 2024 - Shift 1*

The rate of change of the area of a circle with respect to its radius r when r = 6 is

(A) 10π	**(B)** 12π
(C) 8π	**(D)** 11π n n x –2

Q10. *CUSAT CAT 2024 - Shift 1*

If a, b, c and d are sides of a quadrilateral, then the minimum value of is 2 d

(A) 1 1	**(B)** 2 1
(C) 3 1	**(D)** 4

Q11. *CUSAT CAT 2024 - Shift 2*

Let f (x) be differentiable function for all x with f '(x) ≤ 17 and f (2) = 7 . Then the maximum value of f (7) is

(A) 82	**(B)** 92
(C) 102	**(D)** 112

Q12. *CUSAT CAT 2024 - Shift 2*

For real positive numbers a, b, c, the minimum value of + + is a b c

(A) 1	**(B)** 2
(C) 4	**(D)** 6 2 ∑ 5 n 1 2

Q13. *CUSAT CAT 2024 - Shift 4*

Let xy =c . Then the minimum value of a x+b y is

(A) 2abc	**(B)** 2
(C) a 2 b 2 c 2	**(D)** a 2 b 2 c 4

Q14. *CUSAT CAT 2024 - Shift 5*

The minimum value of 4 +4 is

(A) 4	**(B)** 5
(C) 1	**(D)** 8

Q15. *CUSAT CAT 2025 - Shift 1*

A function f is such that f '(a)= f "(a)= f "'(a)=...= f (a)=0 and f has a local maximum value b at x =a if f(x) is

(A) (x–a) 2n+2	**(B)** b–1–(x+1–a) 2n–1
(C) b–(x–a) 2n+2	**(D)** (x–a) 2n+2 –b x4 3 2

Q16. *CUSAT CAT 2025 - Shift 1*

The minimum value of e –x +x is

(A) e	**(B)** e 2
(C) 1 1	**(D)** e dy

Q17. *CUSAT CAT 2025 - Shift 1*

The maximum value of is x

(A) e	**(B)** e e 1
(C) ee 1	**(D)** e

Q18. *CUSAT CAT 2025 - Shift 2*

Subject to the conditions 0≤ x≤10; 0≤ y≤5, the minimum value of the function 4x–5y+10=0 is

(A) 5	**(B)** –15
(C) 10	**(D)** –2 1 1 1 1

Q19. *CUSAT CAT 2025 - Shift 2*

The maximum value of z when z satisfies the condition z+ =2 is z

(A) 3–1	**(B)** 3+1
(C) 3	**(D)** 2 + 3

Q20. *CUSAT CAT 2025 - Shift 3*

The maximum or minimum value of on R is 2 4x +2x+1 4

(A) 3 2	**(B)** 3
(C) 3 3	**(D)** 4

Q21. *CUSAT CAT 2025 - Shift 3*

At present, a firm is manufacturing 2000 items. It is estimated that the rate of change dP of production P w.r.t. additional number of workers x is given by =100–12 x. If dx the firm employs 25 more workers, then the new level of production of items is

(A) 2500	**(B)** 3000
(C) 3500	**(D)** 4500 1

Q22.

CUSAT CAT 2025 - Shift 4

The absolute maximum value of the function f(x)=4x +5x on [1, 2] is

(A) –1	**(B)** 2
(C) 0	**(D)** 1

Q23.

CUSAT CAT 2025 - Shift 4

The minimum value of f(x)=(19–x)e is

(A) e 1	**(B)** e 1
(C) – e 1	**(D)** – 2 e

Q24.

CUSAT CAT 2025 - Shift 4

If Z – =2,then the maximum value of z is equal to z

(A) 5+1	**(B)** 2
(C) 2+ 2	**(D)** 3+1 x x 3 4

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	A	2	B	3	C	4	C	5	B	6	A	7	B	8	B	9	B	10	C
11	B	12	D	13	A	14	A	15	C	16	C	17	C	18	B	19	B	20	A
21	C	22	D	23	C	24	A												

Integrals

56 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Let f(x)=max {x+ x , x–[x]} where [x] denotes the greatest integer less than or 2 equal to x. Then ∫ f(x)dx= –2

(A) 5	**(B)** 3
(C) –2	**(D)** 1 dy

Q2. *CUSAT CAT 2023 - Shift 1*

Let f(x)= . Then ∫ f(x)dx is equal to 1 2 –a

(A) 0 1	**(B)** 2
(C) 3 1	**(D)** – 2 cosθ

Q3. *CUSAT CAT 2023 - Shift 1*

If ∫ dx= f(x)+c, then ∫ dx is equal to 5 5 x+x x+x

(A) log x + f(x)+c	**(B)** log x – f(x)+c
(C) xf(x)+c	**(D)** log f (x)+x+c

Q4. *CUSAT CAT 2023 - Shift 2*

If f(x)=log x ,x≠0 then f '(x) equals 1

(A) x 1	**(B)** – x 1
(C) – x 1	**(D)** x dx

Q5. *CUSAT CAT 2023 - Shift 2*

The value of the integral ∫ dx where C is a closed curve given by C(2z–1)(z–3) z =1 is

(A) πi	**(B)** –πi πi
(C) 5 2πi	**(D)** 5 x+1 y z–3

Q6. *CUSAT CAT 2023 - Shift 3*

The differential equation of all straight lines touching the circle x + y =a is 2 2 dy dy

(A) y– =a 21+ dx dx 2 2 dy dy	**(B)** y–x =a 21+ dx dx dy dy 2
(C) y–x =a 1+ dx dx dy dy 2	**(D)** y– =a 1– dx dx

Q7. *CUSAT CAT 2023 - Shift 3*

If ∫e x2 dx=a, then the value of ∫ log e xdx is 1 e

(A) e4 –e	**(B)** e4–a
(C) 2e4–a	**(D)** 2e4–e–a 2 2 2 2

Q8. *CUSAT CAT 2023 - Shift 3*

The value of ∫ dz, where j = –1 and C is described by x 2 + y 2 =16, is C0.5z–1.5j

(A) 4πj	**(B)** –4πj
(C) 2πj	**(D)** –2πj

Q9. *CUSAT CAT 2023 - Shift 4*

Let g(x) be a function satisfying g'(x)= g(x) with g(0) = 1 and f(x) be a function 1 that satisfies f(x)+g(x)= x 2 . Then the value of the integral ∫ f(x) g(x)dx is 0 e–7

(A) 4 e–3	**(B)** 2 2 e +3
(C) e– 2 2 e +3	**(D)** e+ 2

Q10. *CUSAT CAT 2023 - Shift 4*

If I = ∫ dx and I = ∫ dx, then 1 2 logx x e 1

(A) I =2I 1 2	**(B)** 2I = I 1 2
(C) I +I =0 1 2	**(D)** I –I =0 1 2 2

Q11. *CUSAT CAT 2023 - Shift 4*

If f(x)= A·2 x +B and such that f '(x)=2 and ∫ f(x)dx=7, then B is 0 2 1

(A) 1– log e 2 7 ( ) 2	**(B)** 1– log e 3 2 7 ( ) 2
(C) 1+ log e 3 2 7	**(D)** 1– (log e) 2 3 x 3 2 T T

Q12. *CUSAT CAT 2023 - Shift 4*

If = x+∫y(x) dx, y(0) = 1, then ∫xy(x)dxis dx 0 0 13

(A) 36 19	**(B)** 48 7
(C) 36 101	**(D)** 72 2

Q13. *CUSAT CAT 2023 - Shift 4*

The graph of f(x)= 4–(x–2) is

(A) —	**(B)** —
(C) —	**(D)** ∞ dx

Q14. *CUSAT CAT 2023 - Shift 4*

Let f(x) be a function satisfying f '(x)= f(x) and f(0) = 2. Then ∫ dx is 3+4f(x) equal to 1 (x)

(A) log 3+8e +C 4 1 (x)	**(B)** log 3+5e +C 2 1 (2x)
(C) log 3+4e +C 4 1	**(D)** C+ log(xe) 3 2x 2

Q15. *CUSAT CAT 2023 - Shift 5*

The value of the integral ∫ dx is 1+x 0

(A) 3 log 2	**(B)** 2 log 2 – 1
(C) 3 log 2 + 1	**(D)** 3 log 2 – 1

Q16. *CUSAT CAT 2023 - Shift 5*

If ∫ dx= , then the value of (x 2 +a 2)(x 2 +b 2)(x 2 +c 2) 2(a+b)(b+c)(c+a) 0 ∞ 1 ∫ dx is (2)(2) x +4 x +9 0 π

(A) 60 π	**(B)** 16 π
(C) 12 π	**(D)** 5 (2) (2) log 1+x+x +log 1–x+x

Q17. *CUSAT CAT 2023 - Shift 5*

The difference between the greatest and the least value of the function x F(x)= ∫(t+1)dt on [2, 3] is 0

(A) 3	**(B)** 2 7
(C) 2 3	**(D)** 2 x , x<0

Q18. *CUSAT CAT 2023 - Shift 5*

A particular solution of log =3x+4y, y(0)=0 is dx

(A) e 3x +3e –4y =4	**(B)** 4e 3x –e –4y =3
(C) 3e 3x +5e 4y =7	**(D)** 4e 3x +3e –4y =7 1 1 1

Q19. *CUSAT CAT 2023 - Shift 5*

The function f(x),where f(x)= ∫e x (x–1)(x–2)dx decreases in the interval

(A) (–∞,–2)	**(B)** (–2,–1)
(C) (1, 2)	**(D)** (2, ∞) x–1 y+1 z+10

Q20. *CUSAT CAT 2023 - Shift 5*

If f(x)= lim 2x+4x 3 +...+2nx 2n–1 (0< x<1), then ∫ f(x)dx is equal to n→∞

(A) – 1–x 2 1	**(B)** 2 1–x 1
(C) 2 x –1 1	**(D)** 2 1–x

Q21. *CUSAT CAT 2023 - Shift 5*

∫ x–1dx= –2

(A) 0	**(B)** 5
(C) 1	**(D)** 4

Q22. *CUSAT CAT 2023 - Shift 5*

If ∫ dx=log , then A is equal to e x +e –x 1+ 2 0

(A) e 2 +1	**(B)** e 2 +1
(C) e 2	**(D)** e 2 –1

Q23. *CUSAT CAT 2023 - Shift 5*

If f(x)=(1+tanx)1+tan –x and g(x) is a function with domain R, then 4 1 ∫x 3(g f)(x)dx is 0 1 π

(A) g 2 2 1	**(B)** g(2) 4 1
(C) g(1) 4	**(D)** g(1)

Q24. *CUSAT CAT 2023 - Shift 5*

∫ e –x2 dx=k,then ∫ e –x2 dx = –1 –1

(A) –k –k	**(B)** 2 k
(C) 2	**(D)** 2k 1 x x –1( ) π π

Q25. *CUSAT CAT 2023 - Shift 5*

If ∫ dx= Ax+Blog(x–α)+c, then the value of (A, B) is sin(x–α)

(A) (–sinα,cosα)	**(B)** (cosα,sinα)
(C) (sinα,cosα)	**(D)** (–cosα,sinα) 1

Q26. *CUSAT CAT 2024 - Shift 1*

∫cos 6 x dx= 0 3π

(A) 32 5π	**(B)** 32 7π
(C) 32 π	**(D)** 32 2

Q27. *CUSAT CAT 2024 - Shift 1*

∫ e dx= –log2

(A) 0 1	**(B)** 2
(C) 1	**(D)** –1

Q28. *CUSAT CAT 2024 - Shift 1*

The value of ∫ x+ 12x–36 + x– 12x–36 dx is equal to 3

(A) 4 3	**(B)** 6 3
(C) 12 3	**(D)** 2 3 Subject Name: 101 B TECH 10 MAY S1 2024

Q29. *CUSAT CAT 2024 - Shift 2*

∫xtan –1 xdx= 0

(A) 0	**(B)** π–1 π 1
(C) – 4 2 π	**(D)** e

Q30. *CUSAT CAT 2024 - Shift 2*

A solution of the differential equation –x + y =0 is dx dx

(A) y = 0	**(B)** y = 2x
(C) y = 2x – 4	**(D)** y = 2x 2 – 4 (x) x –x

Q31. *CUSAT CAT 2024 - Shift 3*

If y = e + e + e +...∞ then is dx 1

(A) x 2 e	**(B)** e x x e
(C) 2y–1 x e	**(D)** 2y π

Q32. *CUSAT CAT 2024 - Shift 3*

∫xe x dx= x

(A) (x–1)e	**(B)** x 2 e x 2 x x e
(C) +C 2 x	**(D)** (x–1)e +C 2+3i

Q33. *CUSAT CAT 2024 - Shift 3*

The graph of the polynomial P(x)= x +ax +bx +cx +dx+e has five distinct intercepts, one of which is at (0, 0) . Then the coefficient that cannot be zero is

(A) a	**(B)** b
(C) c	**(D)** d 3

Q34. *CUSAT CAT 2024 - Shift 3*

lim is equal to x→0 x 4

(A) 0	**(B)** 1 1
(C) 6 1	**(D)** – 6 2 dx

Q35. *CUSAT CAT 2024 - Shift 3*

The value of ∫ is equal to 4 x(1+x) 1 1 17

(A) log 4 32 1 32	**(B)** log 4 17 17
(C) log 2 1 17	**(D)** log 4 2

Q36.

CUSAT CAT 2024 - Shift 4

∫ = (x+2) x+3 x+3–1

(A) log x+3+1 x+3+1	**(B)** log x+3–1 x+2–1
(C) log x+2+1 x+2+1	**(D)** log x+2–1 2 2

Q37.

CUSAT CAT 2024 - Shift 4

Let a, b, c, d, e, f be independent of x. If 3 x –4x x+3 x–2 x–2 5x x–1 =ax 5 +bx 4 +cx 3 +dx 2 +ex+ f is true, then the x–3 x+2 4x value of f is

(A) 0	**(B)** 15
(C) 17	**(D)** 19 y

Q38.

CUSAT CAT 2024 - Shift 4

If x y =100, then is equal to dx y(y+xlogx)

(A) – x(ylogx+x) y(x+ y)logx	**(B)** – x(xlog y+ y) y
(C) – x x	**(D)** – y

Q39.

CUSAT CAT 2024 - Shift 4

∫ [x] is equal to –2

(A) 1	**(B)** 2
(C) 4	**(D)** 5

Q40.

CUSAT CAT 2024 - Shift 5

Let o be the origin and p(α,β) where α is the least value of and x2 +1 1 β= ∫ (x–[x]) dx. Then op = –1

(A) 0	**(B)** 1
(C) 2	**(D)** 2

Q41.

CUSAT CAT 2024 - Shift 5

Let y =1+ 1+ ...1+ and x≠0. Then at x=–1 is x x x dx

(A) (n–1)!	**(B)** (–1)n(n–1)!
(C) (–1)nn!	**(D)** n!

Q42.

CUSAT CAT 2024 - Shift 5

lim ∫ dx= x→0dx x2

(A) 0 1	**(B)** 2
(C) 1	**(D)** ∞

Q43. *CUSAT CAT 2024 - Shift 5*

The value of ∫(x–1)(x–2)(x–3)dx is 1 1

(A) 3 9	**(B)** 4 1
(C) 2 9	**(D)** 2 x

Q44. *CUSAT CAT 2024 - Shift 5*

If ∫ f(x)dx= f(x), then ∫(f(x))2 dx is equal to

(A) 1 (f(x))2 2	**(B)** (f(x))2
(C) 1 (f(x))3 3	**(D)** 0

Q45. *CUSAT CAT 2025 - Shift 1*

The value of ∫x(1+x) 88 dx is 0

(A) 0 1	**(B)** 8010 1
(C) – 8010	**(D)** 1

Q46. *CUSAT CAT 2025 - Shift 1*

∫ dx= 2 1+x ( )16 2 x+ 1+x

(A) +c 10 1	**(B)** +c 2 15 1+x +x 15
(C) +c 2 1+x –x ( )15 2 x+ 1+x	**(D)** +c 15

Q47. *CUSAT CAT 2025 - Shift 1*

If ∫ dx= A∫ dx+B∫ ; then 2A+B is 2 2 2 5–4x+x 5–4x+x 5–4x+x

(A) 0	**(B)** 1
(C) 2	**(D)** –1 2n, n=2,4,6,8,...

Q48. *CUSAT CAT 2025 - Shift 2*

If x y =100, then is equal to dx y(x+ y)logx

(A) – x(xlog y+ y) y(y+xlogx)	**(B)** – x(ylogx+x) y
(C) – x x	**(D)** – y 1

Q49. *CUSAT CAT 2025 - Shift 2*

∫ [x]dx is equal to –2

(A) 1	**(B)** 2
(C) 4	**(D)** 5 (2)

Q50. *CUSAT CAT 2025 - Shift 2*

Let f :R→R be a differential function with f(2)=6 and f '(2)= . Then 48 f(x) 3 4t lim ∫ dt = x→2 x–2 0

(A) 24	**(B)** 36
(C) 12	**(D)** 18

Q51. *CUSAT CAT 2025 - Shift 2*

The value of ∫ (x 3 +sinx) dx is –π

(A) 0	**(B)** π
(C) 2π	**(D)** π3

Q52. *CUSAT CAT 2025 - Shift 2*

The value of definite integral ∫ dx is 2 1+x 0 3 π

(A) 72 2 π	**(B)** 32 3 π
(C) 4 3 π	**(D)** 8

Q53. *CUSAT CAT 2025 - Shift 2*

If x =e , then is dx 1

(A) x+1 1–x	**(B)** x x–1
(C) x+1 1	**(D)** x

Q54. *CUSAT CAT 2025 - Shift 3*

The integral of partial fraction ∫ is (x+1)(x+2) x+1

(A) log +c x+2 x+2	**(B)** log +c x+1 x–1
(C) log +c x+2 x+1	**(D)** log +c x–2 1 1 1

Q55. *CUSAT CAT 2025 - Shift 3*

The degree of the differential equation dx 2 = 1+ dx is

(A) 2	**(B)** 3 1
(C) 2	**(D)** 6

Q56. *CUSAT CAT 2025 - Shift 4*

The value of∫e x2 dx 0

(A) > e	**(B)** < e
(C) >2	**(D)** <1

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key

1	A	2	A	3	B	4	D	5	D	6	B	7	D	8	B	9	C	10	D
11	B	12	D	13	D	14	A	15	D	16	A	17	C	18	D	19	C	20	D
21	B	22	B	23	B	24	C	25	B	26	B	27	C	28	B	29	C	30	C
31	C	32	D	33	D	34	C	35	B	36	A	37	C	38	A	39	C	40	C
41	B	42	B	43	C	44	A	45	B	46	D	47	A	48	B	49	C	50	D
51	A	52	B	53	B	54	A	55	D	56	B								

Differential Equations

4 questions

Q1. *CUSAT CAT 2023 - Shift 3*

The order of differential equation x=1+ + + +... is dx 2!dx 3!dx n!dx

(A) 3	**(B)** 1
(C) n	**(D)** not defined ( )

Q2. *CUSAT CAT 2023 - Shift 4*

The differential equation of the family of circles with fixed radius 5 units and center on the line y = 2 is

(A) (x–2)(y') 2 =25–(y–2) 2	**(B)** (y–2)(y') 2 =25–(y–2) 2
(C) (y–2) 2 (y') 2 =25–(y–2) 2	**(D)** (x–2) 2 (y') 2 =25–(y–2) 2

Q3. *CUSAT CAT 2024 - Shift 3*

Let y = y(t) be a solution of the differential equation +ay =ce –bt where dt a >0, b>0 and c>0. Then lim y(t) t→∞

(A) is 0	**(B)** is 1
(C) is –1	**(D)** does not exist

Q4. *CUSAT CAT 2025 - Shift 3*

The integrating factor of the differential equation (ylog y)dx=(log y–x)dy is 1

(A) logy	**(B)** log(log y)
(C) 1+log y	**(D)** log y

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	B	2	C	3	A	4	D												

Vector Algebra & 3D Geometry

18 questions

Q1. *CUSAT CAT 2023 - Shift 4*

Let a = i + j +k ; b = i – j +2k and c = xi +(x–2)j –k . If the vector c lies in the plane of a and b , then x equals to

(A) 0	**(B)** 1
(C) –4	**(D)** –2 (2)80

Q2. *CUSAT CAT 2023 - Shift 5*

The points with position vectors 60i +3j, 40i –8j, ai –52j are collinear if

(A) a = –40	**(B)** a = 40
(C) a = 20	**(D)** a = –20 1 2–x

Q3. *CUSAT CAT 2024 - Shift 1*

Let a and b be two vectors such that a·b = 3 a×b . Then θ is π

(A) 6 π	**(B)** 4 π
(C) 3 π	**(D)** 2

Q4. *CUSAT CAT 2024 - Shift 1*

Which of the following pair of vectors are in same direction? 1 1 1

(A) a =2i + j +k, b = i + j + k 3 3 3 1 1 1	**(B)** a =i + j +k, b = i + j + k 3 3 3 2 1 2
(C) a =i + j +k, b = i + j + k 3 3 3 1 2 1	**(D)** a =i + j +k, b = i + j + k 3 3 3 f

Q5. *CUSAT CAT 2024 - Shift 2*

The points A, B, C with position vectors 2i – j –k, i –3j –5k and 3i –4j –5k are the vertices of

(A) a triangle but not a right angled triangle	**(B)** a right angled triangle
(C) an equilateral triangle	**(D)** an isosceles triangle

Q6. *CUSAT CAT 2024 - Shift 2*

If B =2iˆ+ ˆj–k ˆ and C =iˆ+3k ˆ . If A is a unit vector, then the maximum value of the scalar triple product ABC is

(A) –1	**(B)** 10+ 6
(C) 60	**(D)** 59

Q7. *CUSAT CAT 2024 - Shift 3*

If u =2i +2j –k and v =6i –3j +2k then a unit vector perpendicular to both u and v is

(A) (i –10j –8k) 1	**(B)** (i –2j –18k) 17 1
(C) (7i –10j –18k) 473 1	**(D)** (i –10j –18k) 425 2 2 2 2

Q8. *CUSAT CAT 2024 - Shift 3*

The unit vector in the direction of sum of vectors 3i +5j –4k and i –2j +3k is 4 3 1

(A) i + j + k 26 26 26 –4 3 1	**(B)** i + j + k 26 26 26 4 –3 1
(C) i + j + k 26 26 26 4 3 1	**(D)** i + j – k 26 26 26 3 2

Q9. *CUSAT CAT 2024 - Shift 3*

If the four points with position vectors 3i +6j +9k, i +2j +3k, 2i +3j +k and 4i +6j +λk are coplanar, then the value of λ is

(A) 1	**(B)** 0
(C) –1	**(D)** 2

Q10. *CUSAT CAT 2024 - Shift 3*

Let a and b be two vectors such that a = 14, b = 6 and a×b = 48. Then, 2 ( ) a·b is equal to

(A) 36	**(B)** 6
(C) 12	**(D)** 18

Q11. *CUSAT CAT 2024 - Shift 4*

A vector which makes equal angles with the vectors iˆ–2ˆj+2k ˆ , –4iˆ–3k ˆ , ˆj is 3 5

(A) 5iˆ+ ˆj+5k ˆ	**(B)** –5iˆ+ ˆj+5k ˆ
(C) 5iˆ– ˆj–k ˆ	**(D)** 5iˆ+ ˆj–5k ˆ

Q12. *CUSAT CAT 2024 - Shift 4*

Let a = piˆ+ ˆj–k ˆ and b =2iˆ+ ˆj– pk ˆ , p >0. If the projection of a×b on the vector –iˆ+2ˆj–2k ˆ is 30, then p is equal to

(A) 8 15	**(B)** 2
(C) 7 13	**(D)** 2 1 1

Q13. *CUSAT CAT 2024 - Shift 5*

The unit vector in ZOX plane makes an angle 60° and 45° with a = ˆj–k ˆ and b =2iˆ+2ˆj–k ˆ respectively. Then the unit vector is 1 1

(A) iˆ+ k ˆ 2 2 1 4 1	**(B)** iˆ+ ˆj+ k ˆ 3 2 3 2 3 2 1 1
(C) iˆ– k ˆ 2 2 1 1	**(D)** ˆj– k ˆ 2 2 z–1

Q14. *CUSAT CAT 2025 - Shift 1*

The angle made by the vector i + j + 2k with z-axis is

(A) 1°	**(B)** 30°
(C) 45°	**(D)** 60° 1, x is rational

Q15.

CUSAT CAT 2025 - Shift 1

The number of distinct real values of λ , for which the vectors –λ 2 iˆ+ ˆj+k ˆ , iˆ–λ 2ˆj+k ˆ and iˆ+ ˆj–λ 2 k ˆ are coplanar, is

(A) 0	**(B)** 1
(C) 2	**(D)** 3 z–2

Q16.

CUSAT CAT 2025 - Shift 2

If a,b,c are three non-coplanar mutually perpendicular unit vectors, then a b c is

(A) 2	**(B)** 0
(C) 1	**(D)** 3 2

Q17.

CUSAT CAT 2025 - Shift 3

If the vectors a =iˆ+aˆj+a 2 k ˆ , b =iˆ+bˆj+b 2 k ˆ , c =iˆ+cˆj+c 2 k ˆ are three non-coplanar 2 3 a a 1+a 2 3 vectors and b b 1+b =0, then the value of abc is 2 3 c c 1+c

(A) 0	**(B)** 1
(C) 2	**(D)** –1

Q18.

CUSAT CAT 2025 - Shift 4

The value of λ for which the vectors 3iˆ–6ˆj+k ˆ and 2iˆ–4ˆj+λk ˆ are parallel is 3

(A) 2 2	**(B)** 3 2
(C) 5 5	**(D)** 2

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	A	4	B	5	B	6	D	7	D	8	D	9	D	10	A
11	B	12	C	13	C	14	C	15	C	16	C	17	D	18	B				

Probability & Statistics

52 questions

Q1. *CUSAT CAT 2023 - Shift 1*

Three numbers are chosen at random without replacement from 1,2,3,...,10. The probability that the minimum of the chosen numbers is 4 or their maximum 8, is 1

(A) 40 7	**(B)** 40 3
(C) 40 11	**(D)** 40 x

Q2. *CUSAT CAT 2023 - Shift 1*

If the probability density function of a random variable X is f(x)= , in 0≤ x≤2, 2 then P(X >1.5 X >1) is 7

(A) 16 3	**(B)** 4 7
(C) 12 21	**(D)** 64 2π 4π 6π

Q3. *CUSAT CAT 2023 - Shift 1*

If , , are the probability of three mutually exclusive and exhaustive 2 3 6 events, then the set of all values of ‘p’ is in 1 1

(A) – , 4 3	**(B)** (0, 1) 1
(C) 0, 3 1	**(D)** – ,0 4

Q4. *CUSAT CAT 2023 - Shift 1*

A fair coin is tossed n times. If the probability that head occurs 6 times is equal to the probability that head occurs 8 times, then value of n is

(A) 24	**(B)** 48
(C) 14	**(D)** 16 3 2

Q5. *CUSAT CAT 2023 - Shift 2*

One ticket is selected at random from 50 tickets numbered 00,01,..,49. Then the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals 1

(A) 7 1	**(B)** 14 5
(C) 14 5	**(D)** 7

Q6. *CUSAT CAT 2023 - Shift 2*

A number n is chosen at random from {1,2,3,...,1000}. The probability that n is a number that bears remainder 1 when divided by 7 is 71

(A) 500 143	**(B)** 1000 72
(C) 500 71	**(D)** 1000

Q7. *CUSAT CAT 2023 - Shift 2*

A pair of fair dice is rolled together till a sum of either 5 or 7 is obtained. If P denotes the probability that 7 comes before 5, then 15P is equal to

(A) 8	**(B)** 1
(C) 3	**(D)** 9

Q8. *CUSAT CAT 2023 - Shift 3*

A pair of fair dice is thrown independently three times. The probability of getting a score of exactly a twice is 8

(A) 9 1	**(B)** 729 8
(C) 243 8	**(D)** 729

Q9. *CUSAT CAT 2023 - Shift 3*

If X follows a binomial distribution with parameters n = 100 and p = , then P(X = r) 3 is maximum when r is equal to

(A) 16	**(B)** 32
(C) 33	**(D)** 44 –1( ) –1 – 3

Q10. *CUSAT CAT 2023 - Shift 3*

If m is a natural number such that m ≤ 5, then the probability that the quadratic 1 m 2 equation x +mx+ + =0 has real roots is 2 2 1

(A) 5 2	**(B)** 3 3
(C) 5 4	**(D)** 5

Q11. *CUSAT CAT 2023 - Shift 4*

A fair coin is tossed 100 times. The probability of getting tails an odd number of times is 1

(A) 2 1	**(B)** 8 3
(C) 8	**(D)** None of the above 2

Q12. *CUSAT CAT 2023 - Shift 4*

Three numbers are chosen at random without replacement from {1, 2, ..., 10}. The probability that minimum of the chosen number is 3 or their maximum is 7 is 11

(A) 30 11	**(B)** 40 1
(C) 7 1	**(D)** 8

Q13. *CUSAT CAT 2023 - Shift 4*

A problem in Mathematics is given to three students A, B, C and their respective 1 1 1 probability of solving the problem are , , . Then the probability that the problem 2 3 4 is solved is 3

(A) 4 2	**(B)** 3 1
(C) 2 1	**(D)** 4

Q14. *CUSAT CAT 2023 - Shift 5*

Six boys and six girls sit in a row randomly. The probability that boys and girls sit alternatively is 1

(A) 231 5	**(B)** 462 1
(C) 462 7	**(D)** 101 dy

Q15. *CUSAT CAT 2023 - Shift 5*

There are four machines and it is known that exactly two of them are faulty. They are tested, one by one, in a random order till both the faulty machines are identified. Then the probability that only two tests are needed is 1

(A) 3 1	**(B)** 6 1
(C) 2 1	**(D)** 4

Q16. *CUSAT CAT 2024 - Shift 1*

If A, B, C are three arbitrary events such that P

(A) = P	**(B)** = P
(C) = , 4 1 P(A∩B) = P(B∩C) = 0 and P(A∩C) = , then the probability that at 8 least one of A, B, C occurs is 1	**(D)** 8 1 2 5

Q17. *CUSAT CAT 2024 - Shift 1*

The probability of arranging$\alpha,\alpha,\alpha,\alpha,\beta,\beta$ such an order that no two α's are adjacent is 2

(A) 3 1	**(B)** 3 1
(C) 6!	**(D)** 0 18

Q18. *CUSAT CAT 2024 - Shift 1*

An integer x, satisfying $10 \le x \le 99$, is to be chosen. The probability that at least one digit of x is 7 1

(A) 9 1	**(B)** 5 19
(C) 90 2	**(D)** 9

Q19. *CUSAT CAT 2024 - Shift 1*

The PDF of a continuous random variate is f(x)= , 2≤ x≤4. Then, 18 0, x>4 the probability that x will lie between $2 \le x \le 3$ is 4

(A) 9 2	**(B)** 9 5
(C) 9 –2	**(D)** 9

Q20. *CUSAT CAT 2024 - Shift 2*

In a group of employed persons, 30% are women. 65% of the men and 40% of the women pay income tax. The probability that a randomly chosen person is not a income tax payer is

(A) 0.575	**(B)** 0.525
(C) 0.475	**(D)** 0.425

Q21. *CUSAT CAT 2024 - Shift 2*

4 gentlemen and 4 ladies take seats at random, round a table. The probability that they are sitting alternatively is 4

(A) 35 1	**(B)** 70 1
(C) 35 2	**(D)** 35 x –x 10 –10

Q22. *CUSAT CAT 2024 - Shift 3*

A box consist of 10 white, 4 red and 2 black balls. If two balls are chosen from the box at random, then the probability of getting at least one white ball is 5

(A) 8 3	**(B)** 8 7
(C) 8 9	**(D)** 8

Q23. *CUSAT CAT 2024 - Shift 3*

The probability of getting a sum of 12 in four throws of an ordinary die is 3 15

(A) 66 3 5	**(B)** 6 3 1 5
(C) 366 4 1 5	**(D)** 366

Q24. *CUSAT CAT 2024 - Shift 3*

There are n persons sitting in a row. Two of them are selected at random. The probability that two selected persons are not together is 2

(A) n 2	**(B)** 1 – n n–1
(C) n n	**(D)** 2 1

Q25. *CUSAT CAT 2024 - Shift 3*

The probability that a man will live 10 more years, is and the probability that his 4 1 wife will live 10 more years, is . Then the probability that none of them will be 3 alive after 10 years is 5

(A) 2 1	**(B)** 2 7
(C) 12 11	**(D)** 12

Q26. *CUSAT CAT 2024 - Shift 3*

A Mathematics book contains 200 pages. A page is selected at random. What is the probability that the number on the page selected is a perfect square? 1

(A) 20 7	**(B)** 100 14
(C) 100 7	**(D)** 25 1

Q27. *CUSAT CAT 2024 - Shift 4*

From a pack of 52 cards, a card is drawn at random. The probability of drawing a black card or a queen card is 5

(A) 13 9	**(B)** 13 7
(C) 13 3	**(D)** 13

Q28. *CUSAT CAT 2024 - Shift 4*

Integers a, b, c, d, not necessarily distinct, are chosen independently and at random from 0 to 2007, inclusive. Then the probability that ad – bc is even 7

(A) 16 1	**(B)** 2 9
(C) 16 5	**(D)** 8 2

Q29. *CUSAT CAT 2024 - Shift 4*

One ticket is selected at random from 50 tickets numbered 00, 01, . . . , 50. Then, the probability that the sum of the digits on the selected ticket is 8, given that the product of these digits is zero, equals 1

(A) 14 1	**(B)** 7 5
(C) 14 1	**(D)** 50

Q30. *CUSAT CAT 2024 - Shift 4*

If probability of a defective bolt is 0.1, then mean and standard deviation of distribution of bolts in a total of 400 are

(A) 30, 3	**(B)** 40, 5
(C) 30, 4	**(D)** 40, 6

Q31. *CUSAT CAT 2024 - Shift 4*

Let a dice be rolled n times. The probability of getting odd numbers seven times be equal to the probability of getting odd numbers nine times. If the probability of p getting even number twice is , then p is equal to 15 2

(A) 60	**(B)** 30
(C) 15	**(D)** 90 n

Q32. *CUSAT CAT 2024 - Shift 5*

Three natural numbers are taken at random from the set A={x|1≤ x≤100, x∈ N}. The probability that the AM of the numbers taken is 15, is 44C

(A) 42 100C 3 15C	**(B)** 2 100C 3 44C
(C) 42 100C 15 15C	**(D)** 2 100C 15

Q33. *CUSAT CAT 2024 - Shift 5*

Probability of getting positive integral roots of the equation x2 –n=0 for the integer n, 1≤n≤40 is 1

(A) 5 1	**(B)** 10 3
(C) 20 1	**(D)** 20 π

Q34. *CUSAT CAT 2024 - Shift 5*

The mean and variance of a binomial variable X are 2 and 1 respectively. The probability that X takes values greater than 1, is 5

(A) 16 9	**(B)** 16 11
(C) 16 2	**(D)** 3 1 π

Q35. *CUSAT CAT 2024 - Shift 5*

Two dice are thrown n times in succession. The probability of obtaining a double six at least once is n 1

(A) 36 n 1	**(B)** 12 n 1
(C) 1– 12 n 35	**(D)** 1– 36

Q36. *CUSAT CAT 2024 - Shift 5*

A coin is tossed independently four times. The probability of the event “the number of times head shown up is greater than the number of times tail shown up” is 5

(A) 16 1	**(B)** 16 1
(C) 8 1	**(D)** 4

Q37. *CUSAT CAT 2024 - Shift 5*

A box contains 2 pens, 3 pencils and 4 chocolates. Items are drawn from the box at random one at a time without replacement. The probability of drawing 2 pens first followed by 3 pencils and subsequently the 4 chocolates is 2

(A) 315 1	**(B)** 630 1
(C) 1260 1	**(D)** 2520

Q38. *CUSAT CAT 2024 - Shift 5*

A mapping is selected at random from the set of all the mappings of the set A={1,2,...,n} into itself. The probability that the mapping selected is an injection is 1

(A) n n 1	**(B)** n! (n–1)!
(C) n–1 n n!	**(D)** n–1 n

Q39. *CUSAT CAT 2024 - Shift 5*

If a is an integer lying in [–5, 30] , then the probability that the graph of y = x2 +2(a+4)x–5a+64 is strictly above the x-axis is 1

(A) 6 7	**(B)** 36 2
(C) 9 3	**(D)** 5

Q40. *CUSAT CAT 2025 - Shift 1*

If a die shows an even number, then the probability of getting 4 is 1

(A) 6 1	**(B)** 2 1
(C) 12 1	**(D)** 3 d 2

Q41. *CUSAT CAT 2025 - Shift 1*

A number is palindrome if it remains same when its digits are reversed. A palindrome between 100 and 1000 is chosen at random. The probability that it is divisible by 2 is 2

(A) 45 1	**(B)** 10 1
(C) 9 1	**(D)** 50 2 2 2

Q42. *CUSAT CAT 2025 - Shift 1*

If the probability of a defective bolt is 0.1, then the mean and the standard deviation of the distribution of bolts in a total of 400 are respectively

(A) 30, 3	**(B)** 40, 5
(C) 30, 4	**(D)** 40, 6

Q43. *CUSAT CAT 2025 - Shift 1*

A parents has two children. If one of them is boy, then the probability that the other is, also a boy is 1

(A) 2 1	**(B)** 4 1
(C) 3 2	**(D)** 3 2n

Q44. *CUSAT CAT 2025 - Shift 2*

A fair die is tossed until a number greater than 4 appears. The probability that an even number of tosses shall be required, is 1

(A) 2 1	**(B)** 5 3
(C) 5 2	**(D)** 3

Q45. *CUSAT CAT 2025 - Shift 3*

The probability density function of a continuous random variate is given by 0, x<2 (3+2x) f(x)= , 2≤ x≤4 18 0, x>4 Then, the probability that x will lie between 2≤ x≤3 is 4

(A) 9 2	**(B)** 9 5
(C) 9 –2	**(D)** 9

Q46. *CUSAT CAT 2025 - Shift 3*

There are 12 white and 12 red balls in a bag. Balls are drawn one by one with replacement from the bag. The probability that 7th drawn ball is 4th white is 1

(A) 4 1	**(B)** 8 1
(C) 2 1	**(D)** 3

Q47. *CUSAT CAT 2025 - Shift 4*

When m≤4 is a positive integer, the probability of getting real roots for 2 2x +2mx+1=0 is 3

(A) 5 1	**(B)** 2 3
(C) 4 2	**(D)** 5

Q48. *CUSAT CAT 2025 - Shift 4*

The probability that a man will live 10 more years, is and the probability that his 9 3 wife will live 10 more years, is . Then the probability that none of them will be 5 alive after 10 years is 2

(A) 5 2	**(B)** 9 3
(C) 7 5	**(D)** 9

Q49.

CUSAT CAT 2025 - Shift 4

A committee of 5 is to be chosen from a group of 10 people. The probability that a certain married couple will either serve together or not at all is 1

(A) 2 5	**(B)** 9 4
(C) 9 2	**(D)** 3

Q50.

CUSAT CAT 2025 - Shift 4

A coin with probability p , 0 < p< 1 for getting a head is tossed until a head appears 2 for the first time. If the probability that the number of tosses required is odd is , 5 then p equals 1

(A) 3 2	**(B)** 3 2
(C) 5 3	**(D)** 5 1 1 r

Q51.

CUSAT CAT 2025 - Shift 4

An integer is chosen at random from first two hundred numbers. Then the probability that the integer chosen is divisible by 6 or 8 is 1

(A) 4 2	**(B)** 4 3
(C) 4 1	**(D)** 3

Q52.

CUSAT CAT 2025 - Shift 4

In a deck of 52 cards, two cards are drawn without replacement. Then the probability that both cards are aces is 2

(A) 211 2	**(B)** 221 1
(C) 211 1	**(D)** 221

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	C	3	A	4	C	5	B	6	A	7	D	8	C	9	C	10	C
11	A	12	B	13	A	14	C	15	B	16	D	17	D	18	B	19	A	20	D
21	C	22	C	23	A	24	B	25	B	26	B	27	C	28	D	29	A	30	D
31	A	32	A	33	C	34	C	35	D	36	A	37	C	38	C	39	C	40	D
41	A	42	D	43	C	44	C	45	A	46	C	47	C	48	B	49	C	50	A
51	A	52	D																

Mathematical Reasoning & LP

6 questions

Q1. *CUSAT CAT 2023 - Shift 2*

Which of the following is a contradiction?

(A) $p \vee q$	**(B)** $p \wedge q$
(C) $p \vee \sim p$	**(D)** $p \wedge \sim p$

Q2. *CUSAT CAT 2023 - Shift 4*

Let S: If a number n is even, then n is even. Then the converse of S is 2

(A) if a number n is even, then n is even 2	**(B)** neither n nor n is even 2
(C) if n is not even, then n is not even 2	**(D)** if a number n is not even, then n is even 2 2 2 2

Q3. *CUSAT CAT 2023 - Shift 5*

The contrapositive of $p \rightarrow (q \rightarrow r)$ is logically equivalent to

(A) $p \rightarrow (q \rightarrow r)$	**(B)** $(q \rightarrow r) \rightarrow \sim p$
(C) $p \vee q \rightarrow r$	**(D)** $(q \rightarrow r) \rightarrow p$ 7–i 14

Q4. *CUSAT CAT 2024 - Shift 1*

Consider the problem: Maximize $z = 2x + 3y$ subject to the constraints $x + y \leq 1$, $2x + 2y \geq 6$, $x, y \geq 0$. Then the number of optimal solutions is

(A) 3	**(B)** 2
(C) 1	**(D)** 0

Q5. *CUSAT CAT 2024 - Shift 2*

The negation of $(p \wedge (\sim q)) \vee (\sim p)$ is equivalent to

(A) $p \wedge (q \wedge (\sim p))$	**(B)** $p \wedge (\sim q)$
(C) $p \wedge (q \vee (\sim p))$	**(D)** $p \wedge q$

Q6. *CUSAT CAT 2024 - Shift 3*

Which of the following statements is Tautology?

(A) $p \rightarrow (p \wedge (p \rightarrow q))$	**(B)** $(p \wedge q) \rightarrow (\sim (p) \rightarrow q)$
(C) $(p \wedge (p \rightarrow q)) \sim q$	**(D)** $p \vee (p \wedge q)$

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	D	2	A	3	A	4	D	5	D	6	B								

Mathematics – Miscellaneous

542 questions

Q1. *CUSAT CAT 2023 - Shift 1*

If x +mx+1=0 and (b–c)x +(c–a)x+(a–b)=0 have both roots common, then

(A) m=–2	**(B)** m=–3
(C) a,b,c are in A.P.	**(D)** a,b,c are in G.P.

Q2. *CUSAT CAT 2023 - Shift 1*

If P : P =3:5, then n is equal to n–1 n

(A) 4	**(B)** 3
(C) 6	**(D)** 5 n 2n (2) 2n ∑

Q3. *CUSAT CAT 2023 - Shift 1*

Let n be a positive integer such that 1+x+x =a +a x+...+a x , then a 0 1 2n r r=0

(A) 3 n–2	**(B)** 3 n–1 n 3
(C) 2 n	**(D)** 3

Q4. *CUSAT CAT 2023 - Shift 1*

If = I, then y 0

(A) x=1=2y	**(B)** x= y
(C) x= y 2	**(D)** xy =1 11 1 1 1 1 1 1 1

Q5. *CUSAT CAT 2023 - Shift 1*

If S = + – + + + +··· then S is equal to 22 3 42 2 3 2 62 3 3 3

(A) log1 3	**(B)** log 2 2
(C) log 3 1	**(D)** log 2 2 { 2 2 }

Q6. *CUSAT CAT 2023 - Shift 1*

The domain of the relation R= (x,y):x, y∈Z, x + y ≤4 in Z is

(A) {0,1, 2}	**(B)** {–2, 0}
(C) {–2, –1, 0,1, 2}	**(D)** {–2, –1,1, 2} 1+x

Q7. *CUSAT CAT 2023 - Shift 1*

Let f :R→R be defined by f(x)=log for x∈R. Then f(m)+ f(n)= 1–x

(A) f(m+n) m+n	**(B)** f 1+mn
(C) f(m·n)	**(D)** 0

Q8. *CUSAT CAT 2023 - Shift 1*

The pair of points which lie on the same side of the straight line 3x–8y–7=0 is

(A) (0,–1), (0,0)	**(B)** (–1–1), (3,7)
(C) (–1–1), (3,–7)	**(D)** (0,1), (3,0) 2 2

Q9. *CUSAT CAT 2023 - Shift 1*

Let A=(3,0) and B=(–3,0). Let P be any point on the curve 16x +25y =400. Then PA+PB =

(A) 8	**(B)** 10
(C) 6	**(D)** 12 dy 2π

Q10. *CUSAT CAT 2023 - Shift 1*

lim = x→1 (x–1) 2 1

(A) 3 1	**(B)** 9 2
(C) 3 2	**(D)** 9

Q11. *CUSAT CAT 2023 - Shift 1*

If y(t) is a solution of (1+t) – ty=1 and y(0)=–1, then y(1)= dt 1

(A) e+ 2 1	**(B)** – 2 1
(C) e– 2 1	**(D)** 2 2

Q12. *CUSAT CAT 2023 - Shift 1*

For all x∈R, if kx –9kx+5k+1>0, then k lies in the interval 4

(A) – , 0 61 4	**(B)** 0, – 6 4
(C) 0, 61 4 4	**(D)** – , 61 61

Q13. *CUSAT CAT 2023 - Shift 1*

If z =9y –4–10ix, z =8y –20i and z = z , then z = x+iy is equal to 1 2 1 2

(A) 2+2i	**(B)** –2±2i
(C) –2±i	**(D)** 2–i

Q14. *CUSAT CAT 2023 - Shift 1*

The inequality z–4 < z–2 represents the region given by

(A) Re(z)>0	**(B)** Re(z)<0
(C) Re(z)>2	**(D)** Re(z)>3

Q15.

CUSAT CAT 2023 - Shift 1

Let a,b,c be real numbers with a≠0 and let α,β be the roots of the equation 2 3 2 3 ax +bx+c=0. Then a x +abcx+c =0has roots

(A) α 2 β,αβ 2	**(B)** αβ 2
(C) α 2 β, βα	**(D)** α 3 β,αβ 3

Q16.

CUSAT CAT 2023 - Shift 1

The total number of proper factors of 7875 is

(A) 23	**(B)** 24
(C) 22	**(D)** 21

Q17.

CUSAT CAT 2023 - Shift 1

In a Poisson distribution, if P(X = 2) = P(X = 3), then P(X = 4) is equal to

(A) e –2 –2 27e	**(B)** 4 –3 27e
(C) 8 –3 2e	**(D)** 4 4 4

Q18.

CUSAT CAT 2023 - Shift 1

If A = and A = 2 , then a + b is equal to 0 3 0 b 7

(A) 20 3	**(B)** 20 19
(C) 60 11	**(D)** 20

Q19.

CUSAT CAT 2023 - Shift 1

If the points (x,y), (x',y') and (x–x',y–y') are collinear, then

(A) xy= x'y'	**(B)** xx'= yy'
(C) xy'= x'y	**(D)** x'–y'=1 2 1

Q20.

CUSAT CAT 2023 - Shift 1

a, b, c are three unequal numbers such that a, b, c are in AP and b – a, c – b, a are in GP. Then a : b : c is

(A) 3 : 2 : 1	**(B)** 3 : 1 : 2
(C) 1 : 2 : 3	**(D)** 2 : 1 : 3

Q21.

CUSAT CAT 2023 - Shift 1

The set of points of discontinuity of the function in ■ is log x

(A) {–1, 0, 1}	**(B)** {0}
(C) {0, 1}	**(D)** {0,∞} ( )x ( )x x 4

Q22.

CUSAT CAT 2023 - Shift 1

Let a =2iˆ+3ˆj–k ˆ , b =3iˆ– ˆj+k ˆ and c =iˆ+ ˆj+3k ˆ . If r×b =c×band r·a =0, then r is equal to

(A) 1(iˆ+ ˆj+k ˆ) 2	**(B)** 2 (–iˆ+ ˆj+k ˆ)
(C) 2 (iˆ+ ˆj+k ˆ)	**(D)** 1(iˆ– ˆj+k ˆ) 2 3 2

Q23.

CUSAT CAT 2023 - Shift 1

The roots of the algebraic equation x +x +x+1=0 are

(A) 1,–1,1	**(B)** 0,0,0
(C) –1, j,–j	**(D)** 1, j,–j 2z+1

Q24.

CUSAT CAT 2023 - Shift 1

lim 1– = n→∞ n

(A) 0 –1	**(B)** e 2
(C) e –2	**(D)** 1 2

Q25.

CUSAT CAT 2023 - Shift 1

If g(x)–g(y) ≤ (x– y) ; x,y∈R and g(0)=1, then

(A) g(x) can take any value	**(B)** g(x)<0, x∈R
(C) g(x)>0, x∈R	**(D)** g(x)=0, x∈R 2

Q26.

CUSAT CAT 2023 - Shift 1

All possible values of x –4 lie in

(A) [0,∞)	**(B)** [–2,2]
(C) (0,2]	**(D)** (–∞,∞)

Q27.

CUSAT CAT 2023 - Shift 1

If a,b,p,q are non-zero real numbers, then number of common roots of two 2 2 2 2 2 2 equations 2a x –2abx+b =0 and p x +2pqx+q =0 is

(A) 1	**(B)** 0
(C) 2	**(D)** ∞ (2)

Q28.

CUSAT CAT 2023 - Shift 1

The degree of the polynomial (1+x) (1+x 6)(1+x 11) ... (1+x 101) is

(A) 21	**(B)** 101
(C) 1071	**(D)** 501

Q29. *CUSAT CAT 2023 - Shift 1*

If a,b,c are in A.P., then the straight line ax + by + c = 0 will always pass through the point

(A) (–1,–2)	**(B)** (1,2)
(C) (–1,2)	**(D)** (1,–2)

Q30. *CUSAT CAT 2023 - Shift 1*

Lines r =a +tb and r =a +sb lie on a plane, if 1 1 2 2

(A) a ×a =0 1 2	**(B)** b ×b =0 1 2 ( ) ( )
(C) a –a · b –b =0 1 2 1 2 ( ) ( )	**(D)** a ·a · b ·b =0 1 2 1 2

Q31. *CUSAT CAT 2023 - Shift 1*

Let R be a reflexive relation on a finite set A having n elements and let there be m ordered pairs in R. Then

(A) m ≥ n	**(B)** m ≤ n
(C) m = n	**(D)** m 2

Q32. *CUSAT CAT 2023 - Shift 1*

Let f :R→R and g:R→R be given by f(x)= x and g(x)=[x] for each x∈R. Then { x∈R:g(f(x))≤ f (g(x))} is equal to

(A) ■∪(–∞,0)	**(B)** (–∞,0)
(C) ■	**(D)** ■

Q33. *CUSAT CAT 2023 - Shift 1*

The sum of all natural numbers less than 200, that are divisible neither by 3 nor by 5, is

(A) 10730	**(B)** 10732
(C) 15375	**(D)** 8022 4 5

Q34. *CUSAT CAT 2023 - Shift 1*

The number of words that can be formed with the letters of the word MATHEMATICS by rearranging them is 11!

(A) 2!2! 11!	**(B)** 2!2!2! 11!
(C) 2!	**(D)** 11! n

Q35. *CUSAT CAT 2023 - Shift 1*

If the number of terms in the expansion (x + y + z) is 36, then the value of n is

(A) 8	**(B)** 7
(C) 9	**(D)** 10

Q36.

CUSAT CAT 2023 - Shift 1

If α,β,γ are the roots of x + px +q=0, where q≠0, then Δ= β γ α 1 1 1 γ α β is equal to p

(A) q 1	**(B)** q 2 p
(C) q	**(D)** 0

Q37.

CUSAT CAT 2023 - Shift 1

If one of the lines given by the equation 2x 2 +axy+3y 2 =0 coincide with one of those given by 2x 2 +bxy–3y 2 =0 and the other lines represented by them be perpendicular, then

(A) a = –5, b = 1	**(B)** a = 5, b = –1
(C) a = 5, b = 1	**(D)** a = –5, b = –1 8

Q38.

CUSAT CAT 2023 - Shift 1

The number of natural numbers which are smaller than 2 × 10 and which can be written by means of the digits 1 and 2 is

(A) 766	**(B)** 856
(C) 656	**(D)** 866

Q39.

CUSAT CAT 2023 - Shift 1

If z =4+i 7, then value of z –4z –9z+91 equals

(A) 0	**(B)** 1
(C) –1	**(D)** 2

Q40.

CUSAT CAT 2023 - Shift 1

If S = { x∈■: (log 0.216) log (5–2x)≤0 } then S is equal to 0.6 5

(A) [2.5, ∞)	**(B)** [2, 2.5)
(C) (2, 2.5)	**(D)** (0, 2.5)

Q41.

CUSAT CAT 2023 - Shift 1

If P is a point (x, y) on the line y = –3x such that P and the point (3, 4) are on the opposite sides of the line 3x – 4y = 8, then 8 8

(A) x> , y<– 15 5 8 8	**(B)** x> , y<– 5 15 8 8
(C) x= , y =– 15 5	**(D)** x≠8, y=36

Q42.

CUSAT CAT 2023 - Shift 1

Two integers r and s are drawn one at a time without replacement from the set {1, 2,..., n}. If P = P(r ≤k|s≤k), and n=25, then 8P = k 7

(A) 1	**(B)** 0
(C) 2	**(D)** 4 4 2 a –1 –1

Q43.

CUSAT CAT 2023 - Shift 1

Let f :{2,3,4,5}→{3,4,5,9} and g:{3,4,5,9}→{7,11,15} be functions defined as f (2)=3, f (3)=4, f(4)= f(5)=5 and g(3)= g(4)=7 and g(5)= g(9)=11. Then g f(2,3,4,5)=

(A) (7,7,11,11)	**(B)** (3,5,7,4)
(C) (5,7,11,7)	**(D)** (9,9,15,15)

Q44.

CUSAT CAT 2023 - Shift 2

The number of words that can be formed from the letters a,b,c,d,e, f taken 3 at a time, each word continuing at least one vowel is

(A) 84	**(B)** 96
(C) 102	**(D)** 112 y

Q45.

CUSAT CAT 2023 - Shift 2

The number of ordered triples (x,y,z) such that x,y,z are primes and x +1= z is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q46.

CUSAT CAT 2023 - Shift 2

Let p,q,r be three real numbers. Then the system + + =1, 2 2 2 a b c 2 2 2 2 2 2 x y z x y z – + =1, – + + =1 has 2 2 2 2 2 2 a b c a b c

(A) no solution	**(B)** unique solution
(C) finitely many solution	**(D)** infinitely many solution

Q47.

CUSAT CAT 2023 - Shift 2

The system of equations kx+2y–z =1, (k–1)y–2z =2 and (k+2)z =3 have only one solution when

(A) k =–2	**(B)** k =–1
(C) k =0	**(D)** k =1 ∞ (logx) 2n ∑

Q48.

CUSAT CAT 2023 - Shift 2

If S = , then S is equal to (2n)! n=0

(A) x+x –1	**(B)** x–x –1 1(–1)
(C) x+x 2 1(–1)	**(D)** x–x 2

Q49.

CUSAT CAT 2023 - Shift 2

For p,q∈Z, p|q means p is a factor of q. Then '|'is

(A) reflexive and symmetric	**(B)** symmetric and transitive
(C) reflexive and not transitive	**(D)** reflexive, transitive but not symmetric

Q50. *CUSAT CAT 2023 - Shift 2*

The inverse of the function f(x)= is x –x 3 +3 1 1–x

(A) log 3 2 1+x 1 1–x	**(B)** log e 3 1+x 1 1+x
(C) log 3 2 1–x 1 1+x	**(D)** log e 3 1–x (2) (2) log 1+x+x +log 1–x+x

Q51. *CUSAT CAT 2023 - Shift 2*

The set of values of x for which log(1+x)< x, is

(A) x<0	**(B)** x>0
(C) 0< x<1	**(D)** x>1

Q52. *CUSAT CAT 2023 - Shift 2*

The number of real solutions of the equation =2+ is x 2 –4 x+2

(A) 4	**(B)** 3
(C) 2	**(D)** 1 x+2 1

Q53. *CUSAT CAT 2023 - Shift 2*

The number of integral solutions of > is x 2 +1 2

(A) 1	**(B)** 2
(C) 3	**(D)** 0 100 3 3 49

Q54. *CUSAT CAT 2023 - Shift 2*

If 3 (x+iy)= + i and x=ky, then k is 2 2 1

(A) – 3	**(B)** 3
(C) – 3 1	**(D)** – 3

Q55. *CUSAT CAT 2023 - Shift 2*

The equation NOT representing a circle is given by 1+z

(A) Re =0 1–z	**(B)** zz +iz–iz +1=0 z–1 π
(C) arg = z+1 2 z–1	**(D)** =1 z+1 4 1 1 π 1 1 1

Q56. *CUSAT CAT 2023 - Shift 2*

If + +...+∞= , then + + +...∞ is equal to 1 4 2 4 90 1 4 3 4 5 4 4 π

(A) 96 4 π	**(B)** 45 4 89π
(C) 90 4 π	**(D)** 46 2 2 2 2 2 2

Q57. *CUSAT CAT 2023 - Shift 2*

The sum of first n terms of the series 1 +2·2 +3 +2·4 +5 +2·6 +...is 2 n(n+1) when n is even. When n is odd, the sum is 2 2 n (n+1)

(A) 2 2 n(n+1)	**(B)** 2 2 n(n+1)
(C) 2 n(n+1)	**(D)** 2 3 7 15 31

Q58. *CUSAT CAT 2023 - Shift 2*

The sum of the series 1+ + + + +... to n terms is equal to 2 4 8 16 1

(A) 2n– n 2 1	**(B)** 2+ n 2 1
(C) 2n–1+ n 2 1	**(D)** 2(n–1)+ n–1 2 2 (2) (2)

Q59. *CUSAT CAT 2023 - Shift 2*

If the value of k for which the equation 3x +2 k +1 x+ k –3k+2 =0 possesses roots of opposite sign lies in

(A) (–∞,1)	**(B)** (–∞,0)
(C) (1,2) 3	**(D)** ,2 2 2

Q60. *CUSAT CAT 2023 - Shift 2*

Let αand β be roots of the equation x +x α+β=0. Then

(A) α=1, β=–1	**(B)** α=1,β=–2
(C) α=2, β=1	**(D)** α=2, β=–2

Q61. *CUSAT CAT 2023 - Shift 2*

Consider the region 5x+ y≤100, x+ y≤60, x≥0, y≥0. Then the point (25, 40)

(A) lies outside the region	**(B)** is on the boundary
(C) lies inside the region	**(D)** is the only point in the region 2

Q62. *CUSAT CAT 2023 - Shift 2*

If α and β are the roots of the equation x + x + 1 = 0, then the equation whose roots 19 7 are α and β is

(A) x 2 –x+1=0	**(B)** x 2 +x–1=0
(C) x 2 +x+1=0	**(D)** x 2 –x–1=0

Q63. *CUSAT CAT 2023 - Shift 2*

Let (1 + i)(1 + 2i)(1 + 3i) … (1 + 20i) = a + ib. Then 2.5.10. … .401 is equal to

(A) a – bi	**(B)** ab 2 2
(C) a – b 2 2	**(D)** a + b

Q64. *CUSAT CAT 2023 - Shift 2*

If y = x – x , then the derivative of y with respect to x is 2

(A) 2x – 3x + 1 2	**(B)** 2x + 3x + 1 2
(C) 2x – 3x – 1 2	**(D)** 2x – 6x + 1 x f(x)

Q65. *CUSAT CAT 2023 - Shift 2*

The range of the function e + e = e is

(A) (–∞, 1]	**(B)** [1,∞)
(C) (1,∞)	**(D)** (–∞, 1) x

Q66. *CUSAT CAT 2023 - Shift 2*

The least integer value of m such that (m – 2)x + 8x + m + 4 > 0 for all x ∈ R is

(A) 10	**(B)** 2
(C) 12	**(D)** 5

Q67. *CUSAT CAT 2023 - Shift 2*

The number of variety of salads can be made from cucumber, tomatoes, apples, oranges and bananas is

(A) 16	**(B)** 31
(C) 32	**(D)** 62

Q68. *CUSAT CAT 2023 - Shift 2*

If the three points U(1, 6), V(3, –4) and W(a, b) are collinear, then the equation satisfying by a and b is

(A) 5a+b–11=0	**(B)** 5a+13b+5=0
(C) 5a–13b+5=0	**(D)** 13a–5b+5=0

Q69. *CUSAT CAT 2023 - Shift 2*

A solution to x–5– 9–x >1, x∈■ is

(A) 4	**(B)** 9
(C) 5	**(D)** 8

Q70. *CUSAT CAT 2023 - Shift 2*

If α,β,γ are the roots of the equation x 3 +4x+1=0, then the value of –1 –1 –1 (α+β) +(β+γ) +(γ+α) is

(A) 2	**(B)** –1
(C) 4	**(D)** 0

Q71. *CUSAT CAT 2023 - Shift 2*

The sum of the series is k=1 k k+1+(k+1) k 17

(A) 19 18	**(B)** 19 16
(C) 19 20	**(D)** 19

Q72. *CUSAT CAT 2023 - Shift 2*

Let f(x) be a polynomial of degree 3 such that f(3)=1, f '(3)=–1, f "(3)=0 and f "'(3)=12. Then the value of f '(1) is

(A) 12	**(B)** 23
(C) –13	**(D)** 0 π a+ 2

Q73. *CUSAT CAT 2023 - Shift 2*

The number of solutions of the equation z 2 + z 2 =0, where z∈■, is

(A) one	**(B)** two
(C) three	**(D)** infinitely many

Q74. *CUSAT CAT 2023 - Shift 2*

If n∈■, then 10 n +3 (4 n+2) +5 is divisible by

(A) 7	**(B)** 5
(C) 9	**(D)** 17

Q75. *CUSAT CAT 2023 - Shift 2*

Let f(x+ y)= f(x)f(y) for all x,y where f(0)≠0. If f '(0)=2, then f(x) is equal to

(A) Ae x	**(B)** Ae 2x
(C) 2x	**(D)** Ae –2x

Q76. *CUSAT CAT 2023 - Shift 2*

The exponent of 3 in 100! is

(A) 40	**(B)** 48
(C) 50	**(D)** 45

Q77. *CUSAT CAT 2023 - Shift 2*

Out of 10 consonants and 4 vowels, the number of words that can be formed such that each containing 3 consonants and 2 vowels is

(A) 86500	**(B)** 86800
(C) 86400	**(D)** 86300 ...5 55

Q78. *CUSAT CAT 2023 - Shift 2*

The remainder when x=5 (24 times 5), is divisible by 24, is

(A) 4	**(B)** 24
(C) 5	**(D)** 6

Q79. *CUSAT CAT 2023 - Shift 2*

If ax 2 +bx+c=0, a,b,c∈R has no real roots, and if c < 0, then

(A) a < 0	**(B)** a + b + c > 0
(C) a > 0	**(D)** a=c

Q80. *CUSAT CAT 2023 - Shift 2*

The remainder when 2 is divided by 17 is

(A) 1	**(B)** 2
(C) 8	**(D)** 3 log 24 log 192

Q81. *CUSAT CAT 2023 - Shift 2*

The value of 2 – 2 is log 2 log 2 96 12

(A) 3	**(B)** 0
(C) 2	**(D)** 1 n ( ) ( )

Q82. *CUSAT CAT 2023 - Shift 2*

A line is drawn through the point P(3, 11) to cut the circle x + y =9 at A and B. Then PA·PB is equal to

(A) 9	**(B)** 121
(C) 205	**(D)** 139

Q83. *CUSAT CAT 2023 - Shift 2*

The number of points (p,q) such that p,q∈{1,2,3,4} and the equation 2 px +qx+1=0 has real roots is

(A) 7	**(B)** 8
(C) 9	**(D)** 16 –1 2

Q84. *CUSAT CAT 2023 - Shift 2*

Vertices of a triangle are (0, 0) , (41a, 37) and (–37, 41b) where a and b are the roots 2 of the equation 3x –16x+15=0. The area of the triangle is equal to

(A) 4678	**(B)** 4356
(C) 4887	**(D)** 4879

Q85. *CUSAT CAT 2023 - Shift 3*

Nishi has 5 coins each of the different denomination. The number of different sums of money, she can form, is

(A) 25	**(B)** 30
(C) 31	**(D)** 32 n m

Q86. *CUSAT CAT 2023 - Shift 3*

The number of ordered pairs (m,n), m,n∈{1,2,...,50} such that 6 +9 is a multiple of 5, is

(A) 2500	**(B)** 1500
(C) 1250	**(D)** 750

Q87. *CUSAT CAT 2023 - Shift 3*

In a decimal system of numeration, the number of exactly 6-digit numbers in which the sum of the digits is divisible by 5 is

(A) 180000	**(B)** 210000
(C) 360000	**(D)** 540000 2 1 3 4 3 –4

Q88. *CUSAT CAT 2023 - Shift 3*

Let A= , B = and C = . Then 4 1 2 3 –2 3 ABC A(BC) 2 A(BC) 3 tr

(A) +tr +tr +tr +...∞ where tr	**(B)** 9
(C) 12	**(D)** 16 a b aα+b

Q89. *CUSAT CAT 2023 - Shift 3*

+ + +... will be equal to 3! 5! 7!

(A) 2e –2	**(B)** e –2
(C) e –1	**(D)** 2e –1 4 8 16 –1

Q90. *CUSAT CAT 2023 - Shift 3*

If x=1+2+ + + +..., then x is equal to 2! 3! 4!

(A) e –2	**(B)** e 2 1
(C) e 2 –1	**(D)** e 2 1

Q91. *CUSAT CAT 2023 - Shift 3*

Solutions of the equation 3 +4 =25 are

(A) –1,–2	**(B)** –1,2
(C) 1,–2	**(D)** 1,2 π2

Q92. *CUSAT CAT 2023 - Shift 3*

If log 5=a and log 6=b, then log 2 is equal to 4 5 3 1

(A) 2a+1 1	**(B)** 2b+1
(C) 2ab+1 1	**(D)** 2ab–1 (2 2 2) 1 1 1

Q93. *CUSAT CAT 2023 - Shift 3*

For positive numbers a, b, c, the least value of a +b +c + + is a 2 b 2 c 2

(A) 3	**(B)** 9 27
(C) 4 27	**(D)** 2 334 365 1 3 1 3

Q94. *CUSAT CAT 2023 - Shift 3*

For the equation 3x + px+3=0, p>0, if one of the root is square of the other, then p is equal to 1

(A) 3	**(B)** 1
(C) 3 2	**(D)** 3 3 2

Q95. *CUSAT CAT 2023 - Shift 3*

If the roots of the cubic equation x – px +qx–r =0 are in G.P., then

(A) q 3 = p 3 r	**(B)** p 3 =q 3 r
(C) pq =r	**(D)** pr =q 1 1 3 5 2n–1

Q96. *CUSAT CAT 2023 - Shift 3*

If H =1+ +... , then the value of 1+ + +...+ is n 2 n 2 3 n

(A) H +n n	**(B)** 2n–H n
(C) (n–1)+H n	**(D)** H +2n n

Q97. *CUSAT CAT 2023 - Shift 3*

If the roots of the equation x –2kx+k +k–3=0 are real and less than 3, then

(A) k >4	**(B)** 2≤k ≤3
(C) 3<k ≤4	**(D)** k <2 1 2 3 2

Q98. *CUSAT CAT 2023 - Shift 3*

Let x=2+2 3 +2 3. Then the value of x –6x +6x is

(A) 1	**(B)** 3
(C) 2	**(D)** 4

Q99. *CUSAT CAT 2023 - Shift 3*

The average age of A, B and C, whose ages are integers x, y and z (x ≤ y ≤ z) respectively, is 30. If the age of B is exactly 5 more than that of A, then the minimum possible value of z is

(A) 31	**(B)** 33
(C) 35	**(D)** 37

Q100. *CUSAT CAT 2023 - Shift 3*

Let a, b, c, be positive numbers such that a + b + c = 1 and (1 – a)(1 – b)(1 – c) ≥ kxyz. Then k is equal to

(A) 4	**(B)** 8
(C) 6	**(D)** 0 2 2

Q101. *CUSAT CAT 2023 - Shift 3*

The inverse of the point (1, 2) with respect to the circle x + y – 4x – 6y + 9 = 0, is 1

(A) 1, 2	**(B)** (2, 1)
(C) (0, 1)	**(D)** (1, 0) 2 2 2

Q102. *CUSAT CAT 2023 - Shift 3*

If f(x)= x–2 and g(x)= f [f(x)], then g'(x) for x>20 is

(A) ∞	**(B)** 1
(C) 2	**(D)** –1

Q103. *CUSAT CAT 2023 - Shift 3*

Given that the curves x + y + kx + 4y + 2 = 0 and 2(x + y) – 4x – 3y + k = 0 cut orthogonally. Then the value of k is

(A) 1 1	**(B)** 3 10
(C) 3 10	**(D)** – 3

Q104. *CUSAT CAT 2023 - Shift 3*

Let a, b, c be three positive real numbers such that a + b ≥ c. Then a b c

(A) + ≥ 1+a 1+b 1+c a b c	**(B)** + < 1+a 1+b 1+c a b 1+c
(C) + > 1+a 1+b c a b 1+c	**(D)** + ≤ 1+a 1+b c

Q105. *CUSAT CAT 2023 - Shift 3*

A seven digit number made up of all distinct digits 8, 7, 6, 4, 2, x and y is divisible by 3. Then possible number of ordered pair (x, y) is

(A) 2	**(B)** 4
(C) 6	**(D)** 8 1

Q106. *CUSAT CAT 2023 - Shift 3*

The slopes of the lines represented by x 2 +2hxy+2y 2 =0 are in the ratio 1 : 2. Then the value of h is 3

(A) ± 2	**(B)** ±3
(C) ±1 2	**(D)** ± 3

Q107. *CUSAT CAT 2023 - Shift 3*

The equation of the sphere described on the line joining the points (2, –1, 4) and (–2, 2, –2) as diameter is

(A) 2x 2 –x+z =0	**(B)** x 2 + y 2 +z 2 +x– y+z+7=0
(C) x 2 + y 2 +z 2 – y–2z–14=0	**(D)** x 2 + y 2 +z 2 =0

Q108. *CUSAT CAT 2023 - Shift 3*

Given that two roots of the nonlinear equation x –6x +11x–6=0 are 1 and 3. The third root will be

(A) j	**(B)** –j
(C) 2	**(D)** –2 2 3

Q109. *CUSAT CAT 2023 - Shift 3*

Let A= . If the eigen values of A are 4 and 8, then (x, y) = x y

(A) (4,10)	**(B)** (5,8)
(C) (–3,9)	**(D)** (–4,10) 2 z +8

Q110. *CUSAT CAT 2023 - Shift 3*

If r = xi + yj +zk, then (divr,curlr) is

(A) (0, 1)	**(B)** (3, 0)
(C) (3, 1)	**(D)** (1, 3)

Q111. *CUSAT CAT 2023 - Shift 3*

The value of 6+ 6+ 6+...∞ is

(A) 3	**(B)** 6
(C) –4	**(D)** 2 2

Q112. *CUSAT CAT 2023 - Shift 3*

The minimum height from any point on the curve y = x –4x+6 to the x-axis is

(A) 6	**(B)** 4
(C) 1	**(D)** 2 50

Q113. *CUSAT CAT 2023 - Shift 3*

If (8+i) =3 49(a+ib),then the value of a 2 +b 2 is

(A) 3	**(B)** 9
(C) 27	**(D)** 81

Q114. *CUSAT CAT 2023 - Shift 3*

If f(x) be a differentiable function such that f(xy)= f(x)+ f(y) for all x and y, 1 then f(e)+ f = e

(A) 1	**(B)** 0
(C) –1	**(D)** ∞ 2 3 n dy 1 dy 1 dy 1 dy

Q115. *CUSAT CAT 2023 - Shift 3*

If 25% of the items are less than 20 and 25% are more than 40, the quartile deviation is

(A) 20	**(B)** 30
(C) 40	**(D)** 10 ( ) 2

Q116. *CUSAT CAT 2023 - Shift 3*

If x –x+1=0, then the value of x is

(A) 0	**(B)** –1
(C) 1	**(D)** n

Q117. *CUSAT CAT 2023 - Shift 3*

The remainder when 1! + 2! + 3! + ■ ■ ■ + 49! is divided by 10 is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q118. *CUSAT CAT 2023 - Shift 3*

The number of zeros in the product 5 ·6 ·7 ·8 ·9 ····30 is

(A) 130	**(B)** 132
(C) 137	**(D)** 136

Q119. *CUSAT CAT 2023 - Shift 3*

If A denotes the area enclosed by 3 x +4 y ≤12, then 4A + A+1 is equal to

(A) 2341	**(B)** 2329
(C) 2420	**(D)** 2429 2 8

Q120.

CUSAT CAT 2023 - Shift 3

If log x+log y =2+log 2 and log (x+ y)=2, then 3 3 3 3

(A) x=1,y =8	**(B)** x=1,y =1
(C) x=3,y =6	**(D)** x=9,y =3

Q121.

CUSAT CAT 2023 - Shift 3

The equation of the circle which passes through the origin has its centre on the line 2 2 x+ y =4 and cuts the circle x + y –4x+2y+4 orthogonally, is

(A) x 2 + y 2 –2x–6y =0	**(B)** x 2 + y 2 –6x–3y =0
(C) x 2 + y 2 –4x–4y =0	**(D)** x 2 + y 2 +4x+4y =0 4 4

Q122.

CUSAT CAT 2023 - Shift 3

The number of integral roots of the equation x + x +20 =22 is

(A) 0	**(B)** 2
(C) 4	**(D)** 8

Q123.

CUSAT CAT 2023 - Shift 3

The line x + y = 1 meets the lines represented by the equation 3 2 2 3 y –xy –14x y+24x =0 at the points A,B,C. If O is the origin, 2 2 2 then OA +OB +OC is equal to 22

(A) 9 85	**(B)** 72 181
(C) 72 221	**(D)** 72

Q124.

CUSAT CAT 2023 - Shift 3

By definition a ∗ b = b ∗ a = 1, ∀ a, b ∈ R. Also (a ∗ b) ∗ c = (1 ∗ c) =1 and a ∗ (b ∗ c) = a ∗ (1) = 1, ∀ a, b, c ∈ R. Then R is

(A) only commutative	**(B)** only associative
(C) reflexive and commutative	**(D)** commutative and associative

Q125.

CUSAT CAT 2023 - Shift 4

The value of lim + +...+ is n→∞1–n 4 1–n 4 1–n 4 1

(A) 4 1	**(B)** 8 1
(C) 2 –1	**(D)** 4

Q126.

CUSAT CAT 2023 - Shift 4

Given f(x)= and f(x+5)= f(x) for all x. Then = x+2; 2≤ x≤5 f(11)+ f(–11) –5

(A) 7 –7	**(B)** 5 7
(C) 5 15	**(D)** 7 2 2 2 2 2 2

Q127. *CUSAT CAT 2023 - Shift 4*

Sum of the series S =1 –2 +3 –4 =...–2008 +2009 is

(A) 2019045	**(B)** 1005004
(C) 2000506	**(D)** 2026042

Q128. *CUSAT CAT 2023 - Shift 4*

If log log log x =1, then x is 4 3 2

(A) 2 64	**(B)** 9
(C) 24	**(D)** 2 81

Q129. *CUSAT CAT 2023 - Shift 4*

If a =2, b =5 and a×b =8 then a.b is equal to

(A) 4	**(B)** 6
(C) 5	**(D)** 8

Q130. *CUSAT CAT 2023 - Shift 4*

The difference of a whole number consisting of two digits from the number formed by inter changing the digits is always divisible by

(A) 11	**(B)** 10
(C) 9	**(D)** 6

Q131. *CUSAT CAT 2023 - Shift 4*

Let f :■→■ be any function. Define g:■→■ by g(x)= f(x) for all x. Then which of following is true?

(A) g is 1 – 1 if f is 1 – 1	**(B)** g may be bounded even if f is unbounded
(C) g is continuous if f is continuous	**(D)** g is differentiable if f is differentiable dy x+y y

Q132. *CUSAT CAT 2023 - Shift 4*

Let A be the transpose of A. If A= –3 y –7 and A=–A then x+ y is equal to –2 7 0

(A) 2	**(B)** –1
(C) 0	**(D)** 12 1 1 dy

Q133. *CUSAT CAT 2023 - Shift 4*

Let f(x)= . Then 4 2 x –26x +25

(A) f is continuous on [6, 10]	**(B)** f is continuous on [–2, 2]
(C) f is continuous on [–6, 6]	**(D)** f is continuous on [1, 7] z–5i

Q134. *CUSAT CAT 2023 - Shift 4*

If the points (2a, a), (a, 2a) and (a, a) enclose a triangle of area 72, then the coordinates of the centroid of the triangle is

(A) (16, 16)	**(B)** (4, 4)
(C) (12, 12)	**(D)** (–4, –4)

Q135. *CUSAT CAT 2023 - Shift 4*

The value of k for which the points A = (1, 0, 3), B = (–1, 3, 4), C = (1, 2, 1) and D = (k, 2, 5) are coplanar is

(A) 1	**(B)** 2
(C) 0	**(D)** –1

Q136. *CUSAT CAT 2023 - Shift 4*

The number of real solutions of + = + is x+1 x+5 x+2 x+4

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q137. *CUSAT CAT 2023 - Shift 4*

If the points P(h, k) and Q(k, h) lie on 3x + 2y – 13 = 0 and 4x – y – 5 = 0 respectively, then the equation of PQ is

(A) x– y =6	**(B)** x+ y =5
(C) x– y+5=0	**(D)** 2x+2y =5

Q138. *CUSAT CAT 2023 - Shift 4*

The projection of aon bis twice the projection of b on a if

(A) a =b	**(B)** a = b
(C) a – b = b	**(D)** a – b = a

Q139. *CUSAT CAT 2023 - Shift 4*

A line which is parallel to the x-axis and crosses the curve y = x at an angle of 45° is 1

(A) x= 4 1	**(B)** y = 4 1
(C) y = 2	**(D)** y =1

Q140. *CUSAT CAT 2023 - Shift 4*

A function, which is differentiable everywhere, is

(A) f(x)= x 1	**(B)** f(x)= x
(C) f(x)=(x+3) 4	**(D)** f(x)=lnx x –x

Q141. *CUSAT CAT 2023 - Shift 4*

The orthocentre of the triangle formed by the lines xy = 0 and 2x + 3y – 5 = 0 is

(A) (2, 3)	**(B)** (3, 2)
(C) (0, 0)	**(D)** (5,–5) ( ) ( ) ( )

Q142. *CUSAT CAT 2023 - Shift 4*

The number of times the function y = x +1 is to be differentiated in order to get a polynomial of degree 50 is

(A) 70	**(B)** 80
(C) 110	**(D)** 30

Q143. *CUSAT CAT 2023 - Shift 4*

If the centroid of the triangle with vertices (3c + 2, 2, 0), (2c, –1, –1) and (c + 2, 3c + 1, c + 3) lies in the plane z = c, then the coordinates of the centroid are 2 1 1

(A) – ,– , 3 3 3 10 5	**(B)** , ,1 3 3 4 2 2
(C) , , 3 3 3 2 1 1	**(D)** , ,– 3 3 3

Q144. *CUSAT CAT 2023 - Shift 4*

The length of the perpendicular drawn from the point (3, –1, 11) to the line x y–2 z–3 = = is 2 3 4

(A) 29	**(B)** 33
(C) 53	**(D)** 66

Q145. *CUSAT CAT 2023 - Shift 4*

If f(x+ y)= f(xy) for all x,y∈■ and f(2003)=2003, then f(–2003) equals

(A) 2003	**(B)** 0
(C) –2003	**(D)** 4006

Q146. *CUSAT CAT 2023 - Shift 4*

If x=–2 and Δ= 5x+4y 4x 2x , then the numerical value of Δ is equal to 10x+8y 8x 3x

(A) 8	**(B)** –8
(C) 4	**(D)** –4 2

Q147. *CUSAT CAT 2023 - Shift 4*

If A and B are independent events of a random experiment such that P

(A) —	**(B)** = and 7 ( ) P A∪B =0.8, then P
(C) 0.4	**(D)** 0.3

Q148.

CUSAT CAT 2023 - Shift 4

If 1– + – +...= then the value of + + +... is 3 5 7 4 1.3 5.7 9.11 π

(A) 8 π	**(B)** 6 π
(C) 4 π	**(D)** 34 1 1 1 1

Q149.

CUSAT CAT 2023 - Shift 4

The sum lim S if S = + + +...+ is n n n→∞ 2n 2 2 2 4n –1 4n –4 3n +2n–1 π

(A) 2	**(B)** 2
(C) 1 π	**(D)** 6 101 ∑ k

Q150.

CUSAT CAT 2023 - Shift 4

If z–1 = z–2i = z+1, then the value of z is

(A) 1	**(B)** 2 5
(C) 4 3	**(D)** 4 α 2

Q151.

CUSAT CAT 2023 - Shift 4

Let α,β be roots of the equation x – px+r =0 and ,2β be the roots of the 2 2 equation x –qx+r =0. Then the value of r is 2

(A) (p–q)(2q– p) 9 2	**(B)** (q– p)(2p–q) 9 2
(C) (q–2p)(2q– p) 9 2	**(D)** (2p–q)(2q– p) 9 z–5i

Q152.

CUSAT CAT 2023 - Shift 4

If 4 x –3 2 =3 2 –2 2x–1 , then value of x is 5

(A) 2	**(B)** 2 3
(C) 2	**(D)** 4 7

Q153.

CUSAT CAT 2023 - Shift 4

If A= { 4 n –3n–1:n∈ ■} and B={9n–9,n∈ ■} then A∪B is equal to

(A) B	**(B)** A
(C) ■	**(D)** A∩B 2 2

Q154.

CUSAT CAT 2023 - Shift 4

The number of integral solutions of x + y = x y is

(A) 0	**(B)** 1
(C) infinite	**(D)** None of the above x if (x) is rational

Q155. *CUSAT CAT 2023 - Shift 4*

The order of –3 in the group (Z,+) is

(A) 3	**(B)** 0
(C) 1	**(D)** ∞ 2

Q156. *CUSAT CAT 2023 - Shift 4*

If y =alogx+bx +x has its extreme values at x=1 and x=2,then the values of ‘a’ and ‘b’ are 2 1

(A) a =– , b=– 3 6 1 4	**(B)** a =– , b= 6 3 4 1
(C) a =– , b= 3 6 4 1	**(D)** a = , b=– 3 6

Q157. *CUSAT CAT 2023 - Shift 4*

The angles of elevation of the top of a tower from the top and bottom of a building 100 meters tall are 30° and 45°. The height of the tower is

(A) 100 3meters ( )	**(B)** 50 3+ 3 meters ( )
(C) 50 3+1 meters ( )	**(D)** 100 3+1 meters

Q158. *CUSAT CAT 2023 - Shift 4*

If z ,z ,z are the vertices of an isosceles triangle, right angled at the vertex z , then 1 2 3 2 2 2 ( ) ( ) value of z –z + z –z is 1 2 2 3

(A) –1	**(B)** 0 2
(C) (z –z) 1 3	**(D)** 1

Q159. *CUSAT CAT 2023 - Shift 5*

Let Δ= 1 –2 5 . The solution set of Δ = 0 is 2 1 2x 5x

(A) {–2,3}	**(B)** {–3,4}
(C) {4,–6}	**(D)** {–2,–1} (2) (2) f 3+h – f 3–h

Q160. *CUSAT CAT 2023 - Shift 5*

If f '(3)=2, then lim is h→0 2h 2

(A) 1	**(B)** 2
(C) 3 1	**(D)** 2

Q161. *CUSAT CAT 2023 - Shift 5*

What is the sum of all two digit numbers?

(A) 4900	**(B)** 4895
(C) 4905	**(D)** 4985 2

Q162.

CUSAT CAT 2023 - Shift 5

For any real x, the range of the function f(x)=3x +5x+7 is 59

(A) ,∞ 2 59	**(B)** ,∞ 12 59
(C) –∞, 12 12	**(D)** –∞, 59

Q163.

CUSAT CAT 2023 - Shift 5

If a vertex of a triangle is (1, 1) and the mid points of two sides through this vertex are (–1, 2) and (3, 2) ,then the centroid of the triangle is 7

(A) 1, 3 1 7	**(B)** , 3 3 7
(C) –1, 3 –1 7	**(D)** , 3 3 2

Q164.

CUSAT CAT 2023 - Shift 5

If f(x)=2x –2x+4 and f(2α)=4f(α), then αis equal to

(A) 4	**(B)** 3
(C) 0	**(D)** –2 d (x)

Q165.

CUSAT CAT 2023 - Shift 5

The period of the function f(x)= x + 2x + 3x +....nx – , where n∈N 2 [ ] and denotes the greatest integer function, is

(A) 1	**(B)** n 1
(C) n	**(D)** 2n

Q166.

CUSAT CAT 2023 - Shift 5

If one root of the equation x 2 + px+12=0 is 4 and the equation x 2 + px+q=0 has equal roots, then the value of q is

(A) 3	**(B)** 12 49
(C) 4 39	**(D)** 4

Q167.

CUSAT CAT 2023 - Shift 5

The points (a,b+c),(b,c+a) and (c,a+b) are

(A) vertices of an equilateral triangle	**(B)** concyclic
(C) vertices of a right angled triangle	**(D)** collinear 3x+1 1

Q168.

CUSAT CAT 2023 - Shift 5

If f(x)= , then the roots of the equation f(x)+ f =0are 3x–1 x 1

(A) 3, – 3 1	**(B)** –3, 3 1
(C) 3, 3 1	**(D)** –3, – 3 2 1 a a –bc 2

Q169. *CUSAT CAT 2023 - Shift 5*

Let ∆= 1 b b –ca . Then ∆ is equal to 2 1 c c –ab

(A) 0	**(B)** a+b+c 1(2 2 2)
(C) a +b +c 2	**(D)** 3 1 1 x +2 x +3 x +...+n x x

Q170. *CUSAT CAT 2023 - Shift 5*

lim is x→0 n

(A) n!	**(B)** (n–1)!
(C) n 1	**(D)** (n !) n

Q171. *CUSAT CAT 2023 - Shift 5*

In a network of railways, a small island has 15 stations. The number of different types of tickets to be printed for each class, if every station must have tickets for other station, is

(A) 210	**(B)** 230
(C) 310	**(D)** 340 dy 2y

Q172. *CUSAT CAT 2023 - Shift 5*

Identify the function for the given below: 1

(A) y = x 1	**(B)** y= x–3 1
(C) y= x+3 1	**(D)** y = 2 x 2 x th

Q173. *CUSAT CAT 2023 - Shift 5*

The least value of n so that y = y where y = x +e and y denotes the n n n+1 n order derivative of y, is

(A) 4	**(B)** 3
(C) 5	**(D)** 2

Q174. *CUSAT CAT 2023 - Shift 5*

The ratio in which the line segment joining (2, –3) and (5, 6) is divided by the x-axis is

(A) 3 : 1	**(B)** 3:2
(C) 1 : 2	**(D)** 2 :3 x (x+3)

Q175. *CUSAT CAT 2023 - Shift 5*

Sum of the roots of the equation 4 –3 2 +128=0 is

(A) 5	**(B)** 6
(C) 7	**(D)** 8 b+c q+r y+z x a b

Q176.

CUSAT CAT 2023 - Shift 5

Let Δ= c+a r+ p z+x and Δ = y b q . Then 1 a+b b+q x+ y z c r

(A) Δ=2Δ 1	**(B)** Δ=–2Δ 1
(C) Δ=4Δ 1	**(D)** Δ=–4Δ 1

Q177.

CUSAT CAT 2023 - Shift 5

If the area of triangle formed by the line 4x+3y+c=0 and the axes of co-ordinates is 6, then c is equal to

(A) ±18	**(B)** ±16
(C) ±12	**(D)** ±8

Q178.

CUSAT CAT 2023 - Shift 5

If a, b, c are in A.P., then the line ax + by + c = 0 will always pass through the point

(A) (2, 1)	**(B)** (1, 2)
(C) (1, –2)	**(D)** (–2, –1)

Q179.

CUSAT CAT 2023 - Shift 5

The point(s), at which the function given by f(x)= x is continuous –1, x≥0

(A) x∈R	**(B)** x=0
(C) x∈R–{0}	**(D)** x=–1 and 1 1 2 3 30 31 x

Q180.

CUSAT CAT 2023 - Shift 5

If E = ... =8 , then the value of x is 4 6 8 62 64

(A) –7	**(B)** –9
(C) –10	**(D)** –12 x2 y2

Q181.

CUSAT CAT 2023 - Shift 5

If u=e y2 +e x2 , then x ∂u + y ∂u = ∂x ∂y

(A) u ∂ 2u	**(B)** ∂x∂y 1 1
(C) + x y	**(D)** 0 π

Q182.

CUSAT CAT 2023 - Shift 5

The population p(t)dt time ‘t’ of a certain mouse species satisfies the differential d equation p(t)=0.5p(t)–450. If p(0) = 850, then the time at which the population dt becomes zero is

(A) log9 1	**(B)** log18 2
(C) log18	**(D)** 2log18

Q183. *CUSAT CAT 2023 - Shift 5*

The equation of the plane through (3, 4, –1) which is parallel to the plane r i(2i–3j+5k) +7=0 is

(A) r i(2i–3j+5k) +11=0	**(B)** r i(3i+4j–k) +11=0
(C) r i(3i+4j–k) +7=0	**(D)** r i(2i–3j+5k) –7=0

Q184. *CUSAT CAT 2023 - Shift 5*

The reflection of the point A(1,0,0) in the line = = is 2 –3 8

(A) (3, –4, –2)	**(B)** (5, –8, –4)
(C) (1, –1, –10)	**(D)** (2, –3, 8) m 10 20 p ∑

Q185. *CUSAT CAT 2023 - Shift 5*

The sum (where =0 if p < q) is maximum when m is i=0 i m–i q

(A) 5	**(B)** 10
(C) 15	**(D)** 20 x–1 y–2 z+3

Q186. *CUSAT CAT 2023 - Shift 5*

The number of distinct real values of λ for which the lines = = 1 2 λ2 x–3 y–2 z–1 and = = are coplanar is 1 λ2 2

(A) 2	**(B)** 4
(C) 3	**(D)** 1

Q187. *CUSAT CAT 2023 - Shift 5*

If P

(A) = , P	**(B)** = , P(A∪B)= , then P(A∩B)= 4 2 8 1
(C) 8 5	**(D)** 8 –1 2

Q188. *CUSAT CAT 2023 - Shift 5*

The point on the curve y = x which is closest to the point 4, is 2

(A) (1, 1)	**(B)** (2, 4) 2 4
(C) , 3 9 4 16	**(D)** , 3 9 1

Q189. *CUSAT CAT 2023 - Shift 5*

If A and B are independent events of a random experiment such that P(A∩B)= 6 1 and P(A∩B)= , then P

(A) is equal to 3 1	**(B)** 6 2
(C) 3 1	**(D)** 3

Q190. *CUSAT CAT 2023 - Shift 5*

If letters of the word SACHIN are arranged in all possible ways and are written out as in a dictionary, then the word SACHIN appears at serial number

(A) 603	**(B)** 602
(C) 601	**(D)** 600 1 e x e+ A

Q191. *CUSAT CAT 2023 - Shift 5*

Let f be a nonzero continuous function satisfying f(x+ y)= f(x)f(y) for all x,y∈■. If f(2)=9. Then f(3) is

(A) 1	**(B)** 27
(C) 9	**(D)** 26 π

Q192. *CUSAT CAT 2023 - Shift 5*

If the product of the roots of the equation x 2 –5kx+2e 4lnk –1=0 is 31, then the sum of the roots is

(A) –10	**(B)** 5
(C) –8	**(D)** 10 1 0

Q193. *CUSAT CAT 2023 - Shift 5*

If z+2 z =π+4i, then Im(z) equals

(A) π	**(B)** π
(C) π 2 +16	**(D)** 4

Q194. *CUSAT CAT 2023 - Shift 5*

The inequality z–4 < z–2 represents the region given by

(A) Re(z)≥0	**(B)** Re(z)<3
(C) Re(z)≤0	**(D)** Re(z)>3 x 3 7

Q195. *CUSAT CAT 2023 - Shift 5*

If x=–9 is a root of 2 x 2 =0,then the other two roots are 7 6 x

(A) 3, 7	**(B)** 2, 7
(C) 3, 6	**(D)** 2, 6 z

Q196. *CUSAT CAT 2023 - Shift 5*

If is a root of ax 2 +bx+c=0, (a,b,c∈■), then 3–4i

(A) b+6c=0	**(B)** b=6c
(C) a+25c=0	**(D)** b 2 =c 2 1 ω ω

Q197. *CUSAT CAT 2023 - Shift 5*

The equation e –e =4has

(A) no real roots	**(B)** exactly one real root
(C) exactly four real roots	**(D)** infinite number of real roots π 2 2

Q198. *CUSAT CAT 2023 - Shift 5*

For the equation 3x + px+3=0,p>0,if one of the roots is square of the other, then p is equal to 1

(A) 3	**(B)** 1
(C) 3 2	**(D)** 3

Q199. *CUSAT CAT 2023 - Shift 5*

If z = then z equals 3–4i

(A) 2 7 7	**(B)** 2 i 7
(C) –2 i	**(D)** –2 7

Q200. *CUSAT CAT 2023 - Shift 5*

Value of x= 6+ 6+ 6+...upto ∞ is

(A) 3	**(B)** 2
(C) 1	**(D)** 4 π –1

Q201. *CUSAT CAT 2023 - Shift 5*

The equation z = z has

(A) no solution	**(B)** two solutions
(C) five solutions	**(D)** infinite number of solutions

Q202. *CUSAT CAT 2023 - Shift 5*

The number of points at which the function f(x)= is not continuous is x–[x]

(A) 1	**(B)** 2
(C) 3	**(D)** infinitely many

Q203. *CUSAT CAT 2024 - Shift 1*

The value of 35 7 4 is 17 3 2

(A) 1	**(B)** 0
(C) 2	**(D)** 3 1 1 1

Q204.

CUSAT CAT 2024 - Shift 1

The infinite series + + +... is convergent if p p p 1 2 3

(A) p > 1	**(B)** p = 1
(C) p < 1	**(D)** p ≤ 1

Q205.

CUSAT CAT 2024 - Shift 1

The area between the curvey = x –2x and the straight line y = x, is 3

(A) sq.units 2 5	**(B)** sq.units 2 7
(C) sq.units 2 9	**(D)** sq.units 2 1

Q206.

CUSAT CAT 2024 - Shift 1

The positive integer n so that lim =32 is x→2 x–2

(A) n = 2	**(B)** n = 3
(C) n = 4	**(D)** n = 1 1

Q207.

CUSAT CAT 2024 - Shift 1

If f : R → R is the function defined by f(x)= (3–x 3)3, then (f f)(x) is equal to 1

(A) x3 3	**(B)** x
(C) x (3)	**(D)** 3–x

Q208.

CUSAT CAT 2024 - Shift 1

The area of the triangle whose vertices are (3, 5), (1, 1), (–4, 1) is

(A) –20	**(B)** 0
(C) 10	**(D)** 20

Q209.

CUSAT CAT 2024 - Shift 1

log (e) is e2

(A) 0.4343	**(B)** 1
(C) 0	**(D)** 0.5 3 x

Q210.

CUSAT CAT 2024 - Shift 1

The critical point of f(x)= –4x is at 3

(A) x=4	**(B)** x=3
(C) x=2	**(D)** no point x

Q211.

CUSAT CAT 2024 - Shift 1

1 28 28 = 3 1 29 29

(A) 168	**(B)** 268
(C) 368	**(D)** 468

Q212.

CUSAT CAT 2024 - Shift 1

A circle of radius n units touches both the axes. Then equations of all possible circles formed in the general form 2 2 2

(A) x + y = n 2 2	**(B)** x + y + 2nx + 2ny + n = 0 2 2 2
(C) x + y ± 2nx ± 2ny ± n = 0 2 2 2	**(D)** x + y ± 2nx ± 2ny + n = 0 1 –1

Q213.

CUSAT CAT 2024 - Shift 1

lim xx–1= x→1

(A) ∞	**(B)** e
(C) 1	**(D)** π log2 –x

Q214.

CUSAT CAT 2024 - Shift 1

The mean of number of heads in two tosses of a coin is

(A) 0	**(B)** 1 1
(C) 4 3	**(D)** 4 1

Q215.

CUSAT CAT 2024 - Shift 1

The function f has the property that for each real number x in its domain, is also x 1 in its domain and f(x)+ f = x. Then the largest set of real numbers that can be in x the domain of f is

(A) {x|x≠0}	**(B)** [–1, 1]
(C) {x|x>0}	**(D)** {–1, 1}

Q216.

CUSAT CAT 2024 - Shift 1

If x and y are positive integers for which 2 3 = 1296 , then the value of x + y is

(A) 8	**(B)** 9
(C) 10	**(D)** 11

Q217.

CUSAT CAT 2024 - Shift 1

The total number of proper factors of 2024 is

(A) 14	**(B)** 15
(C) 16	**(D)** 17 8, if n isdivisibleby11and23 11, if n isdivisibleby8and23

Q218.

CUSAT CAT 2024 - Shift 1

For an integer n, with 1≤n≤2024, let a = . n 23, if n isdivisibleby8and11 0, otherwise 2024 ∑ Then a = n n=1

(A) 448	**(B)** 486
(C) 672	**(D)** 2024 24

Q219.

CUSAT CAT 2024 - Shift 1

lim , m∈■ is x→∞ x m

(A) 0	**(B)** 1
(C) e	**(D)** ∞

Q220.

CUSAT CAT 2024 - Shift 1

Integers x and y with x > y > 0 satisfy x + y + xy = 80 . Then x is

(A) 26	**(B)** 27
(C) 80	**(D)** 81

Q221.

CUSAT CAT 2024 - Shift 1

The variance of first 30 natural numbers is

(A) 899 2	**(B)** 899 4
(C) 899 12	**(D)** 899 16

Q222.

CUSAT CAT 2024 - Shift 1

For a positive integer n ≥ 2 satisfying n < 10 , we have 4 n – 2

(A) (n – 2) < 10 4 n – 1	**(B)** (n + 1) < 10 2
(C) n < 10 4 n + 1	**(D)** (n + 1) < 10 1

Q223.

CUSAT CAT 2024 - Shift 1

Solution set of x+ <4 is x ( ) ( )

(A) x∈ 2– 3,2+ 3 ∪ –2– 3,–2+ 3 ( )	**(B)** R– 2– 3,2+ 3 ( )
(C) R– –2– 3,–2+ 3 (( ) ( ))	**(D)** R– 2– 3,2+ 3 ∪ –2– 3,–2+ 3 2 2

Q224.

CUSAT CAT 2024 - Shift 1

Let a= and b= . Then 7+ 5 13+ 11

(A) a = b	**(B)** a < b
(C) a > b	**(D)** a= 2b a–ib

Q225.

CUSAT CAT 2024 - Shift 1

The equation zz +(2–3i)z+(2+3i)z +4=0 represents a circle of radius

(A) 2	**(B)** 3
(C) 4	**(D)** 6

Q226.

CUSAT CAT 2024 - Shift 1

The number of triangles formed by the lines represented by x – x – x – 2 = 0 2 2 and xy + 2xy + 4x – 2y – 4y – 8 = 0 is

(A) One	**(B)** Two
(C) Three	**(D)** Zero 2

Q227.

CUSAT CAT 2024 - Shift 1

The roots of the equation 3 + 3 = 3 + 3 are

(A) 1, –2	**(B)** 1, 2
(C) –1, 2	**(D)** –1, –2 2

Q228.

CUSAT CAT 2024 - Shift 1

The number of roots of the equation | x | = x + x – 4 is

(A) 1	**(B)** 2
(C) 3	**(D)** 4 n P n P n P

Q229.

CUSAT CAT 2024 - Shift 1

If r–1 = r = r+1 , then a b c

(A) a – b + c = 1	**(B)** abc = 1 2
(C) b = a(b + c) 2	**(D)** a = c(a + b) 8 { 6 7 15}

Q230.

CUSAT CAT 2024 - Shift 1

If A= x 1 0 and I = 0 1 0 , then A –3A +3A+I is equal to x x 1 0 0 1

(A) –2I	**(B)** I
(C) 2I	**(D)** 3I 1 1.3 1.3.5

Q231.

CUSAT CAT 2024 - Shift 1

The sum of the series + + +...+∞ is 1.2 1.2.3.4 1.2.3.4.5.6

(A) e	**(B)** e – 1
(C) e – 2	**(D)** e + 2 1+x 1–x

Q232. *CUSAT CAT 2024 - Shift 1*

If f(x)=log , then f(a)+ f(b) is equal to 1–x a+b

(A) f 1–ab a+b	**(B)** f 1+ab
(C) 0	**(D)** 1

Q233. *CUSAT CAT 2024 - Shift 1*

The chances to fail in Physics are 20% and the chances to fail in Mathematics are 10%. The percentage of chance to fail in at least one subject is

(A) 72%	**(B)** 38%
(C) 28%	**(D)** 82% –5 20 (2)40 2 1

Q234. *CUSAT CAT 2024 - Shift 1*

The largest interval for which x –x +x –x+1>0 is

(A) –4< x≤0	**(B)** 0 < x <1
(C) –100 < x < 100	**(D)** –∞< x<∞ 2 2 2 2 2

Q235. *CUSAT CAT 2024 - Shift 1*

Let f : R → R be a function satisfying f (2x + 3) + f (2x + 7) = 2 for all x ∈ R. Then the period of f (x) is

(A) 2	**(B)** 4
(C) 8	**(D)** 12

Q236. *CUSAT CAT 2024 - Shift 1*

Let a, b, x, y be real numbers such that a 2 + b 2 = 25, x 2 + y 2 = 169, and ax + by = 65. If k = ay – bx, then 5

(A) k > 13	**(B)** k =0 5
(C) 0<k ≤ 13 5	**(D)** k = 13

Q237. *CUSAT CAT 2024 - Shift 1*

Let x and y be positive real numbers such that log (x+ y)+log (x– y)=3, and 5 5 log y–log 3. Then xy is equal to 2 2

(A) 150	**(B)** 100
(C) 250	**(D)** 25 ∑∞

Q238. *CUSAT CAT 2024 - Shift 1*

If f(x+ y)= f(x)· f(y) and f(x)=2, x,y∈N, where N is the set of all x=1 f(4) natural numbers, then the value of is f(2) 1

(A) 9 4	**(B)** 9 1
(C) 3 2	**(D)** 3 2x –1

Q239. *CUSAT CAT 2024 - Shift 1*

Area of the parallelogram whose sides are 2x+ y+1=0, 2x+ y+4=0, x–3y–1=0 and x–3y+2=0 is equal to 7

(A) sq.units 9 9	**(B)** sq.units 7 5
(C) sq.units 7 7	**(D)** sq.units 5 n

Q240. *CUSAT CAT 2024 - Shift 1*

The largest term of the sequence given by a = , n∈N is n 4 n +1875 1

(A) 200 1	**(B)** 500 5
(C) 727 1	**(D)** 20 3

Q241. *CUSAT CAT 2024 - Shift 1*

Let the mean of 6 observations 1,2,4,5,x and y be 5 and their variance be 10. Then x is equal to

(A) 4	**(B)** 6
(C) 8	**(D)** 10 –1 –1

Q242. *CUSAT CAT 2024 - Shift 1*

If e –e =4 then the number of real values of x is

(A) 0	**(B)** 1
(C) 2	**(D)** infinite 3 n

Q243. *CUSAT CAT 2024 - Shift 1*

If a is the greatest term in the sequence a = , n=1,2,3,..., then n is equal to n n 4 n +147

(A) 5	**(B)** 6
(C) 4	**(D)** 8

Q244. *CUSAT CAT 2024 - Shift 2*

The value of ω 1 ω is 2 ω ω 1

(A) 0	**(B)** 1
(C) 2	**(D)** 3 1 1 1 1

Q245. *CUSAT CAT 2024 - Shift 2*

The value of lim = x→0 x 3 1

(A) 3 1	**(B)** 4 1
(C) 5 1	**(D)** 6

Q246. *CUSAT CAT 2024 - Shift 2*

The function f : R → R defined by f (x) = x + 1 is

(A) surjective only	**(B)** injective only
(C) both surjective and injective	**(D)** neither surjective nor injective

Q247. *CUSAT CAT 2024 - Shift 2*

The number of common terms in the two sequences 15, 19, 23, 27, …., 415 and 14, 19, 24, 29, …, 464 is

(A) 18	**(B)** 21
(C) 20	**(D)** 19

Q248. *CUSAT CAT 2024 - Shift 2*

If log(xy 3)=1, log(x 2 y)=1, then log(xy) is equal to

(A) 1 3	**(B)** 5 4
(C) 5	**(D)** 2

Q249. *CUSAT CAT 2024 - Shift 2*

Area bounded by the function f(x)=[x], where [x] is the greatest integer less than or equal to x , in the interval [2, 3] is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q250. *CUSAT CAT 2024 - Shift 2*

Consider the system of linear equations x + 2y + z = 7 ; x + y + λz = 3 ; x + 3y – 5z = 5. This system has unique solution for

(A) any real number λ	**(B)** λ = 7
(C) any real number except λ = 7	**(D)** no value of λ 4C 4C 4C 1 2 3

Q251. *CUSAT CAT 2024 - Shift 2*

5C 5C 5C = 1 2 3 6C 6C 6C 1 2 3

(A) 18	**(B)** 19
(C) 20	**(D)** 21

Q252. *CUSAT CAT 2024 - Shift 2*

Let A and B be any two events. Which of the following is not true?

(A) P(B– A)= P(A∩B)	**(B)** P(A∪B)= P
(C) P(A–B)= P(A∪B)	**(D)** P(A∪B)= P

Q253. *CUSAT CAT 2024 - Shift 2*

Kiran ride a bicycle for 30 minutes at a speed of 16 km/hr, then walked for 90 minutes at speed of 4 km/hr. The overall average speed is

(A) 7	**(B)** 9
(C) 10	**(D)** 12 2 2

Q254. *CUSAT CAT 2024 - Shift 2*

In four throws with pair of dice, the chance of throwing doublets at least twice is 5

(A) 432 7	**(B)** 432 11
(C) 432 13	**(D)** 432

Q255. *CUSAT CAT 2024 - Shift 2*

The number of real values of x for which 120– x is an integer, is

(A) 6	**(B)** 9
(C) 10	**(D)** 11 2

Q256. *CUSAT CAT 2024 - Shift 2*

For a positive integer n, let n – 3n + 2 be prime. Then number of possible values of n is

(A) 0	**(B)** 1
(C) 3	**(D)** infinitely many

Q257. *CUSAT CAT 2024 - Shift 2*

Let S (n) equal the sum of the digits of positive integer n. For a particular integer n , let S (n) = 1274 . Then the possible value of S(n + 1) is

(A) 1	**(B)** 3
(C) 1239	**(D)** 1265

Q258. *CUSAT CAT 2024 - Shift 2*

The sum of three whole numbers taken in pairs are 12, 17 and 19 . The middle number is

(A) 4	**(B)** 5
(C) 6	**(D)** 7 1 n!n

Q259. *CUSAT CAT 2024 - Shift 2*

lim equals n n→∞n

(A) e 1	**(B)** e π
(C) 4 4	**(D)** π

Q260.

CUSAT CAT 2024 - Shift 2

The value of x satisfying the condition |x + 3| > |2x – 1| lies in the interval 2

(A) – ,4 3 2	**(B)** –∞, 3
(C) (0, 2)	**(D)** (0, ∞) 1 1 1

Q261.

CUSAT CAT 2024 - Shift 2

For a, b, c > 0, the equations a(1 – b) > , b(1 – c) > and c(1–a)> are 4 4 4

(A) consistent for some values	**(B)** inconsistent
(C) consistent	**(D)** inconsistent for some values b+c c+a a+b

Q262.

CUSAT CAT 2024 - Shift 2

If x – x + 1 = 0, then the value of x + is n x n=1

(A) 8	**(B)** 10
(C) 12	**(D)** 16 z+2i

Q263.

CUSAT CAT 2024 - Shift 2

If Im =0, then z lies on the curve z+2 2 2

(A) x + y + 2x + 2y = 0 2 2	**(B)** x + y – 2x = 0
(C) x + y + 2 = 0 2 2	**(D)** x + y + 2x = 0

Q264.

CUSAT CAT 2024 - Shift 2

In an AP, the notation S denotes the sum of first r terms, S = q and S = p. Then r p q S is equal to p + q

(A) 0	**(B)** –(p + q)
(C) p + q	**(D)** pq m n p

Q265.

CUSAT CAT 2024 - Shift 2

The sum of the series 1 + 2x + 3x + 4x +. . . up to infinity is 1

(A) 1+x 1	**(B)** 1–x 1
(C) 1–2x 1	**(D)** 2 (1–x) 2 x –bx k–1

Q266.

CUSAT CAT 2024 - Shift 2

If the sum of the roots of the equation = , then the value of k is ax–c k+1 a–b

(A) a+b a+b	**(B)** a–b 1
(C) c	**(D)** c 2 4 3

Q267. *CUSAT CAT 2024 - Shift 2*

If x – 1 is a factor of x + ax + 3x – b, then

(A) a = 3, b = –1	**(B)** a = –3, b = 1
(C) a = 3, b = 1	**(D)** a = 4, b = 3 2

Q268. *CUSAT CAT 2024 - Shift 2*

If α is a root of the equation 4x + 2x – 1 = 0, then the other root of the equation is 2

(A) 3α – 4α 3	**(B)** 4α – 3α 2
(C) 3α + 4α 3	**(D)** 4α + 3α 3 1 1

Q269. *CUSAT CAT 2024 - Shift 2*

The number of real roots of the equation x+ +x+ =0 is x x

(A) 0	**(B)** 2
(C) 4	**(D)** 6

Q270. *CUSAT CAT 2024 - Shift 2*

Numbers greater than 1000, but not greater than 4000 that can be formed with the digits 0, 1, 2, 3, 4 (repetition of digits being allowed) is

(A) 375	**(B)** 385
(C) 396	**(D)** 456 th 6

Q271. *CUSAT CAT 2024 - Shift 2*

Let S = where =0 when p<q. The value of m, for i=0 i m–i q which the value of the sum S is maximum, is

(A) 5	**(B)** 10
(C) 15	**(D)** 20

Q272. *CUSAT CAT 2024 - Shift 2*

The number of positive integral solutions of x + y + z = n, n ∈ ■ is n–1

(A) P 2 (n+1)(n+2)(n+3)	**(B)** 6 n–1
(C) C n–4 n–1	**(D)** C 2

Q273. *CUSAT CAT 2024 - Shift 2*

Let [x] be the greatest integer less than or equal to x. If the set of equations x – 3y = 4; 5x + y = 2; [2π]x – [e]y = [2a] are consistent, then a should lie in 7

(A) 3, 2 7	**(B)** 3, 3 7
(C) 3, 3	**(D)** [3,∞) 1 –1 x

Q274.

CUSAT CAT 2024 - Shift 2

If x is real and A= 1 x 1 has no inverse, then x is equal to x –1 1

(A) 2	**(B)** 3
(C) 0	**(D)** 1 1 1 1 2 3 4

Q275.

CUSAT CAT 2024 - Shift 2

The sum of the series 1+ + + +... is 2! 3! 4!

(A) 2e	**(B)** 3e
(C) 5e	**(D)** 6e

Q276.

CUSAT CAT 2024 - Shift 2

The inverse of the function f(x)= , is x –x 10 +10 x –1

(A) f (x)=log 10 2–x 1 x –1	**(B)** f (x)= log 10 2 2–x 1 x –1
(C) f (x)= log 10 2 1–x x –1	**(D)** f (x)=log 10 2+x

Q277.

CUSAT CAT 2024 - Shift 2

In a ΔABC, a = 2b and A–B = , the ∠C is equal to 3 π

(A) 4 π	**(B)** 3 π
(C) 6 π	**(D)** 8 4 2 –1 –1

Q278.

CUSAT CAT 2024 - Shift 2

If (2n + 1) + (2n + 3) + (2n+ 5) +…+ (2n + 47) = 5280, then the value of 1 + 2 + 3 + …+ n is

(A) 1854	**(B)** 1485
(C) 5148	**(D)** 4851 2

Q279.

CUSAT CAT 2024 - Shift 2

Let f(x)=10–4x–x and the range of f(x) be (–∞,b]. Then b+2 is equal to

(A) 0	**(B)** 4
(C) 14	**(D)** 16 (x) (x)

Q280.

CUSAT CAT 2024 - Shift 2

If log 2, log 2 +1 and log 2 +3 are in A.P, then the value of x is 10 10 10

(A) log 5 2	**(B)** log 2 5
(C) log 2 10	**(D)** 1 2

Q281. *CUSAT CAT 2024 - Shift 2*

If 1,ωandω are the cube roots of unity, then (2)(2 4)(4 8)(8 16) 1−ω+ω 1−ω +ω 1−ω +ω 1−ω +ω ...upto 2n factors are

(A) 2n	**(B)** 2 2n
(C) 1	**(D)** −2 2n 680 1 1

Q282. *CUSAT CAT 2024 - Shift 2*

Given E(X +c)=12 and E(X −c)=8 then the value of c is

(A) 4	**(B)** 2
(C) −2	**(D)** 0 a*b

Q283. *CUSAT CAT 2024 - Shift 2*

Let A= a , where a ≠0 for all i, j and A = I. Let a be the sum of all ij ij 2×2 2 2 diagonal elements of A and b= A. Then, 4a +3b is equal to

(A) 4	**(B)** 3
(C) 7	**(D)** 14

Q284. *CUSAT CAT 2024 - Shift 2*

The value of k for which the image of (5,1,−3) by reflection in 2x+ y−3z+k =0 is the point (1,−1,3) is

(A) 3	**(B)** 1
(C) −6	**(D)** 2

Q285. *CUSAT CAT 2024 - Shift 3*

If u n = n2 , the series ∑ u n is convergent, if n l → im ∞ u n n 1 1+ n

(A) < 1	**(B)** = 1
(C) > 1	**(D)** ≤ 1 π dy

Q286. *CUSAT CAT 2024 - Shift 3*

The solution of the system of the equations by Cramer's rule x + y + z = 6, 2x – 3y + 4z = 8 and x – y + 2=5, is

(A) {1, 2, 3}	**(B)** {1, 2, 4}
(C) {1, 3, 5}	**(D)** {1, 3, 4}

Q287. *CUSAT CAT 2024 - Shift 3*

The roots of the equation x – 4x – log A = 0 are real and distinct, where A is a real 2 number, if 1

(A) A< 8 1	**(B)** A< 16 1
(C) A> 16 1	**(D)** A> 8 x, x≥0

Q288.

CUSAT CAT 2024 - Shift 3

Let f(x)= . Then f is 2 x ,x<0

(A) not continuous	**(B)** continuous but not differentiable in ■
(C) both continuous and differentiable in■	**(D)** differentiable only in [–1, 1] 100

Q289.

CUSAT CAT 2024 - Shift 3

Part of the graph f (x) = ax + bx + cx + d is shown in the figure Then b is equal to

(A) 1	**(B)** 2
(C) 0	**(D)** –2 –1

Q290.

CUSAT CAT 2024 - Shift 3

For all integers n≥1, define a(n)=[log (2002)] . Let b=a(2)+a(3)+a(4)+a(5) n and c=a(10)+a(11)+a(12)+a(13)+a(14). Then b–c is equal to

(A) –2	**(B)** –1 1
(C) 2002 1	**(D)** 2

Q291.

CUSAT CAT 2024 - Shift 3

The angle between the planes 2x – y + z = 6 , x + y + 2z = 3 is π

(A) 2 π	**(B)** 3 π
(C) 4 π	**(D)** 5 2 3

Q292.

CUSAT CAT 2024 - Shift 3

If A= then A(adj A) is 5 7 1 0

(A) 0 1 –1 0	**(B)** 0 –1 0 1
(C) 1 0 0 –1	**(D)** –1 0 2

Q293.

CUSAT CAT 2024 - Shift 3

If f(x)=ax+b and f –1 (x)=bx+a for a,b∈■, then a+b=

(A) 1	**(B)** 0
(C) –1	**(D)** –2

Q294.

CUSAT CAT 2024 - Shift 3

If x * y = x – y, then h * (h * h) is equal to

(A) –h	**(B)** 0
(C) h	**(D)** 2h

Q295. *CUSAT CAT 2024 - Shift 3*

Area of the region bounded by the lines 7x – 5y = 35, x-axis, x = –2, x = 3 is 63

(A) 2 96	**(B)** 5 8
(C) 3	**(D)** 12 102 ∑ n

Q296. *CUSAT CAT 2024 - Shift 3*

i = n=1

(A) 0	**(B)** 1 + i
(C) –1 + i	**(D)** 1 – i

Q297. *CUSAT CAT 2024 - Shift 3*

A solution to the system of equations x –x = 3; 2x +3x +4x = 17; x +2x = 7 is 1 2 1 2 3 2 3

(A) (2, 1, 4)	**(B)** (2,–1, 4)
(C) (–2,–1, 4)	**(D)** (2,–1,–4)

Q298. *CUSAT CAT 2024 - Shift 3*

The product of all positive odd integers less than 10000 is 10000!

(A) 2 (5000!) 10000!	**(B)** 50000 2 9999!
(C) 50000 2 10000!	**(D)** 5000 2 ·5000!

Q299. *CUSAT CAT 2024 - Shift 3*

If g(y) = | log y | , then at y = 1 10

(A) g is not continuous	**(B)** g is continuous but not differentiable
(C) g is differentiable	**(D)** derivative is 1

Q300. *CUSAT CAT 2024 - Shift 3*

For how many number of positive integers m, does there exist at least one positive integer n such that m · n ≤ m + n ?

(A) 4	**(B)** 9
(C) 16	**(D)** infinitely many

Q301. *CUSAT CAT 2024 - Shift 3*

The infinite product 10· 10 · 10...is

(A) 10	**(B)** 3 10
(C) 10	**(D)** 10 3 10

Q302. *CUSAT CAT 2024 - Shift 3*

If p(x) is a polynomial such that sum of its co-efficients is zero, then the one of the roots is

(A) –1	**(B)** 0
(C) 1	**(D)** 2 2x

Q303. *CUSAT CAT 2024 - Shift 3*

Let f(x)=1+ x and g(x)= . Then 2 x +1

(A) domain of f +g is (–1,∞)	**(B)** domain of f +g is ■
(C) range of f ∩ range of g is {1}	**(D)** range of f ∪ range of g is [–1,1) 2

Q304. *CUSAT CAT 2024 - Shift 3*

The roots of x – 63x + k = 0 are primes. Then the number of possible values of k is

(A) 0	**(B)** 1
(C) 2	**(D)** more than 3

Q305. *CUSAT CAT 2024 - Shift 3*

Suppose log (x – 3) < 1. Then x lies in e

(A) {0, 3}	**(B)** {0, e}
(C) (3, 3 + e)	**(D)** (0, e + 3) 2 2

Q306. *CUSAT CAT 2024 - Shift 3*

Let t be a real number such that | t | < 1 and x = 6(1 – 2t). Then

(A) –6 < x < 18	**(B)** –18 < x < 6
(C) 0 < x < 18	**(D)** –6 < x < 0

Q307. *CUSAT CAT 2024 - Shift 3*

Let | z | = max{ | z – 1 |, | z + 1 | }. Then 1 1

(A) z+ = z 2	**(B)** z–z =1
(C) z+z =1 1 1	**(D)** z+ = z 4 3 π π4

Q308. *CUSAT CAT 2024 - Shift 3*

The sum of positive terms of the series 10+9 +9 +.... is 7 7 352

(A) 7 437	**(B)** 7 852
(C) 7 347	**(D)** 7

Q309. *CUSAT CAT 2024 - Shift 3*

Number of roots for the equation x+3–4 x–1 + x+8–6 x–1 =1 is

(A) 0	**(B)** 1
(C) 2	**(D)** ∞ n

Q310. *CUSAT CAT 2024 - Shift 3*

If roots of the equation x – 1 = 0 are 1, a , a , . . . , a , then the value of 1 2 n – 1 (1 – a)(1 – a) … (1 – a) is 1 2 n – 1

(A) 0	**(B)** n 2
(C) n n	**(D)** n

Q311. *CUSAT CAT 2024 - Shift 3*

Number of divisors of the form 4n + 2(n ≥ 0) of the integer 240 is

(A) 4	**(B)** 8
(C) 12	**(D)** 16

Q312. *CUSAT CAT 2024 - Shift 3*

The remainder obtained, when 1! + 2! + 3! + . . . 95! is divided by 6 , is

(A) 3	**(B)** 2
(C) 1	**(D)** 0

Q313. *CUSAT CAT 2024 - Shift 3*

The number of five-digit telephone numbers having at least one of their digits repeated is

(A) 90000	**(B)** 100000
(C) 30240	**(D)** 69760

Q314. *CUSAT CAT 2024 - Shift 3*

If A= ab –b 2 bc =ka 2 b 2 c 2 , then the value of k is 2 ac bc –c

(A) 1	**(B)** 2
(C) 3	**(D)** 4 2 2 2 3

Q315. *CUSAT CAT 2024 - Shift 3*

The sum of the series + +...+∞ is 2! 3!

(A) e	**(B)** 2e
(C) 2e + 1	**(D)** 2e – 1 π 1 1 3 5

Q316.

CUSAT CAT 2024 - Shift 3

The value of 2 +2 is 1 1–

(A) ≤2 2 1 1–	**(B)** ≥2 2 1 1+
(C) ≥2 2 1 1+	**(D)** ≤2 2 π

Q317.

CUSAT CAT 2024 - Shift 3

The area bounded by the curves y = x –1and y =– x +1 is

(A) 1	**(B)** 2
(C) 2 2	**(D)** 4

Q318.

CUSAT CAT 2024 - Shift 3

Let f(x)= . Then the derivative of the function f (f (f (x))) is 1–x

(A) 1 1	**(B)** 2
(C) 0	**(D)** 2 2

Q319.

CUSAT CAT 2024 - Shift 3

Let N be the set of natural numbers and R be the binary relation on N defined by R = {(a, b) : a = b – 2, b > 6}. Then

(A) (2, 4) ∈ R	**(B)** (3, 8) ∈ R
(C) (6, 8) ∈ R	**(D)** (8, 7) ∈ R

Q320.

CUSAT CAT 2024 - Shift 3

Let A={3,4,5}, the number of reflexive relations on A is

(A) 1	**(B)** 3
(C) 64	**(D)** 2 9 1 1 1

Q321.

CUSAT CAT 2024 - Shift 3

If a,b,c are positive real numbers, then least value of (a+b+c) + + is a b c

(A) 1	**(B)** 9
(C) 4	**(D)** 8 1

Q322.

CUSAT CAT 2024 - Shift 3

If the coefficients of x and x in (1+x) (1–x) are 4 and –5 respectively, then 2p+3q is equal to

(A) 60	**(B)** 63
(C) 66	**(D)** 69

Q323. *CUSAT CAT 2024 - Shift 3*

The minimum number of elements that must be added to the relation R={(p,q),(q,r)} on the set {p,q,r} so that it becomes symmetric, and transitive is

(A) 7	**(B)** 4
(C) 3	**(D)** 5 2 3 4 x x x π –1

Q324. *CUSAT CAT 2024 - Shift 3*

Let P= 2 2 , A= 1 1 and Q= PAP T if P T Q 2024 P= a b , then 1 3 0 1 c d – 2 2

(A) a =1, b=2024	**(B)** a =1, c=2024
(C) b=1, a =2024	**(D)** b=1, d =2024 dy

Q325. *CUSAT CAT 2024 - Shift 3*

The points z ,z ,z and z in the complex plane are the vertices of a parallelogram 1 2 3 4 taken in order, if and only if

(A) z +z = z +z 1 4 2 3	**(B)** z +z = z +z 1 3 2 4
(C) z +z = z +z 1 2 3 4	**(D)** z –z = z +z 1 2 3 4

Q326. *CUSAT CAT 2024 - Shift 4*

The value of + + +··· is 2.3 4.5 6.7

(A) 1–log2	**(B)** 1+log2
(C) log2	**(D)** 1 4 4 ∑

Q327. *CUSAT CAT 2024 - Shift 4*

The curves y =2x and 2xy =k cut the right angles if k =

(A) 4	**(B)** 6
(C) 8	**(D)** 2

Q328. *CUSAT CAT 2024 - Shift 4*

The value of lim is x→0 x

(A) 1 1	**(B)** 2 1
(C) 3 1	**(D)** 4 π 1 –1

Q329. *CUSAT CAT 2024 - Shift 4*

Two is 10% of x and 20% of y. Then x – y is

(A) 1	**(B)** 5
(C) 10	**(D)** 20 5

Q330. *CUSAT CAT 2024 - Shift 4*

Angle between the line = = and the plane x–2y–4z+7=0 is 2 3 –1

(A) 1 π	**(B)** 2
(C) π	**(D)** 0 1 2 2 2 –1

Q331. *CUSAT CAT 2024 - Shift 4*

If A= 2 1 2 satisfies A –4A–5I =0, then A is 2 2 1 –3 2 2 5 5 5 2 –3 2

(A) 5 5 5 2 2 –3 5 5 5 3 2 2 5 5 5 2 3 2	**(B)** – 5 5 5 2 2 3 5 5 5 1 2 2 5
(C) 2 1 2 5 2 2 1 5 3 2 2 5 5 5 2 3 2	**(D)** 5 5 5 2 2 3 5 5 5 2 2

Q332. *CUSAT CAT 2024 - Shift 4*

The curve of equation 8x + 4xy + 5y – 24x – 24y = 0 is

(A) parabolic	**(B)** elliptic
(C) hyperbolic	**(D)** a circle x+ex+ex+...∞ dy

Q333. *CUSAT CAT 2024 - Shift 4*

The value of c satisfied by Rolle's theorem for the function f(x)=e in [4, 5] is π

(A) 2 3π	**(B)** 2 5π
(C) 2 7π	**(D)** 2

Q334. *CUSAT CAT 2024 - Shift 4*

The two regression lines coincide if and only if

(A) r 2 =±1	**(B)** b xx ·b yy =1
(C) r =±1	**(D)** b xy =b xy

Q335. *CUSAT CAT 2024 - Shift 4*

Which of the following point is farthest from the origin?

(A) i	**(B)** 3
(C) –2 + i	**(D)** 4i

Q336. *CUSAT CAT 2024 - Shift 4*

Number of solutions of the equation z = z is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q337.

CUSAT CAT 2024 - Shift 4

Let * be defined on ■ as a*b= , a,b∈■. Then 2

(A) there exists a , b ∈■ such that a * a ∉■ 1 1 1 2	**(B)** there exists e∈■ such that a * e = a for all a∈■
(C) a * b = b * a for all a, b ∈■	**(D)** a * (b * c) = (a * b) * c for all a, b ∈■

Q338.

CUSAT CAT 2024 - Shift 4

Let f be a linear function for which f (6) – f (2) = 12 . Then f (12) – f (2) is

(A) 12	**(B)** 18
(C) 25	**(D)** 30

Q339.

CUSAT CAT 2024 - Shift 4

The largest integer that is a divisor of (n + 1)(n + 3)(n + 5)(n + 7)(n + 9) for all positive even integers n is

(A) 3	**(B)** 5
(C) 11	**(D)** 15

Q340.

CUSAT CAT 2024 - Shift 4

Define x y = |x – y| for all real numbers x and y . Which of the following is not true?

(A) x y = y x for all x and y	**(B)** 2(x y) = (2x) (2y) for all x and y
(C) x 0 = x for all x	**(D)** x y > 0 if x ≠ y

Q341.

CUSAT CAT 2024 - Shift 4

The set of all solutions of |x – 1| > |x – 3| satisfies

(A) x < 2	**(B)** x ≥ 2
(C) [1, 3]	**(D)** [3, 4] 2 3 4

Q342.

CUSAT CAT 2024 - Shift 4

The greatest value of P = a b c , when a + b + c = 18 is 4 4 4

(A) P = 4 .6 .8 2 3 4	**(B)** P = 4 .6 .8 2 3 4
(C) P = 3 .6 .8 2 2 2	**(D)** P = 4 .6 .8 3n 4π 4π 1+t

Q343.

CUSAT CAT 2024 - Shift 4

24.48.816.....∞ is equal to

(A) 2	**(B)** 1 3
(C) 2 1	**(D)** 2

Q344. *CUSAT CAT 2024 - Shift 4*

The positive integer n for which 2 × 2 + 3 × 2 + 4 × 2 + . . . + n × 2 = 2 is

(A) 510	**(B)** 511
(C) 512	**(D)** 513 1 2

Q345. *CUSAT CAT 2024 - Shift 4*

If x=2+23 +23, then the value of x 3 – 6x 2 + 6x, is

(A) 1	**(B)** 2
(C) 3	**(D)** 4 3x

Q346. *CUSAT CAT 2024 - Shift 4*

The roots of the equation 2 x+2 . 3x+1 =9 are given by 2

(A) log ,–2 2 3	**(B)** 3, –3
(C) –2, 1 – log 3 2 1	**(D)** – ,–2 9

Q347. *CUSAT CAT 2024 - Shift 4*

The number of positive integral solutions of 15 < x + x + x ≤ 20 is equal to 1 2 3

(A) 1150	**(B)** 980
(C) 715	**(D)** 685

Q348. *CUSAT CAT 2024 - Shift 4*

If A= x –1 x–1 8 =ax +bx +cx+d,then the value of c is equal to 2x 3x 0

(A) –1	**(B)** 12
(C) 15	**(D)** 17 3 7 15

Q349. *CUSAT CAT 2024 - Shift 4*

The sum of the series 1+ + + +...+∞ is 2! 3! 4!

(A) e(e+1)	**(B)** e(e–1)
(C) e(1–e)	**(D)** 3e ∞ n 2 +n+1 ∑

Q350. *CUSAT CAT 2024 - Shift 4*

The sum of the series is n! n=1

(A) 3e 3	**(B)** e 2
(C) 2e	**(D)** 4e 2 x

Q351. *CUSAT CAT 2024 - Shift 4*

If the function f : R → A given by f(x)= is a surjection, then A 2 x +1 is equal to

(A) R	**(B)** [0, 1]
(C) [0, 1)	**(D)** (0, 1)

Q352. *CUSAT CAT 2024 - Shift 4*

Consider the points A(–3, 0) and B(0, 4). Let P be a point on the line 2x – 3y – 12 = 0 and such that |PA – PB| is maximum. Then P is

(A) (12,–2)	**(B)** (–2, 12)
(C) (–12,–12)	**(D)** (12, 12)

Q353. *CUSAT CAT 2024 - Shift 4*

Let f(x) and g(x) be two functions such that g(x)= x– and f g(x)= x 3 – . x x 3 Then f '(x) is

(A) x 2 +3 1	**(B)** +2 2 x
(C) 3x 2 +3	**(D)** 3x 2 –3 1( ) 1( )

Q354. *CUSAT CAT 2024 - Shift 4*

Let AB be a diameter of a circle of radius 5 cm. Let P and Q be the two points on the circle so that the length of PB is 6 cm, and the length of AB is twice that of AQ. Then the length, QB is nearest to

(A) 9.1 cm	**(B)** 7.8 cm
(C) 9.3 cm	**(D)** 8.5 cm { } { } 2 2 2 2

Q355. *CUSAT CAT 2024 - Shift 4*

If A= (x,y):x + y =25 and B= (x,y):x +9y =144 then number of points in A∩B is

(A) 1	**(B)** 2
(C) 4	**(D)** 3 2 2 2

Q356. *CUSAT CAT 2024 - Shift 4*

The number of points where the curve f(x)=e –e –3e –e +1, x∈Rcuts x-axis, is equal to

(A) 0	**(B)** 2
(C) 1	**(D)** 8 –2

Q357. *CUSAT CAT 2024 - Shift 4*

If a = then a +a +...+a is equal to n 2 1 2 25 4n –16n+15 51

(A) 144 48	**(B)** 138 50
(C) 141 52	**(D)** 147 2 2

Q358. *CUSAT CAT 2024 - Shift 4*

Let f(x)=2x +λ, λ∈R, n∈N, and f(4)=133, f(5)=255. Then f(2)– f(1)is equal to

(A) 60	**(B)** 15
(C) 12	**(D)** 14

Q359. *CUSAT CAT 2024 - Shift 4*

The solution of equation z –z =1+2i is 3

(A) +2i 2 3	**(B)** –2i 2
(C) 3–2i	**(D)** 3+2i 1 1 1 1 1 1

Q360. *CUSAT CAT 2024 - Shift 4*

lim 22 –2322 –25...22 –22n+1 is equal to n→∞

(A) 0 1	**(B)** 2
(C) 2	**(D)** 1

Q361. *CUSAT CAT 2024 - Shift 4*

The product (16)(16) 3 (16) 9...∞ is equal to

(A) 64	**(B)** 16
(C) 32	**(D)** 0

Q362. *CUSAT CAT 2024 - Shift 5*

The sum of the factors of 8! which are odd and are the form 3m+2, where m is a natural number is

(A) 8	**(B)** 35
(C) 40	**(D)** 45 2 2x n(n+1) 2x

Q363. *CUSAT CAT 2024 - Shift 5*

If x <1,then 1+n + +... is equal to 1+x 2 1+x n 2x

(A) 1+x n 1+x	**(B)** 1–x n 1+x
(C) 2x n 1–x	**(D)** 1+x 1 2 –1

Q364. *CUSAT CAT 2024 - Shift 5*

If A= –1 1 2 , then det (adj (adj

(A))) = 2 –1 1	**(B)** 142
(C) 143	**(D)** 144

Q365.

CUSAT CAT 2024 - Shift 5

If the system of equations ax+ y =3, x+2y =3, 3x+4y =7 is consistent, then the value of a is equal to

(A) 0	**(B)** –1
(C) 1	**(D)** 2 2 1 2 2

Q366.

CUSAT CAT 2024 - Shift 5

Sum of the series 1+ + + +...+∞ is 2! 3! 4!

(A) e2	**(B)** 1(e2 +1) 2
(C) 1(e2 +1) 4	**(D)** e2 –1 1–2x–x2

Q367.

CUSAT CAT 2024 - Shift 5

If a+b+c=6, then 4a+1+ 4b+1+ 4c+1

(A) ≤9	**(B)** ≥9
(C) >9	**(D)** <9

Q368.

CUSAT CAT 2024 - Shift 5

For a,b,x,y∈R, the correct statement is

(A) a >b⇒ax>bx;x≠0	**(B)** x > y ⇒ x> y 1 1
(C) a >b⇒ > a b	**(D)** a>b⇒a+c>b+c

Q369.

CUSAT CAT 2024 - Shift 5

If (a+ib)5 =α+iβ, then (b+ia)5 is equal to

(A) β+iα	**(B)** α–iβ
(C) β–iα	**(D)** –α–iβ 5 5 3+i 3–i

Q370.

CUSAT CAT 2024 - Shift 5

If z = + , then 2 2

(A) Re(z)=0	**(B)** Im(z)=0
(C) Re(z)>0,Im(z)>0	**(D)** Re(z)>0,Im(z)<0

Q371.

CUSAT CAT 2024 - Shift 5

If z = x+iy and 'a' is a real number such that z–ai = z+ai , then the locus of z is

(A) x-axis	**(B)** y-axis
(C) x= y	**(D)** x2 + y2 =1 n

Q372. *CUSAT CAT 2024 - Shift 5*

If (1+x) (1+x2)(1+x4) ... (1+x128) = ∑ xr , then n is r=0

(A) 255	**(B)** 127
(C) 60	**(D)** 90

Q373. *CUSAT CAT 2024 - Shift 5*

If x∈R, the number of solutions of x+1– x–1=1, is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q374. *CUSAT CAT 2024 - Shift 5*

If roots of x2 –(k–3)x+k =0 are such that both of them are greater than 2, then

(A) k∈[7,9]	**(B)** k∈[7,∞)
(C) k∈[9,10)	**(D)** k∈[7,9)

Q375. *CUSAT CAT 2024 - Shift 5*

In ∆ABC, let AB=6 3 cm, AC =12cm and BC =6 cm. Then the angle of B is

(A) 120°	**(B)** 60°
(C) 90°	**(D)** 45° 2 y +2y–11

Q376. *CUSAT CAT 2024 - Shift 5*

For a real y, the expression takes all real values except those which lie y–3 between m and n, then m and n are

(A) 1, 2	**(B)** 4, 12
(C) 4,–12	**(D)** –1,4 3 7 15 31

Q377. *CUSAT CAT 2024 - Shift 5*

The sum of first n terms of the series 1+ + + + +... is equal to 2 4 8 16 1

(A) 2n– n 2 1	**(B)** 2+ n 2 1 n
(C) 2 – n 2 1	**(D)** 2(n–1)+ n–1 2 2 2 2 2

Q378. *CUSAT CAT 2024 - Shift 5*

Which of the following functions from Z to Z is a bijection?

(A) f(x)= x 3	**(B)** f (x) = x+3
(C) f(x) =3x+1	**(D)** f(x)= x 2 +3

Q379. *CUSAT CAT 2024 - Shift 5*

If A and B are two independent events of an experiment such that P(A∪B)= 0.6 and P

(A) = 0.3, then P	**(B)** = 1
(C) 10 2	**(D)** 10 2 2

Q380. *CUSAT CAT 2024 - Shift 5*

Let t = . Then t is equal to n n (n+1)! n=1 20!–1

(A) 20! 21!–1	**(B)** 21! 1
(C) 2(n–1)! 1	**(D)** 2(n+1)! ∂z x y z

Q381. *CUSAT CAT 2024 - Shift 5*

If x y z =a, then is equal to ∂x 1+logx

(A) – 1+logz 1+logx	**(B)** 1+logz 1–logx
(C) – 1–logz 1–logx	**(D)** 1–logz x+1 x –1 –1

Q382. *CUSAT CAT 2024 - Shift 5*

If a and b are roots of the equation y2 –2y+4=0. Then a6 +b3 = A) 60

(A) —	**(B)** 56
(C) 128	**(D)** 256

Q383. *CUSAT CAT 2024 - Shift 5*

Let f(x)= x3 –3, where [.] denotes the greatest integer function. Then the number of points in the interval (1, 2) where f (x) is discontinuous, is

(A) 4	**(B)** 2
(C) 6	**(D)** 5

Q384. *CUSAT CAT 2024 - Shift 5*

If f(x) is an odd periodic function with period 2, then f(4) equals

(A) –4	**(B)** 4
(C) 2	**(D)** 0

Q385. *CUSAT CAT 2024 - Shift 5*

If A={x is a prime number and<30}, then the number of rational numbers whose numerator and denominator belong to A is

(A) 10P 2	**(B)** 10P +1 2
(C) 10P –1 2	**(D)** 10P +2 2

Q386. *CUSAT CAT 2024 - Shift 5*

The number of non negative integer solutions of x + y + z = 10 is

(A) 12C 2	**(B)** 12C 4
(C) 10C 3	**(D)** 11C 3

Q387. *CUSAT CAT 2024 - Shift 5*

If a =99100 and b=10099, then

(A) a > b	**(B)** a = b
(C) a < b 100	**(D)** a + 2b = 101

Q388. *CUSAT CAT 2024 - Shift 5*

The points A(2, 3), B(3, 5), C(7, 7) and D(4, 5) are such that

(A) ABCD is a parallelogram	**(B)** A, B, C, D are colinear
(C) D lies inside the triangle ABC	**(D)** D lies on the boundary of the triangle ABC

Q389. *CUSAT CAT 2024 - Shift 5*

If p and q are distinct primes and x – px + q = 0 has distinct positive integral roots, then p + q equals,

(A) 5	**(B)** 7
(C) 19	**(D)** 40 2

Q390. *CUSAT CAT 2024 - Shift 5*

Let p be a prime number such that p ≥ 23. Let n = p! + 1. The number of primes in the list n+1,n+2,n+3,...,n+ p–1 is

(A) p–1	**(B)** 2
(C) 1	**(D)** 0

Q391. *CUSAT CAT 2024 - Shift 5*

If the slope of one of the lines given by 36x2 +2hxy+72y2 =0 is four times the other, then the value of h2 is equal to

(A) 4050	**(B)** 4020
(C) 3040	**(D)** 3050

Q392. *CUSAT CAT 2024 - Shift 5*

If F =(3,0),F =(–3,0) and P is any point on the curve 16x2 +25y2 =400, then 1 2 PF +PF equals 1 2

(A) 8	**(B)** 6
(C) 10	**(D)** 12

Q393. *CUSAT CAT 2024 - Shift 5*

If α and β are the roots of x2 +x+1=0, then α2020 +β2020 is

(A) –2	**(B)** –1
(C) 1	**(D)** 2 Subject Name: 101 B TECH 12 MAY 2024 - S2 Subject Name: 101 B TECH 12 MAY 2024 - S2

Q394. *CUSAT CAT 2025 - Shift 1*

The possible values of x that satisfies the equation 0 x–3 3 =0 are 0 0 x–5

(A) 1, 2, 3	**(B)** 3, 4, 6
(C) 2, 5, 7	**(D)** 1, 3, 5 x

Q395. *CUSAT CAT 2025 - Shift 1*

Let f :■ →■ be defined by f(x)= , x≠0. Then x

(A) f is continuous in[0,∞)	**(B)** f is continuous
(C) f is continuous in(–1,1)	**(D)** f is not continuous in (–∞,0]

Q396. *CUSAT CAT 2025 - Shift 1*

The local minimum of the function f(x)= x –18x is attained at x=

(A) –3	**(B)** 27
(C) 3	**(D)** –27

Q397. *CUSAT CAT 2025 - Shift 1*

The last digit of M =1!+2!+3!+...2025! is

(A) 0	**(B)** 1
(C) 2	**(D)** 3 x x

Q398. *CUSAT CAT 2025 - Shift 1*

For real numbers a and b, define a◊b=(a–b) 2 . Then (x– y) 2◊(y–x) 2 is equal to

(A) 0	**(B)** 2x 2
(C) 2y 2	**(D)** xy

Q399. *CUSAT CAT 2025 - Shift 1*

The number of values of ‘a’ such that the equation x + y +4ax–2ay+a =0 represents a real circle is

(A) 0	**(B)** 1
(C) 2	**(D)** infinite 3 6

Q400. *CUSAT CAT 2025 - Shift 1*

Which of the following is TRUE?

(A) There exists two distinct irrational numbers such that their sum is rational	**(B)** There exists two distinct irrational numbers such that their difference is rational
(C) There exists two distinct irrational numbers such that their product is rational	**(D)** All the above

Q401. *CUSAT CAT 2025 - Shift 1*

Let S be the set of all points at which the function f(x)= is –1, x is irrational continuous. Then S is equal to

(A) ■	**(B)** [–1,1]
(C) [0,∞]	**(D)** ϕ

Q402. *CUSAT CAT 2025 - Shift 1*

Suppose a circle passes through (0, 2) and (3, 5) and touches the y-axis at Q. If O is the origin, then OQ is equal to

(A) 5	**(B)** 4
(C) 3	**(D)** 2

Q403. *CUSAT CAT 2025 - Shift 1*

If α,β are the roots of the equation x 2 –2x+4=0, then the value of α 5 +β 5 is

(A) 32	**(B)** 64
(C) 128	**(D)** 256 z+2i

Q404. *CUSAT CAT 2025 - Shift 1*

If Im =0, then z lies on the curve z+2

(A) x 2 + y 2 +2x+2y =0	**(B)** x 2 + y 2 –2x=0
(C) x+ y+2=0	**(D)** x 2 + y 2 –2y =0

Q405. *CUSAT CAT 2025 - Shift 1*

The equation zz +(1–4i)z+(1+4i)z +1=0 represents a circle of radius

(A) 3	**(B)** 4
(C) 5	**(D)** 6

Q406. *CUSAT CAT 2025 - Shift 1*

log 2,log 2,log 2 are in 3 6 12

(A) AP but not in GP	**(B)** GP but not in AP
(C) HP	**(D)** Both AP and GP 2 2 3

Q407. *CUSAT CAT 2025 - Shift 1*

The sum of the series (1+3)+(1+3+3)+(1+3+3 +3)+... up to n terms is 3 n+2

(A) 3 –n– 4 n 3 n+2	**(B)** 3 – – 2 4 n n+1
(C) 3 – 4	**(D)** 3 n+1 –1 1 2

Q408. *CUSAT CAT 2025 - Shift 1*

If x=3+33 +33, then the value of x 3 –9x 2 +18x is

(A) 3	**(B)** 16
(C) 21	**(D)** 12

Q409. *CUSAT CAT 2025 - Shift 1*

Number of divisors of the form 4n+2 (n≥0) of the integer 280 is

(A) 3	**(B)** 4
(C) 5	**(D)** 6 3 4 7

Q410. *CUSAT CAT 2025 - Shift 1*

The system of linear equations 4x+3y+2z =7 has 5x+5y+3z =8

(A) infinite number of solutions	**(B)** exactly 3 solutions
(C) a unique solution	**(D)** no solution 1 1 1

Q411. *CUSAT CAT 2025 - Shift 1*

The value of bc ca ab is b+c c+a a+b

(A) 1	**(B)** 0
(C) (a–b)(b–c)(c–a)	**(D)** (a+b)(b+c)(c+a) ∞ 1 m ∑ ∑ n–1

Q412. *CUSAT CAT 2025 - Shift 1*

3 is equal to m=1 m!n=1

(A) e 3 e +e	**(B)** 2
(C) e 3 3 e –e	**(D)** 2

Q413. *CUSAT CAT 2025 - Shift 1*

Let A and B be the sets of all positive divisors of 400 and 1000 respectively including 1. Then the number of elements in A ∩ B is

(A) 4	**(B)** 6
(C) 8	**(D)** 12

Q414. *CUSAT CAT 2025 - Shift 1*

lim is equal to x→0 x 4

(A) 0	**(B)** 1 1
(C) 6 1	**(D)** – 6 x 3

Q415. *CUSAT CAT 2025 - Shift 1*

If 2–i is a root of the equation ax 2 +12x+b–0 (where a and b are real), then the value of ab is equal to

(A) 45	**(B)** 15
(C) –15	**(D)** –45

Q416. *CUSAT CAT 2025 - Shift 1*

For a,b∈R, the equation x 2 +(a–b)x+(1–a–b)=0 has unequal roots for all values of b. Then

(A) a<1	**(B)** a>1
(C) a=2	**(D)** a is real

Q417. *CUSAT CAT 2025 - Shift 1*

A gentle man has 6 friends to invite. In how many ways can he send invitation cards to them, if he has three servants to carry the cards?

(A) 36	**(B)** 63
(C) 8	**(D)** 38

Q418. *CUSAT CAT 2025 - Shift 1*

Define x ■ y to be |x – y| for all real numbers x and y. The value of (1 ■ (2 ■ 3)) – ((1 ■ 2) ■ 3) is

(A) –2	**(B)** –1
(C) 0	**(D)** 1 (x) log (3–x)

Q419. *CUSAT CAT 2025 - Shift 1*

The number of solutions of log 9–2 =10 10 is 2

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q420. *CUSAT CAT 2025 - Shift 1*

If a, b, c are distinct positive real numbers and a2 + b2 + c2 = 1 , then ab + bc + ca is

(A) less than 1	**(B)** equal to 1
(C) greater than 1	**(D)** any real number πx 2

Q421. *CUSAT CAT 2025 - Shift 1*

For all integers n≥9, the value of is always a n!

(A) multiple of 4	**(B)** multiple of 10
(C) prime number	**(D)** perfect square

Q422. *CUSAT CAT 2025 - Shift 1*

In the value of 100!, the number of zeros at the end, is

(A) 11	**(B)** 22
(C) 23	**(D)** 24 n n n

Q423. *CUSAT CAT 2025 - Shift 1*

lim + +...+ == n→an 2 +1 n 2 +2 n 2 +n

(A) 0	**(B)** 1
(C) infinite	**(D)** does not exist ( )15 2 x+ 1+x

Q424. *CUSAT CAT 2025 - Shift 1*

The remainder left when 8 2n –(62) 2n+1 is divided by 9 is

(A) 0	**(B)** 2
(C) 7	**(D)** 8 x

Q425. *CUSAT CAT 2025 - Shift 1*

Let f :R→ R be defined by f(x)= ,x∈R, then the range of f is 2 1+x –1 1

(A) , 2 2	**(B)** R–[–1,1] –1 1
(C) R– , 2 2	**(D)** (–1,1)–{0} x y 1 + p q

Q426. *CUSAT CAT 2025 - Shift 1*

If z = x–iy and z3 = p+iq, then = 2 2 p +q

(A) –2	**(B)** –1
(C) 2	**(D)** 1 4 2

Q427. *CUSAT CAT 2025 - Shift 1*

If the circle x + y –6x–8y+ 25–a =0 touches the axis of x, then a equals to

(A) 0	**(B)** ±4
(C) ±2	**(D)** ±3 2

Q428. *CUSAT CAT 2025 - Shift 1*

If a set A has 2 elements and a set B has 3 elements, then how many relations from A to B are possible?

(A) 6	**(B)** 32
(C) 64	**(D)** 9 2

Q429. *CUSAT CAT 2025 - Shift 1*

The number of real roots of the equation x +5 x +6=0 is

(A) 4	**(B)** 3
(C) 2	**(D)** 0

Q430. *CUSAT CAT 2025 - Shift 1*

The ratio in which the line joining (1,2,3) and (4,5,6) divide xy plane is

(A) 2	**(B)** –2 1
(C) 2 1	**(D)** – 2 –1

Q431. *CUSAT CAT 2025 - Shift 1*

is equal to 2

(A) 1	**(B)** 0
(C) 2	**(D)** –1

Q432. *CUSAT CAT 2025 - Shift 2*

The function f(x)= x–3 at x=3 is

(A) not defined	**(B)** differentiable
(C) not continuous	**(D)** not differentiable x+1, x<2

Q433. *CUSAT CAT 2025 - Shift 2*

If f(x)= , then f is 2x–1, x≥2

(A) continuous except x≠ 2	**(B)** continuous only at (–∞,2]
(C) discontinuous in ■	**(D)** continuous in ■

Q434. *CUSAT CAT 2025 - Shift 2*

A function f satisfies f(x)+2f(6–x)= x for all real numbers x. Then the value of f (1) is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q435. *CUSAT CAT 2025 - Shift 2*

If a line of slope π is reflected over y-axis, then the slope of the reflected line is

(A) π	**(B)** –π π
(C) 2 π	**(D)** – 2 2 4–x

Q436. *CUSAT CAT 2025 - Shift 2*

The largest possible domain of the real valued function f(x)= is 2 x –9

(A) [3, 4]	**(B)** (3, 4]
(C) [3, 4)	**(D)** (3, 4)

Q437. *CUSAT CAT 2025 - Shift 2*

Let P(x) be a polynomial in real variable x of degree 5. Then lim is n→∞ 2 n

(A) 5	**(B)** 1
(C) 0	**(D)** ∞

Q438. *CUSAT CAT 2025 - Shift 2*

lim + + +...+ = n→∞2 4 8 2 n

(A) 0	**(B)** 1
(C) 2	**(D)** ∞

Q439. *CUSAT CAT 2025 - Shift 2*

Let f(n) defined to be the sum of all prime factors of n, where n is any composite positive integer. Then the range of f is the set of all

(A) positive integers	**(B)** composite positive integers
(C) integers greater than 3	**(D)** integers greater than 4 ∞ π ∑ n

Q440. *CUSAT CAT 2025 - Shift 2*

If p(x) is a polynomial such that the sum of co-efficients of p(–x) is zero, then one of the roots of p(x)=0 is

(A) –1	**(B)** 0
(C) 1	**(D)** 2

Q441. *CUSAT CAT 2025 - Shift 2*

The area enclosed within the curve x + y =2 is

(A) 16 sq units	**(B)** 8 sq units
(C) 6 sq units	**(D)** 4 sq units

Q442. *CUSAT CAT 2025 - Shift 2*

If a,b,c are elements of a group (G,*), then the solution of the equation c*y*a =b in G is

(A) b*a –1 *c –1	**(B)** b*c –1 *a –1
(C) a –1 *b*c –1	**(D)** c –1 *b*a –1

Q443. *CUSAT CAT 2025 - Shift 2*

The radius of the circle =5 is z+i 20

(A) 81 40	**(B)** 81
(C) 4	**(D)** 484 1 2

Q444. *CUSAT CAT 2025 - Shift 2*

The maximum sum of the series 30+29 +28 +28+... is 3 3

(A) 690	**(B)** 560
(C) 430	**(D)** 360 2 3 4 n n+6

Q445. *CUSAT CAT 2025 - Shift 2*

The positive integer n for which 2×2 +3×2 +4×2 +...+n×2 =2 , is

(A) 17	**(B)** 21
(C) 31	**(D)** 33

Q446. *CUSAT CAT 2025 - Shift 2*

The number of points (a, b, c) in space, whose each coordinate is a negative integer such that a + b + c + 12 = 0 is

(A) 162	**(B)** 18
(C) 136	**(D)** 153 2x+1 2 x+3 x

Q447. *CUSAT CAT 2025 - Shift 2*

The roots of the equation 4 +4 =4 +4 are

(A) 1, –2	**(B)** 1, 2
(C) –1, 2	**(D)** –1, –2 10 r

Q448. *CUSAT CAT 2025 - Shift 2*

The value of , when the numerator and the denominator take their greatest 11 r value, is 6

(A) 11 5	**(B)** 11 10
(C) 6 10	**(D)** 5 6 { 6 7 15}

Q449. *CUSAT CAT 2025 - Shift 2*

Let N be the set of all natural numbers. IfaN ={an:n∈N} and bN ∩cN =dN, where a,b,c∈N and b,c are coprimes, then

(A) b = cd	**(B)** c = bd
(C) d = bc	**(D)** a = bc

Q450. *CUSAT CAT 2025 - Shift 2*

The equation of the straight line which is parallel to x-axis and crosses the curve y = x at an angle of 45°, is 1

(A) x= 2 1	**(B)** y = 4 1
(C) y = 2	**(D)** y =1 dy y x

Q451. *CUSAT CAT 2025 - Shift 2*

If f :■ → ■ is defined by f(x)= x–[x]– for x∈■ where [x] is the greatest 4 3 integer not exceeding x , then x∈■: f(x)= is equal to 4

(A) ■, the set of all integers	**(B)** ■, the set of all natural numbers
(C) ϕ, the empty set	**(D)** ■, the set of all real numbers x –x 5 –5

Q452. *CUSAT CAT 2025 - Shift 2*

The value of lim is x→0 2x

(A) log5	**(B)** 0
(C) 1	**(D)** 2log5 1 1+2 1+2+3

Q453. *CUSAT CAT 2025 - Shift 2*

If a ,a ,...a are n distinct odd natural numbers not divisible by any prime greater 1 2 n 1 1 1 than 5, then + +...+ is less than a a a 1 2 n 15

(A) 8 16	**(B)** 8 8
(C) 15 15	**(D)** 4 x

Q454. *CUSAT CAT 2025 - Shift 2*

The number of real solutions of the equation e = x is

(A) 0	**(B)** 1
(C) 2	**(D)** infinite x+λ x x

Q455. *CUSAT CAT 2025 - Shift 2*

If f(x)= x x+λ x , then f(3x)– f(x) is equal to x x x+λ

(A) 3xλ 2	**(B)** 6xλ 2
(C) xλ 2	**(D)** 4xλ 2 2

Q456. *CUSAT CAT 2025 - Shift 2*

The minimum number of terms of the series 1 + 3 + 9 + 27 + ... so that the sum may exceed 1000 is

(A) 7	**(B)** 5
(C) 3	**(D)** 8 4

Q457. *CUSAT CAT 2025 - Shift 2*

If 3a–2b+5c=0, the family of lines ax+by+c=0 are always concurrent at a point whose coordinates are –3 2

(A) , 5 5 3 –2	**(B)** , 5 5 3 2
(C) , 5 5 –3 –2	**(D)** , 5 5

Q458. *CUSAT CAT 2025 - Shift 2*

For each n∈■, 2 3n –1 is divisible by

(A) 6	**(B)** 16
(C) 8	**(D)** 7

Q459. *CUSAT CAT 2025 - Shift 2*

The sum of all possible values of k for which the polynomials x2 – 3x + 2 and x2 – 5x + k have a root in common is

(A) 7	**(B)** 8
(C) 9	**(D)** 10

Q460. *CUSAT CAT 2025 - Shift 2*

If a, b, c ∈■ and quadratic equation x2 + (a + b)x + c = 0 has no real roots, then

(A) c(a + b + c) > 0	**(B)** c + c(a + b + c) > 0
(C) c + c(a + b – c) > 0	**(D)** c(a + b – c) > 0 π

Q461. *CUSAT CAT 2025 - Shift 2*

There is a positive integer n such that (n + 1)! + (n + 2)! = n! · 440 . The sum of the digits of n is

(A) 3	**(B)** 8
(C) 10	**(D)** 11

Q462. *CUSAT CAT 2025 - Shift 2*

The number of solutions of the equation 4e –3e +4=0 in [0,2π] is

(A) 1	**(B)** 2
(C) 3	**(D)** 4

Q463. *CUSAT CAT 2025 - Shift 2*

The image of the point (–1, 3, 4) in the plane x–2y =0 is 9 –13

(A) , ,4 5 5 –17 –19	**(B)** , ,1 3 3
(C) (8,4,4) –17 –19	**(D)** , ,4 3 3 π 3π π 3π

Q464. *CUSAT CAT 2025 - Shift 2*

If a and b are real numbers such that (2+α) =a+bα, where α= , then 2 a+b=

(A) 9	**(B)** 24
(C) 33	**(D)** 57

Q465. *CUSAT CAT 2025 - Shift 2*

If the sum of the coefficients of all even powers of x in the product (2 2n)(2 3 2n) 1+x+x +...x 1–x+x –x +...+x is 61, then n is equal to

(A) 30	**(B)** 60
(C) 61	**(D)** 122 2 2 2 2 x y x y 1

Q466. *CUSAT CAT 2025 - Shift 2*

If –2<2x–1<2, then the value of x lies in the interval 1 3

(A) , 2 2 –1 3	**(B)** , 2 2 3 1
(C) , 2 2 3 –1	**(D)** , 2 2 2

Q467. *CUSAT CAT 2025 - Shift 2*

The value of lim is x→0 x 2

(A) 2	**(B)** 1
(C) 4	**(D)** limit does not exist 13π –1

Q468. *CUSAT CAT 2025 - Shift 2*

The principal argument of z =– is 1+ 3i π

(A) 3 2π	**(B)** 3 π
(C) 6 π	**(D)** 4

Q469. *CUSAT CAT 2025 - Shift 2*

There is an equilateral triangle with each side of length 24 cm. The mid-point of its sides is joined to form another triangle. This process is going continuously infinitely many times. Then the sum of perimeter of first 6 triangles (in cm) is

(A) 144	**(B)** 120
(C) 139.5	**(D)** 141.75

Q470. *CUSAT CAT 2025 - Shift 3*

The function f(x)= is continuous in the interval x

(A) (0,∞)	**(B)** (–∞,0)
(C) [–1,1]	**(D)** (–∞,∞)

Q471. *CUSAT CAT 2025 - Shift 3*

Let f be a continuous function on [0, 4]. If f(x) takes rational values for all x and f(4)=2025, then f(π) is

(A) 0	**(B)** 2025
(C) π	**(D)** 1

Q472. *CUSAT CAT 2025 - Shift 3*

The equation 3x +5x –7 =0 has

(A) exactly one real root	**(B)** more than one real root
(C) exactly 13 real roots	**(D)** no real root

Q473. *CUSAT CAT 2025 - Shift 3*

Let a,b,c and d be distinct integers such that (6–a)(6–b)(6–c)(6–d)=36. Then a+b+c+d is equal to

(A) 23	**(B)** 24
(C) 13	**(D)** 14

Q474. *CUSAT CAT 2025 - Shift 3*

The number of transition points for the function f(x)=e is

(A) ∞	**(B)** 2
(C) 1	**(D)** 0

Q475. *CUSAT CAT 2025 - Shift 3*

The least number of positive integers ‘n’ satisfying n n n... =nis

(A) 0	**(B)** 1
(C) 2	**(D)** ∞

Q476. *CUSAT CAT 2025 - Shift 3*

Let G ={(a,b)/a∈■*,b∈■} and be defined as (a, b) ♦ (c,d)=(ac,bc,d). Then which of the property that G does not satisfy under the operation ♦?

(A) Associative	**(B)** Existence of identity
(C) Existence of inverse	**(D)** Commutative

Q477.

CUSAT CAT 2025 - Shift 3

The question paper in a test consists of 15 questions. In how many ways the test can be attempted if each question has four choices?

(A) 154	**(B)** 15
(C) 415	**(D)** 1

Q478.

CUSAT CAT 2025 - Shift 3

The derivative of f(x)= x exists in

(A) ■	**(B)** ■–{0}
(C) (0,∞)	**(D)** [0,∞)

Q479.

CUSAT CAT 2025 - Shift 3

The solution set of log 9–5x >2 is 3 18

(A) ,∞ 5 3 18	**(B)** , 5 5 18
(C) (–∞,0)∪ ,+∞ 5 18	**(D)** –∞,– 5 2

Q480.

CUSAT CAT 2025 - Shift 3

Sum of all terms of an infinite GP is times the sum of odd terms. The common 7 ratio is 6

(A) 7 –1	**(B)** 7 –6
(C) 7 1	**(D)** 7

Q481.

CUSAT CAT 2025 - Shift 3

If x –1 is a factor of x +mx +4x–nk , then

(A) m=4,n=–1	**(B)** m=–4,n=1
(C) m=4,n=1	**(D)** m=–4,n=–1 n P n P n P

Q482.

CUSAT CAT 2025 - Shift 3

If r–1 = r = r+1 , then x y z 1 ∑

(A) =1 x	**(B)** xyz =1
(C) y 2 = x(y+z)	**(D)** x 2 = z(x+ y)

Q483.

CUSAT CAT 2025 - Shift 3

Let X = a:a =1anda is a complex value and { 4 } Y = b:b =1and b is a complex value . Then the set XΔY is equal to

(A) {i,–i}	**(B)** {1,–1,i,–i}
(C) {1,–1}	**(D)** ϕ 1 x

Q484. *CUSAT CAT 2025 - Shift 3*

Let A={1,2,3,4} and r ={(2,2),(3,3),(4,4),(1,2)}. Then this relation R is

(A) reflexive	**(B)** symmetric
(C) transitive	**(D)** anti-symmetric

Q485. *CUSAT CAT 2025 - Shift 3*

The lines x+2y+3=0, x+2y–7=0 and 2x– y–4=0 are the sides of a square. The equation of the remaining side of the square can be

(A) 2x– y+6=0	**(B)** 2x– y+8=0
(C) 2x– y–10=0	**(D)** 2x– y–6=0 2 2

Q486. *CUSAT CAT 2025 - Shift 3*

A polynomial of odd degree with real coefficients must have

(A) at least one real root	**(B)** no real root
(C) only real roots	**(D)** at least one root which is not real

Q487. *CUSAT CAT 2025 - Shift 3*

The sum of the digits in the unit place of all the numbers formed by 3, 4, 5, 8 taken all at a time is

(A) 432	**(B)** 108
(C) 36	**(D)** 18

Q488. *CUSAT CAT 2025 - Shift 3*

Let f be a continuous function on [2, 5] . If f(x) takes rational values for all x and f(4)=8 then f(3.6)=

(A) 0	**(B)** 8
(C) –1	**(D)** 5

Q489. *CUSAT CAT 2025 - Shift 3*

Let M be the least common multiple of all the integers 10 through 30, inclusive. Let N be the least common multiple of M, 32, 33, 34, 35, 36, 37, 38, 39, and 40. What is the N value of ? M

(A) 1	**(B)** 2
(C) 37	**(D)** 74

Q490. *CUSAT CAT 2025 - Shift 3*

The number of ordered pairs (a, b) of integers that the polynomial x3 + ax2 + bx + 6 have 3 distinct integer roots is

(A) 4	**(B)** 5
(C) 6	**(D)** 7

Q491. *CUSAT CAT 2025 - Shift 3*

If lim (1+ax)x =e 4 , where a and b are natural numbers, then x→0

(A) a = 4, b = 2	**(B)** a = 8, b = 4
(C) a = 16, b = 8	**(D)** None of the above

Q492. *CUSAT CAT 2025 - Shift 3*

If AB = A and BA = B , then B2 is equal to

(A) B	**(B)** A
(C) 1	**(D)** 0 5

Q493. *CUSAT CAT 2025 - Shift 3*

The equation of the plane which is parallel to Y- axis and making intercepts of length 3 and 4 on X-axis and Z-axis respectively is

(A) 2x+2z =20	**(B)** 4x+3z=12
(C) 4x–3z=12	**(D)** 46x+13z=15

Q494. *CUSAT CAT 2025 - Shift 3*

The value of is 1–i

(A) –2 15	**(B)** 2 15 i
(C) –2 15 i	**(D)** 6 5 { } { }

Q495. *CUSAT CAT 2025 - Shift 3*

If A= x∈R, x <2 and B= x∈R, x–2 ≥3 , then

(A) A∩B =(–2,–1)	**(B)** B– A= R–(–2,5)
(C) A∪B = R–(–2,5)	**(D)** A–B =[–1,2]

Q496. *CUSAT CAT 2025 - Shift 3*

In how many ways 2 red balls, 3 blue balls and 4 black balls can be arranged if same colour balls are indistinguishable?

(A) 362880	**(B)** 1260
(C) 24	**(D)** 105680 2 2

Q497. *CUSAT CAT 2025 - Shift 3*

What are the critical points of the function f(x)= ? 2 x

(A) 0,1	**(B)** 0,–1
(C) 1,–1	**(D)** 0,1,–1

Q498. *CUSAT CAT 2025 - Shift 3*

Area of the region bounded by the curve y =4x, y-axis and the line y =1 is 1

(A) 3 1	**(B)** 4 1
(C) 12 3	**(D)** 4 x–2 y–3 z+2 x–2 y–3 z+2

Q499. *CUSAT CAT 2025 - Shift 3*

If the lines = = and = = are parallel, then the value k 3 –1 8 6 –2 of k is

(A) –2 1	**(B)** 2
(C) 2	**(D)** 4 (i) (i)

Q500. *CUSAT CAT 2025 - Shift 3*

If a,b,c are in G.P., and ax =b y =cz, then

(A) y= x+z 2	**(B)** y = xz
(C) x+ y+ z =0	**(D)** 2y = x+z a 0 1

Q501. *CUSAT CAT 2025 - Shift 3*

If A= 0 a 1 , then the value of Adj A is 0 0 a

(A) a 3	**(B)** a 9
(C) a 6	**(D)** a 2 3 1 d 2 y dy2

Q502. *CUSAT CAT 2025 - Shift 4*

Which of the following is the equation of a circle?

(A) x 2 +(y+3) 2 +27=0	**(B)** x 2 + y 2 –x+2y+3=0
(C) x 2 + y 2 +x=9	**(D)** x 2 + y 2 + y+3=0 –1

Q503. *CUSAT CAT 2025 - Shift 4*

2020 2019 0 = 2027 2026 1

(A) –5	**(B)** 5
(C) –3	**(D)** 3 2 3

Q504. *CUSAT CAT 2025 - Shift 4*

Which of the following equation is satisfied by A= ? 1 2

(A) A 2 –I =2A	**(B)** A 2 –2A+I =0
(C) A 2 –I =0	**(D)** A 2 –4A+I =0

Q505. *CUSAT CAT 2025 - Shift 4*

The least value of n such that n! is a multiple of 2025 is

(A) 10	**(B)** 25
(C) 81	**(D)** 9 π

Q506. *CUSAT CAT 2025 - Shift 4*

If f,g:■→■ are defined by f(x)= x , g(x)= x 2 , then g f is

(A) x	**(B)** x 2
(C) x	**(D)** 1 2

Q507. *CUSAT CAT 2025 - Shift 4*

The function f defined by f(x)= x is a bijection if the domain and the range are respectively

(A) ■,■	**(B)** ■,(0,∞)
(C) (0,∞),■	**(D)** [0,∞),[0,∞)

Q508. *CUSAT CAT 2025 - Shift 4*

The polynomial f(x)= x +ax +bx +cx+d has real coefficients, and f(3i)= f(3+i)=0. Then the value of a+b+c+d is equal to

(A) 16	**(B)** 25
(C) 36	**(D)** 49 n 1

Q509. *CUSAT CAT 2025 - Shift 4*

lim 1– equals n→∞ n 2

(A) 1 –1	**(B)** e 2
(C) e –2	**(D)** e –1 (2)4

Q510. *CUSAT CAT 2025 - Shift 4*

The total number of proper factors of 2025 is

(A) 13	**(B)** 14
(C) 15	**(D)** 2 dy

Q511. *CUSAT CAT 2025 - Shift 4*

Let A(2, –1) and B(6, 5) be two points. The ratio in which the foot of the perpendicular from (4, 1) divides AB is

(A) 8 : 15	**(B)** 5 : 8
(C) –5 : 8	**(D)** –8 : 5

Q512. *CUSAT CAT 2025 - Shift 4*

A circle touches both the axes. Then the locus of its center is given by

(A) x 2 – y 2 =0	**(B)** x 2 + y 2 =0
(C) x 2 – y 2 =1	**(D)** x 2 + y 2 =1

Q513. *CUSAT CAT 2025 - Shift 4*

The generators of the cyclic group G = 8 |n∈Z are 1

(A) 2 and 2 1	**(B)** 4 and 4 1
(C) 6 and 6 1	**(D)** 8 and 8 a–ib

Q514. *CUSAT CAT 2025 - Shift 4*

The product16(16) 3(16) 9... ∞is equal to

(A) 16	**(B)** 32
(C) 64	**(D)** 0

Q515. *CUSAT CAT 2025 - Shift 4*

The sum of 25 terms of the series 2+ 8+ 18+ 32+... is

(A) 325 2	**(B)** 225 2
(C) 265 2	**(D)** 125 2 2 4 2

Q516. *CUSAT CAT 2025 - Shift 4*

If x –3x+2be one of the factors of the expression x – px +q, then

(A) p=4, q=5	**(B)** p=–5, q=–4
(C) p=5, q=4	**(D)** p=–5, q=4

Q517. *CUSAT CAT 2025 - Shift 4*

How many numbers greater than 2000, but not greater than 3000 can be formed with the digits 0, 1, 2 and 3, where repetition of digits being allowed?

(A) 128	**(B)** 64
(C) 65	**(D)** 52

Q518. *CUSAT CAT 2025 - Shift 4*

The solution set of the equation 2 1 x =0 is 6 7 3

(A) ϕ	**(B)** {0,1}
(C) {1,–1}	**(D)** {1,–3} 2 3 loge loge loge 2 3 4

Q519. *CUSAT CAT 2025 - Shift 4*

loge loge loge is equal to 3 4 5 loge loge loge

(A) 0	**(B)** 1
(C) 4loge	**(D)** 5loge 9 27 81 –1

Q520. *CUSAT CAT 2025 - Shift 4*

If a=1+3+ + + +..., then a is 2! 3! 4!

(A) e –3	**(B)** e 3 1
(C) e3 –1	**(D)** e 3

Q521. *CUSAT CAT 2025 - Shift 4*

If L= lim , then n→∞n n!

(A) L=0	**(B)** L =1
(C) 0< L<1	**(D)** L=∞ π

Q522. *CUSAT CAT 2025 - Shift 4*

If a = b = a+b =1, then a–b is equal to

(A) 1	**(B)** 2
(C) 3	**(D)** 2

Q523. *CUSAT CAT 2025 - Shift 4*

If a+b +c =0 and a =3, b =4 and c = 37,then the angle betweena and b is π

(A) 4 π	**(B)** 2 π
(C) 6 π	**(D)** 3

Q524. *CUSAT CAT 2025 - Shift 4*

If x>0 and log x+log (x) +log (4 x) +log (8 x) +log (16 x) +...=4 then x 2 2 2 2 2 equals

(A) 2	**(B)** 3
(C) 4	**(D)** 5

Q525. *CUSAT CAT 2025 - Shift 4*

If 3f(x)–2f = x, then f '(2)= x 2

(A) 7 1	**(B)** 2
(C) 2 7	**(D)** 2 1

Q526. *CUSAT CAT 2025 - Shift 4*

Which of the following is not divisible by any prime number less than 10 ?

(A) 2606– 1	**(B)** 2606 + 1
(C) 2607– 1	**(D)** 2607 + 1

Q527. *CUSAT CAT 2025 - Shift 4*

If a + b + c = 0 and a, b, c are rational, then the roots of the equation (b + c – a)x2+ (c + a – b)x + (a + b – c) = 0 are

(A) rational	**(B)** irrational
(C) imaginary	**(D)** equal n (2) 2 2n

Q528. *CUSAT CAT 2025 - Shift 4*

If 1–x+x =a +a x+a x +...+a x , then a +a +a +...+a is equal to 0 1 2 2n 0 2 4 2n 1 2

(A) 3n + 2 n 1–3	**(B)** 2 n 3 –1
(C) 2 n 3 +1	**(D)** 2 2 2 x y 2 2

Q529. *CUSAT CAT 2025 - Shift 4*

The ratio of w to x is 4 : 3 , the ratio of y to z is 3 : 2 , and the ratio of z to x is 1 : 6 . Then the ratio of w to y is

(A) 4 : 3	**(B)** 3 : 2
(C) 8 : 3	**(D)** 16 : 3

Q530. *CUSAT CAT 2025 - Shift 4*

The largest term common to the sequences, 1, 11, 21, 31, …..to 100 terms and 31, 36, 41, ….. to 100 terms is

(A) 281	**(B)** 381
(C) 471	**(D)** 521 x a a –x

Q531. *CUSAT CAT 2025 - Shift 4*

If a>0 and lim =–1then a = x→a x x –a a

(A) 0	**(B)** 1
(C) e	**(D)** 2e

Q532. *CUSAT CAT 2025 - Shift 4*

If f(x)+2f =3x, x≠0 andSs={x∈R: f(x)= f(–x)} then the set S x

(A) contains exactly two elements	**(B)** contains more than two elements
(C) is an empty set	**(D)** contains exactly one element 4

Q533. *CUSAT CAT 2025 - Shift 4*

If f(x)= + –1, x∈R then the equation f(x)=0has 5 5

(A) no solution	**(B)** one solution
(C) two solutions	**(D)** more than two solutions 2

Q534. *CUSAT CAT 2025 - Shift 4*

Let f :R→[0,α) be such that lim f(x) exists and lim =0, then x→5 x→5 x–5 lim f(x)= x→5

(A) 0	**(B)** 1
(C) 2	**(D)** 3 n 2n 1 ω ω

Q535. *CUSAT CAT 2025 - Shift 4*

If 1,ω,ω 2 are the cube roots of unity, then Δ= ω n ω 2n 1 = 2n n ω 1 ω

(A) ω 2	**(B)** 0
(C) 1	**(D)** ω

Q536. *CUSAT CAT 2025 - Shift 4*

The number of real values of l for which the system of linear equations 2x+4y–λz =0, 4x+λy+2z =0,λx+2y+2z =0 has infinitely many solutions, is

(A) 0	**(B)** 1
(C) 2	**(D)** 3

Q537. *CUSAT CAT 2025 - Shift 4*

The three points A(1,2,3), B(3,1,2), C(2,3,1) form

(A) an equilateral triangle	**(B)** a right angled triangle
(C) an isosceles triangle	**(D)** a right angled isosceles triangle

Q538. *CUSAT CAT 2025 - Shift 4*

The variance of the first 10 natural numbers is

(A) 7.25	**(B)** 7
(C) 8.25	**(D)** 8 2

Q539. *CUSAT CAT 2025 - Shift 4*

If P

(A) =0.4, P	**(B)** =0.8, P(B| A)=0.6 then P(A∪B) is
(C) 0.48	**(D)** 0.96 z–i

Q540.

CUSAT CAT 2025 - Shift 4

If S =z∈■: ∈■, then z+2i

(A) S contains exactly two elements	**(B)** contains only one element
(C) S is a circle in the complex plane	**(D)** S is a straight line in the complex plane n 5

Q541.

CUSAT CAT 2025 - Shift 4

lim 1+ is equal to n→∞ n

(A) 5loge	**(B)** 5 e
(C) 5e	**(D)** loge 1 1+2 1+2+3

Q542.

CUSAT CAT 2025 - Shift 4

The sum of the series + + +...to ∞equals 1! 2! 3!

(A) e	**(B)** 2e 3e
(C) 2 e	**(D)** 1+ 2 dy

ANSWER KEY

No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key	No.	Key
1	C	2	A	3	D	4	D	5	D	6	C	7	B	8	B	9	B	10	B
11	B	12	C	13	B	14	C	15	A	16	C	17	C	18	A	19	C	20	C
21	A	22	B	23	D	24	C	25	C	26	A	27	B	28	C	29	D	30	C
31	A	32	D	33	B	34	B	35	B	36	D	37	C	38	A	39	C	40	B
41	A	42	A	43	A	44	B	45	B	46	B	47	B	48	C	49	D	50	C
51	B	52	D	53	C	54	D	55	D	56	A	57	A	58	D	59	C	60	B
61	A	62	C	63	D	64	A	65	D	66	D	67	B	68	A	69	B	70	C
71	B	72	B	73	D	74	C	75	B	76	B	77	C	78	C	79	A	80	C
81	A	82	B	83	A	84	C	85	C	86	C	87	A	88	A	89	C	90	A
91	B	92	D	93	B	94	C	95	A	96	B	97	D	98	C	99	B	100	B
101	C	102	B	103	D	104	A	105	D	106	A	107	C	108	C	109	D	110	B
111	A	112	D	113	B	114	B	115	D	116	C	117	A	118	C	119	B	120	C
121	C	122	B	123	D	124	D	125	D	126	A	127	A	128	D	129	B	130	C
131	C	132	C	133	A	134	A	135	D	136	B	137	B	138	C	139	C	140	C
141	C	142	C	143	B	144	C	145	A	146	B	147	B	148	A	149	D	150	D
151	D	152	C	153	A	154	B	155	D	156	A	157	B	158	B	159	D	160	B
161	C	162	B	163	A	164	B	165	A	166	C	167	D	168	D	169	A	170	D
171	A	172	B	173	B	174	C	175	C	176	A	177	C	178	C	179	A	180	D

181	D	182	D	183	A	184	B	185	C	186	C	187	A	188	A	189	D	190	C
191	B	192	D	193	D	194	D	195	B	196	A	197	A	198	C	199	C	200	A
201	C	202	D	203	B	204	A	205	D	206	C	207	C	208	C	209	D	210	C
211	A	212	D	213	B	214	B	215	D	216	A	217	B	218	C	219	D	220	A
221	C	222	D	223	A	224	C	225	B	226	D	227	C	228	B	229	C	230	C
231	B	232	B	233	C	234	D	235	B	236	B	237	A	238	B	239	B	240	B
241	C	242	A	243	A	244	D	245	D	246	C	247	C	248	B	249	B	250	C
251	C	252	C	253	A	254	B	255	D	256	B	257	C	258	D	259	B	260	A
261	B	262	A	263	C	264	B	265	D	266	B	267	B	268	B	269	A	270	A
271	C	272	D	273	A	274	D	275	C	276	A	277	B	278	D	279	B	280	A
281	B	282	B	283	B	284	C	285	A	286	A	287	C	288	B	289	D	290	B
291	B	292	B	293	D	294	C	295	A	296	C	297	B	298	D	299	B	300	D
301	A	302	C	303	C	304	B	305	C	306	A	307	C	308	C	309	D	310	B
311	A	312	A	313	D	314	D	315	B	316	B	317	B	318	A	319	C	320	C
321	B	322	B	323	A	324	A	325	B	326	A	327	C	328	B	329	C	330	D
331	A	332	B	333	B	334	C	335	D	336	D	337	C	338	D	339	D	340	C
341	B	342	B	343	A	344	D	345	B	346	C	347	D	348	D	349	B	350	D
351	C	352	C	353	C	354	A	355	C	356	B	357	C	358	D	359	B	360	A
361	A	362	C	363	B	364	D	365	D	366	C	367	A	368	D	369	A	370	B
371	A	372	A	373	A	374	C	375	C	376	B	377	D	378	B	379	C	380	B
381	A	382	B	383	C	384	D	385	B	386	A	387	A	388	C	389	A	390	D
391	A	392	C	393	B	394	D	395	B	396	C	397	D	398	A	399	A	400	D
401	D	402	C	403	B	404	C	405	B	406	C	407	B	408	D	409	A	410	D
411	C	412	D	413	D	414	C	415	A	416	B	417	A	418	A	419	B	420	A
421	D	422	D	423	B	424	B	425	A	426	A	427	B	428	C	429	D	430	D
431	D	432	D	433	D	434	C	435	B	436	B	437	C	438	B	439	C	440	A
441	B	442	D	443	B	444	A	445	D	446	C	447	C	448	A	449	C	450	C
451	D	452	A	453	A	454	A	455	B	456	A	457	B	458	D	459	D	460	B
461	C	462	B	463	A	464	A	465	A	466	B	467	B	468	B	469	D	470	D

471	B	472	A	473	B	474	D	475	D	476	D	477	C	478	C	479	C	480	C
481	B	482	C	483	A	484	C	485	A	486	A	487	B	488	B	489	D	490	B
491	D	492	A	493	B	494	C	495	B	496	B	497	C	498	C	499	D	500	D
501	C	502	C	503	A	504	D	505	A	506	B	507	D	508	D	509	A	510	A
511	B	512	A	513	D	514	C	515	A	516	C	517	B	518	D	519	A	520	A
521	A	522	C	523	D	524	C	525	B	526	C	527	A	528	D	529	D	530	D
531	B	532	A	533	B	534	D	535	B	536	B	537	A	538	C	539	D	540	D
541	B	542	C																